



PROJECT MANUAL:

# RUNWAY 8-26 REHABILITATION

City of Pensacola Bid No. 25-029

PREPARED FOR:



MARCH 2025  
BID DOCUMENTS

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**TABLE OF CONTENTS**

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**TITLE** **INDEX OF PAGES**

---

Table of Contents..... TOC-1-3

**Bid Information**

Invitation to Bid ..... INV-1

- Appendix A, 44 C.F.R. Part 18 – Certification Regarding Lobbying -  
Certification for Contracts, Grants, Loans, and Cooperative Agreements
- Bidders List Collection Form
- State of Florida Affidavit Regarding the Use of Coercion for Labor and  
Services Pursuant to Florida Statute 787.06(13)
- 52.209-5 FAR Certification Regarding Debarment, Suspension, Proposed  
Debarment, and Other Responsibility Matters
- 52.209-6 FAR Protecting the Government's Interest When Subcontracting  
with Contractors Debarred, Suspended, or Proposed for Debarment
- City of Pensacola Certification for Erosion And Sedimentation  
Compliance
- Statement of No Proposal
- Purchasing General Conditions

Instructions to Bidders..... ITB-1

**Proposal Information**

Proposal Form ..... PF-1

Proposal Affidavit ..... PF-5

Proposal Bond ..... PF-7

Surety's Bond Affidavit ..... PF-9

EEO Report Statement ..... PF-11

Certification of Nonsegregated Facilities ..... PF-13

Sworn Statement on Public Entity Crimes ..... PF-15

Non-Collusion Affidavit ..... PF-19

Drug Free Workplace Certificate ..... PF-21

List of Anticipated Subcontractors ..... PF-23

Statement of Bidder's Qualifications ..... PF-25

Disadvantaged Business Enterprises (DBE) Plan ..... DBE-1

DBE Subcontractor Solicitation ..... DBE-3

DBE Intent to Perform as a Subcontractor ..... DBE-5

DBE Utilization ..... DBE-7

DBE Letter of Intent ..... DBE-9

DBE Exhibit A ..... DBE-11

Insurance Requirements for the Project ..... INS-1

Bid Schedule ..... BS-1

**Contract Forms**

|                               |       |
|-------------------------------|-------|
| Sample Agreement .....        | CF-1  |
| Performance Bond .....        | CF-7  |
| Payment Bond .....            | CF-9  |
| Notice of Award .....         | CF-11 |
| Notice to Proceed .....       | CF-13 |
| Final Release of Claims ..... | CF-15 |

### **Federal Contract Provisions**

|                                                                                                     |       |
|-----------------------------------------------------------------------------------------------------|-------|
| Federal Contract Provisions .....                                                                   | FCP-1 |
| (Including Affirmative Action Requirement, Buy American Preference, Davis-Bacon Requirements, etc.) |       |

|                                                                 |       |
|-----------------------------------------------------------------|-------|
| <b>1 – GENERAL CONTRACT PROVISIONS</b> .....                    | GP 1  |
| Section 10 Definition of Terms .....                            | GP 1  |
| Section 20 Proposal Requirements and Conditions .....           | GP 8  |
| Section 30 Award and Execution of Contract .....                | GP 12 |
| Section 40 Scope of Work .....                                  | GP 14 |
| Section 50 Control of Work .....                                | GP 19 |
| Section 60 Control of Materials .....                           | GP 25 |
| Section 70 Legal Regulations and Responsibility to Public ..... | GP 28 |
| Section 80 Execution and Progress .....                         | GP 35 |
| Section 90 Measurement and Payment .....                        | GP 42 |

|                                                                                                |       |
|------------------------------------------------------------------------------------------------|-------|
| <b>PART 2 – GENERAL CONSTRUCTION ITEMS</b> .....                                               | C-100 |
| Item C-100 Contractor Quality Control Program (CQCP) .....                                     | C-100 |
| Item C-102 Temporary Air and Water Pollution, Soil Erosion, and Siltation Control .....        | C-102 |
| Item C-105 Mobilization .....                                                                  | C-105 |
| Item C-110 Method of Estimating Percentage of Material Within Specification Limits (PWL) ..... | C-110 |

|                                                       |       |
|-------------------------------------------------------|-------|
| <b>PART 3 – SITEWORK</b> .....                        | P-151 |
| Item P-151 Clearing and Grubbing .....                | P-151 |
| Item P-152 Excavation, Subgrade, and Embankment ..... | P-152 |
| Item P-154 Subbase Course .....                       | P-154 |

|                                                |       |
|------------------------------------------------|-------|
| <b>PART 4 –BASE COURSES</b> .....              | P-209 |
| Item P-209 Crushed Aggregate Base Course ..... | P-209 |

|                                                                          |       |
|--------------------------------------------------------------------------|-------|
| <b>PART 5 – FLEXIBLE PAVEMENTS</b> .....                                 | P-401 |
| Item P-401 Asphalt Mix Pavement .....                                    | P-401 |
| Item P-403 Asphalt Mix Pavement Base, Leveling, and Surface Course ..... | P-403 |

|                                                                                            |       |
|--------------------------------------------------------------------------------------------|-------|
| <b>PART 6– MISCELLANEOUS</b> .....                                                         | P-602 |
| Item P-602 Emulsified Asphalt Prime Coat .....                                             | P-602 |
| Item P-603 Emulsified Asphalt Tack Coat .....                                              | P-603 |
| Item P-605 Joint Sealants for Pavements .....                                              | P-605 |
| Item P-606 Adhesive Compounds, Two-Component for Sealing Wire and Lights in Pavement ..... | P-606 |
| Item P-610 Concrete for Miscellaneous Structures .....                                     | P-610 |
| Item P-620 Runway and Taxiway Marking .....                                                | P-620 |

|                                                                                          |                   |
|------------------------------------------------------------------------------------------|-------------------|
| <b>PART 7 – TURFING .....</b>                                                            | <b>T-901</b>      |
| Item T-901 Seeding .....                                                                 | T-901             |
| Item T-904 Sodding .....                                                                 | T-904             |
| Item T-905 Topsoil .....                                                                 | T-905             |
| <br><b>PART 8 – LIGHTING INSTALLATION .....</b>                                          | <br><b>L-108</b>  |
| Item L-108 Underground Power Cable for Airports.....                                     | L-108             |
| Item L-109 Airport Transformer Vault and Vault Equipment .....                           | L-109             |
| Item L-110 Airport Underground Electrical Duct Banks and Conduits.....                   | L-110             |
| Item L-115 Electrical Manholes and Junction Structures .....                             | L-115             |
| Item L-125 Installation of Airport Lighting Systems.....                                 | L-125             |
| <br><b>PART 9 – SPECIAL PROVISIONS .....</b>                                             | <br><b>SP-100</b> |
| Item SP-100 General Provisions and Requirements for Electrical Work.....                 | SP-100            |
| Item SP 104 General Electrical Safety Requirements and Temporary Airfield Lighting ..... | SP-104            |
| Item SP 105 Alterations, Removal and Demolition .....                                    | SP-105            |
| Item SP 106 Submittals, Record Documents and Maintenance Manuals .....                   | SP-106            |
| Item SP-109S Allowances.....                                                             | SP-109S           |
| Item SP-110S Installation of Underground Electrical Directional Bore Duct.....           | SP-110S           |
| Item SP-1111 Airfield Electrical Installation Testing .....                              | SP-111            |

**THE CITY OF PENSACOLA, FLORIDA  
INVITATION TO BID  
BID NO. 25-029**

**RUNWAY 8/26 REHABILITATION AT PENSACOLA INTERNATIONAL AIRPORT**

A sealed, complete hardcopy bid **with original (or electronic) signature**, and **one (1) complete and identical electronic copy (single PDF) on flash drive or CD** will be received on **April 29, 2025, 2:30 P.M.**, local time, at the following location.

**City Hall (lobby)  
222 West Main Street  
Pensacola, Florida, 32502  
Attention: Purchasing**

The OUTER FACE of the sealed submittal package shall **identify the respondent, the bid title, and the bid number** (whether hand-delivered, mailed, or via UPS/Fedex or other courier service). Late submittals will not be accepted. Multiple submittals from the same entity will not be accepted.

Alternatively, electronic responses may be submitted to [Demandstar](#) or [Central Bidding](#). Both sites require registration and varying fees for notifications and document downloads.

NOTE: If submitting an electronic response and a bid/proposal bond is required:

- An insurance company's executed bond **may be included with the electronic submittal**.
- A certified check/cashier's check **must be delivered** to the address above.

Those submittals received will be opened and publicly read the **following business day (April 30, 2025, 10:00 A.M., local time)** via Microsoft Teams at the following link\*: [Microsoft Teams bid reading link](#).

In order to review the project requirements, an **optional** pre-bid meeting will be held at **10:00 A.M., April 11, 2025**, 2025, at the Airport Administration office at Pensacola International Airport, 2430 Airport Boulevard, Suite 225, Pensacola, Florida. The conference is also available via Microsoft Teams at [this link](#)\*. A site tour will follow.

Bidders shall submit a certified check, cashier's check, or insurance company's executed bond, payable to the City of Pensacola in the amount of **five percent (5%) of the base bid** for a period of **one hundred twenty (120) days**.

Specifications will be posted to the City's website at [www.cityofpensacola.com/bids.aspx](http://www.cityofpensacola.com/bids.aspx). Addenda will be posted to the City's website. **Bidders are responsible for obtaining addenda and are advised to check the website frequently**. There is no cost to view or download documents from the City website.

Any questions concerning the bid should be addressed and submitted in writing **no later** than 10:00 A.M., local time, April 17, 2025, to:

**Dedria Lunderman, Purchasing Manager**  
**City of Pensacola**  
**222 W. Main St.**  
**Pensacola, FL 32502**  
[purchasing@cityofpensacola.com](mailto:purchasing@cityofpensacola.com)

A bid tabulation or a Notice of Intent to Award will be posted to the City's website at [www.cityofpensacola.com/bids.aspx](http://www.cityofpensacola.com/bids.aspx).

The City of Pensacola adheres to the Americans with Disabilities Act and will make reasonable accommodations for access to City services, programs, and activities. Please email [ADACoordinator@cityofpensacola.com](mailto:ADACoordinator@cityofpensacola.com) or call (850) 436-5600. Requests must be made at least 48 hours in advance of the event in order to allow the City time to provide the requested services.

The City of Pensacola reserves the right to accept or reject any or all bids, to award bids on a split-order basis by item number when applicable, to waive any bid informalities and to re-advertise for bids when deemed in the best interest of the City of Pensacola.

Attest:  
Ericka L. Burnett  
City Clerk

CITY OF PENSACOLA  
D. C. Reeves  
Mayor

The City of Pensacola provides equal access in employment and public services.

\*Participation in a Microsoft Teams meeting requires a microphone and speakers; however, webcams are optional. Participants may join the meeting either via a PC or Smartphone. Please be sure to check the system requirements at the Microsoft Teams website before the meeting: <https://docs.microsoft.com/en-us/microsoftteams/hardware-requirements-for-the-teams-app>.

***Survey: How did you obtain this solicitation?***

☐ City of Pensacola website      ☐ Central Bidding      ☐ Demandstar

Other: \_\_\_\_\_

**SECURITY NOTICE**

Visitors to City Hall are required to stay in the lobby unless otherwise directed.

**Late submittals will not be accepted.**

**APPENDIX A, 44 C.F.R. PART 18 – CERTIFICATION REGARDING LOBBYING**  
**Certification for Contracts, Grants, Loans, and Cooperative Agreements**

The undersigned certifies, to the best of his or her knowledge and belief, that:

1. No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of an agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
2. If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure Form to Report Lobbying," in accordance with its instructions.
3. The undersigned shall require that the language of this certification be included in the award documents for all subawards at all tiers (including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements) and that all subrecipients shall certify and disclose accordingly.

This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by section 1352, title 31, U.S. Code. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

The Contractor, [REDACTED], certifies or affirms the truthfulness and accuracy of each statement of its certification and disclosure, if any. In addition, the Contractor understands and agrees that the provisions of 31 U.S.C. Chap. 38, Administrative Remedies for False Claims and Statements, apply to this certification and disclosure, if any.

\_\_\_\_\_  
Signature of Contractor's Authorized Official

\_\_\_\_\_  
Date

\_\_\_\_\_  
Name and Title of Contractor's Authorized Official

**THIS FORM MUST BE INCLUDED IN SUBMITTAL.**



## Bidder's List Collection Form

The Pensacola International Airport (PNS) is required per 49 CFR 26. 11 (c) to create and maintain a comprehensive Bidders List. The Bidders List Form will be used to determine the relative availability of Disadvantaged Business Enterprise (DBE) and non-DBEs and will assist with establishing the agency's annual DBE goal. Each Bidders List is a compilation of bidders, proposers, quoters, subcontractors, manufacturers, and suppliers of materials and services who have submitted bids during the advertising period of a specific project. Please provide the following mandatory data:

### Part A: Business Data

Business Name: \_\_\_\_\_

Business Address: \_\_\_\_\_  
Street City State/Zip

Name of Contact Person: \_\_\_\_\_

Phone: (    ) \_\_\_\_\_ Fax: (    ) \_\_\_\_\_

Email Address: \_\_\_\_\_

Is this business certified as a Disadvantaged Business Enterprise?    Yes    No

Business Annual Gross Receipts (circle one):

|                     |                            |                            |
|---------------------|----------------------------|----------------------------|
| Less than \$500,000 | \$500,000 to \$1,000,000   | \$1,000,000 to \$3,000,000 |
|                     | \$3,000,000 to \$6,000,000 | Over \$6,000,000           |

Age of Business: \_\_\_\_\_ Years \_\_\_\_\_ Months

Race and gender information for the firm's majority owner (circle one):

|                   |                        |                             |
|-------------------|------------------------|-----------------------------|
| Black American    | Native American        | Subcontinent Asian American |
| Hispanic American | Asian Pacific American | Other                       |

### Part B: Project and Work Description

RFP # \_\_\_\_\_

Project Name: \_\_\_\_\_

NAICS code(s) applicable to each scope of work: \_\_\_\_\_

Provide brief description of scope of work, services, and/or materials to be performed/furnished:

\_\_\_\_\_

Will you subcontract any of your work?    Yes\*    No

(\* If "Yes," the subcontractor(s) must complete an individual Bidders List Form also.)

### Part C: Signature

**The undersigned declares that the information set forth on this page is current, complete, and accurate.**

Authorized Signature: \_\_\_\_\_ Date: \_\_\_\_\_

Printed Name: \_\_\_\_\_ Title: \_\_\_\_\_

**THIS FORM MUST BE INCLUDED IN SUBMITTAL.**

**State of Florida**  
**Affidavit Regarding the Use of Coercion for Labor and Services**  
**Pursuant to Florida Statute 787.06(13)**

Vendor Name: \_\_\_\_\_

Vendor FEIN: \_\_\_\_\_ Phone Number: \_\_\_\_\_

Vendor's Authorized Representative Name and Title: \_\_\_\_\_

Address: \_\_\_\_\_

City: \_\_\_\_\_ State: \_\_\_\_\_ ZIP: \_\_\_\_\_

Email Address: \_\_\_\_\_

Section 787.06(13) Florida Statutes requires all nongovernmental entities executing, renewing, or extending a contract with a governmental entity to provide an affidavit signed by an officer or representative of the nongovernmental entity under penalty of perjury that the nongovernmental entity does not use coercion for labor or services.

The undersigned hereby certifies that he/she is an officer or authorized representative of the vendor identified above and that said vendor does not use coercion for labor or services as defined in Section 787.06 Florida Statutes.

Florida Statute 787.06

(2)(a) "Coercion" means:

1. Using or threatening to use physical force against any person;
2. Restraining, isolating, or confining or threatening to restrain, isolate, or confine any person without lawful authority and against her or his will;
3. Using lending or other credit methods to establish a debt by any person when labor or services are pledged as a security for the debt, if the value of the labor or services as reasonably assessed is not applied toward the liquidation of the debt, the length and nature of the labor or services are not respectively limited and defined;
4. Destroying, concealing, removing, confiscating, withholding, or possessing any actual or purported passport, visa, or other immigration document, or any other actual or purported government identification document, of any person;
5. Causing or threatening to cause financial harm to any person;
6. Enticing or luring any person by fraud or deceit; or
7. Providing a controlled substance as outlined in Schedule I or Schedule II of s. 893.03 to any person for the purpose of exploitation of that person.

(2)(e) "Labor" means work of economic or financial value. (2)(h) "Services" means any act committed at the behest of, under the supervision of, or for the benefit of another. The term includes, but is not limited to, forced marriage, servitude, or the removal of organs

Under penalties of perjury, I declare that I have read the foregoing document and that the facts stated in it are true.

By: \_\_\_\_\_  
AUTHORIZED SIGNATURE

Print Name and Title: \_\_\_\_\_

Date: \_\_\_\_\_

**THIS FORM MUST BE INCLUDED IN SUBMITTAL.**

**52.209-5 FAR Certification Regarding Debarment, Suspension,  
Proposed Debarment, and Other Responsibility Matters**

The Offeror certifies, to the best of its knowledge and belief, that the Offeror and/or any of its Principals:

- A. Are not presently debarred, suspended, proposed for debarment, or declared ineligible for the award of contracts by any Federal agency.
  - B. Have not, within a three-year period preceding this offer, been convicted of or had a civil judgment rendered against them for: commission of fraud or a criminal offense in connection with obtaining, attempting to obtain, or performing a public (Federal, state, or local) contract or subcontract; violation of Federal or state antitrust statutes relating to the submission of offers; or commission of embezzlement, theft, forgery, bribery, falsification or destruction of records, making false statements, tax evasion, or receiving stolen property; and
  - C. Are not presently indicted for, or otherwise criminally or civilly charged by a governmental entity with, commission of any of the offenses enumerated in paragraph 1-B of this provision.
2. The Offeror has not, within a three-year period preceding this offer, had one or more contracts terminated for default by any Federal agency.
- A. "Principals," for the purposes of this certification, means officers; directors; owners; partners; and, persons having primary management or supervisory responsibilities within a business entity (e.g., general manager; plant manager; head of a subsidiary, division, or business segment, and similar positions).

This Certification Concerns a Matter Within the Jurisdiction of an Agency of the United States and the Making of a False, Fictitious, or Fraudulent Certification May Render the Maker Subject to Prosecution Under Section 1001, Title 18, United States Code.

- B. The Offeror shall provide immediate written notice to the Contracting Officer if, at any time prior to contract award, the Offeror learns that its certification was erroneous when submitted or has become erroneous by reason of changed circumstances.
- C. A certification that any of the items in paragraph (a) of this provision exists will not necessarily result in withholding of an award under this solicitation. However, the certification will be considered in connection with a determination of the Offeror's responsibility. Failure of the Offeror to furnish a certification or provide such additional information as requested by the Contracting Officer may render the Offeror nonresponsible.
- D. Nothing contained in the foregoing shall be construed to require establishment of a system of records in order to render, in good faith, the certification required by paragraph (a) of this provision. The knowledge and information of an Offeror is not required to exceed that which is normally possessed by a prudent person in the ordinary course of business dealings.
- E. The certification in paragraph (a) of this provision is a material representation of fact upon which reliance was placed when making award. If it is later determined that the Offeror knowingly rendered an erroneous certification, in addition to other remedies available to the Government, the Contracting Officer may terminate the contract resulting from this solicitation for default.

Company Name: \_\_\_\_\_

Date: \_\_\_\_\_

Authorized

Signature: \_\_\_\_\_ Printed Name: \_\_\_\_\_

**THIS FORM MUST BE INCLUDED IN SUBMITTAL.**

**52.209-6 FAR Protecting the Government's Interest When Subcontracting with Contractors Debarred, Suspended, or Proposed for Debarment**

1. The Government suspends or debarbs Contractors to protect the Government's interests. The Contractor shall not enter into any subcontract in excess of \$25,000 with a Contractor that is debarred, suspended, or proposed for debarment unless there is a compelling reason to do so.
2. The Contractor shall require each proposed first-tier subcontractor, whose subcontract will exceed \$25,000, to disclose to the Contractor, in writing, whether as of the time of award of the subcontract, the subcontractor, or its principals, is or is not debarred, suspended, or proposed for debarment by the Federal Government.
3. A corporate officer or a designee of the Contractor shall notify the Contracting Officer, in writing, before entering into a subcontract with a party that is debarred, suspended, or proposed for debarment (see FAR 9.404 for information on the Excluded Parties List System). The notice must include the following:
  - A. The name of the subcontractor.
  - B. The Contractor's knowledge of the reasons for the subcontractor being in the Excluded Parties List System.
  - C. The compelling reason(s) for doing business with the subcontractor notwithstanding its inclusion in the Excluded Parties List System.
  - D. The systems and procedures the Contractor has established to ensure that it is fully protecting the Government's interests when dealing with such subcontractor in view of the specific basis for the party's debarment, suspension, or proposed debarment.

\_\_\_\_\_  
Company Name

\_\_\_\_\_  
Authorized Signature

\_\_\_\_\_  
Printed Name

\_\_\_\_\_  
Date

**THIS FORM MUST BE INCLUDED IN SUBMITTAL.**

**City of Pensacola  
Florida**

**CERTIFICATION  
for  
EROSION AND SEDIMENTATION COMPLIANCE**

All site excavation and site disturbance shall comply with the following federal, state and local regulations related to erosion and sedimentation:

- A. Federal    Clean Water Act as amended in 1987
- B. State      Florida Statutes, Chapter 373 and 403, and the rules promulgated thereunder
- C. Local      Code of the City of Pensacola, Chapter 12-9

By signature of its undersigned authorized representative, the Bidder hereby assures the City of Pensacola that any soil-disturbing activities performed by the Bidder will comply with all applicable federal, state, and local regulations.

The cost of compliance with applicable erosion and sedimentation regulations is estimated by the Bidder to be \$ \_\_\_\_\_, which cost is included in the amount of the bid.

The specific methods of compliance with applicable federal, state, and local regulations and the associated costs are as follows:

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\_\_\_\_\_  
Authorized Official

**THIS FORM MUST BE INCLUDED WITH SUBMITTAL.**

## STATEMENT OF NO PROPOSAL

If you are unable to submit a proposal at this time, please complete the following and return by e-mail to [purchasing@cityofpenscola.com](mailto:purchasing@cityofpenscola.com). Your choices or comments below will assist us in properly notifying you of future opportunities.

We have declined to respond to **Bid/RFP/RFQ #25-029** for the following reason(s):

### SPECIFICATIONS

☐ Specifications too "tight" (i.e., limited to one brand or manufacturer)

☐ Specifications unclear

☐ Unable to meet specifications

### NATURE OF AWARD

☐ Insufficient time to respond

☐ Equivalent options not offered

☐ Other (please explain below)

Explanation:

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We request to:

☐ remain in this service category for future City of Pensacola solicitations.

☐ be removed from this service category and remain on the City's solicitation list.

☐ be removed from the City's solicitation list.

Company Name: \_\_\_\_\_

Officer Name: \_\_\_\_\_ Title: \_\_\_\_\_

Address: \_\_\_\_\_

City: \_\_\_\_\_ State: \_\_\_\_\_ Zip Code: \_\_\_\_\_

Office: (\_\_\_\_) \_\_\_\_\_ Fax: (\_\_\_\_) \_\_\_\_\_

Email: \_\_\_\_\_

Signature of Officer: \_\_\_\_\_ Date: \_\_\_\_\_

## **PURCHASING GENERAL CONDITIONS**

To ensure acceptance, all bidders submitting bids to the City of Pensacola shall be governed by the following conditions, attached specifications, and bid form(s) unless otherwise specified. Bids not submitted on the bid form(s) provided shall be rejected, and bids not complying with these conditions will be subject to rejection. **Multiple submittals from the same entity will not be accepted.**

1. **Approved Equivalents or Equals:** Any manufacturer's names, trade names, brand names, model numbers, etc. listed in the specifications are for information only and not intended to limit competition. The Bidder may offer any brand for which he is an authorized representative, which meets or exceeds the specifications as written. If the bid is based on an "approved equivalent or equal" item(s) or service(s), supportive information in the form of the manufacturer's printed literature or brochures, sketches, diagrams, and/or complete specifications must accompany the bid. The bidder must explain in detail the reasons why the proposed equivalent or equal will meet specifications and not be considered an exception thereto. The City of Pensacola reserves the right to determine acceptance of proposed equivalent or equal item(s) or service(s).
2. **Award of Bids:** Recommendations for award of bids are made to the Mayor or City Council based on the lowest and best responsible bidder meeting all conditions and requirements of the specifications.
3. **Bid Bond:** The particular item(s) or service(s) outlined within the attached specifications require(s) that a certified check, cashier's check, or insurance company's executed bond made payable to the City of Pensacola accompany your proposal as specified in the solicitation. To ensure its prompt return, please include the company's name and return address on the face of your good faith check or draft. Checks or drafts accepted as good faith deposits will be retained within the City's Finance Department until award and execution of contract is complete, or until a purchase order is issued to the successful proposer. Any proposer withdrawing his proposal after the proposal opening forfeits the right of return of his good faith deposit.
4. **Bid Withdrawal:** No bid may be withdrawn after closing time for receipt of bids for a period of one hundred twenty (120) days thereafter. The contract award shall be legally binding at the time of award by Mayor or City Council.
5. **Delivery:** Bid quotations shall include all freight costs to Pensacola, Florida to a point(s) specified herein or specified at the time the purchase order is placed. No title to the item(s) or service(s) ordered nor any risk of loss shall be passed to the City of Pensacola until after receipt of delivery has been acknowledged by an authorized representative of the City of Pensacola.
6. **Discounts:** Terms offering a discount for prompt payment will be considered in determining the low bid. The discount period shall begin whenever (1) the conditions of the specifications have been fully met and the product or service judged acceptable to the City of Pensacola or (2) a correct invoice and other required documents have been received, whichever is later. Discounts offered for a period of less than thirty (30) days will not be considered in determining low bid.

7. **E-Verify System (Mandatory):** In compliance with the provisions of F.S. 448.095, the parties to this contract and any subcontractors engaged in the performance of this contract hereby certify that they have registered with and shall use the E-Verify system of the United States Department of Homeland Security to verify the work authorization status of all newly hired employees, within the meaning of the statute.
8. **Exceptions to Specifications:** During the drafting of written specifications, a sincere effort is made to describe products and services best suited to the needs of the City; however, in order that consideration be given in evaluating bids, any exceptions to or deviations from the specifications as written must be noted and fully explained. The Mayor is the final authority in determining the acceptability of any exceptions to specifications.
9. **Governing Law:** The laws of the State of Florida shall be the laws applied in the resolution of any action, claim or other proceeding arising out of this contract.
10. **Identical Tie Bids:** In the event that two or more bids are identical in price, preference shall be given to business with Drug-Free Workplace Programs. A Drug-Free Workplace Certificate is enclosed.
11. **Intent of Specifications:** It is the intent of the specifications attached hereto to set forth and describe a certain item(s) or service(s) to be purchased by the City of Pensacola including all materials, equipment, machinery, tools, apparatus, and means of transportation (including freight costs) necessary to provide the item(s) or service(s).
12. **Interpretations:** All questions concerning the specifications or conditions shall be directed in writing to the Purchasing Office at least ten (10) days prior to submittal deadline, unless otherwise instructed on the Invitation to Bid Page. Inquiries must reference the bid item(s) or service(s) and the date of the bid submittal deadline. Interpretations will be made in the form of an addendum placed on the City's website. The City shall not be responsible for any other explanation or interpretation.
13. **Legal Requirements:** All applicable provisions of Federal, State, County, and local laws including all ordinances, rules, and regulations shall govern the development, submittal and evaluation of all bids received in response to these specifications, and shall govern any and all claims between person(s) submitting a bid response hereto and the City of Pensacola, by and through its officers, employees and authorized representatives. A lack of knowledge by the bidder concerning any of the aforementioned shall not constitute a cognizable defense against the legal effect thereof. The Bidder agrees that it will not discriminate on the basis of race, creed, color, national origin, sex, age or disability.
14. **Licenses, Registration and Certificates:** Each bidder shall possess at the time of submitting its bid all licenses, registration and certificates necessary to engage in the business of contracting (or special contracting if the work to be performed necessitates a particular type of specialty contractor) in the City of Pensacola. Bidder must also possess all licenses, registrations and certificates necessary to comply with federal, state and local laws and regulations. The awarded bidder shall be registered at the time of contract execution as an active vendor with the Florida Department of State, Division of Corporations ([www.sunbiz.org](http://www.sunbiz.org)).



- 15. Mistakes:** Bidders are expected to examine the conditions, scope of work, proposal prices, extensions, and all instructions pertaining to the item(s) or service(s) involved. Failure to do so will be at the bidder's risk. Unit prices bid will govern in award.
- 16. Payment of Invoices:** The City of Pensacola issues checks for payment of invoices on the 10th of each month. The signed receiving copy of the purchase order and a correct invoice must have been received by Accounts Payable Activity prior to the 4th of the month. Item(s) or service(s) received on or after the 4th will be processed in the following month. All invoices are payable by the City under the terms of Florida Prompt Payment Act, Florida Statute §218.70. All purchases are subject to availability of funds in the City's budget.
- 17. Permits and Taxes:** The bidder shall procure all permits, pay all charges, fees, and taxes, and give all notices necessary and incidental to the due and lawful prosecution of the work. Bidders who use public roads of the City of Pensacola, Florida for transport of goods of any kind which said goods were transported from a point without the City of Pensacola, Florida to a point within the City of Pensacola shall obtain a "Use of Streets" permit for a fee not in excess of the license paid for by local licensees engaged in the same business.
- 18. Pre-Bid Meetings:** If a bid requires a mandatory pre-bid meeting, any representative of a firm wishing to submit a bid must sign in with the name of the bidding firm.
- 19. Prohibited Conduct by Bidders:** Upon the publication of any solicitation for sealed bids, requests for proposals, requests for qualifications, or other solicitation of interest or invitation to negotiate by any authorized representative of the City of Pensacola, any party interested in submitting a bid, proposal, or other response reflecting an interest in participating in the purchasing or contracting process shall be prohibited from engaging in any communication **pertaining to formal solicitations** with the Mayor, any member of Pensacola City Council or any member of a selection/evaluation committee for RFPs/RFQs, whether directly or indirectly or through any representative or agent, whether in person, by mail, by facsimile, by telephone, by electronic communications device, or by any other means of communication, until such time as the City has completed all action with respect to the solicitation.
- 20. Protests:** Protests of the plans, specifications, and other requirements of bids and requests for proposals must be received in writing by the Purchasing Office at least ten (10) business days prior to the scheduled bid submittal deadline. A detailed explanation of the reason for the protest must be included. Protests of the intended award of bid or contract must be in writing and received in the Purchasing Office within five (5) business days of the notice of intent to award. A detailed explanation of the protest must be included.
- 21. Public Entity Crimes:** By submitting a proposal each proposer is confirming that the company has not been placed on the convicted vendors list as described in Florida Statute §287.133 (2) (a).
- 22. Public Records:** Any material submitted in response to this Invitation to Bid will become a public document pursuant to Florida Statute §119.07. This includes material which the

responding bidder might consider to be confidential or a trade secret. Any claim of confidentiality is waived upon submission, effective after opening the bid pursuant to Florida Statute §119.07.

**23. Public Records Law:** The Parties shall each comply with Florida Public Records laws. The Parties hereby contractually agree that each Party shall allow public access to all documents, papers, letters, or other public records as defined in Chapter 119, Florida Statutes, made or received by either Party in conjunction with this agreement, or related thereto, unless a statutory exemption from disclosure exists. Notwithstanding any provision to the contrary, it is expressly agreed that Contractor's failure to comply with this provision, within seven (7) days of notice from the City, shall constitute an immediate and material breach of contract for which the City may, in the City's sole discretion, unilaterally terminate this agreement without prejudice to any right or remedy.

**24. Rejection of Bids:** The City of Pensacola reserves the right to accept or reject any or all bids, to award bids on a split-order basis by item or service number, to waive any minor bid irregularities, technicalities, or informalities, and to re-advertise for bids when deemed in the best interest of the City of Pensacola.

**25. Sealed Bids:** The specifications and all executed bid forms must be submitted in a sealed envelope. All bids must be signed by an authorized representative of the bidder. In the event more than one bid submittal deadline is scheduled for the same date and time, do not include bids concerning different sets of specifications within the same envelope. **The face of the sealed envelope shall be plainly marked identifying the bidder, the item(s) or service(s) bid and the bid number.** It shall be the sole responsibility of the bidder to assure receipt of bid at the specified location prior to the published time for the bid submittal deadline. No bid will be accepted after closing time for receipt of bids, **nor will any offers by telephone, fax, internet or email be accepted.**

**26. Tax:** The City of Pensacola is exempt from all State and local sales tax.

**27. Unauthorized Aliens:** The City of Pensacola shall consider the employment by any contracted vendor of unauthorized aliens a violation of Section 274A of the Immigration and Nationality Act. Such violation shall be cause for unilateral termination of this contract.

**28. Venue:** Venue for any claim, action or proceeding arising out of this contract shall be Escambia County, Florida.

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| <b>ANY AND ALL SPECIAL CONDITIONS AND SPECIFICATIONS ATTACHED HERETO WHICH VARY FROM THESE GENERAL CONDITIONS SHALL HAVE PRECEDENCE.</b> |
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## **BID INFORMATION INSTRUCTIONS TO BIDDERS**

### **I. GENERAL**

- A.** Funding for this project is being provided by the Federal Aviation Administration (FAA) Airport Improvement Program, the Florida Department of Transportation (FDOT) and City of Pensacola. Therefore, various federal and state statutes will apply to the project. Award of contract will be subject to the approval of the FAA, the FDOT and the City of Pensacola (hereinafter called "Owner" or "City").
- B.** The Bidder covenants and agrees that he and his agents and employees will comply with all municipal, state and federal laws; applicable national and local codes; and Owner rules and regulations applicable to the work to be conducted under this Agreement and that he shall obtain all necessary permits, pay all required fees and taxes, and otherwise perform these services in a legal manner. The Bidder is expected to be familiar with all federal, state and local laws, ordinances, Owner rules and regulations that in any manner affect the work. Ignorance on the part of the Bidder will in no way relieve him from responsibility.
- C.** Bidder certifies that all material, equipment, etc., contained in his proposal meets all OSHA, FAA, FDOT and Owner requirements.

### **II. PREPARATION AND SUBMISSION OF BID PROPOSAL**

- A.** Complete sets of Bid Documents must be used in preparing a bid and may be obtained from the Issuing Office as stated in the "Invitation to Bid". The Owner assumes no responsibility for errors or misinterpretations resulting from the use of incomplete sets of Bid Documents. The Owner in making copies of Bid Documents available does so only for the purpose of obtaining Bids for the Work and does not confer a license or grant for any other use.
- B.** All Bids will remain subject to acceptance for **one hundred twenty (120) calendar days** after the day of the bid opening, but Owner may, in its sole discretion, release any bid and return the bid security prior to that date.
- C.** Sealed Proposals for the construction of the Project will be received until the time and date, and at the location, stated in the "Invitation to Bid."
- D.** The Bidder must submit his Proposal on the forms furnished by the Owner; no other forms will be accepted. All blank spaces in the Proposal forms must be correctly filled in where indicated and the Bidder must state the price(s) (written in ink) both in words and numerals. The words, unless obviously incorrect, will govern.
- E.** When sent by mail, preferably registered, the sealed proposal, marked as indicated above, shall be enclosed in an additional envelope and sent by registered mail with return receipt requested. The Owner will in no way be responsible for delays caused by the U. S. Postal service or any other deliverer of the Proposal, or for delay received on or before the time and at the place designated in the "Invitation to Bid". Proposals received after the specified opening time shall be returned to the Bidder unopened.

- F.** Submitted envelopes shall contain the signed original and two complete copies of all required documents including:
- Proposal Form
  - Proposal Affidavit
  - Proposal Bond
  - Surety's Bond Affidavit
  - Equal Employment Opportunity Report Statement
  - Certification on Non-Segregated Facilities
  - Sworn Statement on Public Entity Crimes
  - Non-Collusion Affidavit
  - Drug Free Workplace Certificate
  - List of Anticipated Subcontractors
  - Statement of Bidder's Qualifications
  - Disadvantaged Business Enterprises (DBE) Plan
  - DBE Subcontractor Solicitation
  - DBE Intent to Perform as a Subcontractor
  - DBE Utilization
  - Letter of Intent
  - DBE Exhibit A
  - Insurance Requirements for the Project
  - Bid Schedule
- G.** Erasures or other changes in a Proposal shall be explained or noted over the initials or signature of the Bidder.
- H.** Proposals containing reservations, conditions, omissions, unexplained erasures or alternations, items not required in the bid or irregularities of any kind may be rejected by the Owner.
- I.** Each proposal shall indicate the full business name and address of the Bidder and shall be signed by him/her with his/her usual signature.
- J.** A proposal submitted by a partnership shall list the names of all partners and shall be signed in the partnership name by one of the members of the partnership.
- K.** A proposal submitted by a corporation shall be executed in the legal name of the corporation and signed by the President or Vice President. The name of each person signing the Proposal shall be typed or printed below the signature.
- L.** When requested by the Owner, a Power of Attorney or other satisfactory evidence of the authority of the officer signing on behalf of the corporation shall be furnished for the Owner's records.
- M.** The Bidder is responsible to supply all information required.
- N.** Successful Bidders will be required to provide verified breakdown of costs of Work in a manner acceptable to the Engineer and Owner.

### III. ADDENDA

- A. Each Bidder shall carefully examine the Bid Documents, and all addenda or other revisions, and thoroughly familiarize himself with the detailed requirements prior to submitting a proposal. Should a Bidder find discrepancies or ambiguities in, or omissions from, the Bid Documents, or should he be in doubt as to their meaning, he shall at once, and in any event, at least ten (10) days prior to submittal deadline, unless otherwise instructed on the Invitation to Bid (INV-1), submit a question in writing. Questions received after this date may not be answered.
- B. All questions about the meaning or intent of these Bid Documents are to be directed, in writing, to the appropriate contact person identified in the "Invitation to Bid". Interpretations or clarifications considered necessary in response to such questions will be issued by addenda. Only questions answered by formal written addenda will be binding. Oral and other interpretations or clarifications will be without legal effect.
- C. Addenda may also be issued to modify these Bid Documents as deemed necessary by the Owner or the Engineer.
- D. All Addenda issued for the Project will become a part of the Contract Documents.
- E. It is the Bidder's responsibility to contact the City of Pensacola Purchasing Department prior to submitting a bid to ascertain if any addenda have been issued, to obtain all such addenda and return executed addenda with the bid. Acknowledgement of receipt of addenda shall be made in the appropriate place on the Proposal Form. The failure of a Bidder to submit acknowledgement of any addenda that affects the bid price(s) may be considered an irregularity and may be cause for rejection of the bid.

### IV. CONTRACT DURATION

- A. Total contract time established for this Project shall be **300 calendar days**, including 30 calendar days for Mobilization Phase, from Notice to Proceed to Substantial Completion of all Work Areas as defined in the contract drawings. An additional 45 calendar days will be allotted for project closeout and paperwork.

### V. LIQUIDATED DAMAGES

- A. Bidders are notified that the Owner considers time of the essence regarding this Agreement and that Owner will suffer financial loss if the work is not completed within the Contract Duration plus any extensions thereof allowed in accordance with the General Conditions. Accordingly, the Owner has established as liquidated damages for delay that the Contractor shall pay the Owner **\$500.00 for each calendar day** that expires after the time specified for substantial completion that the Project is not complete. These are not penalties for completing the work after the Contract Time but rather a reimbursement for costs lost by the Owner due to the Project not being completed on schedule.

## VI. NONDISCRIMINATION AND SEGREGATED FACILITIES

- A. Each Bidder shall complete, sign and include in his/her Proposal the Equal Employment Opportunity (EEO) Report Statement and Certification of Nonsegregated Facilities. A Proposal may be considered nonresponsive and may be rejected if it fails to furnish required data. When a determination has been made to award a Contract to a specific Contractor, such Contractor shall, prior to award, furnish such other pertinent information regarding his/her own employment policies and practices as the FAA, the FDOT, the Owner, or the Secretary of Labor may require. The Contractor shall comply with Part 152 of the Federal Aviation Regulations (FAR) as amended and specifically FAR 152.411 (c) and (d), incorporated herein by this reference. The Contractor shall require similar compliance with its subcontractors. All such information required of a subcontractor shall be furnished by the Bidder.
- B. The EEO Report Statement, Certification of Nonsegregated Facilities, Equal Opportunity Clause, and all other EEO requirements shall be included in all nonexempt subcontracts entered into by the Bidder. Subcontracts entered into by the Bidder shall also include all other applicable labor provisions. No subcontract shall be awarded to a noncomplying subcontractor.
- C. **Affirmative Action:** If the Contract is an aviation-related activity as defined in 14 CFR Part 152, and is a Construction Contract of \$10,000.00 or more, the Contractor assures that it will undertake an Affirmative Action Program as required by 14 CFR Part 152, Subpart E, to ensure that no person shall, on the grounds of race, creed, color, national origin, or sex, be excluded from participating in or receiving the services or benefits of any program or activity covered by this subpart. Bidder assures that it will require that its covered sub organizations provide assurances to the Bidder that they similarly will undertake Affirmative Action Programs and that they will require assurances from their sub organizations, as required by 14 CFR Part 152, Subpart E to the same effect.
- D. Bidder must comply with the President's Executive Order No. 11246, which prohibits discrimination in employment regarding race, creed, color, sex or national origin.
- E. In addition, the Bidder will also insert in each of its subcontracts a clause requiring the subcontractor(s) to include these provisions in any lower tier subcontracts which they may enter into, together with a clause requiring this insertion in any further subcontracts that may in turn be made.

## VII. DISADVANTAGED BUSINESS ENTERPRISE (DBE)

- A. **General:** Bidders are advised that this project requires the Owner to submit a DBE Program update to the FAA for review and approval.
- B. **DBE Obligation:** The requirements of 49 CFR Part 26, Regulations of the U.S. Department of Transportation, apply to this contract. It is the policy of the City of Pensacola, Florida, to practice nondiscrimination based on race, color, sex, or national origin in the award or performance of this contract. All firms qualifying under this solicitation are encouraged to submit bids/proposals. Award of this contract will be

conditioned upon satisfying the requirements of this bid specification. These requirements apply to all bidders, including those who qualify as a DBE. A DBE contract goal of **12.5 percent** has been established for this contract. The Bidder shall make good faith efforts, as defined in Appendix A, 49 CFR Part 26 (Attachment 1), to meet the contract goal for DBE participation in the performance of this contract.

- C. DBE Expectancies:** Contract expectancies are expectancies established for each specific prime Contract with subcontracting possibilities. The Bidder shall make good faith efforts to subcontract the goal percentage of the dollar value of the prime Contract to small business concerns owned and controlled by socially and economically disadvantaged individuals (DBE). In the event that the Bidder for this solicitation qualifies as a DBE, the Contract goal shall be deemed to have been met.
- D. Definition of DBE:** A DBE is a small business that is both owned and controlled by minorities or both owned and controlled by women. This means that 51 percent of the business must be owned by minorities and/or women and that they must control the management and daily operation of the business.
- E. Certification of Eligibility of DBEs:** To ensure that DBEs are eligible to be considered for fulfillment of the project goal, any DBE that wishes to participate must provide an eligibility certification to the Owner. The City will accept any firm certified as a DBE under an authorized federal, state or local DBE certification program. Any firm having questions regarding its eligibility as a DBE should contact the Owner regarding the certification requirements if it wishes to compete as a bona fide DBE under this Contract.
- F. Bidder's DBE Assurance:** Each Bidder shall submit a written assurance of its ability to meet the prescribed expectancy in its proposal. Proposals received that do not contain this assurance or have not given acceptable reasons for not meeting the expectancy will be considered nonresponsive and may be ineligible for award of the Contract.
- G. Required Submittal:** The Bidder will be required to submit the following information: (1) the names and addresses of DBE firms that will participate in the contract; (2) a description of the work that each DBE firm will perform; (3) the dollar amount of the participation of each DBE firm participating; (4) written documentation of the Bidder's commitment to use a DBE subcontractor whose participation it submits to meet the contract goal; (5) written confirmation from the DBE that it is participating in the contract as provided in the commitment made under (4); and (6) if the contract goal is not met, evidence of good faith efforts. A Proposal that fails to meet these requirements will be considered nonresponsive.
- H. Contractor Efforts to Meet DBE Subcontract Expectancies:**
  - 1. Bidders that have not met the expectancy stated in the DBE Expectancies section but wish to remain in competition must provide the Owner with written statements as to the reasons that the expectancy was not met. Those efforts may include, but are not necessarily limited to, the following:

- a. Whether the Bidder attended any pre-solicitation or pre-bid meetings that were scheduled by the Owner to inform DBEs of contracting and subcontracting opportunities;
- b. Whether the Bidder advertised in general circulation, trade association, and minority-focus media concerning the subcontracting opportunities;
- c. Whether the Bidder provided written notice to a reasonable number of specific DBEs certified by the Owner that their interest in the Contract was being solicited in sufficient time to allow the DBEs to participate effectively;
- d. Whether the Bidder followed up initial solicitations of interest by contacting DBEs certified by the Owner to determine with certainty whether the DBEs were interested;
- e. Whether the Bidder selected portions of the Work to be performed by DBEs certified by the Owner in order to increase the likelihood of meeting the good faith efforts and, in the case of federally funded projects, the DBE expectancy (including, where appropriate, breaking down contracts into economically feasible units to facilitate DBE participation);
- f. Whether the Bidder provided interested DBEs certified by the Owner with adequate information about the Plans, Specifications, or requirements of the Contract;
- g. Efforts made by the Bidder to negotiate with DBEs certified by the Owner for specific subcontract work, including but not limited to:
  - (1) Names, addresses and phone numbers of DBEs that were contacted in this regard.
  - (2) A description of the information provided to DBEs regarding the Contract Documents pertaining to the subcontract work proposed to be performed.
  - (3) A statement as to why additional agreements were not reached with DBE firms.
- h. Whether the Bidder negotiated in good faith with interested DBEs certified by the Owner; not rejecting DBEs as unqualified without sound reasons based on a thorough investigation of their capabilities.
- i. Whether the Bidder made efforts to assist interested DBEs certified by the Owner in obtaining bonding, lines of credit, or insurance required by the recipient or Bidder; and
- j. Whether the Bidder effectively used the services of available minority community organizations; minority contractors' groups; local, state and federal minority business assistance offices; and other organizations that provide assistance in the recruitment and placement of DBEs.



## **VIII. EXAMINATION OF CONDITIONS AFFECTING WORK**

- A.** Prior to submitting a proposal, each Bidder shall examine and thoroughly familiarize himself with all existing conditions, including all applicable laws, codes, ordinances, rules and regulations that will affect his work. Bidders shall visit the project site, examine the grounds and all existing buildings, utilities, pavements and systems and shall ascertain all conditions that will in any manner affect work. Bidders shall ask, in writing, for any additional information deemed necessary for them to be fully informed as to exactly what is to be expected prior to submitting a proposal.
- B.** Due to this airport being a commercial service airport with operations occurring constantly from approximately 5 AM to 11:30 PM, the only authorized site visit allowed will be immediately following the pre-bid conference. This will allow for a single coordinated visit that will be strictly controlled by Airport Operations staff and the air traffic control personnel.

## **IX. ACCURACY OF TECHNICAL DATA**

- A.** Bidders may rely upon the general accuracy of the "technical data" contained in reports of explorations and tests of subsurface conditions provided with the bid documents. However, any interpretations, opinions, or information contained in such reports or otherwise relating to the subsurface conditions at the site shall be utilized at the Bidder's own risk.
- B.** Information and data shown or indicated in these Bid Documents with respect to existing Underground Facilities at or contiguous to the site is based upon information and data furnished to Owner by Owners of such Underground Facilities or others, and the Owner does not assume any responsibility for the accuracy or completeness thereof unless it is expressly provided otherwise in the Project Manual.

## **X. SUBSTITUTIONS**

- A.** The materials, products and equipment described in the Contract Documents establish a standard of required function, dimension, appearance and quality to be met by any proposed substitution. The Bidder is responsible for assuring that all supplies, subcontractors, and vendors conform to the Contract requirements.
- B.** No substitution will be considered prior to receipt of bids unless written request for approval has been submitted in the proper format not less than ten (10) days prior to the receipt of bids. The burden of proof of the merit of the proposed substitution is upon the Bidder. The Engineer's decision of approval or disapproval of a proposed substitution is final.
- C.** In making requests for substitutions, the Bidder shall list the particular system, product, or material he wishes to substitute, and the justification for such a request. Requests submitted shall include any and all adjustments of that and any other work affected thereby.

- D. If the Engineer approves any proposed substitution prior to receipt of bids, such approval will be set forth in an Addenda. Bidders shall not rely on approvals made in any other manner.
- E. No substitutions will be considered after the receipt of bids except as specifically provided for in the Contract Documents.

#### **XI. TAX EXEMPT CERTIFICATE**

- A. If the Bidder desires a tax-exempt certificate, it shall be obtained from the State of Florida and not through the Owner.

#### **XII. DISQUALIFICATION OF BIDDERS**

- A. Bidders will be disqualified and their proposals not considered for any of the following reasons:
  - 1. Where more than one proposal for an individual, firm, partnership or corporation is filed under the same or different names and where such proposals are not identical in every respect;
  - 2. Reasonable grounds for believing that any Bidder is interested in more than one proposal for work contemplated or materials to be furnished;
  - 3. Reason for believing that collusion exists among the Bidders;
  - 4. Where the Bidder has an interest in any litigation or arbitration or other type claim against the Owner or Engineer or has defaulted on a previous Contract;
  - 5. Lack of competency, as revealed by the Statement of Bidder's Qualifications, etc.;
  - 6. Uncompleted work that, in the judgment of the Owner, will hinder or prevent the prompt completion of additional work, if awarded;
  - 7. Failure to pay or satisfactorily settle all bills due for labor and material on former contracts in force at the time of advertisement for bids;
  - 8. Default under previous contract.
  - 9. Previous projects where, in the judgment of the Owner, the Bidder performed unsatisfactorily and did not complete and close out the project in a timely manner resulting in the Owner not being able to close out the project with various funding agencies and resulting in the Owner potentially or actually losing planned funding for other projects.

#### **XIII. SURETY BONDS**

- A. The Proposal Guaranty shall be as specified; only the Proposal Bond and Surety's Bond Affidavit as bound within these documents or a Cashier's Check is acceptable. Each separate proposal shall be accompanied by a Cashier's Check or Proposal Bond on the form provided herein in the amount of 5 percent (5%) of the total amount bid, made payable to the City of Pensacola. If a Proposal Bond is provided in lieu of a Cashier's Check, it must be accompanied by a valid Power of Attorney indicating that the person signing the bond on behalf of the Surety has full legal authority to do so.
- B. The amount of such bond or the check of the Bidder whose proposal is accepted shall be forfeited and paid to the Owner as liquidated damages if said Bidder fails to enter into a Contract with the Owner and to furnish the required executed Contracts, Certificates of Insurance and Performance and Payment Bonds within twenty (20) calendar days after the date of the Notice of Award and Acceptance of the Proposal.
- C. Only the Contract Payment and Performance Bonds and Surety's Bond Affidavit as bound within these Contract Documents are acceptable.
- D. With the execution and delivery of the Contract, the Bidder shall furnish and file with the Owner in the amounts herein required the following surety bonds.
- E. Performance Bond and labor and materials Payment Bond, if the total awarded Contract price exceeds \$25,000.00, in a penal sum not less than the amount of the Contract as awarded, as security for the faithful performance of the Contract and for the payment of all persons, firms or corporations to whom the Bidder may become legally indebted for labor, materials, tools, equipment or services of any nature including utility and transportation services, employed or used by him in performing the work.
- F. The warranty shall begin on the date of official acceptance of the entire completed installation by resolution of the Owner.
- G. No surety will be accepted who is now in default or delinquent on any bond or who is interested in any litigation or arbitration claim against the Owner. All bonds shall be made on forms furnished herein and shall be executed by surety companies licensed to do business in the State of Florida and acceptable to the Owner. Each bond shall be executed by the Bidder and the surety and the bonds shall bear the same date as, or a date subsequent to that of the Contract.
- H. Should any surety on the Contract be determined unsatisfactory at any time by the Owner, notice will be given to the Bidder to that effect and the Bidder shall forthwith substitute a new surety or sureties satisfactory to the Owner. No payment will be made under the Contract until the new surety or sureties, as required, have qualified and been accepted by the Owner. The Contract shall not be operative nor shall any payments be due until approval of the bond has been made by the Owner.
- I. Contractor's Liability Insurance Requirements. The Bidder shall provide with the proposal a listing of general liability, automobile liability, excess liability, worker's compensation, property hazard, and miscellaneous insurance coverage the Bidder currently has and normally has in force, under the column "Bidder's Present Insurance Coverage" along with a copy of a Certificate of Insurance as verification of that coverage. In addition, the Bidder shall provide any additional premium cost for each of

those items listed under the column "Insurance Requirements for this Project" on the Insurance Form issued by the agent or insurance carrier for that coverage.

- J. In the event the Owner determines that the low bidder's coverage in force is inadequate, the Owner may require the low bidder to procure the additional coverage in the amounts specified by the Owner in the Insurance Forms. The cost of premiums for such additional coverage shall be paid by the Owner in the form of a reimbursement under the Contract. The additional premium cost shall be that defined on the Insurance form, TOTAL ADDITIONAL PREMIUM COSTS and that amount shall be inserted in Bid Item I-1, Page P-35 and Bid Item I-1, Page P-36.

#### **XIV. THIRD-PARTY COVERAGE**

- A. The bidding documents require the Bidder to name the Engineer and the Owner as additional insureds. The Bidder shall show the premium cost for the additional insureds in the item defined in the Proposal Form section "Insurance Requirements for this Project."
- B. The Bidder shall submit actual premium costs which do not reflect any overhead, profit or other hidden costs.

#### **XV. EXPARTE COMMUNICATION**

- A. Please note that to insure the proper and fair evaluation of a bid, the City of Pensacola prohibits exparte communication (i.e., unsolicited) initiated by the Bidder to the City Official(s) or Employee(s) evaluating or considering the bids prior to the time a bid decision has been made. Communication between Bidder and the City will be initiated by the appropriate City Official or Employee in order to obtain information or clarification needed to develop a proper and accurate evaluation of the bid. Exparte communication may be grounds for disqualifying the offending Bidder from consideration or award of the bid, then in evaluation, or any future bid.

#### **XVI. CONFLICT OF INTEREST / STATEMENT OF NON-COLLUSION**

- A. The award hereunder is subject to Chapter 112, Florida Statutes. All respondents must disclose with their bid the name of any officer, director, or agent who is also an employee of the City of Pensacola. Further, all respondents must disclose the name of any City of Pensacola employee who owns, directly or indirectly, an interest of five percent (5%) or more of the respondent's firm or any of its branches.
- B. The respondent shall certify that he/she has not, either directly or indirectly, entered into any agreement, participated in any collusion, or otherwise taken any action in restraint of free competitive bidding in connection with the bid and that the respondent is not financially interested in, or otherwise affiliated in a business way with any other respondent on the same land or improvements.

#### **XVII. AWARD OF CONTRACT**

- A.** The City of Pensacola at its sole discretion reserves the right to waive technicalities or irregularities, to reject any or all bids, and/or to accept that bid which is in the best interest of the City. The award of this bid, if made, may be based on considerations other than total cost and may be awarded based on various considerations, including without limitation; Bidder's experience and/or qualifications, past experience, past performance, administrative cost, standardization, technical evaluation and oral and/or written presentations as required. The City reserves the right to accept all or part, or to decline the whole, and to award this bid to one (1) Bidder.

**END OF INSTRUCTIONS TO BIDDERS**

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**CITY OF PENSACOLA  
PROPOSAL FORM**

TO: **City of Pensacola**  
Purchasing Department  
222 West Main Street  
Pensacola, Florida 32502  
City of Pensacola Bid No. 22-040

PROJECT: **Taxiway A Rehabilitation**  
FAA A.I.P. Project No.: **TBD**  
BID No.: 25-029

BIDDER: \_\_\_\_\_

BIDDER'S  
ADDRESS: \_\_\_\_\_

BIDDER'S  
EMAIL: \_\_\_\_\_

DATE: \_\_\_\_\_

BIDDER'S REPRESENTATIVE (to be contacted for additional information on this Proposal):

\_\_\_\_\_  
(Name)

\_\_\_\_\_  
(Telephone Number)

**BIDDER'S DECLARATION AND UNDERSTANDING**

The undersigned declares that no person in the employ of the City of Pensacola, Florida (herein referred to as OWNER) is pecuniarily interested in this proposal, or in the contract or the work which he/she proposed to do; that he has carefully examined the contract and specifications and has informed himself/herself fully in regard to all conditions pertaining to the site where the work is to be done and carefully estimated on the work. He/she understands that the OWNER, its agents and employees, are not to be in any manner held responsible for the accuracy of, or bound by, any estimates or plans of underground structures relating to the work and that if any have been given or made, they are to be considered solely as a base for filling out and preparing several proposals.

The undersigned proposes to furnish all labor, equipment material required for the above outlined construction at the airport known as Pensacola International Airport located in Escambia County, Florida, in accordance with the accompanying specifications and plans prepared for the OWNER for the sums specified herein, subject to additions and deductions according to the specifications and in all respects to the terms thereof.

It is understood that all workmanship and materials under all items of work are guaranteed for one year from the date of final acceptance, unless otherwise specified. It is understood that the OWNER reserves the right to accept or reject any or all bids and to waive any informalities. The Owner reserves the right to award any, all, combination of any, or none of the alternate bid items.

Wages not less than the minimum rates or wages, as pre-determined for this project by the Secretary of the U.S. Department of Labor, were used in the preparation of this proposal. It is agreed that the description under each item, being stated, implies although it does not mention, all incidentals and that the prices stated are intended to cover all such work, materials and incidentals and constitute the Bidder's obligations as described in the specifications, and any details not specifically mentioned, but evidently included in the contract shall be compensated for in the item which most logically includes it.

The Bidder expressly agrees that should environmental permits not be granted for items contingent on their receipt, the OWNER may delete those items from the Contract without Supplemental Agreement or Change Order. In this event, the Bidder agrees that the unit prices proposed for non-environmentally contingent items will remain unchanged and that no claims relating to the deletion of items contingent on environmental permitting will be made.

Enclosed herewith is the Proposal Bond in the form specified which is submitted as a guarantee of the good faith of the Proposal. The Bidder agrees that, upon receipt of notice to award, he will, within 20 calendar days, execute the Contract in accordance with the Proposal as accepted, and satisfy the Contract bonding and insurance requirements stipulated in Bid Documents; and that upon his failure or refusal to do so, the Proposal Bond accompanying his bid shall be forfeited to and become the property of the OWNER as liquidated damages for such failure or refusal.

### **ADDENDA**

The Bidder hereby acknowledges that he has received the following Addenda:

| <u>Addenda No.</u> | <u>Dated</u> |
|--------------------|--------------|
| _____              | _____        |
| _____              | _____        |
| _____              | _____        |

### **TAXES**

The Bidder agrees that any applicable Federal, State and Local sales and use taxes, are included in the stated bid prices. Since often the City of Pensacola, Florida is exempt from taxes for equipment, materials, and services, it is the responsibility of the Contractor to determine whether sales taxes are applicable. The Contractor is liable for any applicable taxes which are not included in the stated bid prices.



NOTE: THE BID PRICES SET FORTH ON THE ATTACHED SHEETS SHALL BE  
CONSIDERED FIRM BIDS NOT SUBJECT TO PRICE ADJUSTMENT.

SIGNATURE ACKNOWLEDGES THAT: (Check One)

\_\_\_\_\_ Bid Proposal is in full compliance with the Specifications.

\_\_\_\_\_ Bid Proposal is NOT in full compliance with the Specifications.

(CORPORATE SEAL)

ATTEST:

BIDDER:

\_\_\_\_\_  
Signature

\_\_\_\_\_  
Signature

By: \_\_\_\_\_

By: \_\_\_\_\_

Title: \_\_\_\_\_

Title: \_\_\_\_\_

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**CITY OF PENSACOLA  
PROPOSAL AFFIDAVIT**

The following affidavit must be executed in order that your Proposal may be considered.

STATE OF \_\_\_\_\_)

COUNTY OF \_\_\_\_\_)

\_\_\_\_\_ of lawful age, being first duly sworn, upon his oath, deposes and says: That he executed the accompanying Proposal on behalf of the Contractor therein named, and that he had lawful authority so to do, and said Contractor has not directly or indirectly, entered into any agreement, expressed or implied, with any Contractor or Contractors, having for its object the controlling of the price or amount of such Proposal or any Proposals, the limiting of the Proposal of Contractors, the parceling or farming out to any Contractor or Contractors, to other persons of any part of the Contract or any of the subject matter of the Proposals, or of the profits thereof, and that he has not and will not divulge the sealed Proposal to any person whomsoever; except those having a partnership or other financial interest with him in said Proposal or Proposals, until after the sealed Proposal or Proposals are opened.

Signed: \_\_\_\_\_

Subscribed and sworn to before me this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

My Commission Expires:

\_\_\_\_\_

\_\_\_\_\_  
Notary Public

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**CITY OF PENSACOLA  
PROPOSAL BOND**

**Bond No.**\_\_\_\_\_

(Not to be filled in if a Cashier's check is submitted)

KNOW ALL MEN BY THESE PRESENTS: That the undersigned Bidder, \_\_\_\_\_, as Principal, and firmly bound unto the **City of Pensacola, Florida** in the sum of \_\_\_\_\_ dollars (\$\_\_\_\_\_), for the payment of which, well and truly to be made, we hereby jointly and severally bind ourselves, our heirs, executors, administrators, successors and assigns.

THE CONDITION OF THIS OBLIGATION is such that if Principal:

1. Does not withdraw the attached Proposal of \_\_\_\_\_ dollars (\$\_\_\_\_\_) for the improvement of **Pensacola International Airport** for a period of one hundred twenty (120) days after the date on which the bids are opened; and
2. Enters into the written contract and furnishes the required Certificates of Insurance, Payment and Performance Bonds, with Surety or Sureties acceptable to the **City of Pensacola, Florida**, within twenty (20) days after notice that the said Proposal is accepted, then this obligation shall be void; otherwise the same shall be in full force and the full amount of this Proposal Bond shall be paid to the **City of Pensacola, Florida** as stipulated herein as liquidated damages.

Signed this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

(PRINCIPAL MUST INDICATE  
WHETHER CORPORATION, PARTNER-  
SHIP, COMPANY OR INDIVIDUAL)

\_\_\_\_\_  
Principal

THE PERSON SIGNING FOR THE  
PRINCIPAL SHALL, IN HIS OWN  
HANDWRITING, SIGN THE PRINCIPAL'S  
NAME, HIS OWN NAME AND HIS TITLE.  
WHERE THE PERSON SIGNING FOR A  
CORPORATION IS OTHER THAN THE  
PRESIDENT OR VICE PRESIDENT, HE  
MUST FURNISH A CORPORATE RESOLUTION  
SHOWING HIS AUTHORITY TO BIND THE  
CORPORATION.

By: \_\_\_\_\_  
Title

(Affix Surety's Corporate Seal)

\_\_\_\_\_  
Surety

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**CITY OF PENSACOLA  
SURETY'S BOND AFFIDAVIT**

STATE OF \_\_\_\_\_)

COUNTY OF \_\_\_\_\_)

Before me the undersigned authority, personally appeared \_\_\_\_\_  
Who, being duly sworn deposes and says that he is a duly authorized (resident) (non-resident)  
insurance agent, properly licensed under the laws of the State of \_\_\_\_\_, and the State of  
Florida, to represent

\_\_\_\_\_, a company authorized to make  
corporate surety bonds under the laws of the State of Florida (the "Surety").

Said \_\_\_\_\_ further certifies that as agent or attorney-in-fact for the said surety, he  
has signed the attached bond in the sum of \_\_\_\_\_  
\_\_\_\_\_ (U.S. \$ \_\_\_\_\_) on behalf of \_\_\_\_\_

to the **City of Pensacola, Florida** covering FAA A.I.P. Project No. **TBD**, Runway 8-26  
Rehabilitation at Pensacola International Airport.

Said \_\_\_\_\_ further certifies that the premium on the said bond is  
\$ \_\_\_\_\_

which will be paid in full directly to the surety or to him as agent or attorney-in-fact, and included  
in his regular commission as agent or attorney-in-fact, for the execution of said bond and that  
his commission will not be divided with anyone except to who is a duly authorized insurance  
agent properly licensed under the laws of the State of Florida.

Countersigned:

\_\_\_\_\_  
Surety

\_\_\_\_\_  
Florida Resident Agent

\_\_\_\_\_  
Attorney-in-Fact

\_\_\_\_\_  
Address of Resident Agent

Acknowledgment for  
Attorney-in-Fact

\_\_\_\_\_  
Address of Bond Company

Sworn to and Subscribed  
Before me this \_\_\_\_ Day of \_\_\_\_\_, 20\_\_\_\_.

\_\_\_\_\_  
Phone Number

Notary Public, State of \_\_\_\_\_

\_\_\_\_\_  
Fax Number

My commission expires: \_\_\_\_\_

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**CITY OF PENSACOLA  
EQUAL EMPLOYMENT OPPORTUNITY REPORT STATEMENT  
as Required by 41 CFR 60-1.7 (b)**

Section 60-1.7(b) of the Regulations of the Secretary of Labor requires each bidder or prospective prime Contractor and proposed Subcontractors, where appropriate, to state in the bid or at the outset of negotiations for the Contract whether it has participated in any previous Contract or subcontracts subject to the equal opportunity clause; and if so, whether it has filed with the Joint Reporting Committee, the Director, an agency, or the former President's Committee on Equal Employment Opportunity all reports due under the applicable filing requirements. In any case in which a bidder or prospective prime Contractor or proposed Subcontractors which participated in a previous Contract subject to Executive Order 10925, 11114, or 111246 has not filed a report due under the applicable filing documents, no Contract or Subcontracts shall be awarded unless such Contractor submits a report covering the delinquent period or such other period specified by the FAA or the Director, OFCCP.

The Bidder shall complete the following statement by checking the appropriate boxes. Failure to complete these blanks may be grounds for rejection of bid.

1. The Bidder has ( ) has not ( ) developed and has on file at each establishment Affirmative Action Programs pursuant to 41 CFR 60-1.4 and 41 CFR 60-2.
2. The Bidder has ( ) has not ( ) participated in any previous Contract or Subcontract subject to the Equal Opportunity Clause prescribed by Executive Order 10925, or Executive Order 11114, or Executive Order 11246.
3. The Bidder has ( ) has not ( ) filed with the Joint Reporting Committee the annual compliance report on Standard Form 100 (EEO-1 Report).
4. The Bidder has ( ) has not ( ) submitted all compliance reports in connection with any such Contract due under the application filing requirements; and that representations indicating submission of required compliance reports signed by proposed Subcontractors will be obtained prior to award of Subcontractors.
5. The Bidder does ( ) does not ( ) employ fifty (50) or more employees.

If the Bidder has participated in a previous Contract subject to the equal opportunity clause and has not submitted compliance reports due under applicable filing requirements, the Bidder shall submit a compliance report on Standard Form 100, "Employee Information EEO-1" prior to the award of Contract.

Standard Form 100 is normally furnished to Contractors annually, based on a mailing list currently maintained by the Joint Reporting Committee. In the event a Contractor has not received the form, he may obtain it by writing to the following address: Joint Reporting Committee, 1800 G Street, Washington, DC 20506.

\_\_\_\_\_  
(Name of Bidder)

Date: \_\_\_\_\_

By: \_\_\_\_\_

Signature\*

\*Must be same signature on Bid Proposal

Title: \_\_\_\_\_

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**CITY OF PENSACOLA  
CERTIFICATION OF NONSEGREGATED FACILITIES**

The federally-assisted construction contractor certifies that she or he does not maintain or provide, for his employees, any segregated facilities at any of his establishments and that she or he does not permit his employees to perform their services at any location, under his control, where segregated facilities are maintained. The federally-assisted construction contractor certifies that she or he will not maintain or provide, for his employees, segregated facilities at any of his establishments and that she or he will not permit his employees to perform their services at any location under his control where segregated facilities are maintained. The federally-assisted construction contractor agrees that a breach of this certification is a violation of the Equal Opportunity Clause in this contract.

As used in this certification, the term "segregated facilities" means any waiting rooms, work areas, restrooms, and washrooms, restaurants and other eating areas, time clocks, locker rooms and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing facilities provided for employees which are segregated by explicit directives or are, in fact, segregated on the basis of race, color, religion, or national origin because of habit, local custom, or any other reason. The federally-assisted construction contractor agrees that (except where she or he has obtained identical certifications from proposed subcontractors for specific time periods) she or he will obtain identical certifications from proposed subcontractors prior to the award of subcontracts exceeding \$10,000 which are not exempt from the provisions of the Equal Opportunity Clause and that she or he will retain such certifications in his files.

**NOTICE TO PROSPECTIVE CONTRACTORS OF REQUIREMENT FOR  
CERTIFICATION OF NONSEGREGATED FACILITIES**

A Certification of Nonsegregated Facilities must be submitted prior to the award of a contract or subcontract exceeding \$10,000 which is not exempt from the provisions of the Equal Opportunity Clause.

Certification--The information above is true and complete to the best of my knowledge and belief.

\_\_\_\_\_  
Name of Signer (Please Print)

\_\_\_\_\_  
Title

\_\_\_\_\_  
Signature of Contractor

\_\_\_\_\_  
Date

NOTE: The penalty for making false statements in offers is prescribed in 18 U.S.C. 1001.

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**SWORN STATEMENT UNDER SECTION 287.113(3)(a),  
FLORIDA STATUTES, ON PUBLIC ENTITY CRIMES**

THIS FORM MUST BE SIGNED IN THE PRESENCE OF A NOTARY PUBLIC OR OTHER OFFICER AUTHORIZED TO ADMINISTER OATHS.

1. This sworn statement is submitted with Bid, Proposal or Contract No. \_\_\_\_\_  
for \_\_\_\_\_.

2. This sworn statement is submitted by \_\_\_\_\_  
**[name of entity submitting sworn statement]**

whose business address is \_\_\_\_\_

\_\_\_\_\_ and (if applicable) its Federal  
Employer

Identification Number (FEIN) is \_\_\_\_\_

(If the entity has no FEIN, include the Social Security number of the individual signing this sworn statement: \_\_\_\_\_.)

3. My name is \_\_\_\_\_ and my relationship  
**[please print name of individual signing]**

to the entity named above is \_\_\_\_\_.

4. I understand that a “public entity crime” as defined in Paragraph 287.133(1)(g), Florida Statutes, means a violation of any state or federal law by a person with respect to and directly related to the transaction of business with any public entity or with an agency or political subdivision of any other state or with the United States, including, but not limited to, any bid or contract for goods or services to be provided to any public entity or an agency or political subdivision of any other state or the United States and involving antitrust, fraud, theft, bribery, collusion, racketeering, conspiracy or material misrepresentation.

5. I understand that “convicted” or “conviction” as defined in Paragraph 287.133(1)(b), Florida Statutes, means a finding of guilt or a conviction of a public entity crime, with or without an adjudication of guilt, in any federal or state trial court of record relating to charges brought by indictment or information after July 1, 1989, as a result of a jury verdict, nonjury trial, or entry of a plea of guilty or nolo contendere.

I understand that an “affiliate” as defined in Paragraph 287.133(1)(a), Florida Statutes, means:

1. A predecessor or successor of a person convicted of a public entity crime: or

2. An entity under the control of any natural person who is active in the management of the entity and who has been convicted of a public entity crime. The term “affiliate” includes those officers, directors, executives, partners, shareholders, employees, members and agents who are active in the management of an affiliate. The ownership by one person of shares constituting a controlling interest in another person, or a pooling of equipment or income among

persons when not for fair market value under an arm's-length agreement, shall be a prima facie case that one person controls another person. A person who knowingly enters into a joint venture with a person who has been convicted of a public entity crime in Florida during the preceding thirty-six (36) months shall be considered an affiliate.

I understand that a "person" as defined in Paragraph 287.133(1)(e), Florida Statutes means any natural person or entity organized under the laws of any state or of the United States, with the legal power to enter into a binding contract and which bids or applies to bid on contracts for the provision of goods or services let by a public entity, or which otherwise transacts or applies to transact business with a public entity. The term "person" includes those officers, directors, executives, partners, shareholders, employees, members and agents who are active in management of an entity.

Based on information and belief, the statement which I have marked below is true in relation to the entity submitting this sworn statement. (Indicate which statement applies.)

\_\_\_\_ Neither the entity submitting this sworn statement, nor any officers, directors, executives, partners, shareholders, employees, members or agents who are active in the management of the entity, nor any affiliate of the entity, have been charged with and convicted of a public entity crime subsequent to July 1, 1989.

\_\_\_\_ The entity submitting this sworn statement, or one or more of its officers, directors, executives, partners, shareholders, employees, members or agents who are active in the management of the entity, or an affiliate of the entity, has been charged with and convicted of a public entity crime subsequent to July 1, 1989, AND [Please indicate which additional statement applies.]

\_\_\_\_ There has been a proceeding concerning the conviction before a hearing officer of the State of Florida, Division of Administrative Hearings. The final order entered by the hearing officer did not place the person or affiliate on the convicted vendor list. [Please attach a copy of the final order.]

\_\_\_\_ The person or affiliate was placed on the convicted vendor list. There has been subsequent proceeding before a hearing officer of the State of Florida, Division of Administrative Hearings. The final order entered by the hearing officer determined that it was in the public interest to remove the person or affiliate from the convicted vendor list. [Please attach a copy of the final order.]

\_\_\_\_ The person or affiliate has not been placed on the convicted vendor list. [Please describe any action taken by or pending with the Department of General Services.]

\_\_\_\_\_  
[Signature]

Date: \_\_\_\_\_

STATE OF \_\_\_\_\_

COUNTY OF \_\_\_\_\_

PERSONALLY APPEARED BEFORE ME, the undersigned authority,

\_\_\_\_\_ who, after first being sworn by me, affixed his/her signature  
[name of individual signing]

in the space provided above on this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

\_\_\_\_\_  
NOTARY PUBLIC

My Commission Expires:

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**CITY OF PENSACOLA  
NON-COLLUSION AFFIDAVIT**

STATE OF FLORIDA

COUNTY OF ESCAMBIA

\_\_\_\_\_, being first duly sworn, deposes and says that:

1. (S)He is \_\_\_\_\_ of \_\_\_\_\_,  
the Bidder that has submitted the attached Bid;

2. (S)He is fully informed respecting the preparation and contents of the attached Bid and  
of all pertinent circumstances respecting such Bid;

3. Such Bid is genuine and is not a collusive or sham Bid;

4. Neither the Bidder nor anyone acting on behalf of the Bidder, including this affiant, has  
in any way colluded, conspired, connived, or agreed, directly or indirectly with any other Bidder,  
firm or person to submit a collusive or sham Bid in connection with the Contract for which the  
attached bid has be submitted or to refrain from bidding in connection with such Contract, or has  
in any manner, directly or indirectly, sought by agreement or collusion or communication or  
conference with any other Bidder, firm or person to fix the price or prices in the attached Bid or of  
any other Bidder, or to fix any overhead, profit or cost element of the Bid price or the Bid price of  
any other Bidder, or to secure through any collusion, conspiracy, connivance or unlawful  
agreement any advantage against the City of Pensacola, Florida or any person interested in the  
proposed Contract; and,

5. The price or prices quoted in the attached Bid are fair and proper and are not tainted by  
any collusion, conspiracy, connivance, or unlawful agreement on the part of the Bidder or anyone  
acting on his/her/its behalf.

\_\_\_\_\_  
(Signature)

\_\_\_\_\_  
(Title)

Subscribed and Sworn to before me of this \_\_ day of \_\_\_\_\_, 20\_\_.

\_\_\_\_\_  
(Notary's Signature)

\_\_\_\_\_  
(Notary's Stamped or Printed Name)

Notary Public, in and for \_\_\_\_\_

\_\_\_\_\_ County,

My commission expires: \_\_\_\_\_

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**CITY OF PENSACOLA  
DRUG-FREE WORKPLACE CERTIFICATE**

**IDENTICAL TIE BIDS** - Pursuant to Section 287.087, Florida Statutes, preference shall be given to business with Drug-Free Workplace Programs. Whenever two or more bids which are equal with respect to price, quality, and service are received for the procurement of commodities or contractual services, a bid received from a business that certifies that it has implemented a Drug-Free Workplace Program shall be given preference in the award process. Established procedures for processing tie bids will be followed if none of the tied vendors have a Drug-Free Workplace Program. In order to have a Drug-Free Workplace Program, a business shall:

- 1) Publish a statement notifying employees that the unlawful manufacture, distribution, dispensing, possession, or use of a controlled substance is prohibited in the workplace and specifying the actions that will be taken against employees for violations of such prohibition.
- 2) Inform employees about the dangers of drug abuse in the workplace, the business's policy of maintaining a Drug-Free Workplace, any available drug counseling, rehabilitation, and employee assistance programs, and the penalties that may be imposed upon employees for drug abuse violations.
- 3) Give each employee engaged in providing the commodities or contractual services that are under bid a copy of the statement specified in subsection (1).
- 4) In the statement specified in subsection (1), notify the employees that, as a condition of working on the commodities or contractual services that are under bid, the employee will abide by the terms of the statement and will notify the employer of any conviction of, or plea of guilty or nolo contendere to, any violation of Chapter 893 or of any controlled substance law of the United States or any state, for a violation occurring in the work place no later than five (5) days after such conviction.
- 5) Impose a sanction on, or require the satisfactory participation in a drug abuse assistance or rehabilitation program if such is available in the employee's community, by any employee who is so convicted.
- 6) Make a good faith effort to continue to maintain a drug-free workplace through implementation of this section.

**AS THE PERSON AUTHORIZED TO SIGN THE STATEMENT, I CERTIFY THAT THIS FIRM COMPLIES FULLY WITH THE ABOVE REQUIREMENTS.**

\_\_\_\_\_  
BIDDER'S/PROPOSER'S SIGNATURE

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The Bidder shall submit, with his Proposal, a list of all subcontractors the Bidder proposes to use on the Work of this Contract and a brief summary of their responsibilities. The Owner reserves the right to approve or reject subcontractors for any Work. After the Owner accepts the Bidder's Proposal and such Bidder is awarded a Contract, the successful Bidder may not substitute a subcontractor listed in the Proposal without the prior written approval of the Owner. Such approval shall be obtained at least ten (10) calendar days prior to the date scheduled for that subcontractor to begin Work.

The Contractor shall be as fully responsible to the Owner for the acts and omissions of his subcontractors, and of persons either directly or indirectly employed by them, or under their control, as he is for the acts and omissions of persons directly employed by him. Nothing contained in the Contract Documents shall create any contractual relationships between any subcontractor and the Owner.

[illegible]

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**CITY OF PENSACOLA  
STATEMENT OF BIDDER'S QUALIFICATIONS**

All questions must be answered and the data given must be clear and comprehensive. This statement must be notarized. If necessary, questions may be answered on separate attached sheets. The Bidder may submit any additional information he desires.

1. Name of Bidder \_\_\_\_\_

2. Permanent main office address \_\_\_\_\_

\_\_\_\_\_

3. When organized \_\_\_\_\_

4. If a corporation, where incorporated \_\_\_\_\_

5. How many years engaged in the business related to this Project under present firm or trade name? Give former names of firm or trade names, with dates of operation under each \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

6. General character of work performed by your company \_\_\_\_\_

\_\_\_\_\_

7. Have you ever failed to complete any work awarded to you? If so, where and why? Give name of Owner \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

8. Have you ever defaulted on a contract? If so, where and why? Give name of Owner. \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

9. Have you ever sued an Owner of a project? If so, where and why? Give name of Owner \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

10. Have you ever sued a Consultant or subconsultant representing the Owner of a project? If so, where and why? Give name of Owner. \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_

11. Have you ever been terminated by the Owner of a project? If so, where and why? Give name of Owner \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_

12. Do you have pending litigation or claims against an Owner? If so, give name of Owner \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_

13. Experience in work similar in importance to this project \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

14. List of equipment that would be available for use on this project \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

15. Background and experience of the principal members of your organization, including the officers \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

16. Give bank references \_\_\_\_\_

\_\_\_\_\_  
\_\_\_\_\_



17. Will you, upon request, fill out a detailed financial statement and furnish any other pertinent information that may be required by the City of Pensacola, Florida? \_\_\_\_\_  
\_\_\_\_\_

18. The undersigned hereby authorizes and requests any persons, firm or corporation to furnish any information requested by the City of Pensacola, Florida, in verification of the recitals comprising this Statement of Bidder's Qualifications.

Dated at \_\_\_\_\_, this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

\_\_\_\_\_  
(Contractor) By: \_\_\_\_\_

Title: \_\_\_\_\_

State of \_\_\_\_\_

License Number \_\_\_\_\_

Date of License \_\_\_\_\_

\_\_\_\_\_, being duly sworn deposes and says that he is \_\_\_\_\_  
\_\_\_\_\_ of \_\_\_\_\_ and that the answers to the  
foregoing questions and all statements therein contained are true and correct.

SUBSCRIBED AND SWORN to before me this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_.

\_\_\_\_\_  
Notary's Signature My commission expires:

\_\_\_\_\_  
Notary's Printed Name

NOTARY PUBLIC, STATE OF \_\_\_\_\_

**END OF PROPOSAL FORM**

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**CITY OF PENSACOLA  
DISADVANTAGED BUSINESS ENTERPRISES (DBE) PLAN**

The **City of Pensacola, Florida** (Owner) in accordance with Title VI Civil Rights Act of 1964, 78 Stat. 262, 42 USC and 49 CFR, Part 21 issued pursuant to such Act, affords Disadvantage Business Enterprises full opportunity to submit an indication of interest in response to this invitation and will not discriminate against any interested firm on the ground of race, creed, color, sex, age or national origin in a contract award. In addition, the Owner has established goals for DBE participation in Owner projects.

**REQUIRED ASSURANCE TO BE INCLUDED IN ALL BID PROPOSALS**

The firm assures that it will utilize not less than **12.5 percent** of Disadvantaged Business participation. Percentage applies to bidder if a DBE or to qualifying subcontractors to be used by Bidder. Upon request, Bidder and/or Contractor, if contract is entered into with Bidder, shall furnish the names of such subcontractors and the anticipated dollar volume to be furnished by Bidder. If the goal for DBE participation is not met, the Bidder is required to submit documentation of efforts it made. The Owner will then determine whether these are "good faith efforts."

CERTIFICATION OF BIDDER for the above:

BIDDER'S NAME: \_\_\_\_\_

ADDRESS: \_\_\_\_\_

FEDERAL EMPLOYER ID NO.: \_\_\_\_\_

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**CITY OF PENSACOLA  
DBE SUBCONTRACTOR SOLICITATION**

Bid Invitation No.: \_\_\_\_\_

Project Name: \_\_\_\_\_

DBE Project Goal: 12.5 %

| <b>* <u>Name, Address, Phone No. of Subcontractor</u></b> | <b><u>Date of Contract</u></b> | <b><u>Result(s) of Contact</u></b> |
|-----------------------------------------------------------|--------------------------------|------------------------------------|
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |
| _____                                                     | _____                          | _____                              |

\_\_\_\_\_  
Title of Official

\_\_\_\_\_  
Company

\_\_\_\_\_  
Date

\_\_\_\_\_  
Title of Official

**THIS FORM MUST BE SUBMITTED WITH THE BID PROPOSAL**

**\* NOTE: Please attach copies of any written solicitation information, including, but not limited to, information regarding follow-up contacts.**

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**CITY OF PENSACOLA  
DBE INTENT TO PERFORM AS A SUBCONTRACTOR**

TO: \_\_\_\_\_, in response to a telephone/written request  
for a quote on \_\_\_\_\_, 20\_\_\_\_ agrees to perform the work outline below in  
the project known as \_\_\_\_\_ for the City  
of Pensacola, Florida.

The work description is as follows:

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The price of said work is \$ \_\_\_\_\_

And \_\_\_\_\_ will enter into a  
formal agreement for the above work conditional upon your entering a contract with the City of  
Pensacola, Florida. Any changes, including the addition or deletion of DBE firms after bid  
opening, must have written approval of the Contractor Coordinator.

\_\_\_\_\_  
(Date)

\_\_\_\_\_  
(Name of DBE Firm)

\*\*\*\*\*

\_\_\_\_\_  
(Name of Prime Contractor)

\_\_\_\_\_  
(Title of Official Signing)

\_\_\_\_\_  
(Signature of Official)

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**CITY OF PENSACOLA  
DISADVANTAGED BUSINESS ENTERPRISE (DBE) UTILIZATION**

The undersigned Bidder has satisfied the requirements of the bid specification in the following manner (please check the appropriate space):

\_\_\_\_\_The Bidder is committed to a minimum of \_\_\_\_\_% DBE utilization on this contract.

\_\_\_\_\_The Bidder (if unable to meet the DBE goal of \_\_\_\_\_%) is committed to a minimum of \_\_\_\_\_% DBE utilization on this contract and submits documentation demonstrating good faith efforts.

Name of Bidder's firm: \_\_\_\_\_

\_\_\_\_\_State Registration No. \_\_\_\_\_

By \_\_\_\_\_  
(Signature)

\_\_\_\_\_  
Title

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**CITY OF PENSACOLA  
DBE LETTER OF INTENT**

Name of Bidder's firm:\_\_\_\_\_

Address:\_\_\_\_\_

City:\_\_\_\_\_ State:\_\_\_\_\_ Zip:\_\_\_\_\_

Name of DBE firm:\_\_\_\_\_

Address:\_\_\_\_\_

City:\_\_\_\_\_ State:\_\_\_\_\_ Zip:\_\_\_\_\_

Telephone No.:\_\_\_\_\_

Description of work to be performed by DBE firm:

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

The Bidder is committed to utilizing the above-named DBE firm for the work described above.  
The estimated dollar value of this work is \$\_\_\_\_\_.

**Affirmation**

The above-named DBE firm affirms that it will perform the portion of the contract for the estimated dollar value as stated above.

By \_\_\_\_\_  
(Signature) (Title)

If the Bidder does not receive award of the prime contract, any and all representations in this Letter of Intent and Affirmation shall be null and void.

(Submit this page for each DBE subcontractor.)

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**CITY OF PENSACOLA  
DBE EXHIBIT A**

These DISADVANTAGED BUSINESS DISCLOSURE REQUIREMENTS must be completed and submitted within **(10) calendar days after Bid Opening** by the apparent low, qualified, responsible, responsive DBE Bidder.

1. Name of Company (firm, etc.): \_\_\_\_\_

2. Principal Office: \_\_\_\_\_  
(Street address or P.O. Box)

\_\_\_\_\_  
(City, state, zip code)

3. Telephone No: \_\_\_\_\_

4. State in which you are authorized/registered to do business: \_\_\_\_\_

5. Nature of Business: \_\_\_\_\_

6. Years in business under the above business name: \_\_\_\_\_

7. Percent of disadvantaged ownership: \_\_\_\_\_  
(owned and/or controlled)

8. List the number of shares of stock issued and subscribed to as of this date:

a. \_\_\_\_\_

b. \_\_\_\_\_

9. List any loans between Owners and between Owners and third parties relative to the business entity: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

10. Identify ownership of your company (all) in the following terms:

| <u>Name/Position</u> | <u>Disadvantaged Status<br/>(if applicable)</u> | <u>% of<br/>Ownership</u> |
|----------------------|-------------------------------------------------|---------------------------|
| _____                |                                                 |                           |
| _____                |                                                 |                           |
| _____                |                                                 |                           |

11. Identify all stock options or other ownership options that are outstanding:

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12. Identify the company/corporation officers as to the following:

a. Name and Title: \_\_\_\_\_

Responsibilities: \_\_\_\_\_

b. Name and Title: \_\_\_\_\_

Responsibilities: \_\_\_\_\_

c. Name and Title: \_\_\_\_\_

Responsibilities: \_\_\_\_\_

13. Name and title of individual who decides what job the firm will take: \_\_\_\_\_

---

14. Name and title of individual who obtains/negotiates and signs for bonds and insurance: \_\_\_\_\_

---

15. Established bonding limitation: \_\_\_\_\_

16. Name, address and telephone number of Bonding Company:

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17. Type and quantity of equipment owned: \_\_\_\_\_

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18. Trade Reference: \_\_\_\_\_

19. Indicate if previously certified as a Disadvantaged Business Enterprise (DBE). By whom: \_\_\_\_\_

---

20. What percentage of work to be performed by your company on this project?

\_\_\_\_\_ (percent)  
\_\_\_\_\_ (location of project)  
\_\_\_\_\_ (dollar value)

21. What work will you perform on this project? \_\_\_\_\_

---

22. List five (5) of your most recent projects which demonstrate your qualifications for the work on this project:

| <u>Type Work</u> | <u>Agency/Company</u> | <u>Date</u> | <u>\$ Value</u> | <u>Location</u> |
|------------------|-----------------------|-------------|-----------------|-----------------|
|                  |                       |             |                 |                 |
|                  |                       |             |                 |                 |
|                  |                       |             |                 |                 |
|                  |                       |             |                 |                 |

23. Attach a copy of the following:

- a. Articles of Incorporation
- b. Business Charter
- c. Complete, current, audited (or unaudited) balance sheets
- d. Company's operating statement
- e. Certificate of minority status with the Florida Department of Transportation and/or the City of Pensacola, Florida.

ALL OF THE FOREGOING INFORMATION should be in affidavit form and should be subscribed and sworn to by the authorized representative of your business and should contain the following statement at the close thereof, above the signature.

The undersigned does hereby declare that the foregoing statements are true and correct and include all material information necessary to identify and explain the operation of \_\_\_\_\_ (company name) thereof. Further, the undersigned does covenant and agree to provide current, complete and accurate information regarding actual work performed, the payment therefore, and any proposed changes in any of the arrangements hereinabove stated and to permit the audit and examination of the books, records, and files of \_\_\_\_\_ (company name) by authorized representatives of the Owner.

I HEREBY DECLARE AND AFFIRM THAT I am a disadvantaged business as defined by the Department of Transportation/Federal Aviation Administration in the specifications for the work and that I will provide information requested by the Owner to document this fact. It is recognized and acknowledged that the statements herein are being given under oath and any material misrepresentation will be grounds for terminating any contract which may be awarded in reliance hereon.

I SOLEMNLY DECLARE AND AFFIRM UNDER THE PENALTIES OF PERJURY THAT THE CONTENTS OF THE FOREGOING ARE TRUE AND CORRECT, AND THAT I AM AUTHORIZED, ON BEHALF OF THE ABOVE FIRM TO MAKE THIS AFFIDAVIT.

\_\_\_\_\_  
(date)

\_\_\_\_\_  
(Affiant-Company's Representative)

STATE OF \_\_\_\_\_

COUNTY OF \_\_\_\_\_

On this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_, before me personally appeared \_\_\_\_\_, known to me to be the person described in the foregoing affidavit and acknowledged that she/he executed the same in the capacity therein stated and for the purpose therein contained.

IN WITNESS THEREOF, I hereunto set my hand and official seal.

NOTARY SEAL

\_\_\_\_\_  
NOTARY PUBLIC

\_\_\_\_\_  
My commission expires

**END OF DBE DOCUMENTS**



## CITY OF PENSACOLA INSURANCE REQUIREMENTS FOR THIS PROJECT

### BIDDER'S PRESENT INSURANCE COVERAGE

#### TYPE OF INSURANCE

|                                                                                                  | <u>Coverage Limits<br/>Each Occurrence</u>     | <u>Coverage Limits<br/>Aggregate</u>     |
|--------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------|
| <u>General Liability:</u>                                                                        |                                                |                                          |
| Bodily Injury, Personal Injury,<br>Sickness, Disease or Death                                    | \$                                             | \$                                       |
| Property Damage                                                                                  | \$                                             | \$                                       |
| BI & PD Combined                                                                                 | \$                                             | \$                                       |
| Does policy provide blanket broad form<br>contractual liability coverage? (Check<br>One)         | Yes _____ No _____                             |                                          |
| Does policy provide coverage for<br>completed operations? (Check One)                            | Yes _____ No _____                             |                                          |
| Does policy have standard collapse<br>and underground hazards exclusions<br>removed? (Check one) | Yes _____ No _____                             |                                          |
| <br><u>Automobile Liability:</u>                                                                 | <br><u>Coverage Limits<br/>Each Occurrence</u> | <br><u>Coverage Limits<br/>Aggregate</u> |
| Bodily Injury<br>(Per person)                                                                    | \$                                             |                                          |
| Bodily Injury<br>(Per Accident)                                                                  | \$                                             |                                          |
| Property Damage                                                                                  | \$                                             |                                          |
| BI & PD Combined                                                                                 | \$                                             |                                          |
| Does policy cover the following<br>vehicles:                                                     |                                                |                                          |
| Owned? (Check One)                                                                               | Yes _____ No _____                             |                                          |
| Non-Owned? (Check One)                                                                           | Yes _____ No _____                             |                                          |
| Hired? (Check One)                                                                               | Yes _____ No _____                             |                                          |

### INSURANCE REQUIREMENTS FOR THIS PROJECT

#### TYPE OF INSURANCE

|                                                                                                 | <u>Coverage Limits<br/>Each Occurrence</u> | <u>Coverage Limits<br/>Aggregate</u>     | <u>Extra Premium<br/>Cost (if any)</u>     |
|-------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------|--------------------------------------------|
| <u>General Liability:</u>                                                                       |                                            |                                          |                                            |
| Bodily Injury, Personal Injury, Sickness,<br>Disease or Death                                   | \$3,000,000.00                             | \$3,000,000.00                           | \$                                         |
| Property Damage                                                                                 | \$3,000,000.00                             | \$3,000,000.00                           | \$                                         |
| BI & PD Combined                                                                                |                                            | \$3,000,000.00                           | \$                                         |
| Policy must provide blanket broad form<br>contractual liability coverage.                       |                                            |                                          | \$                                         |
| Policy must provide coverage for premises,<br>operations, products and completed<br>operations. | \$3,000,000.00                             | \$3,000,000.00                           | \$                                         |
| Policy must have standard collapse and<br>underground hazards exclusions removed.               | \$3,000,000.00                             | \$3,000,000.00                           | \$                                         |
| Independent Contractors                                                                         | \$3,000,000.00                             | \$3,000,000.00                           | \$                                         |
| <br><u>Automobile Liability:</u>                                                                | <br><u>Coverage Limits<br/>CSL</u>         | <br><u>Coverage Limits<br/>Aggregate</u> | <br><u>Extra Premium<br/>Cost (if any)</u> |
| Bodily Injury and Property Damage<br>Combined                                                   | \$3,000,000.00                             |                                          |                                            |
| Policy must cover the following:                                                                |                                            |                                          |                                            |
| Owned Vehicles                                                                                  | \$3,000,000.00                             |                                          | \$                                         |
| Non-Owned Vehicles                                                                              | \$3,000,000.00                             |                                          | \$                                         |
| Hired Vehicles                                                                                  | \$3,000,000.00                             |                                          | \$                                         |

| <u>BIDDER'S PRESENT INSURANCE COVERAGE</u>                                         |                                            |                                      | <u>INSURANCE REQUIREMENTS FOR THIS PROJECT</u> |                                            |                                      |                                            |
|------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------|------------------------------------------------|--------------------------------------------|--------------------------------------|--------------------------------------------|
| <u>TYPE OF INSURANCE</u>                                                           | <u>Coverage Limits<br/>Each Occurrence</u> | <u>Coverage Limits<br/>Aggregate</u> | <u>TYPE OF INSURANCE</u>                       | <u>Coverage Limits<br/>Each Occurrence</u> | <u>Coverage Limits<br/>Aggregate</u> | <u>Extra<br/>Premium<br/>Cost (if any)</u> |
| <u>Excess Liability:</u>                                                           |                                            |                                      | <u>Excess Liability:</u>                       |                                            |                                      |                                            |
| Umbrella Form                                                                      | \$                                         | \$                                   | Following-form Umbrella Policy                 |                                            | \$3,000,000.00                       | \$                                         |
| Other than Umbrella Form                                                           | \$                                         | \$                                   |                                                |                                            |                                      |                                            |
| Does policy follow form with primary<br>general liability coverage? (Check<br>One) | Yes ____ No ____                           |                                      |                                                |                                            |                                      |                                            |

| <u>Worker's Compensation</u>        | <u>Coverage Limits</u>                                                                                                          | <u>Worker's Compensation</u>     | <u>Coverage Limits Each Occurrence</u>                                                                                                                                                                                                                                           |                                         |                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------|
| Each Accident                       | \$                                                                                                                              | Must meet minimum requirements   | Minimum Statutory Limits                                                                                                                                                                                                                                                         |                                         | \$                                  |
| Disease Policy Limit                | \$                                                                                                                              | Accident                         | \$100,00.00                                                                                                                                                                                                                                                                      |                                         |                                     |
| Disease – Each Employee             | \$                                                                                                                              | Disease                          | \$100,000.00                                                                                                                                                                                                                                                                     | \$500,000.00                            |                                     |
|                                     |                                                                                                                                 |                                  | <u>Coverage Limits Each Occurrence</u>                                                                                                                                                                                                                                           | <u>Coverage Limits Aggregate</u>        | <u>Extras Premium Cost (if any)</u> |
| <u>Miscellaneous Requirements:</u>  | <u>Coverage Limits Each Occurrence</u>                                                                                          | <u>Coverage Limits Aggregate</u> | <u>Miscellaneous Requirements:</u>                                                                                                                                                                                                                                               | <u>Coverage Limits Each Occurrence:</u> | <u>Coverage Limits Aggregate:</u>   |
|                                     |                                                                                                                                 |                                  | <u>Companies Affording Coverage:</u><br>All insurance policies must be written by Insurance carriers duly authorized to do business in the State of Florida having a B+ IX rating or better by the most recent edition of the <u>Best's Key Rating Guide, Property-Casualty.</u> |                                         | \$                                  |
|                                     |                                                                                                                                 |                                  | All general liability and automobile liability insurance policies must name the City and Consultant as an additional insured.                                                                                                                                                    |                                         |                                     |
| <u>Companies Affording Coverage</u> | <u>Ratings of Companies Affording Coverage (According to most recent edition of Best's Key Rating Guide, Property Casualty)</u> |                                  | The City shall be provided 30 days prior notice in the event of cancellation, restrictive amendment, or non-renewal of Contractor's insurance coverages.                                                                                                                         |                                         | \$                                  |
| Company A:                          |                                                                                                                                 |                                  |                                                                                                                                                                                                                                                                                  |                                         |                                     |
| Company B:                          |                                                                                                                                 |                                  |                                                                                                                                                                                                                                                                                  |                                         | \$                                  |
| Company C:                          |                                                                                                                                 |                                  | Notice of 10 days if policy is cancelled for non-payment of premium.                                                                                                                                                                                                             |                                         | \$                                  |
| Company D:                          |                                                                                                                                 |                                  |                                                                                                                                                                                                                                                                                  |                                         |                                     |
|                                     |                                                                                                                                 |                                  | TOTAL ADDITIONAL ACTUAL PREMIUM COSTS                                                                                                                                                                                                                                            |                                         |                                     |

**END INSURANCE REQUIREMENTS**

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**CITY OF PENSACOLA  
BID SCHEDULE**

**AIRPORT: PENSACOLA INTERNATIONAL AIRPORT**  
**PROJECT: TAXIWAY A REHABILITATION**  
**FAA AIP No. TBD**  
**CITY OF PENSACOLA BID No. 25-029**

| BASE BID |            |                                                                                                            |                  |      |                       |                       |
|----------|------------|------------------------------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF.  | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                                               | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 1        | C-100-14.1 | Contractor Quality Control Program (CQCP)<br>at _____ dollars<br>_____ cents                               | 1                | LS   | \$ _____              | \$ _____              |
| 2        | C-102-5.1  | Erosion and Turbidity Control<br>at _____ dollars<br>_____ cents                                           | 1                | LS   | \$ _____              | \$ _____              |
| 3        | C-105-1    | Mobilization (MOBILIZATION IS LIMITED TO NO MORE THAN 10% OF BID PRICE)<br>at _____ dollars<br>_____ cents | 1                | LS   | \$ _____              | \$ _____              |
| 4        | C-106-1    | Maintenance of Traffic<br>at _____ dollars<br>_____ cents                                                  | 1                | LS   | \$ _____              | \$ _____              |
| 5        | P-101-5.1a | Cold Milling (0.5 Inches)<br>at _____ dollars<br>_____ cents                                               | 22,267           | SY   | \$ _____              | \$ _____              |

| BASE BID |            |                                                                  |                  |      |                       |                       |
|----------|------------|------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF.  | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                     | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 6        | P-101-5.1b | Cold Milling (0.5-2.5 Inches)<br>at _____ dollars<br>_____ cents | 113,590          | SY   | \$ _____              | \$ _____              |
| 7        | P-101-5.1c | Cold Milling (3 Inches)<br>at _____ dollars<br>_____ cents       | 36,867           | SY   | \$ _____              | \$ _____              |
| 8        | P-151-4.1  | Clearing and Grubbing<br>at _____ dollars<br>_____ cents         | 1                | AC   | \$ _____              | \$ _____              |
| 9        | P-152-4.1  | Unclassified Excavation<br>at _____ dollars<br>_____ cents       | 1,350            | CY   | \$ _____              | \$ _____              |
| 10       | P-152-4.2  | Embankment in Place<br>at _____ dollars<br>_____ cents           | 1,350            | CY   | \$ _____              | \$ _____              |
| 11       | P-154-5.1  | Subbase Course<br>at _____ dollars<br>_____ cents                | 20,247           | SY   | \$ _____              | \$ _____              |
| 12       | P-209-5.1  | Crushed Aggregate Base Course<br>at _____ dollars<br>_____ cents | 20,247           | SY   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                   |                  |      |                       |                       |
|----------|-----------|-------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                      | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 13       | P-401-8.1 | Asphalt Surface Course<br>at _____ dollars<br>_____ cents         | 25,498           | TN   | \$ _____              | \$ _____              |
| 14       | P-403-8.1 | Asphalt Base / Leveling Course<br>at _____ dollars<br>_____ cents | 12,287           | TN   | \$ _____              | \$ _____              |
| 15       | P-602-5.1 | EMULSIFIED ASPHALT PRIME COAT<br>at _____ dollars<br>_____ cents  | 2,817            | GAL  | \$ _____              | \$ _____              |
| 16       | P-603-5.1 | EMULSIFIED ASPHALT TACK COAT<br>at _____ dollars<br>_____ cents   | 9,332            | GAL  | \$ _____              | \$ _____              |
| 17       | P-620-5.1 | RUNWAY AND TAXIWAY MARKING<br>at _____ dollars<br>_____ cents     | 1                | LS   | \$ _____              | \$ _____              |
| 18       | T-901-5.1 | SEEDING<br>at _____ dollars<br>_____ cents                        | 25,000           | SY   | \$ _____              | \$ _____              |
| 19       | T-904-5.1 | SODDING<br>at _____ dollars<br>_____ cents                        | 25,000           | SY   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                                                  |                  |      |                       |                       |
|----------|-----------|------------------------------------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                                                     | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 20       | D-751-5.1 | 10'x10' MANHOLE W/ J-BOTTOM<br>at _____ dollars<br>_____ cents                                                   | 1                | EA   | \$ _____              | \$ _____              |
| 21       | D-701-5.1 | 84", CLASS V, REINFORCED CONCRETE PIPE<br>at _____ dollars<br>_____ cents                                        | 433              | LF   | \$ _____              | \$ _____              |
| 22       | 'SP-104   | TEMPORARY POWER & TEMPORARY AIRFIELD LIGHTING/SIGNAGE/NAVIGATIONAL FACILITIES<br>at _____ dollars<br>_____ cents | 1                | LS   | \$ _____              | \$ _____              |
| 23       | SP-105    | Electrical Demolition<br>at _____ dollars<br>_____ cents                                                         | 16               | EA   | \$ _____              | \$ _____              |
| 24       | SP-111-1  | SHORT CIRCUIT/ COORDINATION/ DEVICE EVALUATION/ ARC FLASH ANALYSIS<br>at _____ dollars<br>_____ cents            | 1                | LS   | \$ _____              | \$ _____              |
| 25       | SP-111-3  | FAA FLIGHT CHECK OF RUNWAY 8-26 LIGHTING<br>at _____ dollars<br>_____ cents                                      | 1                | AL   | \$ _____              | \$ _____              |



| BASE BID |           |                                                                                                                                |                  |      |                       |                       |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                                                                   | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 26       | L-108     | 1/C L-824-TYPE C UNSHIELDED #8 AWG 5 KV STRANDED COPPER CABLE, INSTALLED IN DUCT OR CONDUIT<br>at _____ dollars<br>_____ cents | 120,000          | LF   | \$ _____              | \$ _____              |
| 27       | L-108     | 1/C #2 AWG SOLID COPPER COUNTERPOISE WIRE, INSTALLED OVER DUCT OR CONDUIT<br>at _____ dollars<br>_____ cents                   | 55,700           | LF   | \$ _____              | \$ _____              |
| 28       | L-108     | 0.75" DIAMETER BY 10.00' LONG COPPER CLAD STEEL SECTIONAL GROUND ROD<br>at _____ dollars<br>_____ cents                        | 900              | LF   | \$ _____              | \$ _____              |
| 29       | L-109     | FIELD LIGHTNING ARRESTOR ASSEMBLY<br>at _____ dollars<br>_____ cents                                                           | 45               | EA   | \$ _____              | \$ _____              |
| 30       | L-109     | AIRFIELD LIGHTING CONTROL SYSTEM MODIFICATIONS: VENDOR<br>at _____ dollars<br>_____ cents                                      | 1                | AL   | \$ _____              | \$ _____              |
| 31       | L-109     | AIRFIELD LIGHTING CONTROL SYSTEM MODIFICATIONS: CONTRACTOR<br>at _____ dollars<br>_____ cents                                  | 1                | LS   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                      |                  |      |                       |                       |
|----------|-----------|--------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                         | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 32       | L-109     | AIRFIELD ELECTRICAL VAULT MODIFICATIONS<br>at _____ dollars<br>_____ cents           | 1                | LS   | \$ _____              | \$ _____              |
| 33       | L-109     | 10KW L-829 CONSTANT CURRENT REGULATOR-5 STEP-480V<br>at _____ dollars<br>_____ cents | 1                | EA   | \$ _____              | \$ _____              |
| 34       | L-109     | 15KW L-829 CONSTANT CURRENT REGULATOR-5 STEP-480V<br>at _____ dollars<br>_____ cents | 1                | EA   | \$ _____              | \$ _____              |
| 35       | L-110     | 1 WAY 2" SCHEDULE 40 PVC DIRECT EARTH BURIED DUCT<br>at _____ dollars<br>_____ cents | 22,000           | LF   | \$ _____              | \$ _____              |
| 36       | L-110     | 2 WAY 2" SCHEDULE 40 PVC DIRECT EARTH BURIED DUCT<br>at _____ dollars<br>_____ cents | 800              | LF   | \$ _____              | \$ _____              |
| 37       | L-110     | 4 WAY 2" SCHEDULE 40 PVC DIRECT EARTH BURIED DUCT<br>at _____ dollars<br>_____ cents | 500              | LF   | \$ _____              | \$ _____              |
| 38       | L-110     | 6 WAY 2" SCHEDULE 40 PVC DIRECT EARTH BURIED DUCT<br>at _____ dollars<br>_____ cents | 500              | LF   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                       |                  |      |                       |                       |
|----------|-----------|---------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                          | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 39       | L-110     | 12 WAY 2" SCHEDULE 40 PVC DIRECT EARTH BURIED DUCT<br>at _____ dollars<br>_____ cents | 800              | LF   | \$ _____              | \$ _____              |
| 40       | L-110     | 14 WAY 2" SCHEDULE 40 PVC DIRECT EARTH BURIED DUCT<br>at _____ dollars<br>_____ cents | 500              | LF   | \$ _____              | \$ _____              |
| 41       | L-110     | 1 WAY 2" SCHEDULE 40 PVC CONCRETE ENCASED DUCT<br>at _____ dollars<br>_____ cents     | 22,000           | LF   | \$ _____              | \$ _____              |
| 42       | L-110     | 2 WAY 2" SCHEDULE 40 PVC CONCRETE ENCASED DUCT<br>at _____ dollars<br>_____ cents     | 2,000            | LF   | \$ _____              | \$ _____              |
| 43       | L-110     | 3 WAY 2" SCHEDULE 40 PVC CONCRETE ENCASED DUCT<br>at _____ dollars<br>_____ cents     | 1,000            | LF   | \$ _____              | \$ _____              |
| 44       | L-110     | 4 WAY 2" SCHEDULE 40 PVC CONCRETE ENCASED DUCT<br>at _____ dollars<br>_____ cents     | 4,000            | LF   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                   |                  |      |                       |                       |
|----------|-----------|-----------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                      | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 45       | L-110     | 6 WAY 2" SCHEDULE 40 PVC CONCRETE ENCASED DUCT<br>at _____ dollars<br>_____ cents | 800              | LF   | \$ _____              | \$ _____              |
| 46       | L-110     | 8 WAY 2" SCHEDULE 40 PVC CONCRETE ENCASED DUCT<br>at _____ dollars<br>_____ cents | 800              | LF   | \$ _____              | \$ _____              |
| 47       | L-110     | BASE CAN DRAIN WELL<br>at _____ dollars<br>_____ cents                            | 4                | EA   | \$ _____              | \$ _____              |
| 48       | L-115     | L-867D JUNCTION CAN (NTAC)<br>at _____ dollars<br>_____ cents                     | 18               | EA   | \$ _____              | \$ _____              |
| 49       | L-115     | JUNCTION CAN PLAZA – TWO "D" CANS<br>at _____ dollars<br>_____ cents              | 16               | EA   | \$ _____              | \$ _____              |
| 50       | L-115     | JUNCTION CAN PLAZA – THREE "D" CANS<br>at _____ dollars<br>_____ cents            | 4                | EA   | \$ _____              | \$ _____              |
| 51       | L-115     | JUNCTION CAN PLAZA – FOUR "D" CANS<br>at _____ dollars<br>_____ cents             | 10               | EA   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                      |                  |      |                       |                       |
|----------|-----------|--------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                         | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 52       | L-115     | JUNCTION CAN PLAZA – SIX "D" CANS<br>at _____ dollars<br>_____ cents                 | 3                | EA   | \$ _____              | \$ _____              |
| 53       | L-115     | JUNCTION CAN PLAZA – EIGHT "D" CANS<br>at _____ dollars<br>_____ cents               | 2                | EA   | \$ _____              | \$ _____              |
| 54       | L-116     | JUNCTION CAN PLAZA – TWELVE "D" CANS<br>at _____ dollars<br>_____ cents              | 1                | EA   | \$ _____              | \$ _____              |
| 55       | L-116     | JUNCTION CAN PLAZA – FOURTEEN "D" CANS<br>at _____ dollars<br>_____ cents            | 1                | EA   | \$ _____              | \$ _____              |
| 56       | L-125     | L-804(L) ELEVATED RUNWAY GUARD LIGHT YELLOW (R)<br>at _____ dollars<br>_____ cents   | 2                | EA   | \$ _____              | \$ _____              |
| 57       | L-125     | L-850A RUNWAY CENTERLINE LIGHT - RW 17-35 - (RE)<br>at _____ dollars<br>_____ cents  | 11               | EA   | \$ _____              | \$ _____              |
| 58       | L-125     | L-850C RUNWAY EDGE LIGHT- RW 17-35 RELOCATED (RE)<br>at _____ dollars<br>_____ cents | 2                | EA   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                                   |                  |      |                       |                       |
|----------|-----------|---------------------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                                      | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 59       | L-125     | L-850C(L) RUNWAY EDGE LIGHT-CLEAR/CLEAR (A)<br>at _____ dollars<br>_____ cents                    | 2                | EA   | \$ _____              | \$ _____              |
| 60       | L-125     | L-850C(L) RUNWAY EDGE LIGHT-CLEAR/CLEAR (NE)<br>at _____ dollars<br>_____ cents                   | 2                | EA   | \$ _____              | \$ _____              |
| 61       | L-125     | L-850C(L) RUNWAY EDGE LIGHT-CLEAR/YELLOW (A)<br>at _____ dollars<br>_____ cents                   | 3                | EA   | \$ _____              | \$ _____              |
| 62       | L-125     | L-858(L) SIGN – SINGLE/DOUBLE FACE, LED, SIZE 3, - 1<br>MODULE<br>at _____ dollars<br>_____ cents | 4                | LS   | \$ _____              | \$ _____              |
| 63       | L-125     | L-858(L) SIGN – SINGLE/DOUBLE FACE, LED, SIZE 3, - 2<br>MODULE<br>at _____ dollars<br>_____ cents | 13               | LS   | \$ _____              | \$ _____              |
| 64       | L-125     | L-858(L) SIGN – SINGLE/DOUBLE FACE, LED, SIZE 3, - 3<br>MODULE<br>at _____ dollars<br>_____ cents | 11               | EA   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                                |                  |      |                       |                       |
|----------|-----------|------------------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                                   | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 65       | L-125     | L-858B(L) RUNWAY DISTANCE REMAINING SIGN, LED, SIZE 4<br>at _____ dollars<br>_____ cents       | 5                | EA   | \$ _____              | \$ _____              |
| 66       | L-125     | L-861T(L) OMNIDIRECTIONAL, BLUE, LED TAXIWAY EDGE LIGHT (A)<br>at _____ dollars<br>_____ cents | 31               | EA   | \$ _____              | \$ _____              |
| 67       | L-125     | L-861T(L) OMNIDIRECTIONAL, BLUE, LED TAXIWAY EDGE LIGHT (N)<br>at _____ dollars<br>_____ cents | 116              | LS   | \$ _____              | \$ _____              |
| 68       | L-125     | L-861T(L) OMNIDIRECTIONAL, BLUE, LED TAXIWAY EDGE LIGHT (T)<br>at _____ dollars<br>_____ cents | 32               | LS   | \$ _____              | \$ _____              |
| 69       | L-125     | L-862(L) RUNWAY EDGE LIGHT-CLEAR/CLEAR (A)<br>at _____ dollars<br>_____ cents                  | 22               | EA   | \$ _____              | \$ _____              |
| 70       | L-125     | L-862(L) RUNWAY EDGE LIGHT-CLEAR/YELLOW (A)<br>at _____ dollars<br>_____ cents                 | 41               | EA   | \$ _____              | \$ _____              |

| BASE BID |           |                                                                                     |                  |      |                       |                       |
|----------|-----------|-------------------------------------------------------------------------------------|------------------|------|-----------------------|-----------------------|
| ITEM NO. | SPEC REF. | ITEM DESCRIPTION AND UNIT PRICE BID IN WORDS                                        | APPROX. QUANTITY | UNIT | UNIT PRICE IN NUMBERS | TOTAL AMOUNT PER ITEM |
| 71       | L-125     | L-862E(L) RUNWAY THRESHOLD LIGHT - GREEN/RED (A)<br>at _____ dollars<br>_____ cents | 16               | EA   | \$ _____              | \$ _____              |

TOTAL AMOUNT OF BASE BID (IN NUMBERS) \$ \_\_\_\_\_

TOTAL AMOUNT OF BASE BID (IN WORDS) \$ \_\_\_\_\_



| DEDUCTIVE ALTERNATE |            |                                                                              |        |    |          |          |
|---------------------|------------|------------------------------------------------------------------------------|--------|----|----------|----------|
| 72                  | P-100-14.1 | CONTRACTOR QUALITY CONTROL PROGRAM (CQCP)<br>at _____ dollars<br>_____ cents | 1      | LS | \$ _____ | \$ _____ |
| 73                  | P-102-5.1  | EROSION AND TURBIDITY CONTROL<br>at _____ dollars<br>_____ cents             | 1      | LS | \$ _____ | \$ _____ |
| 74                  | P-105-1    | MOBILIZATION<br>at _____ dollars<br>_____ cents                              | 1      | LS | \$ _____ | \$ _____ |
| 75                  | P-106-1    | MAINTENANCE OF TRAFFIC<br>at _____ dollars<br>_____ cents                    | 1      | LS | \$ _____ | \$ _____ |
| 76                  | P-101-5.1  | COLD MILLING (0.5-2.5 INCHES)<br>at _____ dollars<br>_____ cents             | 44,309 | SY | \$ _____ | \$ _____ |
| 77                  | P-154-5.1  | SUBBASE COURSE<br>at _____ dollars<br>_____ cents                            | 2,236  | SY | \$ _____ | \$ _____ |
| 78                  | P-209-5.1  | CRUSHED AGGREGATE BASE COURSE<br>at _____ dollars<br>_____ cents             | 2,236  | SY | \$ _____ | \$ _____ |

| DEDUCTIVE ALTERNATE |           |                                                                   |       |     |          |          |
|---------------------|-----------|-------------------------------------------------------------------|-------|-----|----------|----------|
| 79                  | P-401-8.1 | ASPHALT SURFACE COURSE<br>at _____ dollars<br>_____ cents         | 6,139 | TN  | \$ _____ | \$ _____ |
| 80                  | P-403-8.1 | ASPHALT BASE / LEVELING COURSE<br>at _____ dollars<br>_____ cents | 1,623 | TN  | \$ _____ | \$ _____ |
| 81                  | P-602-5.1 | EMULSIFIED ASPHALT PRIME COAT<br>at _____ dollars<br>_____ cents  | 250   | GAL | \$ _____ | \$ _____ |
| 82                  | P-603-5.1 | EMULSIFIED ASPHALT TACK COAT<br>at _____ dollars<br>_____ cents   | 1,000 | GAL | \$ _____ | \$ _____ |
| 83                  | P-620-5.1 | RUNWAY AND TAXIWAY MARKING<br>at _____ dollars<br>_____ cents     | 1     | LS  | \$ _____ | \$ _____ |
| 84                  | P-901-5.1 | SEEDING<br>at _____ dollars<br>_____ cents                        | 2,500 | SY  | \$ _____ | \$ _____ |
| 85                  | P-904-5.1 | SODDING<br>at _____ dollars<br>_____ cents                        | 2,500 | SY  | \$ _____ | \$ _____ |

**TOTAL AMOUNT OF DEDUCTIVE ALTERNATE (IN NUMBERS) \$** \_\_\_\_\_

**TOTAL AMOUNT OF DEDUCTIVE ALTERNATE (IN WORDS) \$** \_\_\_\_\_  
\_\_\_\_\_

Name of Bidder (Typed or Printed): \_\_\_\_\_

Signature of Bidder (Same as Proposal Form): \_\_\_\_\_

Title: \_\_\_\_\_

**END OF BID SCHEDULE**

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**CONTRACT BETWEEN CITY OF PENSACOLA AND  
[xxxCONTRACTORxxx]  
BASED UPON INVITATION TO BID #\_\_\_\_\_**

**THIS CONTRACT (“Contract”)** is made this \_\_\_\_ day of \_\_\_\_\_, 20\_\_, by and between the City of Pensacola (“City”), a Florida municipal corporation created and existing under the laws of the State of Florida, located at 222 W. Main Street, Pensacola, Florida 32502, and \_\_\_\_\_, (“Contractor”), a corporation authorized to do business in Florida, located at \_\_\_\_\_, (the City and Contractor collectively referred to hereinafter as the “Parties”).

**WITNESSETH:**

**WHEREAS**, the City solicited for Invitation to Bid #\_\_\_\_\_, on \_\_\_\_\_, 20\_\_ (“Invitation to Bid”), as described in Project Manual/Specifications for \_\_\_\_\_, Bid #\_\_\_\_\_, as modified by any addendum to the Project Manual, all as attached hereto as Exhibit A and incorporated herein by this reference (collectively referred to hereinafter as the “Bid Documents”); and

**WHEREAS**, in response to the Bid Documents, the Contractor submitted to the City a proposal dated \_\_\_\_\_, 20\_\_, (“Proposal”) attached hereto as Exhibit B and incorporated herein by this reference; and

**WHEREAS**, the City has awarded the Contract to the Contractor; and

**WHEREAS**, the Parties desire the Contractor to perform the agreement as described in the Bid Documents and the Proposal and pursuant to the terms and conditions of this Contract; and

**WHEREAS**, the Parties desire to enter into this Contract;

**NOW, THEREFORE**, in consideration of the work to be performed and the payment for the performance of the work, of the mutual covenants and benefits contained herein, and for other good and valuable consideration, the Parties agree as follows:

**Section 1. Recitals.**

The recitals contained above are true and correct and are incorporated into this Contract.

**Section 2. Contractor’s Obligations.**

The Contractor shall perform all work and services described in, and in accordance with, the Contract. The Contractor warrants that all equipment, materials, and workmanship furnished, whether furnished by Contractor or its subcontractors or sub-suppliers, will comply with the Contract and any City specifications, drawings, and other descriptions

supplied or adopted. The Contractor further warrants that the supplies and workmanship will be new, fit, and sufficient for the purpose for which they are intended, of good materials, design, and workmanship, and free from defects or failure. The City or its duly authorized representative shall at all times have full opportunity to inspect the materials to be furnished and the work to be done under this Contract. The Contractor shall comply with all applicable federal, state, and local laws, ordinances, rules, and regulations pertaining to the performance of this Contract. The Contractor is responsible for and shall indemnify the City against all damage or loss caused by fire, theft, or otherwise to materials, tools, equipment, and consumables left on City property by the Contractor.

**Section 3. Term of Contract.**

Subject to the right of termination for cause or convenience, the term of this Contract shall be as specified in the attached Quote Documents and Proposal.

**Section 4. Payment.**

The Contractor agrees to perform all work and services in Section 2 and to furnish all necessary labor, materials, equipment, machinery, tools, apparatus, and means of transportation related to such work and services at Contractor's sole cost and expense, in consideration of the total amount of \_\_\_\_\_ (\$\_\_\_\_\_) to be paid by the City in accordance with the Contract upon the complete performance by Contractor based on unit prices if applicable, or based on partial payments approved by the City, only after written acceptance by the City pursuant to the Contract, and such payment in accordance with the Florida Prompt Payment Act. In the event that the Contractor does not fully perform its obligations under the Contract, the City reserves the right to withhold payments for work not performed, to engage an alternative contractor to complete work not performed, and to withhold such amounts as may be required to hold the City harmless from any claims or damages, direct, indirect or consequential, that may be sustained on account of the Contractor's acts or omissions in the performance of this Contract.

**Section 5. Bond.**

Is a bond required? ( ) Yes ( ) No

If yes: Contractor shall provide all bond(s) as required in the Contract. Should the City in the City's sole discretion at any time deem any of the sureties upon such bond to be unsatisfactory or if for any reason such bond shall cease to be adequate security for the City, the Contractor shall within five (5) days of written notice from the City furnish a new or additional bond in full sum and satisfactory to the City. No payment shall be deemed to be due or to be made to the Contractor unless and until such new or additional bond shall be furnished and approved in writing by the City. The premium and all expenses associated with such new or additional bond shall be paid by, and the sole responsibility of, the Contractor.

**Section 6. Performance Schedule.**

The Contractor shall commence and complete all work and services pursuant to the Contract.

**Section 7. Necessary Approvals.**

Contractor shall procure all permits, licenses, and certificates and any approvals in performance and completion of this Contract as may be required by federal, state, and local laws, ordinances, rules, and regulations, and in accordance with the Contract.

**Section 8. No Waiver.**

No waiver, alterations, consent or modification of any of the provisions of the Contract shall be binding unless in writing and signed by the Mayor or his/her designee.

**Section 9. Governing Law.**

This Contract is governed and construed in accordance with the laws of the State of Florida. The law of the State of Florida shall be the law applied in the resolution of any claim, actions, or proceedings arising out of this Contract.

**Section 10. Venue.**

Venue for any claim, actions, or proceedings arising out of this Contract shall be Escambia County, Florida.

**Section 11. No Discrimination.**

Contractor shall not discriminate on the basis of any class protected by federal, state, or local law in the performance of this Contract.

**Section 12. Assignment.**

The rights and privileges conferred by this Contract shall not be assigned or transferred without the written consent of the City, which consent shall not be unreasonably withheld.

**Section 13. No Other Agreements.**

The Parties agree the Contract contains all the terms and conditions agreed upon by the Parties. No other agreements, oral or otherwise, regarding the subject matter of this Contract shall be deemed to exist or to bind either Party.

**Section 14. Remedies for Failure to Perform or Breach of Contract.**

The City reserves the right to seek all remedies available under law in the event of a failure to perform or other breach of this Contract by the Contractor, and the failure of the City to employ a particular remedy shall not be regarded by the Parties as a waiver of that or any other available remedy.

**Section 15. Termination for Convenience.**

The City may terminate this Contract without cause upon thirty (30) days prior written notice.

**Section 16. Public Records Act.**

The parties acknowledge and agree to fulfill all obligations respecting required contract provisions in any contract entered into or amended after July 1, 2016, in full compliance pursuant to Section 119.0701, *Florida Statutes*, and obligations respecting termination of a contract for failure to provide public access to public records. The parties expressly

agree specifically that the contracting parties hereto shall comply with the requirements within Attachment "A" attached hereto and incorporated by reference.

**Section 17. Mandatory Use of E-Verify System.**

In compliance with the provisions of F.S. 448.095, the parties to this contract and any subcontractors engaged in the performance of this contract hereby certify that they have registered with and shall use the E-Verify system of the United States Department of Homeland Security to verify the work authorization status of all newly hired employees, within the meaning of the statute.

**Section 18. Federal Required Language**

Since the parties anticipate that federal funding will be applied to this Agreement, federal contract clauses contained in Attachment "B" must be complied with, where applicable, in addition to the clauses already mentioned.

**IN WITNESS WHEREOF**, the parties hereto have caused this Contract to be executed and sealed the day and year first above written.

**CONTRACTOR**

**CITY OF PENSACOLA, FLORIDA**

\_\_\_\_\_  
(Contractor's Name)

\_\_\_\_\_  
Mayor, D. C. Reeves

By \_\_\_\_\_  
President

Attest: \_\_\_\_\_  
City Clerk, Ericka L. Burnett

\_\_\_\_\_  
(Printed President's Name)

Approved as to Substance:

Attest \_\_\_\_\_  
Corporate Secretary

\_\_\_\_\_  
Department Director

(CORPORATE SEAL)

Legal in form and execution:

\_\_\_\_\_  
City Attorney



## **Attachment "A"**

**PUBLIC RECORDS:** Contractor shall comply with Chapter 119, Florida Statutes. Specifically, Contractor shall:

- A.** Keep and maintain public records required by the City to perform the service.
- B.** Upon request from the City's custodian of public records, provide the City with a copy of the requested records or allow the records to be inspected or copied within a reasonable time at a cost that does not exceed the cost provided in Chapter 119, Florida Statutes, or as otherwise provided by law.
- C.** Ensure that public records that are exempt or confidential and exempt from public records disclosure requirements are not disclosed except as authorized by law for the duration of the Contract term and following the completion of the Contract if Contractor does not transfer the records to the City.
- D.** Upon completion of the Contract, transfer, at no cost, to the City, all public records in possession of Contractor or keep and maintain public records required by the City to perform the service. If Contractor transfers all public records to the City upon completion of the Contract, Contractor shall destroy any duplicate public records that are exempt or confidential and exempt from public records disclosure requirements. If Contractor keeps and maintains public records upon completion of the Contract, Contractor shall meet all applicable requirements for retaining public records. All records stored electronically must be provided to the City, upon request of the City's custodian of public records, in a format that is compatible with the information technology systems of the City.

Failure by Contractor to comply with Chapter 119, Florida Statutes, shall be grounds for immediate unilateral cancellation of this Contract by the City.

**IF CONTRACTOR HAS QUESTIONS REGARDING THE APPLICATION OF CHAPTER 119, FLORIDA STATUTES, TO THE CONTRACTOR'S DUTY TO PROVIDE PUBLIC RECORDS RELATING TO THIS CONTRACT, CONTACT THE PUBLIC RECORDS COORDINATOR AT:**

**THE OFFICE OF THE CITY CLERK, (850) 435-1715**

**[PUBLCRECORDS@CITYOFPENSACOLA.COM](mailto:PUBLCRECORDS@CITYOFPENSACOLA.COM)**

**222 WEST MAIN STREET, PENSACOLA, FL 32502**

## ATTACHMENT "B"

### FEDERAL CONDITIONS

#### Required Contract Provisions

1. BREACH OF CONTRACT TERMS (2 CFR § 200 Appendix II(A))  
The City reserves the right to seek all remedies available under law in the event of a failure to perform or other breach of this Contract by the Contractor, and the failure of the City to employ a particular remedy shall not be regarded by the Parties as a waiver of that or any other available remedy.
2. TERMINATION FOR CAUSE OR CONVENIENCE(2 CFR § 200 Appendix II(B))  
The City may terminate this Contract for cause or for convenience upon thirty (30) days prior written notice.
3. CLEAN AIR ACT AND FEDERAL WATER POLLUTION CONTROL ACT  
*Clean Air Act*  
Contractor agrees to comply with all applicable standards, orders, or regulations issued pursuant to the Clean Air Act (42 U.S.C. §§ 7401-7671q.) and the Federal Water Pollution Control Act as amended (33 U.S.C. §§ 1251-1387). Violations must be reported to the Regional Office of the Environmental Protection Agency. See 2 C.F.R. Part 200, Appendix II(G).  
The contractor agrees to report each violation to the City of Pensacola and understands and agrees that the City of Pensacola will, in turn, report each violation as required to assure notification to the appropriate Environmental Protection Agency Regional Office.  
The contractor agrees to include these requirements in each subcontract exceeding \$150,000 financed in whole or in part with assistance provided by the federal government.  
  
*Federal Water Pollution Control Act*  
The contractor agrees to comply with all applicable standards, orders, or regulations issued pursuant to the Federal Water Pollution Control Act, as amended, 33 U.S.C. § 1251 et seq.  
The contractor agrees to report each violation to the City of Pensacola and understands and agrees that the City of Pensacola will, in turn, report each violation as required to assure notification to the appropriate Environmental Protection Agency Regional Office.  
The contractor agrees to include these requirements in each subcontract exceeding \$150,000 financed in whole or in part with assistance provided by the federal government.
4. DEBARMENT AND SUSPENSION  
This contract is a covered transaction for purposes of 2 C.F.R. pt. 180 and 2 C.F.R. pt. 3000. As such, the contractor is required to verify that none of the contractor's

principals (defined at 2 C.F.R. § 180.995) or its affiliates (defined at 2 C.F.R. § 180.905) are excluded (defined at 2 C.F.R. § 180.940) or disqualified (defined at 2 C.F.R. § 180.935).

The contractor must comply with 2 C.F.R. pt. 180, subpart C and 2 C.F.R. pt. 3000, subpart C, and must include a requirement to comply with these regulations in any lower tier covered transaction it enters into.

This certification is a material representation of fact relied upon by City of Pensacola. If it is later determined that the contractor did not comply with 2 C.F.R. pt. 180, subpart C and 2 C.F.R. pt. 3000, subpart C, in addition to remedies available to City of Pensacola, the federal government may pursue available remedies, including but not limited to suspension and/or debarment.

The bidder or proposer agrees to comply with the requirements of 2 C.F.R. pt. 180, subpart C and 2 C.F.R. pt. 3000, subpart C while this offer is valid and throughout the period of any contract that may arise from this offer. The bidder or proposer further agrees to include a provision requiring such compliance in its lower tier covered transactions.

5. ACCESS TO RECORDS

The Contractor agrees to provide the City of Pensacola, the federal officials, the Comptroller General of the United States, or any of their authorized representatives access to any books, documents, papers, and records of the Contractor which are directly pertinent to this contract for the purposes of making audits, examinations, excerpts, and transcriptions.

The Contractor agrees to permit any of the foregoing parties to reproduce by any means whatsoever or to copy excerpts and transcriptions as reasonably needed. The Contractor agrees to provide the federal government or his authorized representatives access to construction or other work sites pertaining to the work being completed under the contract.

In compliance with the Disaster Recovery Act of 2018, the City of Pensacola and the Contractor acknowledge and agree that no language in this contract is intended to prohibit audits or internal reviews by the federal government or the Comptroller General of the United States.

6. COMPLIANCE WITH FEDERAL LAW, REGULATIONS, AND EXECUTIVE ORDERS

This is an acknowledgement that federal financial assistance will be used to fund all or a portion of the contract. The contractor will comply with all applicable federal law, regulations, executive orders, federal policies, procedures, and directives.

7. NO OBLIGATION BY FEDERAL GOVERNMENT

The federal government is not a party to this contract and is not subject to any obligations or liabilities to the non-federal entity, contractor, or any other party pertaining to any matter resulting from the contract.

8. PROGRAM FRAUD AND FALSE OR FRAUDULENT STATEMENTS OR

## RELATED ACTS

The Contractor acknowledges that 31 U.S.C. Chap. 38 (Administrative Remedies for False Claims and Statements) applies to the Contractor's actions pertaining to this contract.

### 9. EQUAL EMPLOYMENT OPPORTUNITY

The regulation at 41 C.F.R. Part 60-1.4(b) requires the insertion of the following contract clause: During the performance of this contract, the contractor agrees as follows:

The contractor will not discriminate against any employee or applicant for employment because of race, color, religion, sex, sexual orientation, gender identity, or national origin. The contractor will take affirmative action to ensure that applicants are employed, and that employees are treated during employment without regard to their race, color, religion, sex, sexual orientation, gender identity, or national origin. Such action shall include, but not be limited to the following: Employment, upgrading, demotion, or transfer; recruitment or recruitment advertising; layoff or termination; rates of pay or other forms of compensation; and selection for training, including apprenticeship. The contractor agrees to post in conspicuous places, available to employees and applicants for employment, notices to be provided setting forth the provisions of this nondiscrimination clause. The contractor will, in all solicitations or advertisements for employees placed by or on behalf of the contractor, state that all qualified applicants will receive consideration for employment without regard to race, color, religion, sex, sexual orientation, gender identity, or national origin.

The contractor will not discharge or in any other manner discriminate against any employee or applicant for employment because such employee or applicant has inquired about, discussed, or disclosed the compensation of the employee or applicant or another employee or applicant. This provision shall not apply to instances in which an employee who has access to the compensation information of other employees or applicants as a part of such employee's essential job functions discloses the compensation of such other employees or applicants to individuals who do not otherwise have access to such information, unless such disclosure is in response to a formal complaint or charge, in furtherance of an investigation, proceeding, hearing, or action, including an investigation conducted by the employer, or is consistent with the contractor's legal duty to furnish information.

The contractor will send to each labor union or representative of workers with which he has a collective bargaining agreement or other contract or understanding, a notice to be provided advising the said labor union or workers' representatives of the contractor's commitments under this section, and shall post copies of the notice in conspicuous places available to employees and applicants for employment.

The contractor will comply with all provisions of Executive Order 11246 of

September 24, 1965, and of the rules, regulations, and relevant orders of the Secretary of Labor.

The contractor will furnish all information and reports required by Executive Order 11246 of September 24, 1965, and by rules, regulations, and orders of the Secretary of Labor, or pursuant thereto, and will permit access to his books, records, and accounts by the administering agency and the Secretary of Labor for purposes of investigation to ascertain compliance with such rules, regulations, and orders.

In the event of the contractor's noncompliance with the nondiscrimination clauses of this contract or with any of the said rules, regulations, or orders, this contract may be canceled, terminated, or suspended in whole or in part and the contractor may be declared ineligible for further government contracts or federally assisted construction contracts in accordance with procedures authorized in Executive Order 11246 of September 24, 1965, and such other sanctions may be imposed and remedies invoked as provided in Executive Order 11246 of September 24, 1965, or by rule, regulation, or order of the Secretary of Labor, or as otherwise provided by law.

The contractor will include the portion of the sentence immediately preceding paragraph (1) and the provisions of paragraphs (1) through (8) in every subcontract or purchase order unless exempted by rules, regulations, or orders of the Secretary of Labor issued pursuant to section 204 of Executive Order 11246 of September 24, 1965, so that such provisions will be binding upon each subcontractor or vendor. The contractor will take such action with respect to any subcontract or purchase order as the administering agency may direct as a means of enforcing such provisions, including sanctions for noncompliance:

Provided, however, that in the event a contractor becomes involved in, or is threatened with, litigation with a subcontractor or vendor as a result of such direction by the administering agency, the contractor may request the United States to enter into such litigation to protect the interests of the United States.

The applicant further agrees that it will be bound by the above equal opportunity clause with respect to its own employment practices when it participates in federally assisted construction work: Provided, That if the applicant so participating is a State or local government, the above equal opportunity clause is not applicable to any agency, instrumentality or subdivision of such government which does not participate in work on or under the contract.

The applicant agrees that it will assist and cooperate actively with the administering agency and the Secretary of Labor in obtaining the compliance of contractors and subcontractors with the equal opportunity clause and the rules, regulations, and relevant orders of the Secretary of Labor, that it will furnish the administering agency and the Secretary of Labor such information as they may require for the supervision of such compliance, and that it will otherwise assist the administering agency in the discharge of the agency's primary responsibility for securing compliance.

The applicant further agrees that it will refrain from entering into any contract or contract modification subject to Executive Order 11246 of September 24, 1965, with a contractor debarred from, or who has not demonstrated eligibility for, government contracts and federally assisted construction contracts pursuant to the Executive Order and will carry out such sanctions and penalties for violation of the equal opportunity clause as may be imposed upon contractors and subcontractors by the administering agency or the Secretary of Labor pursuant to Part II, Subpart D of the Executive Order. In addition, the applicant agrees that if it fails or refuses to comply with these undertakings, the administering agency may take any or all of the following actions: Cancel, terminate, or suspend in whole or in part this grant (contract, loan, insurance, guarantee); refrain from extending any further assistance to the applicant under the program with respect to which the failure or refund occurred until satisfactory assurance of future compliance has been received from such applicant; and refer the case to the Department of Justice for appropriate legal proceedings.

10. CONTRACT WORK HOURS AND SAFETY STANDARDS ACT

Compliance with the Contract Work Hours and Safety Standards Act.

Overtime requirements. No contractor or subcontractor contracting for any part of the contract work which may require or involve the employment of laborers or mechanics shall require or permit any such laborer or mechanic in any workweek in which he or she is employed on such work to work in excess of forty hours in such workweek unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of forty hours in such workweek.

Violation; liability for unpaid wages; liquidated damages. In the event of any violation of the clause set forth in paragraph (b)(1) of this section the contractor and any subcontractor responsible therefor shall be liable for the unpaid wages. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such District or to such territory), for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or mechanic, including watchmen and guards, employed in violation of the clause set forth in paragraph (b)(1) of this section, in the sum of \$27 for each calendar day on which such individual was required or permitted to work in excess of the standard workweek of forty hours without payment of the overtime wages required by the clause set forth in paragraph (b)(1) of this section.

Withholding for unpaid wages and liquidated damages. The City of Pensacola shall upon its own action or upon written request of an authorized representative of the Department of Labor withhold or cause to be withheld, from any moneys payable on account of work performed by the contractor or subcontractor under any such contract or any other federal contract with the same prime contractor, or any other federally-assisted contract subject to the Contract Work Hours and Safety

Standards Act, which is held by the same prime contractor, such sums as may be determined to be necessary to satisfy any liabilities of such contractor or subcontractor for unpaid wages and liquidated damages as provided in the clause set forth in paragraph (b)(2) of this section.

Subcontracts. The contractor or subcontractor shall insert in any subcontracts the clauses set forth in paragraph (b)(1) through (4) of this section and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in paragraphs (b)(1) through (4) of this section.

11. PROCUREMENT OF RECOVERED MATERIALS

In the performance of this contract, the Contractor shall make maximum use of products containing recovered materials that are EPA-designated items unless the product cannot be acquired—

- a) Competitively within a timeframe providing for compliance with the contract performance schedule;
- b) Meeting contract performance requirements; or
- c) At a reasonable price.

Information about this requirement, along with the list of EPA- designated items, is available at EPA's Comprehensive Procurement Guidelines web site, <https://www.epa.gov/smm/comprehensive-procurement-guideline-cpg-program>.

The Contractor also agrees to comply with all other applicable requirements of Section 6002 of the Solid Waste Disposal Act."

12. BYRD ANTI-LOBBYING AMENDMENT.

*Byrd Anti-Lobbying Amendment, 31 U.S.C. § 1352 (as amended)*

Contractors who apply or bid for an award of \$100,000 or more shall file the required certification. Each tier certifies to the tier above that it will not and has not used federal appropriated funds to pay any person or organization for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, officer or employee of Congress, or an employee of a Member of Congress in connection with obtaining any federal contract, grant, or any other award covered by 31 U.S.C. § 1352. Each tier shall also disclose any lobbying with non-federal funds that takes place in connection with obtaining any federal award. Such disclosures are forwarded from tier to tier up to the recipient who in turn will forward the certification(s) to the awarding agency.

*Required Certification.* If applicable, contractors must sign and submit to the non-federal entity the attached certification (Appendix A).

## ATTACHEMENT “C”

### DAVIS- BACON REGULATIONS

#### (1) *Minimum wages* —

(i) ***Wage rates and fringe benefits.*** All laborers and mechanics employed or working upon the site of the work (or otherwise working in construction or development of the project under a development statute), will be paid unconditionally and not less often than once a week, and without subsequent deduction or rebate on any account (except such payroll deductions as are permitted by regulations issued by the Secretary of Labor under the Copeland Act (29 CFR part 3)), the full amount of basic hourly wages and bona fide fringe benefits (or cash equivalents thereof) due at time of payment computed at rates not less than those contained in the wage determination of the Secretary of Labor which is attached hereto and made a part hereof, regardless of any contractual relationship which may be alleged to exist between the contractor and such laborers and mechanics. As provided in paragraphs (d) and (e) of this section, the appropriate wage determinations are effective by operation of law even if they have not been attached to the contract. Contributions made or costs reasonably anticipated for bona fide fringe benefits under the Davis-Bacon Act (40 U.S.C. 3141(2)(B)) on behalf of laborers or mechanics are considered wages paid to such laborers or mechanics, subject to the provisions of paragraph (a)(1)(v) of this section; also, regular contributions made or costs incurred for more than a weekly period (but not less often than quarterly) under plans, funds, or programs which cover the particular weekly period, are deemed to be constructively made or incurred during such weekly period. Such laborers and mechanics must be paid the appropriate wage rate and fringe benefits on the wage determination for the classification(s) of work actually performed, without regard to skill, except as provided in paragraph (a)(4) of this section. Laborers or mechanics performing work in more than one classification may be compensated at the rate specified for each classification for the time actually worked therein: *Provided*, That the employer's payroll records accurately set forth the time spent in each classification in which work is performed. The wage determination (including any additional classifications and wage rates conformed under paragraph (a)(1)(iii) of this section) and the Davis-Bacon poster (WH-1321) must be posted at all times by the contractor and its subcontractors at the site of the work in a prominent and accessible place where it can be easily seen by the workers.

#### (ii) ***Frequently recurring classifications.***

(A) In addition to wage and fringe benefit rates that have been determined to be prevailing under the procedures set forth in 29 CFR part 1, a wage determination may contain, pursuant to § 1.3(f), wage and fringe benefit rates for classifications of laborers and mechanics for which conformance requests are regularly submitted pursuant to paragraph (a)(1)(iii) of this section, provided that:



(1) The work performed by the classification is not performed by a classification in the wage determination for which a prevailing wage rate has been determined;

(2) The classification is used in the area by the construction industry; and

(3) The wage rate for the classification bears a reasonable relationship to the prevailing wage rates contained in the wage determination.

(B) The Administrator will establish wage rates for such classifications in accordance with paragraph (a)(1)(iii)(A)(3) of this section. Work performed in such a classification must be paid at no less than the wage and fringe benefit rate listed on the wage determination for such classification.

(iii) **Conformance.**

(A) The contracting officer must require that any class of laborers or mechanics, including helpers, which is not listed in the wage determination and which is to be employed under the contract be classified in conformance with the wage determination. Conformance of an additional classification and wage rate and fringe benefits is appropriate only when the following criteria have been met:

(1) The work to be performed by the classification requested is not performed by a classification in the wage determination; and

(2) The classification is used in the area by the construction industry; and

(3) The proposed wage rate, including any bona fide fringe benefits, bears a reasonable relationship to the wage rates contained in the wage determination.

(B) The conformance process may not be used to split, subdivide, or otherwise avoid application of classifications listed in the wage determination.

(C) If the contractor and the laborers and mechanics to be employed in the classification (if known), or their representatives, and the contracting officer agree on the classification and wage rate (including the amount designated for fringe benefits where appropriate), a report of the action taken will be sent by the contracting officer by email to [DBAconformance@dol.gov](mailto:DBAconformance@dol.gov). The Administrator, or an authorized representative, will approve, modify, or disapprove every additional classification action within 30 days of receipt and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(D) In the event the contractor, the laborers or mechanics to be employed in the classification or their representatives, and the contracting officer do not agree on the proposed classification and wage rate (including the amount designated for fringe benefits, where appropriate), the contracting officer will, by email to [DBAconformance@dol.gov](mailto:DBAconformance@dol.gov), refer the questions, including the views of all interested parties and the recommendation of the contracting officer, to the Administrator for

determination. The Administrator, or an authorized representative, will issue a determination within 30 days of receipt and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(E) The contracting officer must promptly notify the contractor of the action taken by the Wage and Hour Division under paragraphs (a)(1)(iii)(C) and (D) of this section. The contractor must furnish a written copy of such determination to each affected worker or it must be posted as a part of the wage determination. The wage rate (including fringe benefits where appropriate) determined pursuant to paragraph (a)(1)(iii)(C) or (D) of this section must be paid to all workers performing work in the classification under this contract from the first day on which work is performed in the classification.

(iv) ***Fringe benefits not expressed as an hourly rate.*** Whenever the minimum wage rate prescribed in the contract for a class of laborers or mechanics includes a fringe benefit which is not expressed as an hourly rate, the contractor may either pay the benefit as stated in the wage determination or may pay another bona fide fringe benefit or an hourly cash equivalent thereof.

(v) ***Unfunded plans.*** If the contractor does not make payments to a trustee or other third person, the contractor may consider as part of the wages of any laborer or mechanic the amount of any costs reasonably anticipated in providing bona fide fringe benefits under a plan or program, *Provided*, That the Secretary of Labor has found, upon the written request of the contractor, in accordance with the criteria set forth in § 5.28, that the applicable standards of the Davis-Bacon Act have been met. The Secretary of Labor may require the contractor to set aside in a separate account assets for the meeting of obligations under the plan or program.

(vi) ***Interest.*** In the event of a failure to pay all or part of the wages required by the contract, the contractor will be required to pay interest on any underpayment of wages.

## **(2) *Withholding* —**

(i) ***Withholding requirements.*** The [write in name of Federal agency or the recipient of Federal assistance] may, upon its own action, or must, upon written request of an authorized representative of the Department of Labor, withhold or cause to be withheld from the contractor so much of the accrued payments or advances as may be considered necessary to satisfy the liabilities of the prime contractor or any subcontractor for the full amount of wages and monetary relief, including interest, required by the clauses set forth in paragraph (a) of this section for violations of this contract, or to satisfy any such liabilities required by any other Federal contract, or federally assisted contract subject to Davis-Bacon labor standards, that is held by the same prime contractor (as defined in § 5.2). The necessary funds may be withheld from the contractor under this contract, any other Federal contract with the same prime contractor, or any other federally assisted contract that is subject to Davis-Bacon labor standards requirements and is held by the same prime contractor,

regardless of whether the other contract was awarded or assisted by the same agency, and such funds may be used to satisfy the contractor liability for which the funds were withheld. In the event of a contractor's failure to pay any laborer or mechanic, including any apprentice or helper working on the site of the work (or otherwise working in construction or development of the project under a development statute) all or part of the wages required by the contract, or upon the contractor's failure to submit the required records as discussed in paragraph (a)(3)(iv) of this section, the [Agency] may on its own initiative and after written notice to the contractor, sponsor, applicant, owner, or other entity, as the case may be, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds until such violations have ceased.

(ii) **Priority to withheld funds.** The Department has priority to funds withheld or to be withheld in accordance with paragraph (a)(2)(i) or (b)(3)(i) of this section, or both, over claims to those funds by:

(A) A contractor's surety(ies), including without limitation performance bond sureties and payment bond sureties;

(B) A contracting agency for its procurement costs;

(C) A trustee(s) (either a court-appointed trustee or a U.S. trustee, or both) in bankruptcy of a contractor, or a contractor's bankruptcy estate;

(D) A contractor's assignee(s);

(E) A contractor's successor(s); or

(F) A claim asserted under the Prompt Payment Act, 31 U.S.C. 3901–3907.

(3) **Records and certified payrolls —**

(i) **Basic record requirements —**

(A) **Length of record retention.** All regular payrolls and other basic records must be maintained by the contractor and any subcontractor during the course of the work and preserved for all laborers and mechanics working at the site of the work (or otherwise working in construction or development of the project under a development statute) for a period of at least 3 years after all the work on the prime contract is completed.

(B) **Information required.** Such records must contain the name; Social Security number; last known address, telephone number, and email address of each such worker; each worker's correct classification(s) of work actually performed; hourly rates of wages paid (including rates of contributions or costs anticipated for bona fide fringe benefits or cash equivalents thereof of the types described in 40 U.S.C. 3141(2)(B) of the Davis-Bacon Act); daily and weekly number of hours actually

worked in total and on each covered contract; deductions made; and actual wages paid.

(C) **Additional records relating to fringe benefits.** Whenever the Secretary of Labor has found under paragraph (a)(1)(v) of this section that the wages of any laborer or mechanic include the amount of any costs reasonably anticipated in providing benefits under a plan or program described in 40 U.S.C. 3141(2)(B) of the Davis-Bacon Act, the contractor must maintain records which show that the commitment to provide such benefits is enforceable, that the plan or program is financially responsible, and that the plan or program has been communicated in writing to the laborers or mechanics affected, and records which show the costs anticipated or the actual cost incurred in providing such benefits.

(D) **Additional records relating to apprenticeship.** Contractors with apprentices working under approved programs must maintain written evidence of the registration of apprenticeship programs, the registration of the apprentices, and the ratios and wage rates prescribed in the applicable programs.

(ii) **Certified payroll requirements —**

(A) **Frequency and method of submission.** The contractor or subcontractor must submit weekly, for each week in which any DBA- or Related Acts-covered work is performed, certified payrolls to the [write in name of appropriate Federal agency] if the agency is a party to the contract, but if the agency is not such a party, the contractor will submit the certified payrolls to the applicant, sponsor, owner, or other entity, as the case may be, that maintains such records, for transmission to the [write in name of agency]. The prime contractor is responsible for the submission of all certified payrolls by all subcontractors. A contracting agency or prime contractor may permit or require contractors to submit certified payrolls through an electronic system, as long as the electronic system requires a legally valid electronic signature; the system allows the contractor, the contracting agency, and the Department of Labor to access the certified payrolls upon request for at least 3 years after the work on the prime contract has been completed; and the contracting agency or prime contractor permits other methods of submission in situations where the contractor is unable or limited in its ability to use or access the electronic system.

(B) **Information required.** The certified payrolls submitted must set out accurately and completely all of the information required to be maintained under paragraph (a)(3)(i)(B) of this section, except that full Social Security numbers and last known addresses, telephone numbers, and email addresses must not be included on weekly transmittals. Instead, the certified payrolls need only include an individually identifying number for each worker (e.g., the last four digits of the worker's Social Security number). The required weekly certified payroll information may be submitted using Optional Form WH-347 or in any other format desired. Optional Form WH-347 is available for this purpose from the Wage and Hour Division website at <https://www.dol.gov/sites/dolgov/files/WHD/legacy/files/wh3471.pdf> or its successor website. It is not a violation of this section for a prime contractor to

require a subcontractor to provide full Social Security numbers and last known addresses, telephone numbers, and email addresses to the prime contractor for its own records, without weekly submission by the subcontractor to the sponsoring government agency (or the applicant, sponsor, owner, or other entity, as the case may be, that maintains such records).

(C) **Statement of Compliance.** Each certified payroll submitted must be accompanied by a "Statement of Compliance," signed by the contractor or subcontractor, or the contractor's or subcontractor's agent who pays or supervises the payment of the persons working on the contract, and must certify the following:

(1) That the certified payroll for the payroll period contains the information required to be provided under paragraph (a)(3)(ii) of this section, the appropriate information and basic records are being maintained under paragraph (a)(3)(i) of this section, and such information and records are correct and complete;

(2) That each laborer or mechanic (including each helper and apprentice) working on the contract during the payroll period has been paid the full weekly wages earned, without rebate, either directly or indirectly, and that no deductions have been made either directly or indirectly from the full wages earned, other than permissible deductions as set forth in 29 CFR part 3; and

(3) That each laborer or mechanic has been paid not less than the applicable wage rates and fringe benefits or cash equivalents for the classification(s) of work actually performed, as specified in the applicable wage determination incorporated into the contract.

(D) **Use of Optional Form WH-347.** The weekly submission of a properly executed certification set forth on the reverse side of Optional Form WH-347 will satisfy the requirement for submission of the "Statement of Compliance" required by paragraph (a)(3)(ii)(C) of this section.

(E) **Signature.** The signature by the contractor, subcontractor, or the contractor's or subcontractor's agent must be an original handwritten signature or a legally valid electronic signature.

(F) **Falsification.** The falsification of any of the above certifications may subject the contractor or subcontractor to civil or criminal prosecution under 18 U.S.C. 1001 and 31 U.S.C. 3729.

(G) **Length of certified payroll retention.** The contractor or subcontractor must preserve all certified payrolls during the course of the work and for a period of 3 years after all the work on the prime contract is completed.

(iii) **Contracts, subcontracts, and related documents.** The contractor or subcontractor must maintain this contract or subcontract and related documents including, without limitation, bids, proposals, amendments, modifications, and extensions. The contractor or subcontractor must preserve these contracts,

subcontracts, and related documents during the course of the work and for a period of 3 years after all the work on the prime contract is completed.

(iv) ***Required disclosures and access —***

(A) ***Required record disclosures and access to workers.*** The contractor or subcontractor must make the records required under paragraphs (a)(3)(i) through (iii) of this section, and any other documents that the [write the name of the agency] or the Department of Labor deems necessary to determine compliance with the labor standards provisions of any of the applicable statutes referenced by § 5.1, available for inspection, copying, or transcription by authorized representatives of the [write the name of the agency] or the Department of Labor, and must permit such representatives to interview workers during working hours on the job.

(B) ***Sanctions for non-compliance with records and worker access requirements.*** If the contractor or subcontractor fails to submit the required records or to make them available, or refuses to permit worker interviews during working hours on the job, the Federal agency may, after written notice to the contractor, sponsor, applicant, owner, or other entity, as the case may be, that maintains such records or that employs such workers, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds. Furthermore, failure to submit the required records upon request or to make such records available, or to permit worker interviews during working hours on the job, may be grounds for debarment action pursuant to § 5.12. In addition, any contractor or other person that fails to submit the required records or make those records available to WHD within the time WHD requests that the records be produced will be precluded from introducing as evidence in an administrative proceeding under 29 CFR part 6 any of the required records that were not provided or made available to WHD. WHD will take into consideration a reasonable request from the contractor or person for an extension of the time for submission of records. WHD will determine the reasonableness of the request and may consider, among other things, the location of the records and the volume of production.

(C) ***Required information disclosures.*** Contractors and subcontractors must maintain the full Social Security number and last known address, telephone number, and email address of each covered worker, and must provide them upon request to the [write in name of appropriate Federal agency] if the agency is a party to the contract, or to the Wage and Hour Division of the Department of Labor. If the Federal agency is not such a party to the contract, the contractor, subcontractor, or both, must, upon request, provide the full Social Security number and last known address, telephone number, and email address of each covered worker to the applicant, sponsor, owner, or other entity, as the case may be, that maintains such records, for transmission to the [write in name of agency], the contractor, or the Wage and Hour Division of the Department of Labor for purposes of an investigation or other compliance action.

(4) ***Apprentices and equal employment opportunity —***

(i) **Apprentices** —

(A) **Rate of pay.** Apprentices will be permitted to work at less than the predetermined rate for the work they perform when they are employed pursuant to and individually registered in a bona fide apprenticeship program registered with the U.S. Department of Labor, Employment and Training Administration, Office of Apprenticeship (OA), or with a State Apprenticeship Agency recognized by the OA. A person who is not individually registered in the program, but who has been certified by the OA or a State Apprenticeship Agency (where appropriate) to be eligible for probationary employment as an apprentice, will be permitted to work at less than the predetermined rate for the work they perform in the first 90 days of probationary employment as an apprentice in such a program. In the event the OA or a State Apprenticeship Agency recognized by the OA withdraws approval of an apprenticeship program, the contractor will no longer be permitted to use apprentices at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(B) **Fringe benefits.** Apprentices must be paid fringe benefits in accordance with the provisions of the apprenticeship program. If the apprenticeship program does not specify fringe benefits, apprentices must be paid the full amount of fringe benefits listed on the wage determination for the applicable classification. If the Administrator determines that a different practice prevails for the applicable apprentice classification, fringe benefits must be paid in accordance with that determination.

(C) **Apprenticeship ratio.** The allowable ratio of apprentices to journeymen on the job site in any craft classification must not be greater than the ratio permitted to the contractor as to the entire work force under the registered program or the ratio applicable to the locality of the project pursuant to paragraph (a)(4)(i)(D) of this section. Any worker listed on a payroll at an apprentice wage rate, who is not registered or otherwise employed as stated in paragraph (a)(4)(i)(A) of this section, must be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any apprentice performing work on the job site in excess of the ratio permitted under this section must be paid not less than the applicable wage rate on the wage determination for the work actually performed.

(D) **Reciprocity of ratios and wage rates.** Where a contractor is performing construction on a project in a locality other than the locality in which its program is registered, the ratios and wage rates (expressed in percentages of the journeyworker's hourly rate) applicable within the locality in which the construction is being performed must be observed. If there is no applicable ratio or wage rate for the locality of the project, the ratio and wage rate specified in the contractor's registered program must be observed.

(ii) **Equal employment opportunity.** The use of apprentices and journeymen under this part must be in conformity with the equal employment opportunity requirements of Executive Order 11246, as amended, and 29 CFR part 30.

(5) **Compliance with Copeland Act requirements.** The contractor shall comply with the requirements of 29 CFR part 3, which are incorporated by reference in this contract.

(6) **Subcontracts.** The contractor or subcontractor must insert in any subcontracts the clauses contained in paragraphs (a)(1) through (11) of this section, along with the applicable wage determination(s) and such other clauses or contract modifications as the [write in the name of the Federal agency] may by appropriate instructions require, and a clause requiring the subcontractors to include these clauses and wage determination(s) in any lower tier subcontracts. The prime contractor is responsible for the compliance by any subcontractor or lower tier subcontractor with all the contract clauses in this section. In the event of any violations of these clauses, the prime contractor and any subcontractor(s) responsible will be liable for any unpaid wages and monetary relief, including interest from the date of the underpayment or loss, due to any workers of lower-tier subcontractors, and may be subject to debarment, as appropriate.

(7) **Contract termination: debarment.** A breach of the contract clauses in 29 CFR 5.5 may be grounds for termination of the contract, and for debarment as a contractor and a subcontractor as provided in 29 CFR 5.12.

(8) **Compliance with Davis-Bacon and Related Act requirements.** All rulings and interpretations of the Davis-Bacon and Related Acts contained in 29 CFR parts 1, 3, and 5 are herein incorporated by reference in this contract.

(9) **Disputes concerning labor standards.** Disputes arising out of the labor standards provisions of this contract shall not be subject to the general disputes clause of this contract. Such disputes shall be resolved in accordance with the procedures of the Department of Labor set forth in 29 CFR parts 5, 6, and 7. Disputes within the meaning of this clause include disputes between the contractor (or any of its subcontractors) and the contracting agency, the U.S. Department of Labor, or the employees or their representatives.

(10) **Certification of eligibility.**

(i) By entering into this contract, the contractor certifies that neither it nor any person or firm who has an interest in the contractor's firm is a person or firm ineligible to be awarded Government contracts by virtue of 40 U.S.C. 3144(b) or § 5.12(a).

(ii) No part of this contract shall be subcontracted to any person or firm ineligible for award of a Government contract by virtue of 40 U.S.C. 3144(b) or § 5.12(a).

(iii) The penalty for making false statements is prescribed in the U.S. Code, Title 18 Crimes and Criminal Procedure, 18 U.S.C. 1001.

(11) **Anti-retaliation.** It is unlawful for any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate against, or to cause any person to discharge, demote, intimidate, threaten, restrain,



coerce, blacklist, harass, or in any other manner discriminate against, any worker or job applicant for:

- (i) Notifying any contractor of any conduct which the worker reasonably believes constitutes a violation of the DBA, Related Acts, this part, or 29 CFR part 1 or 3;
- (ii) Filing any complaint, initiating or causing to be initiated any proceeding, or otherwise asserting or seeking to assert on behalf of themselves or others any right or protection under the DBA, Related Acts, this part, or 29 CFR part 1 or 3;
- (iii) Cooperating in any investigation or other compliance action, or testifying in any proceeding under the DBA, Related Acts, this part, or 29 CFR part 1 or 3; or
- (iv) Informing any other person about their rights under the DBA, Related Acts, this part, or 29 CFR part 1 or 3.

(b) ***Contract Work Hours and Safety Standards Act (CWHSSA)***. The Agency Head must cause or require the contracting officer to insert the following clauses set forth in paragraphs (b)(1) through (5) of this section in full, or (for contracts covered by the Federal Acquisition Regulation) by reference, in any contract in an amount in excess of \$100,000 and subject to the overtime provisions of the Contract Work Hours and Safety Standards Act. These clauses must be inserted in addition to the clauses required by paragraph (a) of this section or 29 CFR 4.6. As used in this paragraph (b), the terms “laborers and mechanics” include watchpersons and guards.

(1) ***Overtime requirements***. No contractor or subcontractor contracting for any part of the contract work which may require or involve the employment of laborers or mechanics shall require or permit any such laborer or mechanic in any workweek in which he or she is employed on such work to work in excess of forty hours in such workweek unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of forty hours in such workweek.

(2) ***Violation; liability for unpaid wages; liquidated damages***. In the event of any violation of the clause set forth in paragraph (b)(1) of this section the contractor and any subcontractor responsible therefor shall be liable for the unpaid wages and interest from the date of the underpayment. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such District or to such territory), for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or mechanic, including watchpersons and guards, employed in violation of the clause set forth in paragraph (b)(1) of this section, in the sum of \$31 for each calendar day on which such individual was required or permitted to work in excess of the standard workweek of forty hours without payment of the overtime wages required by the clause set forth in paragraph (b)(1).

(3) ***Withholding for unpaid wages and liquidated damages*** —

(i) ***Withholding process.*** The [write in the name of the Federal agency or the recipient of Federal assistance] may, upon its own action, or must, upon written request of an authorized representative of the Department of Labor, withhold or cause to be withheld from the contractor so much of the accrued payments or advances as may be considered necessary to satisfy the liabilities of the prime contractor or any subcontractor for any unpaid wages; monetary relief, including interest; and liquidated damages required by the clauses set forth in this paragraph (b) on this contract, any other Federal contract with the same prime contractor, or any other federally assisted contract subject to the Contract Work Hours and Safety Standards Act that is held by the same prime contractor (as defined in § 5.2). The necessary funds may be withheld from the contractor under this contract, any other Federal contract with the same prime contractor, or any other federally assisted contract that is subject to the Contract Work Hours and Safety Standards Act and is held by the same prime contractor, regardless of whether the other contract was awarded or assisted by the same agency, and such funds may be used to satisfy the contractor liability for which the funds were withheld.

(ii) ***Priority to withheld funds.*** The Department has priority to funds withheld or to be withheld in accordance with paragraph (a)(2)(i) or (b)(3)(i) of this section, or both, over claims to those funds by:

(A) A contractor's surety(ies), including without limitation performance bond sureties and payment bond sureties;

(B) A contracting agency for its procurement costs;

(C) A trustee(s) (either a court-appointed trustee or a U.S. trustee, or both) in bankruptcy of a contractor, or a contractor's bankruptcy estate;

(D) A contractor's assignee(s);

(E) A contractor's successor(s); or

(F) A claim asserted under the Prompt Payment Act, 31 U.S.C. 3901–3907.

(4) ***Subcontracts.*** The contractor or subcontractor must insert in any subcontracts the clauses set forth in paragraphs (b)(1) through (5) of this section and a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor is responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in paragraphs (b)(1) through (5). In the event of any violations of these clauses, the prime contractor and any subcontractor(s) responsible will be liable for any unpaid wages and monetary relief, including interest from the date of the underpayment or loss, due to any workers of lower-tier subcontractors, and associated liquidated damages and may be subject to debarment, as appropriate.

(5) ***Anti-retaliation.*** It is unlawful for any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate

against, or to cause any person to discharge, demote, intimidate, threaten, restrain, coerce, blacklist, harass, or in any other manner discriminate against, any worker or job applicant for:

(i) Notifying any contractor of any conduct which the worker reasonably believes constitutes a violation of the Contract Work Hours and Safety Standards Act (CWHSSA) or its implementing regulations in this part;

(ii) Filing any complaint, initiating or causing to be initiated any proceeding, or otherwise asserting or seeking to assert on behalf of themselves or others any right or protection under CWHSSA or this part;

(iii) Cooperating in any investigation or other compliance action, or testifying in any proceeding under CWHSSA or this part; or

(iv) Informing any other person about their rights under CWHSSA or this part.

(c) **CWHSSA required records clause.** In addition to the clauses contained in paragraph (b) of this section, in any contract subject only to the Contract Work Hours and Safety Standards Act and not to any of the other laws referenced by § 5.1, the Agency Head must cause or require the contracting officer to insert a clause requiring that the contractor or subcontractor must maintain regular payrolls and other basic records during the course of the work and must preserve them for a period of 3 years after all the work on the prime contract is completed for all laborers and mechanics, including guards and watchpersons, working on the contract. Such records must contain the name; last known address, telephone number, and email address; and social security number of each such worker; each worker's correct classification(s) of work actually performed; hourly rates of wages paid; daily and weekly number of hours actually worked; deductions made; and actual wages paid. Further, the Agency Head must cause or require the contracting officer to insert in any such contract a clause providing that the records to be maintained under this paragraph must be made available by the contractor or subcontractor for inspection, copying, or transcription by authorized representatives of the (write the name of agency) and the Department of Labor, and the contractor or subcontractor will permit such representatives to interview workers during working hours on the job.

(d) **Incorporation of contract clauses and wage determinations by reference.** Although agencies are required to insert the contract clauses set forth in this section, along with appropriate wage determinations, in full into covered contracts, and contractors and subcontractors are required to insert them in any lower-tier subcontracts, the incorporation by reference of the required contract clauses and appropriate wage determinations will be given the same force and effect as if they were inserted in full text.

(e) **Incorporation by operation of law.** The contract clauses set forth in this section (or their equivalent under the Federal Acquisition Regulation), along with the correct wage determinations, will be considered to be a part of every prime contract required by the applicable statutes referenced by § 5.1 to include such clauses, and will be effective by operation of law, whether or not they are included or incorporated by reference into such contract, unless the Administrator grants a variance, tolerance, or exemption from the

application of this paragraph. Where the clauses and applicable wage determinations are effective by operation of law under this paragraph, the prime contractor must be compensated for any resulting increase in wages in accordance with applicable law.

**APPENDIX A, 44 C.F.R. PART 18 – CERTIFICATION REGARDING LOBBYING**  
**Certification for Contracts, Grants, Loans, and Cooperative Agreements**

The undersigned certifies, to the best of his or her knowledge and belief, that:

1. No federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of an agency, a member of Congress, an officer or employee of Congress, or an employee of a member of Congress in connection with the awarding of any federal contract, the making of any federal grant, the making of any federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any federal contract, grant, loan, or cooperative agreement.
2. If any funds other than federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a member of Congress, an officer or employee of Congress, or an employee of a member of Congress in connection with this federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure Form to Report Lobbying," in accordance with its instructions.
3. The undersigned shall require that the language of this certification be included in the award documents for all subawards at all tiers (including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements) and that all subrecipients shall certify and disclose accordingly.

This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by section 1352, title 31, U.S. Code. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

The Contractor, [REDACTED], certifies or affirms the truthfulness and accuracy of each statement of its certification and disclosure, if any. In addition, the Contractor understands and agrees that the provisions of 31 U.S.C. Chap. 38, Administrative Remedies for False Claims and Statements, apply to this certification and disclosure, if any.

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Signature of Contractor's Authorized Official

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Date

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Name and Title of Contractor's Authorized Official

**EXHIBIT A**  
**BID DOCUMENTS**

SAMPLE

## **EXHIBIT B**

### **PROPOSAL**

The pages following Exhibit B are the documents comprising the Proposal dated, \_\_\_\_\_, which Contractor submitted in response to the Bid Documents, are hereby incorporated by reference into this Contract. The Proposal includes all attachments and addenda submitted by Contractor in response to the Bid Documents, which are also hereby incorporated into this Contract by reference.

SAMPLE

**CONTRACT FORMS  
PERFORMANCE BOND**

KNOW ALL MEN BY THESE PRESENTS, THAT WE \_\_\_\_\_  
of the State of \_\_\_\_\_ and County of \_\_\_\_\_ hereinafter,  
known as the Principal, and \_\_\_\_\_ a corporation  
chartered and existing under the laws of the State of \_\_\_\_\_ and duly authorized  
to do business in the State of Florida as Surety, are held and firmly bound unto the City of  
Pensacola, Florida hereinafter known as the Owner, in the penal sum of \_\_\_\_\_  
Dollars (\$) to be  
paid to the Owner, for the use and benefit of all persons doing work or furnishing skill, tools,  
machinery or materials, or subcontracting under or for the purpose of the hereinafter named  
contract, for which payment, well and truly to be made, we hereby bind ourselves, our heirs,  
executors, administrators, successors and assignees, jointly and severally, by these presents.

This obligation is, however, subject to the following conditions:

The above bound Principal has entered into a contract with the Owner under which it  
agrees to furnish all the labor and material and do all work necessary to construct all  
improvements described in these contract documents under certain terms, conditions, and  
stipulations and in accordance with the plans and specifications for the project, which are hereto  
attached and made a part of this obligation.

NOW, THEREFORE, the conditions of this obligation are such that if the above bound  
Principal fails to faithfully and fully carry out and comply with the terms and conditions of said  
contract and complete the work therein specified, it shall be the duty of the Surety herein to  
assume the responsibility for the performance of the contract and to complete the work specified  
therein, including, but not limited to, obligations created by way of warranties and/or guarantees  
for workmanship and materials which warranty and/or guarantee may extend for a period of time  
beyond completion of said contract, and such alterations or additions as may be made therein  
or in the plans and specifications, and shall indemnify and save the Engineer and the Owner  
harmless against any claims for using any form of material, process, composition or anything  
which is patented, and likewise indemnify and save the Engineer and the Owner harmless  
against all claims for damages by reason of any default or negligence, want of skill or care on  
the part of said Principal or Agents in and about the performance of said contract, and shall  
comply with all laws pertaining to said work, and shall comply with and perform any and all  
warranties and/or guarantees provided for in said contract. If the Principal faithfully and fully  
carries out and complies with the terms and conditions of said contract, then this obligation shall  
be void; otherwise it shall remain in full force and effect.

And the Surety to this bond, for value received agrees that no change, extensions of  
time, alterations or additions to the terms of the contract or to the work to be performed  
thereunder of the specifications accompanying the same shall in any way affect its obligations  
on this bond, and it does hereby waive notice of any such change, extension of time, alterations  
or additions to the terms of the Contract or the work or to the Plans and Specifications.

It is agreed that this bond is executed pursuant to and in accordance with the provisions  
of Section 255.05 of the Florida Statutes, and is intended to be and shall be construed to be a  
bond in compliance with the requirements thereof.



IN WITNESS WHEREOF, said Principal and Surety have thereunto affixed their hands and seals on this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_, either in person or by agents fully authorized.

As to Principal:

Signed, sealed and delivered in  
the presence of:

\_\_\_\_\_  
Principal

\_\_\_\_\_  
Witness

By: \_\_\_\_\_ (L.S.)

\_\_\_\_\_  
Notary Public

State of \_\_\_\_\_

County of \_\_\_\_\_

As to Surety:

Signed, sealed and delivered in  
the presence of:

\_\_\_\_\_  
Surety

\_\_\_\_\_  
Witness

By: \_\_\_\_\_ (L.S.)

\_\_\_\_\_  
Notary Public

State of \_\_\_\_\_

County of \_\_\_\_\_

Approved as to form:

\_\_\_\_\_

**CONTRACT FORMS  
PAYMENT BOND**

KNOW ALL MEN BY THESE PRESENTS, THAT WE \_\_\_\_\_  
\_\_\_\_\_ of the State of \_\_\_\_\_ and County of \_\_\_\_\_  
hereinafter, known as the Principal, and \_\_\_\_\_, a corporation  
chartered and existing under the laws of the State of \_\_\_\_\_ and \_\_\_\_\_ duly  
authorized to do business in the State of Florida as Surety, are held and firmly bound unto the  
City of Pensacola, Florida hereinafter known as the Owner, in the penal sum of \_\_\_\_\_  
\_\_\_\_\_ Dollars (\$ \_\_\_\_\_) to be  
paid to the Owner, for the use and benefit of all persons doing work or furnishing skill, tools,  
machinery or materials, or subcontracting under or for the purpose of the hereinafter named  
contract, for which payment, well and truly to be made, we hereby bind ourselves, our heirs,  
executors, administrators, successors and assignees, jointly and severally, by these presents.

This obligation is, however, subject to the following conditions:

The above bound Principal has entered into a contract with the Owner under which  
agrees to furnish all the labor and material and do all work necessary to construct all  
improvements described in these contract documents under certain terms, conditions, and  
stipulations and in accordance with the plans and specifications for the project, which are hereto  
attached and made a part of this obligation.

NOW should the above named Principal and all subcontractors, if any, to whom any  
portion of the work provided for in the attached contract is sublet and all assignees of the said  
Principal and of such subcontractors shall promptly make payments to all persons supplying him  
or them with labor, materials, or supplies for or in the prosecution of the work provided for in  
such Contract, or in any amendment or extension of or addition to said contract, and for the  
payment of reasonable attorney's fees, incurred by the claimant or claimants in suits on said  
bond, then the above obligation shall be void; otherwise, to remain in full force and effect.

And the Surety to this bond, for value received agrees that no change, extensions of  
time, alterations or additions to the terms of the contract or to the work to be performed  
thereunder or the specifications accompanying the same shall in any way affect its obligation on  
this bond, and it does hereby waive notice of any such change, extension of time, alterations or  
additions to the terms of the Contract or the work to the Plans and Specifications.

It is agreed that this bond is executed pursuant to and in accordance with the  
provisions of Section 255.05 of the Florida Statutes, and is intended to be and shall be  
construed to be a bond on compliance with the requirements thereof. The payment bond  
required to exempt an Owner under this part shall be furnished by the Contractor in at least the  
amount of the original contract price before commencing the construction of the improvement  
under the direct contract. The bond shall be executed as surety by a surety insurer authorized  
to do business in this state and shall be conditioned that the Contractor shall promptly make  
payments for labor, services, and material to all lienors under the Contractor's direct contract.  
Any form of bond given by a Contractor conditioned to pay for labor, services, and material used  
to improve real property shall be deemed to include the condition of this subsection.

IN WITNESS WHEREOF, said Principal and Surety have thereunto affixed their hands and seals on this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_, either in person or by agents fully authorized.

As to Principal:

Signed, sealed and delivered in  
the presence of:

\_\_\_\_\_  
Principal

\_\_\_\_\_  
Witness

By: \_\_\_\_\_ (L.S.)

\_\_\_\_\_  
Notary Public

State of \_\_\_\_\_

County of \_\_\_\_\_

As to Surety:

Signed, sealed and delivered in  
the presence of:

\_\_\_\_\_  
Surety

\_\_\_\_\_  
Witness

By: \_\_\_\_\_ (L.S.)

\_\_\_\_\_  
Notary Public

State of \_\_\_\_\_

County of \_\_\_\_\_

Approved as to form:

\_\_\_\_\_  
Owner's Attorney

**CONTRACT FORMS  
NOTICE OF AWARD**

STATE OF FLORIDA

COUNTY OF ESCAMBIA

THIS CONTRACT AWARD made this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_,  
by the City of Pensacola, Florida, hereinafter called the OWNER, to \_\_\_\_\_

hereinafter called the CONTRACTOR, is for the completion of a certain project described as:  
FAA AIP Project No. **TBD**, Runway 8-26 Rehabilitation, for the use and benefit of the Owner as  
shown on the plans and described in the specifications as prepared by:

**AVCON**

3298 Summit Blvd., Suite 26  
Pensacola, Florida 32503

The project consists of the Base Bid plus the following Alternates, if applicable:

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The consideration to be paid by the Owner to the Contractor for completion of the project  
in accordance with the contract documents is the sum of \_\_\_\_\_

\_\_\_\_\_ Dollars \$\_\_\_\_\_.

Commencement of work under this contract shall begin not less than five (5) nor more than ten  
(10) days after Contractor's receipt of a Notice to Proceed issued by the Owner and the project  
is to be fully completed on or before 90 calendar days after that specified date unless otherwise  
subsequently agreed.

OWNER: City of Pensacola, Florida

By: \_\_\_\_\_

Name

\_\_\_\_\_  
Signature

\_\_\_\_\_  
Printed Name

\_\_\_\_\_  
Title

Acknowledgement of Receipt of Contract Award by Contractor:

\_\_\_\_\_  
Name

\_\_\_\_\_  
Signature

\_\_\_\_\_  
Printed Name

\_\_\_\_\_  
Title

DATE: \_\_\_\_\_

**CONTACT FORMS  
NOTICE TO PROCEED**

TO: \_\_\_\_\_

DATE: \_\_\_\_\_

\_\_\_\_\_

PROJECT: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

You are hereby notified to commence work in accordance with the Contract dated \_\_\_\_\_, not less than five (5) days nor more than ten (10) days after your receipt of this Notice to Proceed as evidenced by the date of receipt shown on the certified mail return receipt, and you are to complete the WORK within 90 calendar days from the project start date established as set forth herein above.

City of Pensacola, Florida

BY: \_\_\_\_\_

Typed Name

\_\_\_\_\_

Signature

**ACCEPTANCE OF NOTICE:**

Receipt of the above NOTICE TO PROCEED  
is hereby acknowledged by:

\_\_\_\_\_

this \_\_\_\_\_ day of \_\_\_\_\_, 20\_\_\_\_,

By: \_\_\_\_\_  
(Signature)

\_\_\_\_\_

(Printed Name)

\_\_\_\_\_

(Title)

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**CONTRACT FORMS  
FINAL RELEASE OF CLAIMS**

KNOWN ALL MEN BY THESE PRESENTS, that the undersigned, for and in consideration of the payment of the sum of \_\_\_\_\_ Dollars (\$ \_\_\_\_\_), paid by the City of Pensacola, Florida hereinafter referred to as Owner, receipt of which is hereby acknowledged as total compensation for performance of the below-described Contract for Bid Schedule(s) \_\_\_\_\_, does hereby fully and completely discharge and release the Owner from and waives any and all debts, accounts, promises, damages, liens, encumbrances, causes of action, suits, bonds, judgments, claims and demands whatsoever, in law or in equity, which the undersigned ever had, now has or might hereafter have on account of labor performed, material furnished or services rendered, directly or indirectly, for the Contract between the parties dated \_\_\_\_\_, 20 \_\_, known as \_\_\_\_\_ except for those claims, disputes and other matters arising out of or relating to said Contract which have been raised by written demand in accordance with the Contract Documents prior to this date and identified by the Contractor as unsettled in the final Application for Payment and are either in arbitration or court litigation, as the case may be, in accordance with the Contract Documents.

The undersigned further covenants that subcontractors, suppliers, and material suppliers, and any or all other persons supplying materials, supplies, service or labor used directly or indirectly in the prosecution of the work provided for in the Contract, have been paid in full for all work under this contract.

The undersigned agrees to maintain in full force and effect the provisions of the Contract Documents respecting the guaranty against defective work, and any other special guaranties required by the Contract Documents, for the terms provided in the Contract Documents, which terms shall begin to run from the date specified in the Contract Documents.

The undersigned represents and warrants that the statements contained in the foregoing Release are true and correct.

IN WITNESS WHEREOF, I have hereunto set my hand and seal this \_\_\_\_ day of \_\_\_\_\_, 20\_\_.

WITNESSES:

\_\_\_\_\_  
CONTRACTOR

\_\_\_\_\_  
By: \_\_\_\_\_

STATE OF \_\_\_\_\_ Title: \_\_\_\_\_

COUNTY OF \_\_\_\_\_

Sworn to and subscribed before me this \_\_\_\_ day of \_\_\_\_\_, 20\_\_.

(NOTARY SEAL)

\_\_\_\_\_  
NOTARY PUBLIC  
My Commission Expires: \_\_\_\_\_



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**FEDERAL CONTRACT PROVISIONS**  
**ACCESS TO RECORDS AND REPORTS**  
(Reference: 2 CFR § 200.326, 2 CFR § 200.333)

The Contractor must maintain an acceptable cost accounting system. The Contractor agrees to provide the Sponsor, the Federal Aviation Administration, and the Comptroller General of the United States or any of their duly authorized representative's access to any books, documents, papers, and records of the contractor which are directly pertinent to the specific contract for the purpose of making audit, examination, excerpts and transcriptions. The Contractor agrees to maintain all books, records and reports required under this contract for a period of not less than three years after final payment is made and all pending matters are closed.

**FEDERAL CONTRACT PROVISIONS  
AFFIRMATIVE ACTION REQUIREMENT**

(Reference: 41 CFR part 60-4, Executive Order 11246)

**NOTICE OF REQUIREMENT FOR AFFIRMATIVE ACTION to  
ENSURE EQUAL EMPLOYMENT OPPORTUNITY**

1. The Offeror's or Bidder's attention is called to the "Equal Opportunity Clause" and the "Standard Federal Equal Employment Opportunity Construction Contract Specifications" set forth herein.
2. The goals and timetables for minority participation, expressed in percentage terms for the Contractor's aggregate workforce in each trade on all construction work in the covered area, are as follows:

Goals for minority participation for each trade: 12.5%

These goals are applicable to all of the Contractor's construction work (whether or not it is Federal or federally assisted) performed in the covered area. If the Contractor performs construction work in a geographical area located outside of the covered area, it shall apply the goals established for such geographical area where the work is actually performed. With regard to this second area, the Contractor also is subject to the goals for both its federally involved and non-federally involved construction.

The Contractor's compliance with the Executive Order and the regulations in 41 CFR Part 60-4 shall be based on its implementation of the Equal Opportunity Clause, specific affirmative action obligations required by the specifications set forth in 41 CFR 60-4.3(a) and its efforts to meet the goals. The hours of minority and female employment and training must be substantially uniform throughout the length of the contract, and in each trade, and the Contractor shall make a good faith effort to employ minorities and women evenly on each of its projects. The transfer of minority or female employees or trainees from Contractor to Contractor or from project to project for the sole purpose of meeting the Contractor's goals shall be a violation of the contract, the Executive Order and the regulations in 41 CFR Part 60-4. Compliance with the goals will be measured against the total work hours performed.

3. The Contractor shall provide written notification to the Director of the Office of Federal Contract Compliance Programs (OFCCP) within 10 working days of award of any construction subcontract in excess of \$10,000 at any tier for construction work under the contract resulting from this solicitation. The notification shall list the name, address, and telephone number of the subcontractor; employer identification number of the subcontractor; estimated dollar amount of the subcontract; estimated starting and completion dates of the subcontract; and the geographical area in which the subcontract is to be performed.
4. As used in this notice and in the contract resulting from this solicitation, the "covered area" is Florida, Escambia County, Pensacola.

**FEDERAL CONTRACT PROVISIONS**  
**BREACH OF CONTRACT TERMS**  
(Reference 2 CFR § 200 Appendix II(A))

Any violation or breach of terms of this contract on the part of the *Contractor* or its subcontractors may result in the suspension or termination of this contract or such other action that may be necessary to enforce the rights of the parties of this agreement.

Owner will provide *Contractor* written notice that describes the nature of the breach and corrective actions the *Contractor* must undertake in order to avoid termination of the contract. Owner reserves the right to withhold payments to Contractor until such time the Contractor corrects the breach or the Owner elects to terminate the contract. The Owner's notice will identify a specific date by which the *Contractor* must correct the breach. Owner may proceed with termination of the contract if the *Contractor* fails to correct the breach by the deadline indicated in the Owner's notice.

The duties and obligations imposed by the Contract Documents and the rights and remedies available thereunder are in addition to, and not a limitation of, any duties, obligations, rights and remedies otherwise imposed or available by law.

**FEDERAL CONTRACT PROVISIONS**  
**BUY AMERICAN PREFERENCE**  
(Reference: 49 USC § 50101)

The Contractor agrees to comply with 49 USC § 50101, which provides that Federal funds may not be obligated unless all steel and manufactured goods used in AIP funded projects are produced in the United States, unless the Federal Aviation Administration has issued a waiver for the product; the product is listed as an Excepted Article, Material or Supply in Federal Acquisition Regulation subpart 25.108; or is included in the FAA Nationwide Buy American Waivers Issued list.

A bidder or offeror must complete and submit the Buy America certification included herein with their bid or offer. The Owner will reject as nonresponsive any bid or offer that does not include a completed Certificate of Buy American Compliance.

**Certificate of Buy American Compliance for Manufactured Products**  
(Non-building construction projects, equipment acquisition projects)

As a matter of bid responsiveness, the bidder or offeror must complete, sign, date, and submit this certification statement with their proposal. The bidder or offeror must indicate how they intend to comply with 49 USC § 50101 by selecting one on the following certification statements. These statements are mutually exclusive. Bidder must select one or the other (not both) by inserting a checkmark (✓) or the letter "X".

- ☐ Bidder or offeror hereby certifies that it will comply with 49 USC § 50101 by:
- a) Only installing steel and manufactured products produced in the United States;
  - b) Installing manufactured products for which the Federal Aviation Administration (FAA) has issued a waiver as indicated by inclusion on the current FAA Nationwide Buy American Waivers Issued listing; or
  - c) Installing products listed as an Excepted Article, Material or Supply in Federal Acquisition Regulation Subpart 25.108.

By selecting this certification statement, the bidder or offeror agrees:

1. To provide to the Owner evidence that documents the source and origin of the steel and manufactured product.
2. To faithfully comply with providing U.S. domestic product.
3. To furnish U.S. domestic product for any waiver request that the FAA rejects
4. To refrain from seeking a waiver request after establishment of the contract, unless extenuating circumstances emerge that the FAA determines justified.

- ☐ The bidder or offeror hereby certifies it cannot comply with the 100 percent Buy American Preferences of 49 USC § 50101(a) but may qualify for either a Type 3 or Type 4 waiver under 49 USC § 50101(b). By selecting this certification statement, the apparent bidder or offeror with the apparent low bid agrees:
1. To submit to the Owner within 15 calendar days of the bid opening, a formal waiver request and required documentation that supports the type of waiver being requested.
  2. That failure to submit the required documentation within the specified timeframe is cause for a non-responsive determination may result in rejection of the proposal.
  3. To faithfully comply with providing U.S. domestic products at or above the approved U.S. domestic content percentage as approved by the FAA.
  4. To refrain from seeking a waiver request after establishment of the contract, unless extenuating circumstances emerge that the FAA determines justified.

**Required Documentation**

**Type 3 Waiver** – The cost of the item components and subcomponents produced in the United States is more than 60 percent of the cost of all components and subcomponents of the "item". The required documentation for a Type 3 waiver is:

- a) Listing of all product components and subcomponents that are not comprised of 100 percent U.S. domestic content (Excludes products listed on the FAA Nationwide Buy American Waivers Issued listing and products excluded by

Federal Acquisition Regulation Subpart 25.108; products of unknown origin must be considered as non-domestic products in their entirety).

- b) Cost of non-domestic components and subcomponents, excluding labor costs associated with final assembly at place of manufacture.
- c) Percentage of non-domestic component and subcomponent cost as compared to total "item" component and subcomponent costs, excluding labor costs associated with final assembly at place of manufacture.

**Type 4 Waiver** – Total cost of project using U.S. domestic source product exceeds the total project cost using non-domestic product by 25 percent. The required documentation for a Type 4 of waiver is:

- a) Detailed cost information for total project using U.S. domestic product
- b) Detailed cost information for total project using non-domestic product

**False Statements:** Per 49 USC § 47126, this certification concerns a matter within the jurisdiction of the Federal Aviation Administration and the making of a false, fictitious or fraudulent certification may render the maker subject to prosecution under Title 18, United States Code.

\_\_\_\_\_  
Date

\_\_\_\_\_  
Signature

\_\_\_\_\_  
Company Name

\_\_\_\_\_  
Title

**FEDERAL CONTRACT PROVISIONS**

**CIVIL RIGHTS - GENERAL**

(Reference: 49 USC § 47123)

The Contractor agrees that it will comply with pertinent statutes, Executive Orders and such rules as are promulgated to ensure that no person shall, on the grounds of race, creed, color, national origin, sex, age, or handicap be excluded from participating in any activity conducted with or benefiting from Federal assistance.

This provision binds the contractors from the bid solicitation period through the completion of the contract. This provision is in addition to that required of Title VI of the Civil Rights Act of 1964.



**FEDERAL CONTRACT PROVISIONS**  
**CIVIL RIGHTS – TITLE VI ASSURANCES**  
(Reference: 49 USC § 47123, FAA Order 1400.11)

**Title VI Solicitation Notice:**

The City of Pensacola, in accordance with the provisions of Title VI of the Civil Rights Act of 1964 (78 Stat. 252, 42 U.S.C. §§ 2000d to 2000d-4) and the Regulations, hereby notifies all bidders that it will affirmatively ensure that any contract entered into pursuant to this advertisement, disadvantaged business enterprises will be afforded full and fair opportunity to submit bids in response to this invitation and will not be discriminated against on the grounds of race, color, or national origin in consideration for an award.

**Title VI Clauses for Compliance with Nondiscrimination Requirements**

**Compliance with Nondiscrimination Requirements**

During the performance of this contract, the contractor, for itself, its assignees, and successors in interest (hereinafter referred to as the “contractor”) agrees as follows:

During the performance of this contract, the Contractor, for itself, its assignees, and successors in interest (hereinafter referred to as the “Contractor”), agrees as follows:

1. **Compliance with Regulations:** The Contractor (hereinafter includes consultants) will comply with the Title VI List of Pertinent Nondiscrimination Acts and Authorities, as they may be amended from time to time, which are herein incorporated by reference and made a part of this contract.
2. **Nondiscrimination:** The Contractor, with regard to the work performed by it during the contract, will not discriminate on the grounds of race, color, or national origin in the selection and retention of subcontractors, including procurements of materials and leases of equipment. The Contractor will not participate directly or indirectly in the discrimination prohibited by the Nondiscrimination Acts and Authorities, including employment practices when the contract covers any activity, project, or program set forth in Appendix B of 49 CFR part 21.
3. **Solicitations for Subcontracts, including Procurements of Materials and Equipment:** In all solicitations, either by competitive bidding or negotiation made by the Contractor for work to be performed under a subcontract, including procurements of materials, or leases of equipment, each potential subcontractor or supplier will be notified by the Contractor of the contractor’s obligations under this contract and the Nondiscrimination Acts and Authorities on the grounds of race, color, or national origin.
4. **Information and Reports:** The Contractor will provide all information and reports required by the Acts, the Regulations, and directives issued pursuant thereto and will permit access to its books, records, accounts, other sources of information, and its facilities as may be determined by the sponsor or the Federal Aviation Administration to be pertinent to ascertain compliance with such Nondiscrimination Acts and Authorities and instructions. Where any information required of a contractor is in the exclusive possession of another who fails or refuses to furnish the information, the Contractor will

so certify to the sponsor or the Federal Aviation Administration, as appropriate, and will set forth what efforts it has made to obtain the information.

5. **Sanctions for Noncompliance:** In the event of a Contractor's noncompliance with the non-discrimination provisions of this contract, the sponsor will impose such contract sanctions as it or the Federal Aviation Administration may determine to be appropriate, including, but not limited to:
  - a. Withholding payments to the Contractor under the contract until the Contractor complies; and/or
  - b. Cancelling, terminating, or suspending a contract, in whole or in part.
6. **Incorporation of Provisions:** The Contractor will include the provisions of paragraphs one through six in every subcontract, including procurements of materials and leases of equipment, unless exempt by the Acts, the Regulations, and directives issued pursuant thereto. The Contractor will take action with respect to any subcontract or procurement as the sponsor or the Federal Aviation Administration may direct as a means of enforcing such provisions including sanctions for noncompliance. Provided, that if the Contractor becomes involved in, or is threatened with litigation by a subcontractor, or supplier because of such direction, the Contractor may request the sponsor to enter into any litigation to protect the interests of the sponsor. In addition, the Contractor may request the United States to enter into the litigation to protect the interests of the United States.

#### **Title VI Clauses for Deeds Transferring United States Property**

(Source: Appendix B of Appendix 4 of FAA Order 1400.11, Nondiscrimination in Federally-Assisted Programs at the Federal Aviation Administration)

#### **CLAUSES FOR DEEDS TRANSFERRING UNITED STATES PROPERTY**

The following clauses will be included in deeds effecting or recording the transfer of real property, structures, or improvements thereon, or granting interest therein from the United States pursuant to the provisions of the Airport Improvement Program grant assurances.

**NOW, THEREFORE,** the Federal Aviation Administration as authorized by law and upon the condition that the Pensacola International Airport will accept title to the lands and maintain the project constructed thereon in accordance with Federal Aviation Administration, for the AIP Program, and the policies and procedures prescribed by the Federal Aviation Administration of the U.S. Department of Transportation in accordance and in compliance with all requirements imposed by Title 49, Code of Federal Regulations, U.S. Department of Transportation, Subtitle A, Office of the Secretary, Part 21, Non-discrimination in Federally-assisted programs of the U.S. Department of Transportation pertaining to and effectuating the provisions of Title VI of the Civil Rights Act of 1964 (78 Stat. 252; 42 U.S.C. § 2000d to 2000d-4), does hereby remise, release, quitclaim and convey unto the Pensacola International Airport all the right, title and interest of the U.S. Department of Transportation/Federal Aviation Administration in and to said lands described in and made a part hereof.

#### **(HABENDUM CLAUSE)**

**TO HAVE AND TO HOLD** said lands and interests therein unto Pensacola International Airport and its successors forever, subject, however, to the covenants, conditions, restrictions and reservations herein contained as follows, which will remain in effect for the period during which the real property or structures are used for a purpose for which Federal financial assistance is

extended or for another purpose involving the provision of similar services or benefits and will be binding on the Pensacola International Airport, its successors and assigns.

The Pensacola International Airport, in consideration of the conveyance of said lands and interests in lands, does hereby covenant and agree as a covenant running with the land for itself, its successors and assigns, that (1) no person will on the grounds of race, color, or national origin, be excluded from participation in, be denied the benefits of, or be otherwise subjected to discrimination with regard to any facility located wholly or in part on, over, or under such lands hereby conveyed, and (2) that the Pensacola International Airport will use the lands and interests in lands and interests in lands so conveyed, in compliance with all requirements imposed by or pursuant to Title 49, Code of Federal Regulations, U.S. Department of Transportation, Subtitle A, Office of the Secretary, Part 21, Non-discrimination in Federally-assisted programs of the U.S. Department of Transportation, Effectuation of Title VI of the Civil Rights Act of 1964, and as said Regulations and Acts may be amended.

#### **Title VI List of Pertinent Nondiscrimination Authorities**

During the performance of this contract, the Contractor, for itself, its assignees, and successors in interest (hereinafter referred to as the "Contractor") agrees to comply with the following non-discrimination statutes and authorities; including but not limited to:

- Title VI of the Civil Rights Act of 1964 (42 USC § 2000d *et seq.*, 78 stat. 252) (prohibits discrimination on the basis of race, color, national origin);
- 49 CFR part 21 (Non-discrimination in Federally-assisted programs of the Department of Transportation—Effectuation of Title VI of the Civil Rights Act of 1964);
- The Uniform Relocation Assistance and Real Property Acquisition Policies Act of 1970, (42 USC § 4601) (prohibits unfair treatment of persons displaced or whose property has been acquired because of Federal or Federal-aid programs and projects);
- Section 504 of the Rehabilitation Act of 1973 (29 USC § 794 *et seq.*), as amended (prohibits discrimination on the basis of disability); and 49 CFR part 27;
- The Age Discrimination Act of 1975, as amended (42 USC § 6101 *et seq.*) (prohibits discrimination on the basis of age);
- Airport and Airway Improvement Act of 1982 (49 USC § 471, Section 47123), as amended (prohibits discrimination based on race, creed, color, national origin, or sex);
- The Civil Rights Restoration Act of 1987 (PL 100-209) (broadened the scope, coverage and applicability of Title VI of the Civil Rights Act of 1964, the Age Discrimination Act of 1975 and Section 504 of the Rehabilitation Act of 1973, by expanding the definition of the terms "programs or activities" to include all of the programs or activities of the Federal-aid recipients, sub-recipients and contractors, whether such programs or activities are Federally funded or not);
- Titles II and III of the Americans with Disabilities Act of 1990, which prohibit discrimination on the basis of disability in the operation of public entities, public and private transportation systems, places of public accommodation, and certain testing entities (42 USC §§ 12131 – 12189) as implemented by U.S. Department of Transportation regulations at 49 CFR parts 37 and 38;
- The Federal Aviation Administration's Nondiscrimination statute (49 USC § 47123) (prohibits discrimination on the basis of race, color, national origin, and sex);
- Executive Order 12898, Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations, which ensures nondiscrimination against minority populations by discouraging programs, policies, and activities with

disproportionately high and adverse human health or environmental effects on minority and low-income populations;

- Executive Order 13166, Improving Access to Services for Persons with Limited English Proficiency, and resulting agency guidance, national origin discrimination includes discrimination because of limited English proficiency (LEP). To ensure compliance with Title VI, you must take reasonable steps to ensure that LEP persons have meaningful access to your programs (70 Fed. Reg. at 74087 to 74100);
- Title IX of the Education Amendments of 1972, as amended, which prohibits you from discriminating because of sex in education programs or activities (20 USC 1681 et seq).

**FEDERAL CONTRACT PROVISIONS  
CLEAN AIR AND WATER POLLUTION CONTROL**

(Reference: 2 CFR § 200, Appendix II(G))

Contractor agrees to comply with all applicable standards, orders, and regulations issued pursuant to the Clean Air Act (42 USC § 740-7671q) and the Federal Water Pollution Control Act as amended (33 USC § 1251-1387). The Contractor agrees to report any violation to the Owner immediately upon discovery. The Owner assumes responsibility for notifying the Environmental Protection Agency (EPA) and the Federal Aviation Administration.

Contractor must include this requirement in all subcontracts that exceeds \$150,000.

**FEDERAL CONTRACT PROVISIONS**  
**CONTRACT WORKHOURS AND SAFETY STANDARDS ACT REQUIREMENTS**

(Reference: 2 CFR § 200 Appendix II (E))

1. Overtime Requirements.

No contractor or subcontractor contracting for any part of the contract work which may require or involve the employment of laborers or mechanics shall require or permit any such laborer or mechanic, including watchmen and guards, in any workweek in which he or she is employed on such work to work in excess of forty hours in such workweek unless such laborer or mechanic receives compensation at a rate not less than one and one-half times the basic rate of pay for all hours worked in excess of forty hours in such workweek.

2. Violation; Liability for Unpaid Wages; Liquidated Damages.

In the event of any violation of the clause set forth in paragraph (1) of this clause, the Contractor and any subcontractor responsible therefor shall be liable for the unpaid wages. In addition, such contractor and subcontractor shall be liable to the United States (in the case of work done under contract for the District of Columbia or a territory, to such District or to such territory), for liquidated damages. Such liquidated damages shall be computed with respect to each individual laborer or mechanic, including watchmen and guards, employed in violation of the clause set forth in paragraph (1) of this clause, in the sum of \$10 for each calendar day on which such individual was required or permitted to work in excess of the standard workweek of forty hours without payment of the overtime wages required by the clause set forth in paragraph (1) of this clause.

3. Withholding for Unpaid Wages and Liquidated Damages.

The Federal Aviation Administration (FAA) or the Owner shall upon its own action or upon written request of an authorized representative of the Department of Labor withhold or cause to be withheld, from any moneys payable on account of work performed by the contractor or subcontractor under any such contract or any other Federal contract with the same prime contractor, or any other federally assisted contract subject to the Contract Work Hours and Safety Standards Act, which is held by the same prime contractor, such sums as may be determined to be necessary to satisfy any liabilities of such contractor or subcontractor for unpaid wages and liquidated damages as provided in the clause set forth in paragraph (2) of this clause.

4. Subcontractors.

The Contractor or subcontractor shall insert in any subcontracts the clauses set forth in paragraphs (1) through (4) and also a clause requiring the subcontractor to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for compliance by any subcontractor or lower tier subcontractor with the clauses set forth in paragraphs (1) through (4) of this clause.

**FEDERAL CONTRACT PROVISIONS  
COPELAND “ANTI-KICKBACK” ACT**

(Reference: 2 CFR § 200 Appendix II(D), 29 CFR parts 3 & 5)

Contractor must comply with the requirements of the Copeland “Anti-Kickback” Act (18 USC 874 and 40 USC 3145), as supplemented by Department of Labor regulation 29 CFR part 3. Contractor and subcontractors are prohibited from inducing, by any means, any person employed on the project to give up any part of the compensation to which the employee is entitled. The Contractor and each Subcontractor must submit to the Owner, a weekly statement on the wages paid to each employee performing on covered work during the prior week. Owner must report any violations of the Act to the Federal Aviation Administration.

**FEDERAL CONTRACT PROVISIONS**  
**DAVIS-BACON REQUIREMENTS**  
(Reference: 2 CFR § 200 Appendix II(D))

1. Minimum Wages.

(i) All laborers and mechanics employed or working upon the site of the work will be paid unconditionally and not less often than once a week, and without subsequent deduction or rebate on any account (except such payroll deductions as are permitted by the Secretary of Labor under the Copeland Act (29 CFR Part 3)), the full amount of wages and bona fide fringe benefits (or cash equivalent thereof) due at time of payment computed at rates not less than those contained in the wage determination of the Secretary of Labor which is attached hereto and made a part hereof, regardless of any contractual relationship which may be alleged to exist between the Contractor and such laborers and mechanics.

Contributions made or costs reasonably anticipated for bona fide fringe benefits under section 1(b)(2) of the Davis-Bacon Act on behalf of laborers or mechanics are considered wages paid to such laborers or mechanics, subject to the provisions of paragraph (1)(iv) of this section; also, regular contributions made or costs incurred for more than a weekly period (but not less often than quarterly) under plans, funds, or programs which cover the particular weekly period, are deemed to be constructively made or incurred during such weekly period. Such laborers and mechanics shall be paid the appropriate wage rate and fringe benefits on the wage determination for the classification of work actually performed, without regard to skill, except as provided in 29 CFR Part 5.5(a)(4). Laborers or mechanics performing work in more than one classification may be compensated at the rate specified for each classification for the time actually worked therein: *Provided* that the employer's payroll records accurately set forth the time spent in each classification in which work is performed. The wage determination (including any additional classification and wage rates conformed under (1)(ii) of this section) and the Davis-Bacon poster (WH-1321) shall be posted at all times by the Contractor and its subcontractors at the site of the work in a prominent and accessible place where it can easily be seen by the workers.

(ii)(A) The contracting officer shall require that any class of laborers or mechanics, including helpers, which is not listed in the wage determination and which is to be employed under the contract shall be classified in conformance with the wage determination. The contracting officer shall approve an additional classification and wage rate and fringe benefits therefore only when the following criteria have been met:

(1) The work to be performed by the classification requested is not performed by a classification in the wage determination;

(2) The classification is utilized in the area by the construction industry; and

(3) The proposed wage rate, including any bona fide fringe benefits, bears a reasonable relationship to the wage rates contained in the wage determination.

(B) If the Contractor and the laborers and mechanics to be employed in the classification (if known), or their representatives, and the contracting officer agree on the classification and wage rate (including the amount designated for fringe benefits where appropriate), a report of the action taken shall be sent by the contracting officer to the Administrator of the Wage and Hour Division, Employment Standards Administration, U.S. Department of Labor, Washington, DC



20210. The Administrator, or an authorized representative, will approve, modify, or disapprove every additional classification action within 30 days of receipt and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(C) In the event the Contractor, the laborers, or mechanics to be employed in the classification, or their representatives, and the contracting officer do not agree on the proposed classification and wage rate (including the amount designated for fringe benefits where appropriate), the contracting officer shall refer the questions, including the views of all interested parties and the recommendation of the contracting officer, to the Administrator for determination. The Administrator, or an authorized representative, will issue a determination within 30 days of receipt and so advise the contracting officer or will notify the contracting officer within the 30-day period that additional time is necessary.

(D) The wage rate (including fringe benefits where appropriate) determined pursuant to subparagraphs (1)(ii) (B) or (C) of this paragraph, shall be paid to all workers performing work in the classification under this contract from the first day on which work is performed in the classification.

(iii) Whenever the minimum wage rate prescribed in the contract for a class of laborers or mechanics includes a fringe benefit which is not expressed as an hourly rate, the contractor shall either pay the benefit as stated in the wage determination or shall pay another bona fide fringe benefit or an hourly cash equivalent thereof.

(iv) If the Contractor does not make payments to a trustee or other third person, the Contractor may consider as part of the wages of any laborer or mechanic the amount of any costs reasonably anticipated in providing bona fide fringe benefits under a plan or program: *Provided* that the Secretary of Labor has found, upon the written request of the Contractor, that the applicable standards of the Davis-Bacon Act have been met. The Secretary of Labor may require the Contractor to set aside in a separate account assets for the meeting of obligations under the plan or program.

## 2. Withholding.

The Federal Aviation Administration or the sponsor shall upon its own action or upon written request of an authorized representative of the Department of Labor withhold or cause to be withheld from the Contractor under this contract or any other Federal contract with the same prime contractor, or any other federally-assisted contract subject to Davis-Bacon prevailing wage requirements, which is held by the same prime contractor, so much of the accrued payments or advances as may be considered necessary to pay laborers and mechanics, including apprentices, trainees, and helpers, employed by the Contractor or any subcontractor the full amount of wages required by the contract. In the event of failure to pay any laborer or mechanic, including any apprentice, trainee, or helper, employed or working on the site of work, all or part of the wages required by the contract, the Federal Aviation Administration may, after written notice to the Contractor, Sponsor, Applicant, or Owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds until such violations have ceased.

## 3. Payrolls and Basic Records.

(i) Payrolls and basic records relating thereto shall be maintained by the Contractor during the course of the work and preserved for a period of three years thereafter for all laborers and mechanics working at the site of the work. Such records shall contain the name, address, and social security number of each such worker; his or her correct classification; hourly rates of wages paid (including rates of contributions or costs anticipated for bona fide fringe benefits or cash equivalents thereof of the types described in 1(b)(2)(B) of the Davis-Bacon Act); daily and weekly number of hours worked; deductions made; and actual wages paid. Whenever the Secretary of Labor has found under 29 CFR 5.5(a)(1)(iv) that the wages of any laborer or mechanic include the amount of any costs reasonably anticipated in providing benefits under a plan or program described in section 1(b)(2)(B) of the Davis-Bacon Act, the Contractor shall maintain records that show that the commitment to provide such benefits is enforceable, that the plan or program is financially responsible, and that the plan or program has been communicated in writing to the laborers or mechanics affected, and that show the costs anticipated or the actual costs incurred in providing such benefits. Contractors employing apprentices or trainees under approved programs shall maintain written evidence of the registration of apprenticeship programs and certification of trainee programs, the registration of the apprentices and trainees, and the ratios and wage rates prescribed in the applicable programs.

(ii)(A) The Contractor shall submit weekly for each week in which any contract work is performed a copy of all payrolls to the Federal Aviation Administration if the agency is a party to the contract, but if the agency is not such a party, the Contractor will submit the payrolls to the applicant, Sponsor, or Owner, as the case may be, for transmission to the Federal Aviation Administration. The payrolls submitted shall set out accurately and completely all of the information required to be maintained under 29 CFR 5.5(a)(3)(i), except that full social security numbers and home addresses shall not be included on weekly transmittals. Instead the payrolls shall only need to include an individually identifying number for each employee (e.g. the last four digits of the employee's social security number). The required weekly payroll information may be submitted in any form desired. Optional Form WH-347 is available for this purpose from the Wage and Hour Division Web site at [www.dol.gov/whd/forms/wh347instr.htm](http://www.dol.gov/whd/forms/wh347instr.htm) or its successor site. The prime contractor is responsible for the submission of copies of payrolls by all subcontractors. Contractors and subcontractors shall maintain the full social security number and current address of each covered worker and shall provide them upon request to the Federal Aviation Administration if the agency is a party to the contract, but if the agency is not such a party, the Contractor will submit them to the applicant, sponsor, or Owner, as the case may be, for transmission to the Federal Aviation Administration, the Contractor, or the Wage and Hour Division of the Department of Labor for purposes of an investigation or audit of compliance with prevailing wage requirements. It is not a violation of this section for a prime contractor to require a subcontractor to provide addresses and social security numbers to the prime contractor for its own records, without weekly submission to the sponsoring government agency (or the applicant, Sponsor, or Owner).

(B) Each payroll submitted shall be accompanied by a "Statement of Compliance," signed by the Contractor or subcontractor or his or her agent who pays or supervises the payment of the persons employed under the contract and shall certify the following:

(1) The payroll for the payroll period contains the information required to be provided under 29 CFR § 5.5(a)(3)(ii), the appropriate information is being maintained under 29 CFR § 5.5 (a)(3)(i), and that such information is correct and complete;

(2) Each laborer and mechanic (including each helper, apprentice, and trainee) employed on the contract during the payroll period has been paid the full weekly wages earned, without rebate,

either directly or indirectly, and that no deductions have been made either directly or indirectly from the full wages earned, other than permissible deductions as set forth in Regulations 29 CFR Part 3;

(3) Each laborer or mechanic has been paid not less than the applicable wage rates and fringe benefits or cash equivalents for the classification of work performed, as specified in the applicable wage determination incorporated into the contract.

(C) The weekly submission of a properly executed certification set forth on the reverse side of Optional Form WH-347 shall satisfy the requirement for submission of the "Statement of Compliance" required by paragraph (3)(ii)(B) of this section.

(D) The falsification of any of the above certifications may subject the Contractor or subcontractor to civil or criminal prosecution under Section 1001 of Title 18 and Section 231 of Title 31 of the United States Code.

(iii) The Contractor or subcontractor shall make the records required under paragraph (3)(i) of this section available for inspection, copying, or transcription by authorized representatives of the sponsor, the Federal Aviation Administration, or the Department of Labor and shall permit such representatives to interview employees during working hours on the job. If the Contractor or subcontractor fails to submit the required records or to make them available, the Federal agency may, after written notice to the Contractor, Sponsor, applicant, or Owner, take such action as may be necessary to cause the suspension of any further payment, advance, or guarantee of funds. Furthermore, failure to submit the required records upon request or to make such records available may be grounds for debarment action pursuant to 29 CFR 5.12.

#### 4. Apprentices and Trainees.

(i) Apprentices. Apprentices will be permitted to work at less than the predetermined rate for the work they performed when they are employed pursuant to and individually registered in a bona fide apprenticeship program registered with the U.S. Department of Labor, Employment and Training Administration, Bureau of Apprenticeship and Training, or with a State Apprenticeship Agency recognized by the Bureau, or if a person is employed in his or her first 90 days of probationary employment as an apprentice in such an apprenticeship program, who is not individually registered in the program, but who has been certified by the Bureau of Apprenticeship and Training or a State Apprenticeship Agency (where appropriate) to be eligible for probationary employment as an apprentice. The allowable ratio of apprentices to journeymen on the job site in any craft classification shall not be greater than the ratio permitted to the contractor as to the entire work force under the registered program. Any worker listed on a payroll at an apprentice wage rate, who is not registered or otherwise employed as stated above, shall be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any apprentice performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. Where a contractor is performing construction on a project in a locality other than that in which its program is registered, the ratios and wage rates (expressed in percentages of the journeyman's hourly rate) specified in the Contractor's or subcontractor's registered program shall be observed. Every apprentice must be paid at not less than the rate specified in the registered program for the apprentice's level of progress, expressed as a percentage of the journeymen hourly rate specified in the applicable wage determination. Apprentices shall be paid fringe benefits in accordance with the provisions of the apprenticeship program. If the apprenticeship

program does not specify fringe benefits, apprentices must be paid the full amount of fringe benefits listed on the wage determination for the applicable classification. If the Administrator determines that a different practice prevails for the applicable apprentice classification, fringes shall be paid in accordance with that determination. In the event the Bureau of Apprenticeship and Training, or a State Apprenticeship Agency recognized by the Bureau, withdraws approval of an apprenticeship program, the Contractor will no longer be permitted to utilize apprentices at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(ii) Trainees. Except as provided in 29 CFR 5.16, trainees will not be permitted to work at less than the predetermined rate for the work performed unless they are employed pursuant to and individually registered in a program which has received prior approval, evidenced by formal certification by the U.S. Department of Labor, Employment and Training Administration. The ratio of trainees to journeymen on the job site shall not be greater than permitted under the plan approved by the Employment and Training Administration. Every trainee must be paid at not less than the rate specified in the approved program for the trainee's level of progress, expressed as a percentage of the journeyman hourly rate specified in the applicable wage determination. Trainees shall be paid fringe benefits in accordance with the provisions of the trainee program. If the trainee program does not mention fringe benefits, trainees shall be paid the full amount of fringe benefits listed on the wage determination unless the Administrator of the Wage and Hour Division determines that there is an apprenticeship program associated with the corresponding journeyman wage rate on the wage determination that provides for less than full fringe benefits for apprentices. Any employee listed on the payroll at a trainee rate that is not registered and participating in a training plan approved by the Employment and Training Administration shall be paid not less than the applicable wage rate on the wage determination for the classification of work actually performed. In addition, any trainee performing work on the job site in excess of the ratio permitted under the registered program shall be paid not less than the applicable wage rate on the wage determination for the work actually performed. In the event the Employment and Training Administration withdraws approval of a training program, the Contractor will no longer be permitted to utilize trainees at less than the applicable predetermined rate for the work performed until an acceptable program is approved.

(iii) Equal Employment Opportunity. The utilization of apprentices, trainees, and journeymen under this part shall be in conformity with the equal employment opportunity requirements of Executive Order 11246, as amended, and 29 CFR Part 30.

#### 5. Compliance with Copeland Act Requirements.

The Contractor shall comply with the requirements of 29 CFR Part 3, which are incorporated by reference in this contract.

#### 6. Subcontracts.

The Contractor or subcontractor shall insert in any subcontracts the clauses contained in 29 CFR Part 5.5(a)(1) through (10) and such other clauses as the Federal Aviation Administration may by appropriate instructions require, and also a clause requiring the subcontractors to include these clauses in any lower tier subcontracts. The prime contractor shall be responsible for the compliance by any subcontractor or lower tier subcontractor with all the contract clauses in 29 CFR Part 5.5.

#### 7. Contract Termination: Debarment.

A breach of the contract clauses in paragraph 1 through 10 of this section may be grounds for termination of the contract, and for debarment as a contractor and a subcontractor as provided in 29 CFR 5.12.

8. Compliance with Davis-Bacon and Related Act Requirements.

All rulings and interpretations of the Davis-Bacon and Related Acts contained in 29 CFR Parts 1, 3, and 5 are herein incorporated by reference in this contract.

9. Disputes Concerning Labor Standards.

Disputes arising out of the labor standards provisions of this contract shall not be subject to the general disputes clause of this contract. Such disputes shall be resolved in accordance with the procedures of the Department of Labor set forth in 29 CFR Parts 5, 6, and 7. Disputes within the meaning of this clause include disputes between the Contractor (or any of its subcontractors) and the contracting agency, the U.S. Department of Labor, or the employees or their representatives.

10. Certification of Eligibility.

(i) By entering into this contract, the Contractor certifies that neither it (nor he or she) nor any person or firm who has an interest in the Contractor's firm is a person or firm ineligible to be awarded Government contracts by virtue of section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1).

(ii) No part of this contract shall be subcontracted to any person or firm ineligible for award of a Government contract by virtue of section 3(a) of the Davis-Bacon Act or 29 CFR 5.12(a)(1).

(iii) The penalty for making false statements is prescribed in the U.S. Criminal Code, 18 USC 1001.

**FEDERAL CONTRACT PROVISIONS  
DEBARMENT AND SUSPENSION (NON-PROCUREMENT)**

(Reference: 2 CFR part 180 (Subpart C), 2 CFR part 1200, DOT Order 4200.5)

**CERTIFICATION OF OFFER/BIDDER REGARDING DEBARMENT**

By submitting a bid/proposal under this solicitation, the bidder or offeror certifies that at the time the bidder or offeror submits its proposal that neither it nor its principals are presently debarred or suspended by any Federal department or agency from participation in this transaction.

**CERTIFICATION OF LOWER TIER CONTRACTORS REGARDING DEBAMENT**

The successful bidder, by administering each lower tier subcontract that exceeds \$25,000 as a "covered transaction", must verify each lower tier participant of a "covered transaction" under the project is not presently debarred or otherwise disqualified from participation in this federally assisted project. The successful bidder will accomplish this by:

1. Checking the System for Award Management at website: <http://www.sam.gov>
2. Collecting a certification statement similar to the Certificate Regarding Debarment and Suspension (Bidder or Offeror), above.
3. Inserting a clause or condition in the covered transaction with the lower tier contract

If the FAA later determines that a lower tier participant failed to tell a higher tier that it was excluded or disqualified at the time it entered the covered transaction, the FAA may pursue any available remedy, including suspension and debarment.

**FEDERAL CONTRACT PROVISIONS  
DISADVANTAGED BUSINESS ENTERPRISE**

(Reference: 49 CFR part 26)

**Contract Assurance (§ 26.13)** - The contractor or subcontractor shall not discriminate on the basis of race, color, national origin, or sex in the performance of this contract. The contractor shall carry out applicable requirements of 49 CFR Part 26 in the award and administration of DOT assisted contracts. Failure by the contractor to carry out these requirements is a material breach of this contract, which may result in the termination of this contract or such other remedy, as the recipient deems appropriate.

**Prompt Payment (§26.29)**- The prime contractor agrees to pay each subcontractor under this prime contract for satisfactory performance of its contract no later than **30 days** from the receipt of each payment the prime contractor receives from **Owner**. The prime contractor agrees further to return retainage payments to each subcontractor within **30 days** after the subcontractor's work is satisfactorily completed. Any delay or postponement of payment from the above referenced time frame may occur only for good cause following written approval of the **Owner**. This clause applies to both DBE and non-DBE subcontractors.

**FEDERAL CONTRACT PROVISIONS  
DISTRACTED DRIVING**

(Reference Executive Order 13513; DOT Order 3902.10)

**TEXTING WHEN DRIVING**

In accordance with Executive Order 13513, "Federal Leadership on Reducing Text Messaging While Driving", (10/1/2009) and DOT Order 3902.10, "Text Messaging While Driving", (12/30/2009), the Federal Aviation Administration encourages recipients of Federal grant funds to adopt and enforce safety policies that decrease crashes by distracted drivers, including policies to ban text messaging while driving when performing work related to a grant or subgrant.

In support of this initiative, the Owner encourages the Contractor to promote policies and initiatives for its employees and other work personnel that decrease crashes by distracted drivers, including policies that ban text messaging while driving motor vehicles while performing work activities associated with the project. The Contractor must include the substance of this clause in all sub-tier contracts exceeding \$3,500 that involve driving a motor vehicle in performance of work activities associated with the project.



**FEDERAL CONTRACT PROVISIONS**  
**ENERGY CONSERVATION REQUIREMENTS**  
(Reference 2 CFR § 200 Appendix II(H))

The contractor agrees to comply with mandatory standards and policies relating to energy efficiency that are contained in the state energy conservation plan issued in compliance with the Energy Policy and Conservation Act (42 USC 6201 *et seq.*).

**FEDERAL CONTRACT PROVISIONS  
EQUAL OPPORTUNITY CLAUSE AND SPECIFICATIONS**

(Reference 2 CFR 200, Appendix II(c); 41 CFR § 60-1.4; 41 CFR § 60-4.3;  
Executive Order 11246)

During the performance of this contract, the Contractor agrees as follows:

(1) The Contractor will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin. The Contractor will take affirmative action to ensure that applicants are employed, and that employees are treated during employment, without regard to their race, color, religion, sex, sexual orientation, gender identify, or national origin. Such action shall include, but not be limited to, the following: employment, upgrading, demotion, or transfer; recruitment or recruitment advertising; layoff, or termination; rates of pay or other forms of compensation; and selection for training, including apprenticeship. The Contractor agrees to post in conspicuous places, available to employees and applicants for employment, notices to be provided setting forth the provisions of this nondiscrimination clause.

(2) The Contractor will, in all solicitations or advertisements for employees placed by or on behalf of the Contractor, state that all qualified applicants will receive considerations for employment without regard to race, color, religion, sex, or national origin.

(3) The Contractor will send to each labor union or representative of workers with which it has a collective bargaining agreement or other contract or understanding, a notice to be provided advising the said labor union or workers' representatives of the Contractor's commitments under this section and shall post copies of the notice in conspicuous places available to employees and applicants for employment.

(4) The Contractor will comply with all provisions of Executive Order 11246 of September 24, 1965, and of the rules, regulations, and relevant orders of the Secretary of Labor.

(5) The Contractor will furnish all information and reports required by Executive Order 11246 of September 24, 1965, and by rules, regulations, and orders of the Secretary of Labor, or pursuant thereto, and will permit access to his books, records, and accounts by the administering agency and the Secretary of Labor for purposes of investigation to ascertain compliance with such rules, regulations, and orders.

(6) In the event of the Contractor's noncompliance with the nondiscrimination clauses of this contract or with any of the said rules, regulations, or orders, this contract may be canceled, terminated, or suspended in whole or in part and the Contractor may be declared ineligible for further Government contracts or federally assisted construction contracts in accordance with procedures authorized in Executive Order 11246 of September 24, 1965, and such other sanctions may be imposed and remedies invoked as provided in Executive Order 11246 of September 24, 1965, or by rule, regulation, or order of the Secretary of Labor, or as otherwise provided by law.

(7) The Contractor will include the portion of the sentence immediately preceding paragraph (1) and the provisions of paragraphs (1) through (7) in every subcontract or purchase order unless exempted by rules, regulations, or orders of the Secretary of Labor issued pursuant to section 204 of Executive Order 11246 of September 24, 1965, so that such provisions will be binding upon each subcontractor or vendor. The Contractor will take such action with respect to any subcontract or purchase order as the administering agency may direct as a means of enforcing

such provisions, including sanctions for noncompliance: *Provided, however*, that in the event a contractor becomes involved in, or is threatened with, litigation with a subcontractor or vendor as a result of such direction by the administering agency the Contractor may request the United States to enter into such litigation to protect the interests of the United States.

### **STANDARD FEDERAL EQUAL EMPLOYMENT OPPORTUNITY CONSTRUCTION CONTRACT SPECIFICATIONS**

As used in these specifications:

- a. "Covered area" means the geographical area described in the solicitation from which this contract resulted;
- b. "Director" means Director, Office of Federal Contract Compliance Programs (OFCCP), U.S. Department of Labor, or any person to whom the Director delegates authority;
- c. "Employer identification number" means the Federal social security number used on the Employer's Quarterly Federal Tax Return, U.S. Treasury Department Form 941;
- d. "Minority" includes:
  - (1) Black (all persons having origins in any of the Black African racial groups not of Hispanic origin);
  - (2) Hispanic (all persons of Mexican, Puerto Rican, Cuban, Central or South American, or other Spanish culture or origin regardless of race);
  - (3) Asian and Pacific Islander (all persons having origins in any of the original peoples of the Far East, Southeast Asia, the Indian Subcontinent, or the Pacific Islands); and
  - (4) American Indian or Alaskan native (all persons having origins in any of the original peoples of North America and maintaining identifiable tribal affiliations through membership and participation or community identification).

2. Whenever the Contractor, or any subcontractor at any tier, subcontracts a portion of the work involving any construction trade, it shall physically include in each subcontract in excess of \$10,000 the provisions of these specifications and the Notice which contains the applicable goals for minority and female participation and which is set forth in the solicitations from which this contract resulted.

3. If the Contractor is participating (pursuant to 41 CFR part 60-4.5) in a Hometown Plan approved by the U.S. Department of Labor in the covered area either individually or through an association, its affirmative action obligations on all work in the Plan area (including goals and timetables) shall be in accordance with that Plan for those trades which have unions participating in the Plan. Contractors shall be able to demonstrate their participation in and compliance with the provisions of any such Hometown Plan. Each contractor or subcontractor participating in an approved plan is individually required to comply with its obligations under the EEO clause and to make a good faith effort to achieve each goal under the Plan in each trade in which it has employees. The overall good faith performance by other contractors or subcontractors toward a goal in an approved Plan does not excuse any covered contractor's or subcontractor's failure to take good faith efforts to achieve the Plan goals and timetables.

4. The Contractor shall implement the specific affirmative action standards provided in paragraphs 7a through 7p of these specifications. The goals set forth in the solicitation from which this contract resulted are expressed as percentages of the total hours of employment and training of minority and female utilization the Contractor should reasonably be able to achieve in each construction trade in which it has employees in the covered area. Covered construction contractors performing construction work in a geographical area where they do not have a

Federal or federally assisted construction contract shall apply the minority and female goals established for the geographical area where the work is being performed. Goals are published periodically in the Federal Register in notice form, and such notices may be obtained from any Office of Federal Contract Compliance Programs office or from Federal procurement contracting officers. The Contractor is expected to make substantially uniform progress in meeting its goals in each craft during the period specified.

5. Neither the provisions of any collective bargaining agreement nor the failure by a union with whom the Contractor has a collective bargaining agreement to refer either minorities or women shall excuse the Contractor's obligations under these specifications, Executive Order 11246, or the regulations promulgated pursuant thereto.

6. In order for the non-working training hours of apprentices and trainees to be counted in meeting the goals, such apprentices and trainees shall be employed by the Contractor during the training period and the Contractor shall have made a commitment to employ the apprentices and trainees at the completion of their training, subject to the availability of employment opportunities. Trainees shall be trained pursuant to training programs approved by the U.S. Department of Labor.

7. The Contractor shall take specific affirmative actions to ensure equal employment opportunity. The evaluation of the Contractor's compliance with these specifications shall be based upon its effort to achieve maximum results from its actions. The Contractor shall document these efforts fully and shall implement affirmative action steps at least as extensive as the following:

a. Ensure and maintain a working environment free of harassment, intimidation, and coercion at all sites, and in all facilities at which the Contractor's employees are assigned to work. The Contractor, where possible, will assign two or more women to each construction project. The Contractor shall specifically ensure that all foremen, superintendents, and other onsite supervisory personnel are aware of and carry out the Contractor's obligation to maintain such a working environment, with specific attention to minority or female individuals working at such sites or in such facilities.

b. Establish and maintain a current list of minority and female recruitment sources, provide written notification to minority and female recruitment sources and to community organizations when the Contractor or its unions have employment opportunities available, and maintain a record of the organizations' responses.

c. Maintain a current file of the names, addresses, and telephone numbers of each minority and female off-the-street applicant and minority or female referral from a union, a recruitment source, or community organization and of what action was taken with respect to each such individual. If such individual was sent to the union hiring hall for referral and was not referred back to the Contractor by the union or, if referred, not employed by the Contractor, this shall be documented in the file with the reason therefore along with whatever additional actions the Contractor may have taken.

d. Provide immediate written notification to the Director when the union or unions with which the Contractor has a collective bargaining agreement has not referred to the Contractor a minority person or female sent by the Contractor, or when the Contractor has other information that the union referral process has impeded the Contractor's efforts to meet its obligations.

e. Develop on-the-job training opportunities and/or participate in training programs for the area which expressly include minorities and women, including upgrading programs and apprenticeship and trainee programs relevant to the Contractor's employment needs, especially those programs funded or approved by the Department of Labor. The Contractor shall provide notice of these programs to the sources compiled under 7b above.

f. Disseminate the Contractor's EEO policy by providing notice of the policy to unions and training programs and requesting their cooperation in assisting the Contractor in meeting its EEO obligations; by including it in any policy manual and collective bargaining agreement; by publicizing it in the company newspaper, annual report, etc.; by specific review of the policy with all management personnel and with all minority and female employees at least once a year; and by posting the company EEO policy on bulletin boards accessible to all employees at each location where construction work is performed.

g. Review, at least annually, the company's EEO policy and affirmative action obligations under these specifications with all employees having any responsibility for hiring, assignment, layoff, termination, or other employment decisions, including specific review of these items, with onsite supervisory personnel such as superintendents, general foremen, etc., prior to the initiation of construction work at any job site. A written record shall be made and maintained identifying the time and place of these meetings, persons attending, subject matter discussed, and disposition of the subject matter.

h. Disseminate the Contractor's EEO policy externally by including it in any advertising in the news media, specifically including minority and female news media, and providing written notification to and discussing the Contractor's EEO policy with other contractors and subcontractors with whom the Contractor does or anticipates doing business.

i. Direct its recruitment efforts, both oral and written, to minority, female, and community organizations, to schools with minority and female students; and to minority and female recruitment and training organizations serving the Contractor's recruitment area and employment needs. Not later than one month prior to the date for the acceptance of applications for apprenticeship or other training by any recruitment source, the Contractor shall send written notification to organizations, such as the above, describing the openings, screening procedures, and tests to be used in the selection process.

j. Encourage present minority and female employees to recruit other minority persons and women and, where reasonable, provide after school, summer, and vacation employment to minority and female youth both on the site and in other areas of a contractor's workforce.

k. Validate all tests and other selection requirements where there is an obligation to do so under 41 CFR part 60-3.

l. Conduct, at least annually, an inventory and evaluation at least of all minority and female personnel, for promotional opportunities and encourage these employees to seek or to prepare for, through appropriate training, etc., such opportunities.

m. Ensure that seniority practices, job classifications, work assignments, and other personnel practices do not have a discriminatory effect by continually monitoring all personnel and employment related activities to ensure that the EEO policy and the Contractor's obligations under these specifications are being carried out.

n. Ensure that all facilities and company activities are non-segregated except that separate or single user toilet and necessary changing facilities shall be provided to assure privacy between the sexes.

o. Document and maintain a record of all solicitations of offers for subcontracts from minority and female construction contractors and suppliers, including circulation of solicitations to minority and female contractor associations and other business associations.

p. Conduct a review, at least annually, of all supervisor's adherence to and performance under the Contractor's EEO policies and affirmative action obligations.

8. Contractors are encouraged to participate in voluntary associations, which assist in fulfilling one or more of their affirmative action obligations (7a through 7p). The efforts of a contractor association, joint contractor union, contractor community, or other similar groups of which the Contractor is a member and participant may be asserted as fulfilling any one or more of its obligations under 7a through 7p of these specifications provided that the Contractor actively participates in the group, makes every effort to assure that the group has a positive impact on the employment of minorities and women in the industry, ensures that the concrete benefits of the program are reflected in the Contractor's minority and female workforce participation, makes a good faith effort to meet its individual goals and timetables, and can provide access to documentation which demonstrates the effectiveness of actions taken on behalf of the Contractor. The obligation to comply, however, is the Contractor's and failure of such a group to fulfill an obligation shall not be a defense for the Contractor's noncompliance.

9. A single goal for minorities and a separate single goal for women have been established. The Contractor, however, is required to provide equal employment opportunity and to take affirmative action for all minority groups, both male and female, and all women, both minority and non-minority. Consequently, if the particular group is employed in a substantially disparate manner (for example, even though the Contractor has achieved its goals for women generally), the Contractor may be in violation of the Executive Order if a specific minority group of women is underutilized.

10. The Contractor shall not use the goals and timetables or affirmative action standards to discriminate against any person because of race, color, religion, sex, or national origin.

11. The Contractor shall not enter into any subcontract with any person or firm debarred from Government contracts pursuant to Executive Order 11246.

12. The Contractor shall carry out such sanctions and penalties for violation of these specifications and of the Equal Opportunity Clause, including suspension, termination, and cancellation of existing subcontracts as may be imposed or ordered pursuant to Executive Order 11246, as amended, and its implementing regulations, by the Office of Federal Contract Compliance Programs. Any contractor who fails to carry out such sanctions and penalties shall be in violation of these specifications and Executive Order 11246, as amended.

13. The Contractor, in fulfilling its obligations under these specifications, shall implement specific affirmative action steps, at least as extensive as those standards prescribed in paragraph 7 of these specifications, so as to achieve maximum results from its efforts to ensure equal employment opportunity. If the Contractor fails to comply with the requirements of the

Executive Order, the implementing regulations, or these specifications, the Director shall proceed in accordance with 41 CFR part 60-4.8.

14. The Contractor shall designate a responsible official to monitor all employment related activity to ensure that the company EEO policy is being carried out, to submit reports relating to the provisions hereof as may be required by the Government, and to keep records. Records shall at least include for each employee, the name, address, telephone number, construction trade, union affiliation if any, employee identification number when assigned, social security number, race, sex, status (e.g., mechanic, apprentice, trainee, helper, or laborer), dates of changes in status, hours worked per week in the indicated trade, rate of pay, and locations at which the work was performed. Records shall be maintained in an easily understandable and retrievable form; however, to the degree that existing records satisfy this requirement, contractors shall not be required to maintain separate records.

15. Nothing herein provided shall be construed as a limitation upon the application of other laws which establish different standards of compliance or upon the application of requirements for the hiring of local or other area residents (e.g. those under the Public Works Employment Act of 1977 and the Community Development Block Grant Program).

**FEDERAL CONTRACT PROVISIONS**  
**FEDERAL FAIR LABOR STANDARDS ACT (FEDERAL MINIMUM WAGE)**

(Reference: 29 USC § 201, et seq.)

All contracts and subcontracts that result from this solicitation incorporate by reference the provisions of 29 CFR part 201, the Federal Fair Labor Standards Act (FLSA), with the same force and effect as if given in full text. The FLSA sets minimum wage, overtime pay, recordkeeping, and child labor standards for full and part-time workers.

The *Contractor* has full responsibility to monitor compliance to the referenced statute or regulation. The *Contractor* must address any claims or disputes that arise from this requirement directly with the U.S. Department of Labor – Wage and Hour Division.



**FEDERAL CONTRACT PROVISIONS**  
**LOBBYING AND INFLUENCING FEDERAL EMPLOYEES**

(Reference: 49 CFR part 20, Appendix A)

The Bidder or Offeror certifies by signing and submitting this bid or proposal, to the best of his or her knowledge and belief, that:

- (1) No Federal appropriated funds have been paid or will be paid, by or on behalf of the Bidder or Offeror, to any person for influencing or attempting to influence an officer or employee of an agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
- (2) If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure Form to Report Lobbying," in accordance with its instructions.
- (3) The undersigned shall require that the language of this certification be included in the award documents for all sub-awards at all tiers (including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements) and that all sub-recipients shall certify and disclose accordingly.

This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by section 1352, title 31, U.S. Code. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

**FEDERAL CONTRACT PROVISIONS  
PROHIBITION OF SEGREGATED FACILITIES**

(Reference: 41 CFR § 60)

(a) The Contractor agrees that it does not and will not maintain or provide for its employees any segregated facilities at any of its establishments, and that it does not and will not permit its employees to perform their services at any location under its control where segregated facilities are maintained. The Contractor agrees that a breach of this clause is a violation of the Equal Employment Opportunity clause in this contract.

(b) "Segregated facilities," as used in this clause, means any waiting rooms, work areas, rest rooms and wash rooms, restaurants and other eating areas, time clocks, locker rooms and other storage or dressing areas, parking lots, drinking fountains, recreation or entertainment areas, transportation, and housing facilities provided for employees that are segregated by explicit directive or are in fact segregated on the basis of race, color, religion, sex, or national origin because of written or oral policies or employee custom. The term does not include separate or single-user rest rooms or necessary dressing or sleeping areas provided to assure privacy between the sexes.

(c) The Contractor shall include this clause in every subcontract and purchase order that is subject to the Equal Employment Opportunity clause of this contract.

**FEDERAL CONTRACT PROVISIONS**  
**OCCUPATIONAL SAFETY AND HEALTH ACT OF 1970**  
(Reference 20 CFR part 1910)

All contracts and subcontracts that result from this solicitation incorporate by reference the requirements of 29 CFR Part 1910 with the same force and effect as if given in full text. The employer must provide a work environment that is free from recognized hazards that may cause death or serious physical harm to the employee. The employer retains full responsibility to monitor its compliance and their subcontractor's compliance with the applicable requirements of the Occupational Safety and Health Act of 1970 (20 CFR Part 1910). The employer must address any claims or disputes that pertain to a referenced requirement directly with the U.S. Department of Labor – Occupational Safety and Health Administration.

**FEDERAL CONTRACT PROVISIONS  
PROCUREMENT OF RECOVERED MATERIALS**

(Reference 2 CFR § 200.322; 40 CFR Part 247; Solid Waste Disposal Act)

Contractor and subcontractor agree to comply with Section 6002 of the Solid Waste Disposal Act, as amended by the Resource Conservation and Recovery Act, and the regulatory provisions of 40 CFR Part 247. In the performance of this contract and to the extent practicable, the Contractor and subcontractors are to use products containing the highest percentage of recovered materials for items designated by the Environmental Protection Agency (EPA) under 40 CFR Part 247 whenever:

- 1) The contract requires procurement of \$10,000 or more of a designated item during the fiscal year; or
- 2) The contractor has procured \$10,000 or more of a designated item using Federal funding during the previous fiscal year.

The list of EPA-designated items is available at [www.epa.gov/smm/comprehensive-procurement-guidelines-construction-products](http://www.epa.gov/smm/comprehensive-procurement-guidelines-construction-products).

Section 6002(c) establishes exceptions to the preference for recovery of EPA-designated products if the contractor can demonstrate the item is:

- a) Not reasonably available within a timeframe providing for compliance with the contract performance schedule;
- b) Fails to meet reasonable contract performance requirements; or
- c) Is only available at an unreasonable price.

**FEDERAL CONTRACT PROVISIONS  
RIGHT TO INVENTIONS**

(Reference 2 CFR § 200 Appendix II(F); 37 CFR § 401)

Contracts or agreements that include the performance of experimental, developmental, or research work must provide for the rights of the Federal Government and the Owner in any resulting invention as established by 37 CFR part 401, Rights to Inventions Made by Non-profit Organizations and Small Business Firms under Government Grants, Contracts, and Cooperative Agreements. This contract incorporates by reference the patent and inventions rights as specified within 37 CFR §401.14. Contractor must include this requirement in all sub-tier contracts involving experimental, developmental, or research work.

**FEDERAL CONTRACT PROVISIONS**  
**CERTIFICATION OF OFFEROR/BIDDER REGARDING TAX DELINQUENCY AND FELONY**  
**CONVICTIONS**

(Reference Sections 415 and 416 of Title IV, Division L of the Consolidated Appropriations Act, 2014; DOT Order 4200.6)

The applicant must complete the following two certification statements. The applicant must indicate its current status as it relates to tax delinquency and felony conviction by inserting a checkmark (✓) in the space following the applicable response. The applicant agrees that, if awarded a contract resulting from this solicitation, it will incorporate this provision for certification in all lower tier subcontracts.

**Certifications**

- 1) The applicant represents that it is ( ☐ ) is not ( ☐ ) a corporation that has any unpaid Federal tax liability that has been assessed, for which all judicial and administrative remedies have been exhausted or have lapsed, and that is not being paid in a timely manner pursuant to an agreement with the authority responsible for collecting the tax liability.
- 2) The applicant represents that it is ( ☐ ) is not ( ☐ ) is not a corporation that was convicted of a criminal violation under any Federal law within the preceding 24 months.

**Note**

If an applicant responds in the affirmative to either of the above representations, the applicant is ineligible to receive an award unless the sponsor has received notification from the agency suspension and debarment official (SDO) that the SDO has considered suspension or debarment and determined that further action is not required to protect the Government's interests. The applicant therefore must provide information to the owner about its tax liability or conviction to the Owner, who will then notify the FAA Airports District Office, which will then notify the agency's SDO to facilitate completion of the required considerations before award decisions are made.

**Term Definitions**

**Felony conviction:** Felony conviction means a conviction within the preceding twenty-four (24) months of a felony criminal violation under any Federal law and includes conviction of an offense defined in a section of the U.S. code that specifically classifies the offense as a felony and conviction of an offense that is classified as a felony under 18 U.S.C. § 3559.

**Tax Delinquency:** A tax delinquency is any unpaid Federal tax liability that has been assessed, for which all judicial and administrative remedies have been exhausted, or have lapsed, and that is not being paid in a timely manner pursuant to an agreement with the authority responsible for collecting the tax liability.

**FEDERAL CONTRACT PROVISIONS  
TERMINATION OF CONTRACT**

(Reference 2 CFR § 200 Appendix II(B); FAA AC 150/5370-10, Section 80-09)

**TERMINATION FOR CONVENIENCE (CONSTRUCTION & EQUIPMENT CONTRACTS)**

The Owner may terminate this contract in whole or in part at any time by providing written notice to the Contractor. Such action may be without cause and without prejudice to any other right or remedy of Owner. Upon receipt of a written notice of termination, except as explicitly directed by the Owner, the Contractor shall immediately proceed with the following obligations regardless of any delay in determining or adjusting amounts due under this clause:

1. Contractor must immediately discontinue work as specified in the written notice.
2. Terminate all subcontracts to the extent they relate to the work terminated under the notice.
3. Discontinue orders for materials and services except as directed by the written notice.
4. Deliver to the Owner all fabricated and partially fabricated parts, completed and partially completed work, supplies, equipment and materials acquired prior to termination of the work, and as directed in the written notice.
5. Complete performance of the work not terminated by the notice.
6. Take action as directed by the Owner to protect and preserve property and work related to this contract that Owner will take possession.

Owner agrees to pay Contractor for:

- 1) completed and acceptable work executed in accordance with the contract documents prior to the effective date of termination;
- 2) documented expenses sustained prior to the effective date of termination in performing work and furnishing labor, materials, or equipment as required by the contract documents in connection with uncompleted work;
- 3) reasonable and substantiated claims, costs, and damages incurred in settlement of terminated contracts with Subcontractors and Suppliers; and
- 4) reasonable and substantiated expenses to the Contractor directly attributable to Owner's termination action.

Owner will not pay Contractor for loss of anticipated profits or revenue or other economic loss arising out of or resulting from the Owner's termination action.

The rights and remedies this clause provides are in addition to any other rights and remedies provided by law or under this contract.

**TERMINATION FOR DEFAULT (CONSTRUCTION)**

Section 80-09 of FAA Advisory Circular 150/5370-10 establishes conditions, rights, and remedies associated with Owner termination of this contract due to default of the Contractor.

**FEDERAL CONTRACT PROVISIONS  
TRADE RESTRICTION CERTIFICATION**  
(Reference 49 CFR part 30; 49 CFR § 50104)

By submission of an offer, the Offeror certifies that with respect to this solicitation and any resultant contract, the Offeror –

- 1) is not owned or controlled by one or more citizens of a foreign country included in the list of countries that discriminate against U.S. firms as published by the Office of the United States Trade Representative (USTR);
- 2) has not knowingly entered into any contract or subcontract for this project with a person that is a citizen or national of a foreign country included on the list of countries that discriminate against U.S. firms as published by the USTR; and
- 3) has not entered into any subcontract for any product to be used on the Federal project that is produced in a foreign country included on the list of countries that discriminate against U.S. firms published by the USTR.

This certification concerns a matter within the jurisdiction of an agency of the United States of America and the making of a false, fictitious, or fraudulent certification may render the maker subject to prosecution under Title 18 USC Section 1001.

The Offeror/Contractor must provide immediate written notice to the Owner if the Offeror/Contractor learns that its certification or that of a subcontractor was erroneous when submitted or has become erroneous by reason of changed circumstances. The Contractor must require subcontractors provide immediate written notice to the Contractor if at any time it learns that its certification was erroneous by reason of changed circumstances.

Unless the restrictions of this clause are waived by the Secretary of Transportation in accordance with 49 CFR 30.17, no contract shall be awarded to an Offeror or subcontractor:

- 1) who is owned or controlled by one or more citizens or nationals of a foreign country included on the list of countries that discriminate against U.S. firms published by the USTR or
- 2) whose subcontractors are owned or controlled by one or more citizens or nationals of a foreign country on such USTR list or
- 3) who incorporates in the public works project any product of a foreign country on such USTR list.

Nothing contained in the foregoing shall be construed to require establishment of a system of records in order to render, in good faith, the certification required by this provision. The knowledge and information of a contractor is not required to exceed that which is normally possessed by a prudent person in the ordinary course of business dealings.

The Offeror agrees that, if awarded a contract resulting from this solicitation, it will incorporate this provision for certification without modification in all lower tier subcontracts. The Contractor may rely on the certification of a prospective subcontractor that it is not a firm from a foreign



country included on the list of countries that discriminate against U.S. firms as published by USTR, unless the Offeror has knowledge that the certification is erroneous.

This certification is a material representation of fact upon which reliance was placed when making an award. If it is later determined that the Contractor or subcontractor knowingly rendered an erroneous certification, the Federal Aviation Administration (FAA) may direct through the Owner cancellation of the contract or subcontract for default at no cost to the Owner or the FAA.

**FEDERAL CONTRACT PROVISIONS**  
**VETERAN'S PREFERENCE**  
(Reference: 49 USC § 47112(c))

In the employment of labor (excluding executive, administrative, and supervisory positions), the Contractor and all sub-tier contractors must give preference to covered veterans as defined within Title 49 United States Code Section 47112. Covered veterans include Vietnam-era veterans, Persian Gulf veterans, Afghanistan-Iraq war veterans, disabled veterans, and small business concerns (as defined by 15 USC 632) owned and controlled by disabled veterans. This preference only applies when there are covered veterans readily available and qualified to perform the work to which the employment relates.

**END FEDERAL CONTRACT PROVISIONS**

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## SECTION 10 DEFINITION OF TERMS

When the following terms are used in these specifications, in the contract, or in any documents or other instruments pertaining to construction where these specifications govern, the intent and meaning shall be defined as follows:

| <b>Paragraph Number</b> | <b>Term</b>                              | <b>Definition</b>                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10-01</b>            | <b>AASHTO</b>                            | The American Association of State Highway and Transportation Officials.                                                                                                                                                                                                                                                                                                 |
| <b>10-02</b>            | <b>Access Road</b>                       | The right-of-way, the roadway and all improvements constructed thereon connecting the airport to a public roadway.                                                                                                                                                                                                                                                      |
| <b>10-03</b>            | <b>Advertisement</b>                     | A public announcement, as required by local law, inviting bids for work to be performed and materials to be furnished.                                                                                                                                                                                                                                                  |
| <b>10-04</b>            | <b>Airport</b>                           | Airport means an area of land or water which is used or intended to be used for the landing and takeoff of aircraft; an appurtenant area used or intended to be used for airport buildings or other airport facilities or rights of way; airport buildings and facilities located in any of these areas, and a heliport.                                                |
| <b>10-05</b>            | <b>Airport Improvement Program (AIP)</b> | A grant-in-aid program, administered by the Federal Aviation Administration (FAA).                                                                                                                                                                                                                                                                                      |
| <b>10-06</b>            | <b>Air Operations Area (AOA)</b>         | The term air operations area (AOA) shall mean any area of the airport used or intended to be used for the landing, takeoff, or surface maneuvering of aircraft. An air operation area shall include such paved or unpaved areas that are used or intended to be used for the unobstructed movement of aircraft in addition to its associated runway, taxiway, or apron. |
| <b>10-07</b>            | <b>Apron</b>                             | Area where aircraft are parked, unloaded or loaded, fueled and/or serviced.                                                                                                                                                                                                                                                                                             |
| <b>10-08</b>            | <b>ASTM International (ASTM)</b>         | Formerly known as the American Society for Testing and Materials (ASTM).                                                                                                                                                                                                                                                                                                |
| <b>10-09</b>            | <b>Award</b>                             | The Owner's notice to the successful bidder of the acceptance of the submitted bid.                                                                                                                                                                                                                                                                                     |
| <b>10-10</b>            | <b>Bidder</b>                            | Any individual, partnership, firm, or corporation, acting directly or through a duly authorized representative, who submits a proposal for the work contemplated.                                                                                                                                                                                                       |
| <b>10-11</b>            | <b>Building Area</b>                     | An area on the airport to be used, considered, or intended to be used for airport buildings or other airport facilities or rights-of-way together with all airport buildings and facilities located thereon.                                                                                                                                                            |
| <b>10-12</b>            | <b>Calendar Day</b>                      | Every day shown on the calendar.                                                                                                                                                                                                                                                                                                                                        |
| <b>10-13</b>            | <b>Certificate of Analysis (COA)</b>     | The COA is the manufacturer's Certificate of Compliance (COC) including all applicable test results required by the specifications.                                                                                                                                                                                                                                     |

| <b>Paragraph Number</b> | <b>Term</b>                                        | <b>Definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10-14</b>            | <b>Certificate of Compliance (COC)</b>             | The manufacturer's certification stating that materials or assemblies furnished fully comply with the requirements of the contract. The certificate shall be signed by the manufacturer's authorized representative.                                                                                                                                                                                                                                                                                                                            |
| <b>10-15</b>            | <b>Change Order</b>                                | A written order to the Contractor covering changes in the plans, specifications, or proposal quantities and establishing the basis of payment and contract time adjustment, if any, for work within the scope of the contract and necessary to complete the project.                                                                                                                                                                                                                                                                            |
| <b>10-16</b>            | <b>Contract</b>                                    | A written agreement between the Owner and the Contractor that establishes the obligations of the parties including but not limited to performance of work, furnishing of labor, equipment and materials and the basis of payment.<br>The awarded contract includes but may not be limited to: Advertisement, Contract form, Proposal, Performance bond, payment bond, General provisions, certifications and representations, Technical Specifications, Plans, Supplemental Provisions, standards incorporated by reference and issued addenda. |
| <b>10-17</b>            | <b>Contract Item (Pay Item)</b>                    | A specific unit of work for which a price is provided in the contract.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>10-18</b>            | <b>Contract Time</b>                               | The number of calendar days or working days, stated in the proposal, allowed for completion of the contract, including authorized time extensions. If a calendar date of completion is stated in the proposal, in lieu of a number of calendar or working days, the contract shall be completed by that date.                                                                                                                                                                                                                                   |
| <b>10-19</b>            | <b>Contractor</b>                                  | The individual, partnership, firm, or corporation primarily liable for the acceptable performance of the work contracted and for the payment of all legal debts pertaining to the work who acts directly or through lawful agents or employees to complete the contract work.                                                                                                                                                                                                                                                                   |
| <b>10-20</b>            | <b>Contractors Quality Control (QC) Facilities</b> | The Contractor's QC facilities in accordance with the Contractor Quality Control Program (CQCP).                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>10-21</b>            | <b>Contractor Quality Control Program (CQCP)</b>   | Details the methods and procedures that will be taken to assure that all materials and completed construction required by the contract conform to contract plans, technical specifications and other requirements, whether manufactured by the Contractor, or procured from subcontractors or vendors.                                                                                                                                                                                                                                          |

| <b>Paragraph Number</b> | <b>Term</b>                                        | <b>Definition</b>                                                                                                                                                                                                                                                                                                          |
|-------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10-22</b>            | <b>Control Strip</b>                               | A demonstration by the Contractor that the materials, equipment, and construction processes results in a product meeting the requirements of the specification.                                                                                                                                                            |
| <b>10-23</b>            | <b>Construction Safety and Phasing Plan (CSPP)</b> | The overall plan for safety and phasing of a construction project developed by the airport operator, or developed by the airport operator's consultant and approved by the airport operator. It is included in the invitation for bids and becomes part of the project specifications.                                     |
| <b>10-24</b>            | <b>Drainage System</b>                             | The system of pipes, ditches, and structures by which surface or subsurface waters are collected and conducted from the airport area.                                                                                                                                                                                      |
| <b>10-25</b>            | <b>Engineer</b>                                    | The individual, partnership, firm, or corporation duly authorized by the Owner to be responsible for engineering, inspection, and/or observation of the contract work and acting directly or through an authorized representative.                                                                                         |
| <b>10-26</b>            | <b>Equipment</b>                                   | All machinery, together with the necessary supplies for upkeep and maintenance; and all tools and apparatus necessary for the proper construction and acceptable completion of the work.                                                                                                                                   |
| <b>10-27</b>            | <b>Extra Work</b>                                  | An item of work not provided for in the awarded contract as previously modified by change order or supplemental agreement, but which is found by the Owner's Engineer or Resident Project Representative (RPR) to be necessary to complete the work within the intended scope of the contract as previously modified.      |
| <b>10-28</b>            | <b>FAA</b>                                         | The Federal Aviation Administration. When used to designate a person, FAA shall mean the Administrator or their duly authorized representative.                                                                                                                                                                            |
| <b>10-29</b>            | <b>Federal Specifications</b>                      | The federal specifications and standards, commercial item descriptions, and supplements, amendments, and indices prepared and issued by the General Services Administration.                                                                                                                                               |
| <b>10-30</b>            | <b>Force Account</b>                               | <p><b>a.</b> Contract Force Account - A method of payment that addresses extra work performed by the Contractor on a time and material basis.</p> <p><b>b.</b> Owner Force Account - Work performed for the project by the Owner's employees.</p>                                                                          |
| <b>10-31</b>            | <b>Intention of Terms</b>                          | Whenever, in these specifications or on the plans, the words "directed," "required," "permitted," "ordered," "designated," "prescribed," or words of like import are used, it shall be understood that the direction, requirement, permission, order, designation, or prescription of the Engineer and/or Resident Project |

| Paragraph Number | Term                                   | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                        | <p>Representative (RPR) is intended; and similarly, the words “approved,” “acceptable,” “satisfactory,” or words of like import, shall mean approved by, or acceptable to, or satisfactory to the Engineer and/or RPR, subject in each case to the final determination of the Owner.</p> <p>Any reference to a specific requirement of a numbered paragraph of the contract specifications or a cited standard shall be interpreted to include all general requirements of the entire section, specification item, or cited standard that may be pertinent to such specific reference.</p> |
| 10-32            | <b>Lighting</b>                        | A system of fixtures providing or controlling the light sources used on or near the airport or within the airport buildings. The field lighting includes all luminous signals, markers, floodlights, and illuminating devices used on or near the airport or to aid in the operation of aircraft landing at, taking off from, or taxiing on the airport surface.                                                                                                                                                                                                                           |
| 10-33            | <b>Major and Minor Contract Items</b>  | A major contract item shall be any item that is listed in the proposal, the total cost of which is equal to or greater than 20% of the total amount of the award contract. All other items shall be considered minor contract items.                                                                                                                                                                                                                                                                                                                                                       |
| 10-34            | <b>Materials</b>                       | Any substance specified for use in the construction of the contract work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 10-35            | <b>Modification of Standards (MOS)</b> | Any deviation from standard specifications applicable to material and construction methods in accordance with FAA Order 5300.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10-36            | <b>Notice to Proceed (NTP)</b>         | A written notice to the Contractor to begin the actual contract work on a previously agreed to date. If applicable, the Notice to Proceed shall state the date on which the contract time begins.                                                                                                                                                                                                                                                                                                                                                                                          |
| 10-37            | <b>Owner</b>                           | The term “Owner” shall mean the party of the first part or the contracting agency signatory to the contract. Where the term “Owner” is capitalized in this document, it shall mean airport Sponsor only. The Owner for this project is <b>Pensacola International Airport</b> .                                                                                                                                                                                                                                                                                                            |
| 10-38            | <b>Passenger Facility Charge (PFC)</b> | Per 14 Code of Federal Regulations (CFR) Part 158 and 49 United States Code (USC) § 40117, a PFC is a charge imposed by a public agency on passengers enplaned at a commercial service airport it controls.                                                                                                                                                                                                                                                                                                                                                                                |
| 10-39            | <b>Pavement Structure</b>              | The combined surface course, base course(s), and subbase course(s), if any, considered as a single unit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 10-40            | <b>Payment bond</b>                    | The approved form of security furnished by the Contractor and their own surety as a guaranty that the Contractor will pay in full all bills and accounts for                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Paragraph Number | Term                                         | Definition                                                                                                                                                                                                                                                                                                                                                            |
|------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                                              | materials and labor used in the construction of the work.                                                                                                                                                                                                                                                                                                             |
| <b>10-41</b>     | <b>Performance bond</b>                      | The approved form of security furnished by the Contractor and their own surety as a guaranty that the Contractor will complete the work in accordance with the terms of the contract.                                                                                                                                                                                 |
| <b>10-42</b>     | <b>Plans</b>                                 | The official drawings or exact reproductions which show the location, character, dimensions and details of the airport and the work to be done and which are to be considered as a part of the contract, supplementary to the specifications. Plans may also be referred to as 'contract drawings.'                                                                   |
| <b>10-43</b>     | <b>Project</b>                               | The agreed scope of work for accomplishing specific airport development with respect to a particular airport.                                                                                                                                                                                                                                                         |
| <b>10-44</b>     | <b>Proposal</b>                              | The written offer of the bidder (when submitted on the approved proposal form) to perform the contemplated work and furnish the necessary materials in accordance with the provisions of the plans and specifications.                                                                                                                                                |
| <b>10-45</b>     | <b>Proposal guaranty</b>                     | The security furnished with a proposal to guarantee that the bidder will enter into a contract if their own proposal is accepted by the Owner.                                                                                                                                                                                                                        |
| <b>10-46</b>     | <b>Quality Assurance (QA)</b>                | Owner's responsibility to assure that construction work completed complies with specifications for payment.                                                                                                                                                                                                                                                           |
| <b>10-47</b>     | <b>Quality Control (QC)</b>                  | Contractor's responsibility to control material(s) and construction processes to complete construction in accordance with project specifications.                                                                                                                                                                                                                     |
| <b>10-48</b>     | <b>Quality Assurance (QA) Inspector</b>      | An authorized representative of the Engineer and/or Resident Project Representative (RPR) assigned to make all necessary inspections, observations, tests, and/or observation of tests of the work performed or being performed, or of the materials furnished or being furnished by the Contractor.                                                                  |
| <b>10-49</b>     | <b>Quality Assurance (QA) Laboratory</b>     | The official quality assurance testing laboratories of the Owner or such other laboratories as may be designated by the Engineer or RPR. May also be referred to as Engineer's, Owner's, or QA Laboratory.                                                                                                                                                            |
| <b>10-50</b>     | <b>Resident Project Representative (RPR)</b> | The individual, partnership, firm, or corporation duly authorized by the Owner to be responsible for all necessary inspections, observations, tests, and/or observations of tests of the contract work performed or being performed, or of the materials furnished or being furnished by the Contractor, and acting directly or through an authorized representative. |
| <b>10-51</b>     | <b>Runway</b>                                | The area on the airport prepared for the landing and takeoff of aircraft.                                                                                                                                                                                                                                                                                             |



| <b>Paragraph Number</b> | <b>Term</b>                                   | <b>Definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10-52</b>            | <b>Runway Safety Area (RSA)</b>               | A defined surface surrounding the runway prepared or suitable for reducing the risk of damage to aircraft. See the construction safety and phasing plan (CSPP) for limits of the RSA.                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>10-53</b>            | <b>Safety Plan Compliance Document (SPCD)</b> | Details how the Contractor will comply with the CSPP.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>10-54</b>            | <b>Specifications</b>                         | A part of the contract containing the written directions and requirements for completing the contract work. Standards for specifying materials or testing which are cited in the contract specifications by reference shall have the same force and effect as if included in the contract physically.                                                                                                                                                                                                                                                                        |
| <b>10-55</b>            | <b>Sponsor</b>                                | A Sponsor is defined in 49 USC § 47102(24) as a public agency that submits to the FAA for an AIP grant; or a private Owner of a public-use airport that submits to the FAA an application for an AIP grant for the airport.                                                                                                                                                                                                                                                                                                                                                  |
| <b>10-56</b>            | <b>Structures</b>                             | Airport facilities such as bridges; culverts; catch basins, inlets, retaining walls, cribbing; storm and sanitary sewer lines; water lines; underdrains; electrical ducts, manholes, handholes, lighting fixtures and bases; transformers; navigational aids; buildings; vaults; and, other manmade features of the airport that may be encountered in the work and not otherwise classified herein.                                                                                                                                                                         |
| <b>10-57</b>            | <b>Subgrade</b>                               | The soil that forms the pavement foundation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>10-58</b>            | <b>Superintendent</b>                         | The Contractor's executive representative who is present on the work during progress, authorized to receive and fulfill instructions from the RPR, and who shall supervise and direct the construction.                                                                                                                                                                                                                                                                                                                                                                      |
| <b>10-59</b>            | <b>Supplemental Agreement</b>                 | A written agreement between the Contractor and the Owner that establishes the basis of payment and contract time adjustment, if any, for the work affected by the supplemental agreement. A supplemental agreement is required if: (1) in scope work would increase or decrease the total amount of the awarded contract by more than 25%; (2) in scope work would increase or decrease the total of any major contract item by more than 25%; (3) work that is not within the scope of the originally awarded contract; or (4) adding or deleting of a major contract item. |
| <b>10-60</b>            | <b>Surety</b>                                 | The corporation, partnership, or individual, other than the Contractor, executing payment or performance bonds that are furnished to the Owner by the Contractor.                                                                                                                                                                                                                                                                                                                                                                                                            |

| <b>Paragraph Number</b> | <b>Term</b>                               | <b>Definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10-61</b>            | <b>Taxilane</b>                           | A taxiway designed for low speed movement of aircraft between aircraft parking areas and terminal areas.                                                                                                                                                                                                                                                                                                                                                                |
| <b>10-62</b>            | <b>Taxiway</b>                            | The portion of the air operations area of an airport that has been designated by competent airport authority for movement of aircraft to and from the airport's runways, aircraft parking areas, and terminal areas.                                                                                                                                                                                                                                                    |
| <b>10-63</b>            | <b>Taxiway/Taxilane Safety Area (TSA)</b> | A defined surface alongside the taxiway prepared or suitable for reducing the risk of damage to an aircraft. See the construction safety and phasing plan (CSPP) for limits of the TSA.                                                                                                                                                                                                                                                                                 |
| <b>10-64</b>            | <b>Work</b>                               | The furnishing of all labor, materials, tools, equipment, and incidentals necessary or convenient to the Contractor's performance of all duties and obligations imposed by the contract, plans, and specifications.                                                                                                                                                                                                                                                     |
| <b>10-65</b>            | <b>Working day</b>                        | A working day shall be any day other than a legal holiday, Saturday, or Sunday on which the normal working forces of the Contractor may proceed with regular work for at least six (6) hours toward completion of the contract. When work is suspended for causes beyond the Contractor's control, it will not be counted as a working day. Saturdays, Sundays and holidays on which the Contractor's forces engage in regular work will be considered as working days. |
| <b>10-66</b>            | <b>Owner Defined terms</b>                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**END OF SECTION 10**

## SECTION 20 PROPOSAL REQUIREMENTS AND CONDITIONS

### 20-01 ADVERTISEMENT (NOTICE TO BIDDERS). [ ]

**20-02 QUALIFICATION OF BIDDERS.** Each bidder shall submit evidence of competency and evidence of financial responsibility to perform the work to the Owner at the time of bid opening.

Evidence of competency, unless otherwise specified, shall consist of statements covering the bidder's past experience on similar work, and a list of equipment and a list of key personnel that would be available for the work.

Each bidder shall furnish the Owner satisfactory evidence of their financial responsibility. Evidence of financial responsibility, unless otherwise specified, shall consist of a confidential statement or report of the bidder's financial resources and liabilities as of the last calendar year or the bidder's last fiscal year. Such statements or reports shall be certified by a public accountant. At the time of submitting such financial statements or reports, the bidder shall further certify whether their financial responsibility is approximately the same as stated or reported by the public accountant. If the bidder's financial responsibility has changed, the bidder shall qualify the public accountant's statement or report to reflect the bidder's true financial condition at the time such qualified statement or report is submitted to the Owner.

Unless otherwise specified, a bidder may submit evidence that they are prequalified with the State Highway Division and are on the current "bidder's list" of the state in which the proposed work is located. Evidence of State Highway Division prequalification may be submitted as evidence of financial responsibility in lieu of the certified statements or reports specified above.

**20-03 CONTENTS OF PROPOSAL FORMS.** The Owner's proposal forms state the location and description of the proposed construction; the place, date, and time of opening of the proposals; and the estimated quantities of the various items of work to be performed and materials to be furnished for which unit bid prices are asked. The proposal form states the time in which the work must be completed, and the amount of the proposal guaranty that must accompany the proposal. The Owner will accept only those Proposals properly executed on physical forms or electronic forms provided by the Owner. Bidder actions that may cause the Owner to deem a proposal irregular are given in paragraph 20-09 *Irregular proposals*.

**Mobilization is limited to 10 percent of the total project cost.**

**A prebid conference is required on this project to discuss as a minimum, the following items: material requirements; submittals; Quality Control/Quality Assurance requirements; the construction safety and phasing plan including airport access and staging areas; and unique airfield paving construction requirements.**

**20-04 ISSUANCE OF PROPOSAL FORMS.** The Owner reserves the right to refuse to issue a proposal form to a prospective bidder if the bidder is in default for any of the following reasons:

a. Failure to comply with any prequalification regulations of the Owner, if such regulations are cited, or otherwise included, in the proposal as a requirement for bidding.

b. Failure to pay, or satisfactorily settle, all bills due for labor and materials on former contracts in force with the Owner at the time the Owner issues the proposal to a prospective bidder.

c. Documented record of Contractor default under previous contracts with the Owner.

d. Documented record of unsatisfactory work on previous contracts with the Owner.

**20-05 INTERPRETATION OF ESTIMATED PROPOSAL QUANTITIES.** An estimate of quantities of work to be done and materials to be furnished under these specifications is given in the proposal. It is the result of careful calculations and is believed to be correct. It is given only as a basis for comparison of proposals and the award of the contract. The Owner does not expressly, or by implication, agree that the actual quantities involved will correspond exactly therewith; nor shall the bidder plead misunderstanding or deception because of such estimates of quantities, or of the character, location, or other conditions pertaining to the work. Payment to the Contractor will be made only for the actual quantities of work performed or materials furnished in accordance with the plans and specifications. It is understood that the quantities may be increased or decreased as provided in the Section 40, paragraph 40-02, Alteration of Work and Quantities, without in any way invalidating the unit bid prices.

**20-06 EXAMINATION OF PLANS, SPECIFICATIONS, AND SITE.** The bidder is expected to carefully examine the site of the proposed work, the proposal, plans, specifications, and contract forms. Bidders shall satisfy themselves to the character, quality, and quantities of work to be performed, materials to be furnished, and to the requirements of the proposed contract. The submission of a proposal shall be prima facie evidence that the bidder has made such examination and is satisfied to the conditions to be encountered in performing the work and the requirements of the proposed contract, plans, and specifications.

Boring logs and other records of subsurface investigations and tests are available for inspection of bidders. It is understood and agreed that such subsurface information, whether included in the plans, specifications, or otherwise made available to the bidder, was obtained and is intended for the Owner's design and estimating purposes only. Such information has been made available for the convenience of all bidders. It is further understood and agreed that each bidder is solely responsible for all assumptions, deductions, or conclusions which the bidder may make or obtain from their own examination of the boring logs and other records of subsurface investigations and tests that are furnished by the Owner.

**20-07 PREPARATION OF PROPOSAL.** The bidder shall submit their proposal on the forms furnished by the Owner. All blank spaces in the proposal forms, unless explicitly stated otherwise, must be correctly filled in where indicated for each and every item for which a quantity is given. The bidder shall state the price (written in ink or typed) both in words and numerals which they propose for each pay item furnished in the proposal. In case of conflict between words and numerals, the words, unless obviously incorrect, shall govern.

The bidder shall correctly sign the proposal in ink. If the proposal is made by an individual, their name and post office address must be shown. If made by a partnership, the name and post office address of each member of the partnership must be shown. If made by a corporation, the person signing the proposal shall give the name of the state where the corporation was chartered and the name, titles, and business address of the president, secretary, and the treasurer. Anyone signing a proposal as an agent shall file evidence of their authority to do so and that the signature is binding upon the firm or corporation.

**20-08 RESPONSIVE AND RESPONSIBLE BIDDER.** A responsive bid conforms to all significant terms and conditions contained in the Owner's invitation for bid. It is the Owner's responsibility to decide if the exceptions taken by a bidder to the solicitation are material or not and the extent of deviation it is willing to accept.

A responsible bidder has the ability to perform successfully under the terms and conditions of a proposed procurement, as defined in 2 CFR § 200.318(h). This includes such matters as Contractor integrity, compliance with public policy, record of past performance, and financial and technical resources.

**20-09 IRREGULAR PROPOSALS.** Proposals shall be considered irregular for the following reasons:

- a. If the proposal is on a form other than that furnished by the Owner, or if the Owner's form is altered, or if any part of the proposal form is detached.
- b. If there are unauthorized additions, conditional or alternate pay items, or irregularities of any kind that make the proposal incomplete, indefinite, or otherwise ambiguous.
- c. If the proposal does not contain a unit price for each pay item listed in the proposal, except in the case of authorized alternate pay items, for which the bidder is not required to furnish a unit price.
- d. If the proposal contains unit prices that are obviously unbalanced.
- e. If the proposal is not accompanied by the proposal guaranty specified by the Owner.
- f. If the applicable Disadvantaged Business Enterprise information is incomplete.

The Owner reserves the right to reject any irregular proposal and the right to waive technicalities if such waiver is in the best interest of the Owner and conforms to local laws and ordinances pertaining to the letting of construction contracts.

**20-10 BID GUARANTEE.** Each separate proposal shall be accompanied by a bid bond, certified check, or other specified acceptable collateral, in the amount specified in the proposal form. Such bond, check, or collateral, shall be made payable to the Owner.

**20-11 DELIVERY OF PROPOSAL.** See The City of Pensacola, Florida Invitation to Bid

**20-12 WITHDRAWAL OR REVISION OF PROPOSALS.** A bidder may withdraw or revise (by withdrawal of one proposal and submission of another) a proposal provided that the bidder's request for withdrawal is received by the Owner in writing or by email before the time specified for opening bids. Revised proposals must be received at the place specified in the advertisement before the time specified for opening all bids.

**20-13 PUBLIC OPENING OF PROPOSALS.** Proposals shall be opened, and read, publicly at the time and place specified in the advertisement. Bidders, their authorized agents, and other interested persons are invited to attend. Proposals that have been withdrawn (by written or telegraphic request) or received after the time specified for opening bids shall be returned to the bidder unopened.

**20-14 DISQUALIFICATION OF BIDDERS.** A bidder shall be considered disqualified for any of the following reasons:

a. Submitting more than one proposal from the same partnership, firm, or corporation under the same or different name.

b. Evidence of collusion among bidders. Bidders participating in such collusion shall be disqualified as bidders for any future work of the Owner until any such participating bidder has been reinstated by the Owner as a qualified bidder.

c. If the bidder is considered to be in “default” for any reason specified in paragraph 20-04, *Issuance of Proposal Forms*, of this section.

**20-15 Discrepancies and Omissions.** A Bidder who discovers discrepancies or omissions with the project bid documents shall immediately notify the Owner’s Engineer of the matter. A bidder that has doubt as to the true meaning of a project requirement may submit to the Owner’s Engineer a written request for interpretation no later than **10** days prior to bid opening.

Any interpretation of the project bid documents by the Owner’s Engineer will be by written addendum issued by the Owner. The Owner will not consider any instructions, clarifications or interpretations of the bidding documents in any manner other than written addendum.

**END OF SECTION 20**

## SECTION 30 AWARD AND EXECUTION OF CONTRACT

**30-01 CONSIDERATION OF PROPOSALS.** After the proposals are publicly opened and read, they will be compared on the basis of the summation of the products obtained by multiplying the estimated quantities shown in the proposal by the unit bid prices. If a bidder's proposal contains a discrepancy between unit bid prices written in words and unit bid prices written in numbers, the unit bid price written in words shall govern.

Until the award of a contract is made, the Owner reserves the right to reject a bidder's proposal for any of the following reasons:

- a. If the proposal is irregular as specified in Section 20, paragraph 20-09, *Irregular Proposals*.
- b. If the bidder is disqualified for any of the reasons specified Section 20, paragraph 20-14, *Disqualification of Bidders*.

In addition, until the award of a contract is made, the Owner reserves the right to reject any or all proposals, waive technicalities, if such waiver is in the best interest of the Owner and is in conformance with applicable state and local laws or regulations pertaining to the letting of construction contracts; advertise for new proposals; or proceed with the work otherwise. All such actions shall promote the Owner's best interests.

**30-02 AWARD OF CONTRACT.** The award of a contract, if it is to be awarded, shall be made within [ ] calendar days of the date specified for publicly opening proposals, unless otherwise specified herein.

If the Owner elects to proceed with an award of contract, the Owner will make award to the responsible bidder whose bid, conforming with all the material terms and conditions of the bid documents, is the lowest in price.

**30-03 CANCELLATION OF AWARD.** The Owner reserves the right to cancel the award without liability to the bidder, except return of proposal guaranty, at any time before a contract has been fully executed by all parties and is approved by the Owner in accordance with paragraph 30-07 *Approval of Contract*.

**30-04 RETURN OF PROPOSAL GUARANTY.** All proposal guaranties, except those of the two lowest bidders, will be returned immediately after the Owner has made a comparison of bids as specified in the paragraph 30-01, *Consideration of Proposals*. Proposal guaranties of the two lowest bidders will be retained by the Owner until such time as an award is made, at which time, the unsuccessful bidder's proposal guaranty will be returned. The successful bidder's proposal guaranty will be returned as soon as the Owner receives the contract bonds as specified in paragraph 30-05, *Requirements of Contract Bonds*.

**30-05 REQUIREMENTS OF CONTRACT BONDS.** At the time of the execution of the contract, the successful bidder shall furnish the Owner a surety bond or bonds that have been fully executed by the bidder and the surety guaranteeing the performance of the work and the payment of all legal debts that may be incurred by reason of the Contractor's performance of the work. The surety and the form of the bond or bonds shall be acceptable to the Owner. Unless otherwise specified in this subsection, the surety bond or bonds shall be in a sum equal to the full amount of the contract.

**30-06 EXECUTION OF CONTRACT.** The successful bidder shall sign (execute) the necessary agreements for entering into the contract and return the signed contract to the Owner, along with the fully executed surety bond or bonds specified in paragraph 30-05, *Requirements of Contract Bonds*, of this section, within **15** calendar days from the date mailed or otherwise delivered to the successful bidder.

**30-07 APPROVAL OF CONTRACT.** Upon receipt of the contract and contract bond or bonds that have been executed by the successful bidder, the Owner shall complete the execution of the contract in accordance with local laws or ordinances, and return the fully executed contract to the Contractor. Delivery of the fully executed contract to the Contractor shall constitute the Owner's approval to be bound by the successful bidder's proposal and the terms of the contract.

**30-08 FAILURE TO EXECUTE CONTRACT.** Failure of the successful bidder to execute the contract and furnish an acceptable surety bond or bonds within the period specified in paragraph 30-06, *Execution of Contract*, of this section shall be just cause for cancellation of the award and forfeiture of the proposal guaranty, not as a penalty, but as liquidated damages to the Owner.

**END OF SECTION 30**



## SECTION 40 SCOPE OF WORK

**40-01 INTENT OF CONTRACT.** The intent of the contract is to provide for construction and completion, in every detail, of the work described. It is further intended that the Contractor shall furnish all labor, materials, equipment, tools, transportation, and supplies required to complete the work in accordance with the plans, specifications, and terms of the contract.

**40-02 ALTERATION OF WORK AND QUANTITIES.** The Owner reserves the right to make such changes in quantities and work as may be necessary or desirable to complete, in a satisfactory manner, the original intended work. Unless otherwise specified in the Contract, the Owner's Engineer or RPR shall be and is hereby authorized to make, in writing, such in-scope alterations in the work and variation of quantities as may be necessary to complete the work, provided such action does not represent a significant change in the character of the work.

For purpose of this section, a significant change in character of work means: any change that is outside the current contract scope of work; any change (increase or decrease) in the total contract cost by more than 25%; or any change in the total cost of a major contract item by more than 25%.

Work alterations and quantity variances that do not meet the definition of significant change in character of work shall not invalidate the contract nor release the surety. Contractor agrees to accept payment for such work alterations and quantity variances in accordance with Section 90, paragraph 90-03, *Compensation for Altered Quantities*.

Should the value of altered work or quantity variance meet the criteria for significant change in character of work, such altered work and quantity variance shall be covered by a supplemental agreement. Supplemental agreements shall also require consent of the Contractor's surety and separate performance and payment bonds. If the Owner and the Contractor are unable to agree on a unit adjustment for any contract item that requires a supplemental agreement, the Owner reserves the right to terminate the contract with respect to the item and make other arrangements for its completion.

**40-03 OMITTED ITEMS.** The Owner, the Owner's Engineer or the RPR may provide written notice to the Contractor to omit from the work any contract item that does not meet the definition of major contract item. Major contract items may be omitted by a supplemental agreement. Such omission of contract items shall not invalidate any other contract provision or requirement.

Should a contract item be omitted or otherwise ordered to be non-performed, the Contractor shall be paid for all work performed toward completion of such item prior to the date of the order to omit such item. Payment for work performed shall be in accordance with Section 90, paragraph 90-04, *Payment for Omitted Items*.

**40-04 EXTRA WORK.** Should acceptable completion of the contract require the Contractor to perform an item of work not provided for in the awarded contract as previously modified by change order or supplemental agreement, Owner may issue a Change Order to cover the necessary extra work. Change orders for extra work shall contain agreed unit prices for performing the change order work in accordance with the requirements specified in the order, and shall contain any adjustment to the contract time that, in the RPR's opinion, is necessary for completion of the extra work.

When determined by the RPR to be in the Owner's best interest, the RPR may order the Contractor to proceed with extra work as provided in Section 90, paragraph 90-05, *Payment for Extra Work*. Extra work that is necessary for acceptable completion of the project, but is not within the general scope of the work covered by the original contract shall be covered by a supplemental agreement as defined in Section 10, paragraph 10-59, *Supplemental Agreement*.

If extra work is essential to maintaining the project critical path, RPR may order the Contractor to commence the extra work under a Time and Material contract method. Once sufficient detail is available to establish the level of effort necessary for the extra work, the Owner shall initiate a change order or supplemental agreement to cover the extra work.

Any claim for payment of extra work that is not covered by written agreement (change order or supplemental agreement) shall be rejected by the Owner.

**40-05 MAINTENANCE OF TRAFFIC.** It is the explicit intention of the contract that the safety of aircraft, as well as the Contractor's equipment and personnel, is the most important consideration. The Contractor shall maintain traffic in the manner detailed in the Construction Safety and Phasing Plan (CSPP).

a. It is understood and agreed that the Contractor shall provide for the free and unobstructed movement of aircraft in the air operations areas (AOAs) of the airport with respect to their own operations and the operations of all subcontractors as specified in Section 80, paragraph 80-04, *Limitation of Operations*. It is further understood and agreed that the Contractor shall provide for the uninterrupted operation of visual and electronic signals (including power supplies thereto) used in the guidance of aircraft while operating to, from, and upon the airport as specified in Section 70, paragraph 70-15, *Contractor's Responsibility for Utility Service and Facilities of Others*.

b. With respect to their own operations and the operations of all subcontractors, the Contractor shall provide marking, lighting, and other acceptable means of identifying personnel, equipment, vehicles, storage areas, and any work area or condition that may be hazardous to the operation of aircraft, fire-rescue equipment, or maintenance vehicles at the airport in accordance with the construction safety and phasing plan (CSPP) and the safety plan compliance document (SPCD).

c. When the contract requires the maintenance of an existing road, street, or highway during the Contractor's performance of work that is otherwise provided for in the contract, plans, and specifications, the Contractor shall keep the road, street, or highway open to all traffic and shall provide maintenance as may be required to accommodate traffic. The Contractor, at their expense, shall be responsible for the repair to equal or better than preconstruction conditions of any damage caused by the Contractor's equipment and personnel. The Contractor shall furnish, erect, and maintain barricades, warning signs, flag person, and other traffic control devices in reasonable conformity with the Manual on Uniform Traffic Control Devices (MUTCD) (<http://mutcd.fhwa.dot.gov/>), unless otherwise specified. The Contractor shall also construct and maintain in a safe condition any temporary connections necessary for ingress to and egress from abutting property or intersecting roads, streets or highways. **Unless otherwise specified herein, the Contractor will not be required to furnish snow removal for such existing road, street, or highway.**

**40-06 REMOVAL OF EXISTING STRUCTURES.** All existing structures encountered within the established lines, grades, or grading sections shall be removed by the Contractor, unless such existing structures are otherwise specified to be relocated, adjusted up or down, salvaged,

abandoned in place, reused in the work or to remain in place. The cost of removing such existing structures shall not be measured or paid for directly, but shall be included in the various contract items.

Should the Contractor encounter an existing structure (above or below ground) in the work for which the disposition is not indicated on the plans, the Resident Project Representative (RPR) shall be notified prior to disturbing such structure. The disposition of existing structures so encountered shall be immediately determined by the RPR in accordance with the provisions of the contract.

Except as provided in Section 40, paragraph 40-07, *Rights in and Use of Materials Found in the Work*, it is intended that all existing materials or structures that may be encountered (within the lines, grades, or grading sections established for completion of the work) shall be used in the work as otherwise provided for in the contract and shall remain the property of the Owner when so used in the work.

**40-07 RIGHTS IN AND USE OF MATERIALS FOUND IN THE WORK.** Should the Contractor encounter any material such as (but not restricted to) sand, stone, gravel, slag, or concrete slabs within the established lines, grades, or grading sections, the use of which is intended by the terms of the contract to be embankment, the Contractor may at their own option either:

- a. Use such material in another contract item, providing such use is approved by the RPR and is in conformance with the contract specifications applicable to such use; or,
- b. Remove such material from the site, upon written approval of the RPR; or
- c. Use such material for the Contractor's own temporary construction on site; or,
- d. Use such material as intended by the terms of the contract.

Should the Contractor wish to exercise option a., b., or c., the Contractor shall request the RPR's approval in advance of such use.

Should the RPR approve the Contractor's request to exercise option a., b., or c., the Contractor shall be paid for the excavation or removal of such material at the applicable contract price. The Contractor shall replace, at their expense, such removed or excavated material with an agreed equal volume of material that is acceptable for use in constructing embankment, backfills, or otherwise to the extent that such replacement material is needed to complete the contract work. The Contractor shall not be charged for use of such material used in the work or removed from the site.

Should the RPR approve the Contractor's exercise of option a., the Contractor shall be paid, at the applicable contract price, for furnishing and installing such material in accordance with requirements of the contract item in which the material is used.

It is understood and agreed that the Contractor shall make no claim for delays by reason of their own exercise of option a., b., or c.

The Contractor shall not excavate, remove, or otherwise disturb any material, structure, or part of a structure which is located outside the lines, grades, or grading sections established for the work, except where such excavation or removal is provided for in the contract, plans, or specifications.

**40-08 FINAL CLEANUP.** Upon completion of the work and before acceptance and final payment will be made, the Contractor shall remove from the site all machinery, equipment, surplus and discarded materials, rubbish, temporary structures, and stumps or portions of trees. The Contractor shall cut all brush and woods within the limits indicated and shall leave the site in a neat and presentable condition. Material cleared from the site and deposited on adjacent property will not be considered as having been disposed of satisfactorily, unless the Contractor has obtained the written permission of the property Owner.

**END OF SECTION 40**

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## **SECTION 50 CONTROL OF WORK**

**50-01 AUTHORITY OF THE RESIDENT PROJECT REPRESENTATIVE (RPR).** The RPR has final authority regarding the interpretation of project specification requirements. The RPR shall determine acceptability of the quality of materials furnished, method of performance of work performed, and the manner and rate of performance of the work. The RPR does not have the authority to accept work that does not conform to specification requirements.

**50-02 CONFORMITY WITH PLANS AND SPECIFICATIONS.** All work and all materials furnished shall be in reasonably close conformity with the lines, grades, grading sections, cross-sections, dimensions, material requirements, and testing requirements that are specified (including specified tolerances) in the contract, plans, or specifications.

If the RPR finds the materials furnished, work performed, or the finished product not within reasonably close conformity with the plans and specifications, but that the portion of the work affected will, in their opinion, result in a finished product having a level of safety, economy, durability, and workmanship acceptable to the Owner, the RPR will advise the Owner of their determination that the affected work be accepted and remain in place. The RPR will document the determination and recommend to the Owner a basis of acceptance that will provide for an adjustment in the contract price for the affected portion of the work. Changes in the contract price must be covered by contract change order or supplemental agreement as applicable.

If the RPR finds the materials furnished, work performed, or the finished product are not in reasonably close conformity with the plans and specifications and have resulted in an unacceptable finished product, the affected work or materials shall be removed and replaced or otherwise corrected by and at the expense of the Contractor in accordance with the RPR's written orders.

The term "reasonably close conformity" shall not be construed as waiving the Contractor's responsibility to complete the work in accordance with the contract, plans, and specifications. The term shall not be construed as waiving the RPR's responsibility to insist on strict compliance with the requirements of the contract, plans, and specifications during the Contractor's execution of the work, when, in the RPR's opinion, such compliance is essential to provide an acceptable finished portion of the work.

The term "reasonably close conformity" is also intended to provide the RPR with the authority, after consultation with the Sponsor and FAA, to use sound engineering judgment in their determinations to accept work that is not in strict conformity, but will provide a finished product equal to or better than that required by the requirements of the contract, plans and specifications.

The RPR will not be responsible for the Contractor's means, methods, techniques, sequences, or procedures of construction or the safety precautions incident thereto.

**50-03 COORDINATION OF CONTRACT, PLANS, AND SPECIFICATIONS.** The contract, plans, specifications, and all referenced standards cited are essential parts of the contract requirements. If electronic files are provided and used on the project and there is a conflict between the electronic files and hard copy plans, the hard copy plans shall govern. A requirement occurring in one is as binding as though occurring in all. They are intended to be complementary and to describe and provide for a complete work. In case of discrepancy, calculated dimensions will govern over scaled dimensions; contract technical specifications shall govern over contract

general provisions, plans, cited standards for materials or testing, and cited advisory circulars (ACs); contract general provisions shall govern over plans, cited standards for materials or testing, and cited ACs; plans shall govern over cited standards for materials or testing and cited ACs. If any paragraphs contained in the Special Provisions conflict with General Provisions or Technical Specifications, the Special Provisions shall govern.

From time to time, discrepancies within cited testing standards occur due to the timing of the change, edits, and/or replacement of the standards. If the Contractor discovers any apparent discrepancy within standard test methods, the Contractor shall immediately ask the RPR for an interpretation and decision, and such decision shall be final.

The Contractor shall not take advantage of any apparent error or omission on the plans or specifications. In the event the Contractor discovers any apparent error or discrepancy, Contractor shall immediately notify the Owner or the designated representative in writing requesting their written interpretation and decision.

#### **50-04 LIST OF SPECIAL PROVISIONS. [ ]**

**50-05 COOPERATION OF CONTRACTOR.** The Contractor shall be supplied with [ five ] hard copies or an electronic PDF of the plans and specifications. The Contractor shall have available on the construction site at all times one hardcopy each of the plans and specifications. Additional hard copies of plans and specifications may be obtained by the Contractor for the cost of reproduction.

The Contractor shall give constant attention to the work to facilitate the progress thereof, and shall cooperate with the RPR and their inspectors and with other Contractors in every way possible. The Contractor shall have a competent superintendent on the work at all times who is fully authorized as their agent on the work. The superintendent shall be capable of reading and thoroughly understanding the plans and specifications and shall receive and fulfill instructions from the RPR or their authorized representative.

**50-06 COOPERATION BETWEEN CONTRACTORS.** The Owner reserves the right to contract for and perform other or additional work on or near the work covered by this contract.

When separate contracts are let within the limits of any one project, each Contractor shall conduct the work not to interfere with or hinder the progress of completion of the work being performed by other Contractors. Contractors working on the same project shall cooperate with each other as directed.

Each Contractor involved shall assume all liability, financial or otherwise, in connection with their own contract and shall protect and hold harmless the Owner from any and all damages or claims that may arise because of inconvenience, delays, or loss experienced because of the presence and operations of other Contractors working within the limits of the same project.

The Contractor shall arrange their work and shall place and dispose of the materials being used to not interfere with the operations of the other Contractors within the limits of the same project. The Contractor shall join their work with that of the others in an acceptable manner and shall perform it in proper sequence to that of the others.

**50-07 CONSTRUCTION LAYOUT AND STAKES.** The Engineer/RPR shall establish necessary horizontal and vertical control. The establishment of Survey Control and/or reestablishment of

survey control shall be by a State Licensed Land Surveyor. Contractor is responsible for preserving integrity of horizontal and vertical controls established by Engineer/RPR. In case of negligence on the part of the Contractor or their employees, resulting in the destruction of any horizontal and vertical control, the resulting costs will be deducted as a liquidated damage against the Contractor.

Prior to the start of construction, the Contractor will check all control points for horizontal and vertical accuracy and certify in writing to the RPR that the Contractor concurs with survey control established for the project. All lines, grades and measurements from control points necessary for the proper execution and control of the work on this project will be provided to the RPR. The Contractor is responsible to establish all layout required for the construction of the project.

Copies of survey notes will be provided to the RPR for each area of construction and for each placement of material as specified to allow the RPR to make periodic checks for conformance with plan grades, alignments and grade tolerances required by the applicable material specifications. Surveys will be provided to the RPR prior to commencing work items that cover or disturb the survey staking. Survey(s) and notes shall be provided in the following format(s):  
[ ].)

Laser, GPS, String line, or other automatic control shall be checked with temporary control as necessary. In the case of error, on the part of the Contractor, their surveyor, employees or subcontractors, resulting in established grades, alignment or grade tolerances that do not concur with those specified or shown on the plans, the Contractor is solely responsible for correction, removal, replacement and all associated costs at no additional cost to the Owner.

No direct payment will be made, unless otherwise specified in contract documents, for this labor, materials, or other expenses. The cost shall be included in the price of the bid for the various items of the Contract.

**50-08 AUTHORITY AND DUTIES OF QUALITY ASSURANCE (QA) INSPECTORS.** QA inspectors shall be authorized to inspect all work done and all material furnished. Such QA inspection may extend to all or any part of the work and to the preparation, fabrication, or manufacture of the materials to be used. QA inspectors are not authorized to revoke, alter, or waive any provision of the contract. QA inspectors are not authorized to issue instructions contrary to the plans and specifications or to act as foreman for the Contractor.

QA Inspectors are authorized to notify the Contractor or their representatives of any failure of the work or materials to conform to the requirements of the contract, plans, or specifications and to reject such nonconforming materials in question until such issues can be referred to the RPR for a decision.

**50-09 INSPECTION OF THE WORK.** All materials and each part or detail of the work shall be subject to inspection. The RPR shall be allowed access to all parts of the work and shall be furnished with such information and assistance by the Contractor as is required to make a complete and detailed inspection.

If the RPR requests it, the Contractor, at any time before acceptance of the work, shall remove or uncover such portions of the finished work as may be directed. After examination, the Contractor shall restore said portions of the work to the standard required by the specifications. Should the work thus exposed or examined prove acceptable, the uncovering, or removing, and the replacing of the covering or making good of the parts removed will be paid for as extra work; but should the



work so exposed or examined prove unacceptable, the uncovering, or removing, and the replacing of the covering or making good of the parts removed will be at the Contractor's expense.

Provide advance written notice to the RPR of work the Contractor plans to perform each week and each day. Any work done or materials used without written notice and allowing opportunity for inspection by the RPR may be ordered removed and replaced at the Contractor's expense.

Should the contract work include relocation, adjustment, or any other modification to existing facilities, not the property of the (contract) Owner, authorized representatives of the Owners of such facilities shall have the right to inspect such work. Such inspection shall in no sense make any facility owner a party to the contract, and shall in no way interfere with the rights of the parties to this contract.

**50-10 REMOVAL OF UNACCEPTABLE AND UNAUTHORIZED WORK.** All work that does not conform to the requirements of the contract, plans, and specifications will be considered unacceptable, unless otherwise determined acceptable by the RPR as provided in paragraph 50-02, *Conformity with Plans and Specifications*.

Unacceptable work, whether the result of poor workmanship, use of defective materials, damage through carelessness, or any other cause found to exist prior to the final acceptance of the work, shall be removed immediately and replaced in an acceptable manner in accordance with the provisions of Section 70, paragraph 70-14, *Contractor's Responsibility for Work*.

No removal work made under provision of this paragraph shall be done without lines and grades having been established by the RPR. Work done contrary to the instructions of the RPR, work done beyond the lines shown on the plans or as established by the RPR, except as herein specified, or any extra work done without authority, will be considered as unauthorized and will not be paid for under the provisions of the contract. Work so done may be ordered removed or replaced at the Contractor's expense.

Upon failure on the part of the Contractor to comply with any order of the RPR made under the provisions of this subsection, the RPR will have authority to cause unacceptable work to be remedied or removed and replaced; and unauthorized work to be removed and recover the resulting costs as a liquidated damage against the Contractor.

**50-11 LOAD RESTRICTIONS.** The Contractor shall comply with all legal load restrictions in the hauling of materials on public roads beyond the limits of the work. A special permit will not relieve the Contractor of liability for damage that may result from the moving of material or equipment.

The operation of equipment of such weight or so loaded as to cause damage to structures or to any other type of construction will not be permitted. Hauling of materials over the base course or surface course under construction shall be limited as directed. No loads will be permitted on a concrete pavement, base, or structure before the expiration of the curing period. The Contractor, at their own expense, shall be responsible for the repair to equal or better than preconstruction conditions of any damage caused by the Contractor's equipment and personnel.

**50-12 MAINTENANCE DURING CONSTRUCTION.** The Contractor shall maintain the work during construction and until the work is accepted. Maintenance shall constitute continuous and effective work prosecuted day by day, with adequate equipment and forces so that the work is maintained in satisfactory condition at all times.

In the case of a contract for the placing of a course upon a course or subgrade previously constructed, the Contractor shall maintain the previous course or subgrade during all construction operations.

All costs of maintenance work during construction and before the project is accepted shall be included in the unit prices bid on the various contract items, and the Contractor will not be paid an additional amount for such work.

**50-13 FAILURE TO MAINTAIN THE WORK.** Should the Contractor at any time fail to maintain the work as provided in paragraph 50-12, *Maintenance during Construction*, the RPR shall immediately notify the Contractor of such noncompliance. Such notification shall specify a reasonable time within which the Contractor shall be required to remedy such unsatisfactory maintenance condition. The time specified will give due consideration to the exigency that exists.

Should the Contractor fail to respond to the RPR's notification, the Owner may suspend any work necessary for the Owner to correct such unsatisfactory maintenance condition, depending on the exigency that exists. Any maintenance cost incurred by the Owner, shall be recovered as a liquidated damage against the Contractor.

**50-14 PARTIAL ACCEPTANCE.** If at any time during the execution of the project the Contractor substantially completes a usable unit or portion of the work, the occupancy of which will benefit the Owner, the Contractor may request the RPR to make final inspection of that unit. If the RPR finds upon inspection that the unit has been satisfactorily completed in compliance with the contract, the RPR may accept it as being complete, and the Contractor may be relieved of further responsibility for that unit. Such partial acceptance and beneficial occupancy by the Owner shall not void or alter any provision of the contract.

**50-15 FINAL ACCEPTANCE.** Upon due notice from the Contractor of presumptive completion of the entire project, the RPR and Owner will make an inspection. If all construction provided for and contemplated by the contract is found to be complete in accordance with the contract, plans, and specifications, such inspection shall constitute the final inspection. The RPR shall notify the Contractor in writing of final acceptance as of the date of the final inspection.

If, however, the inspection discloses any work, in whole or in part, as being unsatisfactory, the RPR will notify the Contractor and the Contractor shall correct the unsatisfactory work. Upon correction of the work, another inspection will be made which shall constitute the final inspection, provided the work has been satisfactorily completed. In such event, the RPR will make the final acceptance and notify the Contractor in writing of this acceptance as of the date of final inspection.

**50-16 CLAIMS FOR ADJUSTMENT AND DISPUTES.** If for any reason the Contractor deems that additional compensation is due for work or materials not clearly provided for in the contract, plans, or specifications or previously authorized as extra work, the Contractor shall notify the RPR in writing of their intention to claim such additional compensation before the Contractor begins the work on which the Contractor bases the claim. If such notification is not given or the RPR is not afforded proper opportunity by the Contractor for keeping strict account of actual cost as required, then the Contractor hereby agrees to waive any claim for such additional compensation. Such notice by the Contractor and the fact that the RPR has kept account of the cost of the work shall not in any way be construed as proving or substantiating the validity of the claim. When the work on which the claim for additional compensation is based has been completed, the Contractor shall, within 10 calendar days, submit a written claim to the RPR who will present it to the Owner for consideration in accordance with local laws or ordinances.

Nothing in this subsection shall be construed as a waiver of the Contractor's right to dispute final payment based on differences in measurements or computations.

**END OF SECTION 50**

## SECTION 60 CONTROL OF MATERIALS

**60-01 SOURCE OF SUPPLY AND QUALITY REQUIREMENTS.** The materials used in the work shall conform to the requirements of the contract, plans, and specifications. Unless otherwise specified, such materials that are manufactured or processed shall be new (as compared to used or reprocessed).

In order to expedite the inspection and testing of materials, the Contractor shall furnish documentation to the RPR as to the origin, composition, and manufacture of all materials to be used in the work. Documentation shall be furnished promptly after execution of the contract but, in all cases, prior to delivery of such materials.

At the RPR's option, materials may be approved at the source of supply before delivery. If it is found after trial that sources of supply for previously approved materials do not produce specified products, the Contractor shall furnish materials from other sources.

The Contractor shall furnish airport lighting equipment that meets the requirements of the specifications; and is listed in AC 150/5345-53, *Airport Lighting Equipment Certification Program and Addendum*, that is in effect on the date of advertisement.

**60-02 SAMPLES, TESTS, AND CITED SPECIFICATIONS.** All materials used in the work shall be inspected, tested, and approved by the RPR before incorporation in the work unless otherwise designated. Any work in which untested materials are used without approval or written permission of the RPR shall be performed at the Contractor's risk. Materials found to be unacceptable and unauthorized will not be paid for and, if directed by the RPR, shall be removed at the Contractor's expense.

Unless otherwise designated, quality assurance tests will be made by and at the expense of the Owner in accordance with the cited standard methods of ASTM, American Association of State Highway and Transportation Officials (AASHTO), federal specifications, Commercial Item Descriptions, and all other cited methods, which are current on the date of advertisement for bids.

The testing organizations performing on-site quality assurance field tests shall have copies of all referenced standards on the construction site for use by all technicians and other personnel. Unless otherwise designated, samples for quality assurance will be taken by a qualified representative of the RPR. All materials being used are subject to inspection, test, or rejection at any time prior to or during incorporation into the work. Copies of all tests will be furnished to the Contractor's representative at their request after review and approval of the RPR.

A copy of all Contractor QC test data shall be provided to the RPR daily, along with printed reports, in an approved format, on a weekly basis. After completion of the project, and prior to final payment, the Contractor shall submit a final report to the RPR showing all test data reports, plus an analysis of all results showing ranges, averages, and corrective action taken on all failing tests.

**The Contractor shall employ a Quality Control (QC) testing organization to perform all Contractor required QC tests in accordance with Item C-100 Contractor Quality Control Program (CQCP).**

**60-03 CERTIFICATION OF COMPLIANCE/ANALYSIS (COC/COA).** The RPR may permit the use, prior to sampling and testing, of certain materials or assemblies when accompanied by

manufacturer's COC stating that such materials or assemblies fully comply with the requirements of the contract. The certificate shall be signed by the manufacturer. Each lot of such materials or assemblies delivered to the work must be accompanied by a certificate of compliance in which the lot is clearly identified. The COA is the manufacturer's COC and includes all applicable test results.

Materials or assemblies used on the basis of certificates of compliance may be sampled and tested at any time and if found not to be in conformity with contract requirements will be subject to rejection whether in place or not.

The form and distribution of certificates of compliance shall be as approved by the RPR.

When a material or assembly is specified by "brand name or equal" and the Contractor elects to furnish the specified "or equal," the Contractor shall be required to furnish the manufacturer's certificate of compliance for each lot of such material or assembly delivered to the work. Such certificate of compliance shall clearly identify each lot delivered and shall certify as to:

- a. Conformance to the specified performance, testing, quality or dimensional requirements; and,
- b. Suitability of the material or assembly for the use intended in the contract work.

The RPR shall be the sole judge as to whether the proposed "or equal" is suitable for use in the work.

The RPR reserves the right to refuse permission for use of materials or assemblies on the basis of certificates of compliance.

**60-04 PLANT INSPECTION.** The RPR or their authorized representative may inspect, at its source, any specified material or assembly to be used in the work. Manufacturing plants may be inspected from time to time for the purpose of determining compliance with specified manufacturing methods or materials to be used in the work and to obtain samples required for acceptance of the material or assembly.

Should the RPR conduct plant inspections, the following conditions shall exist:

- a. The RPR shall have the cooperation and assistance of the Contractor and the producer with whom the Contractor has contracted for materials.
- b. The RPR shall have full entry at all reasonable times to such parts of the plant that concern the manufacture or production of the materials being furnished.
- c. If required by the RPR, the Contractor shall arrange for adequate office or working space that may be reasonably needed for conducting plant inspections. Place office or working space in a convenient location with respect to the plant.

It is understood and agreed that the Owner shall have the right to retest any material that has been tested and approved at the source of supply after it has been delivered to the site. The RPR shall have the right to reject only material which, when retested, does not meet the requirements of the contract, plans, or specifications.

**60-05 ENGINEER/ RESIDENT PROJECT REPRESENTATIVE (RPR) FIELD OFFICE.** [ The Contractor shall provide dedicated space for the use of the engineer, RPR, and inspectors, as a field office for the duration of the project. This space shall be located conveniently near the construction and shall be separate from any space used by the Contractor. **The Contractor shall furnish water, sanitary facilities, heat, air conditioning, and electricity. An Engineer/RPR field office is not required.**

**60-06 STORAGE OF MATERIALS.** Materials shall be stored to assure the preservation of their quality and fitness for the work. Stored materials, even though approved before storage, may again be inspected prior to their use in the work. Stored materials shall be located to facilitate their prompt inspection. The Contractor shall coordinate the storage of all materials with the RPR. Materials to be stored on airport property shall not create an obstruction to air navigation nor shall they interfere with the free and unobstructed movement of aircraft. Unless otherwise shown on the plans and/or CSPP, the storage of materials and the location of the Contractor's plant and parked equipment or vehicles shall be as directed by the RPR. Private property shall not be used for storage purposes without written permission of the Owner or lessee of such property. The Contractor shall make all arrangements and bear all expenses for the storage of materials on private property. Upon request, the Contractor shall furnish the RPR a copy of the property Owner's permission.

All storage sites on private or airport property shall be restored to their original condition by the Contractor at their expense, except as otherwise agreed to (in writing) by the Owner or lessee of the property.

**60-07 UNACCEPTABLE MATERIALS.** Any material or assembly that does not conform to the requirements of the contract, plans, or specifications shall be considered unacceptable and shall be rejected. The Contractor shall remove any rejected material or assembly from the site of the work, unless otherwise instructed by the RPR.

Rejected material or assembly, the defects of which have been corrected by the Contractor, shall not be returned to the site of the work until such time as the RPR has approved its use in the work.

**60-08 OWNER FURNISHED MATERIALS.** The Contractor shall furnish all materials required to complete the work, except those specified, if any, to be furnished by the Owner. Owner-furnished materials shall be made available to the Contractor at the location specified.

All costs of handling, transportation from the specified location to the site of work, storage, and installing Owner-furnished materials shall be included in the unit price bid for the contract item in which such Owner-furnished material is used.

After any Owner-furnished material has been delivered to the location specified, the Contractor shall be responsible for any demurrage, damage, loss, or other deficiencies that may occur during the Contractor's handling, storage, or use of such Owner-furnished material. The Owner will deduct from any monies due or to become due the Contractor any cost incurred by the Owner in making good such loss due to the Contractor's handling, storage, or use of Owner-furnished materials.

#### END OF SECTION 60

## **SECTION 70 LEGAL REGULATIONS AND RESPONSIBILITY TO PUBLIC**

**70-01 LAWS TO BE OBSERVED.** The Contractor shall keep fully informed of all federal and state laws, all local laws, ordinances, and regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority, which in any manner affect those engaged or employed on the work, or which in any way affect the conduct of the work. The Contractor shall at all times observe and comply with all such laws, ordinances, regulations, orders, and decrees; and shall protect and indemnify the Owner and all their officers, agents, or servants against any claim or liability arising from or based on the violation of any such law, ordinance, regulation, order, or decree, whether by the Contractor or the Contractor's employees.

**70-02 PERMITS, LICENSES, AND TAXES.** The Contractor shall procure all permits and licenses, pay all charges, fees, and taxes, and give all notices necessary and incidental to the due and lawful execution of the work.

**70-03 PATENTED DEVICES, MATERIALS, AND PROCESSES.** If the Contractor is required or desires to use any design, device, material, or process covered by letters of patent or copyright, the Contractor shall provide for such use by suitable legal agreement with the Patentee or Owner. The Contractor and the surety shall indemnify and hold harmless the Owner, any third party, or political subdivision from any and all claims for infringement by reason of the use of any such patented design, device, material or process, or any trademark or copyright, and shall indemnify the Owner for any costs, expenses, and damages which it may be obliged to pay by reason of an infringement, at any time during the execution or after the completion of the work.

**70-04 RESTORATION OF SURFACES DISTURBED BY OTHERS.** The Owner reserves the right to authorize the construction, reconstruction, or maintenance of any public or private utility service, FAA or National Oceanic and Atmospheric Administration (NOAA) facility, or a utility service of another government agency at any time during the progress of the work. To the extent that such construction, reconstruction, or maintenance has been coordinated with the Owner, such authorized work (by others) must be shown on the plans and is indicated as follows: [\_\_\_\_].

Except as listed above, the Contractor shall not permit any individual, firm, or corporation to excavate or otherwise disturb such utility services or facilities located within the limits of the work without the written permission of the RPR.

Should the Owner of public or private utility service, FAA, or NOAA facility, or a utility service of another government agency be authorized to construct, reconstruct, or maintain such utility service or facility during the progress of the work, the Contractor shall cooperate with such Owners by arranging and performing the work in this contract to facilitate such construction, reconstruction or maintenance by others whether or not such work by others is listed above. When ordered as extra work by the RPR, the Contractor shall make all necessary repairs to the work which are due to such authorized work by others, unless otherwise provided for in the contract, plans, or specifications. It is understood and agreed that the Contractor shall not be entitled to make any claim for damages due to such authorized work by others or for any delay to the work resulting from such authorized work.

**70-05 FEDERAL PARTICIPATION.** The United States Government has agreed to reimburse the Owner for some portion of the contract costs. The contract work is subject to the inspection and approval of duly authorized representatives of the FAA Administrator. No requirement of this

contract shall be construed as making the United States a party to the contract nor will any such requirement interfere, in any way, with the rights of either party to the contract.

**70-06 SANITARY, HEALTH, AND SAFETY PROVISIONS.** The Contractor's worksite and facilities shall comply with applicable federal, state, and local requirements for health, safety and sanitary provisions.

**70-07 PUBLIC CONVENIENCE AND SAFETY.** The Contractor shall control their operations and those of their subcontractors and all suppliers, to assure the least inconvenience to the traveling public. Under all circumstances, safety shall be the most important consideration.

The Contractor shall maintain the free and unobstructed movement of aircraft and vehicular traffic with respect to their own operations and those of their own subcontractors and all suppliers in accordance with Section 40, paragraph 40-05, *Maintenance of Traffic*, and shall limit such operations for the convenience and safety of the traveling public as specified in Section 80, paragraph 80-04, *Limitation of Operations*.

The Contractor shall remove or control debris and rubbish resulting from its work operations at frequent intervals, and upon the order of the RPR. If the RPR determines the existence of Contractor debris in the work site represents a hazard to airport operations and the Contractor is unable to respond in a prompt and reasonable manner, the RPR reserves the right to assign the task of debris removal to a third party and recover the resulting costs as a liquidated damage against the Contractor.

**70-08 CONSTRUCTION SAFETY AND PHASING PLAN (CSPP).** The Contractor shall complete the work in accordance with the approved Construction Safety and Phasing Plan (CSPP) developed in accordance with AC 150/5370-2, Operational Safety on Airports During Construction. The CSPP is on sheet(s) [ ] of the project plans.

**70-09 USE OF EXPLOSIVES.** The use of explosives is not permitted on this project.

**70-10 PROTECTION AND RESTORATION OF PROPERTY AND LANDSCAPE.** The Contractor shall be responsible for the preservation of all public and private property, and shall protect carefully from disturbance or damage all land monuments and property markers until the Engineer/RPR has witnessed or otherwise referenced their location and shall not move them until directed.

The Contractor shall be responsible for all damage or injury to property of any character, during the execution of the work, resulting from any act, omission, neglect, or misconduct in manner or method of executing the work, or at any time due to defective work or materials, and said responsibility shall not be released until the project has been completed and accepted.

When or where any direct or indirect damage or injury is done to public or private property by or on account of any act, omission, neglect, or misconduct in the execution of the work, or in consequence of the non-execution thereof by the Contractor, the Contractor shall restore, at their expense, such property to a condition similar or equal to that existing before such damage or injury was done, by repairing, or otherwise restoring as may be directed, or the Contractor shall make good such damage or injury in an acceptable manner.



**70-11 RESPONSIBILITY FOR DAMAGE CLAIMS.** The Contractor shall indemnify and hold harmless the Engineer/RPR and the Owner and their officers, agents, and employees from all suits, actions, or claims, of any character, brought because of any injuries or damage received or sustained by any person, persons, or property on account of the operations of the Contractor; or on account of or in consequence of any neglect in safeguarding the work; or through use of unacceptable materials in constructing the work; or because of any act or omission, neglect, or misconduct of said Contractor; or because of any claims or amounts recovered from any infringements of patent, trademark, or copyright; or from any claims or amounts arising or recovered under the "Workmen's Compensation Act," or any other law, ordinance, order, or decree. Money due the Contractor under and by virtue of their own contract considered necessary by the Owner for such purpose may be retained for the use of the Owner or, in case no money is due, their own surety may be held until such suits, actions, or claims for injuries or damages shall have been settled and suitable evidence to that effect furnished to the Owner, except that money due the Contractor will not be withheld when the Contractor produces satisfactory evidence that he or she is adequately protected by public liability and property damage insurance.

**70-12 THIRD PARTY BENEFICIARY CLAUSE.** It is specifically agreed between the parties executing the contract that it is not intended by any of the provisions of any part of the contract to create for the public or any member thereof, a third-party beneficiary or to authorize anyone not a party to the contract to maintain a suit for personal injuries or property damage pursuant to the terms or provisions of the contract.

**70-13 OPENING SECTIONS OF THE WORK TO TRAFFIC.** If it is necessary for the Contractor to complete portions of the contract work for the beneficial occupancy of the Owner prior to completion of the entire contract, such "phasing" of the work must be specified below and indicated on the approved Construction Safety and Phasing Plan (CSPP) and the project plans. When so specified, the Contractor shall complete such portions of the work on or before the date specified or as otherwise specified.

Upon completion of any portion of work listed above, such portion shall be accepted by the Owner in accordance with Section 50, paragraph 50-14, *Partial Acceptance*.

No portion of the work may be opened by the Contractor until directed by the Owner in writing. Should it become necessary to open a portion of the work to traffic on a temporary or intermittent basis, such openings shall be made when, in the opinion of the RPR, such portion of the work is in an acceptable condition to support the intended traffic. Temporary or intermittent openings are considered to be inherent in the work and shall not constitute either acceptance of the portion of the work so opened or a waiver of any provision of the contract. Any damage to the portion of the work so opened that is not attributable to traffic which is permitted by the Owner shall be repaired by the Contractor at their expense.

The Contractor shall make their own estimate of the inherent difficulties involved in completing the work under the conditions herein described and shall not claim any added compensation by reason of delay or increased cost due to opening a portion of the contract work.

The Contractor must conform to safety standards contained AC 150/5370-2 and the approved CSPP.

Contractor shall refer to the plans, specifications, and the approved CSPP to identify barricade requirements, temporary and/or permanent markings, airfield lighting, guidance signs and other safety requirements prior to opening up sections of work to traffic.

**70-14 CONTRACTOR'S RESPONSIBILITY FOR WORK.** Until the RPR's final written acceptance of the entire completed work, excepting only those portions of the work accepted in accordance with Section 50, paragraph 50-14, *Partial Acceptance*, the Contractor shall have the charge and care thereof and shall take every precaution against injury or damage to any part due to the action of the elements or from any other cause, whether arising from the execution or from the non-execution of the work. The Contractor shall rebuild, repair, restore, and make good all injuries or damages to any portion of the work occasioned by any of the above causes before final acceptance and shall bear the expense thereof except damage to the work due to unforeseeable causes beyond the control of and without the fault or negligence of the Contractor, including but not restricted to acts of God such as earthquake, tidal wave, tornado, hurricane or other cataclysmic phenomenon of nature, or acts of the public enemy or of government authorities.

If the work is suspended for any cause whatever, the Contractor shall be responsible for the work and shall take such precautions necessary to prevent damage to the work. The Contractor shall provide for normal drainage and shall erect necessary temporary structures, signs, or other facilities at their own expense. During such period of suspension of work, the Contractor shall properly and continuously maintain in an acceptable growing condition all living material in newly established planting, seeding, and sodding furnished under the contract, and shall take adequate precautions to protect new tree growth and other important vegetative growth against injury.

**70-15 CONTRACTOR'S RESPONSIBILITY FOR UTILITY SERVICE AND FACILITIES OF OTHERS.** As provided in paragraph 70-04, *Restoration of Surfaces Disturbed by Others*, the Contractor shall cooperate with the owner of any public or private utility service, FAA or NOAA, or a utility service of another government agency that may be authorized by the Owner to construct, reconstruct or maintain such utility services or facilities during the progress of the work. In addition, the Contractor shall control their operations to prevent the unscheduled interruption of such utility services and facilities.

To the extent that such public or private utility services, FAA, or NOAA facilities, or utility services of another governmental agency are known to exist within the limits of the contract work, the approximate locations have been indicated on the plans and/or in the contract documents.

It is understood and agreed that the Owner does not guarantee the accuracy or the completeness of the location information relating to existing utility services, facilities, or structures that may be shown on the plans or encountered in the work. Any inaccuracy or omission in such information shall not relieve the Contractor of the responsibility to protect such existing features from damage or unscheduled interruption of service.

It is further understood and agreed that the Contractor shall, upon execution of the contract, notify the Owners of all utility services or other facilities of their plan of operations. Such notification shall be in writing addressed to "The Person to Contact" as provided in this paragraph and paragraph 70-04, *Restoration of Surfaces Disturbed By Others*. A copy of each notification shall be given to the RPR.

In addition to the general written notification provided, it shall be the responsibility of the Contractor to keep such individual Owners advised of changes in their plan of operations that would affect such Owners.

Prior to beginning the work in the general vicinity of an existing utility service or facility, the Contractor shall again notify each such Owner of their plan of operation. If, in the Contractor's opinion, the Owner's assistance is needed to locate the utility service or facility or the presence of a representative of the Owner is desirable to observe the work, such advice should be included in the notification. Such notification shall be given by the most expeditious means to reach the utility owner's "Person to Contact" no later than two normal business days prior to the Contractor's commencement of operations in such general vicinity. The Contractor shall furnish a written summary of the notification to the RPR.

The Contractor's failure to give the two days' notice shall be cause for the Owner to suspend the Contractor's operations in the general vicinity of a utility service or facility.

Where the outside limits of an underground utility service have been located and staked on the ground, the Contractor shall be required to use hand excavation methods within 3 feet (1 m) of such outside limits at such points as may be required to ensure protection from damage due to the Contractor's operations.

Should the Contractor damage or interrupt the operation of a utility service or facility by accident or otherwise, the Contractor shall immediately notify the proper authority and the RPR and shall take all reasonable measures to prevent further damage or interruption of service. The Contractor, in such events, shall cooperate with the utility service or facility owner and the RPR continuously until such damage has been repaired and service restored to the satisfaction of the utility or facility owner.

The Contractor shall bear all costs of damage and restoration of service to any utility service or facility due to their operations whether due to negligence or accident. The Owner reserves the right to deduct such costs from any monies due or which may become due the Contractor, or their own surety.

**[ 70-15.1 FAA FACILITIES AND CABLE RUNS.** The Contractor is hereby advised that the construction limits of the project include existing facilities and buried cable runs that are owned, operated and maintained by the FAA. The Contractor, during the execution of the project work, shall comply with the following:

**a.** The Contractor shall permit FAA maintenance personnel the right of access to the project work site for purposes of inspecting and maintaining all existing FAA owned facilities.

**b.** The Contractor shall provide notice to the FAA Air Traffic Organization (ATO)/Technical Operations/System Support Center (SSC) Point-of-Contact through the airport [ Owner ] [ operator ] [ manager ] [ ] a minimum of seven (7) calendar days prior to commencement of construction activities in order to permit sufficient time to locate and mark existing buried cables and to schedule any required facility outages.

**c.** If execution of the project work requires a facility outage, the Contractor shall contact the FAA Point-of-Contact a minimum of 72 hours prior to the time of the required outage.

d. Any damage to FAA cables, access roads, or FAA facilities during construction caused by the Contractor's equipment or personnel whether by negligence or accident will require the Contractor to repair or replace the damaged cables, access road, or FAA facilities to FAA requirements. The Contractor shall not bear the cost to repair damage to underground facilities or utilities improperly located by the FAA.

e. If the project work requires the cutting or splicing of FAA owned cables, the FAA Point-of-Contact shall be contacted a minimum of 72 hours prior to the time the cable work commences. The FAA reserves the right to have a FAA representative on site to observe the splicing of the cables as a condition of acceptance. All cable splices are to be accomplished in accordance with FAA specifications and require approval by the FAA Point-of-Contact as a condition of acceptance by the Owner. The Contractor is hereby advised that FAA restricts the location of where splices may be installed. If a cable splice is required in a location that is not permitted by FAA, the Contractor shall furnish and install a sufficient length of new cable that eliminates the need for any splice. ]

**70-16 FURNISHING RIGHTS-OF-WAY.** The Owner will be responsible for furnishing all rights-of-way upon which the work is to be constructed in advance of the Contractor's operations.

**70-17 PERSONAL LIABILITY OF PUBLIC OFFICIALS.** In carrying out any of the contract provisions or in exercising any power or authority granted by this contract, there shall be no liability upon the Engineer, RPR, their authorized representatives, or any officials of the Owner either personally or as an official of the Owner. It is understood that in such matters they act solely as agents and representatives of the Owner.

**70-18 NO WAIVER OF LEGAL RIGHTS.** Upon completion of the work, the Owner will expeditiously make final inspection and notify the Contractor of final acceptance. Such final acceptance, however, shall not preclude or stop the Owner from correcting any measurement, estimate, or certificate made before or after completion of the work, nor shall the Owner be precluded or stopped from recovering from the Contractor or their surety, or both, such overpayment as may be sustained, or by failure on the part of the Contractor to fulfill their obligations under the contract. A waiver on the part of the Owner of any breach of any part of the contract shall not be held to be a waiver of any other or subsequent breach.

The Contractor, without prejudice to the terms of the contract, shall be liable to the Owner for latent defects, fraud, or such gross mistakes as may amount to fraud, or as regards the Owner's rights under any warranty or guaranty.

**70-19 ENVIRONMENTAL PROTECTION.** The Contractor shall comply with all federal, state, and local laws and regulations controlling pollution of the environment. The Contractor shall take necessary precautions to prevent pollution of streams, lakes, ponds, and reservoirs with fuels, oils, asphalts, chemicals, or other harmful materials and to prevent pollution of the atmosphere from particulate and gaseous matter.

**70-20 ARCHAEOLOGICAL AND HISTORICAL FINDINGS.** Unless otherwise specified in this subsection, the Contractor is advised that the site of the work is not within any property, district, or site, and does not contain any building, structure, or object listed in the current National Register of Historic Places published by the United States Department of Interior.

Should the Contractor encounter, during their operations, any building, part of a building, structure, or object that is incongruous with its surroundings, the Contractor shall immediately cease operations in that location and notify the RPR. The RPR will immediately investigate the Contractor's finding and the Owner will direct the Contractor to either resume operations or to suspend operations as directed.

Should the Owner order suspension of the Contractor's operations in order to protect an archaeological or historical finding, or order the Contractor to perform extra work, such shall be covered by an appropriate contract change order or supplemental agreement as provided in Section 40, paragraph 40-04, *Extra Work*, and Section 90, paragraph 90-05, *Payment for Extra Work*. If appropriate, the contract change order or supplemental agreement shall include an extension of contract time in accordance with Section 80, paragraph 80-07, *Determination and Extension of Contract Time*.

**70-21 INSURANCE REQUIREMENTS.** [ Insert local insurance requirements for the project. ]

**END OF SECTION 70**

## SECTION 80 EXECUTION AND PROGRESS

**80-01 SUBLETTING OF CONTRACT.** The Owner will not recognize any subcontractor on the work. The Contractor shall at all times when work is in progress be represented either in person, by a qualified superintendent, or by other designated, qualified representative who is duly authorized to receive and execute orders of the Resident Project Representative (RPR).

The Contractor shall perform, with his organization, an amount of work equal to at least [ ] percent of the total contract cost.

Should the Contractor elect to assign their contract, said assignment shall be concurred in by the surety, shall be presented for the consideration and approval of the Owner, and shall be consummated only on the written approval of the Owner.

**The Contractor shall provide copies of all subcontracts to the RPR 14 days prior to being utilized on the project. As a minimum, the information shall include the following:**

- Subcontractor's legal company name.
- Subcontractor's legal company address, including County name.
- Principal contact person's name, telephone and fax number.
- Complete narrative description, and dollar value of the work to be performed by the subcontractor.
- Copies of required insurance certificates in accordance with the specifications.
- Minority/ non-minority status.

**80-02 NOTICE TO PROCEED (NTP).** The Owners notice to proceed will state the date on which contract time commences. The Contractor is expected to commence project operations within 10 days of the NTP date. The Contractor shall notify the RPR at least **24 hours** in advance of the time contract operations begins. The Contractor shall not commence any actual operations prior to the date on which the notice to proceed is issued by the Owner.

**80-03 EXECUTION AND PROGRESS.** Unless otherwise specified, the Contractor shall submit their coordinated construction schedule showing all work activities for the RPR's review and acceptance at least **10 days** prior to the start of work. The Contractor's progress schedule, once accepted by the RPR, will represent the Contractor's baseline plan to accomplish the project in accordance with the terms and conditions of the Contract. The RPR will compare actual Contractor progress against the baseline schedule to determine that status of the Contractor's performance. The Contractor shall provide sufficient materials, equipment, and labor to guarantee the completion of the project in accordance with the plans and specifications within the time set forth in the proposal.

If the Contractor falls significantly behind the submitted schedule, the Contractor shall, upon the RPR's request, submit a revised schedule for completion of the work within the contract time and modify their operations to provide such additional materials, equipment, and labor necessary to meet the revised schedule. Should the execution of the work be discontinued for any reason, the Contractor shall notify the RPR at least **24 hours** in advance of resuming operations.

The Contractor shall not commence any actual construction prior to the date on which the NTP is issued by the Owner.

The project schedule shall be prepared as a network diagram in Critical Path Method (CPM), Program Evaluation and Review Technique (PERT), or other format, or as otherwise specified. It shall include information on the sequence of work activities, milestone dates, and activity duration. The schedule shall show all work items identified in the project proposal for each work area and shall include the project start date and end date.

The Contractor shall maintain the work schedule and provide an update and analysis of the progress schedule on a **twice** monthly basis, or as otherwise specified in the contract. Submission of the work schedule shall not relieve the Contractor of overall responsibility for scheduling, sequencing, and coordinating all work to comply with the requirements of the contract.

**80-04 LIMITATION OF OPERATIONS.** The Contractor shall control their operations and the operations of their subcontractors and all suppliers to provide for the free and unobstructed movement of aircraft in the air operations areas (AOA) of the airport.

When the work requires the Contractor to conduct their operations within an AOA of the airport, the work shall be coordinated with airport operations (through the RPR) at least [ 48 hours ] prior to commencement of such work. The Contractor shall not close an AOA until so authorized by the RPR and until the necessary temporary marking, signage and associated lighting is in place as provided in Section 70, paragraph 70-08, *Construction Safety and Phasing Plan (CSPP)*.

When the contract work requires the Contractor to work within an AOA of the airport on an intermittent basis (intermittent opening and closing of the AOA), the Contractor shall maintain constant communications as specified; immediately obey all instructions to vacate the AOA; and immediately obey all instructions to resume work in such AOA. Failure to maintain the specified communications or to obey instructions shall be cause for suspension of the Contractor's operations in the AOA until satisfactory conditions are provided. The areas of the AOA identified in the Construction Safety Phasing Plan (CSPP) and as listed below, cannot be closed to operating aircraft to permit the Contractor's operations on a continuous basis and will therefore be closed to aircraft operations intermittently as follows:

*Per phasing plan in contract documents.*

The Contractor shall be required to conform to safety standards contained in AC 150/5370-2, Operational Safety on Airports During Construction and the approved CSPP.

**80-04.1 OPERATIONAL SAFETY ON AIRPORT DURING CONSTRUCTION.** All Contractors' operations shall be conducted in accordance with the approved project Construction Safety and Phasing Plan (CSPP) and the Safety Plan Compliance Document (SPCD) and the provisions set forth within the current version of AC 150/5370-2, Operational Safety on Airports During Construction. The CSPP included within the contract documents conveys minimum requirements for operational safety on the airport during construction activities. The Contractor shall prepare and submit a SPCD that details how it proposes to comply with the requirements presented within the CSPP.

The Contractor shall implement all necessary safety plan measures prior to commencement of any work activity. The Contractor shall conduct routine checks to assure compliance with the safety plan measures.

The Contractor is responsible to the Owner for the conduct of all subcontractors it employs on the project. The Contractor shall assure that all subcontractors are made aware of the requirements of the CSPP and SPCD and that they implement and maintain all necessary measures.

No deviation or modifications may be made to the approved CSPP and SPCD unless approved in writing by the Owner. The necessary coordination actions to review Contractor proposed modifications to an approved CSPP or approved SPCD can require a significant amount of time.

**80-05 CHARACTER OF WORKERS, METHODS, AND EQUIPMENT.** The Contractor shall, at all times, employ sufficient labor and equipment for prosecuting the work to full completion in the manner and time required by the contract, plans, and specifications.

All workers shall have sufficient skill and experience to perform properly the work assigned to them. Workers engaged in special work or skilled work shall have sufficient experience in such work and in the operation of the equipment required to perform the work satisfactorily.

Any person employed by the Contractor or by any subcontractor who violates any operational regulations or operational safety requirements and, in the opinion of the RPR, does not perform his work in a proper and skillful manner or is intemperate or disorderly shall, at the written request of the RPR, be removed immediately by the Contractor or subcontractor employing such person, and shall not be employed again in any portion of the work without approval of the RPR.

Should the Contractor fail to remove such person or persons, or fail to furnish suitable and sufficient personnel for the proper execution of the work, the RPR may suspend the work by written notice until compliance with such orders.

All equipment that is proposed to be used on the work shall be of sufficient size and in such mechanical condition as to meet requirements of the work and to produce a satisfactory quality of work. Equipment used on any portion of the work shall not cause injury to previously completed work, adjacent property, or existing airport facilities due to its use.

When the methods and equipment to be used by the Contractor in accomplishing the work are not prescribed in the contract, the Contractor is free to use any methods or equipment that will accomplish the work in conformity with the requirements of the contract, plans, and specifications.

When the contract specifies the use of certain methods and equipment, such methods and equipment shall be used unless otherwise authorized by the RPR. If the Contractor desires to use a method or type of equipment other than specified in the contract, the Contractor may request authority from the RPR to do so. The request shall be in writing and shall include a full description of the methods and equipment proposed and of the reasons for desiring to make the change. If approval is given, it will be on the condition that the Contractor will be fully responsible for producing work in conformity with contract requirements. If, after trial use of the substituted methods or equipment, the RPR determines that the work produced does not meet contract requirements, the Contractor shall discontinue the use of the substitute method or equipment and shall complete the remaining work with the specified methods and equipment. The Contractor shall remove any deficient work and replace it with work of specified quality, or take such other corrective action as the RPR may direct. No change will be made in basis of payment for the contract items involved nor in contract time as a result of authorizing a change in methods or equipment under this paragraph.



**80-06 TEMPORARY SUSPENSION OF THE WORK.** The Owner shall have the authority to suspend the work wholly, or in part, for such period or periods the Owner may deem necessary, due to unsuitable weather, or other conditions considered unfavorable for the execution of the work, or for such time necessary due to the failure on the part of the Contractor to carry out orders given or perform any or all provisions of the contract.

In the event that the Contractor is ordered by the Owner, in writing, to suspend work for some unforeseen cause not otherwise provided for in the contract and over which the Contractor has no control, the Contractor may be reimbursed for actual money expended on the work during the period of shutdown. No allowance will be made for anticipated profits. The period of shutdown shall be computed from the effective date of the written order to suspend work to the effective date of the written order to resume the work. Claims for such compensation shall be filed with the RPR within the time period stated in the RPR's order to resume work. The Contractor shall submit with their own claim information substantiating the amount shown on the claim. The RPR will forward the Contractor's claim to the Owner for consideration in accordance with local laws or ordinances. No provision of this article shall be construed as entitling the Contractor to compensation for delays due to inclement weather or for any other delay provided for in the contract, plans, or specifications.

If it becomes necessary to suspend work for an indefinite period, the Contractor shall store all materials in such manner that they will not become an obstruction nor become damaged in any way. The Contractor shall take every precaution to prevent damage or deterioration of the work performed and provide for normal drainage of the work. The Contractor shall erect temporary structures where necessary to provide for traffic on, to, or from the airport.

**80-07 DETERMINATION AND EXTENSION OF CONTRACT TIME.** The **300** calendar days shall be stated in the proposal and contract and shall be known as the Contract Time.

If the contract time requires extension for reasons beyond the Contractor's control, it shall be adjusted as follows:

**80-07.1 [ CONTRACT TIME BASED ON WORKING DAYS.** Contract time based on working days shall be calculated weekly by the Resident Project Representative (RPR). The RPR will furnish the Contractor a copy of their weekly statement of the number of working days charged against the contract time during the week and the number of working days currently specified for completion of the contract (the original contract time plus the number of working days, if any, that have been included in approved Change Orders or Supplemental Agreements covering Extra Work).

The weekly statement of contract time charged is based on the following considerations:

(1) Time will be charged for days on which the Contractor could proceed with scheduled work under construction at the time for at least six (6) hours with the normal work force employed on such items. When normal work force is a double-shift, use 12 hours; and when the normal work force is on a triple-shift, use 18 hours. Conditions beyond the Contractor's control such as strikes, lockouts, unusual delays in transportation, temporary suspension of the scheduled work items under construction or temporary suspension of the entire work which have been ordered by the Owner for reasons not the fault of the Contractor, shall not be charged against the contract time.

(2) The RPR will not make charges against the contract time prior to the effective date of the notice to proceed.

(3) The RPR will begin charges against the contract time on the first working day after the effective date of the notice to proceed.

(4) The RPR will not make charges against the contract time after the date of final acceptance as defined in Section 50, paragraph 50-14, *Final Acceptance*.

(5) The Contractor will be allowed one (1) week in which to file a written protest setting forth their own objections to the RPR's weekly statement. If no objection is filed within such specified time, the weekly statement shall be considered as acceptable to the Contractor.

The contract time (stated in the proposal) is based on the originally estimated quantities as described in the Section 20, paragraph 20-05, *Interpretation of Estimated Proposal Quantities*. Should the satisfactory completion of the contract require performance of work in greater quantities than those estimated in the proposal, the contract time shall be increased in the same proportion as the cost of the actually completed quantities bears to the cost of the originally estimated quantities in the proposal. Such increase in contract time shall not consider either the cost of work or the extension of contract time that has been covered by change order or supplemental agreement and shall be made at the time of final payment.

**CONTRACT TIME BASED ON CALENDAR DAYS.** Contract Time based on calendar days shall consist of the number of calendar days stated in the contract counting from the effective date of the Notice to Proceed and including all Saturdays, Sundays, holidays, and non-work days. All calendar days elapsing between the effective dates of the Owner's orders to suspend and resume all work, due to causes not the fault of the Contractor, shall be excluded.

At the time of final payment, the contract time shall be increased in the same proportion as the cost of the actually completed quantities bears to the cost of the originally estimated quantities in the proposal. Such increase in the contract time shall not consider either cost of work or the extension of contract time that has been covered by a change order or supplemental agreement. Charges against the contract time will cease as of the date of final acceptance.

**80-08 FAILURE TO COMPLETE ON TIME.** For each calendar day or working day, as specified in the contract, that any work remains uncompleted after the contract time (including all extensions and adjustments as provided in paragraph 80-07, *Determination and Extension of Contract Time*) the sum specified in the contract and proposal as liquidated damages (LD) will be deducted from any money due or to become due the Contractor or their own surety. Such deducted sums shall not be deducted as a penalty but shall be considered as liquidation of a reasonable portion of damages including but not limited to additional engineering services that will be incurred by the Owner should the Contractor fail to complete the work in the time provided in their contract.

| Liquidated Damages Cost | Allowed Construction Time |
|-------------------------|---------------------------|
| \$500/Day               | 300 Days                  |
|                         |                           |

The maximum construction time allowed for Schedules 300 days. Permitting the Contractor to continue and finish the work or any part of it after the time fixed for its completion, or after the date

to which the time for completion may have been extended, will in no way operate as a waiver on the part of the Owner of any of its rights under the contract.

**80-09 DEFAULT AND TERMINATION OF CONTRACT.** The Contractor shall be considered in default of their contract and such default will be considered as cause for the Owner to terminate the contract for any of the following reasons, if the Contractor:

- a. Fails to begin the work under the contract within the time specified in the Notice to Proceed, or
- b. Fails to perform the work or fails to provide sufficient workers, equipment and/or materials to assure completion of work in accordance with the terms of the contract, or
- c. Performs the work unsuitably or neglects or refuses to remove materials or to perform anew such work as may be rejected as unacceptable and unsuitable, or
- d. Discontinues the execution of the work, or
- e. Fails to resume work which has been discontinued within a reasonable time after notice to do so, or
- f. Becomes insolvent or is declared bankrupt, or commits any act of bankruptcy or insolvency, or
- g. Allows any final judgment to stand against the Contractor unsatisfied for a period of 10 days, or
- h. Makes an assignment for the benefit of creditors, or
- i. For any other cause whatsoever, fails to carry on the work in an acceptable manner.

Should the Owner consider the Contractor in default of the contract for any reason above, the Owner shall immediately give written notice to the Contractor and the Contractor's surety as to the reasons for considering the Contractor in default and the Owner's intentions to terminate the contract.

If the Contractor or surety, within a period of 10 days after such notice, does not proceed in accordance therewith, then the Owner will, upon written notification from the RPR of the facts of such delay, neglect, or default and the Contractor's failure to comply with such notice, have full power and authority without violating the contract, to take the execution of the work out of the hands of the Contractor. The Owner may appropriate or use any or all materials and equipment that have been mobilized for use in the work and are acceptable and may enter into an agreement for the completion of said contract according to the terms and provisions thereof, or use such other methods as in the opinion of the RPR will be required for the completion of said contract in an acceptable manner.

All costs and charges incurred by the Owner, together with the cost of completing the work under contract, will be deducted from any monies due or which may become due the Contractor. If such expense exceeds the sum which would have been payable under the contract, then the Contractor and the surety shall be liable and shall pay to the Owner the amount of such excess.

**80-10 TERMINATION FOR NATIONAL EMERGENCIES.** The Owner shall terminate the contract or portion thereof by written notice when the Contractor is prevented from proceeding with the construction contract as a direct result of an Executive Order of the President with respect to the execution of war or in the interest of national defense.

When the contract, or any portion thereof, is terminated before completion of all items of work in the contract, payment will be made for the actual number of units or items of work completed at the contract price or as mutually agreed for items of work partially completed or not started. No claims or loss of anticipated profits shall be considered.

Reimbursement for organization of the work, and other overhead expenses, (when not otherwise included in the contract) and moving equipment and materials to and from the job will be considered, the intent being that an equitable settlement will be made with the Contractor.

Acceptable materials, obtained or ordered by the Contractor for the work and that are not incorporated in the work shall, at the option of the Contractor, be purchased from the Contractor at actual cost as shown by receipted bills and actual cost records at such points of delivery as may be designated by the RPR.

Termination of the contract or a portion thereof shall neither relieve the Contractor of their responsibilities for the completed work nor shall it relieve their surety of its obligation for and concerning any just claim arising out of the work performed.

**80-11 WORK AREA, STORAGE AREA AND SEQUENCE OF OPERATIONS.** The Contractor shall obtain approval from the RPR prior to beginning any work in all areas of the airport. No operating runway, taxiway, or air operations area (AOA) shall be crossed, entered, or obstructed while it is operational. The Contractor shall plan and coordinate work in accordance with the approved CSPP and SPCD.

**END OF SECTION 80**

## SECTION 90 MEASUREMENT AND PAYMENT

**90-01 MEASUREMENT OF QUANTITIES.** All work completed under the contract will be measured by the RPR, or their authorized representatives, using United States Customary Units of Measurement.

The method of measurement and computations to be used in determination of quantities of material furnished and of work performed under the contract will be those methods generally recognized as conforming to good engineering practice.

Unless otherwise specified, longitudinal measurements for area computations will be made horizontally, and no deductions will be made for individual fixtures (or leave-outs) having an area of 9 square feet (0.8 square meters) or less. Unless otherwise specified, transverse measurements for area computations will be the neat dimensions shown on the plans or ordered in writing by the RPR.

Unless otherwise specified, all contract items which are measured by the linear foot such as electrical ducts, conduits, pipe culverts, underdrains, and similar items shall be measured parallel to the base or foundation upon which such items are placed.

The term "lump sum" when used as an item of payment will mean complete payment for the work described in the contract. When a complete structure or structural unit (in effect, "lump sum" work) is specified as the unit of measurement, the unit will be construed to include all necessary fittings and accessories.

When requested by the Contractor and approved by the RPR in writing, material specified to be measured by the cubic yard (cubic meter) may be weighed, and such weights will be converted to cubic yards (cubic meters) for payment purposes. Factors for conversion from weight measurement to volume measurement will be determined by the RPR and shall be agreed to by the Contractor before such method of measurement of pay quantities is used.

### Measurement and Payment Terms

| Term                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Excavation and Embankment Volume</b>     | In computing volumes of excavation, the average end area method will be used unless otherwise specified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Measurement and Proportion by Weight</b> | The term "ton" will mean the short ton consisting of 2,000 pounds (907 kg) avoirdupois. All materials that are measured or proportioned by weights shall be weighed on accurate, independently certified scales by competent, qualified personnel at locations designated by the RPR. If material is shipped by rail, the car weight may be accepted provided that only the actual weight of material is paid for. However, car weights will not be acceptable for material to be passed through mixing plants. Trucks used to haul material being paid for by weight shall be weighed empty daily at such times as the RPR directs, and each truck shall bear a plainly legible identification mark. |

| Term                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Measurement by Volume</b> | Materials to be measured by volume in the hauling vehicle shall be hauled in approved vehicles and measured therein at the point of delivery. Vehicles for this purpose may be of any size or type acceptable for the materials hauled, provided that the body is of such shape that the actual contents may be readily and accurately determined. All vehicles shall be loaded to at least their water level capacity, and all loads shall be leveled when the vehicles arrive at the point of delivery.                                                                                                                                                                                       |
| <b>Asphalt Material</b>      | Asphalt materials will be measured by the gallon (liter) or ton (kg). When measured by volume, such volumes will be measured at 60°F (16°C) or will be corrected to the volume at 60°F (16°C) using ASTM D1250 for asphalts. Net certified scale weights or weights based on certified volumes in the case of rail shipments will be used as a basis of measurement, subject to correction when asphalt material has been lost from the car or the distributor, wasted, or otherwise not incorporated in the work. When asphalt materials are shipped by truck or transport, net certified weights by volume, subject to correction for loss or foaming, will be used for computing quantities. |
| <b>Cement</b>                | Cement will be measured by the ton (kg) or hundredweight (km).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Structure</b>             | Structures will be measured according to neat lines shown on the plans or as altered to fit field conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Timber</b>                | Timber will be measured by the thousand feet board measure (MFBM) actually incorporated in the structure. Measurement will be based on nominal widths and thicknesses and the extreme length of each piece.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Plates and Sheets</b>     | The thickness of plates and galvanized sheet used in the manufacture of corrugated metal pipe, metal plate pipe culverts and arches, and metal cribbing will be specified and measured in decimal fraction of inch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Miscellaneous Items</b>   | When standard manufactured items are specified such as fence, wire, plates, rolled shapes, pipe conduit, etc., and these items are identified by gauge, unit weight, section dimensions, etc., such identification will be considered to be nominal weights or dimensions. Unless more stringently controlled by tolerances in cited specifications, manufacturing tolerances established by the industries involved will be accepted.                                                                                                                                                                                                                                                          |
| <b>Scales</b>                | <p>Scales must be tested for accuracy and serviced before use. Scales for weighing materials which are required to be proportioned or measured and paid for by weight shall be furnished, erected, and maintained by the Contractor, or be certified permanently installed commercial scales. Platform scales shall be installed and maintained with the platform level and rigid bulkheads at each end.</p> <p>Scales shall be accurate within 0.5% of the correct weight throughout the range of use. The Contractor shall have the scales checked under the observation of the RPR before beginning work and at such other times as</p>                                                      |

| Term                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <p>requested. The intervals shall be uniform in spacing throughout the graduated or marked length of the beam or dial and shall not exceed 0.1% of the nominal rated capacity of the scale, but not less than one pound (454 grams). The use of spring balances will not be permitted.</p> <p>In the event inspection reveals the scales have been “overweighing” (indicating more than correct weight) they will be immediately adjusted. All materials received subsequent to the last previous correct weighting-accuracy test will be reduced by the percentage of error in excess of 0.5%.</p> <p>In the event inspection reveals the scales have been under-weighing (indicating less than correct weight), they shall be immediately adjusted. No additional payment to the Contractor will be allowed for materials previously weighed and recorded.</p> <p>Beams, dials, platforms, and other scale equipment shall be so arranged that the operator and the RPR can safely and conveniently view them.</p> <p>Scale installations shall have available ten standard 50-pound (2.3 km) weights for testing the weighing equipment or suitable weights and devices for other approved equipment.</p> <p>All costs in connection with furnishing, installing, certifying, testing, and maintaining scales; for furnishing check weights and scale house; and for all other items specified in this subsection, for the weighing of materials for proportioning or payment, shall be included in the unit contract prices for the various items of the project.</p> |
| <b>Rental Equipment</b> | <p>Rental of equipment will be measured by time in hours of actual working time and necessary traveling time of the equipment within the limits of the work. Special equipment ordered in connection with extra work will be measured as agreed in the change order or supplemental agreement authorizing such work as provided in paragraph 90-05 <i>Payment for Extra Work</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Pay Quantities</b>   | <p>When the estimated quantities for a specific portion of the work are designated as the pay quantities in the contract, they shall be the final quantities for which payment for such specific portion of the work will be made, unless the dimensions of said portions of the work shown on the plans are revised by the RPR. If revised dimensions result in an increase or decrease in the quantities of such work, the final quantities for payment will be revised in the amount represented by the authorized changes in the dimensions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

**90-02 SCOPE OF PAYMENT.** The Contractor shall receive and accept compensation provided for in the contract as full payment for furnishing all materials, for performing all work under the contract in a complete and acceptable manner, and for all risk, loss, damage, or expense of whatever character arising out of the nature of the work or the execution thereof, subject to the provisions of Section 70, paragraph 70-18, *No Waiver of Legal Rights*.

When the "basis of payment" subsection of a technical specification requires that the contract price (price bid) include compensation for certain work or material essential to the item, this same work or material will not also be measured for payment under any other contract item which may appear elsewhere in the contract, plans, or specifications.

**90-03 COMPENSATION FOR ALTERED QUANTITIES.** When the accepted quantities of work vary from the quantities in the proposal, the Contractor shall accept as payment in full, so far as contract items are concerned, payment at the original contract price for the accepted quantities of work actually completed and accepted. No allowance, except as provided for in Section 40, paragraph 40-02, *Alteration of Work and Quantities*, will be made for any increased expense, loss of expected reimbursement, or loss of anticipated profits suffered or claimed by the Contractor which results directly from such alterations or indirectly from their own unbalanced allocation of overhead and profit among the contract items, or from any other cause.

**90-04 PAYMENT FOR OMITTED ITEMS.** As specified in Section 40, paragraph 40-03, *Omitted Items*, the RPR shall have the right to omit from the work (order nonperformance) any contract item, except major contract items, in the best interest of the Owner.

Should the RPR omit or order nonperformance of a contract item or portion of such item from the work, the Contractor shall accept payment in full at the contract prices for any work actually completed and acceptable prior to the RPR's order to omit or non-perform such contract item.

Acceptable materials ordered by the Contractor or delivered on the work prior to the date of the RPR's order will be paid for at the actual cost to the Contractor and shall thereupon become the property of the Owner.

In addition to the reimbursement hereinbefore provided, the Contractor shall be reimbursed for all actual costs incurred for the purpose of performing the omitted contract item prior to the date of the RPR's order. Such additional costs incurred by the Contractor must be directly related to the deleted contract item and shall be supported by certified statements by the Contractor as to the nature the amount of such costs.

**90-05 PAYMENT FOR EXTRA WORK.** Extra work, performed in accordance with Section 40, paragraph 40-04, *Extra Work*, will be paid for at the contract prices or agreed prices specified in the change order or supplemental agreement authorizing the extra work.

**90-06 PARTIAL PAYMENTS.** Partial payments will be made to the Contractor at least once each month as the work progresses. Said payments will be based upon estimates, prepared by the RPR, of the value of the work performed and materials complete and in place, in accordance with the contract, plans, and specifications. Such partial payments may also include the delivered actual cost of those materials stockpiled and stored in accordance with paragraph 90-07, *Payment for Materials on Hand*. No partial payment will be made when the amount due to the Contractor since the last estimate amounts to less than five hundred dollars.

It is understood and agreed that the Contractor shall not be entitled to demand or receive partial payment based on quantities of work in excess of those provided in the proposal or covered by approved change orders or supplemental agreements, except when such excess quantities have been determined by the RPR to be a part of the final quantity for the item of work in question.



No partial payment shall bind the Owner to the acceptance of any materials or work in place as to quality or quantity. All partial payments are subject to correction at the time of final payment as provided in paragraph 90-09, *Acceptance and Final Payment*.

The Contractor shall deliver to the Owner a complete release of all claims for labor and material arising out of this contract before the final payment is made. If any subcontractor or supplier fails to furnish such a release in full, the Contractor may furnish a bond or other collateral satisfactory to the Owner to indemnify the Owner against any potential lien or other such claim. The bond or collateral shall include all costs, expenses, and attorney fees the Owner may be compelled to pay in discharging any such lien or claim.

**90-07 PAYMENT FOR MATERIALS ON HAND.** Partial payments may be made to the extent of the delivered cost of materials to be incorporated in the work, provided that such materials meet the requirements of the contract, plans, and specifications and are delivered to acceptable sites on the airport property or at other sites in the vicinity that are acceptable to the Owner. Such delivered costs of stored or stockpiled materials may be included in the next partial payment after the following conditions are met:

a. The material has been stored or stockpiled in a manner acceptable to the RPR at or on an approved site.

b. The Contractor has furnished the RPR with acceptable evidence of the quantity and quality of such stored or stockpiled materials.

c. The Contractor has furnished the RPR with satisfactory evidence that the material and transportation costs have been paid.

d. The Contractor has furnished the Owner legal title (free of liens or encumbrances of any kind) to the material stored or stockpiled.

e. The Contractor has furnished the Owner evidence that the material stored or stockpiled is insured against loss by damage to or disappearance of such materials at any time prior to use in the work.

It is understood and agreed that the transfer of title and the Owner's payment for such stored or stockpiled materials shall in no way relieve the Contractor of their responsibility for furnishing and placing such materials in accordance with the requirements of the contract, plans, and specifications.

In no case will the amount of partial payments for materials on hand exceed the contract price for such materials or the contract price for the contract item in which the material is intended to be used.

No partial payment will be made for stored or stockpiled living or perishable plant materials.

The Contractor shall bear all costs associated with the partial payment of stored or stockpiled materials in accordance with the provisions of this paragraph.

**90-08 PAYMENT OF WITHHELD FUNDS.** At the Contractor's option, if an Owner withholds retainage in accordance with the methods described in paragraph 90-06 *Partial Payments*, the

Contractor may request that the Owner deposit the retainage into an escrow account. The Owner's deposit of retainage into an escrow account is subject to the following conditions:

- a. The Contractor shall bear all expenses of establishing and maintaining an escrow account and escrow agreement acceptable to the Owner.
- b. The Contractor shall deposit to and maintain in such escrow only those securities or bank certificates of deposit as are acceptable to the Owner and having a value not less than the retainage that would otherwise be withheld from partial payment.
- c. The Contractor shall enter into an escrow agreement satisfactory to the Owner.
- d. The Contractor shall obtain the written consent of the surety to such agreement.

**90-09 ACCEPTANCE AND FINAL PAYMENT.** When the contract work has been accepted in accordance with the requirements of Section 50, paragraph 50-15, *Final Acceptance*, the RPR will prepare the final estimate of the items of work actually performed. The Contractor shall approve the RPR's final estimate or advise the RPR of the Contractor's objections to the final estimate which are based on disputes in measurements or computations of the final quantities to be paid under the contract as amended by change order or supplemental agreement. The Contractor and the RPR shall resolve all disputes (if any) in the measurement and computation of final quantities to be paid within 30 calendar days of the Contractor's receipt of the RPR's final estimate. If, after such 30-day period, a dispute still exists, the Contractor may approve the RPR's estimate under protest of the quantities in dispute, and such disputed quantities shall be considered by the Owner as a claim in accordance with Section 50, paragraph 50-16, *Claims for Adjustment and Disputes*.

After the Contractor has approved, or approved under protest, the RPR's final estimate, and after the RPR's receipt of the project closeout documentation required in paragraph 90-11, *Contractor Final Project Documentation*, final payment will be processed based on the entire sum, or the undisputed sum in case of approval under protest, determined to be due the Contractor less all previous payments and all amounts to be deducted under the provisions of the contract. All prior partial estimates and payments shall be subject to correction in the final estimate and payment.

If the Contractor has filed a claim for additional compensation under the provisions of Section 50, paragraph 50-16, *Claims for Adjustments and Disputes*, or under the provisions of this paragraph, such claims will be considered by the Owner in accordance with local laws or ordinances. Upon final adjudication of such claims, any additional payment determined to be due the Contractor will be paid pursuant to a supplemental final estimate.

#### **90-10 CONSTRUCTION WARRANTY.**

- a. In addition to any other warranties in this contract, the Contractor warrants that work performed under this contract conforms to the contract requirements and is free of any defect in equipment, material, workmanship, or design furnished, or performed by the Contractor or any subcontractor or supplier at any tier.
- b. This warranty shall continue for a period of one year from the date of final acceptance of the work, except as noted. If the Owner takes possession of any part of the work before final acceptance, this warranty shall continue for a period of one year from the date the Owner takes possession. However, this will not relieve the Contractor from corrective items required by the

final acceptance of the project work. Light Emitting Diode emitting diode (LED) light fixtures with the exception of obstruction lighting, must be warranted by the manufacturer for a minimum of four (4) years after date of installation inclusive of all electronics.

c. The Contractor shall remedy at the Contractor's expense any failure to conform, or any defect. In addition, the Contractor shall remedy at the Contractor's expense any damage to Owner real or personal property, when that damage is the result of the Contractor's failure to conform to contract requirements; or any defect of equipment, material, workmanship, or design furnished by the Contractor.

d. The Contractor shall restore any work damaged in fulfilling the terms and conditions of this clause. The Contractor's warranty with respect to work repaired or replaced will run for one year from the date of repair or replacement.

e. The Owner will notify the Contractor, in writing, within **seven (7)** days after the discovery of any failure, defect, or damage.

f. If the Contractor fails to remedy any failure, defect, or damage within **14** days after receipt of notice, the Owner shall have the right to replace, repair, or otherwise remedy the failure, defect, or damage at the Contractor's expense.

g. With respect to all warranties, express or implied, from subcontractors, manufacturers, or suppliers for work performed and materials furnished under this contract, the Contractor shall: (1) Obtain all warranties that would be given in normal commercial practice; (2) Require all warranties to be executed, in writing, for the benefit of the Owner, as directed by the Owner, and (3) Enforce all warranties for the benefit of the Owner.

h. This warranty shall not limit the Owner's rights with respect to latent defects, gross mistakes, or fraud.

**90-11 CONTRACTOR FINAL PROJECT DOCUMENTATION.** Approval of final payment to the Contractor is contingent upon completion and submittal of the items listed below. The final payment will not be approved until the RPR approves the Contractor's final submittal. The Contractor shall:

a. Provide two (2) copies of all manufacturers warranties specified for materials, equipment, and installations.

b. Provide weekly payroll records (not previously received) from the general Contractor and all subcontractors.

c. Complete final cleanup in accordance with Section 40, paragraph 40-08, *Final Cleanup*.

d. Complete all punch list items identified during the Final Inspection.

e. Provide complete release of all claims for labor and material arising out of the Contract.

f. Provide a certified statement signed by the subcontractors, indicating actual amounts paid to the Disadvantaged Business Enterprise (DBE) subcontractors and/or suppliers associated with the project.

- g.** When applicable per state requirements, return copies of sales tax completion forms.
- h.** Manufacturer's certifications for all items incorporated in the work.
- i.** All required record drawings, as-built drawings or as-constructed drawings.
- j.** Project Operation and Maintenance (O&M) Manual(s).
- k.** Security for Construction Warranty.
- l.** Equipment commissioning documentation submitted, if required.

**END OF SECTION 90**

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## **SECTION C-100 CONTRACTOR QUALITY CONTROL PROGRAM (CQCP)**

**100-1.1 GENERAL.** Quality is more than test results. Quality is the combination of proper materials, testing, workmanship, equipment, inspection, and documentation of the project. Establishing and maintaining a culture of quality is key to achieving a quality project. The Contractor shall establish, provide, and maintain an effective Contractor Quality Control Program (CQCP) that details the methods and procedures that will be taken to assure that all materials and completed construction required by this contract conform to contract plans, technical specifications and other requirements, whether manufactured by the Contractor, or procured from subcontractors or vendors. Although guidelines are established and certain minimum requirements are specified here and elsewhere in the contract technical specifications, the Contractor shall assume full responsibility for accomplishing the stated purpose.

The Contractor shall establish a CQCP that will:

- a. Provide qualified personnel to develop and implement the CQCP.
- b. Provide for the production of acceptable quality materials.
- c. Provide sufficient information to assure that the specification requirements can be met.
- d. Document the CQCP process.

The Contractor shall not begin any construction or production of materials to be incorporated into the completed work until the CQCP has been reviewed and approved by the Resident Project Representative (RPR). No partial payment will be made for materials subject to specific quality control (QC) requirements until the CQCP has been reviewed and approved.

The QC requirements contained in this section and elsewhere in the contract technical specifications are in addition to and separate from the quality assurance (QA) testing requirements. QA testing requirements are the responsibility of the RPR or Contractor as specified in the specifications.

A Quality Control (QC)/Quality Assurance (QA) workshop with the Engineer, Resident Project Representative (RPR), Contractor, subcontractors, testing laboratories, and Owner's representative must be held prior to start of construction. The QC/QA workshop will be facilitated by the Contractor. The Contractor shall coordinate with the Airport and the RPR on time and location of the QC/QA workshop. Items to be addressed, at a minimum, will include:

- a. Review of the CQCP including submittals, QC Testing, Action & Suspension Limits for Production, Corrective Action Plans, Distribution of QC reports, and Control Charts.
- b. Discussion of the QA program.
- c. Discussion of the QC and QA Organization and authority including coordination and information exchange between QC and QA.
- d. Establish regular meetings to discuss control of materials, methods and testing.

- e. Establishment of the overall QC culture.

#### **100-1.2 DESCRIPTION OF PROGRAM.**

**a. General description.** The Contractor shall establish a CQCP to perform QC inspection and testing of all items of work required by the technical specifications, including those performed by subcontractors. The CQCP shall ensure conformance to applicable specifications and plans with respect to materials, off-site fabrication, workmanship, construction, finish, and functional performance. The CQCP shall be effective for control of all construction work performed under this Contract and shall specifically include surveillance and tests required by the technical specifications, in addition to other requirements of this section and any other activities deemed necessary by the Contractor to establish an effective level of QC.

**b. Contractor Quality Control Program (CQCP).** The Contractor shall describe the CQCP in a written document that shall be reviewed and approved by the RPR prior to the start of any production, construction, or off-site fabrication. The written CQCP shall be submitted to the RPR for review and approval at least **10** calendar days before the CQCP Workshop. The Contractor's CQCP and QC testing laboratory must be approved in writing by the RPR prior to the Notice to Proceed (NTP).

The CQCP shall be organized to address, as a minimum, the following:

1. QC organization and resumes of key staff
2. Project progress schedule
3. Submittals schedule
4. Inspection requirements
5. QC testing plan
6. Documentation of QC activities and distribution of QC reports
7. Requirements for corrective action when QC and/or QA acceptance criteria are not met
8. Material quality and construction means and methods. Address all elements applicable to the project that affect the quality of the pavement structure including subgrade, subbase, base, and surface course. Some elements that must be addressed include, but is not limited to mix design, aggregate grading, stockpile management, mixing and transporting, placing and finishing, quality control testing and inspection, smoothness, laydown plan, equipment, and temperature management plan.

The Contractor must add any additional elements to the CQCP that is necessary to adequately control all production and/or construction processes required by this contract.

**100-1.3 CQCP ORGANIZATION.** The CQCP shall be implemented by the establishment of a QC organization. An organizational chart shall be developed to show all QC personnel, their authority,

and how these personnel integrate with other management/production and construction functions and personnel.

The organizational chart shall identify all QC staff by name and function, and shall indicate the total staff required to implement all elements of the CQCP, including inspection and testing for each item of work. If necessary, different technicians can be used for specific inspection and testing functions for different items of work. If an outside organization or independent testing laboratory is used for implementation of all or part of the CQCP, the personnel assigned shall be subject to the qualification requirements of paragraphs 100-03a and 100-03b. The organizational chart shall indicate which personnel are Contractor employees and which are provided by an outside organization.

The QC organization shall, as a minimum, consist of the following personnel:

**a. Program Administrator.** The Contractor Quality Control Program Administrator (CQCPA) must be a full-time on-site employee of the Contractor, or a consultant engaged by the Contractor. The CQCPA must have a minimum of five (5) years of experience in QC pavement construction with prior QC experience on a project of comparable size and scope as the contract.

Included in the five (5) years of paving/QC experience, the CQCPA must meet at least one of the following requirements:

(1) Professional Engineer with one (1) year of airport paving experience.

(2) Engineer-in-training with two (2) years of airport paving experience.

(3) National Institute for Certification in Engineering Technologies (NICET) Civil Engineering Technology Level IV with three (3) years of airport paving experience.

(4) An individual with four (4) years of airport paving experience, with a Bachelor of Science Degree in Civil Engineering, Civil Engineering Technology or Construction.

The CQCPA must have full authority to institute any and all actions necessary for the successful implementation of the CQCP to ensure compliance with the contract plans and technical specifications. The CQCPA authority must include the ability to immediately stop production until materials and/or processes are in compliance with contract specifications. The CQCPA must report directly to a principal officer of the construction firm. The CQCPA may supervise the Quality Control Program on more than one project provided that person can be at the job site within two (2) hours after being notified of a problem.

**b. QC technicians.** A sufficient number of QC technicians necessary to adequately implement the CQCP must be provided. These personnel must be either Engineers, engineering technicians, or experienced craftsman with qualifications in the appropriate field equivalent to NICET Level II in Civil Engineering Technology or higher, and shall have a minimum of two (2) years of experience in their area of expertise.

The QC technicians must report directly to the CQCPA and shall perform the following functions:



(1) Inspection of all materials, construction, plant, and equipment for conformance to the technical specifications, and as required by paragraph 100-6.

(2) Performance of all QC tests as required by the technical specifications and paragraph 100-8.

(3) Performance of tests for the RPR when required by the technical specifications.

Certification at an equivalent level of qualification and experience by a state or nationally recognized organization will be acceptable in lieu of NICET certification.

**c. Staffing levels.** The Contractor shall provide sufficient qualified QC personnel to monitor each work activity at all times. Where material is being produced in a plant for incorporation into the work, separate plant and field technicians shall be provided at each plant and field placement location. The scheduling and coordinating of all inspection and testing must match the type and pace of work activity. The CQCP shall state where different technicians will be required for different work elements.

**100-1.4 PROJECT PROGRESS SCHEDULE.** Critical QC activities must be shown on the project schedule as required by Section 80, paragraph 80-03, *Execution and Progress*.

**100-1.5 SUBMITTALS SCHEDULE.** The Contractor shall submit a detailed listing of all submittals (for example, mix designs, material certifications) and shop drawings required by the technical specifications. The listing can be developed in a spreadsheet format and shall include as a minimum:

- a. Specification item number
- b. Item description
- c. Description of submittal
- d. Specification paragraph requiring submittal
- e. Scheduled date of submittal

**100-1.6 INSPECTION REQUIREMENTS.** QC inspection functions shall be organized to provide inspections for all definable features of work, as detailed below. All inspections shall be documented by the Contractor as specified by paragraph 100-9.

Inspections shall be performed as needed to ensure continuing compliance with contract requirements until completion of the particular feature of work. Inspections shall include the following minimum requirements:

a. During plant operation for material production, QC test results and periodic inspections shall be used to ensure the quality of aggregates and other mix components, and to adjust and control mix proportioning to meet the approved mix design and other requirements of the technical specifications. All equipment used in proportioning and mixing shall be inspected to ensure its proper operating condition. The CQCP shall detail how these and other QC functions will be accomplished and used.

b. During field operations, QC test results and periodic inspections shall be used to ensure the quality of all materials and workmanship. All equipment used in placing, finishing, and compacting shall be inspected to ensure its proper operating condition and to ensure that all such operations are in conformance to the technical specifications and are within the plan dimensions, lines, grades, and tolerances specified. The CQCP shall document how these and other QC functions will be accomplished and used.

#### **100-1.7 CONTRACTOR QC TESTING FACILITY.**

a. For projects that include Item P-401, Item P-403, and Item P-404, the Contractor shall ensure facilities, including all necessary equipment, materials, and current reference standards, are provided that meet requirements in the following paragraphs of ASTM D3666, *Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials*:

- 8.1.3 Equipment Calibration and Checks;
- 8.1.9 Equipment Calibration, Standardization, and Check Records;
- 8.1.12 Test Methods and Procedures

b. For projects that include P-501, the Contractor shall ensure facilities, including all necessary equipment, materials, and current reference standards, are provided that meet requirements in the following paragraphs of ASTM C1077, *Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation*:

- 7 Test Methods and Procedures
- 8 Facilities, Equipment, and Supplemental Procedures

**100-1.8 QC TESTING PLAN.** As a part of the overall CQCP, the Contractor shall implement a QC testing plan, as required by the technical specifications. The testing plan shall include the minimum tests and test frequencies required by each technical specification Item, as well as any additional QC tests that the Contractor deems necessary to adequately control production and/or construction processes.

The QC testing plan can be developed in a spreadsheet fashion and shall, as a minimum, include the following:

- a. Specification item number (e.g., P-401)
- b. Item description (e.g., Hot Mix Asphalt Pavements)
- c. Test type (e.g., gradation, grade, asphalt content)
- d. Test standard (e.g., ASTM or American Association of State Highway and Transportation Officials (AASHTO) test number, as applicable)
- e. Test frequency (e.g., as required by technical specifications or minimum frequency when requirements are not stated)

f. Responsibility (e.g., plant technician)

g. Control requirements (e.g., target, permissible deviations)

The QC testing plan shall contain a statistically-based procedure of random sampling for acquiring test samples in accordance with ASTM D3665. The RPR shall be provided the opportunity to witness QC sampling and testing.

All QC test results shall be documented by the Contractor as required by paragraph 100-9.

**100-1.9 DOCUMENTATION.** The Contractor shall maintain current QC records of all inspections and tests performed. These records shall include factual evidence that the required QC inspections or tests have been performed, including type and number of inspections or tests involved; results of inspections or tests; nature of defects, deviations, causes for rejection, etc.; proposed remedial action; and corrective actions taken.

These records must cover both conforming and defective or deficient features, and must include a statement that all supplies and materials incorporated in the work are in full compliance with the terms of the contract. Legible copies of these records shall be furnished to the RPR daily. The records shall cover all work placed subsequent to the previously furnished records and shall be verified and signed by the CQCPA.

Contractor QC records required for the contract shall include, but are not necessarily limited to, the following records:

**a. Daily inspection reports.** Each Contractor QC technician shall maintain a daily log of all inspections performed for both Contractor and subcontractor operations. These technician's daily reports shall provide factual evidence that continuous QC inspections have been performed and shall, as a minimum, include the following:

- (1) Technical specification item number and description
- (2) Compliance with approved submittals
- (3) Proper storage of materials and equipment
- (4) Proper operation of all equipment
- (5) Adherence to plans and technical specifications
- (6) Summary of any necessary corrective actions
- (7) Safety inspection.
- (8) Photographs and/or video

The daily inspection reports shall identify all QC inspections and QC tests conducted, results of inspections, location and nature of defects found, causes for rejection, and remedial or corrective actions taken or proposed.

The daily inspection reports shall be signed by the responsible QC technician and the CQCPA. The RPR shall be provided at least one copy of each daily inspection report on the work day following the day of record. When QC inspection and test results are recorded and transmitted electronically, the results must be archived.

**b. Daily test reports.** The Contractor shall be responsible for establishing a system that will record all QC test results. Daily test reports shall document the following information:

- (1) Technical specification item number and description
- (2) Test designation
- (3) Location
- (4) Date of test
- (5) Control requirements
- (6) Test results
- (7) Causes for rejection
- (8) Recommended remedial actions
- (9) Retests

Test results from each day's work period shall be submitted to the RPR prior to the start of the next day's work period. When required by the technical specifications, the Contractor shall maintain statistical QC charts. When QC daily test results are recorded and transmitted electronically, the results must be archived.

**100-1.10 CORRECTIVE ACTION REQUIREMENTS.** The CQCP shall indicate the appropriate action to be taken when a process is deemed, or believed, to be out of control (out of tolerance) and detail what action will be taken to bring the process into control. The requirements for corrective action shall include both general requirements for operation of the CQCP as a whole, and for individual items of work contained in the technical specifications.

The CQCP shall detail how the results of QC inspections and tests will be used for determining the need for corrective action and shall contain clear rules to gauge when a process is out of control and the type of correction to be taken to regain process control.

When applicable or required by the technical specifications, the Contractor shall establish and use statistical QC charts for individual QC tests. The requirements for corrective action shall be linked to the control charts.

**100-1.11 INSPECTION AND/OR OBSERVATIONS BY THE RPR.** All items of material and equipment are subject to inspection and/or observation by the RPR at the point of production, manufacture or shipment to determine if the Contractor, producer, manufacturer or shipper maintains an adequate QC system in conformance with the requirements detailed here and the applicable technical specifications and plans. In addition, all items of materials, equipment and

work in place shall be subject to inspection and/or observation by the RPR at the site for the same purpose.

Inspection and/or observations by the RPR does not relieve the Contractor of performing QC inspections of either on-site or off-site Contractor's or subcontractor's work.

#### **100-1.12 NONCOMPLIANCE.**

a. The Resident Project Representative (RPR) will provide written notice to the Contractor of any noncompliance with their CQCP. After receipt of such notice, the Contractor must take corrective action.

b. When QC activities do not comply with either the CQCP or the contract provisions or when the Contractor fails to properly operate and maintain an effective CQCP, and no effective corrective actions have been taken after notification of non-compliance, the RPR will recommend the Owner take the following actions:

(1) Order the Contractor to replace ineffective or unqualified QC personnel or subcontractors and/or

(2) Order the Contractor to stop operations until appropriate corrective actions are taken.

#### **METHOD OF MEASUREMENT**

**100-1.13 BASIS OF MEASUREMENT AND PAYMENT.** [Not Used] [Contractor Quality Control Program (CQCP) is for the personnel, tests, facilities and documentation required to implement the CQCP. The CQCP will be paid as a lump sum with the following schedule of partial payments:

a. With first pay request, 25% with approval of CQCP and completion of the Quality Control (QC)/Quality Assurance (QA) workshop.

b. When 25% or more of the original contract is earned, an additional 25%.

c. When 50% or more of the original contract is earned, an additional 20%.

d. When 75% or more of the original contract is earned, an additional 20%

e. After final inspection and acceptance of project, the final 10%.

#### **BASIS OF PAYMENT**

##### **100-1.14 PAYMENT WILL BE MADE UNDER:**

Item C-100 Contractor Quality Control Program (CQCP)

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

National Institute for Certification in Engineering Technologies (NICET)

ASTM International (ASTM)

|            |                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C1077 | Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation |
| ASTM D3665 | Standard Practice for Random Sampling of Construction Materials                                                                            |
| ASTM D3666 | Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials                              |

**END OF SECTION C-100**

## **SECTION C-102 TEMPORARY AIR AND WATER POLLUTION, SOIL EROSION, AND SILTATION CONTROL**

### **DESCRIPTION**

**102-1.** This item shall consist of temporary control measures as shown on the plans or as ordered by the Resident Project Representative (RPR) during the life of a contract to control pollution of air and water, soil erosion, and siltation through the use of silt fences, berms, dikes, dams, sediment basins, fiber mats, gravel, mulches, grasses, slope drains, and other erosion control devices or methods.

Temporary erosion control shall be in accordance with the approved erosion control plan; the approved Construction Safety and Phasing Plan (CSPP) and AC 150/5370-2, *Operational Safety on Airports During Construction*. The temporary erosion control measures contained herein shall be coordinated with the permanent erosion control measures specified as part of this contract to the extent practical to assure economical, effective, and continuous erosion control throughout the construction period.

Temporary control may include work outside the construction limits such as borrow pit operations, equipment and material storage sites, waste areas, and temporary plant sites.

Temporary control measures shall be designed, installed and maintained to minimize the creation of wildlife attractants that have the potential to attract hazardous wildlife on or near public-use airports.

### **MATERIALS**

**102-2.1 GRASS.** Grass that will not compete with the grasses sown later for permanent cover per Item T-901 shall be a quick-growing species (such as ryegrass, Italian ryegrass, or cereal grasses) suitable to the area providing a temporary cover. Selected grass species shall not create a wildlife attractant.

**102-2.2 MULCHES.** Mulches may be hay, straw, fiber mats, netting, bark, wood chips, or other suitable material reasonably clean and free of noxious weeds and deleterious materials per Item T-908. Mulches shall not create a wildlife attractant.

**102-2.3 FERTILIZER.** Fertilizer shall be a standard commercial grade and shall conform to all federal and state regulations and to the standards of the Association of Official Agricultural Chemists.

**102-2.4 SLOPE DRAINS.** Slope drains may be constructed of pipe, fiber mats, rubble, concrete, asphalt, or other materials that will adequately control erosion.

**102-2.5 SILT FENCE.** Silt fence shall consist of polymeric filaments which are formed into a stable network such that filaments retain their relative positions. Synthetic filter fabric shall contain ultraviolet ray inhibitors and stabilizers to provide a minimum of six months of expected usable construction life. Silt fence shall meet the requirements of ASTM D6461.

**102-2.6 OTHER.** All other materials shall meet commercial grade standards and shall be approved by the RPR before being incorporated into the project.

## **CONSTRUCTION REQUIREMENTS**

**102-3.1 GENERAL.** In the event of conflict between these requirements and pollution control laws, rules, or regulations of other federal, state, or local agencies, the more restrictive laws, rules, or regulations shall apply.

The RPR shall be responsible for assuring compliance to the extent that construction practices, construction operations, and construction work are involved.

**102-3.2 SCHEDULE.** Prior to the start of construction, the Contractor shall submit schedules in accordance with the approved Construction Safety and Phasing Plan (CSPP) and the plans for accomplishment of temporary and permanent erosion control work for clearing and grubbing; grading; construction; paving; and structures at watercourses. The Contractor shall also submit a proposed method of erosion and dust control on haul roads and borrow pits and a plan for disposal of waste materials. Work shall not be started until the erosion control schedules and methods of operation for the applicable construction have been accepted by the RPR.

**102-3.3 CONSTRUCTION DETAILS.** The Contractor will be required to incorporate all permanent erosion control features into the project at the earliest practicable time as outlined in the plans and approved CSPP. Except where future construction operations will damage slopes, the Contractor shall perform the permanent seeding and mulching and other specified slope protection work in stages, as soon as substantial areas of exposed slopes can be made available. Temporary erosion and pollution control measures will be used to correct conditions that develop during construction that were not foreseen during the design stage; that are needed prior to installation of permanent control features; or that are needed temporarily to control erosion that develops during normal construction practices, but are not associated with permanent control features on the project.

Where erosion may be a problem, schedule and perform clearing and grubbing operations so that grading operations and permanent erosion control features can follow immediately if project conditions permit. Temporary erosion control measures are required if permanent measures cannot immediately follow grading operations. The RPR shall limit the area of clearing and grubbing, excavation, borrow, and embankment operations in progress, commensurate with the Contractor's capability and progress in keeping the finish grading, mulching, seeding, and other such permanent control measures current with the accepted schedule. If seasonal limitations make such coordination unrealistic, temporary erosion control measures shall be taken immediately to the extent feasible and justified as directed by the RPR.

The Contractor shall provide immediate permanent or temporary pollution control measures to minimize contamination of adjacent streams or other watercourses, lakes, ponds, or other areas of water impoundment as directed by the RPR. If temporary erosion and pollution control measures are required due to the Contractor's negligence, carelessness, or failure to install permanent controls as a part of the work as scheduled or directed by the RPR, the work shall be performed by the Contractor and the cost shall be incidental to this item.



The RPR may increase or decrease the area of erodible earth material that can be exposed at any time based on an analysis of project conditions.

The erosion control features installed by the Contractor shall be maintained by the Contractor during the construction period.

Provide temporary structures whenever construction equipment must cross watercourses at frequent intervals. Pollutants such as fuels, lubricants, bitumen, raw sewage, wash water from concrete mixing operations, and other harmful materials shall not be discharged into any waterways, impoundments or into natural or manmade channels.

**102-3.4 INSTALLATION, MAINTENANCE AND REMOVAL OF SILT FENCE.** Silt fences shall extend a minimum of 16 inches (41 cm) and a maximum of 34 inches (86 cm) above the ground surface. Posts shall be set no more than 10 feet (3 m) on center. Filter fabric shall be cut from a continuous roll to the length required minimizing joints where possible. When joints are necessary, the fabric shall be spliced at a support post with a minimum 12-inch (300-mm) overlap and securely sealed. A trench shall be excavated approximately 4 inches (100 mm) deep by 4 inches (100 mm) wide on the upslope side of the silt fence. The trench shall be backfilled and the soil compacted over the silt fence fabric. The Contractor shall remove and dispose of silt that accumulates during construction and prior to establishment of permanent erosion control. The fence shall be maintained in good working condition until permanent erosion control is established. Silt fence shall be removed upon approval of the RPR.

## **METHOD OF MEASUREMENT**

**102-4.1** Temporary erosion and pollution control work required will be performed as scheduled or directed by the RPR. Completed and accepted work will be measured as follows:

- a. Temporary seeding and mulching will be measured by the square yard (square meter).
- b. Temporary slope drains will be measured by the linear foot (meter).
- c. Temporary benches, dikes, dams, and sediment basins will be measured by the cubic yard (cubic meter) of excavation performed, including necessary cleaning of sediment basins, and the cubic yard (cubic meter) of embankment placed as directed by the RPR.
- d. All fertilizing will be measured by the ton (kg).
- e. Installation and removal of silt fence will be measured by - lump sum.

**102-4.2** Control work performed for protection of construction areas outside the construction limits, such as borrow and waste areas, haul roads, equipment and material storage sites, and temporary plant sites, will not be measured and paid for directly but shall be considered as a subsidiary obligation of the Contractor.

### **BASIS OF PAYMENT**

**102-5.1** Accepted quantities of temporary water pollution, soil erosion, and siltation control work ordered by the RPR and measured as provided in paragraph 102-4.1 will be paid for under:

|                 |                                                    |                                  |
|-----------------|----------------------------------------------------|----------------------------------|
| Item C-102-5.1a | Temporary seeding and mulching                     | - per square yard (square meter) |
| Item C-102-5.1b | Temporary slope drains                             | - per linear foot (meter)        |
| Item C-102-5.1c | Temporary benches, dikes, dams and sediment basins | - per cubic yard (cubic meter)   |
| Item C-102-5.1d | Fertilizing                                        | - per ton (kg)                   |
| Item C-102-5.1e | Installation and removal of silt fence             | - lump sum                       |

Where other directed work falls within the specifications for a work item that has a contract price, the units of work shall be measured and paid for at the contract unit price bid for the various items.

Temporary control features not covered by contract items that are ordered by the RPR will be paid for in accordance with Section 90, paragraph 90-05 *Payment for Extra Work*.

### **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### Advisory Circulars (AC)

AC 150/5200-33      *Hazardous Wildlife Attractants on or Near Airports*

AC 150/5370-2      *Operational Safety on Airports During Construction*

#### ASTM International (ASTM)

ASTM D6461      *Standard Specification for Silt Fence Materials*

#### United States Department of Agriculture (USDA)

FAA/USDA Wildlife Hazard Management at Airports, A Manual for Airport Personnel

### **END OF ITEM C-102**

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## SECTION C-105 MOBILIZATION

**105-1 DESCRIPTION.** This item of work shall consist of, but is not limited to, work and operations necessary for the movement of personnel, equipment, material and supplies to and from the project site for work on the project except as provided in the contract as separate pay items.

**105-2 MOBILIZATION LIMIT.** Mobilization shall be limited to **10 percent** of the total project cost.

**105-3 POSTED NOTICES.** Prior to commencement of construction activities, the Contractor must post the following documents in a prominent and accessible place where they may be easily viewed by all employees of the prime Contractor and by all employees of subcontractors engaged by the prime Contractor: Equal Employment Opportunity (EEO) Poster "Equal Employment Opportunity is the Law" in accordance with the Office of Federal Contract Compliance Programs Executive Order 11246, as amended; Davis Bacon Wage Poster (WH 1321) - DOL "Notice to All Employees" Poster; and Applicable Davis-Bacon Wage Rate Determination. These notices must remain posted until final acceptance of the work by the Owner.

**105-4 ENGINEER/RPR FIELD OFFICE.** The Contractor shall provide dedicated space for the use of the field RPR and inspectors, as a field office for the duration of the project. This space shall be located conveniently near the construction and shall be separate from any space used by the Contractor. The Contractor shall furnish water, sanitary facilities, heat, air conditioning, and electricity in accordance with local building codes.

## METHOD OF MEASUREMENT

**105-5 BASIS OF MEASUREMENT AND PAYMENT.** Based upon the contract lump sum price for "Mobilization" partial payments will be allowed as follows:

- a. With first pay request, 25%.
- b. When 25% or more of the original contract is earned, an additional 25%.
- c. When 50% or more of the original contract is earned, an additional 40%.
- d. After Final Inspection, Staging area clean-up and delivery of all Project Closeout materials as required by Section 90, paragraph 90-11, *Contractor Final Project Documentation*, the final 10%.

## BASIS OF PAYMENT

### 105-3.1 PAYMENT WILL BE MADE UNDER:

Item C-105      Mobilization

### **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

Office of Federal Contract Compliance Programs (OFCCP)

Executive Order 11246, as amended

EEOC-P/E-1 – Equal Employment Opportunity is the Law Poster

United States Department of Labor, Wage and Hour Division (WHD)

WH 1321 – Employee Rights under the Davis-Bacon Act Poster

**END OF SECTION C-105**

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## **SECTION C-110 METHOD OF ESTIMATING PERCENTAGE OF MATERIAL WITHIN SPECIFICATION LIMITS (PWL)**

**110-1 GENERAL.** When the specifications provide for acceptance of material based on the method of estimating percentage of material within specification limits (PWL), the PWL will be determined in accordance with this section. All test results for a lot will be analyzed statistically to determine the total estimated percent of the lot that is within specification limits. The PWL is computed using the sample average ( $\bar{X}$ ) and sample standard deviation ( $S_n$ ) of the specified number ( $n$ ) of sublots for the lot and the specification tolerance limits,  $L$  for lower and  $U$  for upper, for the particular acceptance parameter. From these values, the respective Quality index,  $Q_L$  for Lower Quality Index and/or  $Q_U$  for Upper Quality Index, is computed and the PWL for the lot for the specified  $n$  is determined from Table 1. All specification limits specified in the technical sections shall be absolute values. Test results used in the calculations shall be to the significant figure given in the test procedure.

There is some degree of uncertainty (risk) in the measurement for acceptance because only a small fraction of production material (the population) is sampled and tested. This uncertainty exists because all portions of the production material have the same probability to be randomly sampled. The Contractor's risk is the probability that material produced at the acceptable quality level is rejected or subjected to a pay adjustment. The Owner's risk is the probability that material produced at the rejectable quality level is accepted.

It is the intent of this section to inform the Contractor that, in order to consistently offset the Contractor's risk for material evaluated, production quality (using population average and population standard deviation) must be maintained at the acceptable quality specified or higher. In all cases, it is the responsibility of the Contractor to produce at quality levels that will meet the specified acceptance criteria when sampled and tested at the frequencies specified.

**110-2 METHOD FOR COMPUTING PWL.** The computational sequence for computing PWL is as follows:

- a. Divide the lot into  $n$  sublots in accordance with the acceptance requirements of the specification.
- b. Locate the random sampling position within the subplot in accordance with the requirements of the specification.
- c. Make a measurement at each location, or take a test portion and make the measurement on the test portion in accordance with the testing requirements of the specification.
- d. Find the sample average ( $\bar{X}$ ) for all subplot test values within the lot by using the following formula:

$$\bar{X} = (x_1 + x_2 + x_3 + \dots + x_n) / n$$

Where:  $\bar{X}$  = Sample average of all subplot test values within a lot

$x_1, x_2, \dots, x_n$  = Individual subplot test values

$n$  = Number of subplot test values

e. Find the sample standard deviation ( $S_n$ ) by use of the following formula:

$$S_n = [(d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2)/(n-1)]^{1/2}$$

Where:  $S_n$  = Sample standard deviation of the number of subplot test values in the set

$d_1, d_2, \dots, d_n$  = Deviations of the individual subplot test values  $x_1, x_2, \dots$  from the average value  $X$

that is:  $d_1 = (x_1 - X), d_2 = (x_2 - X) \dots d_n = (x_n - X)$

$n$  = Number of subplot test values

f. For single sided specification limits (i.e., L only), compute the Lower Quality Index  $Q_L$  by use of the following formula:

$$Q_L = (X - L) / S_n$$

Where: L = specification lower tolerance limit

Estimate the percentage of material within limits (PWL) by entering Table 1 with  $Q_L$ , using the column appropriate to the total number ( $n$ ) of measurements. If the value of  $Q_L$  falls between values shown on the table, use the next higher value of PWL.

g. For double-sided specification limits (i.e., L and U), compute the Quality Indexes  $Q_L$  and  $Q_U$  by use of the following formulas:

$$Q_L = (X - L) / S_n$$

and

$$Q_U = (U - X) / S_n$$

Where: L and U = specification lower and upper tolerance limits

Estimate the percentage of material between the lower (L) and upper (U) tolerance limits (PWL) by entering Table 1 separately with  $Q_L$  and  $Q_U$ , using the column appropriate to the total number ( $n$ ) of measurements, and determining the percent of material above  $P_L$  and percent of material below  $P_U$  for each tolerance limit. If the values of  $Q_L$  fall between values shown on the table, use the next higher value of  $P_L$  or  $P_U$ . Determine the PWL by use of the following formula:

$$PWL = (P_U + P_L) - 100$$

Where:  $P_L$  = percent within lower specification limit

$P_U$  = percent within upper specification limit

### EXAMPLE OF PWL CALCULATION

**Project:** Example Project

**Test Item:** Item P-401, Lot A.

**A. PWL Determination for Mat Density.**



1. Density of four random cores taken from Lot A.

$$A-1 = 96.60$$

$$A-2 = 97.55$$

$$A-3 = 99.30$$

$$A-4 = 98.35$$

$$n = 4$$

2. Calculate average density for the lot.

$$X = (x_1 + x_2 + x_3 + \dots + x_n) / n$$

$$X = (96.60 + 97.55 + 99.30 + 98.35) / 4$$

$$X = 97.95\% \text{ density}$$

3. Calculate the standard deviation for the lot.

$$S_n = [((96.60 - 97.95)^2 + (97.55 - 97.95)^2 + (99.30 - 97.95)^2 + (98.35 - 97.95)^2) / (4 - 1)]^{1/2}$$

$$S_n = [(1.82 + 0.16 + 1.82 + 0.16) / 3]^{1/2}$$

$$S_n = 1.15$$

4. Calculate the Lower Quality Index  $Q_L$  for the lot. ( $L=96.3$ )

$$Q_L = (X - L) / S_n$$

$$Q_L = (97.95 - 96.30) / 1.15$$

$$Q_L = 1.4348$$

5. Determine PWL by entering Table 1 with  $Q_L = 1.44$  and  $n = 4$ .

$$PWL = 98$$

#### **B. PWL Determination for Air Voids.**

1. Air Voids of four random samples taken from Lot A.

$$A-1 = 5.00$$

$$A-2 = 3.74$$

$$A-3 = 2.30$$

$$A-4 = 3.25$$

2. Calculate the average air voids for the lot.

$$X = (x_1 + x_2 + x_3 + \dots + x_n) / n$$

$$X = (5.00 + 3.74 + 2.30 + 3.25) / 4$$

$$X = 3.57\%$$

3. Calculate the standard deviation  $S_n$  for the lot.

$$S_n = [((3.57 - 5.00)^2 + (3.57 - 3.74)^2 + (3.57 - 2.30)^2 + (3.57 - 3.25)^2) / (4 - 1)]^{1/2}$$

$$S_n = [(2.04 + 0.03 + 1.62 + 0.10) / 3]^{1/2}$$

$$S_n = 1.12$$

4. Calculate the Lower Quality Index  $Q_L$  for the lot. ( $L = 2.0$ )

$$Q_L = (X - L) / S_n$$

$$Q_L = (3.57 - 2.00) / 1.12$$

$$Q_L = 1.3992$$

5. Determine  $P_L$  by entering Table 1 with  $Q_L = 1.41$  and  $n = 4$ .

$$P_L = 97$$

6. Calculate the Upper Quality Index  $Q_U$  for the lot. ( $U = 5.0$ )

$$Q_U = (U - X) / S_n$$

$$Q_U = (5.00 - 3.57) / 1.12$$

$$Q_U = 1.2702$$

7. Determine  $P_U$  by entering Table 1 with  $Q_U = 1.29$  and  $n = 4$ .

$$P_U = 93$$

8. Calculate Air Voids PWL

$$PWL = (P_L + P_U) - 100$$

$$PWL = (97 + 93) - 100 = 90$$

### EXAMPLE OF OUTLIER CALCULATION (REFERENCE ASTM E178)

**Project:** Example Project

**Test Item:** Item P-401, Lot A.

#### A. Outlier Determination for Mat Density.

1. Density of four random cores taken from Lot A arranged in descending order.

$$A-3 = 99.30$$

$$A-4 = 98.35$$

$$A-2 = 97.55$$

$$A-1 = 96.60$$

2. From ASTM E178, Table 1, for  $n=4$  an upper 5% significance level, the critical value for test criterion = 1.463.

3. Use average density, standard deviation, and test criterion value to evaluate density measurements.

- a. For measurements greater than the average:

If (measurement - average)/(standard deviation) is less than test criterion, then the measurement is not considered an outlier.

For A-3, check if  $(99.30 - 97.95) / 1.15$  is greater than 1.463.

Since 1.174 is less than 1.463, the value is not an outlier.

- b. For measurements less than the average:

If (average - measurement)/(standard deviation) is less than test criterion, then the measurement is not considered an outlier.

For A-1, check if  $(97.95 - 96.60) / 1.15$  is greater than 1.463.

Since 1.174 is less than 1.463, the value is not an outlier.

**Note:** In this example, a measurement would be considered an outlier if the density were:

Greater than  $(97.95 + 1.463 \times 1.15) = 99.63\%$

OR

less than  $(97.95 - 1.463 \times 1.15) = 96.27\%$ .

**Table 1. Table for Estimating Percent of Lot Within Limits (PWL)**

| Percent Within Limits<br>( $P_L$ and $P_U$ ) | Positive Values of Q ( $Q_L$ and $Q_U$ ) |        |        |        |        |        |        |        |
|----------------------------------------------|------------------------------------------|--------|--------|--------|--------|--------|--------|--------|
|                                              | n=3                                      | n=4    | n=5    | n=6    | n=7    | n=8    | n=9    | n=10   |
| 99                                           | 1.1541                                   | 1.4700 | 1.6714 | 1.8008 | 1.8888 | 1.9520 | 1.9994 | 2.0362 |
| 98                                           | 1.1524                                   | 1.4400 | 1.6016 | 1.6982 | 1.7612 | 1.8053 | 1.8379 | 1.8630 |
| 97                                           | 1.1496                                   | 1.4100 | 1.5427 | 1.6181 | 1.6661 | 1.6993 | 1.7235 | 1.7420 |
| 96                                           | 1.1456                                   | 1.3800 | 1.4897 | 1.5497 | 1.5871 | 1.6127 | 1.6313 | 1.6454 |
| 95                                           | 1.1405                                   | 1.3500 | 1.4407 | 1.4887 | 1.5181 | 1.5381 | 1.5525 | 1.5635 |
| 94                                           | 1.1342                                   | 1.3200 | 1.3946 | 1.4329 | 1.4561 | 1.4717 | 1.4829 | 1.4914 |
| 93                                           | 1.1269                                   | 1.2900 | 1.3508 | 1.3810 | 1.3991 | 1.4112 | 1.4199 | 1.4265 |
| 92                                           | 1.1184                                   | 1.2600 | 1.3088 | 1.3323 | 1.3461 | 1.3554 | 1.3620 | 1.3670 |
| 91                                           | 1.1089                                   | 1.2300 | 1.2683 | 1.2860 | 1.2964 | 1.3032 | 1.3081 | 1.3118 |
| 90                                           | 1.0982                                   | 1.2000 | 1.2290 | 1.2419 | 1.2492 | 1.2541 | 1.2576 | 1.2602 |
| 89                                           | 1.0864                                   | 1.1700 | 1.1909 | 1.1995 | 1.2043 | 1.2075 | 1.2098 | 1.2115 |
| 88                                           | 1.0736                                   | 1.1400 | 1.1537 | 1.1587 | 1.1613 | 1.1630 | 1.1643 | 1.1653 |
| 87                                           | 1.0597                                   | 1.1100 | 1.1173 | 1.1192 | 1.1199 | 1.1204 | 1.1208 | 1.1212 |
| 86                                           | 1.0448                                   | 1.0800 | 1.0817 | 1.0808 | 1.0800 | 1.0794 | 1.0791 | 1.0789 |
| 85                                           | 1.0288                                   | 1.0500 | 1.0467 | 1.0435 | 1.0413 | 1.0399 | 1.0389 | 1.0382 |
| 84                                           | 1.0119                                   | 1.0200 | 1.0124 | 1.0071 | 1.0037 | 1.0015 | 1.0000 | 0.9990 |
| 83                                           | 0.9939                                   | 0.9900 | 0.9785 | 0.9715 | 0.9671 | 0.9643 | 0.9624 | 0.9610 |
| 82                                           | 0.9749                                   | 0.9600 | 0.9452 | 0.9367 | 0.9315 | 0.9281 | 0.9258 | 0.9241 |
| 81                                           | 0.9550                                   | 0.9300 | 0.9123 | 0.9025 | 0.8966 | 0.8928 | 0.8901 | 0.8882 |
| 80                                           | 0.9342                                   | 0.9000 | 0.8799 | 0.8690 | 0.8625 | 0.8583 | 0.8554 | 0.8533 |
| 79                                           | 0.9124                                   | 0.8700 | 0.8478 | 0.8360 | 0.8291 | 0.8245 | 0.8214 | 0.8192 |
| 78                                           | 0.8897                                   | 0.8400 | 0.8160 | 0.8036 | 0.7962 | 0.7915 | 0.7882 | 0.7858 |
| 77                                           | 0.8662                                   | 0.8100 | 0.7846 | 0.7716 | 0.7640 | 0.7590 | 0.7556 | 0.7531 |
| 76                                           | 0.8417                                   | 0.7800 | 0.7535 | 0.7401 | 0.7322 | 0.7271 | 0.7236 | 0.7211 |
| 75                                           | 0.8165                                   | 0.7500 | 0.7226 | 0.7089 | 0.7009 | 0.6958 | 0.6922 | 0.6896 |
| 74                                           | 0.7904                                   | 0.7200 | 0.6921 | 0.6781 | 0.6701 | 0.6649 | 0.6613 | 0.6587 |
| 73                                           | 0.7636                                   | 0.6900 | 0.6617 | 0.6477 | 0.6396 | 0.6344 | 0.6308 | 0.6282 |
| 72                                           | 0.7360                                   | 0.6600 | 0.6316 | 0.6176 | 0.6095 | 0.6044 | 0.6008 | 0.5982 |
| 71                                           | 0.7077                                   | 0.6300 | 0.6016 | 0.5878 | 0.5798 | 0.5747 | 0.5712 | 0.5686 |
| 70                                           | 0.6787                                   | 0.6000 | 0.5719 | 0.5582 | 0.5504 | 0.5454 | 0.5419 | 0.5394 |
| 69                                           | 0.6490                                   | 0.5700 | 0.5423 | 0.5290 | 0.5213 | 0.5164 | 0.5130 | 0.5105 |
| 68                                           | 0.6187                                   | 0.5400 | 0.5129 | 0.4999 | 0.4924 | 0.4877 | 0.4844 | 0.4820 |
| 67                                           | 0.5878                                   | 0.5100 | 0.4836 | 0.4710 | 0.4638 | 0.4592 | 0.4560 | 0.4537 |
| 66                                           | 0.5563                                   | 0.4800 | 0.4545 | 0.4424 | 0.4355 | 0.4310 | 0.4280 | 0.4257 |
| 65                                           | 0.5242                                   | 0.4500 | 0.4255 | 0.4139 | 0.4073 | 0.4030 | 0.4001 | 0.3980 |
| 64                                           | 0.4916                                   | 0.4200 | 0.3967 | 0.3856 | 0.3793 | 0.3753 | 0.3725 | 0.3705 |
| 63                                           | 0.4586                                   | 0.3900 | 0.3679 | 0.3575 | 0.3515 | 0.3477 | 0.3451 | 0.3432 |
| 62                                           | 0.4251                                   | 0.3600 | 0.3392 | 0.3295 | 0.3239 | 0.3203 | 0.3179 | 0.3161 |
| 61                                           | 0.3911                                   | 0.3300 | 0.3107 | 0.3016 | 0.2964 | 0.2931 | 0.2908 | 0.2892 |
| 60                                           | 0.3568                                   | 0.3000 | 0.2822 | 0.2738 | 0.2691 | 0.2660 | 0.2639 | 0.2624 |
| 59                                           | 0.3222                                   | 0.2700 | 0.2537 | 0.2461 | 0.2418 | 0.2391 | 0.2372 | 0.2358 |
| 58                                           | 0.2872                                   | 0.2400 | 0.2254 | 0.2186 | 0.2147 | 0.2122 | 0.2105 | 0.2093 |
| 57                                           | 0.2519                                   | 0.2100 | 0.1971 | 0.1911 | 0.1877 | 0.1855 | 0.1840 | 0.1829 |
| 56                                           | 0.2164                                   | 0.1800 | 0.1688 | 0.1636 | 0.1607 | 0.1588 | 0.1575 | 0.1566 |
| 55                                           | 0.1806                                   | 0.1500 | 0.1406 | 0.1363 | 0.1338 | 0.1322 | 0.1312 | 0.1304 |
| 54                                           | 0.1447                                   | 0.1200 | 0.1125 | 0.1090 | 0.1070 | 0.1057 | 0.1049 | 0.1042 |
| 53                                           | 0.1087                                   | 0.0900 | 0.0843 | 0.0817 | 0.0802 | 0.0793 | 0.0786 | 0.0781 |
| 52                                           | 0.0725                                   | 0.0600 | 0.0562 | 0.0544 | 0.0534 | 0.0528 | 0.0524 | 0.0521 |
| 51                                           | 0.0363                                   | 0.0300 | 0.0281 | 0.0272 | 0.0267 | 0.0264 | 0.0262 | 0.0260 |

| Percent Within Limits<br>(P <sub>L</sub> and P <sub>U</sub> ) | Positive Values of Q (Q <sub>L</sub> and Q <sub>U</sub> ) |        |        |        |        |        |        |        |
|---------------------------------------------------------------|-----------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|
|                                                               | n=3                                                       | n=4    | n=5    | n=6    | n=7    | n=8    | n=9    | n=10   |
| 50                                                            | 0.0000                                                    | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

| Percent Within Limits<br>(P <sub>L</sub> and P <sub>U</sub> ) | Negative Values of Q (Q <sub>L</sub> and Q <sub>U</sub> ) |         |         |         |         |         |         |         |
|---------------------------------------------------------------|-----------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
|                                                               | n=3                                                       | n=4     | n=5     | n=6     | n=7     | n=8     | n=9     | n=10    |
| 49                                                            | -0.0363                                                   | -0.0300 | -0.0281 | -0.0272 | -0.0267 | -0.0264 | -0.0262 | -0.0260 |
| 48                                                            | -0.0725                                                   | -0.0600 | -0.0562 | -0.0544 | -0.0534 | -0.0528 | -0.0524 | -0.0521 |
| 47                                                            | -0.1087                                                   | -0.0900 | -0.0843 | -0.0817 | -0.0802 | -0.0793 | -0.0786 | -0.0781 |
| 46                                                            | -0.1447                                                   | -0.1200 | -0.1125 | -0.1090 | -0.1070 | -0.1057 | -0.1049 | -0.1042 |
| 45                                                            | -0.1806                                                   | -0.1500 | -0.1406 | -0.1363 | -0.1338 | -0.1322 | -0.1312 | -0.1304 |
| 44                                                            | -0.2164                                                   | -0.1800 | -0.1688 | -0.1636 | -0.1607 | -0.1588 | -0.1575 | -0.1566 |
| 43                                                            | -0.2519                                                   | -0.2100 | -0.1971 | -0.1911 | -0.1877 | -0.1855 | -0.1840 | -0.1829 |
| 42                                                            | -0.2872                                                   | -0.2400 | -0.2254 | -0.2186 | -0.2147 | -0.2122 | -0.2105 | -0.2093 |
| 41                                                            | -0.3222                                                   | -0.2700 | -0.2537 | -0.2461 | -0.2418 | -0.2391 | -0.2372 | -0.2358 |
| 40                                                            | -0.3568                                                   | -0.3000 | -0.2822 | -0.2738 | -0.2691 | -0.2660 | -0.2639 | -0.2624 |
| 39                                                            | -0.3911                                                   | -0.3300 | -0.3107 | -0.3016 | -0.2964 | -0.2931 | -0.2908 | -0.2892 |
| 38                                                            | -0.4251                                                   | -0.3600 | -0.3392 | -0.3295 | -0.3239 | -0.3203 | -0.3179 | -0.3161 |
| 37                                                            | -0.4586                                                   | -0.3900 | -0.3679 | -0.3575 | -0.3515 | -0.3477 | -0.3451 | -0.3432 |
| 36                                                            | -0.4916                                                   | -0.4200 | -0.3967 | -0.3856 | -0.3793 | -0.3753 | -0.3725 | -0.3705 |
| 35                                                            | -0.5242                                                   | -0.4500 | -0.4255 | -0.4139 | -0.4073 | -0.4030 | -0.4001 | -0.3980 |
| 34                                                            | -0.5563                                                   | -0.4800 | -0.4545 | -0.4424 | -0.4355 | -0.4310 | -0.4280 | -0.4257 |
| 33                                                            | -0.5878                                                   | -0.5100 | -0.4836 | -0.4710 | -0.4638 | -0.4592 | -0.4560 | -0.4537 |
| 32                                                            | -0.6187                                                   | -0.5400 | -0.5129 | -0.4999 | -0.4924 | -0.4877 | -0.4844 | -0.4820 |
| 31                                                            | -0.6490                                                   | -0.5700 | -0.5423 | -0.5290 | -0.5213 | -0.5164 | -0.5130 | -0.5105 |
| 30                                                            | -0.6787                                                   | -0.6000 | -0.5719 | -0.5582 | -0.5504 | -0.5454 | -0.5419 | -0.5394 |
| 29                                                            | -0.7077                                                   | -0.6300 | -0.6016 | -0.5878 | -0.5798 | -0.5747 | -0.5712 | -0.5686 |
| 28                                                            | -0.7360                                                   | -0.6600 | -0.6316 | -0.6176 | -0.6095 | -0.6044 | -0.6008 | -0.5982 |
| 27                                                            | -0.7636                                                   | -0.6900 | -0.6617 | -0.6477 | -0.6396 | -0.6344 | -0.6308 | -0.6282 |
| 26                                                            | -0.7904                                                   | -0.7200 | -0.6921 | -0.6781 | -0.6701 | -0.6649 | -0.6613 | -0.6587 |
| 25                                                            | -0.8165                                                   | -0.7500 | -0.7226 | -0.7089 | -0.7009 | -0.6958 | -0.6922 | -0.6896 |
| 24                                                            | -0.8417                                                   | -0.7800 | -0.7535 | -0.7401 | -0.7322 | -0.7271 | -0.7236 | -0.7211 |
| 23                                                            | -0.8662                                                   | -0.8100 | -0.7846 | -0.7716 | -0.7640 | -0.7590 | -0.7556 | -0.7531 |
| 22                                                            | -0.8897                                                   | -0.8400 | -0.8160 | -0.8036 | -0.7962 | -0.7915 | -0.7882 | -0.7858 |
| 21                                                            | -0.9124                                                   | -0.8700 | -0.8478 | -0.8360 | -0.8291 | -0.8245 | -0.8214 | -0.8192 |
| 20                                                            | -0.9342                                                   | -0.9000 | -0.8799 | -0.8690 | -0.8625 | -0.8583 | -0.8554 | -0.8533 |
| 19                                                            | -0.9550                                                   | -0.9300 | -0.9123 | -0.9025 | -0.8966 | -0.8928 | -0.8901 | -0.8882 |
| 18                                                            | -0.9749                                                   | -0.9600 | -0.9452 | -0.9367 | -0.9315 | -0.9281 | -0.9258 | -0.9241 |
| 17                                                            | -0.9939                                                   | -0.9900 | -0.9785 | -0.9715 | -0.9671 | -0.9643 | -0.9624 | -0.9610 |
| 16                                                            | -1.0119                                                   | -1.0200 | -1.0124 | -1.0071 | -1.0037 | -1.0015 | -1.0000 | -0.9990 |
| 15                                                            | -1.0288                                                   | -1.0500 | -1.0467 | -1.0435 | -1.0413 | -1.0399 | -1.0389 | -1.0382 |
| 14                                                            | -1.0448                                                   | -1.0800 | -1.0817 | -1.0808 | -1.0800 | -1.0794 | -1.0791 | -1.0789 |
| 13                                                            | -1.0597                                                   | -1.1100 | -1.1173 | -1.1192 | -1.1199 | -1.1204 | -1.1208 | -1.1212 |
| 12                                                            | -1.0736                                                   | -1.1400 | -1.1537 | -1.1587 | -1.1613 | -1.1630 | -1.1643 | -1.1653 |
| 11                                                            | -1.0864                                                   | -1.1700 | -1.1909 | -1.1995 | -1.2043 | -1.2075 | -1.2098 | -1.2115 |
| 10                                                            | -1.0982                                                   | -1.2000 | -1.2290 | -1.2419 | -1.2492 | -1.2541 | -1.2576 | -1.2602 |
| 9                                                             | -1.1089                                                   | -1.2300 | -1.2683 | -1.2860 | -1.2964 | -1.3032 | -1.3081 | -1.3118 |
| 8                                                             | -1.1184                                                   | -1.2600 | -1.3088 | -1.3323 | -1.3461 | -1.3554 | -1.3620 | -1.3670 |
| 7                                                             | -1.1269                                                   | -1.2900 | -1.3508 | -1.3810 | -1.3991 | -1.4112 | -1.4199 | -1.4265 |
| 6                                                             | -1.1342                                                   | -1.3200 | -1.3946 | -1.4329 | -1.4561 | -1.4717 | -1.4829 | -1.4914 |
| 5                                                             | -1.1405                                                   | -1.3500 | -1.4407 | -1.4887 | -1.5181 | -1.5381 | -1.5525 | -1.5635 |
| 4                                                             | -1.1456                                                   | -1.3800 | -1.4897 | -1.5497 | -1.5871 | -1.6127 | -1.6313 | -1.6454 |
| 3                                                             | -1.1496                                                   | -1.4100 | -1.5427 | -1.6181 | -1.6661 | -1.6993 | -1.7235 | -1.7420 |
| 2                                                             | -1.1524                                                   | -1.4400 | -1.6016 | -1.6982 | -1.7612 | -1.8053 | -1.8379 | -1.8630 |

| Percent<br>Within<br>Limits<br>( $P_L$ and $P_U$ ) | Negative Values of Q ( $Q_L$ and $Q_U$ ) |         |         |         |         |         |         |         |
|----------------------------------------------------|------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
|                                                    | n=3                                      | n=4     | n=5     | n=6     | n=7     | n=8     | n=9     | n=10    |
| 1                                                  | -1.1541                                  | -1.4700 | -1.6714 | -1.8008 | -1.8888 | -1.9520 | -1.9994 | -2.0362 |

### REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

ASTM E178                      Standard Practice for Dealing with Outlying Observations

**END OF SECTION C-110**

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## ITEM P-151 CLEARING AND GRUBBING

### DESCRIPTION

**151-1.1** This item shall consist of clearing or clearing and grubbing, including the disposal of materials, for all areas within the limits designated on the plans or as required by the Resident Project Representative (RPR).

**a. Clearing** shall consist of the cutting and removal of all trees, stumps, brush, logs, hedges, the removal of fences and other loose or projecting material from the designated areas. The grubbing of stumps and roots will not be required.

**b. Clearing and grubbing** shall consist of clearing the surface of the ground of the designated areas of all trees, stumps, down timber, logs, snags, brush, undergrowth, hedges, heavy growth of grass or weeds, fences, structures, debris, and rubbish of any nature, natural obstructions or such material which in the opinion of the RPR is unsuitable for the foundation of strips, pavements, or other required structures, including the grubbing of stumps, roots, matted roots, foundations, and the disposal from the project of all spoil materials resulting from clearing and grubbing.

**c. Tree Removal.** Tree Removal shall consist of the cutting and removal of isolated single trees or isolated groups of trees, and the grubbing of stumps and roots. The removal of all the trees of this classification shall be in accordance with the requirements for the particular area being cleared.

### CONSTRUCTION METHODS

**151-2.1 GENERAL.** The areas denoted on the plans to be cleared and grubbed shall be staked on the ground by the Contractor as indicated on the plan.

The removal of existing structures and utilities required to permit orderly progress of work shall be accomplished by local agencies, unless otherwise shown on the plans. Whenever a telephone pole, pipeline, conduit, sewer, roadway, or other utility is encountered and must be removed or relocated, the Contractor shall advise the RPR who will notify the proper local authority or owner to secure prompt action.

**151-2.1.1 Disposal.** All materials removed by clearing or by clearing and grubbing shall be disposed of outside the Airport's limits at the Contractor's responsibility, except when otherwise directed by the RPR. As far as practicable, waste concrete and masonry shall be placed on slopes of embankments or channels. When embankments are constructed of such material, this material shall be placed in accordance with requirements for formation of embankments. Any broken concrete or masonry that cannot be used in construction and all other materials not considered suitable for use elsewhere, shall be disposed of by the Contractor. In no case, shall any discarded materials be left in windrows or piles adjacent to or within the airport limits. The manner and location of disposal of materials shall be subject to the approval of the RPR and shall not create an unsightly or objectionable view. When the Contractor is required to locate a disposal area outside the airport property limits, the Contractor shall obtain and file with the RPR permission in writing from the property owner for the use of private property for this purpose.

**151-2.1.2 Blasting.** Blasting shall not be allowed.

**151-2.2 CLEARING.** The Contractor shall clear the staked or indicated area of all materials as indicated on the plans. Trees unavoidably falling outside the specified clearing limits must be cut up, removed, and disposed of in a satisfactory manner. To minimize damage to trees that are to be left standing, trees shall be felled toward the center of the area being cleared. The Contractor shall preserve and protect from injury all trees not to be removed. The trees, stumps, and brush shall be cut flush with the original ground surface. The grubbing of stumps and roots will not be required.

Fences shall be removed and disposed of as directed by the RPR. Fence wire shall be neatly rolled and the wire and posts stored on the airport if they are to be used again, or stored at a location designated by the RPR if the fence is to remain the property of a local owner or authority.

**151-2.3 CLEARING AND GRUBBING.** In areas designated to be cleared and grubbed, all stumps, roots, buried logs, brush, grass, and other unsatisfactory materials as indicated on the plans, shall be removed, except where embankments exceeding 3-1/2 feet (105 cm) in depth will be constructed outside of paved areas. For embankments constructed outside of paved areas, all unsatisfactory materials shall be removed, but sound trees, stumps, and brush can be cut off flush with the original ground and allowed to remain. Tap roots and other projections over 1-1/2 inches (38 mm) in diameter shall be grubbed out to a depth of at least 18 inches (0.5 m) below the finished subgrade or slope elevation.

Any buildings and miscellaneous structures that are shown on the plans to be removed shall be demolished or removed, and all materials shall be disposed of by removal from the site. The cost of removal is incidental to this item. The remaining or existing foundations, wells, cesspools, and like structures shall be destroyed by breaking down the materials of which the foundations, wells, cesspools, etc., are built to a depth at least 2 feet (60 cm) below the existing surrounding ground. Any broken concrete, blocks, or other objectionable material that cannot be used in backfill shall be removed and disposed of at the Contractor's expense. The holes or openings shall be backfilled with acceptable material and properly compacted.

All holes in embankment areas remaining after the grubbing operation shall have the sides of the holes flattened to facilitate filling with acceptable material and compacting as required in Item P-152. The same procedure shall be applied to all holes remaining after grubbing in areas where the depth of holes exceeds the depth of the proposed excavation.

## **METHOD OF MEASUREMENT**

**151-3.1** The quantities of clearing as shown by the limits on the plans shall be per lump sum of land specifically cleared.

**151-3.2** The quantities of clearing and grubbing as shown by the limits on the plans shall be per lump sum of land specifically cleared and grubbed.

**151-3.3** The quantity of tree removal as shown on the plans shall be the per lump sum of land specifically cleared.



**BASIS OF PAYMENT**

**151-4.1** Payment shall be made at the contract unit price per lump sum for clearing. This price shall be full compensation for furnishing all materials and for all labor, equipment, tools, and incidentals necessary to complete the item.

**151-4.2** Payment shall be made at the contract unit price per lump sum for clearing and grubbing. This price shall be full compensation for furnishing all materials and for all labor, equipment, tools, and incidentals necessary to complete the item.

**151-4.3** Payment shall be made at the contract unit price per lump sum for tree removal. This price shall be full compensation for furnishing all materials and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

|                |                       |            |
|----------------|-----------------------|------------|
| Item P-151-4.1 | Clearing              | - Lump Sum |
| Item P-151-4.2 | Clearing and grubbing | - Lump Sum |
| Item P-152-4.3 | Tree Removal          | - Lump Sum |

**END OF ITEM P-151**

## ITEM P-152 EXCAVATION, SUBGRADE, AND EMBANKMENT

### DESCRIPTION

**152-1.1** This item covers excavation, disposal, placement, and compaction of all materials within the limits of the work required to construct safety areas, runways, taxiways, aprons, and intermediate areas as well as other areas for drainage, building construction, parking, or other purposes in accordance with these specifications and in conformity to the dimensions and typical sections shown on the plans.

**152-1.2 CLASSIFICATION.** All material excavated shall be classified as defined below:

**a. Unclassified excavation.** Unclassified excavation shall consist of the excavation and disposal of all material, regardless of its nature.

**152-1.3 UNSUITABLE EXCAVATION.** Unsuitable material shall be disposed in designated waste areas as shown on the plans. Materials containing vegetable or organic matter, such as muck, peat, organic silt, or sod shall be considered unsuitable for use in embankment construction. Material suitable for topsoil may be used on the embankment slope when approved by the RPR.

### CONSTRUCTION METHODS

**152-2.1 GENERAL.** Before beginning excavation, grading, and embankment operations in any area, the area shall be cleared or cleared and grubbed in accordance with Item P-151.

The suitability of material to be placed in embankments shall be subject to approval by the RPR. All unsuitable material shall be disposed of in waste areas as shown on the plans. All waste areas shall be graded to allow positive drainage of the area and adjacent areas. The surface elevation of waste areas shall be specified on the plans or approved by the RPR.

When the Contractor's excavating operations encounter artifacts of historical or archaeological significance, the operations shall be temporarily discontinued and the RPR notified per Section 70, paragraph 70-20. At the direction of the RPR, the Contractor shall excavate the site in such a manner as to preserve the artifacts encountered and allow for their removal. Such excavation will be paid for as extra work.

Areas outside the limits of the pavement areas where the top layer of soil has become compacted by hauling or other Contractor activities shall be scarified and disked to a depth of 4 inches (100 mm), to loosen and pulverize the soil. Stones or rock fragments larger than 4 inches (100 mm) in their greatest dimension will not be permitted in the top 6 inches (150 mm) of the subgrade.

If it is necessary to interrupt existing surface drainage, sewers or under-drainage, conduits, utilities, or similar underground structures, the Contractor shall be responsible for and shall take all necessary precautions to preserve them or provide temporary services. When such facilities are encountered, the Contractor shall notify the RPR, who shall arrange for their removal if necessary. The Contractor, at their own expense, shall satisfactorily repair or pay the cost of all damage to such facilities or structures that may result from any of the Contractor's operations during the period of the contract.

**a. Blasting.** Blasting shall not be allowed.

**152-2.2 EXCAVATION.** No excavation shall be started until the work has been staked out by the Contractor and the RPR has obtained from the Contractor, the survey notes of the elevations and measurements of the ground surface. The Contractor and RPR shall agree that the original ground lines shown on the original topographic mapping are accurate, or agree to any adjustments made to the original ground lines.

Digital terrain model (DTM) files of the existing surfaces, finished surfaces and other various surfaces were used to develop the design plans.

Existing grades on the design cross sections or DTM's, where they do not match the locations of actual spot elevations shown on the topographic map, were developed by computer interpolation from those spot elevations. Prior to disturbing original grade, Contractor shall verify the accuracy of the existing ground surface by verifying spot elevations at the same locations where original field survey data was obtained as indicated on the topographic map. Contractor shall recognize that, due to the interpolation process, the actual ground surface at any particular location may differ somewhat from the interpolated surface shown on the design cross sections or obtained from the DTM's. Contractor's verification of original ground surface, however, shall be limited to verification of spot elevations as indicated herein, and no adjustments will be made to the original ground surface unless the Contractor demonstrates that spot elevations shown are incorrect. For this purpose, spot elevations which are within 0.1 foot (30 mm) of the stated elevations for ground surfaces, or within 0.04 foot (12 mm) for hard surfaces (pavements, buildings, foundations, structures, etc.) shall be considered "no change". Only deviations in excess of these will be considered for adjustment of the original ground surface. If Contractor's verification identifies discrepancies in the topographic map, Contractor shall notify the RPR in writing at least two weeks before disturbance of existing grade to allow sufficient time to verify the submitted information and make adjustments to the design cross sections or DTM's. Disturbance of existing grade in any area shall constitute acceptance by the Contractor of the accuracy of the original elevations shown on the topographic map for that area.

**a. Selective grading.** When selective grading is indicated on the plans, the more suitable material designated by the RPR shall be used in constructing the embankment or in capping the pavement subgrade. If, at the time of excavation, it is not possible to place this material in its final location, it shall be stockpiled in approved areas until it can be placed. The more suitable material shall then be placed and compacted as specified. Selective grading shall be considered incidental to the work involved. The cost of stockpiling and placing the material shall be included in the various pay items of work involved.

**b. Undercutting.** Rock, shale, hardpan, loose rock, boulders, or other material unsatisfactory for safety areas, subgrades, roads, shoulders, or any areas intended for turf shall be excavated to a minimum depth of 12 inches (300 mm) below the subgrade or to the depth specified by the RPR. Muck, peat, matted roots, or other yielding material, unsatisfactory for subgrade foundation, shall be removed to the depth specified. Unsuitable materials shall be disposed off the airport. The cost is incidental to this item excavated material shall be paid for at the contract unit price per cubic yard (per cubic meter) for unclassified excavation. The excavated area shall be backfilled with suitable material obtained from the grading operations or borrow areas and compacted to specified densities. The necessary backfill will constitute a part of the embankment. Where rock cuts are made, backfill with select material. Any pockets created in the rock surface shall be

drained in accordance with the details shown on the plans. Undercutting will be paid as unclassified excavation.

**c. Over-break.** Over-break, including slides, is that portion of any material displaced or loosened beyond the finished work as planned or authorized by the RPR. All over-break shall be graded or removed by the Contractor and disposed of as directed by the RPR. The RPR shall determine if the displacement of such material was unavoidable and their own decision shall be final. Payment will not be made for the removal and disposal of over-break that the RPR determines as avoidable. Unavoidable over-break will be classified as "Unclassified Excavation."

**d. Removal of utilities.** The removal of existing structures and utilities required to permit the orderly progress of work will be accomplished by the Contractor as indicated on the plans. All existing foundations shall be excavated at least 2 feet (60 cm) below the top of subgrade or as indicated on the plans, and the material disposed of as directed by the RPR. All foundations thus excavated shall be backfilled with suitable material and compacted as specified for embankment or as shown on the plans.

**152-2.3 BORROW EXCAVATION.** There are no borrow sources within the boundaries of the airport property. The Contractor shall locate and obtain borrow sources, subject to the approval of the RPR. The Contractor shall notify the RPR at least 15 days prior to beginning the excavation so necessary measurements and tests can be made by the RPR. All borrow pits shall be opened to expose the various strata of acceptable material to allow obtaining a uniform product. Borrow areas shall be drained and left in a neat, presentable condition with all slopes dressed uniformly. Borrow areas shall not create a hazardous wildlife attractant.

**152-2.4 DRAINAGE EXCAVATION.** Drainage excavation shall consist of excavating drainage ditches including intercepting, inlet, or outlet ditches; or other types as shown on the plans. The work shall be performed in sequence with the other construction. Ditches shall be constructed prior to starting adjacent excavation operations. All satisfactory material shall be placed in embankment fills; unsuitable material shall be placed in designated waste areas or as directed by the RPR. All necessary work shall be performed true to final line, elevation, and cross-section. The Contractor shall maintain ditches constructed on the project to the required cross-section and shall keep them free of debris or obstructions until the project is accepted.

**152-2.5 PREPARATION OF CUT AREAS OR AREAS WHERE EXISTING PAVEMENT HAS BEEN REMOVED.** In those areas on which a subbase or base course is to be placed, the top 12 inches (300 mm) of subgrade shall be compacted to not less than **100 %** of maximum density for non-cohesive soils, and **95%** of maximum density for cohesive soils as determined by ASTM **1557**. As used in this specification, "non-cohesive" shall mean those soils having a plasticity index (PI) of less than 3 as determined by ASTM D4318.

**152-2.6 PREPARATION OF EMBANKMENT AREA.** All sod and vegetative matter shall be removed from the surface upon which the embankment is to be placed. The cleared surface shall be broken up by plowing or scarifying to a minimum depth of 6 inches (150 mm) and shall then be compacted per paragraph 152-2.10.

Sloped surfaces steeper than one (1) vertical to four (4) horizontal shall be plowed, stepped, benched, or broken up so that the fill material will bond with the existing material. When the subgrade is part fill and part excavation or natural ground, the excavated or natural ground portion

shall be scarified to a depth of 12 inches (300 mm) and compacted as specified for the adjacent fill.

No direct payment shall be made for the work performed under this section. The necessary clearing and grubbing and the quantity of excavation removed will be paid for under the respective items of work.

**152-2.7 CONTROL STRIP.** The first half-day of construction of subgrade and/or embankment shall be considered as a control strip for the Contractor to demonstrate, in the presence of the RPR, that the materials, equipment, and construction processes meet the requirements of this specification. The sequence and manner of rolling necessary to obtain specified density requirements shall be determined. The maximum compacted thickness may be increased to a maximum of 12 inches (300 mm) upon the Contractor's demonstration that approved equipment and operations will uniformly compact the lift to the specified density. The RPR must witness this demonstration and approve the lift thickness prior to full production.

Control strips that do not meet specification requirements shall be reworked, re-compacted, or removed and replaced at the Contractor's expense. Full operations shall not begin until the control strip has been accepted by the RPR. The Contractor shall use the same equipment, materials, and construction methods for the remainder of construction, unless adjustments made by the Contractor are approved in advance by the RPR.

**152-2.8 FORMATION OF EMBANKMENTS.** The material shall be constructed in lifts as established in the control strip, but not less than 6 inches (150 mm) nor more than 12 inches (300 mm) of compacted thickness.

When more than one lift is required to establish the layer thickness shown on the plans, the construction procedure described here shall apply to each lift. No lift shall be covered by subsequent lifts until tests verify that compaction requirements have been met. The Contractor shall rework, re-compact and retest any material placed which does not meet the specifications.

The lifts shall be placed, to produce a soil structure as shown on the typical cross-section or as directed by the RPR. Materials such as brush, hedge, roots, stumps, grass and other organic matter, shall not be incorporated or buried in the embankment.

Earthwork operations shall be suspended at any time when satisfactory results cannot be obtained due to rain, freezing, or other unsatisfactory weather conditions in the field. Frozen material shall not be placed in the embankment nor shall embankment be placed upon frozen material. Material shall not be placed on surfaces that are muddy, frozen, or contain frost. The Contractor shall drag, blade, or slope the embankment to provide surface drainage at all times.

The material in each lift shall be within  $\pm 2\%$  of optimum moisture content before rolling to obtain the prescribed compaction. The material shall be moistened or aerated as necessary to achieve a uniform moisture content throughout the lift. Natural drying may be accelerated by blending in dry material or manipulation alone to increase the rate of evaporation.

The Contractor shall make the necessary corrections and adjustments in methods, materials or moisture content to achieve the specified embankment density.

The contractor will take samples of excavated materials which will be used in embankment for testing and develop a Moisture-Density Relations of Soils Report (Proctor) in accordance with ASTM D1557. A new Proctor shall be developed for each soil type based on visual classification.

Density tests will be taken by the contractor for every 3,000 square yards of compacted embankment for each lift which is required to be compacted, or other appropriate frequencies as determined by the RPR.

If the material has greater than 30% retained on the 3/4-inch (19.0 mm) sieve, follow AASHTO T-180 Annex Correction of maximum dry density and optimum moisture for oversized particles.

Rolling operations shall be continued until the embankment is compacted to not less than 100% of maximum density for non-cohesive soils, and 95% of maximum density for cohesive soils as determined by ASTM D1557. Under all areas to be paved, the embankments shall be compacted to a depth of **one foot** and to a density of not less than **100** percent of the maximum density as determined by ASTM D1557. As used in this specification, "non-cohesive" shall mean those soils having a plasticity index (PI) of less than 3 as determined by ASTM D4318.

On all areas outside of the pavement areas, no compaction will be required on the top 4 inches (100 mm) which shall be prepared for a seedbed in accordance with Item T-901.

The in-place field density shall be determined in accordance with ASTM D1556. The Contractor's laboratory shall perform all density tests in the RPR's presence and provide the test results upon completion to the RPR for acceptance. If the specified density is not attained, the area represented by the test or as designated by the RPR shall be reworked and/or re-compacted and additional random tests made. This procedure shall be followed until the specified density is reached.

Compaction areas shall be kept separate, and no lift shall be covered by another lift until the proper density is obtained.

During construction of the embankment, the Contractor shall route all construction equipment evenly over the entire width of the embankment as each lift is placed. Lift placement shall begin in the deepest portion of the embankment fill. As placement progresses, the lifts shall be constructed approximately parallel to the finished pavement grade line.

When rock, concrete pavement, asphalt pavement, and other embankment material are excavated at approximately the same time as the subgrade, the material shall be incorporated into the outer portion of the embankment and the subgrade material shall be incorporated under the future paved areas. Stones, fragmentary rock, and recycled pavement larger than 4 inches (100 mm) in their greatest dimensions will not be allowed in the top 12 inches (300 mm) of the subgrade. Rockfill shall be brought up in lifts as specified or as directed by the RPR and the finer material shall be used to fill the voids forming a dense, compact mass. Rock, cement concrete pavement, asphalt pavement, and other embankment material shall not be disposed of except at places and in the manner designated on the plans or by the RPR.

When the excavated material consists predominantly of rock fragments of such size that the material cannot be placed in lifts of the prescribed thickness without crushing, pulverizing or further breaking down the pieces, such material may be placed in the embankment as directed in lifts not exceeding 2 feet (60 cm) in thickness. Each lift shall be leveled and smoothed with suitable

equipment by distribution of spalls and finer fragments of rock. The lift shall not be constructed above an elevation 4 feet (1.2 m) below the finished subgrade.

There will be no separate measurement of payment for compacted embankment. All costs incidental to placing in lifts, compacting, discing, watering, mixing, sloping, and other operations necessary for construction of embankments will be included in the contract price for excavation, borrow, or other items.

**152-2.9 PROOF ROLLING.** The purpose of proof rolling the subgrade is to identify any weak areas in the subgrade and not for compaction of the subgrade. Before start of embankment, and after compaction is completed, the subgrade area shall be proof rolled with a 20 ton (18.1 metric ton) tandem axle Dual Wheel Dump Truck loaded to the legal limit with tires inflated to 80/100/150 psi in the presence of the RPR. Apply a minimum of **4** coverage, or as specified by the RPR, under pavement areas. A coverage is defined as the application of one tire print over the designated area. Soft areas of subgrade that deflect more than 1 inch (25 mm) or show permanent deformation greater than 1 inch (25 mm) shall be removed and replaced with suitable material or reworked to conform to the moisture content and compaction requirements in accordance with these specifications. Removal and replacement of soft areas is incidental to this item.

**152-2.10 COMPACTION REQUIREMENTS.** The subgrade under areas to be paved shall be compacted to a depth of 12 inches (300 mm) and to a density of not less than 100 percent of the maximum dry density as determined by ASTM D1557. The subgrade in areas outside the limits of the pavement areas shall be compacted to a depth of 12 inches (300 mm) and to a density of not less than 95 percent of the maximum density as determined by ASTM D698.

The material to be compacted shall be within  $\pm 2\%$  of optimum moisture content before being rolled to obtain the prescribed compaction (except for expansive soils). When the material has greater than 30 percent retained on the  $\frac{3}{4}$  inch (19.0 mm) sieve, follow the methods in ASTM D1557. Tests for moisture content and compaction will be taken at a minimum of **500** S.Y. of subgrade. All quality assurance testing shall be done by the Contractor's laboratory in the presence of the RPR, and density test results shall be furnished upon completion to the RPR for acceptance determination.

The in-place field density shall be determined in accordance with ASTM D1556 **or** ASTM D6938 using Procedure A, the direct transmission method, and ASTM D6938 shall be used to determine the moisture content of the material. The machine shall be calibrated in accordance with ASTM D6938 within 12 months prior to its use on this contract. The gage shall be field standardized daily.

Maximum density refers to maximum dry density at optimum moisture content unless otherwise specified.

If the specified density is not attained, the entire lot shall be reworked and/or re-compacted and additional random tests made. This procedure shall be followed until the specified density is reached.

All cut-and-fill slopes shall be uniformly dressed to the slope, cross-section, and alignment shown on the plans or as directed by the RPR and the finished subgrade shall be maintained.

**152-2.11 FINISHING AND PROTECTION OF SUBGRADE.** Finishing and protection of the subgrade is incidental to this item. Grading and compacting of the subgrade shall be performed so that it will drain readily. All low areas, holes or depressions in the subgrade shall be brought to grade. Scarifying, blading, rolling and other methods shall be performed to provide a thoroughly compacted subgrade shaped to the lines and grades shown on the plans. All ruts or rough places that develop in the completed subgrade shall be graded, re-compacted, and retested. The Contractor shall protect the subgrade from damage and limit hauling over the finished subgrade to only traffic essential for construction purposes.

The Contractor shall maintain the completed course in satisfactory condition throughout placement of subsequent layers. No subbase, base, or surface course shall be placed on the subgrade until the subgrade has been accepted by the RPR.

**152-2.12 HAUL.** All hauling will be considered a necessary and incidental part of the work. The Contractor shall include the cost in the contract unit price for the pay of items of work involved. No payment will be made separately or directly for hauling on any part of the work.

The Contractor's equipment shall not cause damage to any excavated surface, compacted lift or to the subgrade as a result of hauling operations. Any damage caused as a result of the Contractor's hauling operations shall be repaired at the Contractor's expense.

The Contractor shall be responsible for providing, maintaining and removing any haul roads or routes within or outside of the work area, and shall return the affected areas to their former condition, unless otherwise authorized in writing by the Owner. No separate payment will be made for any work or materials associated with providing, maintaining and removing haul roads or routes.

**152-2.13 SURFACE TOLERANCES.** In those areas on which a subbase or base course is to be placed, the surface shall be tested for smoothness and accuracy of grade and crown. Any portion lacking the required smoothness or failing in accuracy of grade or crown shall be scarified to a depth of at least 3 inches (75 mm), reshaped and re-compacted to grade until the required smoothness and accuracy are obtained and approved by the RPR. The Contractor shall perform all final smoothness and grade checks in the presence of the RPR. Any deviation in surface tolerances shall be corrected by the Contractor at the Contractor's expense.

- a. **Smoothness.** The finished surface shall not vary more than  $\pm \frac{1}{2}$  inch (12 mm) when tested with a 12-foot (3.7-m) straightedge applied parallel with and at right angles to the centerline. The straightedge shall be moved continuously forward at half the length of the 12-foot (3.7-m) straightedge for the full length of each line on a 50-foot (15-m) grid.
- b. **Grade.** The grade and crown shall be measured on a 50-foot (15-m) grid and shall be within  $\pm 0.05$  feet (15 mm) of the specified grade.

On safety areas, turfed areas and other designated areas within the grading limits where no subbase or base is to be placed, grade shall not vary more than 0.10 feet (30 mm) from specified grade. Any deviation in excess of this amount shall be corrected by loosening, adding or removing materials, and reshaping.

**152-2.14 TOPSOIL.** When topsoil is specified or required as shown on the plans or under Item T-905, it shall be salvaged from stripping or other grading operations. The topsoil shall meet the



requirements of Item T-905. If, at the time of excavation or stripping, the topsoil cannot be placed in its final section of finished construction, the material shall be stockpiled at approved locations. Stockpiles shall be located as shown on the plans and the approved CSPP, and shall not be placed on areas that subsequently will require any excavation or embankment fill. If, in the judgment of the RPR, it is practical to place the salvaged topsoil at the time of excavation or stripping, the material shall be placed in its final position without stockpiling or further re-handling.

Upon completion of grading operations, stockpiled topsoil shall be handled and placed as shown on the plans and as required in Item T-905. Topsoil shall be paid for as provided in Item T-905. No direct payment will be made for topsoil under Item P-152.

## **METHOD OF MEASUREMENT**

**152-3.1** Measurement for payment specified by the cubic yard (cubic meter) shall be computed by the average end areas of design cross sections or the comparison of digital terrain model (DTM) surfaces for computation of neat line design quantities. The end area is that bound by the original ground line established by field cross-sections and the final theoretical pay line established by cross-sections shown on the plans, subject to verification by the RPR.

**152-3.1** The quantity of unclassified excavation to be paid for shall be the number of cubic yards (cubic meters) measured in its original position. Measurement shall not include the quantity of materials excavated without authorization beyond normal slope lines, or the quantity of material used for purposes other than those directed.

**152-3.2** The quantity of embankment in place shall be the number of cubic yards (cubic meters) measured in its final position.

## **BASIS OF PAYMENT**

**152-4.1** Unclassified excavation payment shall be made at the contract unit price per cubic yard (cubic meter). This price shall be full compensation for furnishing all materials, labor, equipment, tools, and incidentals necessary to complete the item.

**152-4.2** For embankment in place, payment shall be made at the contract unit price per cubic yard (cubic meter). This price shall be full compensation for furnishing all materials, labor, equipment, tools, and incidentals necessary to complete the item.

**152-4.3** Stockpiled material shall be paid for on the basis of the number of cubic yards (cubic meters) measured in the stockpiled position.

Payment will be made under:

|                |                     |                                |
|----------------|---------------------|--------------------------------|
| Item P-152-4.1 | Unclassified        | - per Cubic Yard (Cubic Meter) |
| Item P-152-4.2 | Embankment in place | - per Cubic Yard (Cubic Meter) |

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

American Association of State Highway and Transportation Officials (AASHTO)

|              |                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------|
| AASHTO T-180 | Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop |
|--------------|---------------------------------------------------------------------------------------------------------------------------|

ASTM International (ASTM)

|           |                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D698 | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft <sup>3</sup> (600 kN-m/m <sup>3</sup> )) |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| ASTM D1556 | Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method |
|------------|-------------------------------------------------------------------------------------------|

|            |                                                                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D1557 | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft <sup>3</sup> (2700 kN-m/m <sup>3</sup> )) |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| ASTM D6938 | Standard Test Methods for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth) |
|------------|----------------------------------------------------------------------------------------------------------------------------|

Advisory Circulars (AC)

|               |                                                             |
|---------------|-------------------------------------------------------------|
| AC 150/5370-2 | Operational Safety on Airports During Construction Software |
|---------------|-------------------------------------------------------------|

Software

FAARFIELD – FAA Rigid and Flexible Iterative Elastic Layered Design

U.S. Department of Transportation

|              |                                                                 |
|--------------|-----------------------------------------------------------------|
| FAA RD-76-66 | Design and Construction of Airport Pavements on Expansive Soils |
|--------------|-----------------------------------------------------------------|

**END OF ITEM P-152**

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## ITEM P-154 SUBBASE COURSE

### DESCRIPTION

**154-1.1** This item shall consist of a subbase course composed of granular materials constructed on a prepared subgrade or underlying course in accordance with these specifications, and in conformity with the dimensions and typical cross-section shown on the plans.

### MATERIALS

**154-2.1 MATERIALS.** The subbase material shall consist of hard durable particles or fragments of granular aggregates. The material may be obtained from gravel pits, stockpiles, or may be produced from a crushing and screening plant with proper blending. The materials from these sources shall meet the requirements for gradation, quality, and consistency. The material shall be free from vegetative matter, excessive amounts of clay, and other objectionable substances; uniformly blended; and be capable of being compacted into a dense, stable subbase.

The subbase material shall exhibit a California Bearing Ratio (CBR) value of at least 20 when tested in accordance with ASTM D1883. The subbase material shall meet the gradation specified in the table below.

**Subbase Gradation Requirements**

| Sieve designation    | Percentage by weight passing sieves |                                | Contractor's Final Gradation | Job Control Grading Band Tolerances <sup>1</sup> (Percent) |
|----------------------|-------------------------------------|--------------------------------|------------------------------|------------------------------------------------------------|
|                      | Subbase Aggregate                   | Recycled pavement (RAP or RCO) |                              |                                                            |
| 3 inch (75 mm)       | 100                                 |                                |                              | 0                                                          |
| 1 1/2 inch (37.5 mm) |                                     | 100                            |                              | 0                                                          |
| 3/4 inch (19.0 mm)   | 70-100                              | 70-100                         |                              | ±10                                                        |
| No. 10 (2.00 mm)     | 20-100                              | 20-100                         |                              | ±10                                                        |
| No. 40 (425 µm)      | 5-60                                | 5-60                           |                              | ±5                                                         |
| No. 200 (75 µm)      | 0-15                                | 0-15                           |                              | ±5                                                         |

<sup>1</sup>The "Job Control Grading Band Tolerances" shall be applied to "Contractor's Final Gradation" to establish the job control grading band.

The portion of the material passing the No. 40 (425  $\mu$ m) sieve shall have a liquid limit of not more than 25 and a plasticity index of not more than six (6) when tested in accordance with ASTM D4318.

#### **154-2.2 SAMPLING AND TESTING.**

**a. Aggregate base materials.** Samples shall be taken by the Contractor per ASTM D75 for initial aggregate subbase requirements and gradation. Material shall meet the requirements in paragraphs 154-2.1. The Contractor shall submit to the Resident Project Representative (RPR) certified test results showing that the aggregate meets the Material requirements of this section. Tests shall be representative of the material to be used for the project.

**b. Gradation requirements.** The Contractor shall take at least one aggregate subbase sample per day in the presence of the RPR to check the final gradation. Samples shall be taken from the in-place, un-compacted material at sampling locations determined by the RPR on a random basis per ASTM D3665. Sampling shall be per ASTM D75 and tested per ASTM C136 and ASTM C117. Results shall be furnished to the RPR by the Contractor each day during construction. Material shall meet the requirements in paragraph 154-2.1.

**154-2.3 SEPARATION GEOTEXTILE.** Not used.

**154-2.4 GEOGRID.** Not used.

### **CONSTRUCTION METHODS**

**154-3.1 GENERAL.** The subbase course shall be placed where designated on the plans or as directed by the RPR. The material shall be shaped and thoroughly compacted within the tolerances specified.

Granular subbases which, due to grain sizes or shapes, are not sufficiently stable to support the construction equipment without movement, shall be mechanically modified to the depth necessary to provide stability as directed by the RPR. The mechanical modification shall include the addition of a fine-grained medium to bind the particles of the subbase material sufficiently to furnish a bearing strength, so the course will not deform under construction equipment traffic.

**154-3.2 PREPARING UNDERLYING COURSE.** Prior to constructing the subbase course, clean the underlying course or subgrade of all foreign substances. The surface of the underlying course or subgrade shall meet specified compaction and surface tolerances in accordance with Item P-152. Correct ruts, soft yielding spots in the underlying courses, and subgrade areas having inadequate compaction and/or deviations of the surface from the specified requirements, by loosening and removing soft or unsatisfactory material, adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in ASTM D2487, the surface shall be stabilized prior to placement of the overlying course by mixing the overlying course material into the underlying course, and compacting by approved methods. The stabilized material shall be considered as part of the underlying course and shall meet all requirements for the underlying course. The finished underlying course shall not be disturbed by traffic or other operations and shall be maintained in a satisfactory condition until the overlying course is placed. The underlying course shall be checked and accepted by the RPR before placing and spreading operations are started.

To protect the subgrade and to ensure proper drainage, spreading of the subbase shall begin along the centerline of the pavement on a crowned section or on the high side of pavements with a one-way slope.

**154-3.3 CONTROL STRIP.** The first half-day of subbase construction shall be considered as a control strip for the Contractor to demonstrate, in the presence of the RPR, that the materials, equipment, and construction processes meet the requirements of this specification. The sequence and manner of rolling necessary to obtain specified density requirements shall be determined. The maximum compacted thickness may be increased to a maximum of 12 inches (300 mm) upon the Contractor's demonstration that approved equipment and operations will uniformly compact the lift to the specified density. The RPR must witness this demonstration and approve the lift thickness prior to full production.

Control strips that do not meet specification requirements shall be reworked, re-compacted, or removed and replaced at the Contractor's expense. Full operations shall not begin until the control strip has been accepted by the RPR. The Contractor shall use the same equipment, materials, and construction methods for the remainder of construction, unless adjustments made by the Contractor are approved in advance by the RPR.

**154-3.4 PLACEMENT.** The material shall be placed and spread on the prepared underlying layer by spreader boxes or other devices as approved by the RPR, to a uniform thickness and width. The equipment shall have positive thickness controls to minimize the need for additional manipulation of the material. Dumping from vehicles that require re-handling shall not be permitted. Hauling over the uncompacted base course shall not be permitted. The material shall not be placed when the underlying course is soft or yielding.

The material shall meet gradation and moisture requirements prior to compaction. Material may be free-draining and the minimum moisture content shall be established for placement and compaction of the material.

The material shall be constructed in lifts as established in the control strip, but not less than 4 inches (100 mm) nor more than 12 inches (300 mm) of compacted thickness.

When more than one lift is required to establish the layer thickness shown on the plans, the construction procedure described here shall apply to each lift. No lift shall be covered by subsequent lifts until tests verify that compaction requirements have been met. The Contractor shall rework, re-compact and retest any material placed which does not meet the specifications.

**154-3.5 COMPACTION.** The subbase material shall be compacted, adjusting moisture as necessary, to be within  $\pm 2\%$  of optimum moisture. The field density of the compacted material shall be at least **100%** of the maximum density as specified in paragraph 154-3.9a. If the specified density is not attained, the area of the lift represented by the test shall be reworked and/or re-compacted and additional random tests made. This procedure shall be followed until the specified density is reached. Maximum density refers to maximum dry density at optimum moisture content unless otherwise specified.

**154-3.6 WEATHER LIMITATION.** Material shall not be placed unless the ambient air temperature is at least 40°F (4°C) and rising. Work on subbase course shall not be conducted when the subgrade is wet or frozen or the subbase material contains frozen material.

**154-3.7 MAINTENANCE.** No base or surface course shall be placed on the subbase until the subbase has been accepted by the RPR. The Contractor shall maintain the completed course in satisfactory condition throughout placement of subsequent layers. When material has been exposed to excessive rain, snow, or freeze-thaw conditions, the Contractor shall verify that materials still meet all specification requirements before placement of additional material. Equipment may be routed over completed sections of subbase course, provided the equipment does not damage the subbase course and the equipment is routed over the full width of the completed subbase course. Any damage to the subbase course from routing equipment over the subbase course shall be repaired by the Contractor at their expense.

**154-3.8 SURFACE TOLERANCE.** In those areas on which a subbase or base course is to be placed, the surface shall be tested for smoothness and accuracy of grade and crown. Any portion lacking the required smoothness or failing in accuracy of grade or crown shall be scarified to a depth of at least 3 inches (75 mm), reshaped and re-compacted to grade until the required smoothness and accuracy are obtained and approved by the RPR. The Contractor shall perform all final smoothness and grade checks in the presence of the RPR. Any deviation in surface tolerances shall be corrected by the Contractor at the Contractor's expense.

**a. Smoothness.** The finished surface shall not vary more than  $\pm \frac{1}{2}$  inch (12 mm) when tested with a 12-foot (3.7-m) straightedge applied parallel with and at right angles to the centerline. The straightedge shall be moved continuously forward at half the length of the 12-foot (3.7-m) straightedge for the full length of each line on a 50-foot (15-m) grid.

**b. Grade.** The grade and crown shall be measured on a 50-foot (15-m) grid and shall be within  $\pm 0.05$  feet (15 mm) of the specified grade.

**154-3.9 ACCEPTANCE SAMPLING AND TESTING.** The aggregate base course shall be accepted for density and thickness on an area basis. Two test shall be made for density and thickness for each **1200 square yards (1000 square meters)**. Sampling locations will be determined on a random basis per ASTM D3665.

**a. Density.** The Contractor's laboratory shall perform all density tests in the RPR's presence and provide the test results upon completion to the RPR for acceptance.

Each area shall be accepted for density when the field density is at least **100%** of the maximum density of laboratory specimens compacted and tested per ASTM D1557. The in-place field density shall be determined per ASTM D1556. If the specified density is not attained, the area represented by the failed test shall be reworked and/or recompacted and two additional random tests made. This procedure shall be followed until the specified density is reached. Maximum density refers to maximum dry density at optimum moisture content unless otherwise specified.

When the material has greater than 30 percent retained on the  $\frac{3}{4}$  inch (19.0 mm) sieve, use methods in ASTM D1557 and the procedures in AASHTO T180 Annex for correction of maximum dry density and optimum moisture for oversized particles.

**b. Thickness.** The thickness of the base course shall be within  $+0$  and  $-1/2$  inch (12 mm) of the specified thickness as determined by depth tests taken by the Contractor in the presence of the RPR for each area. Where the thickness is deficient by more than  $1/2$ -inch (12 mm), the Contractor shall correct such areas at no additional cost by scarifying to a depth of at least 3 inches (75 mm), adding new material of proper gradation, and the material shall be blended and

recompacted to grade. The Contractor shall replace, at his expense, base material where depth tests have been taken.

## METHOD OF MEASUREMENT

**154-4.1** Subbase course shall be measured by the number of square yards (square meters) of subbase course material placed and compacted to specified density and plan thickness requirements in the completed course. The quantity of subbase course material shall be measured in final position based upon survey of the completed work computed from elevations to the nearest 0.01 foot (3 mm). On individual depth measurements, thicknesses more than 1/2 inch (12 mm) in excess of that shown on the plans shall be considered as the specified thickness plus 1/2 inch (12 mm) in computing the yardage for payment. Subbase materials shall not be included in any other excavation quantities.

## BASIS OF PAYMENT

**154-5.1** Payment shall be made at the contract unit price per **square** yard (square meter) for subbase course. This price shall be full compensation for furnishing all materials; for all preparation, hauling, and placing of these materials; and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

|                |                |                                  |
|----------------|----------------|----------------------------------|
| Item P-154-5.1 | Subbase Course | - per Square Yard (Square Meter) |
|----------------|----------------|----------------------------------|

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C117  | Standard Test Method for Materials Finer than 75- $\mu$ m (No. 200) Sieve in Mineral Aggregates by Washing                                               |
| ASTM C136  | Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates                                                                                    |
| ASTM D75   | Standard Practice for Sampling Aggregates                                                                                                                |
| ASTM D698  | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft <sup>3</sup> (600 kN-m/m <sup>3</sup> )) |
| ASTM D1556 | Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method                                                                |



|            |                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D1557 | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft <sup>3</sup> (2,700 kN-m/m <sup>3</sup> )) |
| ASTM D2487 | Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)                                                |
| ASTM D4253 | Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table                                                           |
| ASTM D4759 | Practice for Determining the Specification Conformance of Geosynthetics                                                                                    |
| ASTM D4318 | Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                                                                       |
| ASTM D6938 | Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)                                  |

American Association of State Highway and Transportation Officials (AASHTO)

|       |                                                   |
|-------|---------------------------------------------------|
| M 288 | Geotextile Specification for Highway Applications |
|-------|---------------------------------------------------|

**END OF ITEM P-154**

### Item P-209 Crushed Aggregate Base Course

#### DESCRIPTION

**209-1.1** This item consists of a base course composed of crushed aggregate base constructed on a prepared course in accordance with these specifications and in conformity to the dimensions and typical cross-sections shown on the plans.

#### MATERIALS

**209-2.1 Crushed aggregate base.** Crushed aggregate shall consist of clean, sound, durable particles of crushed stone, crushed gravel, and shall be free from coatings of clay, silt, organic material, clay lumps or balls or other deleterious materials or coatings. The method used to produce the crushed gravel shall result in the fractured particles in the finished product as consistent and uniform as practicable. Fine aggregate portion, defined as the portion passing the No. 4 (4.75 mm) sieve shall consist of fines from the coarse aggregate crushing operation. The fine aggregate shall be produced by crushing stone, gravel, that meet the coarse aggregate requirements for wear and soundness. Aggregate base material requirements are listed in the following table.

**Crushed Aggregate Base Material Requirements**

| Material Test                                                                | Requirement                                                                                                                | Standard   |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Coarse Aggregate</b>                                                      |                                                                                                                            |            |
| Resistance to Degradation                                                    | Loss: 45% maximum                                                                                                          | ASTM C131  |
| Soundness of Aggregates by Use of Sodium Sulfate <b>or</b> Magnesium Sulfate | Loss after 5 cycles:<br>12% maximum using Sodium sulfate - or -<br>18% maximum using magnesium sulfate                     | ASTM C88   |
| Percentage of Fractured Particles                                            | Minimum 90% by weight of particles with at least two fractured faces and 98% with at least one fractured face <sup>1</sup> | ASTM D5821 |
| Flat Particles, Elongated Particles, or Flat and Elongated Particles         | 10% maximum, by weight, of flat, elongated, or flat and elongated particles <sup>2</sup>                                   | ASTM D4791 |
| <b>Fine Aggregate</b>                                                        |                                                                                                                            |            |
| Liquid limit                                                                 | Less than or equal to 25                                                                                                   | ASTM D4318 |
| Plasticity Index                                                             | Not more than five (5)                                                                                                     | ASTM D4318 |

<sup>1</sup> The area of each face shall be equal to at least 75% of the smallest mid-sectional area of the piece. When two fractured faces are contiguous, the angle between the planes of fractures shall be at least 30 degrees to count as two fractured faces.

<sup>2</sup> A flat particle is one having a ratio of width to thickness greater than five (5); an elongated particle is one having a ratio of length to width greater than five (5).

**209-2.2 Gradation requirements.** The gradation of the aggregate base material shall meet the requirements of the gradation given in the following table when tested per ASTM C117 and ASTM C136. The gradation shall be well graded from coarse to fine and shall not vary from the lower limit on one sieve to the high limit on an adjacent sieve or vice versa.

**Gradation of Aggregate Base**

| Sieve Size                      | Design Range<br>Percentage by Weight<br>passing | Contractor's Final<br>Gradation | Job Control Grading Band<br>Tolerances <sup>1</sup><br>(Percent) |
|---------------------------------|-------------------------------------------------|---------------------------------|------------------------------------------------------------------|
| 2 inch<br>(50 mm)               | 100                                             |                                 | 0                                                                |
| 1-1/2 inch<br>(37.5 mm)         | 95-100                                          |                                 | ±5                                                               |
| 1 inch<br>(25.0 mm)             | 70-95                                           |                                 | ±8                                                               |
| 3/4 inch<br>(19.0 mm)           | 55-85                                           |                                 | ±8                                                               |
| No. 4<br>(4.75 mm)              | 30-60                                           |                                 | ±8                                                               |
| No. 40 <sup>2</sup><br>(425 µm) | 10-30                                           |                                 | ±5                                                               |
| No. 200 <sup>2</sup><br>(75 µm) | 0-10                                            |                                 | ±3                                                               |

<sup>1</sup> The "Job Control Grading Band Tolerances for Contractor's Final Gradation" in the table shall be applied to "Contractor's Final Gradation" to establish a job control grading band. The full tolerance still applies if application of the tolerances results in a job control grading band outside the design range.

<sup>2</sup> The fraction of material passing the No 200 (75 µm) sieve shall not exceed two-thirds the fraction passing the No 40 (425 µm) sieve.

### **209-2.3 Sampling and Testing.**

**a. Aggregate base materials.** The Contractor shall take samples of the aggregate base in accordance with ASTM D75 to verify initial aggregate base requirements and gradation. Material shall meet the requirements in paragraph 209-2.1. This sampling and testing will be the basis for approval of the aggregate base quality requirements.

**b. Gradation requirements.** The Contractor shall take at least **two** aggregate base samples per day in the presence of the Resident Project Representative (RPR) to check the final gradation. Sampling shall be per ASTM D75. Material shall meet the requirements in paragraph 209-2.2. The samples shall be taken from the in-place, un-compacted material at sampling points and intervals designated by the RPR.

**209-2.4 Separation Geotextile.** Not used.

## **CONSTRUCTION METHODS**

**209-3.1 Control strip.** The first half-day of construction shall be considered the control strip. The Contractor shall demonstrate, in the presence of the RPR, that the materials, equipment, and construction processes meet the requirements of the specification. The sequence and manner of rolling necessary to

obtain specified density requirements shall be determined. The maximum compacted thickness may be increased to a maximum of 12 inches (300 mm) upon the Contractor's demonstration that approved equipment and operations will uniformly compact the lift to the specified density. The RPR must witness this demonstration and approve the lift thickness prior to full production.

Control strips that do not meet specification requirements shall be reworked, re-compacted or removed and replaced at the Contractor's expense. Full operations shall not continue until the control strip has been accepted by the RPR. The Contractor shall use the same equipment, materials, and construction methods for the remainder of construction, unless adjustments made by the Contractor are approved by the RPR.

**209-3.2 Preparing underlying subgrade and/or subbase.** The underlying subgrade and/or subbase shall be checked and accepted by the RPR before base course placing and spreading operations begin. Re-proof rolling of the subgrade or proof rolling of the subbase in accordance with Item P-152, at the Contractor's expense, may be required by the RPR if the Contractor fails to ensure proper drainage or protect the subgrade and/or subbase. Any ruts or soft, yielding areas due to improper drainage conditions, hauling, or any other cause, shall be corrected before the base course is placed. To ensure proper drainage, the spreading of the base shall begin along the centerline of the pavement on a crowned section or on the high side of the pavement with a one-way slope.

**209-3.3 Production.** The aggregate shall be uniformly blended and, when at a satisfactory moisture content per paragraph 209-3.5, the approved material may be transported directly to the placement.

**209-3.4 Placement.** The aggregate shall be placed and spread on the prepared underlying layer by spreader boxes or other devices as approved by the RPR, to a uniform thickness and width. The equipment shall have positive thickness controls to minimize the need for additional manipulation of the material. Dumping from vehicles that require re-handling shall not be permitted. Hauling over the uncompacted base course shall not be permitted.

The aggregate shall meet gradation and moisture requirements prior to compaction. The base course shall be constructed in lifts as established in the control strip, but not less than 4 inches (100 mm) nor more than 12 inches (300 mm) of compacted thickness.

When more than one lift is required to establish the layer thickness shown on the plans, the construction procedure described here shall apply to each lift. No lift shall be covered by subsequent lifts until tests verify that compaction requirements have been met. The Contractor shall rework, re-compact and retest any material placed which does not meet the specifications at the Contractor's expense.

**209-3.5 Compaction.** Immediately after completion of the spreading operations, compact each layer of the base course, as specified, with approved compaction equipment. The number, type, and weight of rollers shall be sufficient to compact the material to the required density within the same day that the aggregate is placed on the subgrade.

The field density of each compacted lift of material shall be at least 100% of the maximum density of laboratory specimens prepared from samples of the base material delivered to the jobsite. The laboratory specimens shall be compacted and tested in accordance with ASTM D1557. The moisture content of the material during placing operations shall be within  $\pm 2$  percentage points of the optimum moisture content as determined by ASTM D1557. Maximum density refers to maximum dry density at optimum moisture content unless otherwise specified.

**209-3.6 Weather limitations.** Material shall not be placed unless the ambient air temperature is at least 40°F (4°C) and rising. Work on base course shall not be conducted when the subgrade or subbase is wet or frozen or the base material contains frozen material.

**209-3.7 Maintenance.** The base course shall be maintained in a condition that will meet all specification requirements. When material has been exposed to excessive rain, snow, or freeze-thaw conditions, prior

to placement of additional material, the Contractor shall verify that materials still meet all specification requirements. Equipment may be routed over completed sections of base course, provided that no damage results and the equipment is routed over the full width of the completed base course. Any damage resulting to the base course from routing equipment over the base course shall be repaired by the Contractor at the Contractor's expense.

**209-3.8 Surface tolerances.** After the course has been compacted, the surface shall be tested for smoothness and accuracy of grade and crown. Any portion lacking the required smoothness or failing in accuracy of grade or crown shall be scarified to a depth of at least 3 inches (75 mm), reshaped and recompact to grade until the required smoothness and accuracy are obtained and approved by the RPR. Any deviation in surface tolerances shall be corrected by the Contractor at the Contractor's expense. The smoothness and accuracy requirements specified here apply only to the top layer when base course is constructed in more than one layer.

**a. Smoothness.** The finished surface shall not vary more than 3/8-inch (9 mm) when tested with a 12-foot (3.7-m) straightedge applied parallel with and at right angles to the centerline. The straightedge shall be moved continuously forward at half the length of the 12-foot (3.7-m) straightedge for the full length of each line on a 50-foot (15-m) grid.

**b. Grade.** The grade and crown shall be measured on a 50-foot (15-m) grid and shall be within +0 and -1/2 inch (12 mm) of the specified grade.

**209-3.9 Acceptance sampling and testing.** Crushed aggregate base course shall be accepted for density and thickness on an area basis. Two tests shall be made for density and thickness for each 1200 square yds. Sampling locations will be determined on a random basis per ASTM D3665

**a. Density.** The Contractor's laboratory shall perform all density tests in the RPR's presence and provide the test results upon completion to the RPR for acceptance.

Each area shall be accepted for density when the field density is at least 100% of the maximum density of laboratory specimens compacted and tested per ASTM 1557. The in-place field density shall be determined per ASTM D1556. If the specified density is not attained, the area represented by the failed test must be reworked and/or recompact and two additional random tests made. This procedure shall be followed until the specified density is reached. Maximum density refers to maximum dry density at optimum moisture content unless otherwise specified.

**b. Thickness.** Depth tests shall be made by test holes at least 3 inches (75 mm) in diameter that extend through the base. The thickness of the base course shall be within +0 and -1/2 inch (12 mm) of the specified thickness as determined by depth tests taken by the Contractor in the presence of the RPR for each area. Where the thickness is deficient by more than 1/2-inch (12 mm), the Contractor shall correct such areas at no additional cost by scarifying to a depth of at least 3 inches (75 mm), adding new material of proper gradation, and the material shall be blended and recompact to grade. The Contractor shall replace, at his expense, base material where depth tests have been taken.

## METHOD OF MEASUREMENT

**209-4.1** The quantity of crushed aggregate base course will be determined by measurement of the number of square yards of material actually constructed and accepted by the RPR as complying with the plans and specifications. Base materials shall not be included in any other excavation quantities.

### BASIS OF PAYMENT

**209-5.1** Payment shall be made at the contract unit price per square yard for crushed aggregate base course. This price shall be full compensation for furnishing all materials, for preparing and placing these materials, and for all labor, equipment tools, and incidentals necessary to complete the item.

Payment will be made under:

|                |                                                 |
|----------------|-------------------------------------------------|
| Item P-209-5.1 | Crushed Aggregate Base Course - per square yard |
|----------------|-------------------------------------------------|

### REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C29   | Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate                                                                              |
| ASTM C88   | Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate                                                            |
| ASTM C117  | Standard Test Method for Materials Finer than 75- $\mu$ m (No. 200) Sieve in Mineral Aggregates by Washing                                                |
| ASTM C131  | Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine                       |
| ASTM C136  | Standard Test Method for Sieve or Screen Analysis of Fine and Coarse Aggregates                                                                           |
| ASTM C142  | Standard Test Method for Clay Lumps and Friable Particles in Aggregates                                                                                   |
| ASTM D75   | Standard Practice for Sampling Aggregates                                                                                                                 |
| ASTM D698  | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft <sup>3</sup> (600 kN-m/m <sup>3</sup> ))  |
| ASTM D1556 | Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method                                                                 |
| ASTM D1557 | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft <sup>3</sup> (2700 kN-m/m <sup>3</sup> )) |
| ASTM D2167 | Standard Test Method for Density and Unit Weight of Soil in Place by the Rubber Balloon Method                                                            |
| ASTM D2419 | Standard Test Method for Sand Equivalent Value of Soils and Fine Aggregate                                                                                |
| ASTM D3665 | Standard Practice for Random Sampling of Construction Materials                                                                                           |
| ASTM D4318 | Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                                                                      |
| ASTM D4491 | Standard Test Methods for Water Permeability of Geotextiles by Permittivity                                                                               |

|                                                                             |                                                                                                                                     |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D4643                                                                  | Standard Test Method for Determination of Water Content of Soil and Rock by Microwave Oven Heating                                  |
| ASTM D4751                                                                  | Standard Test Methods for Determining Apparent Opening Size of a Geotextile                                                         |
| ASTM D4791                                                                  | Standard Test Method for Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate                   |
| ASTM D5821                                                                  | Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate                                      |
| ASTM D6938                                                                  | Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)           |
| ASTM D7928                                                                  | Standard Test Method for Particle-Size Distribution (Gradation) of Fine-Grained Soils Using the Sedimentation (Hydrometer) Analysis |
| American Association of State Highway and Transportation Officials (AASHTO) |                                                                                                                                     |
| M288                                                                        | Standard Specification for Geosynthetic Specification for Highway Applications                                                      |

**END OF ITEM P-209**

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## Part 6 – Flexible Pavements

### Item P-401 Asphalt Mix Pavement

#### DESCRIPTION

**401-1.1** This item shall consist of pavement courses composed of mineral aggregate and asphalt binder mixed in a central mixing plant and placed on a prepared base or stabilized course in accordance with these specifications and shall conform to the lines, grades, thicknesses, and typical cross-sections shown on the plans. Each course shall be constructed to the depth, typical section, and elevation required by the plans and shall be rolled, finished, and approved before the placement of the next course.

#### MATERIALS

**401-2.1 Aggregate.** Aggregates shall consist of crushed stone, crushed gravel, crushed slag, screenings, natural sand, and mineral filler, as required. The aggregates should have no known history of detrimental pavement staining due to ferrous sulfides, such as pyrite. Coarse aggregate is the material retained on the No. 4 (4.75 mm) sieve. Fine aggregate is the material passing the No. 4 (4.75 mm) sieve.

**a. Coarse aggregate.** Coarse aggregate shall consist of sound, tough, durable particles, free from films of matter that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances. Coarse aggregate material requirements are given in the table below.

**Coarse Aggregate Material Requirements**

| <b>Material Test</b>                                                  | <b>Requirement</b>                                                                                                                                                                                                   | <b>Standard</b> |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Resistance to Degradation                                             | Loss: 40% maximum                                                                                                                                                                                                    | ASTM C131       |
| Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate | Loss after 5 cycles:<br>12% maximum using Sodium sulfate - or -<br>18% maximum using magnesium sulfate                                                                                                               | ASTM C88        |
| Clay lumps and friable particles                                      | 1.0 % maximum                                                                                                                                                                                                        | ASTM C142       |
| Percentage of Fractured Particles                                     | For pavements designed for aircraft gross weights of 60,000 pounds (27200 kg) or more:<br>Minimum 75% by weight of particles with at least two fractured faces and 85% with at least one fractured face <sup>1</sup> | ASTM D5821      |
|                                                                       | For pavements designed for aircraft gross weights less than 60,000 pounds (27200 kg):<br>Minimum 50% by weight of particles with at least two fractured faces and 65% with at least one fractured face <sup>1</sup>  |                 |
| Flat, Elongated, or Flat and Elongated Particles                      | 8% maximum, by weight, of flat, elongated, or flat and elongated particles at 5:1 <sup>2</sup>                                                                                                                       | ASTM D4791      |
| Bulk density of slag <sup>3</sup>                                     | Weigh not less than 70 pounds per cubic foot (1.12 Mg/cubic meter)                                                                                                                                                   | ASTM C29.       |

<sup>1</sup> The area of each face shall be equal to at least 75% of the smallest mid-sectional area of the piece. When two fractured faces are contiguous, the angle between the planes of fractures shall be at least 30 degrees to count as two fractured faces.

<sup>2</sup> A flat particle is one having a ratio of width to thickness greater than five (5); an elongated particle is one having a ratio of length to width greater than five (5).

<sup>3</sup> Only required if slag is specified.

**b. Fine aggregate.** Fine aggregate shall consist of clean, sound, tough, durable, angular shaped particles produced by crushing stone, slag, or gravel and shall be free from coatings of clay, silt, or other objectionable matter. Natural (non-manufactured) sand may be used to obtain the gradation of the fine aggregate blend or to improve the workability of the mix. Fine aggregate material requirements are listed in the table below.

### Fine Aggregate Material Requirements

| Material Test                                                         | Requirement                                                                                            | Standard   |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------|
| Liquid limit                                                          | 25 maximum                                                                                             | ASTM D4318 |
| Plasticity Index                                                      | 4 maximum                                                                                              | ASTM D4318 |
| Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate | Loss after 5 cycles:<br>10% maximum using Sodium sulfate - or -<br>15% maximum using magnesium sulfate | ASTM C88   |
| Clay lumps and friable particles                                      | 0.3% maximum                                                                                           | ASTM C142  |
| Sand equivalent                                                       | 45 minimum                                                                                             | ASTM D2419 |

**c. Sampling.** ASTM D75 shall be used in sampling coarse and fine aggregate.

**401-2.2 Mineral filler.** Mineral filler (baghouse fines) may be added in addition to material naturally present in the aggregate. Mineral filler shall meet the requirements of ASTM D242.

### Mineral Filler Requirements

| Material Test    | Requirement | Standard   |
|------------------|-------------|------------|
| Plasticity Index | 4 maximum   | ASTM D4318 |

**401-2.3 Asphalt binder.** Asphalt binder shall conform to ASTM D6373 Performance Grade (PG) 76-22.

**401-2.4 Anti-stripping agent.** Any anti-stripping agent or additive (anti-strip) shall be heat stable and shall not change the asphalt binder grade beyond specifications. Anti-strip shall be an approved material of the Department of Transportation of the State in which the project is located.

## COMPOSITION

**401-3.1 Composition of mixture(s).** The asphalt mix shall be composed of a mixture of aggregates, filler and anti-strip agent if required, and asphalt binder. The aggregate fractions shall be sized, handled in separate size groups, and combined in such proportions that the resulting mixture meets the grading requirements of the job mix formula (JMF).

**401-3.2 Job mix formula (JMF) laboratory.** The laboratory used to develop the JMF shall possess a current certificate of accreditation, listing D3666 from a national accrediting authority and all test methods required for developing the JMF; and be listed on the accrediting authority's website. A copy of the laboratory's current accreditation and accredited test methods shall be submitted to the Resident Project Representative (RPR) prior to start of construction.

**401-3.3 Job mix formula (JMF).** No asphalt mixture shall be placed until an acceptable mix design has been submitted to the RPR for review and accepted in writing. The RPR's review shall not relieve the Contractor of the responsibility to select and proportion the materials to comply with this section.

When the project requires asphalt mixtures of differing aggregate gradations and/or binders, a separate JMF shall be submitted for each mix. Add anti-stripping agent to meet tensile strength requirements.

The JMF shall be prepared by an accredited laboratory that meets the requirements of paragraph 401-3.2. The asphalt mixture shall be designed using procedures contained in Asphalt Institute MS-2 Mix Design Manual, 7th Edition. Samples shall be prepared and compacted using the gyratory compactor in accordance with ASTM D6925.

Should a change in sources of materials be made, a new JMF must be submitted to the RPR for review and accepted in writing before the new material is used. After the initial production JMF has been approved by the RPR and a new or modified JMF is required for whatever reason, the subsequent cost of the new or modified JMF, including a new control strip when required by the RPR, will be borne by the Contractor.

The RPR may request samples at any time for testing, prior to and during production, to verify the quality of the materials and to ensure conformance with the applicable specifications.

The JMF shall be submitted in writing by the Contractor at least [ 30 ] days prior to the start of paving operations. The JMF shall be developed within the same construction season using aggregates proposed for project use.

The JMF shall be dated, and stamped or sealed by the responsible professional Engineer of the laboratory and shall include the following items as a minimum:

- Manufacturer's Certificate of Analysis (COA) for the asphalt binder used in the JMF in accordance with paragraph 401-2.3. Certificate of asphalt performance grade is with modifier already added, if used and must indicate compliance with ASTM D6373. For plant modified asphalt binder, certified test report indicating grade certification of modified asphalt binder.
- Manufacturer's Certificate of Analysis (COA) for the anti-stripping agent if used in the JMF in accordance with paragraph 401-2.4.
- Certified material test reports for the course and fine aggregate and mineral filler in accordance with paragraphs 401-2.1.
- Percent passing each sieve size for individual gradation of each aggregate cold feed and/or hot bin; percent by weight of each cold feed and/or hot bin used; and the total combined gradation in the JMF.
- Specific Gravity and absorption of each coarse and fine aggregate.
- Percent natural sand.
- Percent fractured faces.
- Percent by weight of flat particles, elongated particles, and flat and elongated particles (and criteria).
- Percent of asphalt.
- Number of blows or gyrations
- Laboratory mixing and compaction temperatures.
- Supplier-recommended field mixing and compaction temperatures.
- Plot of the combined gradation on a 0.45 power gradation curve.

- Graphical plots of air voids, voids in the mineral aggregate (VMA), and unit weight versus asphalt content. To achieve minimum VMA during production, the mix design needs to account for material breakdown during production.
- Tensile Strength Ratio (TSR).
- Type and amount of Anti-strip agent when used.
- Asphalt Pavement Analyzer (APA) results.
- Date the JMF was developed. Mix designs that are not dated or which are from a prior construction season shall not be accepted.

**Table 1. Asphalt Design Criteria**

| Test Property                                     | Value                                      | Test Method                                                   |
|---------------------------------------------------|--------------------------------------------|---------------------------------------------------------------|
| Number of blows or gyrations                      | 75                                         |                                                               |
| Air voids (%)                                     | 3.5                                        | ASTM D3203                                                    |
| Percent voids in mineral aggregate (VMA), minimum | See Table 2                                | ASTM D6995                                                    |
| Tensile Strength Ratio (TSR) <sup>1</sup>         | not less than 80 at a saturation of 70-80% | ASTM D4867                                                    |
| Asphalt Pavement Analyzer (APA) <sup>2,3]</sup>   | Less than 10 mm @ 4000 passes              | AASHTO T340 at 250 psi hose pressure at 64°C test temperature |

<sup>1</sup> Test specimens for TSR shall be compacted at  $7 \pm 1.0$  % air voids. In areas subject to freeze-thaw, use freeze-thaw conditioning in lieu of moisture conditioning per ASTM D4867.

<sup>2</sup> AASHTO T340 at 100 psi hose pressure at 64°C test temperature may be used in the interim. If this method is used the required Value shall be less than 5 mm @ 8000 passes

The mineral aggregate shall be of such size that the percentage composition by weight, as determined by laboratory sieves, will conform to the gradation or gradations specified in Table 2 when tested in accordance with ASTM C136 and ASTM C117.

The gradations in Table 2 represent the limits that shall determine the suitability of aggregate for use from the sources of supply; be well graded from coarse to fine and shall not vary from the low limit on one sieve to the high limit on the adjacent sieve, or vice versa.

**Table 2. Aggregate - Asphalt Pavements**

| <b>Sieve Size</b>                                           | <b>Percentage by Weight Passing Sieve</b> |
|-------------------------------------------------------------|-------------------------------------------|
| 1 inch (25.0 mm)                                            | --                                        |
| 3/4 inch (19.0 mm)                                          | 100                                       |
| 1/2 inch (12.5 mm)                                          | 90-100                                    |
| 3/8 inch (9.5 mm)                                           | 72-88                                     |
| No. 4 (4.75 mm)                                             | 53-73                                     |
| No. 8 (2.36 mm)                                             | 38-60                                     |
| No. 16 (1.18 mm)                                            | 26-48                                     |
| No. 30 (600 µm)                                             | 18-38                                     |
| No. 50 (300 µm)                                             | 11-27                                     |
| No. 100 (150 µm)                                            | 6-18                                      |
| No. 200 (75 µm)                                             | 3-6                                       |
| <b>Minimum Voids in Mineral Aggregate (VMA)<sup>1</sup></b> | 15.0                                      |
| <b>Asphalt Percent:</b>                                     |                                           |
| Stone or gravel                                             | 5.0-7.5                                   |
| Slag                                                        | 6.5-9.5                                   |
| <b>Recommended Minimum Construction Lift Thickness</b>      | 2                                         |

<sup>1</sup>To achieve minimum VMA during production, the mix design needs to account for material breakdown during production.

The aggregate gradations shown are based on aggregates of uniform specific gravity. The percentages passing the various sieves shall be corrected when aggregates of varying specific gravities are used, as indicated in the Asphalt Institute MS-2 Mix Design Manual, 7th Edition.

**401-3.4 Reclaimed asphalt pavement (RAP).** RAP shall not be used.

**401-3.5 Control Strip.** A control strip is not required.

## **CONSTRUCTION METHODS**

**401-4.1 Weather limitations.** The asphalt shall not be placed upon a wet surface or when the surface temperature of the underlying course is less than specified in Table 4. The temperature requirements may be waived by the RPR, if requested; however, all other requirements including compaction shall be met.

**Table 4. Surface Temperature Limitations of Underlying Course**

| Mat Thickness                                                    | Base Temperature (Minimum) |    |
|------------------------------------------------------------------|----------------------------|----|
|                                                                  | °F                         | °C |
| 3 inches (7.5 cm) or greater                                     | 40 <sup>1</sup>            | 4  |
| Greater than 2 inches (50 mm)<br>but less than 3 inches (7.5 cm) | 45                         | 7  |

**401-4.2 Asphalt plant.** Plants used for the preparation of asphalt shall conform to the requirements of American Association of State Highway and Transportation Officials (AASHTO) M156 including the following items.

**a. Inspection of plant.** The RPR, or RPR's authorized representative, shall have access, at all times, to all areas of the plant for checking adequacy of equipment; inspecting operation of the plant: verifying weights, proportions, and material properties; and checking the temperatures maintained in the preparation of the mixtures.

**b. Storage bins and surge bins.** The asphalt mixture stored in storage and/or surge bins shall meet the same requirements as asphalt mixture loaded directly into trucks. Asphalt mixture shall not be stored in storage and/or surge bins for a period greater than twelve (12) hours. If the RPR determines there is an excessive heat loss, segregation, or oxidation of the asphalt mixture due to temporary storage, temporary storage shall not be allowed.

**401-4.3 Aggregate stockpile management.** Aggregate stockpiles shall be constructed in a manner that prevents segregation and intermixing of deleterious materials. Aggregates from different sources shall be stockpiled, weighed and batched separately at the asphalt batch plant. Aggregates that have become segregated or mixed with earth or foreign material shall not be used.

A continuous supply of materials shall be provided to the work to ensure continuous placement.

**401-4.4 Hauling equipment.** Trucks used for hauling asphalt shall have tight, clean, and smooth metal beds. To prevent the asphalt from sticking to the truck beds, the truck beds shall be lightly coated with a minimum amount of paraffin oil, lime solution, or other material approved by the RPR. Petroleum products shall not be used for coating truck beds. Each truck shall have a suitable cover to protect the mixture from adverse weather. When necessary, to ensure that the mixture will be delivered to the site at the specified temperature, truck beds shall be insulated or heated and covers shall be securely fastened.

**401-4.4.1 Material transfer vehicle (MTV).** Material transfer vehicles used to transfer the material from the hauling equipment to the paver, shall use a self-propelled, material transfer vehicle with a swing conveyor that can deliver material to the paver without making contact with the paver. The MTV shall be able to move back and forth between the hauling equipment and the paver providing material transfer to the paver, while allowing the paver to operate at a constant speed. The Material Transfer Vehicle will have remixing and storage capability to prevent physical and thermal segregation.

**401-4.5 Asphalt pavers.** Asphalt pavers shall be self-propelled with an activated heated screed, capable of spreading and finishing courses of asphalt that will meet the specified thickness, smoothness, and grade. The paver shall have sufficient power to propel itself and the hauling equipment without adversely affecting the finished surface. The asphalt paver shall be

equipped with a control system capable of automatically maintaining the specified screed grade and elevation.

If the spreading and finishing equipment in use leaves tracks or indented areas, or produces other blemishes in the pavement that are not satisfactorily corrected by the scheduled operations, the use of such equipment shall be discontinued.

The paver shall be capable of paving to a minimum width specified in paragraph 401-4.12.

**401-4.6 Rollers.** The number, type, and weight of rollers shall be sufficient to compact the asphalt to the required density while it is still in a workable condition without crushing of the aggregate, depressions or other damage to the pavement surface. Rollers shall be in good condition, clean, and capable of operating at slow speeds to avoid displacement of the asphalt. All rollers shall be specifically designed and suitable for compacting asphalt concrete and shall be properly used. Rollers that impair the stability of any layer of a pavement structure or underlying soils shall not be used.

**401-4.7 Density device.** The Contractor shall have on site a density gauge during all paving operations in order to assist in the determination of the optimum rolling pattern, type of roller and frequencies, as well as to monitor the effect of the rolling operations during production paving. The Contractor shall supply a qualified technician during all paving operations to calibrate the gauge and obtain accurate density readings for all new asphalt. These densities shall be supplied to the RPR upon request at any time during construction. No separate payment will be made for supplying the density gauge and technician.

**401-4.8 Preparation of asphalt binder.** The asphalt binder shall be heated in a manner that will avoid local overheating and provide a continuous supply of the asphalt binder to the mixer at a uniform temperature. The temperature of unmodified asphalt binder delivered to the mixer shall be sufficient to provide a suitable viscosity for adequate coating of the aggregate particles, but shall not exceed 325°F (160°C) when added to the aggregate. The temperature of modified asphalt binder shall be no more than 350°F (175°C) when added to the aggregate.

**401-4.9 Preparation of mineral aggregate.** The aggregate for the asphalt shall be heated and dried. The maximum temperature and rate of heating shall be such that no damage occurs to the aggregates. The temperature of the aggregate and mineral filler shall not exceed 350°F (175°C) when the asphalt binder is added. Particular care shall be taken that aggregates high in calcium or magnesium content are not damaged by overheating. The temperature shall not be lower than is required to obtain complete coating and uniform distribution on the aggregate particles and to provide a mixture of satisfactory workability.

**401-4.10 Preparation of Asphalt mixture.** The aggregates and the asphalt binder shall be weighed or metered and mixed in the amount specified by the JMF. The combined materials shall be mixed until the aggregate obtains a uniform coating of asphalt binder and is thoroughly distributed throughout the mixture. Wet mixing time shall be the shortest time that will produce a satisfactory mixture, but not less than 25 seconds for batch plants. The wet mixing time for all plants shall be established by the Contractor, based on the procedure for determining the percentage of coated particles described in ASTM D2489, for each individual plant and for each type of aggregate used. The wet mixing time will be set to achieve 95% of coated particles. For continuous mix plants, the minimum mixing time shall be determined by dividing the weight of its contents at operating level by the weight of the mixture delivered per second by the mixer. The moisture content of all asphalt upon discharge shall not exceed 0.5%.

**401-4.11 Application of Prime and Tack Coat.** Immediately before placing the asphalt mixture, the underlying course shall be cleaned of all dust and debris.



A prime coat in accordance with Item P-602 shall be applied to aggregate base prior to placing the asphalt mixture.

A tack coat shall be applied in accordance with Item P-603 to all vertical and horizontal asphalt and concrete surfaces prior to placement of the first and each subsequent lift of asphalt mixture.

**401-4.12 Laydown plan, transporting, placing, and finishing.** Prior to the placement of the asphalt, the Contractor shall prepare a laydown plan with the sequence of paving lanes and width to minimize the number of cold joints; the location of any temporary ramps; laydown temperature; and estimated time of completion for each portion of the work (milling, paving, rolling, cooling, etc.). The laydown plan and any modifications shall be approved by the RPR.

Deliveries shall be scheduled so that placing and compacting of asphalt is uniform with minimum stopping and starting of the paver. Hauling over freshly placed material shall not be permitted until the material has been compacted, as specified, and allowed to cool to approximately ambient temperature. The Contractor, at their expense, shall be responsible for repair of any damage to the pavement caused by hauling operations.

Contractor shall survey each lift of asphalt surface course and certify to RPR that every lot of each lift meets the grade tolerances of paragraph 401-6.2d before the next lift can be placed.

Edges of existing asphalt pavement abutting the new work shall be saw cut and the cut off material and laitance removed. Apply a tack coat in accordance with P-603 before new asphalt material is placed against it.

The speed of the paver shall be regulated to eliminate pulling and tearing of the asphalt mat. Placement of the asphalt mix shall begin along the centerline of a crowned section or on the high side of areas with a one way slope unless shown otherwise on the laydown plan as accepted by the RPR. The asphalt mix shall be placed in consecutive adjacent lanes having a minimum width of 8 feet (2.44 m) except where edge lanes require less width to complete the area. Additional screed sections attached to widen the paver to meet the minimum lane width requirements must include additional auger sections to move the asphalt mixture uniformly along the screed extension.

The longitudinal joint in one course shall offset the longitudinal joint in the course immediately below by at least one foot (30 cm); however, the joint in the surface top course shall be at the centerline of crowned pavements. Transverse joints in one course shall be offset by at least 10 feet (3 m) from transverse joints in the previous course. Transverse joints in adjacent lanes shall be offset a minimum of 10 feet (3 m). On areas where irregularities or unavoidable obstacles make the use of mechanical spreading and finishing equipment impractical, the asphalt may be spread and luted by hand tools.

The RPR may at any time, reject any batch of asphalt, on the truck or placed in the mat, which is rendered unfit for use due to contamination, segregation, incomplete coating of aggregate, or overheated asphalt mixture. Such rejection may be based on only visual inspection or temperature measurements. In the event of such rejection, the Contractor may take a representative sample of the rejected material in the presence of the RPR, and if it can be demonstrated in the laboratory, in the presence of the RPR, that such material was erroneously rejected, payment will be made for the material at the contract unit price.

Areas of segregation in the surface course, as determined by the RPR, shall be removed and replaced at the Contractor's expense. The area shall be removed by saw cutting and milling a minimum of the construction lift thickness as specified in paragraph 401-3.3, Table 2 for the approved mix design. The area to be removed and replaced shall be a minimum width of the paver and a minimum of 10 feet (3 m) long.

**401-4.13 Compaction of asphalt mixture.** After placing, the asphalt mixture shall be thoroughly and uniformly compacted by self-propelled rollers. The surface shall be compacted as soon as possible when the asphalt has attained sufficient stability so that the rolling does not cause undue displacement, cracking or shoving. The sequence of rolling operations and the type of rollers used shall be at the discretion of the Contractor. The speed of the roller shall, at all times, be sufficiently slow to avoid displacement of the hot mixture and be effective in compaction. Any surface defects and/or displacement occurring as a result of the roller, or from any other cause, shall be corrected at the Contractor's expense.

Sufficient rollers shall be furnished to handle the output of the plant. Rolling shall continue until the surface is of uniform texture, true to grade and cross-section, and the required field density is obtained. To prevent adhesion of the asphalt to the roller, the wheels shall be equipped with a scraper and kept moistened with water as necessary.

In areas not accessible to the roller, the mixture shall be thoroughly compacted with approved power tampers.

Any asphalt that becomes loose and broken, mixed with dirt, contains check-cracking, or in any way defective shall be removed and replaced with fresh hot mixture and immediately compacted to conform to the surrounding area. This work shall be done at the Contractor's expense. Skin patching shall not be allowed.

**401-4.14 Joints.** The formation of all joints shall be made to ensure a continuous bond between the courses and obtain the required density. All joints shall have the same texture as other sections of the course and meet the requirements for smoothness and grade.

The roller shall not pass over the unprotected end of the freshly laid asphalt except when necessary to form a transverse joint. When necessary to form a transverse joint, it shall be made by means of placing a bulkhead or by tapering the course. The tapered edge shall be cut back to its full depth and width on a straight line to expose a vertical face prior to placing the adjacent lane. In both methods, all contact surfaces shall be coated with an asphalt tack coat before placing any fresh asphalt against the joint.

Longitudinal joints which have been left exposed for more than four (4) hours; the surface temperature has cooled to less than 175°F (80°C); or are irregular, damaged, uncompacted or otherwise defective shall be cut back with a cutting wheel or pavement saw a maximum of 3 inches (75 mm) to expose a clean, sound, uniform vertical surface for the full depth of the course. All cutback material and any laitance produced from cutting joints shall be removed from the project. Asphalt tack coat in accordance with P-603 shall be applied to the clean, dry joint prior to placing any additional fresh asphalt against the joint. The cost of this work shall be considered incidental to the cost of the asphalt.

**401-4.15 Saw-cut grooving.** Saw-cut grooves shall be provided as specified in Item P-621.

**401-4.16 Diamond grinding.** Diamond grinding shall be completed prior to pavement grooving. Diamond grinding shall be accomplished by sawing with saw blades impregnated with industrial diamond abrasive.

Diamond grinding shall be performed with a machine designed specifically for diamond grinding capable of cutting a path at least 3 feet (0.9 m) wide. The saw blades shall be 1/8-inch (3-mm) wide with a sufficient number of blades to create grooves between 0.090 and 0.130 inches (2 and 3.5 mm) wide; and peaks and ridges approximately 1/32 inch (1 mm) higher than the bottom of the grinding cut. The actual number of blades will be determined by the Contractor and depend on the hardness of the aggregate. Equipment or grinding procedures that cause ravels, aggregate fractures, spalls or disturbance to the pavement will not be permitted.

Contractor shall demonstrate to the RPR that the grinding equipment will produce satisfactory results prior to making corrections to surfaces. Grinding will be tapered in all directions to provide smooth transitions to areas not requiring grinding. The slurry resulting from the grinding operation shall be continuously removed and the pavement left in a clean condition. The Contractor shall apply a surface treatment per P-608 to all areas that have been subject to grinding.

**401-4.17 Nighttime paving requirements.** The Contractor shall provide adequate lighting during any nighttime construction. A lighting plan shall be submitted by the Contractor and approved by the RPR prior to the start of any nighttime work. All work shall be in accordance with the approved CSPP and lighting plan.

## **CONTRACTOR QUALITY CONTROL (CQC)**

**401-5.1 General.** Contractor shall develop a Contractor Quality Control Program (CQCP) in accordance with Item C-100. No partial payment will be made for materials without an approved CQCP.

**401-5.2 Contractor quality control (QC) facilities.** The Contractor shall provide or contract for testing facilities in accordance with Item C-100. The RPR shall be permitted unrestricted access to inspect the Contractor's QC facilities and witness QC activities. The RPR will advise the Contractor in writing of any noted deficiencies concerning the QC facility, equipment, supplies, or testing personnel and procedures. When the deficiencies are serious enough to be adversely affecting the test results, the incorporation of the materials into the work shall be suspended immediately and will not be permitted to resume until the deficiencies are satisfactorily corrected.

**401-5.3 Contractor QC testing.** The Contractor shall perform all QC tests necessary to control the production and construction processes applicable to these specifications

**a. Asphalt content.** A minimum of two tests shall be performed per day in accordance with ASTM D6307 or ASTM D2172 for determination of asphalt content. When using ASTM D6307, the correction factor shall be determined as part of the first test performed at the beginning of plant production; and as part of every tenth test performed thereafter. The asphalt content for the day will be determined by averaging the test results.

**b. Gradation.** Aggregate gradations shall be determined a minimum of twice per day from mechanical analysis of extracted aggregate in accordance with ASTM D5444, ASTM C136, and ASTM C117.

**c. Moisture content of aggregate.** The moisture content of aggregate used for production shall be determined a minimum of once per day in accordance with ASTM C566.

**d. Moisture content of asphalt.** The moisture content shall be determined once per day in accordance with AASHTO T329 or ASTM D1461.

**e. Temperatures.** Temperatures shall be checked, at least four times per day, at necessary locations to determine the temperatures of the dryer, the asphalt binder in the storage tank, the asphalt at the plant, and the asphalt at the job site.

**f. In-place density monitoring.** The Contractor shall conduct any necessary testing to ensure that the specified density is being achieved. A nuclear gauge may be used to monitor the pavement density in accordance with ASTM D2950.

**g. Smoothness for Contractor Quality Control.** The Contractor shall perform smoothness testing in transverse and longitudinal directions daily to verify that the construction processes are producing pavement with variances less than 1/4 inch in 12 feet, identifying areas that may pond water which could lead to hydroplaning of aircraft. If the smoothness criteria is not met, appropriate changes and corrections to the construction process shall be made by the Contractor before construction continues.

The Contractor may use a 12-foot (3.7 m) "straightedge, a rolling inclinometer meeting the requirements of ASTM E2133 or rolling external reference device that can simulate a 12-foot (3.7m) straightedge approved by the RPR. Straight-edge testing shall start with one-half the length of the straightedge at the edge of pavement section being tested and then moved ahead one-half the length of the straightedge for each successive measurement. Testing shall be continuous across all joints. The surface irregularity shall be determined by placing the freestanding (unleveled) straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length, and measuring the maximum gap between the straightedge and the pavement surface in the area between the two high points. If the rolling inclinometer or external reference device is used, the data may be evaluated using either the FAA profile program, ProFAA, or FHWA ProVal, using the 12-foot straightedge simulation function.

Smoothness readings shall not be made across grade changes or cross slope transitions. The transition between new and existing pavement shall be evaluated separately for conformance with the plans.

**(1) Transverse measurements.** Transverse measurements shall be taken for each day's production placed. Transverse measurements shall be taken perpendicular to the pavement centerline each 50 feet (15 m) or more often as determined by the RPR. The joint between lanes shall be tested separately to facilitate smoothness between lanes.

**(2) Longitudinal measurements.** Longitudinal measurements shall be taken for each day's production placed. Longitudinal tests shall be parallel to the centerline of paving; at the center of paving lanes when widths of paving lanes are less than 20 feet (6 m); and at the third points of paving lanes when widths of paving lanes are 20 ft (6 m) or greater. When placement abuts previously placed material the first measurement shall start with one half the length of the straight edge on the previously placed material.

Deviations on the final surface course in either the transverse or longitudinal direction that will trap water greater than 1/4 inch (6 mm) shall be corrected with diamond grinding per paragraph 401-4.16 or by removing and replacing the surface course to full depth. Grinding shall be tapered in all directions to provide smooth transitions to areas not requiring grinding. All areas in which diamond grinding has been performed shall be subject to the final pavement thickness tolerances specified in paragraph 401-6.1d(3). Areas that have been ground shall be sealed with a surface treatment in accordance with Item P-608. To avoid the surface treatment creating any conflict with runway or taxiway markings, it may be necessary to seal a larger area.

Control charts shall be kept to show area of each day's placement and the percentage of corrective grinding required. Corrections to production and placement shall be initiated when corrective grinding is required. If the Contractor's machines and/or methods produce significant areas that need corrective actions in excess of 10 percent of a day's production, production shall be stopped until corrective measures are implemented by the Contractor.

**h. Grade.** Grade shall be evaluated daily to allow adjustments to paving operations when grade measurements do not meet specifications. As a minimum, grade shall be evaluated prior to and after the placement of the first lift and after placement of the surface lift.

Measurements will be taken at appropriate gradelines (as a minimum at center and edges of paving lane) and longitudinal spacing as shown on cross-sections and plans. The final surface of the pavement will not vary from the gradeline elevations shown on the plans by more than 1/2 inch (12 mm) vertically. The documentation will be provided by the Contractor to the RPR.

Areas with humps or depressions that exceed grade or smoothness criteria and that retain water on the surface must be ground off provided the course thickness after grinding is not more than 1/2 inch (12 mm) less than the thickness specified on the plans. Grinding shall be in accordance with paragraph 401-4.16.

The Contractor shall repair low areas or areas that cannot be corrected by grinding by removal of deficient areas to the depth of the final course plus 1/2 inch and replacing with new material. Skin patching is not allowed.

**401-5.4 Sampling.** When directed by the RPR, the Contractor shall sample and test any material that appears inconsistent with similar material being sampled, unless such material is voluntarily removed and replaced or deficiencies corrected by the Contractor. All sampling shall be in accordance with standard procedures specified.

**401-5.5 Control charts.** The Contractor shall maintain linear control charts for both individual measurements and range (i.e. difference between highest and lowest measurements) for aggregate gradation, asphalt content, and VMA. The VMA for each day will be calculated and monitored by the QC laboratory.

Control charts shall be posted in a location satisfactory to the RPR and kept current. As a minimum, the control charts shall identify the project number, the contract item number, the test number, each test parameter, the Action and Suspension Limits applicable to each test parameter, and the Contractor's test results. The Contractor shall use the control charts as part of a process control system for identifying potential problems and assignable causes before they occur. If the Contractor's projected data during production indicates a problem and the Contractor is not taking satisfactory corrective action, the RPR may suspend production or acceptance of the material.

**a. Individual measurements.** Control charts for individual measurements shall be established to maintain process control within tolerance for aggregate gradation, asphalt content, and VMA. The control charts shall use the job mix formula target values as indicators of central tendency for the following test parameters with associated Action and Suspension Limits:

**Control Chart Limits for Individual Measurements**

| Sieve                  | Action Limit | Suspension Limit |
|------------------------|--------------|------------------|
| 3/4 inch (19.0 mm)     | ±6%          | ±9%              |
| 1/2 inch (12.5 mm)     | ±6%          | ±9%              |
| 3/8 inch (9.5 mm)      | ±6%          | ±9%              |
| No. 4 (4.75 mm)        | ±6%          | ±9%              |
| No. 16 (1.18 mm)       | ±5%          | ±7.5%            |
| No. 50 (300 µm)        | ±3%          | ±4.5%            |
| No. 200 (75 µm)        | ±2%          | ±3%              |
| <b>Asphalt Content</b> | ±0.45%       | ±0.70%           |
| <b>Minimum VMA</b>     | -0.5%        | -1.0%            |

**b. Range.** Control charts shall be established to control gradation process variability. The range shall be plotted as the difference between the two test results for each control parameter. The Suspension Limits specified below are based on a sample size of  $n = 2$ . Should the Contractor elect to perform more than two tests per lot, the Suspension Limits shall be adjusted by multiplying the Suspension Limit by 1.18 for  $n = 3$  and by 1.27 for  $n = 4$ .

**Control Chart Limits Based on Range**

| Sieve                  | Suspension Limit |
|------------------------|------------------|
| 1/2 inch (12.5 mm)     | 11%              |
| 3/8 inch (9.5 mm)      | 11%              |
| No. 4 (4.75 mm)        | 11%              |
| No. 16 (1.18 mm)       | 9%               |
| No. 50 (300 µm)        | 6%               |
| No. 200 (75 µm)        | 3.5%             |
| <b>Asphalt Content</b> | 0.8%             |

**c. Corrective Action.** [ The CQCP shall indicate that appropriate action shall be taken when the process is believed to be out of tolerance. The Plan shall contain rules to gauge when a process is out of control and detail what action will be taken to bring the process into control. As a minimum, a process shall be deemed out of control and production stopped and corrective action taken, if:

- (1) One point falls outside the Suspension Limit line for individual measurements or range; or
- (2) Two points in a row fall outside the Action Limit line for individual measurements. ]

**401-5.6 QC reports.** The Contractor shall maintain records and shall submit reports of QC activities daily.

## MATERIAL ACCEPTANCE

**401-6.1 Acceptance sampling and testing.** Unless otherwise specified, all acceptance sampling and testing necessary to determine conformance with the requirements specified in this section will be performed by the RPR at no cost to the Contractor except that coring as required in this section shall be completed and paid for by the Contractor.

**a. Quality assurance (QA) testing laboratory.** The QA testing laboratory performing these acceptance tests will be accredited in accordance with ASTM D3666. The QA laboratory accreditation will be current and listed on the accrediting authority's website. All test methods required for acceptance sampling and testing will be listed on the lab accreditation.

**b. Lot size.** A standard lot will be equal to one day's production divided into approximately equal sublots of between 400 to 600 tons. When only one or two sublots are produced in a day's production, the sublots will be combined with the production lot from the previous or next day.

Where more than one plant is simultaneously producing asphalt for the job, the lot sizes will apply separately for each plant.

**c. Asphalt air voids.** Plant-produced asphalt will be tested for air voids on a subplot basis.

**(1) Sampling.** Material from each subplot shall be sampled in accordance with ASTM D3665. Samples shall be taken from material deposited into trucks at the plant or at the job site in accordance with ASTM D979. The sample of asphalt may be put in a covered metal tin and placed in an oven for not less than 30 minutes nor more than 60 minutes to maintain the material at or above the compaction temperature as specified in the JMF.

**(2) Testing.** Air voids will be determined for each subplot in accordance with ASTM D3203 for a set of three compacted specimens prepared in accordance with ASTM D6925.

**d. In-place asphalt mat and joint density.** Each subplot will be tested for in-place mat and joint density as a percentage of the theoretical maximum density (TMD).

**(1) Sampling.** The RPR will cut minimum 5 inch (125 mm) diameter samples in accordance with ASTM D5361. The Contractor shall furnish all tools, labor, and materials for cleaning, and filling the cored pavement. Laitance produced by the coring operation shall be removed immediately after coring, and core holes shall be filled within one day after sampling in a manner acceptable to the RPR.

**(2) Bond.** Each lift of asphalt shall be bonded to the underlying layer. If cores reveal that the surface is not bonded, additional cores shall be taken as directed by the RPR to determine the extent of unbonded areas. Unbonded areas shall be removed by milling and replaced at no additional cost as directed by the RPR.

**(3) Thickness.** Thickness of each lift of surface course will be evaluated by the RPR for compliance to the requirements shown on the plans after any necessary corrections for grade. Measurements of thickness will be made using the cores extracted for each subplot for density measurement. The maximum allowable deficiency at any point will not be more than 1/4 inch (6 mm) less than the thickness indicated for the lift. Average thickness of lift, or combined lifts, will not be less than the indicated thickness. Where the thickness tolerances are not met, the lot or subplot shall be corrected by the Contractor at his expense by removing the deficient area and replacing with new pavement. The Contractor, at his expense, may take additional cores as approved by the RPR to circumscribe the deficient area.

**(4) Mat density.** One core shall be taken from each subplot. Core locations will be determined by the RPR in accordance with ASTM D3665. Cores for mat density shall not be taken closer than one foot (30 cm) from a transverse or longitudinal joint. The bulk specific

gravity of each cored sample will be determined in accordance with ASTM D2726. The percent compaction (density) of each sample will be determined by dividing the bulk specific gravity of each subplot sample by the TMD for that subplot.

**(5) Joint density.** One core centered over the longitudinal joint shall be taken for each subplot that has a longitudinal joint. Core locations will be determined by the RPR in accordance with ASTM D3665. The bulk specific gravity of each core sample will be determined in accordance with ASTM D2726. The percent compaction (density) of each sample will be determined by dividing the bulk specific gravity of each joint density sample by the average TMD for the lot. The TMD used to determine the joint density at joints formed between lots will be the lower of the average TMD values from the adjacent lots.

#### **401-6.2 Acceptance criteria.**

**a. General.** Acceptance will be based on the implementation of the Contractor Quality Control Program (CQCP) and the following characteristics of the asphalt and completed pavements: air voids, mat density, joint density, and grade.

**b. Air Voids and Mat density.** Acceptance of each lot of plant produced material for mat density and air voids will be based on the percentage of material within specification limits (PWL). If the PWL of the lot equals or exceeds 90%, the lot will be acceptable. Acceptance and payment will be determined in accordance with paragraph 401-8.1.

**c. Joint density.** Acceptance of each lot of plant produced asphalt for joint density will be based on the PWL. If the PWL of the lot is equal to or exceeds 90%, the lot will be considered acceptable. If the PWL is less than 90%, the Contractor shall evaluate the reason and act accordingly. If the PWL is less than 80%, the Contractor shall cease operations and until the reason for poor compaction has been determined. If the PWL is less than 71%, the pay factor for the lot used to complete the joint will be reduced by five (5) percentage points. This lot pay factor reduction will be incorporated and evaluated in accordance with paragraph 401-8.1.

**d. Grade.** The final finished surface of the pavement shall be surveyed to verify that the grade elevations and cross-sections shown on the plans do not deviate more than 1/2 inch (12 mm) vertically.

Cross-sections of the pavement shall be taken at a minimum 50-foot (15-m) longitudinal spacing, at all longitudinal grade breaks, and at start and end of each lane placed. Minimum cross-section grade points shall include grade at centerline,  $\pm$  10 feet of centerline, and edge of runway and taxiway pavement.

The survey and documentation shall be stamped and signed by a licensed surveyor. Payment for sublots that do not meet grade for over 25% of the subplot shall not be more than 95%.

**e. Profilograph roughness for QA Acceptance.** Not used.

**401-6.3 Percentage of material within specification limits (PWL).** The PWL will be determined in accordance with procedures specified in Item C-110. The specification tolerance limits (L) for lower and (U) for upper are contained in Table 5.



**Table 5. Acceptance Limits for Air Voids and Density**

| Test Property                         | Pavements Specification<br>Tolerance Limits |     |
|---------------------------------------|---------------------------------------------|-----|
|                                       | L                                           | U   |
| <b>Air Voids Total Mix (%)</b>        | 2.0                                         | 5.0 |
| <b>Surface Course Mat Density (%)</b> | 92.8                                        | -   |
| <b>Base Course Mat Density (%)</b>    | 92.0                                        | -   |
| <b>Joint density (%)</b>              | 90.5                                        | --  |

**a. Outliers.** All individual tests for mat density and air voids will be checked for outliers (test criterion) in accordance with ASTM E178, at a significance level of 5%. Outliers will be discarded, and the PWL will be determined using the remaining test values. The criteria in Table 5 is based on production processes which have a variability with the following standard deviations: Surface Course Mat Density (%), 1.30; Base Course Mat Density (%), 1.55; Joint Density (%), 1.55.

The Contractor should note that (1) 90 PWL is achieved when consistently producing a surface course with an average mat density of at least 94.5% with 1.30% or less variability, (2) 90 PWL is achieved when consistently producing a base course with an average mat density of at least 94.0% with 1.55% or less variability, and (3) 90 PWL is achieved when consistently producing joints with an average joint density of at least 92.5% with 1.55% or less variability.

#### **401-6.4 Resampling pavement for mat density.**

**a. General.** Resampling of a lot of pavement will only be allowed for mat density, and then, only if the Contractor requests same, in writing, within 48 hours after receiving the written test results from the RPR. A retest will consist of all the sampling and testing procedures contained in paragraphs 401-6.1d and 401-6.2b. Only one resampling per lot will be permitted.

(1) A redefined PWL will be calculated for the resampled lot. The number of tests used to calculate the redefined PWL will include the initial tests made for that lot plus the retests.

(2) The cost for resampling and retesting shall be borne by the Contractor.

**b. Payment for resampled lots.** The redefined PWL for a resampled lot will be used to calculate the payment for that lot in accordance with Table 6.

**c. Outliers.** Check for outliers in accordance with ASTM E178, at a significance level of 5%.

**401-6.5 Leveling course.** The leveling course is the first variable thickness lift placed to correct surface irregularities prior to placement of subsequent courses. The leveling course shall meet the aggregate gradation in Table 2, paragraph 401-3.3. The leveling course shall meet the requirements of paragraph 401-3.3, 401-6.2b for air voids, but shall not be subject to the density requirements of paragraph 401-6.2b for mat density and 401-6.2c for joint density. The leveling course shall be compacted with the same effort used to achieve density of the control strip. The leveling course shall not exceed the lift thickness associated with each gradation in Table 2, paragraph 401-3.3.

## METHOD OF MEASUREMENT

**401-7.1 Measurement.** Asphalt shall be measured by the number of tons (kg) of asphalt used in the accepted work. Batch weights or truck scale weights will be used to determine the basis for the tonnage.

## BASIS OF PAYMENT

**401-8.1 Payment.** Payment for a lot of asphalt meeting all acceptance criteria as specified in paragraph 401-6.2 shall be made based on results of tests for mat density and air voids. Payment for acceptable lots shall be adjusted according to paragraph 401-8.1c for mat density and air voids; and paragraph 401-6.2c for joint density, subject to the limitation that:

**a.** The total project payment for plant mix asphalt pavement shall not exceed **100** percent of the product of the contract unit price and the total number of tons (kg) of asphalt used in the accepted work.

**b.** The price shall be compensation for furnishing all materials, for all preparation, mixing, and placing of these materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

**c. Basis of adjusted payment.** The pay factor for each individual lot shall be calculated in accordance with Table 6. A pay factor shall be calculated for both mat density and air voids. The lot pay factor shall be the higher of the two values when calculations for both mat density and air voids are 100% or higher. The lot pay factor shall be the product of the two values when only one of the calculations for either mat density or air voids is 100% or higher. The lot pay factor shall be the lower of the two values when calculations for both mat density and air voids are less than 100%. If PWL for joint density is less than 71% then the lot pay factor shall be reduced by 5% but be no higher than 95%.

For each lot accepted, the adjusted contract unit price shall be the product of the lot pay factor for the lot and the contract unit price. Payment shall be subject to the total project payment limitation specified in paragraph 401-8.1a. Payment in excess of 100% for accepted lots of asphalt shall be used to offset payment for accepted lots of asphalt pavement that achieve a lot pay factor less than 100%.

Payment for sublots which do not meet grade in accordance with paragraph 401-6.2d after correction for over 25% of the subplot shall be reduced by 5%.

**Table 6. Price adjustment schedule<sup>1</sup>**

| <b>Percentage of material within specification limits (PWL)</b> | <b>Lot pay factor (percent of contract unit price)</b> |
|-----------------------------------------------------------------|--------------------------------------------------------|
| 96 – 100                                                        | 106                                                    |
| 90 – 95                                                         | PWL + 10                                               |
| 75 – 89                                                         | 0.5 PWL + 55                                           |
| 55 – 74                                                         | 1.4 PWL – 12                                           |
| Below 55                                                        | Reject <sup>2</sup>                                    |

<sup>1</sup> Although it is theoretically possible to achieve a pay factor of 106% for each lot, actual payment above 100% shall be subject to the total project payment limitation specified in paragraph 401-8.1a.

<sup>2</sup> The lot shall be removed and replaced. However, the RPR may decide to allow the rejected lot to remain. In that case, if the RPR and Contractor agree in writing that the lot shall not be removed, it

shall be paid for at 50% of the contract unit price and the total project payment shall be reduced by the amount withheld for the rejected lot.

**d. Profilograph Roughness.** Not used.

**401-8.1 Payment.**

Payment will be made under:

|            |                                     |
|------------|-------------------------------------|
| Item P-401 | 3" Asphalt Surface Course - per ton |
|------------|-------------------------------------|

**REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C29   | Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate                                                        |
| ASTM C88   | Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate                                      |
| ASTM C117  | Standard Test Method for Materials Finer than 75- $\mu$ m (No. 200) Sieve in Mineral Aggregates by Washing                          |
| ASTM C127  | Standard Test Method for Density, Relative Density (Specific Gravity) and Absorption of Coarse Aggregate                            |
| ASTM C131  | Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine |
| ASTM C136  | Standard Test Method for Sieve or Screen Analysis of Fine and Coarse Aggregates                                                     |
| ASTM C142  | Standard Test Method for Clay Lumps and Friable Particles in Aggregates                                                             |
| ASTM C566  | Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying                                                   |
| ASTM D75   | Standard Practice for Sampling Aggregates                                                                                           |
| ASTM D242  | Standard Specification for Mineral Filler for Bituminous Paving Mixtures                                                            |
| ASTM D946  | Standard Specification for Penetration-Graded Asphalt Cement for Use in Pavement Construction                                       |
| ASTM D979  | Standard Practice for Sampling Asphalt Paving Mixtures                                                                              |
| ASTM D1073 | Standard Specification for Fine Aggregate for Asphalt Paving Mixtures                                                               |
| ASTM D1188 | Standard Test Method for Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Coated Samples                    |

|            |                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| ASTM D2172 | Standard Test Method for Quantitative Extraction of Bitumen from Asphalt Paving Mixtures                          |
| ASTM D1461 | Standard Test Method for Moisture or Volatile Distillates in Asphalt Paving Mixtures                              |
| ASTM D2041 | Standard Test Method for Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures           |
| ASTM D2419 | Standard Test Method for Sand Equivalent Value of Soils and Fine Aggregate                                        |
| ASTM D2489 | Standard Practice for Estimating Degree of Particle Coating of Bituminous-Aggregate Mixtures                      |
| ASTM D2726 | Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures        |
| ASTM D2950 | Standard Test Method for Density of Bituminous Concrete in Place by Nuclear Methods                               |
| ASTM D3203 | Standard Test Method for Percent Air Voids in Compacted Dense and Open Bituminous Paving Mixtures                 |
| ASTM D3381 | Standard Specification for Viscosity-Graded Asphalt Cement for Use in Pavement Construction                       |
| ASTM D3665 | Standard Practice for Random Sampling of Construction Materials                                                   |
| ASTM D3666 | Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials     |
| ASTM D4318 | Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                              |
| ASTM D4552 | Standard Practice for Classifying Hot-Mix Recycling Agents                                                        |
| ASTM D4791 | Standard Test Method for Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate |
| ASTM D4867 | Standard Test Method for Effect of Moisture on Asphalt Concrete Paving Mixtures                                   |
| ASTM D5361 | Standard Practice for Sampling Compacted Asphalt Mixtures for Laboratory Testing                                  |
| ASTM D5444 | Standard Test Method for Mechanical Size Analysis of Extracted Aggregate                                          |
| ASTM D5821 | Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate                    |
| ASTM D6084 | Standard Test Method for Elastic Recovery of Bituminous Materials by Durometer                                    |
| ASTM D6307 | Standard Test Method for Asphalt Content of Hot Mix Asphalt by Ignition Method                                    |
| ASTM D6373 | Standard Specification for Performance Graded Asphalt Binder                                                      |

|                                                                             |                                                                                                                                                                 |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D6752                                                                  | Standard Test Method for Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Automatic Vacuum Sealing Method                               |
| ASTM D6925                                                                  | Standard Test Method for Preparation and Determination of the Relative Density of Hot Mix Asphalt (HMA) Specimens by Means of the SuperPave Gyratory Compactor. |
| ASTM D6926                                                                  | Standard Practice for Preparation of Bituminous Specimens Using Marshall Apparatus                                                                              |
| ASTM D6927                                                                  | Standard Test Method for Marshall Stability and Flow of Bituminous Mixtures                                                                                     |
| ASTM D6995                                                                  | Standard Test Method for Determining Field VMA based on the Maximum Specific Gravity of the Mix (Gmm)                                                           |
| ASTM E11                                                                    | Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves                                                                                          |
| ASTM E178                                                                   | Standard Practice for Dealing with Outlying Observations                                                                                                        |
| ASTM E1274                                                                  | Standard Test Method for Measuring Pavement Roughness Using a Profilograph                                                                                      |
| ASTM E950                                                                   | Standard Test Method for Measuring the Longitudinal Profile of Traveled Surfaces with an Accelerometer Established Inertial Profiling Reference                 |
| ASTM E2133                                                                  | Standard Test Method for Using a Rolling Inclinator to Measure Longitudinal and Transverse Profiles of a Traveled Surface                                       |
| American Association of State Highway and Transportation Officials (AASHTO) |                                                                                                                                                                 |
| AASHTO M156                                                                 | Standard Specification for Requirements for Mixing Plants for Hot-Mixed, Hot-Laid Bituminous Paving Mixtures.                                                   |
| AASHTO T329                                                                 | Standard Method of Test for Moisture Content of Hot Mix Asphalt (HMA) by Oven Method                                                                            |
| AASHTO T324                                                                 | Standard Method of Test for Hamburg Wheel-Track Testing of Compacted Asphalt Mixtures                                                                           |
| AASHTO T 340                                                                | Standard Method of Test for Determining the Rutting Susceptibility of Hot Mix Asphalt (APA) Using the Asphalt Pavement Analyzer (APA)                           |
| Asphalt Institute (AI)                                                      |                                                                                                                                                                 |
| Asphalt Institute Handbook MS-26, Asphalt Binder                            |                                                                                                                                                                 |
| Asphalt Institute MS-2Mix Design Manual, 7th Edition                        |                                                                                                                                                                 |
| AI State Binder Specification Database                                      |                                                                                                                                                                 |
| Federal Highway Administration (FHWA)                                       |                                                                                                                                                                 |
| Long Term Pavement Performance Binder Program                               |                                                                                                                                                                 |
| Advisory Circulars (AC)                                                     |                                                                                                                                                                 |
| AC 150/5320-6                                                               | Airport Pavement Design and Evaluation                                                                                                                          |

FAA Orders

5300.1

Modifications to Agency Airport Design, Construction, and  
Equipment Standards

Software

FAARFIELD

**END OF ITEM P-401**

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### Item P-403 Asphalt Mix Pavement

#### DESCRIPTION

**403-1.1** This item shall consist of pavement courses composed of mineral aggregate and asphalt binder mixed in a central mixing plant and placed on a prepared course in accordance with these specifications and shall conform to the lines, grades, thicknesses, and typical cross-sections shown on the plans. Each course shall be constructed to the depth, typical section, and elevation required by the plans and shall be rolled, finished, and approved before the placement of the next course.

#### MATERIALS

**403-2.1 Aggregate.** Aggregates shall consist of crushed stone, crushed gravel, crushed slag, screenings, natural sand and mineral filler, as required. The aggregates should have no known history of detrimental pavement staining due to ferrous sulfides, such as pyrite. ***A geotechnical or petrography report is required with the submittal.*** Coarse aggregate is the material retained on the No. 4 sieve. Fine aggregate is the material passing the No. 4 sieve.

a. Coarse aggregate. Coarse aggregate shall consist of sound, tough, durable particles, free from films of matter that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances. Coarse aggregate material requirements are given in the table below.



### Coarse Aggregate Material Requirements

| Material Test                                                         | Requirement                                                                                                                                                                                                             | Standard   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Resistance to Degradation                                             | leveling course Loss: 50% maximum for base course                                                                                                                                                                       | ASTM C131  |
| Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate | Loss after 5 cycles:<br>12% maximum using Sodium sulfate - or -<br>18% maximum using magnesium sulfate                                                                                                                  | ASTM C88   |
| Clay lumps and friable particles                                      | 1.0 % maximum                                                                                                                                                                                                           | ASTM C142  |
| Percentage of Fractured Particles                                     | For pavements designed for aircraft gross weights of 60,000 pounds (27200 kg) or more:                                                                                                                                  | ASTM D5821 |
|                                                                       | Minimum 75% by weight of particles with at least two fractured faces and 85% with at least one fractured face <sup>1</sup>                                                                                              |            |
|                                                                       | For pavements designed for aircraft gross weights less than 60,000 pounds (27200 kg):<br><br>Minimum 50% by weight of particles with at least two fractured faces and 65% with at least one fractured face <sup>1</sup> |            |
| Flat, Elongated, or Flat and Elongated Particles                      | 8% maximum, by weight, of flat, elongated, or flat and elongated particles with a value of 5:1 <sup>2</sup>                                                                                                             | ASTM D4791 |
| Bulk density of slag <sup>3</sup>                                     | Weigh not less than 70 pounds per cubic foot                                                                                                                                                                            | ASTM C29.  |

<sup>1</sup> The area of each face shall be equal to at least 75% of the smallest mid-sectional area of the piece. When two fractured faces are contiguous, the angle between the planes of fractures shall be at least 30 degrees to count as two fractured faces.

<sup>2</sup> A flat particle is one having a ratio of width to thickness greater than five (5); an elongated particle is one having a ratio of length to width greater than five (5).

<sup>3</sup> Only required if slag is specified.

**b. Fine aggregate.** Fine aggregate shall consist of clean, sound, tough, durable, angular shaped particles produced by crushing stone, slag, or gravel and shall be free from coatings of clay, silt, or other objectionable matter. Natural (non-manufactured) sand may be used to obtain the gradation of the aggregate blend or to improve the workability of the mix. Fine aggregate material requirements are listed in the table below.

### Fine Aggregate Material Requirements

| Material Test                                                         | Requirement                                                                                               | Standard   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------|
| Liquid limit                                                          | 25 maximum                                                                                                | ASTM D4318 |
| Plasticity Index                                                      | 4 maximum                                                                                                 | ASTM D4318 |
| Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate | Loss after 5 cycles:<br>10% maximum using Sodium sulfate -<br>or -<br>15% maximum using magnesium sulfate | ASTM C88   |
| Clay lumps and friable particles                                      | 1.0 % maximum                                                                                             | ASTM C142  |
| Sand equivalent                                                       | 45 minimum                                                                                                | ASTM D2419 |
| Natural Sand                                                          | 0 to 15% maximum by weight of total aggregate                                                             | ASTM D1073 |

**c. Sampling.** ASTM D75 shall be used in sampling coarse and fine aggregate, and ASTM C183 shall be used in sampling mineral filler.

**403-2.2 Mineral filler.** Mineral filler (baghouse fines) may be added in addition to material naturally present in the aggregate. Mineral filler shall meet the requirements of ASTM D242.

### Mineral filler Requirements

| Material Test    | Requirement | Standard   |
|------------------|-------------|------------|
| Plasticity Index | 4 maximum   | ASTM D4318 |

**403-2.3 Asphalt binder.** Asphalt binder shall conform to ASTM D6373 Performance Grade (PG) 76-22.

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### Asphalt Binder PG Plus Test Requirements

| Material Test    | Requirement | Standard                |
|------------------|-------------|-------------------------|
| Elastic Recovery | 75% minimum | ASTM D6084 <sup>1</sup> |

**403-2.4 Anti-stripping agent.** Any anti-stripping agent or additive (anti-strip) shall be heat stable and shall not change the asphalt binder grade beyond specifications. Anti-strip shall be an approved material of the Florida Department of Transportation.

## COMPOSITION

**403-3.1 Composition of mixture.** The asphalt plant mix shall be composed of a mixture of well-graded aggregate, filler and anti-strip agent if required, and asphalt binder. The several aggregate fractions shall be sized, handled in separate size groups, and combined in such

proportions that the resulting mixture meets the grading requirements of the job mix formula (JMF).

**403-3.2 Job mix formula (JMF) laboratory.** The laboratory used to develop the JMF shall possess a current certificate of accreditation, listing D3666 from a national accrediting authority and all test methods required for developing the JMF, and listed on the accrediting authority's website. A copy of the laboratory's current accreditation and accredited test methods shall be submitted to the RPR prior to start of construction.

**403-3.3 Job mix formula (JMF).** No asphalt mixture shall be placed until an acceptable mix design has been submitted to the RPR for review and accepted in writing. The RPR's review shall not relieve the Contractor of the responsibility to select and proportion the materials to comply with this section.

When the project requires asphalt mixtures of differing aggregate gradations and/or binders, a separate JMF shall be submitted for each mix. Add anti-stripping agent to meet tensile strength requirements.

The JMF shall be prepared by an accredited laboratory that meets the requirements of paragraph 403-3.2. The asphalt mixture shall be designed using procedures contained in Asphalt Institute MS-2 Mix Design Manual, 7th Edition. Samples shall be prepared and compacted using the gyratory compactor in accordance with ASTM D6925.

Should a change in sources of materials be made, a new JMF must be submitted to the RPR for review and accepted in writing before the new material is used. After the initial production JMF has been approved by the RPR and a new or modified JMF is required for whatever reason, the subsequent cost of the new or modified JMF, including a new control strip when required by the RPR, will be borne by the Contractor.

The RPR may request samples at any time for testing, prior to and during production, to verify the quality of the materials and to ensure conformance with the applicable specifications.

The JMF shall be submitted in writing by the Contractor at least 30 days prior to the start of paving operations. The JMF shall be developed within the same construction season using aggregates proposed for project use.

The submitted JMF shall be dated, and stamped or sealed by the responsible professional Engineer of the laboratory and shall include the following items as a minimum:

- Manufacturer's Certificate of Analysis (COA) for the asphalt binder used in the JMF in accordance with paragraph 403-2.3. Certificate of asphalt performance grade is with modifier already added, if used and must indicate compliance with ASTM D6373. For plant modified asphalt binder, certified test report indicating grade certification of modified asphalt binder.
- Manufacturer's Certificate of Analysis (COA) for the anti-stripping agent if used in the JMF in accordance with paragraph 403-2.4.
- Certified material test reports for the course and fine aggregate and mineral filler in accordance with paragraphs 403-2.1 and 403-2.2.
- Percent passing each sieve size for individual gradation of each aggregate cold feed and/or hot bin; percent by weight of each cold feed and/or hot bin used; and the total combined gradation in the JMF.
- Specific Gravity and absorption of each course and fine aggregate.
- Percent natural sand.

- Percent fractured faces.
- Percent by weight of flat particles, elongated particles, and flat and elongated particles (and criteria).
- Percent of asphalt.
- Number of blows or gyrations.
- Laboratory mixing and compaction temperatures.
- Supplier recommended mixing and compaction temperatures.
- Plot of the combined gradation on the 0.45 power gradation curve.
- Graphical plots of air voids, voids in the mineral aggregate (VMA), and unit weight versus asphalt content. To achieve minimum VMA during production, the mix design needs to account for material breakdown during production.
- Tensile Strength Ratio (TSR).
- Type and amount of Anti-strip agent when used.
- Asphalt Pavement Analyzer (APA) results.
- Date the JMF was developed. Mix designs that are not dated or which are from a prior construction season shall not be accepted.

**Table 1. Asphalt Design Criteria**

| Test Property                                     | Value                                      | Test Method |
|---------------------------------------------------|--------------------------------------------|-------------|
| Number of blows/gyrations                         | <b>50</b>                                  |             |
| Air voids (%)                                     | 3.5                                        | ASTM D3203  |
| Percent voids in mineral aggregate (VMA), minimum | See Table 2                                | ASTM D6995  |
| TSR <sup>1</sup>                                  | not less than 80 at a saturation of 70-80% | ASTM D4867  |

<sup>1</sup> Test specimens for TSR shall be compacted at  $7 \pm 1.0$  % air voids. In areas subject to freeze-thaw, use freeze-thaw conditioning in lieu of moisture conditioning per ASTM D4867.

<sup>2</sup> AASHTO T340 at 100 psi hose pressure at 64°C test temperature may be used in the interim. If this method is used the required Value shall be less than 5 mm @ 8000 passes

The mineral aggregate shall be of such size that the percentage composition by weight, as determined by laboratory sieves, will conform to the gradation or gradations specified in Table 2 when tested in accordance with ASTM C136 and ASTM C117.

The gradations in Table 2 represent the limits that shall determine the suitability of aggregate for use from the sources of supply, be well graded from coarse to fine and shall not vary from the low limit on one sieve to the high limit on the adjacent sieve, or vice versa.

**Table 2. Aggregate - Asphalt Pavements**

| <b>Sieve Size</b>                                      | <b>Percentage by Weight Passing Sieve</b> |
|--------------------------------------------------------|-------------------------------------------|
| 1 inch (25.0 mm)                                       | --                                        |
| 3/4 inch (19.0 mm)                                     | --                                        |
| 1/2 inch (12.5 mm)                                     | 100                                       |
| 3/8 inch (9.5 mm)                                      | 90-100                                    |
| No. 4 (4.75 mm)                                        | 58-78                                     |
| No. 8 (2.36 mm)                                        | 40-60                                     |
| No. 16 (1.18 mm)                                       | 28-48                                     |
| No. 30 (600 µm)                                        | 18-38                                     |
| No. 50 (300 µm)                                        | 11-27                                     |
| No. 100 (150 µm)                                       | 6-18                                      |
| No. 200 (75 µm)                                        | 3-6                                       |
| <b>Voids in Mineral Aggregate (VMA)<sup>1</sup></b>    | 16                                        |
| <b>Asphalt Percent:</b>                                |                                           |
| Stone or gravel                                        | 5.5-8.0                                   |
| Slag                                                   | 7.0-10.5                                  |
| <b>Recommended Minimum Construction Lift Thickness</b> | 1 1/2" inch                               |

<sup>1</sup>To achieve minimum VMA during production, the mix design needs to account for material breakdown during production.

The aggregate gradations shown are based on aggregates of uniform specific gravity. The percentages passing the various sieves shall be corrected when aggregates of varying specific gravities are used, as indicated in the Asphalt Institute MS-2 Mix Design Manual, 7th Edition.

**Table 2. Aggregate - Asphalt Pavements**

| Sieve Size                                         | Percentage by Weight Passing Sieves |             |                          |
|----------------------------------------------------|-------------------------------------|-------------|--------------------------|
|                                                    | Gradation 1                         | Gradation 2 | Gradation 3 <sup>1</sup> |
| 1 inch (25.0 mm)                                   | 100                                 | --          | --                       |
| 3/4 inch (19.0 mm)                                 | 90-100                              | 100         | --                       |
| 1/2 inch (12.5 mm)                                 | 68-88                               | 90-100      | 100                      |
| 3/8 inch (9.5 mm)                                  | 60-82                               | 72-88       | 90-100                   |
| No. 4 (4.75 mm)                                    | 45-67                               | 53-73       | 58-78                    |
| No. 8 (2.36 mm)                                    | 32-54                               | 38-60       | 40-60                    |
| No. 16 (1.18 mm)                                   | 22-44                               | 26-48       | 28-48                    |
| No. 30 (600 µm)                                    | 15-35                               | 18-38       | 18-38                    |
| No. 50 (300 µm)                                    | 9-25                                | 11-27       | 11-27                    |
| No. 100 (150 µm)                                   | 6-18                                | 6-18        | 6-18                     |
| No. 200 (75 µm)                                    | 3-6                                 | 3-6         | 3-6                      |
| Voids in Mineral Aggregate (VMA)                   | 14                                  | 15          | 16                       |
| <b>Asphalt percent by total weight of mixture:</b> |                                     |             |                          |
| Stone or gravel                                    | 4.5-7.0                             | 5.0-7.5     | 5.5-8.0                  |
| Slag                                               | 5.0-7.5                             | 6.5-9.5     | 7.0-10.5                 |
| Recommended Minimum Construction Lift Thickness    | 3 inch                              | 2 inch      | 1 1/2 inch               |

<sup>1</sup>Gradation 3 is intended for leveling courses. FAA approval is required for use in other locations.

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**403-3.4 Reclaimed Asphalt Pavement (RAP).** RAP shall not be used.

**403-3.5 Control strip.** Full production shall not begin until an acceptable control strip has been constructed and accepted in writing by the RPR. The Contractor shall prepare and place a quantity of asphalt according to the JMF. The underlying grade or pavement structure upon which the control strip is to be constructed shall be the same as the remainder of the course represented by the control strip.

The Contractor will not be allowed to place the control strip until the Contractor quality control program (CQCP), showing conformance with the requirements of paragraph 403-5.1, has been accepted, in writing, by the RPR.

The control strip will consist of at least 250 tons (227 metric tons) or 1/2 subplot, whichever is greater. The control strip shall be placed in two lanes of the same width and depth to be used in production with a longitudinal cold joint. The cold joint must be cut back in accordance with paragraph 403-4.13 using the same procedure that will be used during production. The cold joint for the control strip will be an exposed construction joint at least four (4) hours old or when

the mat has cooled to less than 160°F (71°C). The equipment used in construction of the control strip shall be the same type, configuration and weight to be used on the project.

The control strip shall be evaluated for acceptance as a single lot in accordance with the acceptance criteria in paragraph 403-6.1 and 403-6.2.

The control strip will be considered acceptable by the RPR if the gradation, asphalt content, and VMA are within the action limits specified in paragraph 403-5.5a; and Mat density greater than or equal to 94%, air voids 3.5% +/- 1%, and joint density greater than or equal to 92%.

If the control strip is unacceptable, necessary adjustments to the JMF, plant operation, placing procedures, and/or rolling procedures shall be made and another control strip shall be placed. Unacceptable control strips shall be removed at the Contractor's expense.

The control strip will be considered one lot for payment based upon the average of a minimum of 3 samples (no sublots required for control strip). Payment will only be made for an acceptable control strip in accordance with paragraph 403-8.1.

**403-3.6 PAVING PLAN.** *The contractor shall prepare and submit a paving plan to the engineer at least 10 days prior to the start of paving. The plan shall include test strip location, paving lane locations, paving lengths and widths, (start and stop stations), the direction of pulls, partially loaded truck plan and/or temporary haul road (to preclude damaging underlying base/subgrade) and proposed haul routes.*

## CONSTRUCTION METHODS

**403-4.1 Weather limitations.** The asphalt shall not be placed upon a wet surface or when the surface temperature of the underlying course is less than specified in Table 4. The temperature requirements may be waived by the RPR, if requested; however, all other requirements including compaction shall be met.

**Table 4. Surface Temperature Limitations of Underlying Course**

| Mat Thickness                                                    | Base Temperature (Minimum) |           |
|------------------------------------------------------------------|----------------------------|-----------|
|                                                                  | Degrees F                  | Degrees C |
| 3 inches (7.5 cm) or greater                                     | 40                         | 4         |
| Greater than 2 inches (50 mm)<br>but less than 3 inches (7.5 cm) | 45                         | 7         |

**403-4.2 Asphalt plant.** Plants used for the preparation of asphalt shall conform to the requirements of American Association of State Highway and Transportation Officials (AASHTO) M156 including the following items:

**a. Inspection of plant.** The RPR, or RPR's authorized representative, shall have access, at all times, to all areas of the plant for checking adequacy of equipment; inspecting operation of the plant: verifying weights, proportions, and material properties; and checking the temperatures maintained in the preparation of the mixtures.

**b. Storage bins and surge bins.** The asphalt mixture stored in storage and/or surge bins shall meet the same requirements as asphalt mixture loaded directly into trucks. Asphalt

mixture shall not be stored in storage and/or surge bins for a period greater than twelve (12) hours. If the RPR determines there is an excessive heat loss, segregation or oxidation of the asphalt mixture due to temporary storage, temporary storage shall not be allowed.

**403-4.3 Aggregate stockpile management.** Aggregate stockpiles shall be constructed in such a manner that prevents segregation and intermixing of deleterious materials. Aggregates from different sources shall be stockpiled, weighed and batched separately at the concrete batch plant. Aggregates that have become segregated or mixed with earth or foreign material shall not be used.

A continuous supply of materials shall be provided to the work to ensure continuous placement.

**403-4.4 Hauling equipment.** Trucks used for hauling asphalt shall have tight, clean, and smooth metal beds. To prevent the asphalt from sticking to the truck beds, the truck beds shall be lightly coated with a minimum amount of paraffin oil, lime solution, or other material approved by the RPR. Petroleum products shall not be used for coating truck beds. Each truck shall have a suitable cover to protect the mixture from adverse weather. When necessary, to ensure that the mixture will be delivered to the site at the specified temperature, truck beds shall be insulated or heated and covers shall be securely fastened.

**403-4.4.1 Material transfer vehicle (MTV).** Material transfer Vehicles shall be required due to the improvement in smoothness and decrease in both physical and thermal segregation. To transfer the material from the hauling equipment to the paver, use a self-propelled, material transfer vehicle with a swing conveyor that can deliver material to the paver without making contact with the paver. The MTV shall be able to move back and forth between the hauling equipment and the paver providing material transfer to the paver, while allowing the paver to operate at a constant speed. The Material Transfer Vehicle will have remixing and storage capability to prevent physical and thermal segregation.

**403-4.5 Asphalt pavers.** Asphalt pavers shall be self-propelled with an activated heated screed, capable of spreading and finishing courses of asphalt that will meet the specified thickness, smoothness, and grade. The paver shall have sufficient power to propel itself and the hauling equipment without adversely affecting the finished surface. The asphalt paver shall be equipped with a control system capable of automatically maintaining the specified screed grade and elevation. ***The paver shall have GPS grade controls and enough base stations shall be utilized for accurate grade control over long lanes. External grade controls shall also be used as a backup and for the owner's representative independent checks. External controls shall be surveyed grade points along the paving lane.***

If the spreading and finishing equipment in use leaves tracks or indented areas, or produces other blemishes in the pavement that are not satisfactorily corrected by the scheduled operations, the use of such equipment shall be discontinued.

The paver shall be capable of paving to a minimum width specified in paragraph 401-4.11.

**403-4.6 Rollers.** The number, type, and weight of rollers shall be sufficient to compact the asphalt to the required density while it is still in a workable condition without crushing of the aggregate, depressions or other damage to the pavement surface. Rollers shall be in good condition, capable of operating at slow speeds to avoid displacement of the asphalt. All rollers shall be specifically designed and suitable for compacting asphalt concrete and shall be properly used. Rollers that impair the stability of any layer of a pavement structure or underlying soils shall not be used.

**403-4.6.1 Density device.** The Contractor shall have on site a density gauge during all paving operations in order to assist in the determination of the optimum rolling pattern, type of roller



and frequencies, as well as to monitor the effect of the rolling operations during production paving. The Contractor shall also supply a qualified technician during all paving operations to calibrate the density gauge and obtain accurate density readings for all new asphalt. These densities shall be supplied to the RPR upon request at any time during construction. No separate payment will be made for supplying the density gauge and technician.

**403-4.7 Preparation of asphalt binder.** The asphalt binder shall be heated in a manner that will avoid local overheating and provide a continuous supply of the asphalt material to the mixer at a uniform temperature. The temperature of the unmodified asphalt binder delivered to the mixer shall be sufficient to provide a suitable viscosity for adequate coating of the aggregate particles, but shall not exceed 325°F (160°C) when added to the aggregate. The temperature of modified asphalt binder shall be no more than 350°F (175°C) when added to the aggregate.

**403-4.8 Preparation of mineral aggregate.** The aggregate for the asphalt shall be heated and dried. The maximum temperature and rate of heating shall be such that no damage occurs to the aggregates. The temperature of the aggregate and mineral filler shall not exceed 350°F (175°C) when the asphalt binder is added. Particular care shall be taken that aggregates high in calcium or magnesium content are not damaged by overheating. The temperature shall not be lower than is required to obtain complete coating and uniform distribution on the aggregate particles and to provide a mixture of satisfactory workability.

**403-4.9 Preparation of asphalt mixture.** The aggregates and the asphalt binder shall be weighed or metered and introduced into the mixer in the amount specified by the JMF. The combined materials shall be mixed until the aggregate obtains a uniform coating of asphalt binder and is thoroughly distributed throughout the mixture. Wet mixing time shall be the shortest time that will produce a satisfactory mixture, but not less than 25 seconds for batch plants. The wet mixing time for all plants shall be established by the Contractor, based on the procedure for determining the percentage of coated particles described in ASTM D2489, for each individual plant and for each type of aggregate used. The wet mixing time will be set to achieve 95% of coated particles. For continuous mix plants, the minimum mixing time shall be determined by dividing the weight of its contents at operating level by the weight of the mixture delivered per second by the mixer. The moisture content of all asphalt upon discharge shall not exceed 0.5%.

**403-4.10 Application of Prime and Tack Coat.** Immediately before placing the asphalt mixture, the underlying course shall be cleaned of all dust and debris.

A prime coat in accordance with Item P-602 shall be applied to aggregate base prior to placing the asphalt mixture.

A tack coat shall be applied in accordance with Item P-603 to all vertical and horizontal asphalt and concrete surfaces prior to placement of the first and each subsequent lift of asphalt mixture.

**403-4.11 Laydown plan, transporting, placing, and finishing.** Prior to the placement of the asphalt, the Contractor shall prepare a laydown plan with the sequence of paving lanes and width to minimize the number of cold joints; the location of any temporary ramps; laydown temperature; and estimated time of completion for each portion of the work (milling, paving, rolling, cooling, etc.). The laydown plan and any modifications shall be approved by the RPR.

Deliveries shall be scheduled so that placing and compacting of asphalt is uniform with minimum stopping and starting of the paver. Hauling over freshly placed material shall not be permitted until the material has been compacted, as specified, and allowed to cool to approximately ambient temperature. The Contractor, at their expense, shall be responsible for repair of any damage to the pavement caused by hauling operations.

Contractor shall survey each lift of asphalt surface course and certify to RPR that every lot of each lift meets the grade tolerances of paragraph 401-6.2e before the next lift can be placed.

***The survey shall be by a licensed Florida surveyor in accordance with Item C-104. Surveyed grades of each lift of asphalt shall be on the same 50' x 50' grid, saved in in AutoCAD sent via email to the construction manager prior to starting the next lift. The lift name and sequence shall be identified in the CAD file. The contractor's licensed surveyor shall also paint grades on the surface ahead of the paver, on a 50' x 50' grid, before every lift so the owner's representative can verify the previous lift is to the correct grade and check the depth of the subsequent lift.***

Edges of existing asphalt pavement abutting the new work shall be saw cut and the cut off material and laitance removed. Apply a tack coat in accordance with P-603 before new asphalt material is placed against it.

The speed of the paver shall be regulated to eliminate pulling and tearing of the asphalt mat. Placement of the asphalt mix shall begin along the centerline of a crowned section or on the high side of areas with a one way slope unless shown otherwise on the laydown plan as accepted by the RPR. The asphalt mix shall be placed in consecutive adjacent lanes having a minimum width of **10** feet except where edge lanes require less width to complete the area. Additional screed sections attached to widen the paver to meet the minimum lane width requirements must include additional auger sections to move the asphalt mixture uniformly along the screed extension ***without resulting in aggregate segregation.***

The longitudinal joint in one course shall offset the longitudinal joint in the course immediately below by at least 1 foot ; however, the joint in the surface top course shall be at the centerline of crowned pavements. Transverse joints in one course shall be offset by at least 10 feet (3 m) from transverse joints in the previous course. Transverse joints in adjacent lanes shall be offset a minimum of 10 feet. On areas where irregularities or unavoidable obstacles make the use of mechanical spreading and finishing equipment impractical, the asphalt may be spread and luted by hand tools.

The RPR may at any time, reject any batch of asphalt, on the truck or placed in the mat, which is rendered unfit for use due to contamination, segregation, incomplete coating of aggregate, or overheated asphalt mixture. Such rejection may be based on only visual inspection or temperature measurements. In the event of such rejection, the Contractor may take a representative sample of the rejected material in the presence of the RPR, and if it can be demonstrated in the laboratory, in the presence of the RPR, that such material was erroneously rejected, payment will be made for the material at the contract unit price.

Areas of segregation in the surface course, as determined by the RPR, shall be removed and replaced at the Contractor's expense. The area shall be removed by saw cutting and milling a minimum of the construction lift thickness as specified in paragraph 401-3.3, Table 2 for the approved mix design. The area to be removed and replaced shall be a minimum width of the paver and a minimum of 10 feet long.

**403-4.12 Compaction of asphalt mixture.** After placing, the asphalt mixture shall be thoroughly and uniformly compacted by self-propelled rollers. The surface shall be compacted as soon as possible when the asphalt has attained sufficient stability so that the rolling does not cause undue displacement, cracking or shoving. The sequence of rolling operations and the type of rollers used shall be at the discretion of the Contractor. The speed of the roller shall, at all times, be sufficiently slow to avoid displacement of the hot mixture and be effective in compaction. Any surface defects, ***mat tearing*** and/or displacement occurring as a result of the roller, ***saw cut, pizza cut,*** or from any other cause, shall be corrected at the Contractor's expense.

Sufficient rollers shall be furnished to handle the output of the plant. Rolling shall continue until the surface is of uniform texture, true to grade and cross-section, and the required field density is obtained. To prevent adhesion of the asphalt to the roller, the wheels shall be equipped with a scraper and kept moistened with water as necessary.

In areas not accessible to the roller, the mixture shall be thoroughly compacted with approved power tampers.

Any asphalt that becomes loose and broken, mixed with dirt, contains check-cracking, or in any way defective shall be removed and replaced with fresh hot mixture and immediately compacted to conform to the surrounding area. This work shall be done at the Contractor's expense. Skin patching shall not be allowed.

**403-4.13 Joints.** The formation of all joints shall be made in such a manner as to ensure a continuous bond between the courses and obtain the required density. All joints shall have the same texture as other sections of the course and meet the requirements for smoothness and grade.

The roller shall not pass over the unprotected end of the freshly laid asphalt except when necessary to form a transverse joint. When necessary to form a transverse joint, it shall be made by means of placing a bulkhead or by tapering the course. The tapered edge shall be cut back to its full depth and width on a straight line to expose a vertical face prior to placing the adjacent lane. In both methods, all contact surfaces shall be coated with an asphalt tack coat before placing any fresh asphalt against the joint.

Longitudinal joints which have been left exposed for more than four (4) hours; the surface temperature has cooled to less than 175°F (80°C); or are irregular, damaged, uncompacted or otherwise defective shall be cut back with a cutting wheel or pavement saw a maximum of 3 inches (75 mm) to expose a clean, sound, uniform vertical surface for the full depth of the course. All cutback material and any laitance produced from cutting joints shall be removed from the project. An asphalt tack coat or other product approved by the RPR shall be applied to the clean, dry joint prior to placing any additional fresh asphalt against the joint. The cost of this work shall be considered incidental to the cost of the asphalt.

**403-4.14 Saw-cut grooving.** Saw-cut grooving is not required.

**403-4.15 Diamond grinding.** Diamond grinding shall be completed prior to pavement grooving. Diamond grinding shall be accomplished by sawing with saw blades impregnated with industrial diamond abrasive.

Diamond grinding shall be performed with a machine designed specifically for diamond grinding capable of cutting a path at least 3 feet (0.9 m) wide. The saw blades shall be 1/8-inch (3-mm) wide with a minimum of 55 to 60 blades per 12 inches (300 mm) of cutting head width; grooves between 0.090 and 0.130 inches (2 and 3.5 mm) wide; and peaks and ridges approximately 1/32 inch (1 mm) higher than the bottom of the grinding cut. The actual number of blades will be determined by the Contractor and depend on the hardness of the aggregate. Equipment or grinding procedures that causes ravels, aggregate fractures, spalls or disturbance to the pavement will not be permitted.

Grinding will be tapered in all directions to provide smooth transitions to areas not requiring grinding. The slurry resulting from the grinding operation shall be continuously removed and the pavement left in a clean condition. The Contractor shall apply a surface treatment per P-608 to all areas that have been subject to grinding.

**403-4.16 Nighttime Paving Requirements.** The Contractor shall provide adequate lighting during any nighttime construction. A lighting plan shall be submitted by the Contractor and

approved by the RPR prior to the start of any nighttime work. All work shall be in accordance with the approved CSPP and lighting plan.

**403-5.1 General.** The Contractor shall develop a CQCP in accordance with Item C-100. No partial payment will be made for materials that are subject to specific QC requirements without an approved CQCP.

**403-5.2 Contractor quality control (QC) facilities.** The Contractor shall provide or contract for testing facilities in accordance with Item C-100. The RPR shall be permitted unrestricted access to inspect the Contractor's QC facilities and witness QC activities. The RPR will advise the Contractor in writing of any noted deficiencies concerning the QC facility, equipment, supplies, or testing personnel and procedures. When the deficiencies are serious enough to be adversely affecting the test results, the incorporation of the materials into the work shall be suspended immediately and will not be permitted to resume until the deficiencies are satisfactorily corrected.

**403-5.3 Quality Control (QC) testing.** The Contractor shall perform all QC tests necessary to control the production and construction processes applicable to these specifications and as set forth in the approved CQCP. The testing program shall include, but not necessarily be limited to, tests for the control of asphalt content, aggregate gradation, temperatures, aggregate moisture, field compaction, and surface smoothness. A QC Testing Plan shall be developed as part of the CQCP.

**a. Asphalt content.** A minimum of two tests shall be performed per day in accordance with ASTM D6307 or ASTM D2172 for determination of asphalt content. When using ASTM D6307, the correction factor shall be determined as part of the first test performed at the beginning of plant production; and as part of every tenth test performed thereafter. The asphalt content for the day will be determined by averaging the test results.

**b. Gradation.** Aggregate gradations shall be determined a minimum of twice per lot from mechanical analysis of extracted aggregate in accordance with ASTM D5444 and ASTM C136, and ASTM C117.

**c. Moisture content of aggregate.** The moisture content of aggregate used for production shall be determined a minimum of once per lot in accordance with ASTM C566.

**d. Moisture content of asphalt.** The moisture content of the asphalt shall be determined once per lot in accordance with AASHTO T329 or ASTM D1461.

**e. Temperatures.** Temperatures shall be checked, at least four times per lot, at necessary locations to determine the temperatures of the dryer, the asphalt binder in the storage tank, the asphalt at the plant, and the asphalt at the job site.

**f. In-place density monitoring.** The Contractor shall conduct any necessary testing to ensure that the specified density is being achieved. A nuclear gauge may be used to monitor the pavement density in accordance with ASTM D2950.

**g. Smoothness for Contractor Quality Control.**

The Contractor shall perform smoothness testing in transverse and longitudinal directions daily (**on all lifts of asphalt**) to verify that the construction processes are producing pavement with variances less than ¼ inch in 12 feet, identifying areas that may pond water which could lead to hydroplaning of aircraft. If the smoothness criteria is not met, **the area shall be corrected by grinding or removal and** appropriate changes and corrections to the construction process shall be made by the Contractor before construction continues

The Contractor may use a 12-foot (3.7 m) "straightedge, a rolling inclinometer meeting the requirements of ASTM E2133 or rolling external reference device that can simulate a 12-foot (3.7m) straightedge approved by the RPR. Straight-edge testing shall start with one-half the length of the straightedge at the edge of pavement section being tested and then moved ahead one-half the length of the straightedge for each successive measurement. Testing shall be continuous across all joints. The surface irregularity shall be determined by placing the freestanding (unleveled) straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length, and measuring the maximum gap between the straightedge and the pavement surface in the area between the two high points. If the rolling inclinometer or external reference device is used, the data may be evaluated using the FAA profile program, ProFAA, using the 12-foot straightedge simulation function.

Smoothness readings shall not be made across grade changes or cross slope transitions. The transition between new and existing pavement and between the start and stop of lanes place shall be evaluated separately for conformance with the plans.

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**Include detail for transition between new and existing pavement including smoothness and grade limitations.**

\*\*\*\*\*

**(1) Transverse measurements.** Transverse measurements shall be taken for each day's production placed. Transverse measurements will be taken perpendicular to the pavement centerline each 50 feet (15 m) or more often as determined by the RPR. The joint between lanes shall be tested separately to facilitate smoothness between lanes.

**(2) Longitudinal measurements.** Longitudinal measurements shall be taken for each day's production placed. Longitudinal tests will be parallel to the centerline of paving; at the center of paving lanes when widths of paving lanes are less than 20 feet (6 m); and at the third points of paving lanes when widths of paving lanes are 20 ft (6 m) or greater. When placement abuts previously placed material the first measurement shall start with one half the length of the straight edge on the previously placed material.

Deviations on the final surface course in either the transverse or longitudinal direction that will trap water greater than 1/4 inch (6 mm) shall be corrected with diamond grinding per paragraph 403-4.15 or by removing and replacing the surface course to full depth. Grinding shall be tapered in all directions to provide smooth transitions to areas not requiring grinding. All areas in which diamond grinding has been performed shall be subject to the final pavement thickness tolerances specified in paragraph 401-6.1d(3) Areas that have been ground shall be sealed with a surface treatment in accordance with Item P-608. To avoid the surface treatment creating any conflict with runway or taxiway markings, it may be necessary to seal a larger area.

Control charts shall be kept to show area of each day's placement and the percentage of corrective grinding required. Corrections to production and placement shall be initiated when corrective grinding is required. If the Contractor's machines and/or methods produce significant areas that need corrective actions in excess of 10 percent of a day's production, production shall be stopped until corrective measures are implemented by the Contractor.

**h. Grade.** Grade shall be evaluated daily to allow adjustments to paving operations when grade measurements do not meet specifications. As a minimum, grade shall be evaluated prior to the placement of the first lift and then prior to all lifts.

Measurements will be taken at appropriate gradelines (as a minimum at center and edges of paving lane) and longitudinal spacing as shown on cross-sections and plans. The final surface

of the pavement will not vary from the gradeline elevations and cross-sections shown on the plans by more than 1/2 inch (12 mm) vertically and 0.1 feet laterally. The documentation will be provided by the Contractor to the RPR within 24 hours.

Areas with humps or depressions that exceed grade or smoothness criteria and that retain water on the surface must be ground off provided the course thickness after grinding is not more than 1/2 inch (12 mm) less than the thickness specified on the plans. Grinding shall be in accordance with paragraph 403-4.15.

The Contractor shall repair low areas or areas that cannot be corrected by grinding by removal of deficient areas to the depth of the final course plus 1/2 inch and replacing with new material. Skin patching is not allowed.

**403-5.4 Sampling.** When directed by the RPR, the Contractor shall sample and test any material that appears inconsistent with similar material being sampled, unless such material is voluntarily removed and replaced or deficiencies corrected by the Contractor. All sampling shall be in accordance with standard procedures specified.

**403-5.5 Control charts.** The Contractor shall maintain linear control charts both for individual measurements and range (i.e., difference between highest and lowest measurements) for aggregate gradation, asphalt content, and VMA. The VMA for each day shall be calculated and monitored by the QC laboratory.

Control charts shall be posted in a location satisfactory to the RPR and kept current. As a minimum, the control charts shall identify the project number, the contract item number, the test number, each test parameter, the Action and Suspension Limits applicable to each test parameter, and the Contractor's test results. The Contractor shall use the control charts as part of a process control system for identifying potential problems and assignable causes before they occur. If the Contractor's projected data during production indicates a problem and the Contractor is not taking satisfactory corrective action, the RPR may suspend production or acceptance of the material.

**a. Individual measurements.** Control charts for individual measurements shall be established to maintain process control within tolerance for aggregate gradation, asphalt content, and VMA. The control charts shall use the JMF target values as indicators of central tendency for the following test parameters with associated Action and Suspension Limits:

**Control Chart Limits for Individual Measurements**

| Sieve              | Action Limit | Suspension Limit |
|--------------------|--------------|------------------|
| 3/4 inch (19.0 mm) | ±6%          | ±9%              |
| 1/2 inch (12.5 mm) | ±6%          | ±9%              |
| 3/8 inch (9.5 mm)  | ±6%          | ±9%              |
| No. 4 (4.75 mm)    | ±6%          | ±9%              |
| No. 16 (1.18 mm)   | ±5%          | ±7.5%            |
| No. 50 (300 µm)    | ±3%          | ±4.5%            |
| No. 200 (75 µm)    | ±2%          | ±3%              |
| Asphalt Content    | ±0.45%       | ±0.70%           |
| Minimum VMA        | -0.5%        | -1.0%            |

**b. Range.** Control charts for range shall be established to control process variability for the test parameters and Suspension Limits listed below. The range shall be computed for each lot as the difference between the two test results for each control parameter. The Suspension Limits specified below are based on a sample size of  $n = 2$ . Should the Contractor elect to perform more than two tests per lot, the Suspension Limits shall be adjusted by multiplying the Suspension Limit by 1.18 for  $n = 3$  and by 1.27 for  $n = 4$ .

**Control Chart Limits Based on Range  
( $n = 2$ )**

| Sieve                | Suspension Limit |
|----------------------|------------------|
| 1/2 inch (12.5 mm)   | 11%              |
| 3/8 inch (9.5 mm)    | 11%              |
| No. 4 (4.75 mm)      | 11%              |
| No. 16 (1.18 mm)     | 9%               |
| No. 50 (300 $\mu$ m) | 6%               |
| No. 200 (75 $\mu$ m) | 3.5%             |
| Asphalt Content      | 0.8%             |

**c. Corrective action.** [ The CQCP shall indicate that appropriate action shall be taken when the process is believed to be out of tolerance. The Plan shall contain sets of rules to gauge when a process is out of control and detail what action will be taken to bring the process into control. As a minimum, a process shall be deemed out of control and production stopped and corrective action taken, if:

- (1) One point falls outside the Suspension Limit line for individual measurements or range; or
- (2) Two points in a row fall outside the Action Limit line for individual measurements. ]

**403-5.6 Quality control (QC) reports.** The Contractor shall maintain records and shall submit reports of QC activities daily in accordance with the CQCP described in Item C-100.

## MATERIAL ACCEPTANCE

**403-6.1. Quality Assurance Acceptance sampling and testing.** Unless otherwise specified, all acceptance sampling and testing necessary to determine conformance with the requirements specified in this section will be performed by the RPR at no cost to the Contractor except that coring as required in this section shall be completed and paid for by the Contractor.

**a. Quality Assurance (QA) testing laboratory.** The QA testing laboratory performing these acceptance tests will be accredited in accordance with ASTM D3666. The QA laboratory accreditation will be current and listed on the accrediting authority's website. All test methods required for acceptance sampling and testing will be listed on the lab accreditation.

**b. Lot Size.** A standard lot will be equal to one day's production divided into approximately equal sublots of between 400 to 600 tons. When only one or two sublots are produced in a day's production, the sublots will be combined with the production lot from the previous or next day.

Where more than one plant is simultaneously producing asphalt for the job, the lot sizes will apply separately for each plant.

**c. Asphalt air voids.** Plant-produced asphalt will be tested for air voids on a subplot basis.

**(1) Sampling.** Material from each subplot shall be sampled in accordance with ASTM D3665. Samples shall be taken from material deposited into trucks at the plant or at the job site in accordance with ASTM D979. The sample of asphalt may be put in a covered metal tin and placed in an oven for not less than 30 minutes nor more than 60 minutes to maintain the material at or above the compaction temperature as specified in the JMF.

**(2) Testing.** Air voids will be determined for each subplot in accordance with ASTM D3203 for a set of three compacted specimens prepared in accordance with ASTM D6925.

**d. In-place asphalt mat and joint density.** Each subplot will be tested for in-place mat and joint density as a percentage of the theoretical maximum density (TMD).

**(1) Sampling.** The Contractor will cut minimum 5 inches (125 mm) diameter samples in accordance with ASTM D5361. The Contractor shall furnish all tools, labor, and materials for cleaning, and filling the cored pavement. Laitance produced by the coring operation shall be removed immediately after coring, and core holes shall be filled within one day after sampling in a manner acceptable to the RPR.

**(2) Bond.** Each lift of asphalt shall be bonded to the underlying layer. If cores reveal that the surface is not bonded, additional cores shall be taken as directed by the RPR to determine the extent of unbonded areas. Unbonded areas shall be removed by milling and replaced at no additional cost as directed by the RPR.

**(3) Thickness.** Thickness of each lift of surface course will be evaluated by the RPR for compliance to the requirements shown on the plans after any necessary corrections for grade. Measurements of thickness will be made using the cores extracted for each subplot for density measurement. The maximum allowable deficiency at any point will not be more than 1/4 inch (6 mm) less than the thickness indicated for the lift. Average thickness of lift, or combined lifts, will not be less than the indicated thickness. Where the thickness tolerances are not met, the lot or subplot shall be corrected by the Contractor at his expense by removing the deficient area and replacing with new pavement. The Contractor, at his expense, may take additional cores as approved by the RPR to circumscribe the deficient area.

**(4) Mat density.** One core shall be taken from each subplot. Core locations will be determined by the RPR in accordance with ASTM D3665. Cores for mat density shall not be taken closer than one foot (30 cm) from a transverse or longitudinal joint. The bulk specific gravity of each cored sample will be determined in accordance with ASTM D2726. The percent compaction (density) of each sample will be determined by dividing the bulk specific gravity of each subplot sample by the TMD for that subplot.

**(5) Joint density.** One core centered over the longitudinal joint shall be taken for each subplot which contains a longitudinal joint. Core locations will be determined by the RPR in accordance with ASTM D3665. The bulk specific gravity of each core sample will be determined in accordance with ASTM D2726. The percent compaction (density) of each sample will be determined by dividing the bulk specific gravity of each joint density sample by the average TMD for the lot. The TMD used to determine the joint density at joints formed between lots will be the lower of the average TMD values from the adjacent lots.

#### **403-6.2 Acceptance criteria.**

**a. General.** Acceptance will be based on the implementation of the Contractor Quality Control Program (CQCP) and the following characteristics of the asphalt and completed pavements: air voids, mat density, joint density, grade and Profilograph smoothness.

**b. Air voids.** Acceptance of each lot of plant produced material for air voids will be based upon the average air void from the sublots. If the average air voids of the lot are equal to or



greater than 2% and equal to or less than 5%, then the lot will be acceptable. If the average is below 2% or greater than 5%, the lot shall be removed and replaced at the Contractor's expense.

**c. Mat density.** Acceptance of each lot of plant produced material for mat density will be based on the average of all of the densities taken from the sublots. If the average mat density of the lot so established equals or exceeds 94%, the lot will be acceptable. If the average mat density of the lot is below 94%, the lot shall be removed and replaced at the Contractor's expense.

**d. Joint density.** Acceptance of each lot of plant produced asphalt for joint density will be based on the average of all of the joint densities taken from the sublots. If the average joint density of the lot so established equals or exceeds 92%, the lot will be acceptable. If the average joint density of the lot is less than 92%, the Contractor shall stop production and evaluate the method of compacting joints. Production may resume once the reason for poor compaction has been determined and appropriate measures have been taken to ensure proper compaction.

**e. Grade.** The final finished surface of the pavement of the completed project shall be surveyed to verify that the grade elevations and cross-sections shown on the plans do not deviate more than 1/2 inch (12 mm) vertically or 0.1 feet (30 mm) laterally.

Cross-sections of the pavement shall be taken at a minimum 50-foot (15-m) longitudinal spacing and at all longitudinal grade breaks. Minimum cross-section grade points shall include grade at centerline,  $\pm 10$  feet of centerline, and edge of runway and taxiway pavement.

The survey and documentation shall be stamped and signed by a licensed surveyor. Payment for sublots that do not meet grade for over 25% of the subplot shall not be more than 95%.

#### **403-6.3 Resampling Pavement for Mat Density.**

**a. General.** Resampling of a lot of pavement will only be allowed for mat density and then, only if the Contractor requests same in writing, within 48 hours after receiving the written test results from the RPR. A retest will consist of all the sampling and testing procedures contained in paragraphs 403-6.1. Only one resampling per lot will be permitted.

(1) A redefined mat density will be calculated for the resampled lot. The number of tests used to calculate the redefined mat density will include the initial tests made for that lot plus the retests.

(2) The cost for resampling and retesting shall be borne by the Contractor.

**b. Payment for resampled lots.** The redefined mat density for a resampled lot will be used to evaluate the acceptance of that lot in accordance with paragraph 403-6.2.

**c. Outliers.** Check for outliers in accordance with ASTM E178, at a significance level of 5%. Outliers will be discarded and density determined using the remaining test values.

### **METHOD OF MEASUREMENT**

**403-7.1 Measurement.** Plant mix asphalt mix pavement shall be measured by the number of tons (kg) of asphalt pavement used in the accepted work. Recorded batch weights or truck scale weights will be used to determine the basis for the tonnage.

## **BASIS OF PAYMENT**

**403-8.1 Payment.** Payment for a lot of asphalt mixture meeting all acceptance criteria as specified in paragraph 403-6.2 shall be made at the contract unit price per ton (kg) for asphalt. The price shall be compensation for furnishing all materials, for all preparation, mixing, and placing of these materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

Item P-403                      Asphalt Mixture Surface Course - per ton

## **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C29   | Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate                                                        |
| ASTM C88   | Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate                                      |
| ASTM C117  | Standard Test Method for Materials Finer than 75- $\mu$ m (No. 200) Sieve in Mineral Aggregates by Washing                          |
| ASTM C127  | Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate                           |
| ASTM C131  | Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine |
| ASTM C136  | Standard Test Method for Sieve or Screen Analysis of Fine and Coarse Aggregates                                                     |
| ASTM C142  | Standard Test Method for Clay Lumps and Friable Particles in Aggregates                                                             |
| ASTM C183  | Standard Practice for Sampling and the Amount of Testing of Hydraulic Cement                                                        |
| ASTM C566  | Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying                                                   |
| ASTM D75   | Standard Practice for Sampling Aggregates                                                                                           |
| ASTM D242  | Standard Specification for Mineral Filler for Bituminous Paving Mixtures                                                            |
| ASTM D946  | Standard Specification for Penetration-Graded Asphalt Cement for Use in Pavement Construction                                       |
| ASTM D979  | Standard Practice for Sampling Bituminous Paving Mixtures                                                                           |
| ASTM D1073 | Standard Specification for Fine Aggregate for Bituminous Paving Mixtures                                                            |

|            |                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------|
| ASTM D1074 | Standard Test Method for Compressive Strength of Bituminous Mixtures                                                           |
| ASTM D1461 | Standard Test Method for Moisture or Volatile Distillates in Bituminous Paving Mixtures                                        |
| ASTM D2041 | Standard Test Method for Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures                        |
| ASTM D2172 | Standard Test Method for Quantitative Extraction of Bitumen from Bituminous Paving Mixtures                                    |
| ASTM D2419 | Standard Test Method for Sand Equivalent Value of Soils and Fine Aggregate                                                     |
| ASTM D2489 | Standard Practice for Estimating Degree of Particle Coating of Bituminous-Aggregate Mixtures                                   |
| ASTM D2726 | Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures                     |
| ASTM D2950 | Standard Test Method for Density of Bituminous Concrete in Place by Nuclear Methods                                            |
| ASTM D3203 | Standard Test Method for Percent Air Voids in Compacted Dense and Open Bituminous Paving Mixtures                              |
| ASTM D3381 | Standard Specification for Viscosity-Graded Asphalt Cement for Use in Pavement Construction                                    |
| ASTM D3665 | Standard Practice for Random Sampling of Construction Materials                                                                |
| ASTM D3666 | Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials                  |
| ASTM D4125 | Standard Test Methods for Asphalt Content of Bituminous mixtures by the Nuclear Method                                         |
| ASTM D4318 | Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                                           |
| ASTM D4552 | Standard Practice for Classifying Hot-Mix Recycling Agents                                                                     |
| ASTM D4791 | Standard Test Method for Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate              |
| ASTM D4867 | Standard Test Method for Effect of Moisture on Asphalt Concrete Paving Mixtures                                                |
| ASTM D5444 | Standard Test Method for Mechanical Size Analysis of Extracted Aggregate                                                       |
| ASTM D5581 | Standard Test Method for Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus (6 inch-Diameter Specimen) |
| ASTM D5821 | Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate                                 |
| ASTM D6307 | Standard Test Method for Asphalt Content of Hot-Mix Asphalt by Ignition Method                                                 |

|                                                                             |                                                                                                                                                                |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D6373                                                                  | Standard Specification for Performance Graded Asphalt Binder                                                                                                   |
| ASTM D6752                                                                  | Standard Test Method for Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Automatic Vacuum Sealing Method                              |
| ASTM D6925                                                                  | Standard Test Method for Preparation and Determination of the Relative Density of Hot Mix Asphalt (HMA) Specimens by Means of the SuperPave Gyratory Compactor |
| ASTM D6926                                                                  | Standard Practice for Preparation of Bituminous Specimens Using Marshall Apparatus                                                                             |
| ASTM D6927                                                                  | Standard Test Method for Marshall Stability and Flow of Bituminous Mixtures                                                                                    |
| ASTM D6995                                                                  | Standard Test Method for Determining Field VMA based on the Maximum Specific Gravity of the Mix (Gmm)                                                          |
| ASTM E11                                                                    | Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves                                                                                         |
| ASTM E178                                                                   | Standard Practice for Dealing with Outlying Observations                                                                                                       |
| ASTM E2133                                                                  | Standard Test Method for Using a Rolling Inclinator to Measure Longitudinal and Transverse Profiles of a Traveled Surface                                      |
| American Association of State Highway and Transportation Officials (AASHTO) |                                                                                                                                                                |
| AASHTO M156                                                                 | Standard Specification for Requirements for Mixing Plants for Hot-Mixed, Hot-Laid Bituminous Paving Mixtures                                                   |
| AASHTO T329                                                                 | Standard Method of Test for Moisture Content of Hot Mix Asphalt (HMA) by Oven Method                                                                           |
| AASHTO T 340                                                                | Standard Method of Test for Determining the Rutting Susceptibility of Hot Mix Asphalt (APA) Using the Asphalt Pavement Analyzer (APA)                          |
| Asphalt Institute (AI)                                                      |                                                                                                                                                                |
| MS-2                                                                        | Mix Design Manual, 7th Edition                                                                                                                                 |
| MS-26                                                                       | Asphalt Binder Handbook<br>AI State Binder Specification Database                                                                                              |
| FAA Orders                                                                  |                                                                                                                                                                |
| 5300.1                                                                      | Modifications to Agency Airport Design, Construction, and Equipment Standards                                                                                  |
| Federal Highway Administration (FHWA)                                       |                                                                                                                                                                |
| Long Term Pavement Performance Binder program                               |                                                                                                                                                                |
| Software                                                                    |                                                                                                                                                                |
| FAARFIELD                                                                   |                                                                                                                                                                |

**END OF ITEM P-403**



## ITEM P-602 EMULSIFIED ASPHALT PRIME COAT

### DESCRIPTION

**602-1.1** This item shall consist of an application of emulsified asphalt material on the prepared base course in accordance with these specifications and in reasonably close conformity to the lines shown on the plans.

### MATERIALS

**602-2.1 EMULSIFIED ASPHALT MATERIAL.** The emulsified asphalt material shall be as specified in ASTM D3628 for use as a prime coat appropriate to local conditions. The Contractor shall provide a copy of the manufacturer's Certificate of Analysis (COA) for the emulsified asphalt material. The COA shall be provided to and approved by the Resident Project Representative (RPR) before the emulsified asphalt material is applied. The furnishing of the COA for the emulsified asphalt material shall not be interpreted as a basis for final acceptance. The manufacturer's COA may be subject to verification by testing the material delivered for use on the project.

### CONSTRUCTION METHODS

**602-3.1 WEATHER LIMITATIONS.** The emulsified asphalt prime coat shall be applied only when the existing surface is dry; the atmospheric temperature is 50°F (10°C) or above, and the temperature has not been below 35°F (2°C) for the 12 hours prior to application; and when the weather is not foggy or rainy. The temperature requirements may be waived when directed by the RPR.

**602-3.2 EQUIPMENT.** The equipment shall include a self-powered pressure asphalt material distributor and equipment for heating asphalt material.

Provide a distributor with pneumatic tires of such size and number that the load produced on the base surface does not exceed 65.0 psi (4.5 kg/sq cm) of tire width to prevent rutting, shoving or otherwise damaging the base, surface or other layers in the pavement structure. Design and equip the distributor to spray the asphalt material in a uniform coverage at the specified temperature, at readily determined and controlled rates from 0.05 to 1.0 gallons per square yard (0.23 to 4.5 L/square meter), with a pressure range of 25 to 75 psi (172.4 to 517.1 kPa) and with an allowable variation from the specified rate of not more than ±5%, and at variable widths. Include with the distributor equipment a separate power unit for the bitumen pump, full-circulation spray bars, tachometer, pressure gauges, volume-measuring devices, adequate heaters for heating of materials to the proper application temperature, a thermometer for reading the temperature of tank contents, and a hand hose attachment suitable for applying asphalt material manually to areas inaccessible to the distributor. Equip the distributor to circulate and agitate the asphalt material during the heating process. If the distributor is not equipped with an operable quick shutoff valve, the prime operations shall be started and stopped on building paper.

A power broom and power blower suitable for cleaning the surfaces to which the asphalt coat is to be applied shall be provided.

Asphalt distributors must be calibrated annually in accordance with ASTM D2995. The Contractor must furnish a current calibration certification for the asphalt distributor truck from any State or other agency as approved by the RPR.

**602-3.3 APPLICATION OF EMULSIFIED ASPHALT MATERIAL.** Immediately before applying the prime coat, the full width of the surface to be primed shall be swept with a power broom to remove all loose dirt and other objectionable material.

The asphalt emulsion material shall be uniformly applied with an asphalt distributor at the rate of 0.15 to 0.30 gallons per square yard (0.68 to 1.36 liters per square meter) depending on the base course surface texture. The type of asphalt material and application rate shall be approved by the RPR prior to application.

Following application of the emulsified asphalt material and prior to application of the succeeding layer of pavement, allow the asphalt coat to cure and to obtain evaporation of any volatiles or moisture. Maintain the coated surface until the succeeding layer of pavement is placed, by protecting the surface against damage and by repairing and recoating deficient areas. Allow the prime coat to cure without being disturbed for a period of at least 48 hours or longer, as may be necessary to attain penetration into the treated course. Furnish and spread sand to effectively blot up and cure excess asphalt material. The Contractor shall remove blotting sand prior to asphalt concrete lay down operations at no additional expense to the Owner. Keep traffic off surfaces freshly treated with asphalt material. Provide sufficient warning signs and barricades so that traffic will not travel over freshly treated surfaces.

**602-3.4 TRIAL APPLICATION RATES.** The Contractor shall apply a minimum of three lengths of at least 100 feet (30 m) for the full width of the distributor bar to evaluate the amount of emulsified asphalt material that can be satisfactorily applied with the equipment. Apply three different application rates of emulsified asphalt materials within the application range specified in paragraph 602-3.3. Other trial applications can be made using various amounts of material as directed by the RPR. The trial application is to demonstrate the equipment can uniformly apply the emulsified asphalt material within the rates specified and determine the application rate for the project.

**602-3.5 FREIGHT AND WAYBILLS.** The Contractor shall submit waybills and delivery tickets during the progress of the work. Before the final estimate is allowed, file with the RPR certified waybills and certified delivery tickets for all emulsified asphalt materials used in the construction of the pavement covered by the contract. Do not remove emulsified asphalt material from storage until the initial outage and temperature measurements have been taken. The delivery or storage units will not be released until the final outage has been taken.

## **METHOD OF MEASUREMENT**

**602-4.1** The emulsified asphalt material for prime coat shall be measured by the gallon. Volume shall be corrected to the volume at 60°F (16°C) in accordance with ASTM D4311. The emulsified asphalt material paid for will be the measured quantities used in the accepted work, provided that the measured quantities are not 10% over the specified application rate. Any amount of emulsified asphalt material more than 10% over the specified application rate for each application will be deducted from the measured quantities, except for irregular areas where hand spraying of the emulsified asphalt material is necessary. Water added to emulsified asphalt will not be measured for payment.

## **BASIS OF PAYMENT**

**602-5.1** Payment shall be made at the contract unit price per gallon for emulsified asphalt prime coat. This price shall be full compensation for furnishing all materials and for all preparation, delivering, and applying the materials, and for all labor, equipment, tools, and incidentals necessary to complete this item.

Payment will be made under:

|                |                                            |
|----------------|--------------------------------------------|
| Item P-602-5.1 | Emulsified Asphalt Prime Coat - per gallon |
|----------------|--------------------------------------------|

## **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------|
| ASTM D2995 | Standard Practice for Estimating Application Rate and Residual Application Rate of Bituminous Distributors |
| ASTM D3628 | Standard Practice for Selection and Use of Emulsified Asphalts                                             |

**END OF ITEM P-602**



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## ITEM P-603 EMULSIFIED ASPHALT TACK COAT

### DESCRIPTION

**603-1.1** This item shall consist of preparing and treating an asphalt or concrete surface with asphalt material in accordance with these specifications and in reasonably close conformity to the lines shown on the plans.

### MATERIALS

**603-2.1 ASPHALT MATERIALS.** The asphalt material shall be an emulsified asphalt as specified in ASTM D3628 as an asphalt application for tack coat appropriate to local conditions. The emulsified asphalt shall not be diluted. The Contractor shall provide a copy of the manufacturer's Certificate of Analysis (COA) for the asphalt material to the Resident Project Representative (RPR) before the asphalt material is applied for review and acceptance. The furnishing of COA for the asphalt material shall not be interpreted as a basis for final acceptance. The manufacturer's COA may be subject to verification by testing the material delivered for use on the project.

### CONSTRUCTION METHODS

**603-3.1 WEATHER LIMITATIONS.** The tack coat shall be applied only when the existing surface is dry and the atmospheric temperature is 50°F (10°C) or above; the temperature has not been below 35°F (2°C) for the 12 hours prior to application; and when the weather is not foggy or rainy. The temperature requirements may be waived when directed by the RPR.

**603-3.2 EQUIPMENT.** The Contractor shall provide equipment for heating and applying the emulsified asphalt material. The emulsion shall be applied with a manufacturer-approved computer rate-controlled asphalt distributor. The equipment shall be in good working order and contain no contaminants or diluents in the tank. Spray bar tips must be clean, free of burrs, and of a size to maintain an even distribution of the emulsion. Any type of tip or pressure source is suitable that will maintain predetermined flow rates and constant pressure during the application process with application speeds under eight (8) miles per hour (13 km per hour) or seven (700) feet per minute (213 m per minute).

The equipment will be tested under pressure for leaks and to ensure proper set-up before use to verify truck set-up (via a test-shot area), including but not limited to, nozzle tip size appropriate for application, spray-bar height and pressure and pump speed, evidence of triple-overlap spray pattern, lack of leaks, and any other factors relevant to ensure the truck is in good working order before use.

The distributor truck shall be equipped with a minimum 12-foot (3.7-m) spreader spray bar with individual nozzle control with computer-controlled application rates. The distributor truck shall have an easily accessible thermometer that constantly monitors the temperature of the emulsion, and have an operable mechanical tank gauge that can be used to cross-check the computer accuracy. If the distributor is not equipped with an operable quick shutoff valve, the prime operations shall be started and stopped on building paper.

The distributor truck shall be equipped to effectively heat and mix the material to the required temperature prior to application as required. Heating and mixing shall be done in accordance with the manufacturer's recommendations. Do not overheat or over mix the material.

The distributor shall be equipped with a hand sprayer.

Asphalt distributors must be calibrated annually in accordance with ASTM D2995. The Contractor must furnish a current calibration certification for the asphalt distributor truck from any State or other agency as approved by the RPR.

A power broom and/or power blower suitable for cleaning the surfaces to which the asphalt tack coat is to be applied shall be provided.

**603-3.3 APPLICATION OF EMULSIFIED ASPHALT MATERIAL.** The emulsified asphalt shall not be diluted. Immediately before applying the emulsified asphalt tack coat, the full width of surface to be treated shall be swept with a power broom and/or power blower to remove all loose dirt and other objectionable material.

The emulsified asphalt material shall be uniformly applied with an asphalt distributor at the rates appropriate for the conditions and surface specified in the table below. The type of asphalt material and application rate shall be approved by the RPR prior to application.

**Emulsified Asphalt**

| <b>Surface Type</b>     | <b>Residual Rate, gal/SY<br/>(L/square meter)</b> | <b>Emulsion Application Bar Rate,<br/>gal/SY (L/square meter)</b> |
|-------------------------|---------------------------------------------------|-------------------------------------------------------------------|
| <b>New asphalt</b>      | 0.02-0.05 (0.09-0.23)                             | 0.03-0.07 (0.13-0.32)                                             |
| <b>Existing asphalt</b> | 0.04-0.07 (0.18-0.32)                             | 0.06-0.11 (0.27-0.50)                                             |
| <b>Milled Surface</b>   | 0.04-0.08 (0.18-0.36)                             | .0.06-0.12 (0.27-0.54)                                            |
| <b>Concrete</b>         | 0.03-0.05 (0.13-0.23)                             | 0.05-0.08 (0.23-0.36)                                             |

After application of the tack coat, the surface shall be allowed to cure without being disturbed for the period of time necessary to permit drying and setting of the tack coat. This period shall be determined by the RPR. The Contractor shall protect the tack coat and maintain the surface until the next course has been placed. When the tack coat has been disturbed by the Contractor, tack coat shall be reapplied at the Contractor's expense.

**603-3.4 FREIGHT AND WAYBILLS.** The Contractor shall submit waybills and delivery tickets, during progress of the work. Before the final statement is allowed, file with the RPR certified waybills and certified delivery tickets for all emulsified asphalt materials used in the construction of the pavement covered by the contract. Do not remove emulsified asphalt material from storage until the initial outage and temperature measurements have been taken. The delivery or storage units will not be released until the final outage has been taken.

## **METHOD OF MEASUREMENT**

**603-4.1** The emulsified asphalt material for tack coat shall be measured by the gallon. Volume shall be corrected to the volume at 60°F (16°C) in accordance with ASTM D1250. The emulsified asphalt material paid for will be the measured quantities used in the accepted work, provided that

the measured quantities are not 10% over the specified application rate. Any amount of emulsified asphalt material more than 10% over the specified application rate for each application will be deducted from the measured quantities, except for irregular areas where hand spraying of the emulsified asphalt material is necessary. Water added to emulsified asphalt will not be measured for payment.

### **BASIS OF PAYMENT**

**603.5-1** Payment shall be made at the contract unit price per gallon of emulsified asphalt material. This price shall be full compensation for furnishing all materials, for all preparation, delivery, and application of these materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

|                |                                           |
|----------------|-------------------------------------------|
| Item P-603-5.1 | Emulsified Asphalt Tack Coat - per gallon |
|----------------|-------------------------------------------|

### **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------|
| ASTM D1250 | Standard Guide for Use of the Petroleum Measurement Tables                                                 |
| ASTM D2995 | Standard Practice for Estimating Application Rate and Residual Application Rate of Bituminous Distributors |
| ASTM D3628 | Standard Practice for Selection and Use of Emulsified Asphalts                                             |

**END ITEM P-603**

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## ITEM P-605 JOINT SEALANTS FOR PAVEMENTS

### DESCRIPTION

**605-1.1** This item shall consist of providing and installing a resilient and adhesive joint sealing material capable of effectively sealing joints in pavement; joints between different types of pavements; and cracks in existing pavement.

### MATERIALS

**605-2.1 JOINT SEALANTS.** Joint sealant materials shall meet the requirements of ASTM D5893.

Each lot or batch of sealant shall be delivered to the jobsite in the manufacturer's original sealed container. Each container shall be marked with the manufacturer's name, batch or lot number, the safe heating temperature, and shall be accompanied by the manufacturer's certification stating that the sealant meets the requirements of this specification.

**605-2.2 BACKER ROD.** The material furnished shall be a compressible, non-shrinking, non-staining, non-absorbing material that is non-reactive with the joint sealant in accordance with ASTM D5249. The backer-rod material shall be  $25\% \pm 5\%$  larger in diameter than the nominal width of the joint.

**605-2.3 BOND BREAKING TAPES.** Provide a bond breaking tape or separating material that is a flexible, non-shrinkable, non-absorbing, non-staining, and non-reacting adhesive-backed tape. The material shall have a melting point at least 5°F (3°C) greater than the pouring temperature of the sealant being used when tested in accordance with ASTM D789. The bond breaker tape shall be approximately 1/8 inch (3 mm) wider than the nominal width of the joint and shall not bond to the joint sealant.

### CONSTRUCTION METHODS

**605-3.1 TIME OF APPLICATION.** Joints shall be sealed as soon after completion of the curing period as feasible and before the pavement is opened to traffic, including construction equipment. The pavement temperature shall be 50°F (10°C) and rising at the time of application of the poured joint sealing material. Do not apply sealant if moisture is observed in the joint.

**605-3.2 EQUIPMENT.** Machines, tools, and equipment used in the performance of the work required by this section shall be approved before the work is started and maintained in satisfactory condition at all times. Submit a list of proposed equipment to be used in performance of construction work including descriptive data, 10 days prior to use on the project.

**a. Tractor-mounted routing tool.** Provide a routing tool, used for removing old sealant from the joints, of such shape and dimensions and so mounted on the tractor that it will not damage the sides of the joints. The tool shall be designed so that it can be adjusted to remove the old material to varying depths as required. The use of V-shaped tools or rotary impact routing devices will not be permitted. Hand-operated spindle routing devices may be used to clean and enlarge random cracks.

**b. Concrete saw.** Provide a self-propelled power saw, with water-cooled diamond or abrasive saw blades, for cutting joints to the depths and widths specified.

**c. Sandblasting equipment.** Sandblasting is not allowed.

**d. Waterblasting equipment.** The Contractor must demonstrate waterblasting equipment including the pumps, hose, guide and nozzle size, under job conditions, before approval in accordance with paragraph 605-3.3. The Contractor shall demonstrate, in the presence of the RPR, that the method cleans the joint and does not damage the joint.

**e. Hand tools.** Hand tools may be used, when approved, for removing defective sealant from a crack and repairing or cleaning the crack faces. Hand tools should be carefully evaluated for potential spalling effects prior to approval for use.

**f. Hot-poured sealing equipment.** The unit applicators used for heating and installing ASTM D6690 joint sealant materials shall be mobile and shall be equipped with a double-boiler, agitator-type kettle with an oil medium in the outer space for heat transfer; a direct-connected pressure-type extruding device with a nozzle shaped for inserting in the joint to be filled; positive temperature devices for controlling the temperature of the transfer oil and sealant; and a recording type thermometer for indicating the temperature of the sealant. The applicator unit shall be designed so that the sealant will circulate through the delivery hose and return to the inner kettle when not in use.

**g. Cold-applied, single-component sealing equipment.** The equipment for installing ASTM D5893 single component joint sealants shall consist of an extrusion pump, air compressor, following plate, hoses, and nozzle for transferring the sealant from the storage container into the joint opening. The dimension of the nozzle shall be such that the tip of the nozzle will extend into the joint to allow sealing from the bottom of the joint to the top. Maintain the initially approved equipment in good working condition, serviced in accordance with the supplier's instructions, and unaltered in any way without obtaining prior approval. Small hand-held air-powered equipment (i.e., caulking guns) may be used for small applications.

**605-3.3 PREPARATION OF JOINTS.** Pavement joints for application of material in this specification must be dry, clean of all scale, dirt, dust, curing compound, and other foreign matter. The Contractor shall demonstrate, in the presence of the RPR, that the method cleans the joint and does not damage the joint.

**a. Sawing.** All joints shall be sawed in accordance with specifications and plan details. Immediately after sawing the joint, the resulting slurry shall be completely removed from joint and adjacent area by flushing with a jet of water, and by use of other tools as necessary.

**b. Sealing.** Immediately before sealing, the joints shall be thoroughly cleaned of all remaining laitance, curing compound, filler, protrusions of hardened concrete, old sealant and other foreign material from the sides and upper edges of the joint space to be sealed. Cleaning shall be accomplished by concrete saw as specified in paragraph 605-3.2. The newly exposed concrete joint faces and the pavement surface extending a minimum of 1/2 inch (12 mm) from the joint edge shall be sandblasted clean. Sandblasting shall be accomplished in a minimum of two passes. One pass per joint face with the nozzle held at an angle directly toward the joint face and not more than 3 inches (75 mm) from it. After final cleaning and immediately prior to sealing, blow out the joints with compressed air and leave them completely free of debris and water. The joint faces shall be surface dry when the seal is applied.

**c. Backer Rod.** When the joint opening is of a greater depth than indicated for the sealant depth, plug or seal off the lower portion of the joint opening using a backer rod in accordance with paragraph 605-2.2 to prevent the entrance of the sealant below the specified depth. Take care to ensure that the backer rod is placed at the specified depth and is not stretched or twisted during installation.

**d. Bond-breaking tape.** Where inserts or filler materials contain bitumen, or the depth of the joint opening does not allow for the use of a backup material, insert a bond-separating tape breaker in accordance with paragraph 605-2.3 to prevent incompatibility with the filler materials and three-sided adhesion of the sealant. Securely bond the tape to the bottom of the joint opening so it will not float up into the new sealant.

**605-3.4 INSTALLATION OF SEALANTS.** Joints shall be inspected for proper width, depth, alignment, and preparation, and shall be approved by the RPR before sealing is allowed. Sealants shall be installed in accordance with the following requirements:

Immediately preceding, but not more than 50 feet (15 m) ahead of the joint sealing operations, perform a final cleaning with compressed air. Fill the joints from the bottom up to 1/4 inch below the top of pavement surface; or bottom of groove for grooved pavement. Remove and discard excess or spilled sealant from the pavement by approved methods. Install the sealant in such a manner as to prevent the formation of voids and entrapped air. In no case shall gravity methods or pouring pots be used to install the sealant material. Traffic shall not be permitted over newly sealed pavement until authorized by the RPR. When a primer is recommended by the manufacturer, apply it evenly to the joint faces in accordance with the manufacturer's instructions. Check the joints frequently to ensure that the newly installed sealant is cured to a tack-free condition within the time specified.

**605-3.5 INSPECTION.** The Contractor shall inspect the joint sealant for proper rate of cure and set, bonding to the joint walls, cohesive separation within the sealant, reversion to liquid, entrapped air and voids. Sealants exhibiting any of these deficiencies at any time prior to the final acceptance of the project shall be removed from the joint, wasted, and replaced as specified at no additional cost to the airport.

**605-3.6 CLEAN-UP.** Upon completion of the project, remove all unused materials from the site and leave the pavement in a clean condition.

## **METHOD OF MEASUREMENT**

**605-4.1** Joint sealing material shall be incidental to the work.

## **BASIS OF PAYMENT**

**605-5.1** Joint sealing material shall be incidental to the work.

## **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.



ASTM International (ASTM)

|            |                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D789  | Standard Test Method for Determination of Relative Viscosity of Polyamide (PA)                                                              |
| ASTM D5249 | Standard Specification for Backer Material for Use with Cold- and Hot-Applied Joint Sealants in Portland-Cement Concrete and Asphalt Joints |
| ASTM D5893 | Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements  |

Advisory Circulars (AC)

|                |                                                         |
|----------------|---------------------------------------------------------|
| AC 150/5340-30 | Design and Installation Details for Airport Visual Aids |
|----------------|---------------------------------------------------------|

**END OF ITEM P-605**

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## ITEM P-606 ADHESIVE COMPOUNDS, TWO-COMPONENT FOR SEALING WIRE AND LIGHTS IN PAVEMENT

### DESCRIPTION

**606-1.1** This specification covers two types of material; a liquid suitable for sealing electrical wire in saw cuts in pavement and for sealing light fixtures or bases in pavement, and a paste suitable for embedding light fixtures in the pavement. Both types of material are two-component filled formulas with the characteristics specified in paragraph 606-2.4. Materials supplied for use with asphalt and/or concrete pavements must be formulated so they are compatible with the asphalt and/or concrete.

### MATERIALS

**606-2.1 CURING.** When pre-warmed to 77°F (25°C), mixed, and placed in accordance with manufacturer's directions, the materials shall cure at temperatures of 45°F (7°C) or above without the application of external heat.

**606-2.2 STORAGE.** The adhesive components shall not be stored at temperatures over 86°F (30°C), unless otherwise specified by the manufacturer.

**606-2.3 CAUTION.** Installation and use shall be in accordance with the manufacturer's recommended procedures. Avoid prolonged or repeated contact with skin. In case of contact, wash with soap and flush with water. If taken internally, call doctor. Keep away from heat or flame. Avoid vapor. Use in well-ventilated areas. Keep in cool place. Keep away from children.

**606-2.4 CHARACTERISTICS.** When mixed and cured in accordance with the manufacturer's directions, the materials shall have the following properties shown in Table 1.

**Table 1. Property Requirements**

| Physical or Electrical Property        | Minimum                    | Maximum               | ASTM Method |
|----------------------------------------|----------------------------|-----------------------|-------------|
| Tensile                                |                            |                       |             |
| Portland cement concrete               | 1,000 psi<br>(70 kg/sq cm) |                       | D 638       |
| Asphalt concrete                       | 500 psi<br>(35 kg/sq cm)   |                       |             |
| Elongation                             |                            |                       |             |
| Portland cement concrete               |                            | See note <sup>1</sup> | D 638       |
| Asphalt concrete                       | 50%                        |                       | D 638       |
| Coef. of cub. exp.<br>cu. cm/cu. cm/°C | 0.00090                    | 0.00120               | D 1168      |
| Coef. of lin. exp.<br>cm/cm/°C         | 0.000030                   | 0.000040              | D 1168      |

| Physical or Electrical Property      | Minimum                    | Maximum | ASTM Method |
|--------------------------------------|----------------------------|---------|-------------|
| Dielectric strength, short time test | 350 volts/mil.             |         | D 149       |
| Arc resistance                       | 125 sec                    |         |             |
| Pull-off                             |                            |         |             |
| Adhesion to steel                    | 1,000 psi<br>(70 kg/sq cm) |         |             |
| Adhesion to Portland cement concrete | 200 psi<br>(14 kg/sq cm)   |         |             |
| Adhesion to asphalt concrete         | No test available.         |         |             |
| Adhesion to aluminum                 | 250 psi                    |         |             |

<sup>1</sup> 20% or more (without filler) for formulations to be supplied for areas subject to freezing.

## SAMPLING, INSPECTION, AND TEST PROCEDURES

**606-3.1 TENSILE PROPERTIES.** Tests for tensile strength and elongation shall be conducted in accordance with ASTM D638.

**606-3.2 EXPANSION.** Tests for coefficients of linear and cubical expansion shall be conducted in accordance with, Method B, except that mercury shall be used instead of glycerine. The test specimen shall be mixed in the proportions specified by the manufacturer, and cured in a glass tub approximately 2 inch (50 mm) long by 3/8 inch (9 mm) in diameter. The interior of the tube shall be precoated with a silicone mold release agent. The hardened sample shall be removed from the tube and aged at room temperature for one (1) week before conducting the test. The test temperature range shall be from 35°F (2°C) to 140°F (60°C).

**606-3.3 TEST FOR DIELECTRIC STRENGTH.** Test for dielectric strength shall be conducted in accordance with ASTM D149 for sealing compounds to be furnished for sealing electrical wires in pavement.

**606-3.4 TEST FOR ARC RESISTANCE.** Test for arc resistance shall be conducted for sealing compounds to be furnished for sealing electrical wires in pavement.

**606-3.5 TEST FOR ADHESION TO STEEL.** The ends of two smooth, clean, steel specimens of convenient size (1 inch by 1 inch by 6 inch) (25 mm by 25 mm by 150 mm) would be satisfactory when bonded together with adhesive mixture and allowed to cure at room temperature for a period of time to meet formulation requirements and then tested to failure on a Riehle (or similar) tensile tester. The thickness of adhesive to be tested shall be 1/4 inch (6 mm).

### 606-3.6 ADHESION TO PORTLAND CEMENT CONCRETE.

**a. Concrete test block preparation.** The aggregate grading shall be as shown in Table 2.

The coarse aggregate shall consist of crushed rock having a minimum of 75% of the particles with at least one fractured face and having a water absorption of not more than 1.5%. The fine aggregate shall consist of crushed sand manufactured from the same parent rock as the coarse

aggregate. The concrete shall have a water-cement ratio of 5.5 gallons (21 liters) of water per bag of cement, a cement factor of 6,  $\pm 0.5$ , bags of cement per cubic yard (0.76 cubic meter) of concrete, and a slump of 2-1/2 inch (60 mm),  $\pm 1/2$  inch (60 mm  $\pm 12$  mm). The ratio of fine aggregate to total aggregate shall be approximately 40% by solid volume. The air content shall be 5.0%,  $\pm 0.5\%$ , and it shall be obtained by the addition to the batch of an air-entraining admixture such as Vinsol® resin. The mold shall be of metal and shall be provided with a metal base plate.

Means shall be provided for securing the base plate to the mold. The assembled mold and base plate shall be watertight and shall be oiled with mineral oil before use. The inside measurement of the mold shall be such that several one inch (25 mm) by 2-inch (75 mm) by 3-inch (25 mm by 50 mm by 75 mm) test blocks can be cut from the specimen with a concrete saw having a diamond blade. The concrete shall be prepared and cured in accordance with ASTM C192.

**Table 2. Aggregate for Bond Test Blocks**

| Type                    | Sieve Size            | Percent Passing |
|-------------------------|-----------------------|-----------------|
| <b>Coarse Aggregate</b> | 3/4 inch (19.0 mm)    | 97 to 100       |
|                         | 1/2 inch (12.5 mm)    | 63 to 69        |
|                         | 3/8 inch (9.5 mm)     | 30 to 36        |
|                         | No. 4 (4.75 mm)       | 0 to 3          |
| <b>Fine Aggregate</b>   | No. 4 (4.75 mm)       | 100             |
|                         | No. 8 (2.36 mm)       | 82 to 88        |
|                         | No. 16 (1.18 mm)      | 60 to 70        |
|                         | No. 30 (600 $\mu$ m)  | 40 to 50        |
|                         | No. 50 (300 $\mu$ m)  | 16 to 26        |
|                         | No. 100 (150 $\mu$ m) | 5 to 9          |

**b. Bond test.** Prior to use, oven-dry the test blocks to constant weight at a temperature of 220°F to 230°F (104°C to 110°C), cool to room temperature, 73.4°F  $\pm 3^\circ$ F (23°C  $\pm 1.6^\circ$ C), in a desiccator, and clean the surface of the blocks of film or powder by vigorous brushing with a stiff-bristled fiber brush. Two test blocks shall be bonded together on the one inch by 3 inch (25 mm by 75 mm) sawed face with the adhesive mixture and allowed to cure at room temperature for a period of time to meet formulation requirements and then tested to failure in a Riehle (or similar) tensile tester. The thickness of the adhesive to be tested shall be 1/4 inch (6 mm).

**606-3.7 COMPATIBILITY WITH ASPHALT MIX.** Test for compatibility with asphalt in accordance with ASTM D5329.

**606-3.8 ADHESIVE COMPOUNDS - CONTRACTOR'S RESPONSIBILITY.** The Contractor shall furnish the vendor's certified test reports for each batch of material delivered to the project. The report shall certify that the material meets specification requirements and is suitable for use with asphalt concrete pavements. The report shall be provided to and accepted by the Resident Project Representative (RPR) before use of the material. In addition, the Contractor shall obtain a statement from the supplier or manufacturer that guarantees the material for one year. The supplier or manufacturer shall furnish evidence that the material has performed satisfactorily on other projects.



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## ITEM P-610 CONCRETE FOR MISCELLANEOUS STRUCTURES

### DESCRIPTION

**610-1.1** This item shall consist of concrete and reinforcement, as shown on the plans, prepared and constructed in accordance with these specifications. This specification shall be used for all concrete other than airfield pavement which are cast-in-place.

### MATERIALS

**610-2.1 GENERAL.** Only approved materials, conforming to the requirements of these specifications, shall be used in the work. Materials may be subject to inspection and tests at any time during their preparation or use. The source of all materials shall be approved by the Resident Project Representative (RPR) before delivery or use in the work. Representative preliminary samples of the materials shall be submitted by the Contractor, when required, for examination and test. Materials shall be stored and handled to ensure preservation of their quality and fitness for use and shall be located to facilitate prompt inspection. All equipment for handling and transporting materials and concrete must be clean before any material or concrete is placed in them.

The use of pit-run aggregates shall not be permitted unless the pit-run aggregate has been screened and washed, and all fine and coarse aggregates stored separately and kept clean. The mixing of different aggregates from different sources in one storage stockpile or alternating batches of different aggregates shall not be permitted.

**a. Reactivity.** Fine aggregate and coarse aggregates to be used in all concrete shall have been tested separately within six months of the project in accordance with ASTM C1260. Test results shall be submitted to the RPR. The aggregate shall be considered innocuous if the expansion of test specimens, tested in accordance with ASTM C1260, does not exceed 0.08% at 14 days (16 days from casting). If the expansion either or both test specimen is greater than 0.08% at 14 days, but less than 0.20%, a minimum of 25% of Type F fly ash, or between 40% and 55% of slag cement shall be used in the concrete mix.

If the expansion is greater than 0.20% the aggregates shall not be used, and test results for other aggregates must be submitted for evaluation; or aggregates that meet P-501 reactivity test requirements may be utilized.

**610-2.2 COARSE AGGREGATE.** The coarse aggregate for concrete shall meet the requirements of ASTM C33 and the requirements of Table 4, Class Designation 5S; and the grading requirements shown below, as required for the project.



**Coarse Aggregate Grading Requirements**

| <b>Maximum Aggregate Size</b> | <b>ASTM C33, Table 3 Grading Requirements (Size No.)</b> |
|-------------------------------|----------------------------------------------------------|
| 1 1/2 inch (37.5 mm)          | 467 or<br>4 and 67                                       |
| 1 inch (25 mm)                | 57                                                       |
| 3/4 inch (19 mm)              | 67                                                       |
| 1/2 inch (12.5 mm)            | 7                                                        |

**610-2.2.1 Coarse Aggregate susceptibility to durability (D) cracking.** Not used.

**610-2.3 FINE AGGREGATE.** The fine aggregate for concrete shall meet all fine aggregate requirements of ASTM C33.

**610-2.4 CEMENT.** Cement shall conform to the requirements of ASTM C150 Type **I, II or III**.

**610-2.5 CEMENTITIOUS MATERIALS.**

**a. Fly ash.** Fly ash shall meet the requirements of ASTM C618, with the exception of loss of ignition, where the maximum shall be less than 6%. Fly ash shall have a Calcium Oxide (CaO) content of less than **15%** and a total available alkali content less than 3% per ASTM C311. Fly ash produced in furnace operations using liming materials or soda ash (sodium carbonate) as an additive shall not be acceptable. The Contractor shall furnish the previous three most recent, consecutive ASTM C618 reports for each source of fly ash proposed in the concrete mix, and shall furnish each additional report as they become available during the project. The reports can be used for acceptance or the material may be tested independently by the RPR.

**b. Slag cement (ground granulated blast furnace (GGBF)).** Slag cement shall conform to ASTM C989, Grade 100 or Grade 120. Slag cement shall be used only at a rate between 25% and 55% of the total cementitious material by mass.

**610-2.6 WATER.** Water used in mixing or curing shall be from potable water sources. Other sources shall be tested in accordance with ASTM C1602 prior to use.

**610-2.7 ADMIXTURES.** The Contractor shall submit certificates indicating that the material to be furnished meets all of the requirements indicated below. In addition, the RPR may require the Contractor to submit complete test data from an approved laboratory showing that the material to be furnished meets all of the requirements of the cited specifications. Subsequent tests may be made of samples taken by the RPR from the supply of the material being furnished or proposed for use on the work to determine whether the admixture is uniform in quality with that approved.

**a. Air-entraining admixtures.** Air-entraining admixtures shall meet the requirements of ASTM C260 and shall consistently entrain the air content in the specified ranges under field conditions. The air-entrainment agent and any water reducer admixture shall be compatible.

**b. Water-reducing admixtures.** Water-reducing admixture shall meet the requirements of ASTM C494, Type A, B, or D. ASTM C494, Type F and G high range water reducing admixtures and ASTM C1017 flowable admixtures shall not be used.

**c. Other chemical admixtures.** The use of set retarding, and set-accelerating admixtures shall be approved by the RPR. Retarding shall meet the requirements of ASTM C494, Type A, B, or D and set-accelerating shall meet the requirements of ASTM C494, Type C. Calcium chloride and admixtures containing calcium chloride shall not be used.

**610-2.8 PREMOLDED JOINT MATERIAL.** Premolded joint material for expansion joints shall meet the requirements of ASTM D1751 or D1752.

**610-2.9 JOINT FILLER.** The filler for joints shall meet the requirements of Item P-605, unless otherwise specified.

**610-2.10 STEEL REINFORCEMENT.** Reinforcing shall consist of Reinforcing Steel conforming to the requirements of ASTM A615, ASTM A706, ASTM A775, or ASTM A934.

**610-2.11 MATERIALS FOR CURING CONCRETE.** Curing materials shall conform to White-pigmented Liquid Membrane-Forming Compound, Type 2, Class B – ASTM C309.

## CONSTRUCTION METHODS

**610-3.1 GENERAL.** The Contractor shall furnish all labor, materials, and services necessary for, and incidental to, the completion of all work as shown on the drawings and specified here. All machinery and equipment used by the Contractor on the work, shall be of sufficient size to meet the requirements of the work. All work shall be subject to the inspection and approval of the RPR.

**610-3.2 CONCRETE MIXTURE.** The concrete shall develop a compressive strength of 4000 psi in 28 days as determined by test cylinders made in accordance with ASTM C31 and tested in accordance with ASTM C39. The concrete shall contain not less than 470 pounds of cementitious material per cubic yard (280 kg per cubic meter). The water cementitious ratio shall not exceed 0.45 by weight. The air content of the concrete shall be 5% +/- 1.2% as determined by ASTM C231 and shall have a slump of not more than 4 inches (100 mm) as determined by ASTM C143.

**610-3.3 MIXING.** Concrete may be mixed at the construction site, at a central point, or wholly or in part in truck mixers. The concrete shall be mixed and delivered in accordance with the requirements of ASTM C94 or ASTM C685.

The concrete shall be mixed only in quantities required for immediate use. Concrete shall not be mixed while the air temperature is below 40°F (4°C) without the RPRs approval. If approval is granted for mixing under such conditions, aggregates or water, or both, shall be heated and the concrete shall be placed at a temperature not less than 50°F (10°C) nor more than 100°F (38°C). The Contractor shall be held responsible for any defective work, resulting from freezing or injury in any manner during placing and curing, and shall replace such work at his expense.

Retempering of concrete by adding water or any other material is not permitted.

The rate of delivery of concrete to the job shall be sufficient to allow uninterrupted placement of the concrete.

**610-3.4 FORMS.** Concrete shall not be placed until all the forms and reinforcements have been inspected and approved by the RPR. Forms shall be of suitable material and shall be of the type, size, shape, quality, and strength to build the structure as shown on the plans. The forms shall be true to line and grade and shall be mortar-tight and sufficiently rigid to prevent displacement and sagging between supports. The surfaces of forms shall be smooth and free from irregularities, dents, sags, and holes. The Contractor shall be responsible for their adequacy.

The internal form ties shall be arranged so no metal will show in the concrete surface or discolor the surface when exposed to weathering when the forms are removed. All forms shall be wetted with water or with a non-staining mineral oil, which shall be applied immediately before the concrete is placed. Forms shall be constructed so they can be removed without injuring the concrete or concrete surface.

**610-3.5 PLACING REINFORCEMENT.** All reinforcement shall be accurately placed, as shown on the plans, and shall be firmly held in position during concrete placement. Bars shall be fastened together at intersections. The reinforcement shall be supported by approved metal chairs. Shop drawings, lists, and bending details shall be supplied by the Contractor when required.

**610-3.6 EMBEDDED ITEMS.** Before placing concrete, all embedded items shall be firmly and securely fastened in place as indicated. All embedded items shall be clean and free from coating, rust, scale, oil, or any foreign matter. The concrete shall be spaded and consolidated around and against embedded items. The embedding of wood shall not be allowed.

**610-3.7 CONCRETE CONSISTENCY.** The Contractor shall monitor the consistency of the concrete delivered to the project site; collect each batch ticket; check temperature; and perform slump tests on each truck at the project site in accordance with ASTM C143.

**610-3.8 PLACING CONCRETE.** All concrete shall be placed during daylight hours, unless otherwise approved. The concrete shall not be placed until the depth and condition of foundations, the adequacy of forms and falsework, and the placing of the steel reinforcing have been approved by the RPR. Concrete shall be placed as soon as practical after mixing, but in no case later than one (1) hour after water has been added to the mix. The method and manner of placing shall avoid segregation and displacement of the reinforcement. Troughs, pipes, and chutes shall be used as an aid in placing concrete when necessary. The concrete shall not be dropped from a height of more than 5 feet (1.5 m). Concrete shall be deposited as nearly as practical in its final position to avoid segregation due to rehandling or flowing. Do not subject concrete to procedures which cause segregation. Concrete shall be placed on clean, damp surfaces, free from running water, or on a properly consolidated soil foundation.

**610-3.9 VIBRATION.** Vibration shall follow the guidelines in American Concrete Institute (ACI) Committee 309R, Guide for Consolidation of Concrete.

**610-3.10 JOINTS.** Joints shall be constructed as indicated on the plans.

**610-3.11 FINISHING.** All exposed concrete surfaces shall be true, smooth, and free from open or rough areas, depressions, or projections. All concrete horizontal plane surfaces shall be brought flush to the proper elevation with the finished top surface struck-off with a straightedge and floated.

**610-3.12 CURING AND PROTECTION.** All concrete shall be properly cured in accordance with the recommendations in American Concrete Institute (ACI) 308R, Guide to External Curing of Concrete. The concrete shall be protected from damage until project acceptance.

**610-3.13 COLD WEATHER PLACING.** When concrete is placed at temperatures below 40°F (4°C), follow the cold weather concreting recommendations found in ACI 306R, Cold Weather Concreting.

**610-3.14 HOT WEATHER PLACING.** When concrete is placed in hot weather greater than 85°F (30 °C), follow the hot weather concreting recommendations found in ACI 305R, Hot Weather Concreting.

## **QUALITY ASSURANCE (QA)**

**610-4.1 QUALITY ASSURANCE SAMPLING AND TESTING.** Concrete for each day's placement will be accepted on the basis of the compressive strength specified in paragraph 610-3.2. The RPR will sample the concrete in accordance with ASTM C172; test the slump in accordance with ASTM C143; **test air content in accordance with ASTM C231**; make and cure compressive strength specimens in accordance with ASTM C31; and test in accordance with ASTM C39. The QA testing agency will meet the requirements of ASTM C1077.

The Contractor shall provide adequate facilities for the initial curing of cylinders.

**610-4.2 DEFECTIVE WORK.** Any defective work that cannot be satisfactorily repaired as determined by the RPR, shall be removed and replaced at the Contractor's expense. Defective work includes, but is not limited to, uneven dimensions, honeycombing and other voids on the surface or edges of the concrete.

## **METHOD OF MEASUREMENT**

**610-5.1** Concrete shall be considered incidental and no separate measurement shall be made of concrete complete in place and accepted.

## **BASIS OF PAYMENT**

**610-6.1** Payment shall be made at the contract price concrete shall be considered incidental and no separate payment shall be made. This price shall be full compensation for furnishing all materials including reinforcement and embedded items and for all preparation, delivery, installation, and curing of these materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

|                |          |                                |
|----------------|----------|--------------------------------|
| Item P-610-6.1 | Concrete | incidental to other work items |
|----------------|----------|--------------------------------|

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

### ASTM International (ASTM)

|            |                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------|
| ASTM A184  | Standard Specification for Welded Deformed Steel Bar Mats for Concrete Reinforcement                         |
| ASTM A615  | Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement                   |
| ASTM A704  | Standard Specification for Welded Steel Plain Bar or Rod Mats for Concrete Reinforcement                     |
| ASTM A706  | Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement                |
| ASTM A775  | Standard Specification for Epoxy-Coated Steel Reinforcing Bars                                               |
| ASTM A884  | Standard Specification for Epoxy-Coated Steel Wire and Welded Wire Reinforcement                             |
| ASTM A934  | Standard Specification for Epoxy-Coated Prefabricated Steel Reinforcing Bars                                 |
| ASTM A1064 | Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete |
| ASTM C31   | Standard Practice for Making and Curing Concrete Test Specimens in the Field                                 |
| ASTM C33   | Standard Specification for Concrete Aggregates                                                               |
| ASTM C39   | Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens                              |
| ASTM C94   | Standard Specification for Ready-Mixed Concrete                                                              |
| ASTM C136  | Standard Test Method for Sieve or Screen Analysis of Fine and Coarse Aggregates                              |
| ASTM C114  | Standard Test Methods for Chemical Analysis of Hydraulic Cement                                              |
| ASTM C136  | Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates                                        |
| ASTM C143  | Standard Test Method for Slump of Hydraulic-Cement Concrete                                                  |
| ASTM C150  | Standard Specification for Portland Cement                                                                   |

|            |                                                                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C171  | Standard Specification for Sheet Materials for Curing Concrete                                                                                            |
| ASTM C172  | Standard Practice for Sampling Freshly Mixed Concrete                                                                                                     |
| ASTM C231  | Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method                                                                     |
| ASTM C260  | Standard Specification for Air-Entraining Admixtures for Concrete                                                                                         |
| ASTM C309  | Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete                                                                          |
| ASTM C311  | Standard Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use in Portland-Cement Concrete                                           |
| ASTM C494  | Standard Specification for Chemical Admixtures for Concrete                                                                                               |
| ASTM C618  | Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete                                                          |
| ASTM C666  | Standard Test Method for Resistance of Concrete to Rapid Freezing and Thawing                                                                             |
| ASTM C685  | Standard Specification for Concrete Made by Volumetric Batching and Continuous Mixing                                                                     |
| ASTM C989  | Standard Specification for Slag Cement for Use in Concrete and Mortars                                                                                    |
| ASTM C1017 | Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete                                                                      |
| ASTM C1077 | Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation                |
| ASTM C1157 | Standard Performance Specification for Hydraulic Cement                                                                                                   |
| ASTM C1260 | Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)                                                                    |
| ASTM C1365 | Standard Test Method for Determination of the Proportion of Phases in Portland Cement and Portland-Cement Clinker Using X-Ray Powder Diffraction Analysis |
| ASTM C1602 | Standard Specification for Mixing Water Used in the Production of Hydraulic Cement Concrete                                                               |

ASTM D1751      Standard Specification for Preformed Expansion Joint Filler for  
Concrete Paving and Structural Construction (Nonextruding and  
Resilient Asphalt Types)

ASTM D1752      Standard Specification for Preformed Sponge Rubber Cork and  
Recycled PVC Expansion Joint Fillers for Concrete Paving and  
Structural Construction

American Concrete Institute (ACI)

ACI 305R      Hot Weather Concreting

ACI 306R      Cold Weather Concreting

ACI 308R      Guide to External Curing of Concrete

ACI 309R      Guide for Consolidation of Concrete

**END OF ITEM P-610**

## ITEM P-620 RUNWAY AND TAXIWAY MARKING

### DESCRIPTION

**620-1.1** This item shall consist of the preparation and painting of numbers, markings, and stripes on the surface of runways, taxiways, and aprons, in accordance with these specifications and at the locations shown on the plans, or as directed by the Resident Project Representative (RPR). The terms “paint” and “marking material” as well as “painting” and “application of markings” are interchangeable throughout this specification.

### MATERIALS

**620-2.1 MATERIALS ACCEPTANCE.** The Contractor shall furnish manufacturer’s certified test reports, for materials shipped to the project. The certified test reports shall include a statement that the materials meet the specification requirements. This certification along with a copy of the paint manufacturer’s surface preparation; marking materials, including adhesion, flow promoting and/or floatation additive; and application requirements must be submitted and approved by the Resident Project Representative (RPR) prior to the initial application of markings. The reports can be used for material acceptance or the RPR may perform verification testing. The reports shall not be interpreted as a basis for payment. The Contractor shall notify the RPR upon arrival of a shipment of materials to the site. All material shall arrive in sealed containers that are easily quantifiable for inspection by the RPR.

#### 620-2.2 MARKING MATERIALS.

**Table 1. Marking Materials**

| Paint <sup>1</sup> |        |                     |                          | Glass Beads <sup>2</sup> |                          |
|--------------------|--------|---------------------|--------------------------|--------------------------|--------------------------|
| Type               | Color  | Fed Std. 595 Number | Application Rate Maximum | Type                     | Application Rate Minimum |
| III                | White  | 37925               | 90 ft <sup>2</sup> /gal  | III                      | 8 lb/gal                 |
| III                | Black  | 37038               | 90 ft <sup>2</sup> /gal  | III                      | 8 lb/gal                 |
| III                | Red    | 31136               | 90 ft <sup>2</sup> /gal  | III                      | 8 lb/gal                 |
| III                | Yellow | 33538 or 33655      | 90 ft <sup>2</sup> /gal  | III                      | 8 lb/gal                 |

<sup>1</sup> See paragraph 620-2.2a

<sup>2</sup> See paragraph 620-2.2b

**a. Paint.** Paint shall be waterborne in accordance with the requirements of this paragraph. Paint colors shall comply with Federal Standard No. 595 as indicated in Table 1.

**Waterborne.** Paint shall meet the requirements of Federal Specification TT-P-1952F, Type III. The non-volatile portion of the vehicle for all paint types shall be composed of a 100% acrylic polymer as determined by infrared spectral analysis. The acrylic resin used for Type III shall be 100% cross linking acrylic as evidenced by infrared peaks at wavelengths 1568, 1624, and 1672 cm-l with intensities equal to those produced by an acrylic resin known to be 100% cross linking.



**b. Reflective media.** Glass beads for white and yellow paint shall meet the requirements for Federal Specification TT-B-1325D Type III.

Glass beads for red and pink paint shall meet the requirements for Type IV, Gradation A.

Glass beads shall be treated with all compatible coupling agents recommended by the manufacturers of the paint and reflective media to ensure adhesion and embedment.

Glass beads shall not be used in black and green paint.

Type III glass beads shall not be used in red and pink paint.

## CONSTRUCTION METHODS

**620-3.1 WEATHER LIMITATIONS.** Painting shall only be performed when the surface is dry, and the ambient temperature and the pavement surface temperature meet the manufacturer's recommendations in accordance with paragraph 620-2.1. Painting operations shall be discontinued when the ambient or surface temperatures does not meet the manufacturer's recommendations. Markings shall not be applied when the wind speed exceeds 10 mph unless windscreens are used to shroud the material guns. Markings shall not be applied when weather conditions are forecasts to not be within the manufacturers' recommendations for application and dry time.

**620-3.2 EQUIPMENT.** Equipment shall include the apparatus necessary to properly clean the existing surface, a mechanical marking machine, a bead dispensing machine, and such auxiliary hand-painting equipment as may be necessary to satisfactorily complete the job.

The mechanical marker shall be an atomizing spray-type or airless type marking machine with automatic glass bead dispensers suitable for application of traffic paint. It shall produce an even and uniform film thickness and appearance of both paint and glass beads at the required coverage and shall apply markings of uniform cross-sections and clear-cut edges without running or spattering and without over spray. The marking equipment for both paint and beads shall be calibrated daily.

**620-3.3 PREPARATION OF SURFACES.** Immediately before application of the paint, the surface shall be dry and free from dirt, grease, oil, laitance, or other contaminants that would reduce the bond between the paint and the pavement. Use of any chemicals or impact abrasives during surface preparation shall be approved in advance by the RPR. After the cleaning operations, sweeping, blowing, or rinsing with pressurized water shall be performed to ensure the surface is clean and free of grit or other debris left from the cleaning process.

**a. Preparation of new pavement surfaces.** The area to be painted shall be cleaned by broom, blower, water blasting, or by other methods approved by the RPR to remove all contaminants, including PCC curing compounds, minimizing damage to the pavement surface.

**b. Preparation of pavement to remove existing markings.** Existing pavement markings shall be removed by rotary grinding, water blasting, or by other methods approved by the RPR minimizing damage to the pavement surface. The removal area may need to be larger than the area of the markings to eliminate ghost markings. After removal of markings on asphalt

pavements, apply a fog seal or seal coat to 'block out' the removal area to eliminate 'ghost' markings.

**c. Preparation of pavement markings prior to remarking.** Prior to remarking existing markings, loose existing markings must be removed minimizing damage to the pavement surface, with a method approved by the RPR. After removal, the surface shall be cleaned of all residue or debris.

Prior to the application of markings, the Contractor shall certify in writing that the surface is dry and free from dirt, grease, oil, laitance, or other foreign material that would prevent the bond of the paint to the pavement or existing markings. This certification along with a copy of the paint manufactures application and surface preparation requirements must be submitted to the RPR prior to the initial application of markings.

**620-3.4 LAYOUT OF MARKINGS.** The proposed markings shall be laid out in advance of the paint application. The locations of markings to receive glass beads shall be shown on the plans.

**620-3.5 APPLICATION.** A period of **30** days shall elapse between placement of surface course or seal coat and application of the permanent paint markings. Paint shall be applied at the locations and to the dimensions and spacing shown on the plans. Paint shall not be applied until the layout and condition of the surface has been approved by the RPR.

The edges of the markings shall not vary from a straight line more than 1/2 inch (12 mm) in 50 feet (15 m), and marking dimensions and spacing shall be within the following tolerances:

**Marking Dimensions and Spacing Tolerance**

| <b>Dimension and Spacing</b>                      | <b>Tolerance</b>  |
|---------------------------------------------------|-------------------|
| 36 inch (910 mm) or less                          | ±1/2 inch (12 mm) |
| greater than 36 inch to 6 feet (910 mm to 1.85 m) | ±1 inch (25 mm)   |
| greater than 6 feet to 60 feet (1.85 m to 18.3 m) | ±2 inch (50 mm)   |
| greater than 60 feet (18.3 m)                     | ±3 inch (76 mm)   |

The paint shall be mixed in accordance with the manufacturer's instructions and applied to the pavement with a marking machine at the rate shown in Table 1. The addition of thinner will not be permitted.

Glass beads shall be distributed upon the marked areas at the locations shown on the plans to receive glass beads immediately after application of the paint. A dispenser shall be furnished that is properly designed for attachment to the marking machine and suitable for dispensing glass beads. Glass beads shall be applied at the rate shown in Table 1. Glass beads shall not be applied to black paint or green paint. Glass beads shall adhere to the cured paint or all marking operations shall cease until corrections are made. Different bead types shall not be mixed. Regular monitoring of glass bead embedment and distribution should be performed.

**620-3.6 APPLICATION--PREFORMED THERMOPLASTIC AIRPORT PAVEMENT MARKINGS.**

Preformed thermoplastic pavement markings not used.

**620-3.7 CONTROL STRIP.** Prior to the full application of airfield markings, the Contractor shall prepare a control strip in the presence of the RPR. The Contractor shall demonstrate the surface preparation method and all striping equipment to be used on the project. The marking equipment must achieve the prescribed application rate of paint and population of glass beads (per Table 1) that are properly embedded and evenly distributed across the full width of the marking. Prior to acceptance of the control strip, markings must be evaluated during darkness to ensure a uniform appearance.

**620-3.8 RETRO-REFLECTANCE.** Reflectance shall be measured with a portable retro-reflectometer meeting ASTM E1710 (or equivalent). A total of 6 readings shall be taken over a 6 square foot area with 3 readings taken from each direction. The average shall be equal to or above the minimum levels of all readings which are within 30% of each other.

**Minimum Retro-Reflectance Values**

| Material                                          | Retro-reflectance mcd/m <sup>2</sup> /lux |        |     |
|---------------------------------------------------|-------------------------------------------|--------|-----|
|                                                   | White                                     | Yellow | Red |
| Initial Type I                                    | 300                                       | 175    | 35  |
| Initial Type III                                  | 600                                       | 300    | 35  |
| Initial Thermoplastic                             | 225                                       | 100    | 35  |
| All materials, remark when less than <sup>1</sup> | 100                                       | 75     | 10  |

<sup>1</sup> Prior to remarking, determine if removal of contaminants on markings will restore retro-reflectance

**620-3.9 PROTECTION AND CLEANUP.** After application of the markings, all markings shall be protected from damage until dry. All surfaces shall be protected from excess moisture and/or rain and from disfiguration by spatter, splashes, spillage, or drippings. The Contractor shall remove from the work area all debris, waste, loose reflective media, and by-products generated by the surface preparation and application operations to the satisfaction of the RPR. The Contractor shall dispose of these wastes in strict compliance with all applicable state, local, and federal environmental statutes and regulations.

## METHOD OF MEASUREMENT

**620-4.1a** The quantity of surface preparation shall be measured by lump sum.

**620-4.1b** The quantity of markings shall be paid for shall be measured by the number of square feet (square meters) of painting.

**620-4.1c** The quantity of reflective media shall be paid for by lump sum of reflective media.

**620-4.1d** The quantity of temporary markings to be paid for shall be lump sum price performed in accordance with the specifications and accepted by the RPR. Temporary marking includes surface preparation, application and complete removal of the temporary marking.

## **BASIS OF PAYMENT**

**620-5.1** This price shall be full compensation for furnishing all materials and for all labor, equipment, tools, and incidentals necessary to complete the item complete in place and accepted by the RPR in accordance with these specifications.

**620-5.1a** Payment for surface preparation shall be made at the contract price for lump sum.

**620-5.2b** Payment for markings shall be made at the contract price for by the number of square feet (square meters) of painting.

**620-5.3c** Payment for reflective media shall be made at the contract unit price for lump sum.

**620-5.4d** Payment for temporary markings shall be made at the contract price for lump sum price. This price shall be full compensation for furnishing all materials and for all labor, equipment, tools, and incidentals necessary to complete the item.

## **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|            |                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D476  | Standard Classification for Dry Pigmentary Titanium Dioxide Products                                                                                |
| ASTM D968  | Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive                                                               |
| ASTM D1652 | Standard Test Method for Epoxy Content of Epoxy Resins                                                                                              |
| ASTM D2074 | Standard Test Method for Total, Primary, Secondary, and Tertiary Amine Values of Fatty Amines by Alternative Indicator Method                       |
| ASTM D2240 | Standard Test Method for Rubber Property - Durometer Hardness                                                                                       |
| ASTM D7585 | Standard Practice for Evaluating Retroreflective Pavement Markings Using Portable Hand-Operated Instruments                                         |
| ASTM E303  | Standard Test Method for Measuring Surface Frictional Properties Using the British Pendulum Tester                                                  |
| ASTM E1710 | Standard Test Method for Measurement of Retroreflective Pavement Marking Materials with CEN-Prescribed Geometry Using a Portable Retroreflectometer |

ASTM E2302      Standard Test Method for Measurement of the Luminance Coefficient Under Diffuse Illumination of Pavement Marking Materials Using a Portable Reflectometer

ASTM G154      Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp Apparatus for Exposure of Nonmetallic Materials

Code of Federal Regulations (CFR)

40      CFR      Part      60,      Appendix      A-7,      Method      24  
Determination of volatile matter content, water content, density, volume solids, and weight solids of surface coatings

29 CFR Part 1910.1200      Hazard Communication

Federal Specifications (FED SPEC)

FED SPEC TT-B-1325D      Beads (Glass Spheres) Retro-Reflective

FED SPEC TT-P-1952F      Paint, Traffic and Airfield Marking, Waterborne

FED STD 595      Colors used in Government Procurement

Commercial Item Description

A-A-2886B      Paint, Traffic, Solvent Based

Advisory Circulars (AC)

AC 150/5340-1      Standards for Airport Markings

AC 150/5320-12      Measurement, Construction, and Maintenance of Skid Resistant Airport Pavement Surfaces

**END OF ITEM P-620**

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## ITEM T-901 SEEDING

### DESCRIPTION

**901-1.1** This item shall consist of soil preparation, seeding and fertilizing the areas shown on the plans or as directed by the RPR in accordance with these specifications.

### MATERIALS

**901-2.1 Seed.** The species and application rates of grass, legume, and cover-crop seed furnished shall be those stipulated herein. Seed shall conform to the requirements of Federal Specification JJJ-S-181, Federal Specification, Seeds, Agricultural.

Seed shall be furnished separately or in mixtures in standard containers labeled in conformance with the Agricultural Marketing Service (AMS) Seed Act and applicable state seed laws with the seed name, lot number, net weight, percentages of purity and of germination and hard seed, and percentage of maximum weed seed content clearly marked for each kind of seed. The Contractor shall furnish the RPR duplicate signed copies of a statement by the vendor certifying that each lot of seed has been tested by a recognized laboratory for seed testing within six (6) months of date of delivery. This statement shall include: name and address of laboratory, date of test, lot number for each kind of seed, and the results of tests as to name, percentages of purity and of germination, and percentage of weed content for each kind of seed furnished, and, in case of a mixture, the proportions of each kind of seed. Wet, moldy, or otherwise damaged seed will be rejected.

Seeds shall be applied as follows:

#### Seed Properties and Rate of Application

| Seed  | Minimum Seed Purity (Percent) | Minimum Germination (Percent) | Rate of Application lb/1,000 S.F. |
|-------|-------------------------------|-------------------------------|-----------------------------------|
| Bahia | 95%                           | 80%                           | 3                                 |

Seeding shall be performed during the period between March and November inclusive, unless otherwise approved by the RPR.

**901-2.2 Lime.** Not required.

**901-2.3 Fertilizer.** Fertilizer shall be standard commercial fertilizers supplied separately or in mixtures containing the percentages of total nitrogen, available phosphoric acid, and water-soluble potash. They shall be applied at the rate and to the depth specified, and shall meet the requirements of applicable state laws. They shall be furnished in standard containers with name, weight, and guaranteed analysis of contents clearly marked thereon. No cyanamide compounds or hydrated lime shall be permitted in mixed fertilizers.

The fertilizers may be supplied in one of the following forms:

- a. A dry, free-flowing fertilizer suitable for application by a common fertilizer spreader;

- b. A finely-ground fertilizer soluble in water, suitable for application by power sprayers; or
- c. A granular or pellet form suitable for application by blower equipment.

**901-2.4 Soil for repairs.** The soil for fill and topsoiling of areas to be repaired shall be at least of equal quality to that which exists in areas adjacent to the area to be repaired. The soil shall be relatively free from large stones, roots, stumps, or other materials that will interfere with subsequent sowing of seed, compacting, and establishing turf, and shall be approved by the RPR before being placed.

## CONSTRUCTION METHODS

**901-3.1 Advance preparation and cleanup.** After grading of areas has been completed and before applying fertilizer and ground limestone, areas to be seeded shall be raked or otherwise cleared of stones larger than 2 inches (50 mm) in any diameter, sticks, stumps, and other debris that might interfere with sowing of seed, growth of grasses, or subsequent maintenance of grass-covered areas. If any damage by erosion or other causes has occurred after the completion of grading and before beginning the application of fertilizer and ground limestone, the Contractor shall repair such damage include filling gullies, smoothing irregularities, and repairing other incidental damage.

An area to be seeded shall be considered a satisfactory seedbed without additional treatment if it has recently been thoroughly loosened and worked to a depth of not less than 5 inches (125 mm) as a result of grading operations and, if immediately prior to seeding, the top 3 inches (75 mm) of soil is loose, friable, reasonably free from large clods, rocks, large roots, or other undesirable matter, and if shaped to the required grade.

When the area to be seeded is sparsely sodded, weedy, barren and unworked, or packed and hard, any grass and weeds shall first be cut or otherwise satisfactorily disposed of, and the soil then scarified or otherwise loosened to a depth not less than 5 inches (125 mm). Clods shall be broken and the top 3 inches (75 mm) of soil shall be worked into a satisfactory seedbed by discing, or by use of cultipackers, rollers, drags, harrows, or other appropriate means.

### **901-3.2 Dry application method.**

**a. Liming.** Not required.

**b. Fertilizing.** Following advance preparations and cleanup fertilizer shall be uniformly spread at the rate that will provide not less than the minimum quantity stated in paragraph 901-2.3.

**c. Seeding.** Grass seed shall be sown at the rate specified in paragraph 901-2.1 immediately after fertilizing. The fertilizer and seed shall be raked within the depth range stated in the special provisions. Seeds of legumes, either alone or in mixtures, shall be inoculated before mixing or sowing, in accordance with the instructions of the manufacturer of the inoculant. When seeding is required at other than the seasons shown on the plans or in the special provisions, a cover crop shall be sown by the same methods required for grass and legume seeding.

**d. Rolling.** After the seed has been properly covered, the seedbed shall be immediately compacted by means of an approved lawn roller, weighing 40 to 65 pounds per foot (60 to 97 kg



per meter) of width for clay soil (or any soil having a tendency to pack), and weighing 150 to 200 pounds per foot (223 to 298 kg per meter) of width for sandy or light soils.

### **901-3.3 Wet application method.**

**a. General.** The Contractor may elect to apply seed and fertilizer (and lime, if required) by spraying them on the previously prepared seedbed in the form of an aqueous mixture and by using the methods and equipment described herein. The rates of application shall be as specified in the special provisions.

**b. Spraying equipment.** The spraying equipment shall have a container or water tank equipped with a liquid level gauge calibrated to read in increments not larger than 50 gallons (190 liters) over the entire range of the tank capacity, mounted so as to be visible to the nozzle operator. The container or tank shall also be equipped with a mechanical power-driven agitator capable of keeping all the solids in the mixture in complete suspension at all times until used.

The unit shall also be equipped with a pressure pump capable of delivering 100 gallons (380 liters) per minute at a pressure of 100 lb / sq inches (690 kPa). The pump shall be mounted in a line that will recirculate the mixture through the tank whenever it is not being sprayed from the nozzle. All pump passages and pipe lines shall be capable of providing clearance for 5/8 inch (16 mm) solids. The power unit for the pump and agitator shall have controls mounted so as to be accessible to the nozzle operator. There shall be an indicating pressure gauge connected and mounted immediately at the back of the nozzle.

The nozzle pipe shall be mounted on an elevated supporting stand in such a manner that it can be rotated through 360 degrees horizontally and inclined vertically from at least 20 degrees below to at least 60 degrees above the horizontal. There shall be a quick-acting, three-way control valve connecting the recirculating line to the nozzle pipe and mounted so that the nozzle operator can control and regulate the amount of flow of mixture delivered to the nozzle. At least three different types of nozzles shall be supplied so that mixtures may be properly sprayed over distance varying from 20 to 100 feet (6 to 30 m). One shall be a close-range ribbon nozzle, one a medium-range ribbon nozzle, and one a long-range jet nozzle. For case of removal and cleaning, all nozzles shall be connected to the nozzle pipe by means of quick-release couplings.

In order to reach areas inaccessible to the regular equipment, an extension hose at least 50 feet (15 m) in length shall be provided to which the nozzles may be connected.

**c. Mixtures.** Lime, if required, shall be applied separately, in the quantity specified, prior to the fertilizing and seeding operations. Not more than 220 pounds (100 kg) of lime shall be added to and mixed with each 100 gallons (380 liters) of water. Seed and fertilizer shall be mixed together in the relative proportions specified, but not more than a total of 220 pounds (100 kg) of these combined solids shall be added to and mixed with each 100 gallons (380 liters) of water.

All water used shall be obtained from fresh water sources and shall be free from injurious chemicals and other toxic substances harmful to plant life. The Contractor shall identify to the RPR all sources of water at least two (2) weeks prior to use. The RPR may take samples of the water at the source or from the tank at any time and have a laboratory test the samples for chemical and saline content. The Contractor shall not use any water from any source that is disapproved by the RPR following such tests.

All mixtures shall be constantly agitated from the time they are mixed until they are finally applied to the seedbed. All such mixtures shall be used within two (2) hours from the time they were mixed or they shall be wasted and disposed of at approved locations.

**d. Spraying.** Lime, if required, shall be sprayed only upon previously prepared seedbeds. After the applied lime mixture has dried, the lime shall be worked into the top 3 inches (75 mm), after which the seedbed shall again be properly graded and dressed to a smooth finish.

Mixtures of seed and fertilizer shall only be sprayed upon previously prepared seedbeds on which the lime, if required, shall already have been worked in. The mixtures shall be applied by means of a high-pressure spray that shall always be directed upward into the air so that the mixtures will fall to the ground like rain in a uniform spray. Nozzles or sprays shall never be directed toward the ground in such a manner as might produce erosion or runoff.

Particular care shall be exercised to ensure that the application is made uniformly and at the prescribed rate and to guard against misses and overlapped areas. Proper predetermined quantities of the mixture in accordance with specifications shall be used to cover specified sections of known area.

Checks on the rate and uniformity of application may be made by observing the degree of wetting of the ground or by distributing test sheets of paper or pans over the area at intervals and observing the quantity of material deposited thereon.

On surfaces that are to be mulched as indicated by the plans or designated by the RPR, seed and fertilizer applied by the spray method need not be raked into the soil or rolled. However, on surfaces on which mulch is not to be used, the raking and rolling operations will be required after the soil has dried.

**901-3.4 Maintenance of seeded areas.** The Contractor shall protect seeded areas against traffic or other use by warning signs or barricades, as approved by the RPR. Surfaces gullied or otherwise damaged following seeding shall be repaired by regrading and reseeding as directed. The Contractor shall mow, water as directed, and otherwise maintain seeded areas in a satisfactory condition until final inspection and acceptance of the work.

When either the dry or wet application method outlined above is used for work done out of season, it will be required that the Contractor establish a good stand of grass of uniform color and density to the satisfaction of the RPR. A grass stand shall be considered adequate when bare spots are one square foot (0.01 sq m) or less, randomly dispersed, and do not exceed 3% of the area seeded.

## **METHOD OF MEASUREMENT**

**901-4.1** The quantity of seeding to be paid for shall be the number of units 1,000 square feet measured on the ground surface, completed and accepted.

## **BASIS OF PAYMENT**

**901-5.1** Payment shall be made at the contract unit price per 1,000 square feet or fraction thereof, which price and payment shall be full compensation for furnishing and placing all material and for all labor, equipment, tools, and incidentals necessary to complete the work prescribed in this item.

Payment will be made under:

|              |                                 |
|--------------|---------------------------------|
| Item 901-5.1 | Seeding - per 1,000 square feet |
|--------------|---------------------------------|

## **REFERENCES**

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|           |                                                          |
|-----------|----------------------------------------------------------|
| ASTM C602 | Standard Specification for Agricultural Liming Materials |
|-----------|----------------------------------------------------------|

Federal Specifications (FED SPEC)

|          |                                                       |
|----------|-------------------------------------------------------|
| FED SPEC | JJJ-S-181, Federal Specification, Seeds, Agricultural |
|----------|-------------------------------------------------------|

Advisory Circulars (AC)

|                |                                                    |
|----------------|----------------------------------------------------|
| AC 150/5200-33 | Hazardous Wildlife Attractants on or Near Airports |
|----------------|----------------------------------------------------|

FAA/United States Department of Agriculture

|                                                                        |
|------------------------------------------------------------------------|
| Wildlife Hazard Management at Airports, A Manual for Airport Personnel |
|------------------------------------------------------------------------|

**END OF ITEM T-901**

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## ITEM T-904 SODDING

### DESCRIPTION

**904-1.1** This item shall consist of furnishing, hauling, and placing approved live sod on prepared areas in accordance with this specification at the locations shown on the plans or as directed by the RPR.

### MATERIALS

**904-2.1 SOD.** Sod furnished by the Contractor shall have a good cover of living or growing grass. This shall be interpreted to include grass that is seasonally dormant during the cold or dry seasons and capable of renewing growth after the dormant period. All sod shall be obtained from areas where the soil is reasonably fertile and contains a high percentage of loamy topsoil. Sod shall be cut or stripped from living, thickly matted turf relatively free of weeds or other undesirable foreign plants, large stones, roots, or other materials that might be detrimental to the development of the sod or to future maintenance. At least 70% of the plants in the cut sod shall be composed of the species stated in the special provisions, and any vegetation more than 6 inches (150 mm) in height shall be mowed to a height of 3 inches (75 mm) or less before sod is lifted. Sod, including the soil containing the roots and the plant growth showing above, shall be cut uniformly to a thickness not less than that stated in the special provisions.

**904-2.2 LIME.** Not required.

**904-2.3 FERTILIZER.** Fertilizer shall be standard commercial fertilizers supplied separately or in mixtures containing the percentages of total nitrogen, available phosphoric acid, and water-soluble potash. They shall be applied at the rate and to the depth specified, and shall meet the requirements of applicable state laws. They shall be furnished in standard containers with name, weight, and guaranteed analysis of contents clearly marked thereon. No cyanamide compounds or hydrated lime shall be permitted in mixed fertilizers.

The fertilizers may be supplied in one of the following forms:

- a. A dry, free-flowing fertilizer suitable for application by a common fertilizer spreader;
- b. A finely-ground fertilizer soluble in water, suitable for application by power sprayers; or
- c. A granular or pellet form suitable for application by blower equipment.

Fertilizers shall be approved by the RPR and spread at the rate as specified by the manufacturer.

**904-2.4 WATER.** The water shall be sufficiently free from oil, acid, alkali, salt, or other harmful materials that would inhibit the growth of grass.

**904-2.5 SOIL FOR REPAIRS.** The soil for fill and topsoiling of areas to be repaired shall be at least of equal quality to that which exists in areas adjacent to the area to be repaired. The soil shall be relatively free from large stones, roots, stumps, or other materials that will interfere with subsequent sowing of seed, compacting, and establishing turf, and shall be approved by the RPR before being placed.

## CONSTRUCTION METHODS

**904-3.1 GENERAL.** Areas to be solid, strip, or spot sodded shall be shown on the plans. Areas requiring special ground surface preparation such as tilling and those areas in a satisfactory condition that are to remain undisturbed shall also be shown on the plans.

Suitable equipment necessary for proper preparation of the ground surface and for the handling and placing of all required materials shall be on hand, in good condition, and shall be approved by the RPR before the various operations are started. The Contractor shall demonstrate to the RPR before starting the various operations that the application of required materials will be made at the specified rates.

**904-3.2 PREPARING THE GROUND SURFACE.** After grading of areas has been completed and before applying fertilizer and limestone, areas to be sodded shall be raked or otherwise cleared of stones larger than 2 inches (50 mm) in any diameter, sticks, stumps, and other debris which might interfere with sodding, growth of grasses, or subsequent maintenance of grass-covered areas. If any damage by erosion or other causes occurs after grading of areas and before beginning the application of fertilizer and ground limestone, the Contractor shall repair such damage. This may include filling gullies, smoothing irregularities, and repairing other incidental damage.

**904-3.3 APPLYING FERTILIZER AND GROUND LIMESTONE.** Following ground surface preparation, fertilizer shall be uniformly spread at a rate which will provide not less than the minimum quantity of each fertilizer ingredient, as stated in the special provisions. If use of ground limestone is required, it shall then be spread at a rate that will provide not less than the minimum quantity stated in the special provisions. These materials shall be incorporated into the soil to a depth of not less than 2 inches (50 mm) by discing, raking, or other suitable methods. Any stones larger than 2 inches (50 mm) in any diameter, large clods, roots, and other litter brought to the surface by this operation shall be removed.

**904-3.4 OBTAINING AND DELIVERING SOD.** After inspection and approval of the source of sod by the RPR, the sod shall be cut with approved sod cutters to such a thickness that after it has been transported and placed on the prepared bed, but before it has been compacted, it shall have a uniform thickness of not less than 2 inches (50 mm). Sod sections or strips shall be cut in uniform widths, not less than 10 inches (250 mm), and in lengths of not less than 18 inches (0.5 m), but of such length as may be readily lifted without breaking, tearing, or loss of soil. Where strips are required, the sod must be rolled without damage with the grass folded inside. The Contractor may be required to mow high grass before cutting sod.

The sod shall be transplanted within 24 hours from the time it is stripped, unless circumstances beyond the Contractor's control make storing necessary. In such cases, sod shall be stacked, kept moist, and protected from exposure to the air and sun and shall be kept from freezing. Sod shall be cut and moved only when the soil moisture conditions are such that favorable results can be expected. Where the soil is too dry, approval to cut sod may be granted only after it has been watered sufficiently to moisten the soil to the depth the sod is to be cut.

**904-3.5 LAYING SOD.** Sodding shall be performed only during the seasons when satisfactory results can be expected. Frozen sod shall not be used and sod shall not be placed upon frozen soil. Sod may be transplanted during periods of drought with the approval of the RPR, provided the sod bed is watered to moisten the soil to a depth of at least 4 inches (100 mm) immediately prior to laying the sod.

The sod shall be moist and shall be placed on a moist earth bed. Pitch forks shall not be used to handle sod, and dumping from vehicles shall not be permitted. The sod shall be carefully placed by hand, edge to edge and with staggered joints, in rows at right angles to the slopes, commencing at the base of the area to be sodded and working upward. The sod shall immediately be pressed firmly into contact with the sod bed by tamping or rolling with approved equipment to provide a true and even surface, and ensure knitting without displacement of the sod or deformation of the surfaces of sodded areas. Where the sod may be displaced during sodding operations, the workmen, when replacing it, shall work from ladders or treaded planks to prevent further displacement. Screened soil of good quality shall be used to fill all cracks between sods. The quantity of the fill soil shall not cause smothering of the grass. Where the grades are such that the flow of water will be from paved surfaces across sodded areas, the surface of the soil in the sod after compaction shall be set approximately one inch (25 mm) below the pavement edge. Where the flow will be over the sodded areas and onto the paved surfaces around manholes and inlets, the surface of the soil in the sod after compaction shall be placed flush with pavement edges.

On slopes steeper than one (1) vertical to 2-1/2 horizontal and in v-shaped or flat-bottom ditches or gutters, the sod shall be pegged with wooden pegs not less than 12 inches (300 mm) in length and have a cross-sectional area of not less than 3/4 sq inch (18 sq mm). The pegs shall be driven flush with the surface of the sod.

**904-3.6 WATERING.** Adequate water and watering equipment must be on hand before sodding begins, and sod shall be kept moist until it has become established and its continued growth assured. In all cases, watering shall be done in a manner that will avoid erosion from the application of excessive quantities and will avoid damage to the finished surface.

**904-3.7 ESTABLISHING TURF.** The Contractor shall provide general care for the sodded areas as soon as the sod has been laid and shall continue until final inspection and acceptance of the work. All sodded areas shall be protected against traffic or other use by warning signs or barricades approved by the RPR. The Contractor shall mow the sodded areas with approved mowing equipment, depending upon climatic and growth conditions and the needs for mowing specific areas. Weeds or other undesirable vegetation shall be mowed and the clippings raked and removed from the area.

**904-3.8 REPAIRING.** When the surface has become gullied or otherwise damaged during the period covered by this contract, the affected areas shall be repaired to re-establish the grade and the condition of the soil, as directed by the RPR, and shall then be sodded as specified in paragraph 904-3.5.

## **METHOD OF MEASUREMENT**

**904-4.1** This item shall be measured on the basis of the area in square yards (square meters) of the surface covered with sod and accepted.

## **BASIS OF PAYMENT**

**904-5.1** This item will be paid for on the basis of the contract unit price per square yard (square meter) for sodding, which price shall be full compensation for all labor, equipment, material, staking, and incidentals necessary to satisfactorily complete the items as specified.

Item T-904-5.1 Sodding - per Square Yard (Square Meter)

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM C602      Standard Specification for Agricultural Liming Materials

AC 150/5200-33 Hazardous Wildlife Attractants on or Near Airports

# Wildlife Hazard Management at Airports, A Manual for Airport Personnel

MARCH 2025



## ITEM T-905 TOPSOIL

### DESCRIPTION

**905-1.1** This item shall consist of preparing the ground surface for topsoil application, removing topsoil from designated stockpiles or areas to be stripped on the site or from approved sources off the site, and placing and spreading the topsoil on prepared areas in accordance with this specification at the locations shown on the plans or as directed by the RPR.

### MATERIALS

**905-2.1 TOPSOIL.** Topsoil shall be the surface layer of soil with no admixture of refuse or any material toxic to plant growth, and it shall be reasonably free from subsoil and stumps, roots, brush, stones (2 inches (50 mm) or more in diameter), and clay lumps or similar objects. Brush and other vegetation that will not be incorporated with the soil during handling operations shall be cut and removed. Ordinary sod and herbaceous growth such as grass and weeds are not to be removed, but shall be thoroughly broken up and intermixed with the soil during handling operations. Heavy sod or other cover, which cannot be incorporated into the topsoil by discing or other means, shall be removed. The topsoil or soil mixture, unless otherwise specified or approved, shall have a pH range of approximately 5.5 pH to 7.6 pH, when tested in accordance with the methods of testing of the Association of Official Agricultural Chemists in effect on the date of invitation of bids. The organic content shall be not less than 3% nor more than 20% as determined by the wet-combustion method (chromic acid reduction). There shall be not less than 20% nor more than 80% of the material passing the 200 mesh (75  $\mu$ m) sieve as determined by the wash test in accordance with ASTM C117.

Natural topsoil may be amended by the Contractor with approved materials and methods to meet the above specifications.

**905-2.2 INSPECTION AND TESTS.** Within 10 days following acceptance of the bid, the RPR shall be notified of the source of topsoil to be furnished by the Contractor. The topsoil shall be inspected to determine if the selected soil meets the requirements specified and to determine the depth to which stripping will be permitted. At this time, the Contractor may be required to take representative soil samples from several locations within the area under consideration and to the proposed stripping depths, for testing purposes as specified in paragraph 905-2.1.

### CONSTRUCTION METHODS

**905-3.1 GENERAL.** Areas to be topsoiled shall be shown on the plans. If topsoil is available on the site, the location of the stockpiles or areas to be stripped of topsoil and the stripping depths shall be shown on the plans.

Suitable equipment necessary for proper preparation and treatment of the ground surface, stripping of topsoil, and for the handling and placing of all required materials shall be on hand, in good condition, and approved by the RPR before the various operations are started.

**905-3.2 PREPARING THE GROUND SURFACE.** Immediately prior to dumping and spreading the topsoil on any area, the surface shall be loosened by discs or spike-tooth harrows, or by other

means approved by the RPR, to a minimum depth of 2 inches (50 mm) to facilitate bonding of the topsoil to the covered subgrade soil. The surface of the area to be topsoiled shall be cleared of all stones larger than 2 inches (50 mm) in any diameter and all litter or other material which may be detrimental to proper bonding, the rise of capillary moisture, or the proper growth of the desired planting. Limited areas, as shown on the plans, which are too compact to respond to these operations shall receive special scarification.

Grades on the area to be topsoiled, which have been established by others as shown on the plans, shall be maintained in a true and even condition. Where grades have not been established, the areas shall be smooth-graded and the surface left at the prescribed grades in an even and compacted condition to prevent the formation of low places or pockets where water will stand.

**905-3.3 OBTAINING TOPSOIL.** Prior to the stripping of topsoil from designated areas, any vegetation, briars, stumps and large roots, rubbish or stones found on such areas, which may interfere with subsequent operations, shall be removed using methods approved by the RPR. Heavy sod or other cover, which cannot be incorporated into the topsoil by discing or other means shall be removed.

When suitable topsoil is available on the site, the Contractor shall remove this material from the designated areas and to the depth as directed by the RPR. The topsoil shall be spread on areas already tilled and smooth-graded, or stockpiled in areas approved by the RPR. Any topsoil stockpiled by the Contractor shall be rehandled and placed without additional compensation. Any topsoil that has been stockpiled on the site by others, and is required for topsoil purposes, shall be removed and placed by the Contractor. The sites of all stockpiles and areas adjacent thereto which have been disturbed by the Contractor shall be graded if required and put into a condition acceptable for seeding.

When suitable topsoil is secured off the airport site, the Contractor shall locate and obtain the supply, subject to the approval of the RPR. The Contractor shall notify the RPR sufficiently in advance of operations in order that necessary measurements and tests can be made. The Contractor shall remove the topsoil from approved areas and to the depth as directed. The topsoil shall be hauled to the site of the work and placed for spreading, or spread as required. Any topsoil hauled to the site of the work and stockpiled shall be rehandled and placed without additional compensation.

**905-3.4 PLACING TOPSOIL.** The topsoil shall be evenly spread on the prepared areas to a uniform depth of 2 inches (50 mm) after compaction, unless otherwise shown on the plans or stated in the special provisions. Spreading shall not be done when the ground or topsoil is frozen, excessively wet, or otherwise in a condition detrimental to the work. Spreading shall be carried on so that turving operations can proceed with a minimum of soil preparation or tilling.

After spreading, any large, stiff clods and hard lumps shall be broken with a pulverizer or by other effective means, and all stones or rocks (2 inches (50 mm) or more in diameter), roots, litter, or any foreign matter shall be raked up and disposed of by the Contractor. After spreading is completed, the topsoil shall be satisfactorily compacted by rolling with a cultipacker or by other means approved by the RPR. The compacted topsoil surface shall conform to the required lines, grades, and cross-sections. Any topsoil or other dirt falling upon pavements as a result of hauling or handling of topsoil shall be promptly removed.

## METHOD OF MEASUREMENT

**905-4.1** Topsoil obtained on the site shall be measured by the number of cubic yards (cubic meters) of topsoil measured in its original position and stripped or excavated. Topsoil stockpiled by others and removed for topsoil by the Contractor shall be measured by the number of cubic yards (cubic meters) of topsoil measured in the stockpile. Topsoil shall be measured by volume in cubic yards (cubic meters) computed by the method of end areas.

**905-4.2** Topsoil obtained off the site shall be measured by the number of cubic yards (cubic meters) of topsoil measured in its original position and stripped or excavated. Topsoil shall be measured by volume in cubic yards (meters) computed by the method of end areas.

## BASIS OF PAYMENT

**905-5.1** Payment will be made at the contract unit price per cubic yard (cubic meter) for topsoil (obtained on the site). This price shall be full compensation for furnishing all materials and for all preparation, placing, and spreading of the materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

**905-5.2** Payment will be made at the contract unit price per cubic yard (cubic meter) for topsoil (obtained off the site). This price shall be full compensation for furnishing all materials and for all preparation, placing, and spreading of the materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

|                |                                                                                    |
|----------------|------------------------------------------------------------------------------------|
| Item T-905-5.1 | Topsoil (Obtained on Site or Removed from Stockpile - per cubic yard (cubic meter) |
| Item T-905-5.2 | Topsoil (Furnished from Off the Site) - per cubic yard (cubic meter)               |

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM International (ASTM)

|           |                                                                                        |
|-----------|----------------------------------------------------------------------------------------|
| ASTM C117 | Materials Finer than 75 $\mu\text{m}$ (No. 200) Sieve in Mineral Aggregates by Washing |
|-----------|----------------------------------------------------------------------------------------|

Advisory Circulars (AC)

|                |                                                    |
|----------------|----------------------------------------------------|
| AC 150/5200-33 | Hazardous Wildlife Attractants on or Near Airports |
|----------------|----------------------------------------------------|

FAA/United States Department of Agriculture

Wildlife Hazard Management at Airports, A Manual for Airport Personnel

PENSACOLA INTERNATIONAL AIRPORT  
RUNWAY 8-26 REHABILITATION

TOPSOIL  
ITEM T-905

**END OF ITEM T-905**

## **ITEM L-108 UNDERGROUND POWER CABLE FOR AIRPORTS**

### **DESCRIPTION**

**108-1.1** This item shall consist of furnishing and installing power cables that are direct buried and furnishing and/or installing power cables within conduit or duct banks per these specifications at the locations shown on the plans. It includes excavation and backfill of trench for direct-buried cables only. Also included are the installation of counterpoise wires, ground wires, ground rods and connections, cable splicing, cable marking, cable testing, and all incidentals necessary to place the cable in operating condition as a completed unit to the satisfaction of the RPR. This item shall not include the installation of duct banks or conduit, trenching and backfilling for duct banks or conduit, or furnishing or installation of cable for FAA owned/operated facilities.

### **EQUIPMENT AND MATERIALS**

#### **108-2.1 GENERAL.**

**a.** Airport lighting equipment and materials covered by advisory circulars (AC) shall be approved under the Airport Lighting Equipment Certification Program per AC 150/5345-53, current version.

**b.** All other equipment and materials covered by other referenced specifications shall be subject to acceptance through manufacturer's certification of compliance with the applicable specification, when requested by the RPR.

**c.** Manufacturer's certifications shall not relieve the Contractor of the responsibility to provide materials per these specifications. Materials supplied and/or installed that do not comply with these specifications shall be removed (when directed by the RPR) and replaced with materials that comply with these specifications at the Contractor's cost.

**d.** All materials and equipment used to construct this item shall be submitted to the RPR for approval prior to ordering the equipment. Submittals consisting of marked catalog sheets or shop drawings shall be provided. Submittal data shall be presented in a clear, precise and thorough manner. Original catalog sheets are preferred. Photocopies are acceptable provided they are as good a quality as the original. Clearly and boldly mark each copy to identify products or models applicable to this project. Indicate all optional equipment and delete any non-pertinent data. Submittals for components of electrical equipment and systems shall identify the equipment to which they apply on each submittal sheet. Markings shall be made bold and clear with arrows or circles (highlighting is not acceptable). The Contractor is solely responsible for delays in the project that may accrue directly or indirectly from late submissions or resubmissions of submittals.

**e.** The data submitted shall be sufficient, in the opinion of the RPR, to determine compliance with the plans and specifications. The Contractor's submittals shall be electronically submitted in pdf format. The RPR reserves the right to reject any and all equipment, materials, or procedures that do not meet the system design and the standards and codes, specified in this document.

**f.** All equipment and materials furnished and installed under this section shall be guaranteed against defects in materials and workmanship for at least twelve (12) months from the date of final acceptance by the Owner. The defective materials and/or equipment shall be repaired or replaced, at the Owner's discretion, with no additional cost to the Owner. The Contractor shall

maintain a minimum insulation resistance in accordance with paragraph 108-3.10e with isolation transformers connected in new circuits and new segments of existing circuits through the end of the contract warranty period when tested in accordance with AC 150/5340-26, *Maintenance Airport Visual Aid Facilities*, paragraph 5.1.3.1, Insulation Resistance Test.

**Only Third Party certified manufacturers, listed in AC 150/5345-53, Appendix 3 Addendum (as required) and meeting the BUY AMERICAN preference requirements can provide equipment and materials specified in the Contract Documents. Documentation certifying compliance with the BUY AMERICAN preference rules for Airport Improvement Program (AIP) cited in 49 USC §50101) shall be included with each equipment and material submittal.**

**108-2.2 CABLE.** Underground cable for airfield lighting facilities (runway and taxiway lights and signs) shall conform to the requirements of AC 150/5345-7, Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits latest edition. Conductors for use on 6.6 ampere primary airfield lighting series circuits shall be single conductor, seven strand, #8 American wire gauge (AWG), L-824 Type C, 5,000 volts, non-shielded, with cross-linked polyethylene insulation. L-824 conductors for use on the L-830 secondary of airfield lighting series circuits shall be sized in accordance with the manufacturer's recommendations. All other conductors shall comply with FAA and National Electric Code (NEC) requirements. Conductor sizes noted above shall not apply to leads furnished by manufacturers on airfield lighting transformers and fixtures.

Wire for electrical circuits up to 600 volts shall comply with Specification L-824 and/or Commercial Item Description A-A-59544A and shall be type THWN-2, 75°C for installation in conduit and RHW-2, 75°C for direct burial installations. Conductors for parallel (voltage) circuits shall be type and size and installed in accordance with NFPA-70, National Electrical Code.

Unless noted otherwise, all 600-volt and less non-airfield lighting conductor sizes are based on a 75°C, THWN-2, 600-volt insulation, copper conductors, not more than three single insulated conductors, in raceway, in free air. The conduit/duct sizes are based on the use of THWN-2, 600-volt insulated conductors. The Contractor shall make the necessary increase in conduit/duct sizes for other types of wire insulation. In no case shall the conduit/duct size be reduced. The minimum power circuit wire size shall be #12 AWG.

Conductor sizes may have been adjusted due to voltage drop or other engineering considerations. Equipment provided by the Contractor shall be capable of accepting the quantity and sizes of conductors shown in the Contract Documents. All conductors, pigtails, cable step-down adapters, cable step-up adapters, terminal blocks and splicing materials necessary to complete the cable termination/splice shall be considered incidental to the respective pay items provided.

Cable type, size, number of conductors, strand and service voltage shall be as specified in the Contract Document.

**108-2.3 BARE COPPER WIRE (COUNTERPOISE, BARE COPPER WIRE GROUND AND GROUND RODS).** Wire for counterpoise or ground installations for airfield lighting systems shall be No. 2 AWG bare solid copper wire for counterpoise and/or No. 6 AWG insulated stranded for grounding bond wire per ASTM B3 and ASTM B8, and shall be bare copper wire per ASTM B33. For voltage powered circuits, the equipment grounding conductor shall comply with NEC Article 250.

Ground rods shall be sectional copper-clad steel. The ground rods shall be of the length and diameter specified on the plans, but in no case be less than 10 feet (2.54 m) long and 3/4 inch (19 mm) in diameter.

**108-2.4 Cable connections.** In-line connections or splices of underground primary cables shall be of the type called for on the plans, and shall be one of the types listed below. No separate payment will be made for cable connections.

**a. The cast splice.** A cast splice, employing a plastic mold and using epoxy resin equivalent to that manufactured by 3M™ Company, "Scotchcast" Kit No. 82-B, or an approved equivalent, used for potting the splice is acceptable.

**b. The field-attached plug-in splice.** Field attached plug-in splices shall be installed as shown on the plans. The Contractor shall determine the outside diameter of the cable to be spliced and furnish appropriately sized connector kits and/or adapters. Tape or heat shrink tubing with integral sealant shall be in accordance with the manufacturer's requirements. Primary Connector Kits manufactured by Amerace, "Super Kit", Integro "Complete Kit", or approved equal is acceptable. Connector kits shall be colored in accordance with Pensacola International Airport color coding criteria per circuit type.

**c. The factory-molded plug-in splice.** Specification for L-823 Connectors, Factory-Molded to Individual Conductors, is acceptable.

**d. The taped or heat-shrink splice.** Taped splices employing field-applied rubber, or synthetic rubber tape covered with plastic tape is acceptable. The rubber tape should meet the requirements of ASTM D4388 and the plastic tape should comply with Military Specification MIL-I-24391 or Commercial Item Description A-A-55809. Heat shrinkable tubing shall be heavy-wall, self-sealing tubing rated for the voltage of the wire being spliced and suitable for direct-buried installations. The tubing shall be factory coated with a thermoplastic adhesive-sealant that will adhere to the insulation of the wire being spliced forming a moisture- and dirt-proof seal. Additionally, heat shrinkable tubing for multi-conductor cables, shielded cables, and armored cables shall be factory kits that are designed for the application. Heat shrinkable tubing and tubing kits shall be manufactured by Tyco Electronics/ Raychem Corporation, Energy Division, or approved equivalent.

In all the above cases, connections of cable conductors shall be made using crimp connectors using a crimping tool designed to make a complete crimp before the tool can be removed. All L-823/L-824 splices and terminations shall be made per the manufacturer's recommendations and listings.

All connections of counterpoise, grounding conductors and ground rods shall be made by the exothermic process or approved equivalent, except that a light base ground clamp connector shall be used for attachment to the light base. All exothermic connections shall be made per the manufacturer's recommendations and listings.

**108-2.5 SPLICER QUALIFICATIONS.** Every airfield lighting cable splicer shall be qualified in making airport cable splices and terminations on cables rated at or above 5,000 volts AC. The Contractor shall submit to the RPR proof of the qualifications of each proposed cable splicer for the airport cable type and voltage level to be worked on. Cable splicing/terminating personnel

shall have a minimum of three (3) years continuous experience in terminating/splicing medium voltage cable.

**108-2.6 CONCRETE.** Concrete shall be proportioned, placed, and cured per Item P-610, Concrete for Miscellaneous Structures.

**108-2.7 FLOWABLE BACKFILL.** Flowable material used to backfill trenches for power cable trenches shall conform to the requirements of Item P-153, Controlled Low Strength Material.

**108-2.8 CABLE IDENTIFICATION TAGS.** Cable identification tags shall be made from a non-corrosive material with the circuit identification stamped or etched onto the tag. The tags shall be of the type as detailed on the plans.

**108-2.9 TAPE.** Electrical tapes shall be Scotch™ Electrical Tapes –Scotch™ 88 (1-1/2 inch (38 mm) wide) and Scotch™ 130C® linerless rubber splicing tape (2-inch (50 mm) wide), as manufactured by the Minnesota Mining and Manufacturing Company (3M™), or an approved equivalent.

**108-2.10 ELECTRICAL COATING.** Electrical coating shall be Scotchkote™ as manufactured by 3M™, or an approved equivalent.

**108-2.11 EXISTING CIRCUITS.** Whenever the scope of work requires connection to an existing circuit, the existing circuit's insulation resistance shall be tested, in the presence of the RPR. The test shall be performed per this item and prior to any activity that will affect the respective circuit. The Contractor shall record the results on forms acceptable to the RPR. When the work affecting the circuit is complete, the circuit's insulation resistance shall be checked again, in the presence of the RPR. The Contractor shall record the results on forms acceptable to the RPR. The second reading shall be equal to or greater than the first reading or the Contractor shall make the necessary repairs to the existing circuit to bring the second reading above the first reading. All repair costs including a complete replacement of the L-823 connectors, L-830 transformers and L-824 cable, if necessary, shall be borne by the Contractor. All test results shall be submitted in the Operation and Maintenance (O&M) Manual. [ ]

**108-2.12 DETECTABLE WARNING TAPE.** Plastic, detectable, American Public Works Association (APWA) Red (electrical power lines, cables, conduit and lighting cable) with continuous legend tape shall be polyethylene film with a metalized foil core and shall be 3-6 inches (75-150 mm) wide. Detectable tape is incidental to the respective bid item. Detectable warning tape for communication cables shall be orange. Detectable warning tape color code shall comply with the APWA Uniform Color Code.

## CONSTRUCTION METHODS

**108-3.1 GENERAL.** The Contractor shall install the specified cable at the approximate locations indicated on the plans. Unless otherwise shown on the plans, all cable required to cross under pavements expected to carry aircraft loads shall be installed in concrete encased duct banks. Cable shall be run without splices, from fixture to fixture.



Cable connections between lights will be permitted only at the light locations for connecting the underground cable to the primary leads of the individual isolation transformers. The Contractor shall be responsible for providing cable in continuous lengths for home runs or other long cable runs without connections unless otherwise authorized in writing by the RPR or shown on the plans.

In addition to connectors being installed at individual isolation transformers, L-823 cable connectors for maintenance and test points shall be installed at locations shown on the plans. Cable circuit identification markers shall be installed on both sides of the L-823 connectors installed and on both sides of slack loops where a future connector would be installed.

Provide not less than 3 feet (1 m) of cable slack on each side of all connections, isolation transformers, light units, and at points where cable is connected to field equipment. Where provisions must be made for testing or for future above grade connections, provide enough slack to allow the cable to be extended at least one foot (30 cm) vertically above the top of the access structure. This requirement also applies where primary cable passes through empty light bases, junction boxes, and access structures to allow for future connections, or as designated by the RPR.

Primary airfield lighting cables installed shall have cable circuit identification markers attached on both sides of each L-823 connector and on each airport lighting cable entering or leaving cable access points, such as manholes, hand holes, pull boxes, junction boxes, etc. Markers shall be of sufficient length for imprinting the cable circuit identification legend on one line, using letters not less than 1/4 inch (6 mm) in size. The cable circuit identification shall match the circuits noted on the construction plans.

**108-3.2 INSTALLATION IN DUCT BANKS OR CONDUITS.** This item includes the installation of the cable in duct banks or conduit per the following paragraphs. The maximum number and voltage ratings of cables installed in each single duct or conduit, and the current-carrying capacity of each cable shall be per the latest version of the National Electric Code, or the code of the local agency or authority having jurisdiction.

The Contractor shall make no connections or splices of any kind in cables installed in conduits or duct banks.

Unless otherwise designated in the plans, where ducts are in tiers, use the lowest ducts to receive the cable first, with spare ducts left in the upper levels. Check duct routes prior to construction to obtain assurance that the shortest routes are selected and that any potential interference is avoided.

Duct banks or conduits shall be installed as a separate item per Item L-110, Airport Underground Electrical Duct Banks and Conduit. The Contractor shall run a mandrel through duct banks or conduit prior to installation of cable to ensure that the duct bank or conduit is open, continuous and clear of debris. The mandrel size shall be compatible with the conduit size. The Contractor shall swab out all conduits/ducts and clean light bases, manholes, etc., interiors immediately prior to pulling cable. Once cleaned and swabbed, the light bases and all accessible points of entry to the duct/conduit system shall be kept closed except when installing cables. Cleaning of ducts, light bases, manholes, etc., is incidental to the pay item of the item being cleaned. All raceway systems left open, after initial cleaning, for any reason shall be re-cleaned at the Contractor's

expense. The Contractor shall verify existing ducts proposed for use in this project as clear and open. The Contractor shall notify the RPR of any blockage in the existing ducts.

The cable shall be installed in a manner that prevents harmful stretching of the conductor, damage to the insulation, or damage to the outer protective covering. The ends of all cables shall be sealed with moisture-seal tape providing moisture-tight mechanical protection with minimum bulk, or alternately, heat shrinkable tubing before pulling into the conduit and it shall be left sealed until connections are made. Where more than one cable is to be installed in a conduit, all cable shall be pulled in the conduit at the same time. The pulling of a cable through duct banks or conduits may be accomplished by hand winch or power winch with the use of cable grips or pulling eyes. Maximum pulling tensions shall not exceed the cable manufacturer's recommendations. A non-hardening cable-pulling lubricant recommended for the type of cable being installed shall be used where required.

The Contractor shall submit the recommended pulling tension values to the RPR prior to any cable installation. If required by the RPR, pulling tension values for cable pulls shall be monitored by a dynamometer in the presence of the RPR. Cable pull tensions shall be recorded by the Contractor and reviewed by the RPR. Cables exceeding the maximum allowable pulling tension values shall be removed and replaced by the Contractor at the Contractor's expense.

The manufacturer's minimum bend radius or NEC requirements (whichever is more restrictive) shall apply. Cable installation, handling and storage shall be per manufacturer's recommendations. During cold weather, particular attention shall be paid to the manufacturer's minimum installation temperature. Cable shall not be installed when the temperature is at or below the manufacturer's minimum installation temperature. At the Contractor's option, the Contractor may submit a plan, for review by the RPR, for heated storage of the cable and maintenance of an acceptable cable temperature during installation when temperatures are below the manufacturer's minimum cable installation temperature.

Cable shall not be dragged across base can or manhole edges, pavement or earth. When cable must be coiled, lay cable out on a canvas tarp or use other appropriate means to prevent abrasion to the cable jacket.

**108-3.3 INSTALLATION OF DIRECT-BURIED CABLE IN TRENCHES.** Unless otherwise specified, the Contractor shall not use a cable plow for installing the cable. Cable shall be unreeled uniformly in place alongside or in the trench and shall be carefully placed along the bottom of the trench. The cable shall not be unreeled and pulled into the trench from one end. Slack cable sufficient to provide strain relief shall be placed in the trench in a series of S curves. Sharp bends or kinks in the cable shall not be permitted.

Where cables must cross over each other, a minimum of 3 inches (75 mm) vertical displacement shall be provided with the topmost cable depth at or below the minimum required depth below finished grade.

**a. Trenching.** Where turf is well established and the sod can be removed, it shall be carefully stripped and properly stored. Trenches for cables may be excavated manually or with mechanical trenching equipment. Walls of trenches shall be essentially vertical so that a minimum of surface is disturbed. Graders shall not be used to excavate the trench with their blades. The bottom surface of trenches shall be essentially smooth and free from coarse aggregate. Unless otherwise

specified, cable trenches shall be excavated to a minimum depth of 18 inches (0.5 m) below finished grade per NEC Table 300.5, except as follows:

- When off the airport or crossing under a roadway or driveway, the minimum depth shall be 36 inches (91 cm) unless otherwise specified.
- Minimum cable depth when crossing under a railroad track, shall be 42 inches (1 m) unless otherwise specified.

The Contractor shall excavate all cable trenches to a width not less than 6 inches (150 mm). Unless otherwise specified on the plans, all cables in the same location and running in the same general direction shall be installed in the same trench.

When rock is encountered, the rock shall be removed to a depth of at least 3 inches (75 mm) below the required cable depth and it shall be replaced with bedding material of earth or sand containing no mineral aggregate particles that would be retained on a 1/4-inch (6.3 mm) sieve. Flowable backfill material may alternatively be used.

Duct bank or conduit markers temporarily removed for trench excavations shall be replaced as required.

It is the Contractor's responsibility to locate existing utilities within the work area prior to excavation. Where existing active cables cross proposed installations, the Contractor shall ensure that these cables are adequately protected. Where crossings are unavoidable, no splices will be allowed in the existing cables, except as specified on the plans. Installation of new cable where such crossings must occur shall proceed as follows:

(1) Existing cables shall be located manually. Unearthed cables shall be inspected to assure absolutely no damage has occurred.

(2) Trenching, etc., in cable areas shall then proceed, with approval of the RPR, with care taken to minimize possible damage or disruption of existing cable, including careful backfilling in area of cable.

In the event that any previously identified cable is damaged during the course of construction, the Contractor shall be responsible for the complete repair or replacement.

**b. Backfilling.** After the cable has been installed, the trench shall be backfilled. The first layer of backfill in the trench shall encompass all cables ; be 3 inches (75 mm) deep, loose measurement; and shall be either earth or sand containing no mineral aggregate particles that would be retained on a 1/4-inch (6.3 mm) sieve. This layer shall not be compacted. The second layer shall be 5 inches (125 mm) deep, loose measurement, and shall contain no particles that would be retained on a one inch (25.0 mm) sieve. The remaining third and subsequent layers of backfill shall not exceed 8 inches (20 cm) of loose measurement and be excavated or imported material and shall not contain stone or aggregate larger than 4 inches (100 mm) maximum diameter.

The second and subsequent layers shall be thoroughly tamped and compacted to at least the density of the adjacent material. If the cable is to be installed in locations or areas where other compaction requirements are specified (under pavements, embankments, etc.) the backfill

compaction shall be to a minimum of 100 percent of ASTM D1557 in areas of turf and backfill with controlled low strength material (CLSM) in accordance with P-153 in areas under pavement.

Trenches shall not contain pools of water during backfilling operations. The trench shall be completely backfilled and tamped level with the adjacent surface, except that when turf is to be established over the trench, the backfilling shall be stopped at an appropriate depth consistent with the type of turbing operation to be accommodated. A proper allowance for settlement shall also be provided. Any excess excavated material shall be removed and disposed of per the plans and specifications.

Underground electrical warning (caution) tape shall be installed in the trench above all direct-buried cable. Contractor shall submit a sample of the proposed warning tape for acceptance by the RPR. If not shown on the plans, the warning tape shall be located 6 inches (150 mm) above the direct-buried cable or the counterpoise wire if present. A 3-6 inch (75 - 150 mm) wide polyethylene film detectable tape, with a metalized foil core, shall be installed above all direct buried cable or counterpoise. The tape shall be of the color and have a continuous legend as indicated on the plans. The tape shall be installed 8 inches (200 mm) minimum below finished grade.

**c. Restoration.** Following restoration of all trenching near airport movement surfaces, the Contractor shall visually inspect the area for foreign object debris (FOD) and remove any that is found. Where soil and sod has been removed, it shall be replaced as soon as possible after the backfilling is completed. All areas disturbed by work shall be restored to its original condition. The restoration shall include the sodding as shown on the plans. The Contractor shall be held responsible for maintaining all disturbed surfaces and replacements until final acceptance. When trenching is through paved areas, restoration shall be equal to existing conditions. If the cable is to be installed in locations or areas where other compaction requirements are specified (under pavements, embankments, etc.) the backfill compaction shall be to a minimum of 100 percent of ASTM D1557 in areas of turf and backfill with controlled low strength material (CLSM) in accordance with P-153 areas under pavement. Restoration shall be considered incidental to the pay item of which it is a component part.

**108-3.4 CABLE MARKERS FOR DIRECT-BURIED CABLE.** The location of direct buried circuits shall be marked by a concrete slab marker, 2 feet (60 cm) square and 4-6 inch (10 - 15 cm) thick, extending approximately one inch (25 mm) above the surface. Each cable run from a line of lights and signs to the equipment vault shall be marked at approximately every 200 feet (61 m) along the cable run, with an additional marker at each change of direction of cable run. All other direct-buried cable shall be marked in the same manner. Cable markers shall be installed directly above the cable. The Contractor shall impress the word "CABLE" and directional arrows on each cable marking slab. The letters shall be approximately 4 inches (100 mm) high and 3 inches (75 mm) wide, with width of stroke 1/2 inch (12 mm) and 1/4 inch (6 mm) deep. Stencils shall be used for cable marker lettering; no hand lettering shall be permitted.

At the location of each underground cable connection/splice, except at lighting units, or isolation transformers, a concrete marker slab shall be installed to mark the location of the connection/splice. The Contractor shall impress the word "SPLICE" on each slab. The Contractor also shall impress additional circuit identification symbols on each slab as directed by the RPR. All cable markers and splice markers shall be painted international orange. Paint shall be specifically manufactured for uncured exterior concrete. After placement, all cable or splice

markers shall be given one coat of high-visibility aviation orange paint as approved by the RPR. Furnishing and installation of cable markers is incidental to the respective cable pay item.

**108-3.5 SPLICING.** Connections of the type shown on the plans shall be made by experienced personnel regularly engaged in this type of work and shall be made as follows:

**a. Cast splices.** These shall be made by using crimp connectors for jointing conductors. Molds shall be assembled, and the compound shall be mixed and poured per the manufacturer's instructions and to the satisfaction of the RPR.

**b. Field-attached plug-in splices.** These shall be assembled per the manufacturer's instructions. These splices shall be made by plugging directly into mating connectors. The joint where the connectors come together shall be finished by one of the following methods: (1) wrapped with at least one layer of rubber or synthetic rubber tape and one layer of plastic tape, one-half lapped, extending at least 1-1/2 inches (38 mm) on each side of the joint (2) Covered with heat shrinkable tubing with integral sealant extending at least 1-1/2 inches (38 mm) on each side of the joint or (3) On connector kits equipped with water seal flap; roll-over water seal flap to sealing position on mating connector.

**c. Factory-molded plug-in splices.** These shall be made by plugging directly into mating connectors. The joint where the connectors come together shall be finished by one of the following methods: (1) Wrapped with at least one layer of rubber or synthetic rubber tape and one layer of plastic tape, one-half lapped, extending at least 1-1/2 inches (38 mm) on each side of the joint. (2) Covered with heat shrinkable tubing with integral sealant extending at least 1-1/2 inches (38 mm) on each side of the joint. or (3) On connector kits so equipped with water seal flap; roll-over water seal flap to sealing position on mating connector.

**d. Taped or heat-shrink splices.** A taped splice shall be made in the following manner:

Bring the cables to their final position and cut so that the conductors will butt. Remove insulation and jacket allowing for bare conductor of proper length to fit compression sleeve connector with 1/4 inch (6 mm) of bare conductor on each side of the connector. Prior to splicing, the two ends of the cable insulation shall be penciled using a tool designed specifically for this purpose and for cable size and type. Do not use emery paper on splicing operation since it contains metallic particles. The copper conductors shall be thoroughly cleaned. Join the conductors by inserting them equidistant into the compression connection sleeve. Crimp conductors firmly in place with crimping tool that requires a complete crimp before tool can be removed. Test the crimped connection by pulling on the cable. Scrape the insulation to assure that the entire surface over which the tape will be applied (plus 3 inches (75 mm) on each end) is clean. After scraping, wipe the entire area with a clean lint-free cloth. Do not use solvents.

Apply high-voltage rubber tape one-half lapped over bare conductor. This tape should be tensioned as recommended by the manufacturer. Voids in the connector area may be eliminated by highly elongating the tape, stretching it just short of its breaking point. The manufacturer's recommendation for stretching tape during splicing shall be followed. Always attempt to exactly half-lap to produce a uniform buildup. Continue buildup to 1-1/2 times cable diameter over the body of the splice with ends tapered a distance of approximately one inch (25 mm) over the original jacket. Cover rubber tape with two layers of vinyl pressure-sensitive tape one-half lapped. Do not use glyptol or lacquer over vinyl tape as they react as solvents to the tape. No further cable covering or splice boxes are required.

Heat shrinkable tubing shall be installed following manufacturer's instructions. Direct flame heating shall not be permitted unless recommended by the manufacturer. Cable surfaces within the limits of the heat-shrink application shall be clean and free of contaminants prior to application.

**e. Assembly.** Surfaces of equipment or conductors being terminated or connected shall be prepared in accordance with industry standard practice and manufacturer's recommendations. All surfaces to be connected shall be thoroughly cleaned to remove all dirt, grease, oxides, nonconductive films, or other foreign material. Paints and other nonconductive coatings shall be removed to expose base metal. Clean all surfaces at least 1/4 inch (6.4 mm) beyond all sides of the larger bonded area on all mating surfaces. Use a joint compound suitable for the materials used in the connection. Repair painted/coated surface to original condition after completing the connection.

**108-3.6 BARE COUNTERPOISE WIRE INSTALLATION FOR LIGHTNING PROTECTION AND GROUNDING.** If shown on the plans or included in the job specifications, bare solid [ #6 AWG ] copper counterpoise wire shall be installed for lightning protection of the underground cables. The RPR shall select one of two methods of lightning protection for the airfield lighting circuit based upon sound engineering practice and lightning strike density.

**a. Equipotential.** The counterpoise size is as shown on the plans. The equipotential method is applicable to all airfield lighting systems; i.e. runway, taxiway, apron – touchdown zone, centerline, edge, threshold and approach lighting systems. The equipotential method is also successfully applied to provide lightning protection for power, signal and communication systems. The light bases, counterpoise, etc. – all components - are bonded together and bonded to the vault power system ground loop/electrode.

Counterpoise wire shall be installed in the same trench for the entire length of buried cable, conduits and duct banks that are installed to contain airfield cables. The counterpoise is centered over the cable/conduit/duct to be protected.

The counterpoise conductor shall be installed no less than 8 inches (200 mm) minimum or 12 inches (300 mm) maximum above the raceway or cable to be protected, except as permitted below:

(1) The minimum counterpoise conductor height above the raceway or cable to be protected shall be permitted to be adjusted subject to coordination with the airfield lighting and pavement designs.

(2) The counterpoise conductor height above the protected raceway(s) or cable(s) shall be calculated to ensure that the raceway or cable is within a 45-degree area of protection, (45 degrees on each side of vertical creating a 90 degree angle).

The counterpoise conductor shall be bonded to each metallic light base, mounting stake, and metallic airfield lighting component.

All metallic airfield lighting components in the field circuit on the output side of the constant current regulator (CCR) or other power source shall be bonded to the airfield lighting counterpoise system.

All components rise and fall at the same potential; with no potential difference, no damaging arcing and no damaging current flow.

See AC 150/5340-30, Design and Installation Details for Airport Visual Aids and NFPA 780, Standard for the Installation of Lightning Protection Systems, Chapter 11, for a detailed description of the Equipotential Method of lightning protection.

Reference FAA STD-019E, Lightning and Surge Protection, Grounding Bonding and Shielding Requirements for Facilities and Electronic Equipment, Part 4.1.1.7.] [not used]

**b. Isolation.** Not used

**c. Common Installation requirements.** When a metallic light base is used, the grounding electrode shall be bonded to the metallic light base or mounting stake with a No. 6 AWG bare, annealed or soft drawn, solid copper conductor.

Grounding electrodes may be rods, ground dissipation plates, radials, or other electrodes listed in the NFPA 70 (NEC) or NFPA 780.

Where raceway is installed by the directional bore, jack and bore, or other drilling method, the counterpoise conductor shall be permitted to be installed concurrently with the directional bore, jack and bore, or other drilling method raceway, external to the raceway or sleeve.

The counterpoise wire shall also be exothermically welded to ground rods installed as shown on the plans but not more than 500 feet (150 m) apart around the entire circuit. The counterpoise system shall be continuous and terminate at the transformer vault or at the power source. It shall be securely attached to the vault or equipment external ground ring or other made electrode-grounding system. The connections shall be made as shown on the plans and in the specifications.

Where an existing airfield lighting system is being extended or modified, the new counterpoise conductors shall be interconnected to existing counterpoise conductors at each intersection of the new and existing airfield lighting counterpoise systems.

**d. Parallel Voltage Systems.** Provide grounding and bonding in accordance with NFPA 70, National Electrical Code.

**108-3.7 COUNTERPOISE INSTALLATION ABOVE MULTIPLE CONDUITS AND DUCT BANKS.** Counterpoise wires shall be installed above multiple conduits/duct banks for airfield lighting cables, with the intent being to provide a complete area of protection over the airfield lighting cables. When multiple conduits and/or duct banks for airfield cable are installed in the same trench, the number and location of counterpoise wires above the conduits shall be adequate to provide a complete area of protection measured 45 degrees each side of vertical.

Where duct banks pass under pavement to be constructed in the project, the counterpoise shall be placed above the duct bank. Reference details on the construction plans.

**108-3.8 COUNTERPOISE INSTALLATION AT EXISTING DUCT BANKS.** When airfield lighting cables are indicated on the plans to be routed through existing duct banks, the new counterpoise wiring shall be terminated at ground rods at each end of the existing duct bank where the cables

being protected enter and exit the duct bank. The new counterpoise conductor shall be bonded to the existing counterpoise system.

**108-3.9 EXOTHERMIC BONDING.** Bonding of counterpoise wire shall be by the exothermic welding process or equivalent method accepted by the RPR. Only personnel experienced in and regularly engaged in this type of work shall make these connections.

Contractor shall demonstrate to the satisfaction of the RPR, the welding kits, materials and procedures to be used for welded connections prior to any installations in the field. The installations shall comply with the manufacturer's recommendations and the following:

a. All slag shall be removed from welds.

b. Using an exothermic weld to bond the counterpoise to a lug on a galvanized light base is not recommended unless the base has been specially modified. Consult the manufacturer's installation directions for proper methods of bonding copper wire to the light base. See AC 150/5340-30 for galvanized light base exception.

c. If called for in the plans, all buried copper and weld material at weld connections shall be thoroughly coated with 6 mm of 3M™ Scotchkote™, or approved equivalent, or coated with coal tar Bitumastic® material to prevent surface exposure to corrosive soil or moisture.

**108-3.10 TESTING.** The Contractor shall furnish all necessary equipment and appliances for testing the airport electrical systems and underground cable circuits before and after installation. The Contractor shall perform all tests in the presence of the RPR. The Contractor shall demonstrate the electrical characteristics to the satisfaction of the RPR. All costs for testing are incidental to the respective item being tested. For phased projects, the tests must be completed by phase. The Contractor must maintain the test results throughout the entire project as well as during the warranty period that meet the following:

a. Earth resistance testing methods shall be submitted to the RPR for approval. Earth resistance testing results shall be recorded on an approved form and testing shall be performed in the presence of the RPR. All such testing shall be at the sole expense of the Contractor.

b. Should the counterpoise or ground grid conductors be damaged or suspected of being damaged by construction activities the Contractor shall test the conductors for continuity with a low resistance ohmmeter. The conductors shall be isolated such that no parallel path exists and tested for continuity. The RPR shall approve of the test method selected. All such testing shall be at the sole expense of the Contractor.

After installation, the Contractor shall test and demonstrate to the satisfaction of the RPR the following:

c. That all affected lighting power and control circuits (existing and new) are continuous and free from short circuits.

d. That all affected circuits (existing and new) are free from unspecified grounds.



e. That the insulation resistance to ground of all new non-grounded high voltage series circuits or cable segments is not less than 500 megohms. Verify continuity of all series airfield lighting circuits prior to energization.

f. That the insulation resistance to ground of all new non-grounded conductors of new multiple circuits or circuit segments is not less than 100 megohms.

g. That all affected circuits (existing and new) are properly connected per applicable wiring diagrams.

h. That all affected circuits (existing and new) are operable. Tests shall be conducted that include operating each control not less than 10 times and the continuous operation of each lighting and power circuit for not less than 1/2 hour.

i. That the impedance to ground of each ground rod does not exceed 10 ohms prior to establishing connections to other ground electrodes. The fall-of-potential ground impedance test shall be used, as described by American National Standards Institute/Institute of Electrical and Electronic Engineers (ANSI/IEEE) Standard 81, to verify this requirement. As an alternate, clamp-on style ground impedance test meters may be used to satisfy the impedance testing requirement. Test equipment and its calibration sheets shall be submitted for review and approval by the RPR prior to performing the testing.

Two copies of tabulated results of all cable tests performed shall be supplied by the Contractor to the RPR. Where connecting new cable to existing cable, insulation resistance tests shall be performed on the new cable prior to connection to the existing circuit.

There are no approved "repair" procedures for items that have failed testing other than complete replacement.

## **METHOD OF MEASUREMENT**

**108-4.1** The cost of all excavation, backfill, dewatering and restoration regardless of the type of material encountered shall be included in the unit price bid for the work

**108-4.2** Cable or counterpoise wire installed in trench, duct bank or conduit shall be measured by the number of linear feet (meters) installed and grounding connectors, and trench marking tape ready for operation, and accepted as satisfactory. Separate measurement shall be made for each cable or counterpoise wire installed in trench, duct bank or conduit. The measurement for this item shall include additional quantities required for slack.

**108-4.3** Ground rods shall be measured by each 10-foot section installed complete.

## **BASIS OF PAYMENT**

**108-5.1** Payment will be made at the contract unit price for trenching, cable and bare counterpoise wire installed in trench (direct-buried), or cable and equipment ground installed in duct bank or conduit, in place by the Contractor and accepted by the RPR. This price shall be full compensation for furnishing all materials and for all preparation and installation of these materials, and for all labor, equipment, tools, and incidentals, including ground rods and ground connectors and trench marking tape, necessary to complete this item.

Payment will be made under:

|                |                                                                                                                                                                                                        |                         |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Item L-108-5.1 | <del>Trenching for direct-buried cable, 18-inch minimum depth -- per linear foot (meter)</del>                                                                                                         |                         |
| Item L-108-5.2 | <del>[ No. 8 AWG ] [ No. 6 AWG ], [ 5 kV ] [ 600V ], L-824, [ Type C ] [ Type B ] Cable, Installed in Trench, Duct Bank or Conduit -- per linear foot (meter)</del>                                    |                         |
| Item L-108-5.3 | <del>No. [ 6 ] [ 4 ] [ 2 ] AWG, Solid, Bare Copper Counterpoise Wire, Installed [ in Trench ], [ Above the Duct Bank or Conduit ], Including Connections/Terminations -- per linear foot (meter)</del> |                         |
| Item L-108-5.4 | <del>No. [ 6 ] [ 4 ] [ 2 ] AWG, [ Bare ] [ Insulated ], Stranded Equipment [ bonding ] [ Ground ], Installed in Duct Bank or Conduit -- per linear foot (meter).</del>                                 |                         |
| Item L-108-1   | 1/C L-824-Type C Unshielded #8 AWG 5 KV Stranded Copper Cable, Installed in Duct or Conduit                                                                                                            | -- per Linear Foot (LF) |
| Item L-108-2   | 1/C #2 AWG Solid Copper Counterpoise Cable, Installed over Duct or Conduit                                                                                                                             | -- per Linear Foot (LF) |
| Item L-108-3   | 0.75" Diameter by 10.00' Long Copper Clad Steel Sectional Ground Rod                                                                                                                                   | -- per Each (EA)        |
| Item L-108-4   | Field Lightning Arrestor Assembly                                                                                                                                                                      | -- per Each (EA)        |

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

### Advisory Circulars (AC)

|                |                                                                                    |
|----------------|------------------------------------------------------------------------------------|
| AC 150/5340-26 | Maintenance of Airport Visual Aid Facilities                                       |
| AC 150/5340-30 | Design and Installation Details for Airport Visual Aids                            |
| AC 150/5345-7  | Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits |
| AC 150/5345-26 | Specification for L-823 Plug and Receptacle, Cable Connectors                      |
| AC 150/5345-53 | Airport Lighting Equipment Certification Program                                   |

Commercial Item Description

|            |                                                                   |
|------------|-------------------------------------------------------------------|
| A-A-59544A | Cable and Wire, Electrical (Power, Fixed Installation)            |
| A-A-55809  | Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic |

ASTM International (ASTM)

|            |                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------|
| ASTM B3    | Standard Specification for Soft or Annealed Copper Wire                                          |
| ASTM B8    | Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft |
| ASTM B33   | Standard Specification for Tin-Coated Soft or Annealed Copper Wire for Electrical Purposes       |
| ASTM D4388 | Standard Specification for Nonmetallic Semi-Conducting and Electrically Insulating Rubber Tapes  |

Mil Spec

|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| MIL-PRF-23586F | Performance Specification: Sealing Compound (with Accelerator), Silicone Rubber, Electrical |
| MIL-I-24391    | Insulation Tape, Electrical, Plastic, Pressure Sensitive                                    |

National Fire Protection Association (NFPA)

|          |                                                               |
|----------|---------------------------------------------------------------|
| NFPA-70  | National Electrical Code (NEC)                                |
| NFPA-780 | Standard for the Installation of Lightning Protection Systems |

American National Standards Institute (ANSI)/Institute of Electrical and Electronics Engineers (IEEE)

|                  |                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------|
| ANSI/IEEE STD 81 | IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System |
|------------------|---------------------------------------------------------------------------------------------------------------|

Federal Aviation Administration Standard

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| FAA STD-019E | Lightning and Surge Protection, Grounding Bonding and Shielding Requirements for Facilities and Electronic Equipment |
|--------------|----------------------------------------------------------------------------------------------------------------------|

**END OF ITEM L-108**

## **ITEM L-109 AIRPORT TRANSFORMER VAULT AND VAULT EQUIPMENT**

### **DESCRIPTION**

**109-1.1** This item shall consist of modifications to an existing vault is to be used shall be the furnishing of new vault equipment, wiring, electrical buses, cable, conduit, and grounding systems. Included in this work shall be the modification of the existing ADB-Safegate L-890 Airfield Lighting Control System (ALCS). This work shall also include the painting of equipment and conduit; the marking and labeling of equipment and the labeling or tagging of wires; the testing of the installation; and the furnishing of all incidentals necessary to place it in operating condition as a completed unit to the satisfaction of the RPR.

### **EQUIPMENT AND MATERIALS**

#### **109-2.1 GENERAL.**

**a.** Airport lighting equipment and materials covered by advisory circulars (AC) shall be certified in AC 150/5345-53, Airport Lighting Equipment Certification Program (ALECP) and listed in the ALECP Addendum.

**b.** All other equipment and materials covered by other referenced specifications shall be subject to acceptance through manufacturer's certification of compliance with the applicable specification when requested by the RPR.

**c.** Manufacturer's certifications shall not relieve the Contractor of the responsibility to provide materials per these specifications. Materials supplied and/or installed that do not comply with these specifications shall be removed (when directed by the RPR) and replaced with materials that comply with these specifications at the Contractor's cost.

**d.** All materials and equipment used to construct this item shall be submitted to the RPR for approval prior to ordering the equipment. Submittals consisting of marked catalog sheets or shop drawings shall be provided. Submittal data shall be presented in a clear, precise and thorough manner. Original catalog sheets are preferred. Photocopies are acceptable provided they are as good a quality as the original. Clearly and boldly mark each copy to identify products or models applicable to this project. Indicate all optional equipment and delete any non-pertinent data. Submittals for components of electrical equipment and systems shall identify the equipment to which they apply on each submittal sheet. Markings shall be made bold and clear with arrows or circles (highlighting is not acceptable). The Contractor is solely responsible for delays in the project that may accrue directly or indirectly from late submissions or resubmissions of submittals.

**e.** The data submitted shall be sufficient, in the opinion of the RPR, to determine compliance with the plans and specifications. The Contractor's submittals shall be provided in electronic pdf format, tabbed by specification section. The RPR reserves the right to reject any and all equipment, materials or procedures that do not meet the system design and the standards and codes, specified in this document.

**f.** All equipment and materials furnished and installed under this section shall be guaranteed against defects in materials and workmanship for a period of at least twelve (12) months from final acceptance by the Owner. The defective materials and/or equipment shall be repaired or replaced, at the Owner's discretion, with no additional cost to the Owner.

## **MODIFICATIONS OF EXISTING VAULT**

**109-3.1 ELECTRICAL VAULT BUILDING.** The electrical vault building must comply with NEC Article 110.31, Enclosure for Electrical Installations, Item (A) Electrical Vaults. Construct the building of materials having adequate structural strength for the conditions and installed location, has a minimum fire rating of two or three hours as determined by the authority having jurisdiction (AHJ), and is bullet resistant to minimum UL 752 Level 4.

**109-3.2 CONCRETE.** Concrete shall be proportioned, placed, and cured per Item P-610, Concrete for Miscellaneous Structures.

**109-3.3 PRECAST CONCRETE STRUCTURES.** Precast concrete structures shall be furnished by a plant meeting National Precast Concrete Association Plant Certification Program or another RPR approved third party certification program. Precast concrete structures shall conform to ASTM C478.

**109-3.4 REINFORCING STEEL.** Reinforcing steel bars shall be intermediate or structural grade deformed-type bars and shall be per ASTM A615.

**109-3.5 BRICK.** Not used.

**109-3.6 RIGID STEEL CONDUIT.** Rigid steel conduit and fittings shall be per Underwriters Laboratories Standards 6 and 514B.

**109-3.7 PLASTIC CONDUIT AND FITTINGS.** Plastic Conduit and fittings shall conform to the requirements of UL-651 and UL-654 schedule 40 polyvinyl chloride (PVC) suitable for use above or below ground.

**109-3.8 LIGHTING.** Not used.

**109-3.9 OUTLETS.** Convenience outlets shall be heavy-duty duplex units designed for industrial service.

**109-3.10 SWITCHES.** Vault or metal-housing light switches shall be single-pole switches.

### **109-3.11 PAINT.**

a. Priming paint for non-galvanized metal surfaces shall be a high solids alkyd primer compatible with the manufacturer's recommendations for the intermediate or topcoat.

b. White paint for body and finish coats on metal and wood surfaces shall be ready-mixed paint conforming to the Master Painter's Institute (MPI), Reference #9, Exterior Alkyd, Gloss.

c. Priming paint for wood surfaces shall be mixed on the job by thinning the specified white paint by adding 1/2 pint (0.24 liter) of raw linseed oil to each gallon (liter).

d. Paint for the floor, ceiling, and inside walls shall be per Porter Paint Company 69, 71, and 79 or equivalent. Walls and ceiling shall be light gray and the floor shall be medium gray.

e. The roof coating shall be hot asphalt material per ASTM D2823. Asbestos-free roof coating per ASTM D4479 may be substituted if required by local codes.

**109-3.12 GROUND BUS.** Ground bus shall be 1/8 × 3/4 inch (3 × 19 mm) minimum copper bus bar.

**109-3.13 SQUARE DUCT.** Duct shall be square similar to that manufactured by the Square D Company (or equivalent), or the Trumbull Electric Manufacturing Company (or equivalent). The entire front of the duct on each section shall consist of hinged or removable cover for ready access to the interior. The cross-section of the duct shall be not less than 4 × 4 inch (100 × 100 mm) except where otherwise shown in the plans.

**109-3.14 GROUND RODS.** Ground rods shall be in accordance with Item L-108.

**109-3.15 VAULT PREFABRICATED METAL HOUSING.** Not used.

**109-3.16 FAA-APPROVED EQUIPMENT.** Certain items of airport lighting equipment installed in vaults are covered by individual ACs listed below:

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| AC 150/5345-3  | Specification for L-821, Panels for Remote Control of Airport Lighting                                   |
| AC 150/5345-5  | Circuit Selector Switch                                                                                  |
| AC 150/5345-7  | Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits                       |
| AC 150/5345-10 | Specification for Constant Current Regulators and Regulator Monitors                                     |
| AC 150/5345-13 | Specification for L-841 Auxiliary Relay Cabinet Assembly for Pilot Control of Airport Lighting Circuits. |
| AC 150/5345-49 | Specification for L-854, Radio Control Equipment                                                         |
| AC 150/5345-56 | Specification for L-890 Airport Lighting Control and Monitoring System (ALCMS)                           |

**109-3.17 OTHER ELECTRICAL EQUIPMENT.** Distribution transformers, cutouts, relays, terminal blocks, transfer relays, circuit breakers, and all other regularly used commercial items of electrical equipment not covered by FAA equipment specifications and ACs shall conform to the applicable rulings and standards of the Institute of Electrical and Electronic Engineers (IEEE) or the National Electrical Manufacturers Association (NEMA). When specified, test reports from a testing laboratory indicating that the equipment meets the specifications shall be supplied. In all cases, equipment shall be new and a first-grade product. This equipment shall be supplied in the quantities required for the specific project and shall incorporate the electrical and mechanical characteristics specified in the proposal and plans. Equipment selected and installed by the Contractor shall maintain the interrupting current rating of the existing systems or specified rating whichever is greater.

**109-3.18 WIRE.** Wire (in conduit) rated up to 5,000 volts shall be per AC 150/5345-7, Specification for L-824 Underground Electrical Cables for Airport Lighting Circuits. For ratings up to 600 volts, moisture and heat resistant thermoplastic wire conforming to Commercial Item Description A-A-59544A Type THWN-2 shall be used. The wires shall be of the type, size, number of conductors, and voltage shown in the plans or in the proposal.

**a. Control circuits.** Unless otherwise indicated on the plans, wire shall be not less than No. 12 American wire gauge (AWG) and shall be insulated for 600 volts. If telephone control cable is specified, No. 19 AWG telephone cable per ANSI/Insulated Cable Engineers Association (ICEA) S-85-625 specifications shall be used.

**b. Power circuits.**

(1) 600 volts maximum – Wire shall be No. 6 AWG or larger and insulated for at least 600 volts.

(2) 3,000 volts maximum – Wire shall be No. 6 AWG or larger and insulated for at least 3,000 volts.

(3) Over 3,000 volts-Wire shall be No. 6 AWG or larger and insulated for at least the circuit voltage.

**109-3.19 SHORT CIRCUIT / COORDINATION / DEVICE EVALUATION / ARC FLASH ANALYSIS.** The Contractor shall, based upon the equipment provided, include as a part of the submittal process the electrical system “Short Circuit / Coordination / Device evaluation / Arc Flash Analysis”. The analysis shall be performed by the equipment manufacturer and submitted in a written report. The analysis shall be signed and sealed by a registered professional Engineer from the state in which the project is located. The analysis shall comply with NFPA-70E and IEEE 1584.

The analysis will include: one line diagrams, short circuit analysis, coordination analysis, equipment evaluation, arc flash analysis and arc flash labels containing at a minimum, equipment name, voltage/current rating, available incident energy and flash protection boundary.

The selected firms field service Engineer shall perform data gathering for analysis completion and device settings, perform device setting as recommended by the analysis and will furnish and install the arc flash labels. The components worst case incident energy will be considered the available arc flash energy at that specific point in the system. Submit three written copies and one electronic copy of the report.

**CONSTRUCTION METHODS**

**CONSTRUCTION OF VAULT AND PREFABRICATED METAL HOUSING, NOT USED**

**109-4.1 GENERAL.** Not used.

**109-4.2 FOUNDATION AND WALLS. Not used.**

- a. **Reinforced concrete construction.** Not used.
- b. **Brick and concrete construction.** Not used.
- c. **Concrete masonry construction.** Not used.

**109-4.3 ROOF.** Not used.

**109-4.4 FLOOR.** Not used.

**109-4.5 FLOOR DRAIN.** No

**109-4.6 CONDUITS IN FLOOR AND FOUNDATION.** Not used.

**109-4.7 DOORS.** Not used.

**109-4.8 PAINTING.** Not used.

**109-4.9 LIGHTS AND SWITCHES.** Not used.

**INSTALLATION OF EQUIPMENT IN VAULT**

**109-5.1 GENERAL.** The Contractor shall furnish, install, and connect all equipment, equipment accessories, conduit, cables, wires, buses, grounds, and support necessary to ensure a complete and operable electrical distribution center for the airport lighting system as specified herein and shown in the plans. When specified, an emergency power supply and transfer switch shall be provided and installed.

The equipment installation and mounting shall comply with the requirements of the National Electrical Code and local code agency having jurisdiction. All electrical work shall comply with the NEC and local code agency having jurisdiction including the separation of under 600V work from 5,000V work."

**109-5.2 POWER SUPPLY EQUIPMENT.** Transformers, regulators, booster transformers, and other power supply equipment items shall be furnished and installed at the location shown in the plans or as directed by the RPR. The power supply equipment shall be set on steel "H" sections, "I" beams, channels, or concrete blocks to provide a minimum space of 1-1/2 inch (38 mm) between the equipment and the floor. The equipment shall be placed so as not to obstruct the oil-sampling plugs of the oil-filled units; and name-plates shall, so far as possible, not be obscured.

If specified in the plans and specifications, equipment for an alternate power source or an emergency power generator shall be furnished and installed. The alternate power supply installation shall include all equipment, accessories, an automatic changeover switch, and all necessary wiring and connections. The emergency power generator set shall be the size and type specified.

**109-5.3 SWITCHGEAR AND PANELS.** Oil switches, fused cutouts, relays, transfer switches, panels, panel boards, and other similar items shall be furnished and installed at the location shown



in the plans or as directed by the RPR. Wall or ceiling mounted items shall be attached to the wall or ceiling with galvanized bolts of not less than 3/8-inch (9 mm) diameter engaging metal expansion shields or anchors in masonry or concrete vaults.

**109-5.4 DUCT AND CONDUIT.** The Contractor shall furnish and install square-type exposed metallic ducts with hinged covers for the control circuits in the vault. These shall be mounted along the walls behind all floor-mounted equipment and immediately below all wall-mounted equipment. The hinged covers shall be placed to open from the front side with the hinges at the front bottom.

Wall brackets for square ducts shall be installed at all joints 2 feet (60 cm) or more apart with intermediate brackets as specified. Conduit shall be used between square ducts and equipment or between different items of equipment when the equipment is designed for conduit connection. When the equipment is not designed for conduit connection, conductors shall enter the square-type control duct through insulating bushings in the duct or on the conduit risers.

**109-5.5 WIRING AND CONNECTIONS.** The Contractor shall make all necessary electrical connections in the vault per the wiring diagrams furnished and as directed by the RPR. In wiring to the terminal blocks, the Contractor shall leave sufficient extra length on each control lead to make future changes in connections at the terminal block. This shall be accomplished by running each control lead the longest way around the box to the proper terminal. Leads shall be neatly laced in place.

**109-5.6 MARKING AND LABELING.** All equipment, control wires, terminal blocks, etc., shall be tagged, marked, or labeled as specified below:

**a. Wire identification.** The Contractor shall furnish and install self-sticking wire labels or identifying tags on all control wires at the point where they connect to the control equipment or to the terminal blocks. Wire labels, if used, shall be of the self-sticking preprinted type and of the manufacturer's recommended size for the wire involved. Identification -markings designated in the plans shall be followed. Tags, if used, shall be of fiber not less than 3/4 inch (19 mm) in diameter and not less than 1/32 inch (1 mm) thick. Identification markings designated in the plans shall be stamped on tags by means of small tool dies. Each tag shall be securely tied to the proper wire by a nonmetallic cord.

**b. Labels.** The Contractor shall stencil identifying labels on the cases of regulators, breakers, and distribution and control relay cases with white oil paint as designated by the RPR. The letters and numerals shall be not less than one inch (25 mm) in height and shall be of proportionate width. The Contractor shall also mark the correct circuit designations per the wiring diagram on the terminal marking strips, which are a part of each terminal block.

## **METHOD OF MEASUREMENT**

**109-6.1** The quantity of vaults to be paid for under this item shall consist of the number of vaults constructed in place and accepted as a complete unit.

**109-6.2** The quantity of prefabricated metal housings to be paid for under this item shall consist of the number of housings constructed in place and accepted as a complete unit.

**109-6.3** The quantity of equipment to be paid for under this item shall consist of all equipment installed, connected and accepted as a complete unit ready for operation within an existing vault or prefabricated metal housing.

## **BASIS OF PAYMENT**

**109-7.1** Payment will be made at the contract unit price for each completed and accepted vault or prefabricated metal housing equipment installation. This price shall be full compensation for furnishing all materials and for all preparation, assembly, and installation of these materials, and for all labor, equipment, tools, and incidentals necessary to complete the item.

Payment will be made under:

~~Item L-109-7.1 — Construction of Airport Transformer Vault in Place — per unit~~

~~Item L-109-7.2 — Installation of Airport Transformer Vault Equipment in Place — per unit~~

~~Item L-109-7.3 — Construction of [ Prefabricated Metal Housing ] [ Prefabricated Concrete Building ] and Foundation in Place — per unit~~

~~Item L-109-7.4 — Installation of Equipment with in existing vault or prefabricated metal housing in Place — per unit~~

**Item L-109-1      Airfield Lighting Control System Modifications: Vendor      -- Allowance (AL)**

Payment shall be made at the contract unit price to enable the Contractor to obtain, all necessary equipment, specialty materials and technical support from the OEM for the ALCS modifications. The Contractor will utilize the OEM ALCS furnished materials and will pay for the ALCS components within the allowance included in the bid documents (including sales tax). Documentation, meeting time and commissioning of the ALCS are also included. This pay item also includes modifications by the manufacturer for any graphics revisions of the airfield lighting control panel to update the airfield geometry, lighting circuit display and control. The Owner shall determine the precise menu of services and associated cost for payment by the Contractor from the set aside allowance – Payment shall be made in accordance with completed vendor invoices.

**Item L-109-2      Airfield Lighting Control System Modifications: Contractor      -- Lump Sum (LS)**

Payment shall be made at the contract unit price to include all coordination, materials, labor, supervision, escort of manufacturer representatives, and all incidentals required to install the OEM furnished materials and to modify the existing Airfield Lighting Control System (ALCS) including the L-890 control panel modifications in the Airfield Lighting Vault and Air Traffic Control Tower (ATCT), interconnection of the existing tower, vault computers and new Constant Current Regulators (CCR), provide and install all controls and monitoring and all modifications necessary for a complete and operational Airfield Lighting Control System to provide a fully functional system in accordance with the Contract Documents.

**Item L-109-3      Airfield Electrical Vault Modifications      -- Lump Sum (LS)**

Payment shall be made at the contract unit price to include all transportation, coordination, materials, labor, supervision, and all incidentals required to furnish and install all equipment, raceway, conductors, lightning protection, alternator, utility services, feeders, branch circuits, grounding, bonding, coordination study, overcurrent protection, terminations, etc. required for the new airfield lighting equipment and revisions to the power distribution system as shown in the Contract Documents, install all controls and monitoring and all appurtenances necessary for a complete and functional electrical system in accordance with the Contract Documents.

**Item L-109-4      10 KW L-829 Constant Current Regulator- 5 Step - 480V      -- per Each (EA)**

L-829 Constant current regulator, air cooled, ferro-resonant, dry type, with 120 Volt internal control, **480** Volt single phase input, **6.6** amp output, **5** step brightness control. The units shall have integral primary switches and input and output lightning protection. Each shall have local control switches for on/off and all brightness steps, and shall have a face plate digital output RMS ammeter and face plate digital output RMS voltmeter. Each CCR shall have an internally mounted SCO cut out installed in the interior of the CCR and be labeled as such on the exterior of the CCR. This item includes installation and testing of the unit, 5kv pigtails and L-823 connectors, grounding and all incidentals.

**Item L-109-5      15 KW L-829 Constant Current Regulator- 5 Step - 480V      -- per Each (EA)**

L-829 Constant current regulator, air cooled, ferro-resonant, dry type, with 120 Volt internal control, **480** Volt single phase input, **6.6** amp output, **5** step brightness control. The units shall have integral primary switches and input and output lightning protection. Each shall have local control switches for on/off and all brightness steps, and shall have a face plate digital output RMS ammeter and face plate digital output RMS voltmeter. Each CCR shall have an internally mounted SCO cut out installed in the interior of the CCR and be labeled as such on the exterior of the CCR. This item includes installation and testing of the unit, 5kv pigtails and L-823 connectors, grounding and all incidentals.

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

### Advisory Circulars (AC)

|                |                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------|
| AC 150/5340-30 | Design and Installation Details for Airport Visual Aids                                                 |
| AC 150/5345-3  | Specification for L-821, Panels for Remote Control of Airport Lighting                                  |
| AC 150/5345-5  | Circuit Selector Switch                                                                                 |
| AC 150/5345-7  | Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits                      |
| AC 150/5345-10 | Specification for Constant Current Regulators and Regulator Monitors                                    |
| AC 150/5345-13 | Specification for L-841 Auxiliary Relay Cabinet Assembly for Pilot Control of Airport Lighting Circuits |
| AC 150/5345-49 | Specification L-854, Radio Control Equipment;                                                           |
| AC 150/5345-53 | Airport Lighting Equipment Certification Program                                                        |

### American National Standards Institute / Insulated Cable Engineers Association (ANSI/ICEA)

ANSI/ICEA S-85-625 Standard for Telecommunications Cable Aircore, Polyolefin Insulated, Copper Conductor Technical Requirements

ASTM International (ASTM)

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| ASTM A615  | Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement |
| ASTM C62   | Standard Specification for Building Brick (Solid Masonry Units Made from Clay or Shale)    |
| ASTM C90   | Standard Specification for Loadbearing Concrete Masonry Units                              |
| ASTM D2823 | Standard Specification for Asphalt Roof Coatings, Asbestos Containing                      |
| ASTM D4479 | Standard Specification for Asphalt Roof Coatings – Asbestos-Free                           |

Commercial Item Description (CID)

|           |                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| A-A 59544 | Cable and Wire, Electrical (Power, Fixed Installation)<br>Institute of Electrical and Electronic Engineers (IEEE) |
| IEEE 1584 | Guide for Performing Arc-Flash Hazard Calculations                                                                |

Master Painter's Institute (MPI)

|                  |                                            |
|------------------|--------------------------------------------|
| MPI Reference #9 | Alkyd, Exterior, Gloss (MPI Gloss Level 6) |
|------------------|--------------------------------------------|

Underwriters Laboratories (UL)

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| UL Standard 6    | Electrical Rigid Metal Conduit – Steel                        |
| UL Standard 514B | Conduit, Tubing, and Cable Fittings                           |
| UL Standard 514C | Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers      |
| UL Standard 651  | Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings |
| UL Standard 651A | Type EB and A Rigid PVC Conduit and HDPE Conduit              |

National Fire Protection Association (NFPA)

|          |                                                               |
|----------|---------------------------------------------------------------|
| NFPA-70  | National Electrical Code (NEC)                                |
| NFPA-70E | Standard for Electrical Safety in the Workplace               |
| NFPA-780 | Standard for the Installation of Lightning Protection Systems |

**END OF ITEM L-109**

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## ITEM L-110 AIRPORT UNDERGROUND ELECTRICAL DUCT BANKS AND CONDUITS

### DESCRIPTION

**110-1.1** This item shall consist of underground electrical conduits and duct banks (single or multiple conduits encased in concrete or buried in sand) installed per this specification at the locations and per the dimensions, designs, and details shown on the plans. This item shall include furnishing and installing of all underground electrical duct banks and individual and multiple underground conduits. It shall also include all turbing trenching, backfilling, removal, and restoration of any paved or turfed areas; concrete encasement, mandrelling, pulling lines, duct markers, plugging of conduits, and the testing of the installation as a completed system ready for installation of cables per the plans and specifications. This item shall also include furnishing and installing conduits and all incidentals for providing positive drainage of the system. Verification of existing ducts is incidental to the pay items provided in this specification.

### EQUIPMENT AND MATERIALS

#### 110-2.1 GENERAL.

a. All equipment and materials covered by referenced specifications shall be subject to acceptance through manufacturer's certification of compliance with the applicable specification when requested by the RPR.

b. Manufacturer's certifications shall not relieve the Contractor of the responsibility to provide materials per these specifications and acceptable to the RPR. Materials supplied and/or installed that do not comply with these specifications shall be removed, when directed by the RPR and replaced with materials, that comply with these specifications, at the Contractor's cost.

c. All materials and equipment used to construct this item shall be submitted to the RPR for approval prior to ordering the equipment. Submittals consisting of marked catalog sheets or shop drawings shall be provided. Submittal data shall be presented in a clear, precise and thorough manner. Original catalog sheets are preferred. Photocopies are acceptable provided they are as good a quality as the original. Clearly and boldly mark each copy to identify products or models applicable to this project. Indicate all optional equipment and delete non-pertinent data. Submittals for components of electrical equipment and systems shall identify the equipment for which they apply on each submittal sheet. Markings shall be made bold and clear with arrows or circles (highlighting is not acceptable). The Contractor is solely responsible for delays in project that accrue directly or indirectly from late submissions or resubmissions of submittals.

d. The data submitted shall be sufficient, in the opinion of the RPR, to determine compliance with the plans and specifications. The Contractor's submittals shall be electronically submitted in pdf format, tabbed by specification section. The RPR reserves the right to reject any and all equipment, materials or procedures that do not meet the system design and the standards and codes specified in this document.

e. All equipment and materials furnished and installed under this section shall be guaranteed against defects in materials and workmanship for a period of at least twelve (12) months from final acceptance by the Owner. The defective materials and/or equipment shall be repaired or replaced, at the Owner's discretion, with no additional cost to the Owner.

**110-2.2 STEEL CONDUIT.** Rigid galvanized steel (RGS) conduit and fittings shall be hot dipped galvanized inside and out and conform to the requirements of Underwriters Laboratories Standards 6, 514B, and 1242. All RGS conduits or RGS elbows installed below grade, in concrete, permanently wet locations or other similar environments shall be painted with a 10-mil thick coat of asphaltum sealer or shall have a factory-bonded polyvinyl chloride (PVC) cover. Any exposed galvanizing or steel shall be coated with 10 mils of asphaltum sealer. When using PVC coated RGS conduit, care shall be exercised not to damage the factory PVC coating. Damaged PVC coating shall be repaired per the manufacturer's written instructions. In lieu of PVC coated RGS, corrosion wrap tape shall be permitted to be used where RGS is in contact with direct earth."

**110-2.3 PLASTIC CONDUIT.** Plastic conduit and fittings shall conform to the following requirements:

- UL 514B covers W-C-1094-Conduit fittings all types, classes 1 thru 3 and 6 thru 10.
- UL 514C covers W-C-1094- all types, Class 5 junction box and cover in plastic (PVC).
- UL 651 covers W-C-1094-Rigid PVC Conduit, types I and II, Class 4.
- UL 651A covers W-C-1094-Rigid PVC Conduit and high-density polyethylene (HDPE) Conduit type III and Class 4.

Underwriters Laboratories Standards UL-651 and Article 352 of the current National Electrical Code shall be one of the following, as shown on the plans:

- a. Type I—Schedule 40 and Schedule 80 PVC suitable for underground use either direct-buried or encased in concrete.
- b. Type II—Schedule 40 PVC suitable for either above ground or underground use.
- c. Type III – Schedule 80 PVC suitable for either above ground or underground use either direct-buried or encased in concrete.
- d. Type III –HDPE pipe, minimum standard dimensional ratio (SDR) 11, suitable for placement with directional boring under pavement.

The type of solvent cement shall be as recommended by the conduit/fitting manufacturer.

**110-2.4 SPLIT CONDUIT.** Split conduit shall be pre-manufactured for the intended purpose and shall be made of steel or plastic.

**110-2.5 CONDUIT SPACERS.** Conduit spacers shall be prefabricated interlocking units manufactured for the intended purpose. They shall be of double wall construction made of high grade, high density polyethylene complete with interlocking cap and base pads. They shall be designed to accept No. 4 reinforcing bars installed vertically.

**110-2.6 CONCRETE.** Concrete shall be proportioned, placed, and cured per Item P-610, Concrete for Miscellaneous Structures.



**110-2.7 PRECAST CONCRETE STRUCTURES.** Precast concrete structures shall be furnished by a plant meeting National Precast Concrete Association Plant Certification Program or another RPR approved third party certification program. Precast concrete structures shall conform to ASTM C478.

**110-2.8 FLOWABLE BACKFILL.** Flowable material used to back fill conduit and duct bank trenches shall conform to the requirements of Item P-153, Controlled Low Strength Material.

**110-2.9 DETECTABLE WARNING TAPE.** Plastic, detectable, American Public Works Association (APWA) red (electrical power lines, cables, conduit and lighting cable), orange (telephone/fiber optic cabling) with continuous legend magnetic tape shall be polyethylene film with a metallized foil core and shall be 3-6 inches (75-150 mm) wide. Detectable tape is incidental to the respective bid item.

## **CONSTRUCTION METHODS**

**110-3.1 GENERAL.** The Contractor shall install underground duct banks and conduits at the approximate locations indicated on the plans. The RPR shall indicate specific locations as the work progresses, if required to differ from the plans. Duct banks and conduits shall be of the size, material, and type indicated on the plans or specifications. Where no size is indicated on the plans or in the specifications, conduits shall be not less than 2 inches (50 mm) inside diameter or comply with the National Electrical Code based on cable to be installed, whichever is larger. All duct bank and conduit lines shall be laid so as to grade toward access points and duct or conduit ends for drainage. Unless shown otherwise on the plans, grades shall be at least 3 inches (75 mm) per 100 feet (30 m). On runs where it is not practicable to maintain the grade all one way, the duct bank and conduit lines shall be graded from the center in both directions toward access points or conduit ends, with a drain into the storm drainage system. Pockets or traps where moisture may accumulate shall be avoided. Under pavement, the top of the duct bank shall not be less than 18 inches (0.5 m) below the subgrade; in other locations, the top of the duct bank or underground conduit shall be not less than 18 inches (0.5 m) below finished grade.

The Contractor shall mandrel each individual conduit whether the conduit is direct-buried or part of a duct bank. An iron-shod mandrel, not more than 1/4 inch (6 mm) smaller than the bore of the conduit shall be pulled or pushed through each conduit. The mandrel shall have a leather or rubber gasket slightly larger than the conduit hole.

The Contractor shall swab out all conduits/ducts and clean base can, manhole, pull boxes, etc., interiors immediately prior to pulling cable. Once cleaned and swabbed the light bases, manholes, pull boxes, etc., and all accessible points of entry to the duct/conduit system shall be kept closed except when installing cables. Cleaning of ducts, base cans, manholes, etc., is incidental to the pay item of the item being cleaned. All raceway systems left open, after initial cleaning, for any reason shall be recleaned at the Contractor's expense. All accessible points shall be kept closed when not installing cable. The Contractor shall verify existing ducts proposed for use in this project as clear and open. The Contractor shall notify the RPR of any blockage in the existing ducts.

For pulling the permanent wiring, each individual conduit, whether the conduit is direct-buried or part of a duct bank, shall be provided with a 200-pound (90 kg) test polypropylene pull rope. The ends shall be secured and sufficient length shall be left in access points to prevent it from slipping back into the conduit. Where spare conduits are installed, as indicated on the plans, the open ends shall be plugged with removable tapered plugs, designed for this purpose.

All conduits shall be securely fastened in place during construction and shall be plugged to prevent contaminants from entering the conduits. Any conduit section having a defective joint shall not be installed. Ducts shall be supported and spaced apart using approved spacers at intervals not to exceed 5 feet (1.5 m).

Unless otherwise shown on the plans, concrete encased duct banks shall be used when crossing under pavements expected to carry aircraft loads, such as runways, taxiways, taxilanes, ramps and aprons. When under paved shoulders and other paved areas, conduit and duct banks shall be encased using flowable fill for protection.

All conduits within concrete encasement of the duct banks shall terminate with female ends for ease in current and future use. Install factory plugs in all unused ends. Do not cover the ends or plugs with concrete.

Where turf is well established and the sod can be removed, it shall be carefully stripped and properly stored.

Trenches for conduits and duct banks may be excavated manually or with mechanical trenching equipment unless in pavement, in which case they shall be excavated with mechanical trenching equipment. Walls of trenches shall be essentially vertical so that a minimum of shoulder surface is disturbed. Blades of graders shall not be used to excavate the trench.

When rock is encountered, the rock shall be removed to a depth of at least 3 inches (75 mm) below the required conduit or duct bank depth and it shall be replaced with bedding material of earth or sand containing no mineral aggregate particles that would be retained on a 1/4-inch (6.3 mm) sieve. Flowable backfill may alternatively be used

Underground electrical warning (Caution) tape shall be installed in the trench above all underground duct banks and conduits in unpaved areas. Contractor shall submit a sample of the proposed warning tape for approval by the RPR. If not shown on the plans, the warning tape shall be located 6 inches above the duct/conduit or the counterpoise wire if present.

Joints in plastic conduit shall be prepared per the manufacturer's recommendations for the particular type of conduit. Plastic conduit shall be prepared by application of a plastic cleaner and brushing a plastic solvent on the outside of the conduit ends and on the inside of the couplings. The conduit fitting shall then be slipped together with a quick one-quarter turn twist to set the joint tightly. Where more than one conduit is placed in a single trench, or in duct banks, joints in the conduit shall be staggered a minimum of 2 feet (60 cm).

Changes in direction of runs exceeding 10 degrees, either vertical or horizontal, shall be accomplished using manufactured sweep bends.

Whether or not specifically indicated on the drawings, where the soil encountered at established duct bank grade is an unsuitable material, as determined by the RPR, the unsuitable material shall be removed per Item P-152 and replaced with suitable material. Additional duct bank supports shall be installed, as approved by the RPR.

All excavation shall be unclassified and shall be considered incidental to Item L-110. Dewatering necessary for duct installation, and erosion per federal, state, and local requirements is incidental to Item L-110.

Unless otherwise specified, excavated materials that are deemed by the RPR to be unsuitable for use in backfill or embankments shall be removed and disposed of offsite.

Any excess excavation shall be filled with suitable material approved by the RPR and compacted per Item P-152.

It is the Contractor's responsibility to locate existing utilities within the work area prior to excavation. Where existing active cables cross proposed installations, the Contractor shall ensure that these cables are adequately protected. Where crossings are unavoidable, no splices will be allowed in the existing cables, except as specified on the plans. Installation of new cable where such crossings must occur shall proceed as follows:

a. Existing cables shall be located manually. Unearthed cables shall be inspected to assure absolutely no damage has occurred

b. Trenching, etc., in cable areas shall then proceed with approval of the RPR, with care taken to minimize possible damage or disruption of existing cable, including careful backfilling in area of cable.

In the event that any previously identified cable is damaged during the course of construction, the Contractor shall be responsible for the complete repair.

**110-3.2 DUCT BANKS.** Unless otherwise shown in the plans, duct banks shall be installed so that the top of the concrete envelope is not less than 18 inches (0.5 m) below the bottom of the base or stabilized base course layers where installed under runways, taxiways, aprons, or other paved areas, and not less than 18 inches (0.5 m) below finished grade where installed in unpaved areas.

Unless otherwise shown on the plans, duct banks under paved areas shall extend at least 3 feet (1 m) beyond the edges of the pavement or 3 feet (1 m) beyond any under drains that may be installed alongside the paved area. Trenches for duct banks shall be opened the complete length before concrete is placed so that if any obstructions are encountered, provisions can be made to avoid them. Unless otherwise shown on the plans, all duct banks shall be placed on a layer of concrete not less than 3 inches (75 mm) thick prior to its initial set. The Contractor shall space the conduits not less than 3 inches (75 mm) apart (measured from outside wall to outside wall). All such multiple conduits shall be placed using conduit spacers applicable to the type of conduit. As the conduit laying progresses, concrete shall be placed around and on top of the conduits not less than 3 inches (75 mm) thick unless otherwise shown on the plans. All conduits shall terminate with female ends for ease of access in current and future use. Install factory plugs in all unused ends. Do not cover the ends or plugs with concrete.

Conduits forming the duct bank shall be installed using conduit spacers. No. 4 reinforcing bars shall be driven vertically into the soil a minimum of 6 inches (150 mm) to anchor the assembly into the earth prior to placing the concrete encasement. For this purpose, the spacers shall be fastened down with locking collars attached to the vertical bars. Spacers shall be installed at 5-foot (1.5-m) intervals. Spacers shall be in the proper sizes and configurations to fit the conduits. Locking collars and spacers shall be submitted to the RPR for review prior to use.

When specified, the Contractor shall reinforce the bottom side and top of encasements with steel reinforcing mesh or fabric or other approved metal reinforcement. When directed, the Contractor

shall supply additional supports where the ground is soft and boggy, where ducts cross under roadways, or where shown on the plans. Under such conditions, the complete duct structure shall be supported on reinforced concrete footings, piers, or piles located at approximately 5-foot (1.5-m) intervals.

All pavement surfaces that are to have ducts installed therein shall be neatly saw cut to form a vertical face. All excavation shall be included in the contract with price for the duct.

Install a plastic, detectable, color as noted, 3 to 6 inches (75 to 150 mm) wide tape, 8 inches (200 mm) minimum below grade above all underground conduit or duct lines not installed under pavement. Utilize the 3-inch (75-mm) wide tape only for single conduit runs. Utilize the 6-inch (150-mm) wide tape for multiple conduits and duct banks. For duct banks equal to or greater than 24 inches (600 mm) in width, utilize more than one tape for sufficient coverage and identification of the duct bank as required.

When existing cables are to be placed in split duct, encased in concrete, the cable shall be carefully located and exposed by hand tools. Prior to being placed in duct, the RPR shall be notified so that he may inspect the cable and determine that it is in good condition. Where required, split duct shall be installed as shown on the drawings or as required by the RPR.

**110-3.3 CONDUITS WITHOUT CONCRETE ENCASEMENT.** Trenches for single-conduit lines shall be not less than 6 inches (150 mm) nor more than 12 inches (300 mm) wide. The trench for 2 or more conduits installed at the same level shall be proportionately wider. Trench bottoms for conduits without concrete encasement shall be made to conform accurately to grade so as to provide uniform support for the conduit along its entire length.

Unless otherwise shown on the plans, a layer of fine earth material, at least 4 inches (100 mm) thick (loose measurement) shall be placed in the bottom of the trench as bedding for the conduit. The bedding material shall consist of soft dirt, sand or other fine fill, and it shall contain no particles that would be retained on a 1/4-inch (6.3 mm) sieve. The bedding material shall be tamped until firm. Flowable backfill may alternatively be used.

Unless otherwise shown on plans, conduits shall be installed so that the tops of all conduits within the Airport's secured area where trespassing is prohibited are at least 18 inches (0.5 m) below the finished grade. Conduits outside the Airport's secured area shall be installed so that the tops of the conduits are at least 24 inches (60 cm) below the finished grade per National Electric Code (NEC), Table 300.5.

When two or more individual conduits intended to carry conductors of equivalent voltage insulation rating are installed in the same trench without concrete encasement, they shall be spaced not less than 3 inches (75 mm) apart (measured from outside wall to outside wall) in a horizontal direction and not less than 6 inches (150 mm) apart in a vertical direction. Where two or more individual conduits intended to carry conductors of differing voltage insulation rating are installed in the same trench without concrete encasement, they shall be placed not less than 3 inches (75 mm) apart (measured from outside wall to outside wall) in a horizontal direction and not less than 6 inches (150 mm) apart in a vertical direction.

Trenches shall be opened the complete length between normal termination points before conduit is installed so that if any unforeseen obstructions are encountered, proper provisions can be made to avoid them.

Conduits shall be installed using conduit spacers. No. 4 reinforcing bars shall be driven vertically into the soil a minimum of 6 inches (150 mm) to anchor the assembly into the earth while backfilling. For this purpose, the spacers shall be fastened down with locking collars attached to the vertical bars. Spacers shall be installed at 5-foot (1.5-m) intervals. Spacers shall be in the proper sizes and configurations to fit the conduits. Locking collars and spacers shall be submitted to the RPR for review prior to use.

**110-3.4 MARKERS.** The location of each end and of each change of direction of conduits and duct banks shall be marked by a concrete slab marker 2 feet (60 cm) square and 4 - 6 inches (100 - 150 mm) thick extending approximately one inch (25 mm) above the surface. The markers shall also be located directly above the ends of all conduits or duct banks, except where they terminate in a junction/access structure or building. Each cable or duct run from a line of lights and signs to the equipment vault must be marked at approximately every 200 feet (61 m) along the cable or duct run, with an additional marker at each change of direction of cable or duct run.

The Contractor shall impress the word "DUCT" or "CONDUIT" on each marker slab. Impression of letters shall be done in a manner, approved by the RPR, for a neat, professional appearance. All letters and words must be neatly stenciled. After placement, all markers shall be given one coat of high-visibility orange paint, as approved by the RPR. The Contractor shall also impress on the slab the number and size of conduits beneath the marker along with all other necessary information as determined by the RPR. The letters shall be 4 inches (100 mm) high and 3 inches (75 mm) wide with width of stroke 1/2 inch (12 mm) and 1/4 inch (6 mm) deep or as large as the available space permits. Furnishing and installation of duct markers is incidental to the respective duct pay item.

**110-3.5 BACKFILLING FOR CONDUITS.** For conduits, 8 inches (200 mm) of sand, soft earth, or other fine fill (loose measurement) shall be placed around the conduits ducts and carefully tamped around and over them with hand tampers. The remaining trench shall then be backfilled and compacted per Item P-152 except that material used for back fill shall be select material not larger than 4 inches (100 mm) in diameter.

Flowable backfill may alternatively be used.

Trenches shall not contain pools of water during back filling operations.

The trench shall be completely backfilled and tamped level with the adjacent surface; except that, where sod is to be placed over the trench, the backfilling shall be stopped at a depth equal to the thickness of the sod to be used, with proper allowance for settlement.

Any excess excavated material shall be removed and disposed of per instructions issued by the RPR.

**110-3.6 BACKFILLING FOR DUCT BANKS.** After the concrete has cured, the remaining trench shall be backfilled and compacted per Item P-152 "Excavation and Embankment" except that the material used for backfill shall be select material not larger than 4 inches (100 mm) in diameter. In addition to the requirements of Item P-152, where duct banks are installed under pavement, one moisture/density test per lift shall be made for each 250 linear feet (76 m) of duct bank or one work period's construction, whichever is less.

Flowable backfill may alternatively be used.

Trenches shall not contain pools of water during backfilling operations.

The trench shall be completely backfilled and tamped level with the adjacent surface; except that, where sod is to be placed over the trench, the backfilling shall be stopped at a depth equal to the thickness of the sod to be used, with proper allowance for settlement.

Any excess excavated material shall be removed and disposed of per instructions issued by the RPR.

**110-3.7 RESTORATION.** Where sod has been removed, it shall be replaced as soon as possible after the backfilling is completed. All areas disturbed by the work shall be restored to its original condition. The restoration shall include sodding shown on the plans. The Contractor shall be held responsible for maintaining all disturbed surfaces and replacements until final acceptance. All restoration shall be considered incidental to the respective L-110 pay item. Following restoration of all trenching near airport movement surfaces, the Contractor shall thoroughly visually inspect the area for foreign object debris (FOD), and remove any such FOD that is found. This FOD inspection and removal shall be considered incidental to the pay item of which it is a component part.

**110-3.8 OWNERSHIP OF REMOVED CABLE.** All removed cable shall become the property of the contractor and shall be legally disposed of off airport property

#### **METHOD OF MEASUREMENT**

**110-4.1** Underground conduits and duct banks shall be measured by the linear feet (meter) of conduits and duct banks installed, including encasement, locator tape, trenching and backfill with designated material, and restoration, and for drain lines, the termination at the drainage structure, all measured in place, completed, and accepted. Separate measurement shall be made for the various types and sizes.

#### **BASIS OF PAYMENT**

**110-5.1** Payment will be made at the contract unit price per linear foot for each type and size of conduit and duct bank completed and accepted, including trench and backfill with the designated material, and, for drain lines, the termination at the drainage structure. This price shall be full compensation for removal and disposal of existing duct banks and conduits as shown on the plans, furnishing all materials and for all preparation, assembly, and installation of these materials, and for all labor, equipment, tools, and incidentals necessary to complete this item per the provisions and intent of the plans and specifications.

Payment will be made under:

~~Item L-110-5.1 [ Concrete Encased ] [ Non-Encased ] Electrical Duct Bank, [ # and Size ] - per linear foot (meter)~~

~~Item L-110-5.2 [ Concrete Encased ] [ Non-Encased ] Electrical Conduit, [ # and Size ] - per linear foot (meter)~~

|               |                                                    |                        |
|---------------|----------------------------------------------------|------------------------|
| Item L-110-1  | 1 Way 2" Schedule 40 PVC Direct Earth Buried Duct  | - per Linear Foot (LF) |
| Item L-110-2  | 2 Way 2" Schedule 40 PVC Direct Earth Buried Duct  | - per Linear Foot (LF) |
| Item L-110-3  | 4 Way 2" Schedule 40 PVC Direct Earth Buried Duct  | - per Linear Foot (LF) |
| Item L-110-4  | 6 Way 2" Schedule 40 PVC Direct Earth Buried Duct  | - per Linear Foot (LF) |
| Item L-110-5  | 12 Way 2" Schedule 40 PVC Direct Earth Buried Duct | - per Linear Foot (LF) |
| Item L-110-6  | 14 Way 2" Schedule 40 PVC Direct Earth Buried Duct | - per Linear Foot (LF) |
| Item L-110-7  | 1 Way 2" Schedule 80 PVC Concrete Encased Duct     | - per Linear Foot (LF) |
| Item L-110-8  | 2 Way 2" Schedule 80 PVC Concrete Encased Duct     | - per Linear Foot (LF) |
| Item L-110-9  | 3 Way 2" Schedule 80 PVC Concrete Encased Duct     | - per Linear Foot (LF) |
| Item L-110-10 | 4 Way 2" Schedule 80 PVC Concrete Encased Duct     | - per Linear Foot (LF) |
| Item L-110-11 | 6 Way 2" Schedule 80 PVC Concrete Encased Duct     | - per Linear Foot (LF) |
| Item L-110-12 | 8 Way 2" Schedule 80 PVC Concrete Encased Duct     | - per Linear Foot (LF) |
| Item L-110-13 | Base Can Drain Well                                | - per Each (EA)        |

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

Advisory Circular (AC)

AC 150/5340-30      Design and Installation Details for Airport Visual Aids

AC 150/5345-53      Airport Lighting Equipment Certification Program

ASTM International (ASTM)

ASTM A615      Standard Specification for Deformed and Plain Carbon-Steel Bars  
for Concrete Reinforcement

National Fire Protection Association (NFPA)

NFPA-70      National Electrical Code (NEC)

Underwriters Laboratories (UL)

UL Standard 6      Electrical Rigid Metal Conduit - Steel

UL Standard 514B      Conduit, Tubing, and Cable Fittings

UL Standard 514C      Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers

UL Standard 1242      Electrical Intermediate Metal Conduit Steel

UL Standard 651      Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings

UL Standard 651A      Type EB and A Rigid PVC Conduit and HDPE Conduit

**END OF ITEM L-110**



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## ITEM L-115 ELECTRICAL MANHOLES AND JUNCTION STRUCTURES

### DESCRIPTION

**115-1.1** This item shall consist of electrical manholes and junction structures (hand holes, pull boxes, junction cans, etc.) installed per this specification, at the indicated locations and conforming to the lines, grades and dimensions shown on the plans or as required by the RPR. This item shall include the installation of each electrical manhole and/or junction structures with all associated excavation, backfilling, sheeting and bracing, concrete, reinforcing steel, ladders, appurtenances, testing, dewatering and restoration of surfaces to the satisfaction of the RPR.

### EQUIPMENT AND MATERIALS

#### 115-2.1 GENERAL.

**a.** All equipment and materials covered by referenced specifications shall be subject to acceptance through manufacturer's certification of compliance with the applicable specification when so requested by the RPR.

**b.** Manufacturer's certifications shall not relieve the Contractor of the responsibility to provide materials per these specifications. Materials supplied and/or installed that do not comply with these specifications shall be removed (when directed by the RPR) and replaced with materials that comply with these specifications at the Contractor's cost.

**c.** All materials and equipment used to construct this item shall be submitted to the RPR for approval prior to ordering the equipment. Submittals consisting of marked catalog sheets or shop drawings shall be provided. Submittal data shall be presented in a clear, precise and thorough manner. Original catalog sheets are preferred. Photocopies are acceptable provided they are as good a quality as the original. Clearly and boldly mark each copy to identify products or models applicable to this project. Indicate all optional equipment and delete any non-pertinent data. Submittals for components of electrical equipment and systems shall identify the equipment to which they apply on each submittal sheet. Markings shall be made bold and clear with arrows or circles (highlighting is not acceptable). The Contractor is solely responsible for delays in the project that may accrue directly or indirectly from late submissions or resubmissions of submittals.

**d.** The data submitted shall be sufficient, in the opinion of the RPR, to determine compliance with the plans and specifications. The Contractor's submittals shall be electronically submitted in pdf format, tabbed by specification section. The RPR reserves the right to reject any and all equipment, materials or procedures that do not meet the system design and the standards and codes, specified in this document.

**e.** All equipment and materials furnished and installed under this section shall be guaranteed against defects in materials and workmanship for a period of at least twelve (12) months from the date of final acceptance by the Owner. The defective materials and/or equipment shall be repaired or replaced, at the Owner's discretion, with no additional cost to the Owner.

***All submittals shall comply with Item SP-106 "Submittals, Record Documents and Maintenance Manuals" and other requirements of the Contract Documents. Where other parts of the Contract Documents apply, the stricter item shall govern.***

**115-2.2 CONCRETE STRUCTURES.** Concrete shall be proportioned, placed, and cured per Item P-610, Concrete for Miscellaneous Structures. Cast-in-place concrete structures shall be as shown on the plans.

**115-2.3 PRECAST CONCRETE STRUCTURES.** Precast concrete structures shall be furnished by a plant meeting National Precast Concrete Association Plant Certification Program or another engineer approved third party certification program. Provide precast concrete structures where shown on the plans.

Precast concrete structures shall be an approved standard design of the manufacturer. Precast units shall have mortar or bitumastic sealer placed between all joints to make them watertight. The structure shall be designed to withstand 100,000 lb. aircraft loads, unless otherwise shown on the plans. Openings or knockouts shall be provided in the structure as detailed on the plans.

Threaded inserts and pulling eyes shall be cast in as shown on the plans.

If the Contractor chooses to propose a different structural design, signed and sealed shop drawings, design calculations, and other information requested by the RPR shall be submitted by the Contractor to allow for a full evaluation by the RPR. The RPR shall review per the process defined in the General Provisions.

**115-2.4 JUNCTION BOXES.** Junction boxes shall be L-867 Class 1 (non-load bearing) or L-868 Class 1 (load bearing) airport light bases that are encased in concrete. The light bases shall have a L-894 blank cover, gasket, and stainless steel hardware. All bolts, studs, nuts, lock washers, and other similar fasteners used for the light fixture assemblies must be fabricated from 316L (equivalent to EN 1.4404), 18-8, 410, or 416 stainless steel. If 18-8, 410, or 416 stainless steel is utilized it shall be passivated and be free from any discoloration. Covers shall be 3/8-inch (9-mm) thickness for L-867 and 3/4-inch (19-mm) thickness for L-868. All junction boxes shall be provided with both internal and external ground lugs.

**115-2.5 MORTAR.** The mortar shall be composed of one part of cement and two parts of mortar sand, by volume. The cement shall be per the requirements in ASTM C150, Type I. The sand shall be per the requirements in ASTM C144. Hydrated lime may be added to the mixture of sand and cement in an amount not to exceed 15% of the weight of cement used. The hydrated lime shall meet the requirements of ASTM C206. Water shall be potable, reasonably clean and free of oil, salt, acid, alkali, sugar, vegetable, or other substances injurious to the finished product.

**115-2.6 CONCRETE.** Concrete shall be proportioned, placed and cured per Item P-610, Concrete for Miscellaneous Structures.

**115-2.7 FRAMES AND COVERS.** The frames shall conform to one of the following requirements:

- a. ASTM A48      Gray iron castings
- b. ASTM A47      Malleable iron castings
- c. ASTM A27      Steel castings
- d. ASTM A283, Grade D   Structural steel for grates and frames

e. ASTM A536 Ductile iron castings

f. ASTM A897 Austempered ductile iron castings

All castings specified shall withstand a maximum tire pressure of **250** psi and maximum load of **100,000** lbs.

All castings or structural steel units shall conform to the dimensions shown on the plans and shall be designed to support the loadings specified.

Each frame and cover unit shall be provided with fastening members to prevent it from being dislodged by traffic, but which will allow easy removal for access to the structure.

All castings shall be thoroughly cleaned. After fabrication, structural steel units shall be galvanized to meet the requirements of ASTM A123.

Each cover shall have the word "ELECTRIC" or other approved designation cast on it. Each frame and cover shall be as shown on the plans or approved equivalent. No cable notches are required.

Each manhole shall be provided with a "DANGER -- PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" safety warning sign as detailed in the Contract Documents and in accordance with OSHA 1910.146 (c)(2).

**115-2.8 LADDERS.** Not used.

**115-2.9 REINFORCING STEEL.** All reinforcing steel shall be deformed bars of new billet steel meeting the requirements of ASTM A615, Grade 60.

**115-2.10 BEDDING/SPECIAL BACKFILL.** Bedding or special backfill shall be as shown on the plans.

**115-2.11 FLOWABLE BACKFILL.** Flowable material used to backfill shall conform to the requirements of Item P-153, Controlled Low Strength Material.

**115-2.12 CABLE TRAYS.** Not Used.

**115-2.13 PLASTIC CONDUIT.** Plastic conduit shall comply with Item L-110, Airport Underground Electrical Duct Banks and Conduits.

**115-2.14 CONDUIT TERMINATORS.** Conduit terminators shall be pre-manufactured for the specific purpose and sized as required or as shown on the plans.

**115-2.15 PULLING-IN IRONS.** Not used.

**115-2.16 GROUND RODS.** Ground rods shall be one piece, copper clad steel. The ground rods shall be of the length and diameter specified on the plans, but in no case shall they be less than 10 feet long nor less than 3/4 inch in diameter.

## CONSTRUCTION METHODS

**115-3.1 UNCLASSIFIED EXCAVATION.** It is the Contractor's responsibility to locate existing utilities within the work area prior to excavation. Damage to utility lines, through lack of care in excavating, shall be repaired or replaced to the satisfaction of the RPR without additional expense to the Owner.

The Contractor shall perform excavation for structures and structure footings to the lines and grades or elevations shown on the plans or as staked by the RPR. The excavation shall be of sufficient size to permit the placing of the full width and length of the structure or structure footings shown.

All excavation shall be unclassified and shall be considered incidental to Item L-115. Dewatering necessary for structure installation and erosion per federal, state, and local requirements is incidental to Item L-115.

Boulders, logs and all other objectionable material encountered in excavation shall be removed. All rock and other hard foundation material shall be cleaned of all loose material and cut to a firm surface either level, stepped or serrated, as directed by the RPR. All seams, crevices, disintegrated rock and thin strata shall be removed. When concrete is to rest on a surface other than rock, special care shall be taken not to disturb the bottom of the excavation. Excavation to final grade shall not be made until just before the concrete or reinforcing is to be placed.

The Contractor shall provide all bracing, sheeting and shoring necessary to implement and protect the excavation and the structure as required for safety or conformance to governing laws. The cost of bracing, sheeting and shoring shall be included in the unit price bid for the structure.

Unless otherwise provided, bracing, sheeting and shoring involved in the construction of this item shall be removed by the Contractor after the completion of the structure. Removal shall be effected in a manner that will not disturb or mar finished masonry. The cost of removal shall be included in the unit price bid for the structure.

After each excavation is completed, the Contractor shall notify the RPR. Structures shall be placed after the RPR has approved the depth of the excavation and the suitability of the foundation material.

Prior to installation the Contractor shall provide a minimum of 6 inches (150 mm) of sand or a material approved by the RPR as a suitable base to receive the structure. The base material shall be compacted and graded level and at proper elevation to receive the structure in proper relation to the conduit grade or ground cover requirements, as indicated on the plans.

**115-3.2 CONCRETE STRUCTURES.** Concrete structures shall be built on prepared foundations conforming to the dimensions and form indicated on the plans. The concrete and construction methods shall conform to the requirements specified in Item P-610. Any reinforcement required shall be placed as indicated on the plans and shall be approved by the RPR before the concrete is placed.

**115-3.3 PRECAST UNIT INSTALLATIONS.** Precast units shall be installed plumb and true. Joints shall be made watertight by use of sealant at each tongue-and-groove joint and at roof of

manhole. Excess sealant shall be removed and severe surface projections on exterior of neck shall be removed.

**115-3.4 PLACEMENT AND TREATMENT OF CASTINGS, FRAMES AND FITTINGS.** All castings, frames and fittings shall be placed in the positions indicated on the Plans or as directed by the RPR and shall be set true to line and to correct elevation. If frames or fittings are to be set in concrete or cement mortar, all anchors or bolts shall be in place and position before the concrete or mortar is placed. The unit shall not be disturbed until the mortar or concrete has set.

Field connections shall be made with bolts, unless indicated otherwise. Welding will not be permitted unless shown otherwise on the approved shop drawings and written approval is granted by the casting manufacturer. Erection equipment shall be suitable and safe for the workman. Errors in shop fabrication or deformation resulting from handling and transportation that prevent the proper assembly and fitting of parts shall be reported immediately to the RPR and approval of the method of correction shall be obtained. Approved corrections shall be made at Contractor's expense.

Anchor bolts and anchors shall be properly located and built into connection work. Bolts and anchors shall be preset by the use of templates or such other methods as may be required to locate the anchors and anchor bolts accurately.

Pulling-in irons shall be located opposite all conduit entrances into structures to provide a strong, convenient attachment for pulling-in blocks when installing cables. Pulling-in irons shall be set directly into the concrete walls of the structure.

**115-3.5 INSTALLATION OF LADDERS.** Not used.

**115-3.6 REMOVAL OF SHEETING AND BRACING.** In general, all sheeting and bracing used to support the sides of trenches or other open excavations shall be withdrawn as the trenches or other open excavations are being refilled. That portion of the sheeting extending below the top of a structure shall be withdrawn, unless otherwise directed, before more than 6 inches (150 mm) of material is placed above the top of the structure and before any bracing is removed. Voids left by the sheeting shall be carefully refilled with selected material and rammed tight with tools especially adapted for the purpose or otherwise as may be approved.

The RPR may direct the Contractor to delay the removal of sheeting and bracing if, in his judgment, the installed work has not attained the necessary strength to permit placing of backfill.

**115-3.7 BACKFILLING.** After a structure has been completed, the area around it shall be backfilled in horizontal layers not to exceed 6 inches (150 mm) in thickness measured after compaction to the density requirements in Item P-152. Each layer shall be deposited all around the structure to approximately the same elevation. The top of the fill shall meet the elevation shown on the plans or as directed by the RPR.

Backfill shall not be placed against any structure until approval is given by the RPR. In the case of concrete, such approval shall not be given until tests made by the laboratory under supervision of the RPR establish that the concrete has attained sufficient strength to provide a factor of safety against damage or strain in withstanding any pressure created by the backfill or the methods used in placing it.

Where required, the RPR may direct the Contractor to add, at his own expense, sufficient water during compaction to assure a complete consolidation of the backfill. The Contractor shall be responsible for all damage or injury done to conduits, duct banks, structures, property or persons due to improper placing or compacting of backfill.

**115-3.8 CONNECTION OF DUCT BANKS.** To relieve stress of joint between concrete-encased duct banks and structure walls, reinforcement rods shall be placed in the structure wall and shall be formed and tied into duct bank reinforcement at the time the duct bank is installed.

**115-3.9 GROUNDING.** A ground rod shall be installed in the floor of all concrete structures so that the top of rod extends 6 inches (150 mm) above the floor. The ground rod shall be installed within one foot (30 cm) of a corner of the concrete structure. Ground rods shall be installed prior to casting the bottom slab. Where the soil condition does not permit driving the ground rod into the earth without damage to the ground rod, the Contractor shall drill a 4-inch (100 mm) diameter hole into the earth to receive the ground rod. The hole around the ground rod shall be filled throughout its length, below slab, with Portland cement grout. Ground rods shall be installed in precast bottom slab of structures by drilling a hole through bottom slab and installing the ground rod. Bottom slab penetration shall be sealed watertight with Portland cement grout around the ground rod.

A grounding bus of 4/0 bare stranded copper shall be exothermically bonded to the ground rod and loop the concrete structure walls. The ground bus shall be a minimum of one foot (30 cm) above the floor of the structure and separate from other cables. No. 2 American wire gauge (AWG) bare copper pigtailed shall bond the grounding bus to all cable trays and other metal hardware within the concrete structure. Connections to the grounding bus shall be exothermic. If an exothermic weld is not possible, connections to the grounding bus shall be made by using connectors approved for direct burial in soil or concrete per UL 467. Hardware connections may be mechanical, using a lug designed for that purpose.

**115-3.10 CLEANUP AND REPAIR.** After erection of all galvanized items, damaged areas shall be repaired by applying a liquid cold-galvanizing compound per MIL-P-21035. Surfaces shall be prepared and compound applied per the manufacturer's recommendations.

Prior to acceptance, the entire structure shall be cleaned of all dirt and debris.

**115-3.11 RESTORATION.** After the backfill is completed, the Contractor shall dispose of all surplus material, dirt and rubbish from the site. The Contractor shall restore all disturbed areas equivalent to or better than their original condition. All sodding, grading and restoration shall be considered incidental to the respective Item L-115 pay item.

The Contractor shall grade around structures as required to provide positive drainage away from the structure.

Areas with special surface treatment, such as roads, sidewalks, or other paved areas shall have backfill compacted to match surrounding areas, and surfaces shall be repaired using materials comparable to original materials.

Following restoration of all trenching near airport movement surfaces, the Contractor shall thoroughly visually inspect the area for foreign object debris (FOD), and remove any such FOD

that is found. This FOD inspection and removal shall be considered incidental to the pay item of which it is a component part.

After all work is completed, the Contractor shall remove all tools and other equipment, leaving the entire site free, clear and in good condition.

**115-3.12 INSPECTION.** Prior to final approval, the electrical structures shall be thoroughly inspected for conformance with the plans and this specification. Any indication of defects in materials or workmanship shall be further investigated and corrected. The earth resistance to ground of each ground rod shall not exceed 25 ohms. Each ground rod shall be tested using the fall-of-potential ground impedance test per American National Standards Institute / Institute of Electrical and Electronic Engineers (ANSI/IEEE) Standard 81. This test shall be performed prior to establishing connections to other ground electrodes.

**115-3.13 MANHOLE ELEVATION ADJUSTMENTS.** ~~The Contractor shall adjust the tops of existing manholes in areas designated in the Contract Documents to the new elevations shown. The Contractor shall be responsible for determining the exact height adjustment required to raise or lower the top of each manhole to the new elevations. The existing top elevation of each manhole to be adjusted shall be determined in the field and subtracted/added from the proposed top elevation.~~

~~The Contractor shall remove/extend the existing top section or ring and cover on the manhole structure or manhole access. The Contractor shall install precast concrete sections or grade rings of the required dimensions to adjust the manhole top to the new proposed elevation or shall cut the existing manhole walls to shorten the existing structure, as required by final grades. The Contractor shall reinstall the manhole top section or ring and cover on top and check the new top elevation.~~

~~The Contractor shall construct a concrete slab around the top of adjusted structures located in graded areas that are not to be paved. The concrete slab shall conform to the dimensions shown on the plans.~~

**115-3.14 DUCT EXTENSION TO EXISTING DUCTS.** Where existing concrete encased ducts are to be extended, the duct extension shall be concrete encased plastic conduit. The fittings to connect the ducts together shall be standard manufactured connectors designed and approved for the purpose. The duct extensions shall be installed according to the concrete encased duct detail and as shown on the plans.

## **METHOD OF MEASUREMENT**

**115-4.1** Electrical junction structures shall be measured by each unit completed in place and accepted. The following items shall be included in the price of each unit: All required excavation and dewatering; sheeting and bracing; all required backfilling with on-site materials; restoration of



all surfaces and finished grading and turfing; all required connections; temporary cables and connections; and ground rod testing

~~115-4.2 Manhole elevation adjustments shall be measured by the completed unit installed, in place, completed, and accepted. Separate measurement shall not be made for the various types and sizes.~~

## **BASIS OF PAYMENT**

**115-5.1** The accepted quantity of junction structures will be paid for at the Contract unit price per each, complete and in place. This price shall be full compensation for furnishing all materials and for all preparation, excavation, backfilling and placing of the materials, furnishing and installation of appurtenances and connections to duct banks and other structures as may be required to complete the item as shown on the plans and for all labor, equipment, tools and incidentals necessary to complete the structure.

Payment will be made under:

~~Item L-115-5.1 Electrical Manhole [ size and type ] Per Each~~

~~Item L-115-5.2 Electrical Junction Structure [ size and type ] Per Each~~

~~Item L-115-5.3 Existing Electrical Manhole/Junction Structure Elevation Adjustment  
[ size and type ] Per Each~~

~~Item L-115-5.4 Electrical Handhole [ Size and Type ] Per Each~~

**Item L-115-1 L-867D Junction Can (NTAC) -- per Each (EA)**

L-867D 16" Diameter 24" deep light base with 1/2" galvanized steel blank cover and gasket secured with stainless steel bolts installed in turf or pavement, with concrete encasement all incidentals. All light bases installed in asphalt pavement shall be installed using a telescopic light base. All light bases installed in PCC pavement or turf shall be single piece light bases. Align and level as required.

**Item L-115-2 Junction Can Plaza – Two L-867D Cans -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with four light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

**Item L-115-3      Junction Can Plaza – Three L-867D Cans      -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with four light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

**Item L-115-4      Junction Can Plaza – Four L-867D Cans      -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with four light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

**Item L-115-5      Junction Can Plaza – Six L-867D Cans      -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with six light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

**Item L-115-5      Junction Can Plaza – Eight L-867D Cans      -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with six light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

**Item L-115-6      Junction Can Plaza – Twelve L-867D Cans      -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with twelve light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

**Item L-115-6      Junction Can Plaza – Fourteen L-867D Cans      -- per Each (EA)**

This item includes all materials, labor, transportation, supervision, services and all incidentals required to provide and install a junction can plaza with L-867D, 24" deep light bases. Conduit, ground rods and counterpoise wire are paid under their respective pay items. This item includes all concrete, rebar, excavation, forming, backfill, sod, light bases, cover plates, hardware, identification tags, etc., and all appurtenances to construct a junction can plaza with twelve light bases in accordance with the Contract Documents and to the satisfaction of the Engineer.

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

American National Standards Institute / Insulated Cable Engineers Association (ANSI/ICEA)

ANSI/IEEE STD 81      IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

Advisory Circular (AC)

AC 150/5345-7      Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits

AC 150/5345-26      Specification for L-823 Plug and Receptacle, Cable Connectors

AC 150/5345-42      Specification for Airport Light Bases, Transformer Housings, Junction Boxes, and Accessories

AC 150/5340-30      Design and Installation Details for Airport Visual Aids

AC 150/5345-53      Airport Lighting Equipment Certification Program

Commercial Item Description (CID)

A-A 59544 Cable and Wire, Electrical (Power, Fixed Installation)

ASTM International (ASTM)

ASTM A27 Standard Specification for Steel Castings, Carbon, for General Application

ASTM A47 Standard Specification for Ferritic Malleable Iron Castings

ASTM A48 Standard Specification for Gray Iron Castings

ASTM A123 Standard Specification for Zinc (Hot Dip Galvanized) Coatings on Iron and Steel Products

ASTM A283 Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates

ASTM A536 Standard Specification for Ductile Iron Castings

ASTM A615 Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

ASTM A897 Standard Specification for Austempered Ductile Iron Castings

ASTM C144 Standard Specification for Aggregate for Masonry Mortar

ASTM C150 Standard Specification for Portland Cement

ASTM C206 Standard Specification for Finishing Hydrated Lime

FAA Engineering Brief (EB)

EB #83 In Pavement Light Fixture Bolts

Mil Spec

MIL-P-21035 Paint High Zinc Dust Content, Galvanizing Repair

National Fire Protection Association (NFPA)

NFPA-70 National Electrical Code (NEC)

**END OF ITEM L-115**

## **ITEM L-125 INSTALLATION OF AIRPORT LIGHTING SYSTEMS**

### **DESCRIPTION**

**125-1.1** This item shall consist of airport lighting systems furnished and installed in accordance with this specification, the referenced specifications, and the applicable advisory circulars (ACs). The systems shall be installed at the locations and in accordance with the dimensions, design, and details shown in the plans. This item shall include the furnishing of all equipment, materials, services, and incidentals necessary to place the systems in operation as completed units to the satisfaction of the RPR.

### **EQUIPMENT AND MATERIALS**

#### **125-2.1 GENERAL.**

a. Airport lighting equipment and materials covered by Federal Aviation Administration (FAA) specifications shall be certified under the Airport Lighting Equipment Certification Program in accordance with AC 150/5345-53, current version. FAA certified airfield lighting shall be compatible with each other to perform in compliance with FAA criteria and the intended operation. If the Contractor provides equipment that does not perform as intended because of incompatibility with the system, the Contractor assumes all costs to correct the system for to operate properly.

b. Manufacturer's certifications shall not relieve the Contractor of their responsibility to provide materials in accordance with these specifications and acceptable to the RPR. Materials supplied and/or installed that do not comply with these specifications shall be removed, when directed by the RPR and replaced with materials, which do comply with these specifications, at the sole cost of the Contractor.

c. All materials and equipment used shall be submitted to the RPR for approval prior to ordering the equipment. Submittals consisting of marked catalog sheets or shop drawings shall be provided. Clearly mark each copy to identify pertinent products or models applicable to this project. Indicate all optional equipment and delete non-pertinent data. Submittals for components of electrical equipment and systems shall identify the equipment for which they apply on each submittal sheet. Markings shall be clearly made with arrows or circles (highlighting is not acceptable). The Contractor shall be responsible for delays in the project accruing directly or indirectly from late submissions or resubmissions of submittals.

d. The data submitted shall be sufficient, in the opinion of the RPR, to determine compliance with the plans and specifications. The Contractor's submittals shall be submitted in electronic PDF format, tabbed by specification section. The RPR reserves the right to reject any or all equipment, materials or procedures, which, in the RPR's opinion, does not meet the system design and the standards and codes, specified herein.

e. All equipment and materials furnished and installed under this section shall be guaranteed against defects in materials and workmanship for a period of at least twelve (12) months from final acceptance by the Owner. The defective materials and/or equipment shall be repaired or replaced, at the Owner's discretion, with no additional cost to the Owner.

**f. All LED light fixtures, with the exception of obstruction lighting (AC 150/5345-43) must be warranted by the manufacturer for a minimum of 4 years after date of installation inclusive of all electronics.” Obstruction lighting warranty is set by the individual manufacturer.**

## **EQUIPMENT AND MATERIALS**

**125-2.2 CONDUIT/DUCT.** Conduit shall conform to Specification Item L-110 Airport Underground Electrical Duct Banks and Conduits.

**125-2.3 CABLE AND COUNTERPOISE.** Cable and Counterpoise shall conform to Item L-108 Underground Power Cable for Airports.

**125-2.4 TAPE.** Rubber and plastic electrical tapes shall be Scotch Electrical Tape Numbers 23 and 88 respectively, as manufactured by 3M Company or an approved equal.

**125-2.5 CABLE CONNECTIONS.** Cable Connections shall conform to Item L-108 Installation of Underground Cable for Airports.

**125-2.6 RETROREFLECTIVE MARKERS.** - Not required.

**125-2.7 RUNWAY AND TAXIWAY LIGHTS.** Runway and taxiway lights shall conform to the requirements of AC 150/5345-46. Lamps shall be of size and type indicated, or as required by fixture manufacturer for each lighting fixture required under this contract. Filters shall be of colors conforming to the specification for the light concerned or to the standard referenced.

### **Lights**

| Type      | Class | Mode | Style | Option | Base     | Filter          | Transformer | Notes   |
|-----------|-------|------|-------|--------|----------|-----------------|-------------|---------|
| L-804(L)  |       | 1    |       |        | L-867B/D | UNI-YELLOW      | L-830       | 26" MAX |
| L-850A    | 2     | 1    | 3     |        | L-868B   | BI-CLEAR/RED    | L-830       |         |
| L-850A    | 2     | 1    | 3     |        | L-868B   | BI-CLEAR/CLEAR  | L-830       |         |
| L-850C    | 2     | 1    | 3     |        | L-868B   | BI-CLEAR/YELLOW | L-830       |         |
| L-850C(L) | 2     | 1    | 3     |        | L-868B   | BI-CLEAR/CLEAR  | L-830       |         |
| L-861T(L) |       | 1    |       |        | L-867B/D | BI-BLUE         | L-830       | 14" MAX |
| L-862(L)  |       | 1    |       |        | L-867B/D | BI-CLEAR/CLEAR  | L-830       | 14" MAX |
| L-862(L)  |       | 1    |       |        | L-867B/D | BI-CLEAR/YELLOW | L-830       | 14" MAX |
| L-862E(L) |       | 1    |       |        | L-867B/D | BI-GREEN/RED    | L-830       | 14" MAX |

**125-2.8 RUNWAY AND TAXIWAY SIGNS.** Runway and Taxiway Guidance Signs should conform to the requirements of AC 150/5345-44.

### Signs

| Type           | Size | Style | Class | Mode | Notes |
|----------------|------|-------|-------|------|-------|
| L-858Y,R, L(L) | 3    | 5     | 1     | 2    |       |
| L-858B(L)      | 4    | 5     | 1     | 2    |       |

**125-2.9 RUNWAY END IDENTIFIER LIGHT (REIL).** - Not required.

**125-2.10 PRECISION APPROACH PATH INDICATOR (PAPI).** - Not required.

**125-2.11 CIRCUIT SELECTOR CABINET.** – Not required.

**125-2.12 LIGHT BASE AND TRANSFORMER HOUSINGS.** Light Base and Transformer Housings should conform to the requirements of AC 150/5345-42. Light bases shall be Type L-867B, L-868, Class 1A , Size B shall be provided as indicated or as required to accommodate the fixture or device installed thereon. Base plates, cover plates, and adapter plates shall be provided to accommodate various sizes of fixtures.

**125-2.13 ISOLATION TRANSFORMERS.** Isolation Transformers shall be Type L-830, size as required for each installation. Transformer shall conform to AC 150/5345-47.

### INSTALLATION

**125-3.1 INSTALLATION.** The Contractor shall furnish, install, connect and test all equipment, accessories, conduit, cables, wires, buses, grounds and support items necessary to ensure a complete and operable airport lighting system as specified here and shown in the plans.

The equipment installation and mounting shall comply with the requirements of the National Electrical Code and state and local code agencies having jurisdiction.

The Contractor shall install the specified equipment in accordance with the applicable advisory circulars and the details shown on the plans.

**125-3.2 TESTING.** All lights shall be fully tested by continuous operation for not less than 24 hours as a completed system prior to acceptance. The test shall include operating the constant current regulator in each step not less than 10 times at the beginning and end of the 24-hour test. The fixtures shall illuminate properly during each portion of the test.

**125-3.3 SHIPPING AND STORAGE.** Equipment shall be shipped in suitable packing material to prevent damage during shipping. Store and maintain equipment and materials in areas protected from weather and physical damage. Any equipment and materials, in the opinion of the RPR, damaged during construction or storage shall be replaced by the Contractor at no additional cost to the owner. Painted or galvanized surfaces that are damaged shall be repaired in accordance with the manufacturer's recommendations.

**125-3.4 ELEVATED AND IN-PAVEMENT LIGHTS.** Water, debris, and other foreign substances shall be removed prior to installing fixture base and light.

A jig or holding device shall be used when installing each light fixture to ensure positioning to the proper elevation, alignment, level control, and azimuth control. Light fixtures shall be oriented with the light beams parallel to the runway or taxiway centerline and facing in the required direction.

The outermost edge of fixture shall be level with the surrounding pavement. Surplus sealant or flexible embedding material shall be removed. The holding device shall remain in place until sealant has reached its initial set.

#### METHOD OF MEASUREMENT

**125-4.1** Runway and taxiway lights will be measured by the number of each type installed as completed units in place, ready for operation, and accepted by the RPR. Guidance signs will be measured by the number of each type and size installed as completed units, in place, ready for operation, and accepted by the RPR.

#### BASIS OF PAYMENT

**125-5.1** Payment will be made at the Contract unit price for each complete runway or taxiway light, guidance sign, installed by the Contractor and accepted by the RPR. This payment will be full compensation for furnishing all materials and for all preparation, assembly, and installation of these materials, and for all labor, equipment, tools and incidentals necessary to complete this item.

Payment will be made under:

| Pay Item     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Item L-125-1 | <b>L-804(L) Elevated Runway Guard Light Yellow (R)</b> -- per Each (EA)<br><br>L-804(L), Existing Unidirectional, yellow runway guard light, <b>LED</b> , wig-wag, current-driven, column-mounted, with existing transformer shall be relocated and installed on a new L-867B deep light base, with telescopic top section for adjustment to proper elevation installed in asphalt shoulder pavement, with a heavy duty base plate, gasket, tether, and L-823 connectors and all incidentals. The Existing L-804(L) fixture and transformer shall be removed from the existing turf installation, preserved and protected for installation on a new light base in new asphalt shoulder pavement installation. "R" = Relocate existing fixture on new light base in new asphalt shoulder pavement. The overall height shall not exceed 26". |



| Pay Item     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Item L-125-2 | <p><b>L-850A Runway Centerline Light – RW 17-35 - (RE)</b></p> <p>L-850A, Relocate an existing Bi-directional, R/W 17-35 centerline light mounted on an existing L-868B deep light base with a new appropriately sized top can section, <math>\frac{3}{4}</math>" nominal spacer ring (w/concrete ring), spacers and all incidentals. Preserve and protect, to be reused in this project, the existing L-850A fixture, transformer, conductor, connector kit, and bottom can section. Existing L-868B light base is within an area of mill and overlay activities and will require an elevation adjustment using new top can section, <math>\frac{3}{4}</math>" nominal spacer ring (w/concrete ring) and spacers.</p> <p>"RE" = Relocate an existing fixture onto an existing light base that will remain in place and be extended to proposed final grade of asphalt pavement. Align and level as required</p> | -- per Each (EA) |
| Item L-125-3 | <p><b>L-850C Runway Edge Light - RW 17-35 Relocated (RE)</b></p> <p>L-850C, Relocate an existing Bi-directional, R/W 17-35 edge light mounted on an existing L-868B deep light base with a new appropriately sized top can section, <math>\frac{3}{4}</math>" nominal spacer ring (w/concrete ring), spacers and all incidentals. <b>Preserve and protect</b> to be reused in this project the existing L-850A fixture, transformer, conductor, and connector kit. Existing L-868B light base is within an area of mill and overlay activities and will require an elevation adjustment using new top can section, and spacers.</p> <p>"RE" = Relocate an existing fixture onto an existing light base that will remain in place and be extended to proposed final grade of asphalt pavement. Align and level as required</p>                                                                                    | -- per Each (EA) |
| Item L-125-4 | <p><b>L-850C(L) Runway Edge Light-Clear/Clear (A)</b></p> <p>L-850C(L), Bi-directional, 360 degree clear, LED, HIRL, R/W edge light, style 3, L-868B deep light base with a new appropriately sized top can section, <math>\frac{3}{4}</math>" nominal spacer ring (w/concrete ring), spacers, properly sized L-830 transformer, L-823 connectors, and all incidentals.</p> <p>"A" = Installed in existing asphalt shoulder pavement. Align and level as required</p>                                                                                                                                                                                                                                                                                                                                                                                                                                            | -- per Each (EA) |

| <b>Pay Item</b>     | <b>Description</b>                                                                                                                                                                                                                                                                               |                         |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Item L-125-4</b> | <b>L-850C(L) Runway Edge Light-Clear/Clear (NE)</b>                                                                                                                                                                                                                                              | <b>-- per Each (EA)</b> |
|                     | L-850C(L), Bi-directional, 360 degree clear, LED, HIRL, R/W edge light, style 3, L-868B deep light base with a new appropriately sized top can section, 3/4" nominal spacer ring (w/concrete ring), spacers, properly sized L-830 transformer, L-823 connectors, and all incidentals.            |                         |
|                     | "NE" = Install a fixture on an existing base can in areas where no paving activities occur or where seal coating occurs. The existing base cans in these areas do not require height extensions.                                                                                                 |                         |
| <b>Item L-125-5</b> | <b>L-850C(L) Runway Edge Light-Clear/Yellow (A)</b>                                                                                                                                                                                                                                              | <b>-- per Each (EA)</b> |
|                     | L-850C(L), Bidirectional, 180 degree clear, 180 degree yellow, LED, HIRL, R/W edge light, style 3, mounted on L-868B deep light base with a properly sized top can section, 3/4" nominal spacer ring (w/concrete ring), properly sized L-830 transformer, L-823 connectors, and all incidentals. |                         |
|                     | "A" = Installed in existing asphalt shoulder pavement. Align and level as required.                                                                                                                                                                                                              |                         |
| <b>Item L-125-6</b> | <b>L-858(L) Sign – Single/Double Face, LED, Size 3 - 1 Module</b>                                                                                                                                                                                                                                | <b>-- per Each (EA)</b> |
|                     | L-858(L) Y,R,L sign, size 3 with LED installed on a new concrete base with a L-867B 24" deep light base, with 1/2" thick galvanized steel cover plate and gasket, properly sized L-830 transformer(s), L-823 connectors, and all incidentals. Align and level as required.                       |                         |
| <b>Item L-125-7</b> | <b>L-858(L) Sign – Single/Double Face, LED, Size 3 - 2 Module</b>                                                                                                                                                                                                                                | <b>-- per Each (EA)</b> |
|                     | L-858(L), Y,R,L sign, with LED installed on a new concrete base with a L-867B 24" deep light base, with 1/2" thick galvanized steel cover plate and gasket, properly sized L-830 transformer(s), L-823 connectors, and all incidentals. Align and level as required.                             |                         |
| <b>Item L-125-8</b> | <b>L-858(L) Sign – Single/Double Face, LED, Size 3 - 3 Module</b>                                                                                                                                                                                                                                | <b>-- per Each (EA)</b> |
|                     | L-858(L) Y,R,L sign, with LED installed on a new concrete base with a L-867B 24" deep light base, with 1/2" thick galvanized steel cover plate and gasket, properly sized L-830 transformer(s), L-823 connectors, and all incidentals. Align and level as required.                              |                         |

| <b>Pay Item</b>      | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Item L-125-9</b>  | <b>L-858(L) Sign – Single/Double Face, LED, Size 3 - 4 Module</b><br>L-858(L) Y,R,L sign, with LED installed on a new concrete base with a L-867B 24" deep light base, with 1/2" thick galvanized steel cover plate and gasket, properly sized L-830 transformer(s), L-823 connectors, and all incidentals. Align and level as required.                                                                                                                                      | <b>-- per Each (EA)</b> |
| <b>Item L-125-10</b> | <b>L-858B(L) Runway Distance Remaining Sign, LED, Size 4</b><br>L-858B(L), R/W distance remaining sign, double sided sign, with black background/white letters and LED Light Source, installed on a new concrete base with L-867B 24" deep light base, with 1/2" thick galvanized steel cover plate and gasket, properly sized L-830 transformer(s), L-823 connectors, and all incidentals. Align and level as required.                                                      | <b>-- per Each (EA)</b> |
| <b>Item L-125-11</b> | <b>L-861T(L) Omnidirectional, Blue, LED, Taxiway Edge Light (A)</b><br>L-861T(L), Omni-directional, blue, LED, MITL T/W edge light, column mounted on a two piece telescopic L-867B/D deep light base with a "corten" or accepted equivalent base plate with gasket and properly sized L-830 transformer, L-823 connectors, and all incidentals. Align and level as required.<br><br>"A" = Installed in existing asphalt shoulder pavement. The overall height shall be 14".  | <b>-- per Each (EA)</b> |
| <b>Item L-125-12</b> | <b>L-861T(L) Omnidirectional, Blue, LED, Taxiway Edge Light (N)</b><br>L-861T(L), Omni-directional, blue, LED, MITL T/W edge light, column mounted on a two piece telescopic L-867B/D deep light base with a "corten" or accepted equivalent base plate with gasket and properly sized L-830 transformer, L-823 connectors, and all incidentals. Align and level as required.<br><br>"N" = Installed in newly constructed shoulder pavement. The overall height shall be 14". | <b>-- per Each (EA)</b> |
| <b>Item L-125-13</b> | <b>L-861T(L) Omnidirectional, Blue, LED, Taxiway Edge Light (T)</b><br>L-861T(L), Omni-directional, blue, LED, MITL T/W edge light, column mounted on a single piece L-867B/D deep light base with a "corten" or accepted equivalent base plate with gasket and properly sized L-830 transformer, L-823 connectors, and all incidentals. Align and level as required.<br><br>"T" = Installed in Turf. The overall height shall be 14".                                        | <b>-- per Each (EA)</b> |

| <b>Pay Item</b>      | <b>Description</b>                                                                                                                                                                                                                                                                                               |                         |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Item L-125-14</b> | <b>L-862(L) Runway Edge Light - Clear/Clear (A)</b>                                                                                                                                                                                                                                                              | <b>-- per Each (EA)</b> |
|                      | L-862(L), Bi-directional, 360 clear, LED, HIRL R/W edge light, column mounted on a L-867B/D deep light base with a "corten" or accepted equivalent base plate with gasket and properly sized L-830 transformer, L-823 connectors, and all incidentals. Align and level as required.                              |                         |
|                      | "A" = Installed in existing asphalt shoulder pavement. The overall height shall be 14".                                                                                                                                                                                                                          |                         |
| <b>Item L-125-15</b> | <b>L-862(L) Runway Edge Light - Clear/Yellow (A)</b>                                                                                                                                                                                                                                                             | <b>-- per Each (EA)</b> |
|                      | L-862(L), Bi-directional, 180 degree yellow, 180 degree clear, LED, HIRL R/W edge light, column mounted on a L-867B/D deep light base with a "corten" or accepted equivalent base plate with gasket and properly sized L-830 transformer, L-823 connectors, and all incidentals. Align and level as required.    |                         |
|                      | "A" = Installed in existing asphalt shoulder pavement. The overall height shall be 14".                                                                                                                                                                                                                          |                         |
| <b>Item L-125-16</b> | <b>L-862E(L) Runway Threshold Light - Green/Red (A)</b>                                                                                                                                                                                                                                                          | <b>-- per Each (EA)</b> |
|                      | L-862E(L), Bi-directional, 180 degree green, 180 degree red, LED, HIRL R/W threshold light, column mounted on a L-867B/D deep light base with a "corten" or accepted equivalent base plate with gasket and properly sized L-830 transformer, L-823 connectors, and all incidentals. Align and level as required. |                         |
|                      | "A" = Installed in existing asphalt shoulder pavement. The overall height shall be 14".                                                                                                                                                                                                                          |                         |

## REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

### Advisory Circulars (AC)

|                |                                                         |
|----------------|---------------------------------------------------------|
| AC 150/5340-18 | Standards for Airport Sign Systems                      |
| AC 150/5340-26 | Maintenance of Airport Visual Aid Facilities            |
| AC 150/5340-30 | Design and Installation Details for Airport Visual Aids |
| AC 150/5345-5  | Circuit Selector Switch                                 |

|                        |                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------|
| AC 150/5345-7          | Specification for L-824 Underground Electrical Cable for Airport Lighting Circuits            |
| AC 150/5345-26         | Specification for L-823 Plug and Receptacle, Cable Connectors                                 |
| AC 150/5345-28         | Precision Approach Path Indicator (PAPI) Systems                                              |
| AC 150/5345-39         | Specification for L-853, Runway and Taxiway Retroreflective Markers                           |
| AC 150/5345-42         | Specification for Airport Light Bases, Transformer Housings, Junction Boxes, and Accessories  |
| AC 150/5345-44         | Specification for Runway and Taxiway Signs                                                    |
| AC 150/5345-46         | Specification for Runway and Taxiway Light Fixtures                                           |
| AC 150/5345-47         | Specification for Series to Series Isolation Transformers for Airport Lighting Systems        |
| AC 150/5345-51         | Specification for Discharge-Type Flashing Light Equipment                                     |
| AC 150/5345-53         | Airport Lighting Equipment Certification Program                                              |
| Engineering Brief (EB) |                                                                                               |
| EB No. 67              | Light Sources Other than Incandescent and Xenon for Airport and Obstruction Lighting Fixtures |

**END OF ITEM L-125**

## **ITEM SP-100 GENERAL PROVISIONS AND REQUIREMENTS FOR ELECTRICAL WORK**

### **GENERAL**

#### **SPECIAL REQUIREMENTS FOR ELECTRICAL WORK.**

a. This item shall apply to all electrical work contained within the Contract Documents. Where the contract general conditions, special provisions or other parts of the Contract Documents also apply, the stricter of the documents shall govern.

#### **100-1.1 AUXILIARIES AND ACCESSORIES.**

a. Include all auxiliaries and accessories for a complete, fully functional and properly operating system in accordance with the Contract Documents.

b. Provide and install all electrical systems and any necessary appurtenances as per Federal Aviation Administration (FAA) Advisory Circulars and FAA Orders, NFPA 70 - National Electrical Code (NEC), federal, state and local codes whether specified or shown in the Contract Documents or not. The content of the Contract Documents in general only refers to work required above and beyond the requirements of the FAA, NEC and applicable federal, state and local codes.

#### **100-1.2 PROJECT PAY ITEMS.**

a. The project pay items are provided to be inclusive of all work to be performed as shown in the Contract Documents. All work not identified with a specific pay item shall be considered required work to complete the project and shall be considered subsidiary to the cost of the project pay items provided.

#### **100-1.3 CONTRACTOR EXPERIENCE.**

a. All Contractors used to perform work of this contract shall be licensed to perform electrical work in the state in which the project is located at the time bids are submitted. At least one member (qualifier) of the electrical contracting firm shall hold a state certified (unlimited, unrestricted) electrical contractor's license.

#### **100-1.4 EXPERIENCE AND SUPERVISION OF CONTRACTOR PERSONNEL.**

a. Provide a resident on-site electrical supervisor to manage work at the job site whenever electrical work is being performed or affected. Electrical work is defined as any work described, included, depicted or contained within the L-XXX specification series, SP-XXX specification series, E- series drawings, and all other work within the Contract Documents dealing with the distribution of electrical power/light, electrical/electronic/light monitoring, communication and control.

**b.** When work is being performed within the Aircraft Operations Area, the on-site supervisor shall also be within the Aircraft Operations Area, except for incidental absences. Incidental absences shall not exceed 45 minutes in length. The Contractor is encouraged to keep the same on-site supervisor(s) from commencement to completion of work.

**c.** Each on-site supervisor shall have a minimum of 5 years (2,000 hours per year) of successful supervisory experience either constructing or maintaining airfield lighting systems. At least 3 of these years of experience shall have occurred in the last 5 years on projects of comparable size and complexity. On-site supervisor shall be certified as a "Qualified Person". The definition of a Qualified Person is defined in OSHA 1910.399 and NFPA 70E (Article 100 & 110.2(1)). The certification shall be signed by the Contractor's Electrical License qualifier. Equivalent experience and training may be considered by the Engineer. On-Site supervisor may substitute one year of education in qualified electrical curriculum for one-half year electrical experience for a maximum of 2,000 hours of the required 10,000 hours of experience. Records of on-site supervisor's experience demonstrating compliance with these requirements shall be submitted to the Engineer for verification and review prior to starting work. Experience records shall include contact information for references (name, title and phone number) from airports where the proposed on-site supervisor has completed the required electrical experience. Only on-site supervisors whose experience records have been confirmed by the Engineer shall supervise work. Work shall not start prior to approval of the on-site supervisor, unless under the direct supervision of the Contractor's Electrical License qualifier. Multiple certified on-site supervisors may be required to support multiple shifts or around the clock activities.

**d.** If at any time the Resident Project Representative (RPR) believes that work is being performed without the presence of an on-site supervisor, the RPR will request that the workman in question demonstrate that the on-site supervisor is present. If the on-site supervisor's presence cannot be verified within 45 minutes the RPR may direct the Contractor to suspend work until proper supervision is made available. The Contractor shall not be remunerated for costs associated with suspension of work under these conditions.

**e.** It is agreed and understood that, if requested in writing by the RPR, the Contractor shall replace any on-site supervisor with another meeting the required qualifications within seven (7) days of receipt of request.

**f.** Electricians are tasked with installing raceways and fittings; electrical distribution and supply equipment, pulling, splicing and terminating wire/cable; grounding, lightning protection, installing light bases, light fixtures; connecting signs; performing electrical testing or other work contained in the Contract Documents or as described in 1.5(A) above. Electricians performing work shall have a minimum of 4 years (2,000 hours per year) of successful experience either constructing or maintaining airfield lighting systems and shall be certified a "Qualified Person". A copy of each Electrician's experience shall be submitted to the RPR. Each electrician's experience submittal shall comply with the above requirements. An apprentice/helper/laborer may assist the electrician with the above tasks as directed by the on-site supervisor. An electrician may supervise a maximum of four apprentices/helpers/laborers.

**g.** Electrical apprentices, helpers, laborers, shall be qualified persons for the work they are performing and shall be under the supervision of the on-site supervisor or electrician.

h. Other helpers, laborers, equipment operators, concrete workers and other crafts persons (skilled or unskilled) shall be qualified persons for the work they are performing and shall be under the supervision of the on-site supervisor and are not subject to the four person limitation in 1.5.f.

i. Any person who does not have the required experience or who demonstrates a lack of knowledge appropriate for the task they are performing shall be dismissed from the project for remainder of the contract.

## **SUMMARY OF WORK**

### **100-2.1 GENERAL.**

a. In support of this project this contract includes installation of airfield lighting, counterpoise, signage, power distribution, standby generator, lightning protection and other electrical work as shown in the Contract Documents. The above is not intended to be all inclusive but is provided as an aid to the Contractor. It is the intent of the Contract Documents that the Contractor is to furnish and install all materials and provide the necessary supervision, transportation, preparation, assembly, survey and installation of these materials, and for all labor, equipment, tools, incidentals, and appurtenances necessary to provide a fully functional and operational system as described by these Contract Documents. All items are new and unused unless specifically noted otherwise.

### **100-2.2 RECORD DOCUMENTS.**

a. The Contractor shall maintain the Contract Documents, Shop Drawings and samples at the site, in good order and annotated daily to show all changes made during the construction process per Item SP-106. These shall be available to the RPR for examination.

### **100-2.3 ENGINEERING OBSERVATIONS.**

a. Items noted by Engineer during construction and before final acceptance, which does not comply with the Contract, will be listed in accordance with the general conditions. These items will be sent to the Contractor for action. The Contractor shall have these items corrected.

b. The Contractor shall check items noted after acceptance during the warranty/guarantee period in the same manner as above.

### **100-2.4 EXISTING CONDITIONS.**

a. Investigate site thoroughly and reroute all conduit/duct and wiring in area of new construction in order to maintain continuity of existing circuitry. Existing conduit/duct shown in the Contract Documents show approximate locations only. Contractor must verify and coordinate existing site conduit/ducts, wiring and piping. The bids shall include hand digging and all required rerouting in areas of existing conduit/ducts and/or pipes.



**b.** Check site and existing conditions thoroughly before bidding. Advise Engineer of discrepancies or questions noted. If any difference is discovered between the existing conditions and the Contract Documents, the Engineer shall be notified in writing immediately.

**c.** The Electrical Contractor shall investigate all existing systems within the construction limits and complete the performance verification prior to the beginning of any work, which could affect these existing systems.

**d.** Advise Engineer in writing of all deficiencies in existing system(s) prior to work. All systems will be assumed to be fully operational if written notification is not received by Engineer prior to work on system.

**e.** The work involved is in connection with existing systems/facilities, which must remain in operation while work is being performed. Work must be done in accordance with the schedule specified in the Contract Documents. Schedule all work to minimize the period when systems are inoperable. Request written permission and receive written approval from Engineer a minimum of 2 weeks in advance of any shutdown of existing systems. (FAA requires 45 days minimum notice.) Perform work required at other than standard working hours where outages cannot be accepted during regular working hours. Protect existing buildings and equipment during construction as required.

**f.** There may be piping, fixtures or other items, which must be removed or relocated in order to perform the alteration work. All conduit/duct, wiring, boxes, etc. that do not comply with the Contract Documents shall be removed or corrected to comply with the Contract Documents. All unused conduit/duct not removed shall be identified and a pull line shall be installed. Bid shall include all removal and relocation required for completion of the alterations and the new construction.

**g.** Special attention is directed to relocation of the existing power conduits and duct bank and communication systems. Cable counts, location, and function must be field verified before starting work.

**h.** The Contractor shall verify the areas, conditions, and features necessary to tie into existing construction. This verification shall be done prior to submittal of shop drawings, fabrication or erection, construction or installation. The Contractor shall be responsible for the accurate tie-in of the new work to existing facilities. This requirement assumes that existing items comply with current codes and standards.

#### **100-2.5 SYSTEMS WARRANTY.**

**a.** The work required under this division shall include a one (1) year warranty from the date of final acceptance unless required otherwise by the Contract Documents. This warranty shall be by the Contractor/Manufacturer to the Owner for any defective workmanship or material, which has been furnished under this contract. The Contractor/Manufacturer at no cost to the Owner shall perform replacement of defective workmanship or material for a period of one (1) year (or greater if noted in the Contract Documents) from the date of final acceptance of the system. This warranty shall not include light bulbs in service after one (1) month from date of substantial completion of the system. Explain the provisions of the warranty to the Owner during project close out.

**b.** In accordance with EB-67D, Light Sources Other Than Incandescent and Xenon for Airport and Obstruction Lighting Fixtures; all LED light fixtures with the exception of obstruction lighting (AC 150/5345-43) shall be warranted by the manufacturer for a minimum of four (4) years after date of installation inclusive of all electronics. This warranty shall be by the Contractor/Manufacturer to the Owner for any defective workmanship or material, which has been furnished under this contract. The Contractor/Manufacturer at no cost to the Owner shall perform replacement of defective LED fixture workmanship or material for a period of four (4) years after date of installation.

#### **100-2.6 SUBSTANTIAL COMPLETION.**

**a.** All specified work shall be complete prior to final observation of work, and all forms and other information requested, including Operation and Maintenance Manuals, shall be submitted to the Engineer for approval one (1) week before the request for substantial completion of the work.

#### **100-2.7 CONTRACTORS RESPONSIBILITIES.**

**a.** Provide necessary layout, labor, material, equipment, tools, transportation, full time supervision and services required for the satisfactory and timely completion of the work in accordance with the Contract Documents.

**b.** Unload, store, protect and re-handle the materials required for this Contract until such time that material is in place. Provide protection of materials required of this contract after installation.

**c.** Provide all required transportation, erection, maintenance, dismantling and removal of temporary facilities and equipment required by this contract.

**d.** Provide all transportation, unloading, distribution, hoisting, rigging, material handling and scaffolding required installing the work of this contract.

**e.** Provide all engineering and layout required to perform the work from established base lines and benchmarks.

**f.** The Electrical Contractor shall coordinate the work of all the Electrical Contractors vendors and subcontractors to provide a fully functional and operating system.

**g.** Provide temporary electrical power and temporary water and sanitary sewer for this Contractor's field office, if required. Pay all utility company charges. Provide temporary power required for this Contractor's work.

**h.** Prior to start of work this Contractor is to inspect work performed by others on which this work is to be placed on or adjacent to, and report in writing to the Engineer, any condition found to be unacceptable. Failure to make said report shall constitute acceptance of the conditions found and any claims made thereafter due the unacceptable conditions will not be considered by the Engineer.

**i.** Provide all required coordination and supervision where work connects to or is affected by work of others and comply with all requirements affecting this work. Work required under

other items of the Contract Documents to be performed by this Contractor shall be coordinated with the respective Contractor, and such work performed at no additional cost to the Owner including but not limited to electrical work in support of other trades as shown in the Contract Documents.

j. It is the responsibility of the Contractor to coordinate the exact required location of any electrical or electronic equipment, system, or cabinets to be installed or relocated. No existing equipment may be relocated without prior coordination and with approval of the Engineer.

k. Provide and pay for all permits, licenses, fees and inspections required for the performance of the work. The Contractor shall pay all sales, consumer, use and other taxes required in accordance with the law of the place of the project.

l. Provide all tests as required, per the Contract Documents and submit all test reports to the Engineer.

m. Provide all excavation, backfill, compaction, shoring and dewatering required for performance of the work.

n. Provide sleeves for all conduit/duct required as specified.

o. Protect all work of this Contract from damage and intrusion of dirt and foreign objects. Close off open ends of conduit/duct and sleeves on work, which is to be completed at a later date. Remove closure material prior to continuance of work.

p. Prior to Final Inspection, submit to the Engineer, all Record Drawings and Operation and Maintenance Manuals as specified. Instruct Owner's maintenance personnel in the operation and maintenance of the systems as required by the Contract Documents.

q. The above is not all inclusive of the work described by the Contract Documents, which form the basis for this Contract, but is presented for the Contractor's convenience.

#### **100-2.8 EXISTING UTILITIES.**

a. The Contractor is solely responsible for the location of all existing utilities within the project limits. Please refer to SP-104 for additional requirements.

b. The Contractor is solely responsible for the protection of all existing utilities within the project limits. All manholes, hand holes, light bases, conduits and ducts shall be kept tightly closed and sealed during all phases of construction. The items may be opened in support of the work but shall be resealed or closed or covered immediately upon completion of the specific task that required them to be opened.

### **BASIC MATERIAL & METHODS**

#### **100-3.1 REQUIREMENTS OF BASIC MATERIALS AND METHODS.**

a. The work shall include the furnishing of the systems, equipment and material specified in this Division and as called for in the Contract Documents, to include supervision, operations,

methods and labor for the fabrication, installation, start-up and tests for the complete electrical installation. Provide the necessary inter-trade/Contractor coordination for the installation to be in a neat and work person like manner.

**b.** Drawings for the work are diagrammatic, intended to convey the Scope of the Work and to indicate the general arrangement and locations of the work. The drawings shall not be scaled for exact sizes or locations. Because of the scale of the drawings, certain basic items such as: conduit/duct fittings, access panels, sleeves, pull and junction boxes may not be shown. Where such items are required by Code or by other items or where they are required for proper installation of the work, such items shall be included. Coordinate final equipment locations with governing architectural, civil and structural drawings as applicable. Layout equipment before installation so that all trades may install equipment in the space available.

**c.** Equipment Specifications may not deal with minute items such as components, parts, controls and devices, which may be required to produce the equipment performance specified, or as required to meet the equipment warranties. Where such items are required, the Contractor/Supplier of the equipment shall include them, whether specifically called for or not.

**d.** Conduit/duct routed through the building/structure that interferes with other equipment and construction shall not constitute a reason for an extra charge. Equipment, conduit/duct, and fixtures shall fit into available spaces in the building/structure; do not introduce these into the building/structure at such times or in such manner as to cause damage to the structure. Equipment that requires servicing shall be readily accessible.

**e.** Locate all openings required for work performed under this item. Provide sleeves, guards or other acceptable methods to allow passage of items installed under this item.

**f.** Keep cutting and patching to a minimum. Insofar as possible, determine in advance the proper chase size and openings necessary for the work.

**g.** Where cutting and patching are required due to an error of the Contractor, or where the Contractor has not given enough advance notice of the need for holes, recesses, and chases, patching shall be performed by those trades skilled in the use of the materials involved and shall be done at the Contractor's expense.

**h.** Any cutting of work in place shall be patched and decorated by such mechanics and in such a manner that the quality of workmanship and finish shall be compatible with that of adjacent construction.

**i.** The location of electrical work is indicated on the drawings. The Engineer may adjust exact locations of new work as work progresses. Before any item is installed, the indicated location may be changed by twenty-five (25) feet in any direction without additional cost.

**j.** All penetrations through roof, wall or floor shall maintain structure fire rating and environmental barriers. Any roof penetrations shall be made by a state licensed roofing Contractor and shall maintain any in place roofing warranty.

**k.** Surface mounted fixtures, outlets, cabinets, conduit/duct, panels, etc., shall have finish or shall be painted as directed by Engineer. Paint shall be in accordance with other applicable items of the Contract Documents.

**l.** All materials utilized shall be suitable for the environment encountered. No combination of materials shall be used that forms an electrolytic coupling of such nature that in the presence of moisture corrosion is accelerated.

**m.** In general, all relays, contactors, panelboards, dry type transformers, disconnect switches, and circuit breakers are to be supplied and manufactured by the same manufacturer and shall be submitted and accepted equivalent to that specified.

**n.** Where the requirements of another division, item, or part of these Contract Documents exceed the requirements of this division, those requirements shall govern.

**o.** Make electrical connections to regulators, transformers, instruments, mechanical equipment, controls and at other locations as required with approximately 3 feet (12" minimum) of liquid tight flexible metal conduit. Grounding type liquid tight flexible metallic conduit connectors shall be used. Determine the requirements from the Contract Documents and the accepted manufacturer drawings.

**p.** Provide inserts, hangers, supports, braces, and anchor bolts as necessary for all work called for under the Contract Documents.

**q.** Provide equipment grounding conductor in all conduit/ducts, for all circuits, except series airfield lighting.

**r.** All galvanized materials shall be hot-dip galvanized after fabrication, conforming to ASTM A-386 and/or A-153, unless noted otherwise (UNO).

**s.** Unless noted otherwise, all panelboards, junction boxes, wire ways, etc., shall be spaced off the concrete structure by using a Unistrut P-1060 series square washer or accepted equivalent between the mounting surface and the equipment at each mounting point. Equipment as listed above, mounted on Unistrut or accepted equivalent shall have Unistrut P-1060 series square washer, Unistrut P-28XX flat plate fitting or accepted equivalent installed between the Unistrut channel or accepted equivalent and the equipment at each mounting point.

**t.** All bolted connections and equipment mountings shall utilize two flat washers, lock washer and hex head bolting hardware, unless noted otherwise.

**u.** Unless noted otherwise, all wire sizes are based on a 90 Deg. C., THWN-2, 600-volt insulation, copper conductors, not more than three single insulated conductors, in raceway, in free air. The conduit/duct sizes are based on the use of THWN-2, 600-volt insulated conductors. The Contractor shall make the necessary increase in conduit/duct sizes for other types of wire insulation. In no case shall the conduit/duct size be reduced. The minimum power circuit wire size shall be #12 AWG.

**v.** Conductor sizes may have been adjusted due to voltage drop or other engineering considerations. Equipment provided by the Contractor shall be capable of accepting the quantity and sizes of conductors shown in the Contract Documents. All conductor pigtails, cable step-down adapters, cable step-up adapters, terminal blocks and splicing materials necessary to complete the cable termination/splice shall be considered incidental to the respective pay items provided.

**w.** All equipment enclosures, junction boxes, wire ways, pull boxes, etc. utilized in damp, wet or exterior locations shall be NEMA 4X stainless steel. All mounting channel, hardware, nuts, bolts, lock washers, etc. utilized in damp, wet or exterior locations shall be stainless steel.

**x.** All electrical conductors, windings, bus bars, etc. shall be high conductivity (98% conductivity) copper.

**y.** This Contractor shall furnish and install all required overcurrent protection required by the NEC and the Contract Documents. The overcurrent protection shall be sized according to the equipment nameplate data.

**z.** Powder-actuated fasteners will not be allowed without express written approval of the Engineer. No fasteners shall pierce the structure until accepted by the Engineer.

**aa.** Use Ideal NOALOX antioxidant compound or accepted equivalent on all aluminum to aluminum mechanical or electrical connections. Use Thomas and Betts KOPR-SHIELD or accepted equivalent on all rigid galvanized steel conduit connections and all copper grounding connections using mechanical lugs. Use an approved anti-seize compound on all bolted connections.

### **100-3.2 ELECTRICAL REFERENCE SYMBOLS.**

**a.** Symbols used in the Contract Documents are defined in the Electrical Legend. Not necessarily will all symbols defined be applicable to the project.

### **100-3.3 ACTIVE SERVICES.**

**a.** Existing active services, water, gas, sewer, electric, communication, jet fuel, etc. when encountered, shall be protected against damage. Do not prevent or disturb operation of active services, which are to remain. If active services are encountered which require relocation, make request to Authorities Having Jurisdiction (AHJ) for determination of procedures. Where existing services are to be abandoned, they shall be terminated in conformance with requirements of the AHJ, Utility or Municipality having jurisdiction.

### **100-3.4 CODES AND STANDARDS.**

**a.** Install in accordance with latest edition of the National Electrical Code and the regulations of governing Federal, State, County, local and other applicable codes, including the Utility Companies. Where a conflict in code requirements occurs the most stringent requirement shall govern. The Contractor shall be responsible to obtain and pay for all required licenses, fees and inspections including meter installation fee. The cost for such shall be included in the bid price.

**b.** The work shall meet the requirements and recommendations of applicable portions of the latest editions of these laws, codes and standards:

- 1.** OSHA 1910 & 1926
- 2.** NFPA 70, National Electrical Code

3. NFPA 70E, Standard for Electrical Safety in the Workplace
4. NFPA 101, Life Safety Code
5. NFPA 409, Standard on Aircraft Hangars
6. NFPA 780, Standard for the Installation of Lightning Protection Systems (UL 96, UL 96A)
7. FAA Advisory Circulars, Engineering Briefs and Orders
8. Applicable local building code
9. Florida Building Code Project State Specific
10. ANSI C2, National Electrical Safety Code
11. NEMA Standards (NEMA)
12. Underwriter's Laboratories (UL)
13. Institute of Electrical and Electronics Engineers (IEEE)
14. FDOT - Standard Specifications for Road and Bridge Construction Project State Specific
15. FDOT – Design Standards Index Project State Specific
16. AWS D1.1, Structural Welding Code – Steel
17. Merriam-Webster Dictionary

The above is not all inclusive of applicable codes and standards but is presented for the Contractors convenience.

The advisory circulars are available online at:

[https://www.faa.gov/regulations\\_policies/advisory\\_circulars/index.cfm/go/document.list?display=current&q=](https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.list?display=current&q=)

### **100-3.5 LISTED AND LABELED.**

a. All materials shall be new and free of defects and shall be U.L. listed, bear the U.L. label or be labeled or listed with an accepted, nationally recognized Electrical Testing Agency. Where no labeling or listing service is available for certain types of equipment, test data shall be submitted to prove to the Engineer that equipment meets or exceeds available standards. All U.L. listed material shall be used only for the intended purpose.

**100-3.6 UTILITY COMPANY FEES, CHARGES, AND COSTS.**

a. It is the Contractor's responsibility to contact the applicable Utility Company(s) to determine if any fees, charges or costs will be due the Utility Company(s) as required by the Utility Company(s) for temporary power, installations, hook-ups, etc. This fee, charge or cost shall be included in this Contractor's price.

**100-3.7 TESTING – QUALITY CONTROL.**

a. Systems shall be tested by the Contractor, in the presence of the RPR, and placed in proper working order prior to demonstrating systems to the Owner. Refer to the requirements in each item for other applicable standards.

b. Provide all tests as required, per the Contract Documents and submit all test reports to the RPR. Costs of all testing shall be borne by the Contractor and shall be considered incidental to the respective pay item of the tested item. The Contractor shall perform all Quality Control in accordance with the Contract Documents and the Contractor's Quality Control Plan General Conditions. The Owner will perform Quality Assurance/Acceptance.

c. After work is completed a load balance test shall be made to demonstrate that with full lighting and mechanical load the balance between phases is within 5%. Unbalance beyond this limit shall be corrected.

d. Perform such tests as required by AHJ over the site.

e. Testing methods shall be acceptable to the Engineer and shall be submitted to the RPR for review, a minimum of thirty (30) days prior to the scheduled test.

**100-3.8 LAMINATED PHENOLIC PLASTIC NAMEPLATES.**

a. Contractor shall provide nameplates for wiring systems and equipment as called for herein. All nameplates shall have beveled edges and one-half inch (1/2") lettering. If equipment is smaller than ten inches by six inches (10"x 6"), one-quarter inch (1/4") lettering may be used. Smaller lettering may be used with permission of the RPR.

b. Nameplates shall be laminated phenolic plastic, black front and back with white core, with lettering etched through the outer covering. White engraved letters on black background. Emergency systems shall use red front and back with white core for nameplates. Attach nameplates with stainless steel pop rivets, sealed with clear silicon caulk. Adhesive may be used only on panelboard interiors. Where conditions do not warrant piercing the enclosure "LOCTITE" brand adhesive or accepted equivalent may be used with permission of the RPR.

c. Coordinate nameplates with Contract Documents and submit 1 to 1 scale detail of all nameplates to RPR for review.

d. All electrical equipment, enclosures, pull boxes and all similar items in the sole opinion of the RPR shall be equipped with nameplates. Examples are all constant current regulators, motors, pushbutton stations, hand holes, pull boxes, junction boxes, manholes, control panels, system cabinets, terminal cabinets, time switches, disconnect switches, panelboards, circuit



breakers, contactors, relays, high voltage boxes and cabinets. Special electrical systems shall be identified at junction and pull boxes, terminal cabinets and equipment racks.

e. Nameplates shall adequately describe the function of the particular equipment involved. Where nameplates are detailed in the Contract Documents, inscription and size of letters shall be as shown. Nameplates for panelboards and switchboards shall include the panel designation, panel name, source(s) of power and voltage and phase of the supply. For example, "Equip YY, Panel A, fed from Panel XYZ, 480/277V, 3-phase, 4-wire." The unique name of a particular piece of equipment shall be used consistently on all nameplates referencing that particular piece of equipment. Equipment name/numbers utilized shall be consistent throughout the entire system. When equipment has multiple sources of power, all sources of power shall appear on the nameplate. Power source descriptions shall contain as a minimum the information listed above. Nameplates shall include as a minimum the following.

1. Equipment Number
2. Equipment Name
3. Power Source(s)
4. Voltage Level and number of phases (for each power source)

f. Panelboard and distribution panels shall have each breaker position identified with phenolic labels.

g. All major pull and junction boxes in service areas, tunnels, above accessible ceilings and in accessible chases shall have nameplates identifying the feeder or system. This shall include the access cover on lighting pole bases.

h. Systems with conductors exceeding 100 volts to ground shall have voltage identification nameplates with one-half inch (1/2") high letters on all panels, switches, pull boxes and junction boxes.

### **100-3.9 CONCRETE WORK.**

a. Concrete bases and pads for all equipment furnished by Contractor shall be the responsibility of Contractor unless noted otherwise.

b. Furnish and install reinforced concrete pads for transformers, switchgear, and motor control centers, etc. of size as shown in the Contract Documents. Chamfer the top edges 1". Trowel all surfaces smooth, provide broom finish where noted. Reinforce the pads as shown in the Contract Documents.

c. Concrete equipment pads shall be installed by tradespersons skilled in concrete work. Equipment pads shall be level within one-eighth inch (1/8") in 10 feet (10'). Equipment pads not meeting the previous requirements shall be removed and replaced at the Contractor's expense.

d. The Contractor shall furnish all equipment anchor bolts and shall be responsible for their proper installation and accurate location.

### 100-3.10 EXCAVATING, TRENCHING AND BACKFILLING.

a. This Contractor shall do excavating necessary for light bases, underground wiring, conduit, duct banks and other underground work and shall backfill trenches and excavations after work has been inspected. Care shall be taken in excavating that walls and footings and adjacent load bearing soils are not disturbed in any way, except where lines must cross under a footing, pad or other structural item. Where a line must pass under a footing, pad or other structural item the crossing shall be made by the smallest possible trench to accommodate the conduit/duct. Excavations shall be kept free from water. No greater length of trench shall be left open in advance of conduit/duct installation than that, which is authorized or directed by the RPR.

b. Roots shall be removed to a level of eighteen (18") below finished grades and deeper as required for conduit/duct runs, manholes and light pole bases. No roots shall be allowed to remain under the work.

c. Backfill about the work shall be placed, where practical, as the work of construction progresses. Backfilling on or against concrete work shall be done only when directed. Backfilling of conduit/duct lines shall progress as rapidly as the testing and acceptance of the finished items of the work will permit and shall be carried to a crown approximately six inches (6") above the existing grades. In backfilling around conduit/duct lines, selected material shall be compacted firmly around the conduit/duct. Fill and backfill shall be clean and free from vegetable matter and refuse.

d. All loose materials shall be removed from all excavations for electrical items. The bottom of the excavation shall be compacted to the following criteria:

| SOIL TYPE                 | DESIGN AIRCRAFT GROSS WEIGHT | TEST STANDARD | UNDER PAVEMENT | IN TURF |
|---------------------------|------------------------------|---------------|----------------|---------|
| <30% RETAINED ON ¾" SIEVE | >60,000 LBS                  | ASTM D1557    | 100%           | 95%     |
| <30% RETAINED ON ¾" SIEVE | ≤60,000 LBS                  | ASTM D698     | 100%           | 98%     |
| >30% RETAINED ON ¾" SIEVE | >60,000 LBS                  | AASHTO T-180  | 100%           | 95%     |
| >30% RETAINED ON ¾" SIEVE | ≤60,000 LBS                  | AASHTO T-99   | 100%           | 95%     |

Backfill shall comply with the same compaction requirements.

e. All trenches and other excavation left open by necessity shall be barricaded and guarded as required by OSHA or applicable codes and regulations.

### 100-3.12 WELDING.

a. All welding and weld procedures shall be in accordance with AWS D1.1, Latest Edition. Qualifications of welders and welding operators shall be in accordance with AWS D1.1, Latest

Edition. The welder qualification test shall be performed on a 1" A-36 Test Coupon in the 3G and 4G positions. The welder qualification shall be current within 12 months of the work being performed. Weld inspections shall be per the criteria set forth in AWS D1.1 for visual weld inspection.

### **100-3.13 ELECTRONIC DRAWING FILES.**

a. Electronic versions of the Contract Documents may be available for the Contractors use. Please be advised that Engineer cannot be held responsible for third party use of this information, and furthermore, that this information is supplied as a professional courtesy. This applies to all electronic data files transmitted on behalf of Engineer to third parties.

b. The files are without warranty as to performance, accuracy, or freedom from error, or as to any results generated through their use at any time. Written dimensions in file, details, specifications take precedence.

c. The files being transmitted are the intellectual property of Engineer and as such are protected by the copyright laws of the United States. The files are being transmitted for the single use on this project only.

d. The electronic files provided are non-sealed and unofficial versions of the Contract Documents that have been created by the Engineer. The electronic files are provided only for the convenience of explicitly authorized users.

e. The hard copy Conformed Documents that have been provided are the only official documents that may be relied upon for accurate presentation. The electronic files provided will be at the authorized user's risk and does not pose any risk or liability to the Owner or Engineer.

f. Engineer does not guarantee or warrant the authenticity of electronic files. The authorized users agree and recognize that the electronic files may unintentionally become corrupted or modified without the authorization of the Engineer.

g. Written dimensions will govern over scaled values.

h. AutoCAD files were developed using base files provided by others.

i. Variations in printer drivers may cause the electronic file presentation to vary considerably.

## **DESIGNATION OF MATERIALS**

### **100-4.1 DESIGNATION OF MATERIALS.**

a. Where equipment or materials are specified using the name and catalog number of one manufacturer, that equipment or material specified shall set the quality and functionality of the item. Equivalents accepted by the Engineer will be acceptable.

**b.** Where equipment or materials are specified using the name and catalog number of more than one manufacturer, that equipment or material shall be one of those specified. No alternative will be acceptable.

**c.** Where no brand name is specified, the source and quality shall be subject to the Engineer's review and acceptance.

**d.** When a product is specified to be in accordance with a trade association or government standard, at the request of Engineer, Contractor shall furnish a certificate that the product complies with the referenced standard. Upon request of the Engineer, the Contractor shall submit supporting test data to substantiate compliance.

**e.** The Engineer shall be the sole judge of whether the proposed "equivalent" is suitable for use in the work.

**f.** The Engineer reserves the right to reject any and all equipment, materials or procedures, which, in the Engineer's opinion, does not meet the system design and the standards and codes, shown in the Contract Documents.

**g.** Each Bidder represents their bid is based upon the materials and equipment described in the Contract Documents.

## **PROTECTION OF MATERIALS, EQUIPMENT AND WORK**

### **100-5.1 PROTECTION OF MATERIALS, EQUIPMENT AND WORK.**

**a.** Materials shall be stored to assure the preservation of their quality and fitness for the work. Stored materials, even though accepted before storage, shall be subject to reinspection prior to their use in the work. The Contractor shall coordinate the storage of all materials with the RPR.

**b.** The Contractor shall protect electrical raceway, cable, lighting fixtures, associated support systems, and all other electrical items against damage from movement of equipment and material, welding, flame cutting, and other construction damage. Raceways and supporting structures for raceway and lighting fixtures shall not be used as access scaffolding at any time. Whenever welding or flame cutting operations occur above or near raceways, cables lighting fixtures, etc not shielded from such operations by concrete floor or other protective covers, the Contractor shall protect the items from damage by means of fireproof boards or blankets. Damaged materials shall be repaired or replaced, by and at the Contractor's expense, subject to the RPR's direction and acceptance.

**c.** Surfaces of most equipment, such as panels, switchgear, transformers, constant current regulators (CCR) and circuit breakers, are finished at the factory. Great care shall be exercised to prevent damage to this original finish during installation of the equipment and during construction work.

**d.** If the factory finish is damaged during construction, the entire surface of the damaged component shall be refinished by and at the expense of the Contractor.

e. The refinished surface shall be equivalent in every respect to the original surface, including color, texture and smoothness. Refinishing paint, if furnished with the equipment, may be used; otherwise, the paint shall be obtained from the equipment manufacturer.

f. All cut edges of galvanized materials and marred or scratched galvanized surfaces shall be repaired using LPS-1G cold galvanizing compound or accepted equivalent.

## **GENERAL CONSTRUCTION REQUIREMENTS**

### **100-6.1 GENERAL CONSTRUCTION REQUIREMENTS.**

a. Provide the bracing, shoring, rails, guards, and covers necessary to prevent damage or injury. Do not leave energized electrical items exposed or unprotected. Protect personnel from exposure to contact with electricity. Deliver equipment and materials to the job site in their original, unopened, labeled containers. Store ferrous materials to prevent rusting. Store finished materials and equipment to prevent staining and discoloring.

b. Maintain separation between stainless steel items and carbon steel items to prevent contamination of stainless-steel item and subsequent rust stains on stainless steel item.

c. All sheeting, shoring, dewatering and cleaning necessary to keep trenches and their grades in proper condition for the work to be continued, including the removal of water by mechanical means, shall be the Contractor's responsibility.

## **METHOD OF MEASUREMENT**

### **100-7.1 METHOD OF MEASUREMENT.**

a. The items described in Item SP-100 are incidental to other pay items and shall not be measured for payment.

## **BASIS OF PAYMENT**

### **100-8.1 BASIS OF PAYMENT**

a. No direct payment shall be made for the work described in Item SP-100. The work described in Item SP-100 is incidental to other items and shall be paid for in the respective bid item of which it is a component part.

## **END OF ITEM SP-100**

**ITEM SP-104 GENERAL ELECTRICAL SAFETY REQUIREMENTS AND TEMPORARY  
AIRFIELD LIGHTING**

**DESCRIPTION**

**104-1.1 PURPOSE:**

**a.** The purpose of this item is to assist in establishing the proper safety guidelines necessary to protect aircraft, passengers, crews, the general public, Owner, Engineer, all workers and vehicles involved in their daily tasks.

**b.** **The Contractor is solely responsible for all issues related to the safety program and guidelines and implementation of such programs and guidelines necessary to protect aircraft, passengers, crews, the general public, Owner, Engineer, all workers and vehicles involved in their daily tasks.**

**c.** The Contractor shall comply with FAA Advisory Circulars, FAA Engineering Bulletins, specific airport requirements, OSHA 29CFR1910, OSHA 29CFR1926 and NFPA 70E as minimum requirements.

**104-1.2 REFERENCED STANDARDS AND CODES:**

**a.** All applicable requirements of the below listed advisory circulars, codes, standards and related reading, as a minimum, shall be compiled with:

- 1.** AC 150/5200-18, Airport Safety Self-Inspection
- 2.** AC 150/5210-5, Painting, Marking and Lighting of Vehicles
- 3.** AC 150/5340-1, Standards for Airport Markings
- 4.** AC 150/5340-18, Standards for Airport Sign Systems
- 5.** AC 150/5340-26, Maintenance of Airport Visual Aid Facilities
- 6.** AC 150/5340-30, Design and Installation Details for Airport Visual Aids
- 7.** AC 150/5370-2, Operational Safety On Airports During Construction
- 8.** 29 CFR Part 1926/1910, Occupational Safety and Health Standards
- 9.** NFPA 70, National Electrical Code
- 10.** NFPA 70E, Standard for Electrical Safety in the Workplace
- 11.** ANSI C2, National Electrical Safety Code

The Contractor is responsible for obtaining and using the latest edition of the referenced documents. This list is not all inclusive but is offered as a convenience to the Contractor. The advisory circulars are available online at:

[https://www.faa.gov/regulations\\_policies/advisory\\_circulars/index.cfm/go/document.list?display=current&q=](https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.list?display=current&q=)

#### **104-1.3 GENERAL SAFETY PROVISIONS:**

**a.** Contractor shall take safety and health measures in performing work under this contract. Contractor shall meet with the Engineer and Owner to develop a mutual understanding relative to administration of the safety plan. The Contractor is subject to and shall comply with all applicable federal, state and local laws, regulations, ordinances, codes and orders relating to safety and health in effect on the date of this contract. Where the contract general conditions, special provisions or other parts of the Contract Documents also apply, the stricter of the documents shall govern.

**b.** As a minimum workplace safety shall comply with NFPA 70E, Standard for Electrical Safety in the Workplace, OSHA, federal, state and local requirements. Where a conflict in requirements occurs the most stringent requirement shall govern.

**c.** During the performance of work under this contract, the Contractor shall comply with procedures prescribed for control and safety of persons visiting the project site.

**d.** Contractor is responsible for the Contractor's personnel and for familiarizing each of the Subcontractors with safety requirements.

**e.** Contractor shall advise the Engineer of any special safety restrictions established so that Owner and Engineer personnel can be notified of these restrictions.

**f.** The Contractor shall be solely and completely responsible for initiating, maintaining and supervising all safety precautions and programs in connection with the work. The Contractor shall take all necessary precautions for the safety of, and shall provide the necessary protection to prevent damage, injury or loss to:

**1.** All employees on the work and all other persons (including but not limited to the general public) who may be affected thereby,

**2.** All the work and all materials or equipment to be incorporated therein, whether in storage on or off the site,

**3.** Other property at the site, adjacent thereto, or utilized by the Contractor including but not limited to trees, shrubs, lawns, walks, pavements, structures, underground facilities, and other utilities not designated for removal, relocation or replacement in the course of construction regardless of whether or not such other property is indicated in the Contract Documents.

**g.** The installation and/or removal of lighting equipment may be critical to airport operations; therefore, the Contractor shall follow work schedules established in the Contract Documents or as directed by the RPR. The system shall be installed in accordance with the Contract Documents, FAA criteria, the National Electrical Code and/or federal, state, or local code requirements.

**h.** The Contractor shall generate and submit for approval A Maintenance of Power Plan for all electrical services impacted by this project. The Contractor shall always maintain power (permanent or temporary) to all FAA and airport owned facilities during this project. Power outages may be scheduled at the FAA and Owners convenience. Scheduling and duration of any power outage is at the sole discretion of the power system owner. Cost of maintenance of power and all necessary materials, labor transportation, supervision, coordination and all appurtenances shall be considered incidental to the SP-104-1 or SP-104-2 pay items.

**i.** The Contractor shall provide temporary wiring as required to reconnect existing circuits to provide guidance for aircraft to pass through the construction areas on those taxiways/runways, which must remain open. The Contractor, in conjunction with the Owner's Maintenance Department, shall perform a visual lighting check of all lighting circuits within the Contractor's area of responsibility before dark each day (or at additional times as designated by the RPR) to assure that they are fully functional. In the event of failure, the Contractor shall immediately take steps to restore operation.

#### **104-1.4 FIRE PREVENTION AND PROTECTION:**

**a.** All tools producing sparks or heat, open-flame heating devices, operations utilizing such devices, etc. shall be in accordance with the airport fire department and airport "Burn Permit Procedures". Work shall not start until all requirements of the burn permit procedures are met

**b.** Open fires, salamanders and open-flame heating devices will not be permitted except by special approval in writing. Such permission will not be granted unless the Contractor has taken reasonable precautions to make such devices safe. Burning trash, brush or wood on the project site will not be permitted. Approval for open fires and open-flame heating devices will in no way relieve the Contractor from the responsibility for any damage incurred because of fires.

**c.** Flammable liquids shall be stored and handled in accordance with NFPA and OSHA requirements.

**d.** Smoking shall not be permitted within the AOA or in areas such as paint storage, fuel storage, and posted no smoking areas.

**e.** Welding, flame cutting, melting and other such operations in all operating areas, shall not be permitted until approved at the beginning of each workday by the RPR. The RPR may approve longer periods of time for welding and burning in some operating areas if the detailed safety procedures are established beforehand. Operating open flame devices shall not be left unattended in any area.

**f.** Contractor shall provide the necessary firefighting equipment, fire prevention methods and fire watch.



**g.** A Contractor's employee shall be assigned as fire watch for every welding and burning operation. The fire watch shall be properly trained in firefighting techniques and the operation of the fire extinguishers. The fire watch shall be equipped with two full 15-pound Class A, B & C fire extinguishers and shall check all areas around and below the welding or burning operation for fires. The fire watch shall continue this check for at least 60 minutes after the completion of the welding or burning operation.

**h.** Contractor shall discontinue all burning, welding, or cutting operations, one hour prior to the end of the normal workday. Contractor shall provide a fire watch to remain at the site for one hour after discontinuing these operations. This fire watch shall make a thorough inspection of the area for possible sources of latent combustion. Any unsafe conditions shall be corrected.

**i.** During operations involving possible fire hazard, the Contractor shall notify the RPR and not proceed until clearance is obtained in writing. The RPR may request a standby from the Fire Station. However, this does not relieve the Contractor of responsibility for welding and cutting safety.

**j.** Clean up of scrap materials and waste of this Contractor shall be completed as a minimum daily or more often as required to maintain airport operations.

#### **104-1.5 TEMPORARY EXITS AND ENTRANCES:**

**a.** Such passageways shall provide adequate fire protection and safety of Owner personnel and representatives.

#### **104-1.6 SWITCHING:**

**a.** Electrical switching required for clearance to work on equipment operating from electrical circuits will be performed only by Owner personnel authorized as safety operators for the specific equipment or as specifically authorized by the Owner.

#### **104-1.7 REMOVAL OF EQUIPMENT:**

**a.** When permanently removing equipment, the electrical wiring, conduit/duct and control boxes shall be removed to the source of feed, unless otherwise specified or indicated.

**b.** After equipment has been removed, the electrical wiring diagrams, schematics, etc., shall be marked to show the change.

**c.** Conduit/duct not removed shall have a pull string installed.

#### **104-1.8 OTHER SAFETY REQUIREMENTS:**

**a.** Temporary wiring shall comply with NEC. Indiscriminate use of extension cords, portable cable or junction boxes creating tripping hazards as well as overloaded circuits will not be permitted.

**b.** Unplug or disconnect energy source of power tools when not in use.

- c. Before maintaining or repairing any electrical equipment, it shall be disconnected from the power source.
- d. Do not use any equipment that has frayed cords or three-wire plugs that have had the grounding prongs removed. Faulty equipment and tools shall be repaired by a qualified person.
- e. Do not use metal ladders when working on electrical equipment.

#### **104-1.9 EXISTING UTILITIES:**

- a. The Contractor is solely responsible for the location of all existing utilities within the project limits. Prior to initiation of any construction in the field, the Contractor shall provide a written notice (return receipt requested) to each of the utility companies along with the various airport departments and **Florida** Sunshine One Call as prescribed by state law. The Contractor shall provide the Owner and Engineer with a copy of the receipt of said written notification to each of the utility companies or airport departments. This requirement is in addition to any other state laws regarding public notification prior to excavation.

### **EXCAVATION**

#### **104-2.1 EXCAVATION OPERATIONS:**

- a. Prior to any excavation or demolition, the Contractor shall obtain an "Airport Dig Permit".
- b. Methods of excavation, means of earth support, and manner of backfill shall be conducted with consideration for the safety of persons and work, and prevention of damage to adjacent pavement, utilities, structures and other facilities, due to settlement, lateral movement, undermining and washout. Minimum requirements for trench shoring and bracing shall be in accordance with OSHA, State and local requirements. Excavation shall be performed in a manner to prevent surface water and subsurface or groundwater from flowing into excavations, and to prevent water from flooding conduit/duct trench and adjacent or surrounding area.

### **PROTECTION OF WORK**

#### **104-3.1 PROTECTION OF WORK:**

- a. Provide adequate stand-by mechanical equipment for emergency use.
- b. Excavations shall have substantial barricades and be posted with warning signs for the safety of persons. Warning lights shall be provided during hours of darkness.
- c. Barricades shall be erected immediately around manhole openings when covers are opened/removed.
- d. Prior to performing any excavation work or any surface penetrations 6 inches or deeper (such as driving stakes more than 6 inches in the ground) on any ground surface, for personnel

safety and to prevent possible interruption of major utility services encountered during excavation, the following procedures shall be followed:

1. Refer to 104-1.9.
2. The Contractor shall obtain from the RPR, local utilities, etc., the current up-to-date subsurface utility drawing of the particular area to be worked on.
3. All agencies/utilities, etc. that may be affected by the excavating shall be contacted by the Contractor so that all lines, pipes, etc. can be marked/staked. The Contractor shall be responsible to maintain the utility locate markings.
4. The Contractor shall stake out subsurface high voltage cables, communication cables, pipelines, all subsurface utilities, etc., indicated within the scope of the work contemplated. All subsurface utilities shall be located by hand digging; hand digging shall extend for 5 feet on both sides of the subsurface utility. Hand digging and protection of existing systems shall not be paid for separately but shall be considered incidental to the item being installed.
5. After hand exposure of cable or pipelines, the Contractor shall obtain agreement from the RPR, AHJ, Agency/Utility on how much closer to cable or pipe the excavations can be permitted.
6. Detectable printed mylar foil warning tape, color coded, printed with the words "CAUTION ELECTRIC LINE BELOW," red, "CAUTION COMMUNICATION LINE BELOW," orange, "CAUTION WATER LINE BELOW," blue, "CAUTION FUEL LINE BELOW," yellow, etc., shall be installed over the buried utilities. Material and installation of detectable tape shall be in accordance with L-110.
7. Contractor shall notify the RPR, 72 hours prior to the start of excavation work or surface penetration, to enable the RPR to review measures being taken to prevent hazard to employees and to prevent possible damage to subsurface utilities. Where emergency conditions preclude the 72 hours advance notification, Contractor shall nevertheless inform the RPR of the Contractor's intention to initiate work prior to actual start of activity.
8. After all underground utilities have been located and staked, the Contractor shall proceed with excavating work, or other surface penetration work. The Contractor, however, shall temporarily halt any machine excavation work or other surface penetration when approaching within 5 feet of the staked-out cable or pipeline, until Contractor has hand excavated down to expose the underground utility to exactly fix its location.
9. No digging, dirt moving, or other heavy equipment shall physically enter any construction area before all FAA utilities, airport utilities and all other underground utilities have been located and properly staked out. It is the Contractor's responsibility to locate all utilities before digging, sawing, coring, boring, etc. Any damage caused by digging, sawing, boring coring, etc., is the Contractor's responsibility for repair. Any damage must be reported immediately to the RPR. No repair shall be attempted without approval.
10. All electrical cables shall be disconnected before excavation is attempted.

**11.** Provide minimum of 48" undisturbed earth cover or 1" thick steel plate over electrical conduit/duct, cables or other utilities when heavy equipment is being used in the area to protect underground systems/utilities. In addition, protect utilities in accordance with AHJ requirements.

**12.** The requirements listed above shall be considered incidental to the item for which the excavation is required.

**13.** A written description of the utility protection measures shall be submitted to the RPR for review by the Owner, FAA and AHJ.

## **SAFETY TAGGING AND LOCKOUT**

### **104-4.1 SAFETY WITH ELECTRICAL CIRCUITS AND EQUIPMENT:**

**a.** No one may work on an energized circuit without written permission from the Contractor's On-Site Supervisor. The Contractor's On-Site Supervisor shall review the circumstances and the necessary safety precautions with the RPR, prior to giving permission for the energized work. The Contractor assumes all liability in connection with any work on energized circuits.

**b.** No one may disconnect or cause to be disconnected any electrical circuit before permission is requested from and granted by the Owner or authorized representative through the RPR.

**c.** Identification markings on light and power circuits shall not be relied on for establishing electrically safe work conditions. Always verify the proper safe "deenergized" conditions with properly operating test equipment.

**d.** Before any circuit supplying radar, ILS, weather, VORTAC, airport beacon, runway/taxiway lighting equipment or any other equipment is disconnected, permission must first be granted by the systems Owner or authorized representative, and, if applicable, FAA Sector Field Office and the Airport's Air Traffic Control Tower. After permission is granted, the following requirements must be compiled with. **THE FAA REQUIRES 45 DAY NOTICE PRIOR TO WORKING ON OR DISTURBING FAA CABLES OR EQUIPMENT.**

**e.** As a minimum the LOCK OUT/TAG OUT (LOTO) procedure shall comply with the Owner's requirements, OSHA1910.147, OSHA1910.147 Appendix A and NFPA 70E. If conflicts between standards arise the stricter requirement shall apply.

**f.** Comply with OSHA 1910.333, Selection and Use of Work Practices.

**g.** A qualified person shall perform a hazard risk assessment, job safety plan, and job briefing in accordance with NFPA 70E.

**h.** A qualified person shall create an Electrically Safe Work Condition in accordance with NFPA 70E.

i. Restoration shall be accomplished in a safe manner and locks/tags removed only by the Electrical Contractor's qualified person who placed the locks and tags in the presence of the Owner or authorized representative. Should the qualified person who placed the lock/tag not be available, see OSHA 1910.333(b)(2)(v)(C).

j. The Owner or authorized representative shall record time, date and operational status of circuit after restoration.

k. Electrical circuits and apparatus shall be operated at these facilities only by qualified persons and authorized personnel.

l. The RPR shall be provided contact names and phone numbers for two qualified persons (one as primary, one as backup) to provide for removal or addition of any Contractor LOTO work during off hours. The qualified person(s) shall respond onsite within one hour of initial contact.

#### **104-4.2 TEMPORARY AIRFIELD LIGHTING AND SIGNAGE:**

a. Temporary airfield electrical fixtures and conductors are allowable when necessary, but shall be installed as follows:

1. Temporary lights shall be bolted to the pavement in a manner rendering the light stationery and allowing space for conductors to enter or exit and to be spliced.

2. When the above is not practical, lights shall be fastened to a weighted object adaptable for the purpose and of sufficient weight to inhibit movement by jet engine blast.

3. Temporary conductors supplying temporary lights shall be installed in a rigid galvanized steel conduit system and secured every five feet to prevent movement by jet engine blast. See temporary airfield lighting details.

4. All joints or splices in temporary conductors shall have heat shrink tubing with integral sealant or tape applied to secure mechanical and electrical connection and prevent water entry.

5. All temporary plug-in connections shall comply with the permanent wiring detailed in L-108 and the wiring details.

6. Temporary lights and signage shall conform as closely as possible to permanent locations normally on the taxiway or runway but shall guide aircraft in a safe path. Please refer to the temporary airfield lighting details provided in the Contract Documents for additional requirements.

b. Closed taxiways and runways shall be blacked out in a manner acceptable to FAA, RPR and Airport Operations and said black/out/marking/lighting shall be kept in acceptable condition.

1. The required method of temporarily bypassing a fixture is to unplug and bypass the primary side of the L-830 isolation transformer.

2. Alternately, pipe sleeves may be used to obscure the elevated edge lights. Pipe sleeve length shall be of sufficient length to block all light output and to prevent any light leakage.

3. In extreme circumstances and with written permission of the Engineer, no more than 5% of the fixtures in each airfield lighting circuit may be unplugged from the L-830 isolation transformer secondary or have the lamp removed. If at any time a larger number of fixtures are required to be bypassed (or have the lamp removed); the L-830 isolation transformer secondary shall have a shorting device installed.

4. During runway closures the respective runway's mandatory signs shall remain operational.

5. During runway closures the respective runway's approach lighting systems (PAPI, REIL, MALSR, ALSF, etc.) shall be turned off.

6. Airfield signs providing directions to closed areas shall be covered (black out). No stray light may be emitted from the covered portions of the sign. The signs shall be covered with 3/8" thick exterior plywood panels. The plywood panels shall be painted black and shall be banded to the sign. Please refer to the temporary airfield lighting details provided in the Contract Documents for additional requirements.

7. Areas closed to aircraft shall not be lighted (black out). The Contractor shall ensure that lighting in the closed areas cannot be inadvertently energized.

8. Adequate lighting in the opinion of the Engineer or Airport Operations shall be provided to delineate the active and closed areas of the AOA.

c. **CAUTION:** The series lighting circuit must always be complete before a regulator is energized. Normal circuit voltage is less than 5,000 volts; open circuit voltage can approach 10,000 volts. All personnel shall be instructed to protect the integrity of the lighting circuit. Turn off and lock out/tag out the regulator at the vault before opening the circuit. Continuity of the circuit shall be checked before the regulator is reconnected and reenergized.

d. The installation and/or removal of lighting equipment is critical to airport operations; therefore, the Contractor shall follow work schedules established in the Contract Documents or as directed by the RPR. The Contractor shall attend weekly coordination meetings with Airport Operations for the duration of the contract. The system shall be installed in accordance with applicable FAA Advisory Circulars, the National Electrical Code and/or other federal, state or local requirements.

e. The Contractor shall provide temporary wiring as required to reconnect existing circuits to provide guidance for aircraft to pass through the construction areas on those taxiways/runways, which must remain open. The Contractor shall bypass or otherwise render inoperative those portions of the runway/taxiway lighting circuits in closed areas of the airfield.

f. All runways/taxiways not closed for construction shall have fully operable airfield lighting systems using the airport airfield lighting control system. The Contractor shall coordinate the temporary airfield lighting requirements for each phase with the RPR. The Contractor shall submit

a temporary airfield lighting plan to the RPR two weeks prior to the need for the temporary airfield lighting. The RPR will coordinate the plan with the airport and FAA. The Contractor shall make revisions/corrections to the temporary airfield lighting plan as required by the RPR.

**g.** Existing conditions may vary from the existing information shown in the Contract Documents. The Contractor's means and methods and sequence of construction within a specific phase is determined by the Contractor. Therefore, the temporary airfield lighting plan in 4.2.f above is the Contractor's responsibility. The Owner and Engineer will provide reasonable assistance to the Contractor.

**h.** It shall be the Contractor's responsibility to determine that all airfield lighting circuits, except those that are serving closed taxiways or runways, are fully functional, using tower controls, at the end of each work shift and shall so certify to the RPR at the end of each shift. The Contractor shall not leave the work area until the RPR has confirmed circuit operation. The certification shall be provided to the RPR at 4:00 PM and one hour before sunset. Second shift report shall report one hour before sunset. Any other shift shall report one hour prior to the need for airfield lighting or as determined by the RPR. Should bad weather cause poor visibility the RPR may require additional status reports of system operability and may call for the operation of the airfield lighting system at any time. In the event of airfield lighting system failure, the Contractor shall immediately take the necessary steps to restore proper operation of the airfield lighting system.

**i.** During runway closures the respective runway's mandatory signs shall remain operational.

**j.** It shall be the Contractor's responsibility to determine that all airfield lighting and signage circuits serving closed taxiways or runways (except runway mandatory signs) are turned off, bypassed as previously described or completely obscured at the end of each work shift and shall so certify to the RPR at the end of each shift. The Contractor shall not leave the work area until the RPR has confirmed the closed area is properly blacked out.

**k.** The Contractor shall check the status of the temporary airfield lighting/signage and black out equipment on a daily basis. The Contractor shall provide temporary airfield lighting/signage and black out maintenance and support on a 24 hour a day, seven days a week basis for the duration of the project. The RPR shall be provided contact names and phone numbers for two persons (one as primary, one as backup) to provide maintenance or to meet temporary airfield lighting/signage and black out needs during off hours. The designated person shall respond onsite within one hour of initial contact.

**l.** Whenever the scope of work requires connection to an existing circuit, or should the scope of work involve working adjacent to or near an existing circuit, and the work could have any impact on the existing circuit, the circuit's insulation resistance shall be tested, in the presence of the RPR. The test shall be performed in accordance with Item SP-111 and prior to any activity affecting or impacting the respective circuit. The Contractor shall record the results on forms included in SP-111. When the work affecting the circuit is complete (or at end of shift prior to energization), the circuit's insulation resistance shall be checked again, in the presence of the RPR. The Contractor shall record the results on forms included in SP-111. The second reading shall be equal to or greater than the first reading or the Contractor shall make the necessary repairs to the circuit to bring the second reading above the first reading. All repair costs including a complete replacement of the L-823 connectors, L-830 transformers and L-824 cable, if

necessary, shall be borne by the Contractor. The Contractor shall measure and record the insulation resistance of **ALL** airfield lighting circuits prior to starting any work. All test results shall be submitted in the Operation and Maintenance (O&M) Manual.

**m.** Should the Contractor fail to perform the tests prior to starting work, the Contractor shall be responsible to bring the impacted circuits to a minimum 50 megohm insulation resistance value or higher value in conformance with any previous airport test records.

**n.** Provide temporary wiring to permit operation of lighting circuits associated with each phase of construction. Temporary cable connections shall be made in airfield lighting circuits when permanent wiring cannot be completed during daylight hours.

**o.** The Contractor shall furnish, install and maintain hazard marking and lighting to identify closure areas, open manholes, small areas under repair, stockpiles, and other construction related areas hazardous to aircraft or personnel.

**p.** The Contractor is responsible for operating all temporary circuits within their rated capacity.

**q.** The cost of temporary and reconnected airfield lighting/power shall be considered incidental to the pay items provided within the Contract Documents.

#### **104-4.3 TEMPORARY ELECTRICAL AND LIGHTING INSTALLATION:**

**a.** Temporary electrical and/or lighting fixtures shall be provided in operational areas of buildings where required to maintain public safety and continued airport operations.

**b.** Provide temporary roadway lighting on airport roads and detours during construction. Design installation and removal of temporary systems shall be the responsibility of the Contractor. Light levels shall be 1.0 maintained foot-candles on the traveled part of the roadway. Lighting under the bridges during construction shall have a metal halide or other white light source, providing a maintained light level of 2.0 foot-candles. Maximum to minimum uniformity ratios shall be 10:1.

**c.** Temporary lighting must be installed to ANSI/OSHA standards for impacted area. Light levels shall also meet the requirements of the Illuminating Engineering Society for the type of task being performed.

**d.** Temporary installations shall comply with the NEC and shall be acceptable to the Owner, RPR or authorized representative.

**e.** The cost of temporary and reconnected area/building lighting/power shall be considered incidental to the pay items provided within the Contract Documents.

#### **104-4.4 MISCELLANEOUS REGULATIONS:**

**a.** Electrical main distribution panels, branch circuits panels, control panels, etc., may have their front covers removed for access to live parts. Access to live parts shall be guarded/barricaded in accordance with NFPA 70E.



- b.** Draw-out type breakers, regardless of operating voltage must be drawn completely out to open position and tagged and locked out per 104-4.1.
- c.** In hazardous locations, regardless of class, all electrical tools and extension cords shall be of a type accepted for use in such areas.
- d.** No counterpoise/grounding conductors may be joined, connected, or affixed to any terminal, grounding electrode, or other point of attachment by any method except exothermic welding, or accepted equivalent. See Item L-108.
- e.** All counterpoise or grounding systems, when damaged, shall be immediately reported to the RPR. Once the repair/replace procedure has been accepted by the RPR the Contractor shall immediately repair/replace the counterpoise or grounding components.
- f.** No high voltage switch shall be engaged or disengaged under load.
- g.** All backhoes, cranes, etc., shall be enclosed by safety pylon or other markers and rope festooned by accepted safety tapes.
- h.** All security gates in use by Contractors are the responsibility of the Contractor and must be used in a fully secure manner. Any damage to a security gate shall be reported immediately to the RPR.

#### **104-4.5 TEMPORARY COMMUNICATIONS/CONTROL:**

- a.** Temporary communication and control systems shall be provided in operational areas of buildings where required to maintain public safety and continued airport operations. Temporary communication and control systems shall be as described in the Contract Documents.
- b.** The cost of temporary and reconnected communication and control shall be considered incidental to the pay items provided within the Contract Documents.

### **METHOD OF MEASUREMENT**

#### **104-5.1 METHOD OF MEASUREMENT:**

- a.** Except for the two pay items described below all other items shall be considered incidental to other items and shall be included with its applicable system.
- b.** Temporary Power & Temporary Airfield Lighting/Signage/Navigational Facilities shall be measured as a lump sum item.

**BASIS OF PAYMENT**

**104-6.1 BASIS OF PAYMENT:**

a. Except for pay item, as described below, no direct payment shall be made for the work described in this section. The work described in this section is incidental to other items and shall be paid for the respective bid item of which it is a component part.

b. Temporary Power & Temporary Airfield Lighting/Signage/Navigational Facilities payment shall be made at the contract lump sum price for the required work. This item shall include all testing, troubleshooting, maintenance and other consequential work required to maintain the temporary power, and temporary airfield lighting, signage, and Navigational Facilities as impacted by this Contract. This price shall be full compensation for all planning, supervision, furnishing all material, labor, transportation, services, equipment, utility company fees and for all preparation, removal of the temporary materials and equipment and for all labor, equipment, tools and all incidentals required to furnish and install Temporary Power and Temporary Airfield Lighting, Signage, and Navigational Facilities in accordance with the Contract Documents. This item includes all excavation, backfill, cable, conduit and other materials and appurtenances necessary to complete these items.

Payment will be made under:

|                      |                                                                                              |                         |
|----------------------|----------------------------------------------------------------------------------------------|-------------------------|
| <b>Item SP-104-1</b> | <b>Temporary Power &amp; Temporary Airfield<br/>Lighting/Signage/Navigational Facilities</b> | <b>-- Lump Sum (LS)</b> |
|----------------------|----------------------------------------------------------------------------------------------|-------------------------|

**END OF ITEM SP-104**

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## **ITEM SP-105 ALTERATIONS, REMOVAL AND DEMOLITION**

### **PART 1 – GENERAL**

#### **105-1.1 GENERAL:**

**a.** This item shall apply to all electrical work contained within the Contract Documents. Where the contract general conditions, special provisions or other parts of the Contract Documents also apply, the stricter of the documents shall govern.

**b.** Alterations shall mean any change or rearrangement in the component parts, including structural, mechanical, and electrical systems, or internal or external arrangements of an existing item, component, system or structure.

**c.** Removal shall mean the dismantling and disposal of existing materials, components, equipment, and utilities. Removed items shall be handled, prepared for storage, transported to storage areas, or disposed of as specified.

**d.** Demolition shall mean abandonment, removal and disposal, or salvage and delivery to the Owner as required by the Contract Documents.

**e.** The installation and/or removal of lighting equipment may be critical to airport operations; therefore, the Contractor shall follow work schedule established in the Contract Documents or as directed by the Engineer.

**f.** Temporary power and lighting shall be accomplished in accordance with the Contract Documents.

#### **105-1.2 CONDITION OF EXISTING FACILITIES:**

**a.** The Contractor shall verify the areas, conditions, and features necessary to tie into existing construction. This verification shall be done prior to submittal of shop drawings, fabrication or erection, construction or installation. The Contractor shall be responsible for the accurate tie-in of the new work to existing facilities.

**b.** The Contractor shall protect all existing utilities from damage resulting from alteration, removal and demolition activities.

**c. Any hazardous waste issue shall be immediately reported to the Engineer.**

**d.** Whenever the scope of work requires, connection to an existing circuit, or should the scope of work involve working adjacent or near to an existing circuit and the work could have an adverse impact on the existing circuit, the circuit's insulation resistance shall be tested, in the presence of the Engineer. The test shall be performed in accordance with Item SP-111 and prior to any activity affecting or impacting the respective circuit. The Contractor shall record the results on forms included in Item SP-111. When the work affecting the circuit is complete (or at end of shift prior to energization), the circuit's insulation resistance shall be checked again, in the presence of the Engineer. The Contractor shall record the results on forms included in Item SP-111. The second reading shall be equal to or greater than the first reading or the Contractor

shall make the necessary repairs to the circuit to bring the second reading above the first reading. All repair costs including a complete replacement of the L-823 connectors, L-830 transformers and L-824 cable, if necessary, shall be borne by the Contractor. The Contractor shall measure and record the insulation resistance of **ALL** airfield lighting circuits prior to starting any work. All test results shall be submitted in the Operation and Maintenance (O&M) Manual.

e. Should the Contractor fail to perform the insulation resistance tests prior to starting work, the Contractor shall be responsible to bring the impacted circuits to a minimum 50 megohm earth resistance value or higher value in conformance with any previous airport test records.

#### **105-1.3 OCCUPANCY AND USE OF EXISTING FACILITIES:**

a. The Owner will occupy and use the facilities within the areas of work during the entire construction period. The Contractor shall be required to plan and coordinate the Contractor's activities in order to provide all necessary controls for the abatement of dust, noise, and inconvenience to Owner personnel during all phases of the work.

#### **105-1.4 VACATING OCCUPIED AREAS:**

a. The Owner will remove all portable items of furniture, equipment, and fixtures prior to the start of work.

#### **105-1.5 SAFETY REQUIREMENTS:**

a. The Contractor shall conduct all operations in a manner that will ensure the safety of persons and property in accordance with the OSHA requirements of CFR 29 Part 1926/1910.

b. The Contractor shall comply with the Airport Maintenance Department's LOCK OUT/TAG OUT (LOTO) procedure and NFPA 70E.

#### **105-1.6 CLASSIFICATION OF REMOVED ITEMS:**

a. Existing materials and equipment indicated to be removed will be classified as "salvageable" and shall remain the property of the Owner or will be classified as "demolition or debris" and shall be legally disposed.

##### **b. Salvageable Items:**

1. Salvaged materials and equipment shall be reused in the work only in construction of temporary electrical facilities, unless noted otherwise.
2. Properly box, protect and ready for transport salvageable items. Retained salvaged items shall be stored on Owner property.
3. Salvaged materials and equipment to be retained by the Owner but not reused in the work shall be turned over to the Owner at a site at the facility to be determined by the Owner.

4. Airfield lighting fixtures, signs and isolation transformers designated to be removed and reinstalled by the Contractor shall be stored indoors in a conditioned space. The indoor conditioned space shall be provided by the Contractor and considered incidental to the pay items provided. As each item designated for removal and reinstallation is removed the Contractor and Engineer shall immediately inspect the item for appropriateness for reinstallation. Any damage not reported in writing to the Engineer within two hours of removal shall become the Contractor's responsibility for replacement/repair of the material as directed by the Engineer. The Contractor is asked to pay special attention to the protection of the semiflush fixture leads. The semiflush fixture lead insulation is easily damaged which results in water infiltration into the fixture.

**c. Demolition or Debris:**

1. Items classified as demolition or debris shall be legally disposed of off the Owner's property. The cost of such disposal shall be included in the bid price.

**105-1.7 TEMPORARY PROTECTION:**

- a. Contractor shall protect persons and property throughout the progress of the work in accordance with the Contract Documents.

**105-1.8 DEMOLITION TASK PLAN AND SCHEDULE:**

- a. The Contractor shall submit to the Engineer a Demolition Task Plan and Schedule for all existing utility relocation(s) and demolition work. The task plan shall be submitted within 30 days of notice to proceed but no later than 30 days prior to the start of the relocation/demolition work.

- b. The task plan and schedule shall detail the phasing of the utility relocation(s) and demolition activities with the balance of the project phasing plan. The task plan and schedule shall detail the Contractor's plan for protection of the existing utilities.

- c. The task plan and schedule shall demonstrate preliminary coordination has taken place with the FAA, Airport, communications and power utilities and also demonstrates compliance with the Contract Documents.

- d. No work shall be initiated until the task plan and schedule has been submitted to the Engineer and accepted by the utilities. No demolition, alteration or relocation work shall be started until the permanent work replacing the demolition, alteration or relocation is complete, in place, accepted and ready for operation; or until other temporary means have been approved and accepted by the Engineer.

**PART 2 - EXECUTION**

**105-2.1 DISCONNECTING UTILITIES:**

- a. Prior to the start of work, the Contractor shall coordinate with the Owner and effected utility companies to have each utility placed in a safe work condition.

b. Each utility effected shall be locked out and tagged by the Contractor. Proper safety grounds or equivalent shall be placed for personnel protection.

**105-2.2 TEMPORARY UTILITY SERVICES:**

a. Contractor shall install temporary utility services in satisfactory operating condition before disconnecting existing utilities. Such temporary services shall be maintained during the period of construction and removed only after new permanent services have been tested and are in operation. Please refer to Item L-104.

**105-2.3 REMOVAL WORK:**

a. The Contractor shall not disturb the existing construction beyond that indicated or necessary for installation of new work. Temporary shoring and bracing for support of building components to prevent settlement or other movement shall be as required to protect the work and existing facilities.

b. The Contractor shall provide protective measures to control accumulation and migration of dust and dirt in all areas of work, particularly those adjacent to occupied areas. The Contractor shall remove dust, dirt, and debris from the areas of work daily.

**105-2.4 SALVAGEABLE MATERIALS AND EQUIPMENT:**

a. The Contractor shall remove all salvageable materials and equipment in a manner that will cause the least possible damage thereto. Salvageable items, which are to be retained by the Owner, shall be carefully handled, stored, and protected.

b. The Contractor shall provide identification tags on all salvageable items boxed or placed in containers, indicating the type, size, and quantity of materials.

c. The Contractor shall exercise extreme care when removing airfield lighting bolting hardware. The existing bolting hardware may have been installed in excess of twenty-five years. The Contractor shall be responsible for the proper removal/repair/replacement of all stripped/broken/damaged fixture bolts or light base bolts or tapped holes in light bases and/or appurtenances encountered during construction. Contractor shall use drilling/tapping/inserts fixture/jigs/tooling and methods approved by the light base manufacturer or an accepted equivalent. Payment for this work shall be considered incidental to the respective fixture and light base pay items provided in the Contract Documents.

**105-2.5 BUILDINGS AND STRUCTURES:**

a. The Contractor shall perform removal operations in existing areas/buildings/structures as indicated and as otherwise required to complete the work.

b. Existing concrete shall be demolished, removed, and disposed. Square, straight edges shall be provided where existing concrete adjoins new work and at other locations where indicated. Existing steel reinforcement shall be protected where indicated; otherwise, it shall be cut off flush with face of concrete.

c. The Contractor shall dismantle structural steel components at field connections and in a manner that will prevent bending or damage.

d. The use of flame-cutting torches will be permitted only when other methods of dismantling are not practical, and when approved in writing by the Engineer.

#### **105-2.6 ELECTRICAL EQUIPMENT AND FIXTURES:**

a. Wiring systems and components shall be salvaged. Loose items shall be boxed and tagged for identification.

b. Primary, secondary, control, communication, and signal circuits shall be disconnected at the point of attachment to their distribution system.

c. The Contractor shall remove and salvage electrical fixtures. Incandescent lamps, mercury-vapor lamps, and fluorescent lamps shall be salvaged, boxed and tagged for identification, and protected from breakage. The fixtures shall be turned over to the Owners Maintenance Department. This Contractor shall legally dispose of concrete bases, etc. off the Airport property.

d. The Contractor shall remove and salvage switches, receptacles, fixtures, transformers, regulators, meters, instruments, cover plates, circuit breakers, panelboards, outlet boxes, and similar items; these items shall be boxed, and tagged for identification according to type and size.

e. The Contractor shall remove and salvage cable tray, wiring ducts and troughs.

f. The Contractor shall remove and dispose of conductors, including insulated wire, nonmetallic sheathed and flexible armored cable.

g. All removed or demolished items not identified for reuse in the work or to be turned over to the Owner, unless noted otherwise, shall be properly and legally disposed of off the airport site.

#### **105-2.7 DEMOLITION OPERATIONS:**

a. Demolition operations shall be conducted to ensure the safe passage of persons to and from facilities occupied and used by the Owner.

b. The sequence of operations shall be such that maximum protection from inclement weather will be provided for materials and equipment in partially dismantled state.

#### **105-2.8 MAINTAINING TRAFFIC:**

a. Demolition operations and removal of debris to disposal areas shall be conducted to ensure minimum interference with roads, streets, walks, and other facilities occupied and used by the Owner.

b. Streets, walks, runways, taxiways and other facilities occupied and used by the Owner shall not be closed or obstructed without written permission.



**105-2.9 UTILITY ADJUSTMENTS:**

a. The Contractor shall adjust utility elevations and or locations as described in the Contract Documents.

**PART 3 – DISPOSAL OF REMOVED/DEMOLISHED MATERIALS**

**105-3.1 GENERAL:**

a. The Contractor shall dispose of debris, rubbish, scrap, and other non-salvageable materials resulting from demolition/removal operations. Removed/demolished materials shall not be stored or disposed of on the project site.

b. Removed/demolished materials shall be properly and legally disposed of off airport property.

c. Removed/demolished debris shall be transported from Owner property and legally disposed at no additional cost to the Owner. The Contractor shall pay permits and fees for disposal.

**105-3.2 RACEWAY DEMOLITION:**

a. Pull wire out of existing raceways. Underground conduit/duct may be abandoned in place when not in an area of excavation. When abandoned in place install waterproof caps/plugs and concrete end of duct markers.

b. Raceways that interfere with new work shall be completely removed.

c. Install 200 pound test pull rope in all conduits/ducts to remain in place. Conduit/duct banks may be abandoned in place if they are not within 24" of any new subgrade material.

d. Dispose of raceway as debris.

**105-3.3 AIRFIELD LIGHTING AND SIGNS:**

a. Carefully remove existing airfield lighting, signs, etc. as noted in the Contract Documents. Properly store all signs, fixtures, base plates and spacer rings for reinstallation or turn over to Owner as noted in the Contract Documents.

b. There may be concrete foundations (anchors) around light bases that extend into the existing asphalt surface as well as concrete backfilled trenches, conduits and duct banks within the asphalt paved areas that may be encountered during the required milling process. To the extent known, these duct banks and conduits are shown on the Contract Documents under/within the paved areas. The milling of any such concrete around the light bases, or over trenches, ducts and conduits shall be considered incidental to the milling pay item.

- c. Existing concrete sign pads, light bases, and raceways shall be demolished, transported from Owner property and legally disposed.
- d. If required by the Contract Documents reinstall signs and fixtures.
- e. Turn over to the Owner all signs, fixtures, base plates and spacer rings not reinstalled as a part of this contract.
- f. Remove existing light bases, L-824 cable, L-830 isolation transformers and associated foundations and legally dispose.

**105-3.4 MANHOLE/HAND HOLE DEMOLITION:**

- a. Manholes, hand holes, pull boxes, etc. shall be removed and disposed of as demolition debris.
- b. Demolished manholes, hand holes, pull boxes, etc. below the new full strength pavement shall be completely removed.
- c. Demolished manholes, hand holes, pull boxes, etc. below shoulder pavement or in turfed areas shall be cut down to a depth of 48" min. below finished grade, filled and compacted in accordance with the Contract Documents.

**105-3.5 GENERAL – CUTTING AND PATCHING:**

- a. Cutting, patching, repairing, and other alteration work shall be done by mechanics skilled in the particular trade or work required.
- b. Salvaged items for reuse shall be as accepted by the Engineer.
- c. Where required to patch or extend existing construction, or both, such alteration work shall match existing exposed surface materials in finish, color, texture, and pattern.

**105-3.6 UTILITY ADJUSTMENTS:**

- a. The Contractor shall provide for all materials, labor, transportation, supervision, services and all incidentals required to adjust the elevation of existing electrical, communication and fiber optic utilities as shown in the Contract Documents. This work shall include hand excavation for the entire length of the utility adjustment described in the Contract Documents and if required the excavation of an additional 250 linear feet on each side of the area shown for adjustment within the Contract Documents. This item includes all concrete, rebar, excavation, forming, backfill, sod, restoration etc., and all materials and appurtenances not specifically identified, but necessary to complete the adjustment to the elevation of an existing utility in accordance with the Contract Documents.

**105-3.7 REQUIRED DEMOLITION OF UNFORESEEN CONDITIONS:**

- a. During the course of construction, unforeseen or unanticipated abandoned electrical bases, structures or components (items) remaining from previous construction and for which no record was available or for which no field observation can confirm, may be uncovered. The

items may be constructed of reinforced concrete, steel or other durable materials. These items may be under the ground surface or may be discovered during the milling process. An example of such a structure would be an old military concrete light base remaining from a previous project, located under shoulder pavement that requires full depth removal. These items are over and above the items shown on the Existing Condition and Demolition drawings which denote demolition. This work shall provide for the removal of the item, backfill and restoration of the surrounding excavation or disturbed areas caused by the removal of the item. The Engineer shall make the final determination if the item in question is unforeseen or unanticipated. The Engineer may adjust the Contract Documents to relocate the new design element to avoid the conflict with the unforeseen or unanticipated item.

#### **PART 4 - METHOD OF MEASUREMENT**

##### **105-4.1 METHOD OF MEASUREMENT:**

a. All Electrical Demolition covered by this item will be paid for at the contract lump sum price, as set forth below, for demolition, alteration and removal complete and in accordance with the Contract Documents.

b. All utility elevation adjustments work covered by this item will be paid for at the contract lump sum price, as set forth below for utility elevation adjustments, complete and in accordance with the Contract Documents and accepted and ready for operation.

c. L-867 and L-868 light bases, not currently designated for demolition/removal, but required to be removed to support paving operations shall be measured by each unit removed, completed and accepted. No separate measurement will be made for the various types and sizes. The Engineer shall designate, as construction progresses which additional light bases are required to be removed to support milling/paving operations.

d. Demolition of unforeseen or unanticipated items shall be measured by the entire item removal, backfill, and restoration, completed, and accepted. Separate measurement will be made for the various types and sizes. Items 250 cubic feet and larger in size shall be paid for in multiples of the two pay items provided. The multiples selected for the oversized unforeseen or unanticipated items shall be those most advantageous to the Owner.

#### **PART 5 - BASIS OF PAYMENT**

##### **105-5.1 BASIS OF PAYMENT:**

a. Payment will be made at the contract lump sum price for the required demolition, alteration and removal. This price shall be full compensation for furnishing all material, equipment and for all preparation, removal of the demolished materials and equipment and for all materials, labor, equipment, tools, services and incidentals necessary or required for the electrical demolition as shown in the Contract Documents. It is the intent of the demolition pay item that all equipment, devices, raceway, pull boxes, manholes, fixtures, wiring, materials, systems, appurtenances, etc., which are no longer required as a result of the project be removed. All waste materials shall be removed from the airport property and legally disposed of by the Contractor. This item includes all backfill, P-610, sod and other materials and

appurtenances not specifically identified, but necessary to complete these items in accordance with the Contract Documents.

**b.** Payment will be made at the contract lump sum price for the required utility elevation adjustments. This price shall be full compensation for furnishing all material, equipment and for all preparation, removal of the demolished materials and equipment and for all labor, equipment, tools and incidentals necessary to complete this item. It is the intent of this pay item that all materials and appurtenances not specifically identified, but necessary to complete the adjustment of the elevation of an existing utility in accordance with the Contract Documents and to the satisfaction of the Utility/Engineer is included. This item includes all backfill, P-610, sod, restoration and other materials and appurtenances necessary to complete these items in accordance with the Contract Documents.

**c.** Payment shall be made at the contract unit price for each additional L-867 or L-868 light base not currently designated on the plans for removal/demolition that is required to be removed. The Contractor shall include in the bid price the necessary hand work required to work around light bases designated to remain in place. Included in this pay item is the uncovering and preparation for extension of the existing raceway and counterpoise connected to the exiting light base. The Engineer shall designate which additional light bases (if any) warrant removal to support milling/paving operations. Additional light bases shall be cored out; alternate method of removal shall be subject to the Engineer's acceptance. This item includes all materials, supervision, labor, transportation, incidentals and services required for the demolition and coring out of additionally designated existing light bases from existing asphalt or concrete pavement in support of the installation of a new light base and fixture. The Engineer shall make final determination if an additional light base shall be removed or if additional handwork is required to complete the item under the respective pay item furnished.

**d.** Payment shall be made at the contract unit price for each unforeseen or unanticipated item requiring demolition not specified or shown on the Contract Documents and approved by the Engineer. This item includes all materials, supervision, labor, transportation, incidentals and services required for the demolition and restoration of the unforeseen or unanticipated item. The Engineer shall make the final determination if the item in question is unforeseen or unanticipated.

Payment will be made under:

|               |                       |                  |
|---------------|-----------------------|------------------|
| Item SP-105-1 | Electrical Demolition | -- Lump Sum (LS) |
|---------------|-----------------------|------------------|

**END OF ITEM SP-105**

## ITEM SP-106 SUBMITTALS, RECORD DOCUMENTS AND MAINTENANCE MANUALS

### PART 1 – DESCRIPTION

#### 106-1.1 GENERAL:

a. The items described in this item are applicable to all electrical work. Electrical work is defined as any work described, included, depicted or contained within the Construction Specifications Institute (CSI) Division 16, 26, 27, 28, 33 or 34 specifications, L-XXX specification series, L- series drawings, E- series drawings, ED- series drawings and all other work within the Contract Documents dealing with the distribution of electrical power/light, electrical/electronic/light monitoring, communication and control.

b. Only Third Party Certified manufacturers, listed in AC 150/5345-53, Appendix 3 Addendum (as required) and meeting the BUY AMERICAN preference requirements can provide equipment and materials specified in the Contract Documents. Documentation certifying compliance with the BUY AMERICAN preference rules for AIP (in accordance with Title 49, United State Code [USC], 1990; later designated as 49 USC §50101) shall be included with each equipment and material submittal.

c. The requirements for submittals, record drawings and Operation and Maintenance (O&M) Manuals supplement the requirements and are in addition to the requirements of the general conditions, special provisions or other parts of the Contract Documents. Where the contract general conditions, special provisions or other parts of the Contract Documents also apply, the stricter of the documents shall govern.

#### 106-1.2 SCOPE:

a. This item includes the requirements for submittals, record documents and operation and maintenance (O&M) manuals. All submittals and O & M Manuals shall be submitted in book form as described by this item.

### PART 2 – REQUIREMENTS FOR SUBMITTALS

#### 106-2.1 DATA, SHOP DRAWINGS AND SAMPLES:

a. Shop drawings are drawings, diagrams, illustrations, schedules, performance charts, brochures and other data which are prepared by the Contractor or any subcontractor, manufacturer, supplier or distributor, and which illustrate some portion of the work.

b. Submittal data for electrical materials and equipment shall consist of shop drawings and/or catalog cuts showing technical data as necessary to evaluate the material or equipment, to include dimensions, wiring diagrams, performance curves, ratings, control sequence and other descriptive data necessary to describe fully the item proposed and its operating characteristics.

c. Prior to the installation of any material or equipment and **within 30 days of notice to proceed**, the Contractor shall provide to the Engineer all submittals as required by the Contract

Documents. The submittals shall be complete and made in **ONE** submission in the format required by this item. Partial submission will only be considered under special circumstances.

d. The Contractor shall perform no portion of the work, nor purchase any materials, requiring submission until the Engineer has accepted the relevant submittal/shop drawing/sample. Should the Contractor commence any work or install any materials prior to the receipt of an accepted submittal/shop drawing/sample the Contractor is deemed to be proceeding at the Contractor's own risk.

e. Samples are physical examples furnished by the Contractor to illustrate materials, equipment or workmanship, and to establish standards by which the work will be judged. Each sample shall be accompanied by the manufacturer's instructions regarding installation, operation and maintenance and shall be identified by item number, and specification.

f. By approving and submitting Shop Drawings and Samples, the Contractor thereby represents that the Contractor has determined and verified all field measurements, field construction criteria, materials, catalog numbers and similar data and that the Contractor, has checked and coordinated each Shop Drawing and Sample with the requirements of the work of the Contract Documents.

g. Unless otherwise stated in the Contract Documents the Engineer will review Data, Shop Drawings and Samples within thirty (30) days after receipt, but only for general conformance with the design concept of the project and general compliance with the information given in the Contract Documents. The Engineer's review of a separate item shall not indicate approval of an assembly in which the item functions.

h. The Engineer reserves the right to reject any and all equipment, materials and procedures, which, in the Engineer's opinion, does not comply with the system design and the standards and codes specified herein.

i. The Engineer shall be the sole judge of whether the proposed "acceptable equivalent" is suitable for use in the work.

j. The Contractor shall make any corrections required by the Engineer and shall resubmit the required number of corrected shop drawings or new samples until accepted. The Contractor shall direct specific attention in writing on resubmitted shop drawings to revisions other than the corrections requested by the Engineer on previous submissions.

k. If the Contractor desires to substitute materials or deviate from the requirements of the Contract Documents, the Contractor shall separately submit all substitute materials and deviations to the Contract Documents. Such submittals shall be clearly and boldly marked "**SUBSTITUTION**" or "**DEVIATION TO CONTRACT DOCUMENTS**". If the substitution or deviation is accepted, the Contractor will not be excused from producing work in conformity with the Contract Documents. If a trial use establishes the work does not comply with the Contract Documents, the Contractor shall take such action as the Engineer determines necessary.

l. The Engineer's approval of shop drawings or samples shall not relieve the Contractor of responsibility for any deviation from the requirements of the Contract Documents unless the Contractor has informed the Engineer in writing of such deviation at the time of submission and the Engineer has provided written acceptance to the specific deviation. The Engineer's

approval shall not relieve the Contractor from responsibility for errors or omissions in the shop drawings or samples.

**m.** The submittals will be checked only for general conformance with the design concept of the project and general compliance with the information given in the Contract Documents. Any action noted by the Engineer on the submittal is subject to the requirements of the Contract Documents.

**n.** The Contractor is responsible for dimensions, quantities, fabrication processes and methods of construction, coordination of the Contractor's work with that of all trades. The Contractor shall be responsible for satisfactory performance of the Contractor's work and supplying a complete and fully functional system. The Contractor shall confirm and correlate dimensions at the jobsite.

**o.** Samples shall be submitted upon request or as required by the Contract Documents.

**p.** The Contractor shall maintain an accurate updated "SUBMITTAL REGISTER" containing the status of all submittals. The log shall be updated on a weekly basis. The Contractor shall bring the submittal register to each progress meeting. The submittal register shall contain the following information:

- 1.** Submittal description and number assigned.
- 2.** Contract Document required submittal date.
- 3.** Submittal electrical system type; airfield lighting or airfield lighting control or airfield lighting vault power system.
- 4.** Reference Item # (L-108, 26 05 33.13, etc).
- 5.** Date sent to Engineer.
- 6.** Date returned to Contractor from Engineer.
- 7.** Submittal status.
- 8.** Date of resubmittal and return – as applicable.
- 9.** Date material released for fabrication.
- 10.** Projected date of fabrication.
- 11.** Projected date of delivery to site.

#### **106-2.2 ELECTRONIC SUBMITTAL:**

**a.** Electronic submittal shall consist of a single PDF document separated and book marked per each electrical system, per specification section as described in section 106-2.2.d. containing all materials for each item to be installed in the project.

**b.** Electronic submittals larger than 10 megabyte shall be separated into smaller PDF files for transmission to make up a single PDF document. Each separate PDF shall be book marked per specification section containing all materials for each item to be installed in the project.

**c.** Each PDF for electronic transmission shall be clearly labeled as transmittal X of Y per submittal section.

**d.** Provide separate submittal PDF's for each of the following electrical system types:

1. One PDF submittal set for the Airfield Lighting System. All components to be included in the contract for lighting, conduit, grounding and miscellaneous systems shall be included.
2. One PDF submittal set for the Airfield Lighting Control System.
3. One PDF submittal set for the Airfield Lighting Vault Modifications,

#### **106-2.3 SUBMITTAL CONTENTS:**

a. First sheet(s) in shall be a cover sheet containing the project information and location. See appendix A at the end of SP-106 specification.

b. The second sheet shall be a table of contents identifying the specification section(s) being submitted.

c. Third sheet shall be prepared and filled out by Contractor and shall list project address of airport, Engineer, Contractor & subcontractor(s).

d. Fourth sheet shall also be filled out by Contractor and list project schedule information.

e. Submittals consisting of marked catalog sheets or shop drawings shall be inserted in the PDF in specification number/section order. Submittal data shall be presented in a clear and thorough manner. **Original PDF catalog sheets are preferred. Scans of catalog sheets are acceptable provided they are equivalent in quality to the original document.**

f. Clearly and boldly mark each page to identify pertinent products or models applicable to this project. Indicate all optional equipment and delete non-pertinent data. **Markings shall be boldly and clearly made with arrows or circles (highlighting is not acceptable).**

g. All sheets within the submittal shall be oriented in a portrait setting. Mixing of portrait and landscape oriented standard 8.5 X 11 sheets shall not be permitted.

h. All sheets contained within the PDF shall be straight. Sheets scanned on a drastic angle making the submittal page difficult to read shall not be permitted and shall result in the submittal being rejected.

i. Some catalog sheets have colored backgrounds or colored text. When scanned these items appear black and obscure the printed text and markings by the contractor. For this reason all scans of sheets with colored backgrounds or text shall be made in color. Sheets that are not readable due to obscured information shall be rejected.

j. Submittals for components of electrical equipment and systems shall identify the equipment for which they are a component part on each submittal sheet.

k. Shop Drawings: Drawings to include identification of project and names of Engineer(s), General Contractor, subcontractor and supplier, data, number sequentially and indicate the following:

1. Fabrication and erection dimensions.
2. Arrangements and sectional views.



3. Necessary details, including complete information for making connections with other work.
4. Kinds of materials and finishes.
5. Descriptive names of equipment.
6. Modifications and options to standard equipment required by the work.
7. Leave blank area, size approximately 5 x 5 inches, near title block for the Engineer's stamp imprint.
8. Point-to-point wiring diagrams.
9. Conduit/duct/raceway rough-in drawings.
10. See specific items of the Contract Documents for further requirements.

I. Submittal Product Data - Technical data is required for all items as called for in the Contract Documents regardless if item furnished is as specified.

1. Submit technical data verifying that the item submitted complies with the requirements of the Contract Documents. Technical data shall include manufacturer's name and model number, dimensions, weights, electrical characteristics, and clearances required. Indicate all optional equipment and changes from the standard item as called for in the Contract Documents. Furnish drawings, or diagrams, dimensioned and in correct scale, covering equipment, showing arrangement of components and overall coordination.
2. In order to facilitate review of product data, insofar as practicable, they shall be noted, indicating by cross reference the Contract Documents, note, drawing number and/or specification number and item paragraph numbers where item(s) occur in the Contract Documents.
3. See specific items in Contract Documents for further requirements.

#### **106-2.4 PROCESSING SUBMITTALS:**

- a. Submit the PDF submittal and shop drawings per Specification section
- b. The Contractor shall review, stamp with the Contractor's approval and submit to the Engineer, eight (8) prints of shop drawings, eight (8) copies of submittal books and three (3) sets of samples where required, as described in this item,
- c. The General Contractor shall review the submittal books before submitting to the Engineer. No request for payment will be considered until the submittal book has been reviewed and submitted for review by the Engineer.
- d. Product Data: For standard manufactured materials, products and items, submit one (1) copy or sets of data (per book).
- e. Shop Drawings for custom fabricated items and systems:
  1. Initially submit a transparency (suitable for reproduction) together with four (4) prints made the transparency.
  2. When submittal is acceptable, furnish one (1) print per book made from the accepted transparency.
- f. Acceptance: When returned to Contractor, the front of each submittal will be marked with the Engineer's stamp. If box marked "Submit Specified Item", or "Rejected" or "Revise and

Resubmit" is checked, submittal is not accepted and Contractor is to correct and resubmit as noted. If submittal is marked "Make Correction Noted" Contractor may begin construction utilizing the submitted item with corrections made. **However, the corrected submittal must be resubmitted for record keeping purposes.** Contractor is to comply with notation making necessary corrections on submittal and resubmit for final record.

g. If the submittal is marked "No Exception Taken" the Engineer took no exceptions to the submitted items.

h. Corrections or comments made on the shop drawings during this review do not relieve Contractor from compliance with requirements of the Contract Documents. This check is only for review of the general conformance with the design concept of the project and general compliance with the information given in the Contract Documents. The Contractor is responsible for; confirming and correlating all quantities and dimensions; selecting fabrication processes and techniques of construction; coordinating the Contractor's work with that of all other trades and performing all work in a safe and satisfactory manner.

i. Note that the acceptance of shop drawings or other information submitted in accordance with the requirements contained in this item, does not assure that the Engineer or any other Owner's Representative attests:

1. To the dimensional accuracy or dimensional suitability of the material or equipment involved.
2. The ability of the material or equipment involved to perform as designed.
3. The mechanical or electrical performance of equipment.

j. Approval of shop drawings does not invalidate the Contract Documents if in conflict; unless a letter requesting such a change is submitted and accepted on the Engineer's letterhead.

#### **106-2.5 DELAYS:**

a. Contractor is responsible for delays in job progress accruing directly or indirectly from late submissions or resubmissions of shop drawings, or product data.

#### **106-2.6 RE-SUBMITTALS:**

a. The Engineer shall be reimbursed for engineering costs to review resubmittals subsequent to the second submittal.

### **PART 3 – RECORD DOCUMENTS**

#### **106-3.1 PROGRESS AND RECORD DRAWINGS:**

a. Keep two sets of blue line prints on the job and neatly mark up design drawings each day as components are installed. **Red pencil shall be used for additions and green pencil for deletions. Use of "white out" or similar products is not permitted.**

**b.** All items on record drawings shall be shown in actual location installed. Drawings shall be inspected weekly, by the Engineer, for compliance and accuracy. Project payments shall be withheld if the marked-up drawings are not current.

**c.** Change the equipment schedules to agree with items actually furnished.

**d.** All hand holes, pull boxes, junction can plazas, manholes, underground ducts, conduits, force mains, drains, ground grids, etc. (all underground utilities) installed by this Contractor or located by this Contractor during the construction of this project shall be surveyed and recorded on the record documents. Record utility locations in state plane grid coordinates. Provide north-south, east-west coordinate and elevation.

**e.** At the end of the project, all changes shall be reviewed by the Electrical Contractor. The drawings shall be marked "As Built" and dated and signed by the Electrical Contractor's Superintendent.

**f.** Prior to request for final payment, furnish a set of "As Built" prints to the Engineer for review. The final prints shall be professionally drafted to indicate "As Built" conditions to the Engineer. The prints shall be stamped "As Built" signed and dated by the Contractor. The Contractor's signature shall certify to the Engineer that the "As Built" drawings are accurate and complete.

**g.** The Contractor's failure to produce representative "As Built" drawings in accordance with requirements of this item, shall be cause for the Engineer to produce such "As-built" drawings and the Contractor shall reimburse Engineer for all costs to produce a representative set of "As Built" drawings.

## **PART 4 – OPERATION AND MAINTENANCE MANUALS**

### **106-4.1 REQUIREMENTS FOR OPERATION AND MAINTENANCE MANUALS:**

**a.** Two (2) CD's shall be provided with all O & M Manual information included in PDF format. The Operation and Maintenance Manual shall be submitted as a PDF document. As a minimum, the Operation and Maintenance Manual shall contain the following:

- 1.** Safety precautions used while maintaining the equipment.
- 2.** Theory of circuit and system operation.
- 3.** Complete schematic and interconnecting wiring diagrams.
- 4.** Complete parts list with each circuit component keyed to designations assigned on schematics and wiring diagrams. Complete information shall be given for each part to permit ordering for replacement purposes. This information shall include the components rating, name of manufacturer and the manufacturer's part number.
- 5.** Recommended preventative maintenance, including care, cleaning lubrication, service intervals, etc.
- 6.** Troubleshooting procedures.
- 7.** Physical characteristics (weight, size, mounting dimensions, etc.).
- 8.** Installation instructions.
- 9.** Operating instruction.

10. Recommended spare parts and usage for a one (1) year period.
11. Submit for checking purposes a specific set of written operating instructions on each item, which requires instructions to operate. After acceptance, provide one copy for insertion in each Operation and Maintenance Manual.
12. Submit for acceptance maintenance information consisting of manufacturer's printed instructions and parts list for each major item of equipment. After acceptance, insert information in each Operations and Maintenance Manual. Detailed schematic diagrams shall be furnished for all electrical/electronic equipment.
13. Bill of materials.
14. Physical layout plans.
15. Equipment supplier list.
16. Panel schedules shall be submitted with the respective panel data.
17. Special instructions
18. Service maintenance contracts including the name, address and 24 hour phone number and contact of manufacturer's authorized Repair Company.

There shall be no "Black Boxes" for which there are no schematic/wiring diagrams.

#### **106-4.2 OPERATION AND MAINTENANCE MANUALS:**

a. O & M Manuals shall consist of a PDF document. The manual shall be bookmarked into sections by Table of Contents, Project Contacts, and each technical specification section. Shop drawings, product data and/or catalog sheets shall be inserted into the O & M PDF by technical specification.

b. Provide correct designation of project on first page of each technical specification submittal section. Description sheet shall be white with black letters, minimum of 11" high and full width of page. Description shall describe project and match project Contract Document description. Description to include submittal type.

c. All documents inserted into the O & M Manual PDF, including but not limited to catalog cut sheets, theory of operation, and all other related product data, shall be the original documentation from the manufacturer.

d. Any scanned documentation shall be legible, and have the correct orientation in the pdf so the document can be read horizontally, from left to right, in line with all other pages in the O & M Manual. Failure to meet these requirements will result in the O & M Manual's rejection. Upon rejection any errata shall be fixed and resubmitted to the engineer for approval.

e. Provide one O & M Manual for the following electrical systems required by the contract documents:

1. System Descriptions to be included in future submittal.

#### **106-4.3 OPERATION AND MAINTENANCE MANUAL CONTENTS:**

a. Operation & Maintenance Manuals shall include:

1. First sheet in PDF shall be the cover sheet.

2. The second sheet shall be a table of contents.
3. The third sheet shall be filled out by Contractor and shall list project address of airport, Engineer, Contractor & subcontractor(s).
4. The fourth sheet shall also be filled out by Contractor and list project schedule information showing project "as completed".
5. Shop Drawings: Shop drawings shall be a copy of the final and accepted shop drawings submitted as required in Item SP-106-2. These shall be inserted into the PDF in specification number/section order. Each catalog sheet shall clearly identify where the product is used and the Contract Document identification for equipment.
6. Product data and/or catalog sheets shall be a copy of the final and accepted submittal submitted as required in Item SP-106-2. These shall be inserted into the PDF in specification number/section order. Each catalog sheet shall clearly identify where the product is used and the Contract Document identification for equipment.
7. Warranty/Guarantee: Provide copy of warranty/guarantee and letters of certification, in respective location in O & M Manual PDF. Original warranty/guarantee shall be incorporated into separate project warranty PDF with warranty/guarantees provided for other items and submitted for Engineer review.

**106-4.4 PROCESSING OPERATION AND MAINTENANCE MANUALS:**

a. The General Contractor shall review the manuals before submitting to the Engineer. Submit one CD containing the PDF set of O& M Manuals to the owner and one CD containing the PDF set to the Engineer.

**106-4.5 DELAYS:**

a. Contractor is responsible for delays in job progress accruing directly or indirectly from late submissions or resubmissions of the Operation and Maintenance Manuals.

**106-4.6 RE-SUBMITTALS:**

a. The Engineer shall be reimbursed the cost to review Operation and Maintenance Manual re-submittals subsequent to the second submittal.

**PART 5 – METHOD OF MEASUREMENT**

**106-5.1 METHOD OF MEASUREMENT**

a. The items described in Item SP-106 are incidental to other pay items and shall not be measured for payment.

## **PART 6 – BASIS OF PAYMENT**

### **106-6.1 BASIS OF PAYMENT**

a. No direct payment shall be made for the work described in Item SP-106. The work described in Item SP-106 is incidental to other items and shall be paid for in the respective bid item of which it is a component part.

## **PART 7 – SUBMITTAL LIST**

### **106-7.1 SUBMITTAL LIST**

a. The following is a list of materials expected by the Engineer to be incorporated within the project. This list is not all inclusive as the Contractor's chosen means and methods will have a direct impact on the materials used.

b. It is the intent of this item that all materials used or incorporated into this project is submitted to the Engineer for review.

c. The Contractor shall maintain a submittal register in accordance with Part 2 of this item. The Submittal register shall include all items used or incorporated into the project.

#### **LIST TO FOLLOW:**

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                   |
|-----------------------------|------------------------------|--------------------------------------------------------------------------------|
| SP-100                      | 1.4-A                        | CERTIFICATION OF ELECTRICAL CONTRACTOR'S EXPERIENCE                            |
| SP-100                      | 1.4-A                        | COPY OF ELECTRICAL CONTRACTOR'S STATE ELECTRICAL LICENSE                       |
| SP-100                      | 1.5-C                        | CERTIFICATION OF ELECTRICAL SUPERINTENDENT'S EXPERIENCE                        |
| SP-100                      | 1.5-C                        | ELECTRICAL SUPERINTENDENT'S RESUME' AND COPY OF JOURNEYMAN ELECTRICIAN LICENSE |
| SP-100                      | 1.5-C                        | ELECTRICAL SUPERINTENDENT'S REFERENCES, AIRPORT NAME, CONTACT AND PHONE NUMBER |
| SP-100                      | 1.5-F                        | COPY OF EACH JOURNEYMAN ELECTRICIAN'S LICENSE                                  |
| SP-100                      | 2.4-B                        | EXISTING FACILITIES INVESTIGATION MEMO                                         |
| SP-100                      | 2.4-E                        | SCHEDULED SHUTDOWNS                                                            |
| SP-100                      | 2.5-A                        | WARRANTIES                                                                     |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                                                                                                             |
|-----------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SP-100                      | 2.6-A                        | O&M MANUALS                                                                                                                                                              |
| SP-100                      | 2.7-L                        | TEST REPORTS                                                                                                                                                             |
| SP-100                      | 2.7-P                        | RECORD DOCUMENTS                                                                                                                                                         |
| SP-100                      | 3.1-A                        | GENERAL LIGHTING PROVISIONS SUPERVISION & ATTENDANCE                                                                                                                     |
| SP-100                      | 3.1-K                        | COLOR CODE PAINT                                                                                                                                                         |
| SP-100                      | 3.7-C                        | LOAD BALANCE TEST                                                                                                                                                        |
| SP-100                      | 3.8-B                        | PHENOLIC NAMEPLATES 1 TO 1 SCALE DETAIL OF EACH NAMEPLATE                                                                                                                |
| SP-100                      | 3.12-A                       | WELDER QUALIFICATIONS                                                                                                                                                    |
| SP-100                      | 3.12-A                       | WELDING PROCEDURES                                                                                                                                                       |
| SP-100                      | 3.12-A                       | WELDER QUALIFICATION TEST                                                                                                                                                |
| SP-100                      | GENERIC REQUIREMENT          | PERMANENT BLACK MARKER                                                                                                                                                   |
| SP-100                      | GENERIC REQUIREMENT          | SELF-ADHESIVE CLEAR PRINTED LABELS W/ BLACK TYPED LETTERS                                                                                                                |
| SP-104                      | 1.3-A                        | SAFETY PLAN                                                                                                                                                              |
| SP-104                      | 1.9-A                        | WRITTEN VERIFICATION PROVIDING PROOF OF CORRESPONDENCE WITH REPRESENTATIVES OF ALL UTILITIES/AGENCIES TO LOCATE ALL EXISTING UTILITIES/SYSTEMS WITHIN THE PROJECT LIMITS |
| SP-104                      | 4.1-A                        | TAGGING AND LOCKOUT PROCEDURE                                                                                                                                            |
| SP-104                      | 4.1-D                        | FAA 45 DAY WRITTEN NOTICE OF WORK                                                                                                                                        |
| SP-104                      | 4.2-F                        | TEMPORARY AIRFIELD LIGHTING/POWER PLAN                                                                                                                                   |
| SP-104                      | 4.2-L                        | INSULATION RESISTANCE TESTS ON EXISTING CIRCUITS TO BE MODIFIED BY CONTRACTOR                                                                                            |
| SP-105                      | 1.2-A                        | VERIFICATION OF EXISTING CONDITIONS (SD-21)                                                                                                                              |
| SP-105                      | 1.2-D                        | INSULATION RESISTANCE TESTS ON EXISTING CIRCUITS TO BE MODIFIED BY CONTRACTOR                                                                                            |
| SP-105                      | 1.8-A                        | DEMOLITION TASK PLAN                                                                                                                                                     |
| SP-105                      | 1.8-B                        | EXISTING CABLE PROTECTION PLAN                                                                                                                                           |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                                                            |
|-----------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| SP-105                      | 2.4-B                        | IDENTIFICATION TAGS FOR SALVAGEABLE ITEMS                                                                               |
| L-107                       | 2.1-D                        | MATERIALS AND EQUIPMENT FOR CONSTRUCTION                                                                                |
| L-107                       | 2.2                          | L-807(L) MAIN WINDCONE STYLE 1B, SIZE 2, LED INTERNALLY LIGHTED WIND CONE, WITH L-810(L) OBSTRUCTION LIGHT              |
| L-107                       | 2.2                          | SUPPLEMENTAL WIND CONE TYPE L-806(L), STYLE 1A, SIZE 1 – WITH ALL INSTALLATION COMPONENTS, ACCESSORIES AND INCIDENTALS. |
| L-107                       | 2.7                          | PAINTING COLORS AND PROCEDURES                                                                                          |
| L-107                       | 3.8                          | CHAIN AND PADLOCK                                                                                                       |
| L-108                       | 2.1-D                        | ALL MATERIALS AND EQUIPMENT USED TO CONSTRUCT THIS ITEM SHALL BE SUBMITTED TO THE ENGINEER                              |
| L-108                       | 2.1-F                        | WARRANTY                                                                                                                |
| L-108                       | 2.2                          | #8 AWG, 5KV L-824 AIRFIELD LIGHTING CABLE (6.6A)                                                                        |
| L-108                       | 2.2                          | L-824 SECONDARY CONDUCTORS, EXTENSION CORDS, ETC.                                                                       |
| L-108                       | 2.2                          | CERTIFIED COPY OF FACTORY TEST REPORTS FOR ALL CABLE PROVIDED                                                           |
| L-108                       | 2.2                          | THWN-2 WIRE FOR CIRCUITS UP TO 600V                                                                                     |
| L-108                       | 2.2                          | COPPER BUS BAR BY SIZE, TYPE AND USE                                                                                    |
| L-108                       | 2.2                          | CONTROL CABLE                                                                                                           |
| L-108                       | 2.2                          | FIBER OPTIC CABLE                                                                                                       |
| L-108                       | 2.2                          | ALCS CONTROL CABLE                                                                                                      |
| L-108                       | 2.3                          | #2 AWG SOLID COPPER COUNTERPOISE                                                                                        |
| L-108                       | 2.3                          | 3/4" DIAMETER BY 10' LONG COPPER CLAD STEEL SECTIONAL GROUND ROD                                                        |
| L-108                       | 2.3                          | GROUND ROD COUPLINGS                                                                                                    |
| L-108                       | 2.3                          | GROUND ROD DRIVING STUDS                                                                                                |
| L-108                       | 2.3                          | GROUNDING CONDUCTORS FOR VOLTAGE CIRCUITS                                                                               |



| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                                      |
|-----------------------------|------------------------------|---------------------------------------------------------------------------------------------------|
| L-108                       | 2.4-A                        | CAST SPLICE INCLUDING BLOCKING COMPOUND, CONNECTORS, SPLICE AND CRIMPING TOOL                     |
| L-108                       | 2.4-B                        | FIELD-ATTACHED PLUG-IN SPLICE                                                                     |
| L-108                       | 2.4-D                        | TAPED OR HEAT-SHRINKED SPLICE METHODS                                                             |
| L-108                       | 2.4-D                        | LIST OF APPROVED CRIMP CONNECTORS AND CRIMPING TOOLS                                              |
| L-108                       | 2.4-D                        | COMPRESSION BUTT SPLICES INCLUDING MANUFACTURER'S INSTRUCTIONS                                    |
| L-108                       | 2.4-D                        | COMPRESSION LUGS INCLUDING MANUFACTURER'S INSTRUCTIONS                                            |
| L-108                       | 2.4-D                        | C-TAPS INCLUDING MANUFACTURER'S INSTRUCTIONS                                                      |
| L-108                       | 2.4-D                        | COMPRESSION TOOLING, CALIBRATION CERTIFICATE, PROCEDURES AND MANUFACTURER'S RECOMMENDED PRACTICES |
| L-108                       | 2.4-D                        | EXOTHERMIC CONNECTIONS INCLUDING ALL MOLDS, SHOTS AND MANUFACTURER'S INSTRUCTIONS                 |
| L-108                       | 2.4-D                        | BASE CAN GROUND CLAMPS                                                                            |
| L-108                       | 2.5                          | AIRFIELD LIGHTING CABLE SPLICER QUALIFICATIONS                                                    |
| L-108                       | 2.5                          | AIRFIELD LIGHTING CABLE SPLICING DEMONSTRATIONS                                                   |
| L-108                       | 2.5                          | LIST OF APPROVED AIRFIELD LIGHTING CABLE SPLICERS AND ASSIGNED SPLICER IDENTIFICATION NUMBER.     |
| L-108                       | 2.5                          | EXOTHERMIC WELDER QUALIFICATIONS                                                                  |
| L-108                       | 2.5                          | SAMPLE EXOTHERMIC WELD DEMONSTRATIONS                                                             |
| L-108                       | 2.5                          | LIST OF APPROVED EXOTHERMIC WELDERS.                                                              |
| L-108                       | 2.5                          | ALL GROUNDING AND BONDING TOOLS AND EQUIPMENT                                                     |
| L-108                       | 2.5                          | GROUNDING AND BONDING QUALIFICATIONS                                                              |
| L-108                       | 2.5                          | SAMPLE CONNECTOR/LUG DEMONSTRATIONS                                                               |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                    |
|-----------------------------|------------------------------|---------------------------------------------------------------------------------|
| L-108                       | 2.5                          | LIST OF APPROVED GROUNDING AND BONDING PERSONNEL                                |
| L-108                       | 2.8                          | CABLE IDENTIFICATION TAGS AND TYRAP                                             |
| L-108                       | 2.8                          | CABLE ID TAG STAMPED SAMPLES                                                    |
| L-108                       | 2.9                          | ALL ELECTRICAL TAPES                                                            |
| L-108                       | 2.9                          | HEAT SHRINK TUBING                                                              |
| L-108                       | 2.10                         | ELECTRICAL COATING                                                              |
| L-108                       | 2.11                         | INSULATION RESISTANCE TESTS OF EXISTING CIRCUITS                                |
| L-108                       | 2.12                         | FIELD LIGHTING ARRESTOR ASSEMBLY CONNECTIONS                                    |
| L-108                       | 2.13                         | DETECTABLE WARNING TAPE                                                         |
| L-108                       | 3.1                          | L-823 CONNECTORS INCLUDING MANUFACTURER'S INSTRUCTIONS                          |
| L-108                       | 3.1                          | ADDITIONAL L-823 CONNECTORS TURNED OVER TO OWNER                                |
| L-108                       | 3.2                          | PULL ROPE                                                                       |
| L-108                       | 3.2                          | MOISTURE-SEAL TAPE                                                              |
| L-108                       | 3.2                          | CABLE PULLING LUBRICANT                                                         |
| L-108                       | 3.2                          | CABLE PULLING TENSION VALUES                                                    |
| L-108                       | 3.5-D                        | CABLE STRIPPING AND PENCILING TOOLS                                             |
| L-108                       | 3.9                          | EXOTHERMIC WELD PHOTOGRAPHIC LOG AND CD                                         |
| L-108                       | 3.9                          | EXOTHERMIC WELD MANUFACTURER'S INSPECTION / ACCEPTANCE CRITERIA                 |
| L-108                       | 3.10-H                       | LIGHTING AND POWER CIRCUIT CONTINUOUS OPERATION TEST                            |
| L-108                       | 3.10                         | COUNTERPOISE CONTINUITY TEST                                                    |
| L-108                       | 4.3                          | GROUND ROD INSPECTION PITS                                                      |
| L-108                       | GENERIC REQUIREMENT          | MECHANICAL LUGS AND TORQUING REQUIREMENTS                                       |
| L-108                       | GENERIC REQUIREMENT          | TORQUE WRENCH, CALIBRATION CERTIFICATE AND MANUFACTURER'S RECOMMENDED PRACTICES |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                             |
|-----------------------------|------------------------------|------------------------------------------------------------------------------------------|
| L-108                       | GENERIC REQUIREMENT          | WIRE NUTS                                                                                |
| L-108                       | GENERIC REQUIREMENT          | TERMINAL BLOCKS                                                                          |
| L-108                       | GENERIC REQUIREMENT          | SS WIRE MESH STRAIN RELIEF BASKETS FOR 5 KV CABLES                                       |
| L-108                       | GENERIC REQUIREMENT          | ELECTRICAL JOINT COMPOUND                                                                |
| L-110                       | 2.1                          | SPECIFICATIONS AND DATA                                                                  |
| L-110                       | 2.2                          | RIGID GALVANIZED STEEL (RGS) CONDUIT                                                     |
| L-110                       | 2.3-B                        | SCHEDULE 40 PVC CONDUIT (2",3",4")                                                       |
| L-110                       | 2.3-B                        | SCHEDULE 40 PVC END BELLS, FITTINGS, TERMINATIONS, CLEANER AND SOLVENT CEMENT (2",3",4") |
| L-110                       | 2.4                          | SPLIT CONDUIT (4")                                                                       |
| L-110                       | 2.5                          | CONDUIT SPACERS AND DUCT PLUGS                                                           |
| L-110                       | 2.6                          | CONCRETE                                                                                 |
| L-110                       | 2.7                          | PRECAST CONCRETE STRUCTURES                                                              |
| L-110                       | 2.8                          | FLOWABLE BACKFILL                                                                        |
| L-110                       | 2.9                          | DETECTABLE MARKER TAPE                                                                   |
| L-110                       | 2.9                          | LONG RADIUS RGS BENDS                                                                    |
| L-110                       | GENERIC REQUIREMENT          | WEATHERPROOF CONDUIT HUBS                                                                |
| L-110                       | GENERIC REQUIREMENT          | LOCKNUTS                                                                                 |
| L-110                       | GENERIC REQUIREMENT          | GROUNDING BUSHINGS W/ INSULATED THROAT                                                   |
| L-110                       | GENERIC REQUIREMENT          | BUSHINGS W/ INSULATED THROAT                                                             |
| L-110                       | GENERIC REQUIREMENT          | CONDULETS, COVERS AND GASKETS                                                            |
| L-110                       | GENERIC REQUIREMENT          | EXPANSION FITTINGS                                                                       |
| L-110                       | GENERIC REQUIREMENT          | CABLE RACKS, SUPPORTS, TIES AND STRAPS                                                   |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                                                                     |
|-----------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| L-110                       | GENERIC REQUIREMENT          | CONDUIT THREAD COMPOUND                                                                                                          |
| L-110                       | GENERIC REQUIREMENT          | COLD GALVANIZING COMPOUND                                                                                                        |
| L-110                       | GENERIC REQUIREMENT          | ASPHALTUM PAINT                                                                                                                  |
| L-110                       | GENERIC REQUIREMENT          | EXPANSION FITTINGS                                                                                                               |
| SP-111                      | 2.1-A                        | SUBMIT ALL MATERIALS, TEST EQUIPMENT, WRITTEN PROCEDURES, EQUIPMENT CALIBRATION CERTIFICATES FOR PERFORMING THE FOLLOWING TESTS: |
| SP-111                      | 2.1-C                        | FIELD TEST PLAN                                                                                                                  |
| SP-111                      | 2.3-A                        | QUALIFICATIONS OF TESTING PERSONNEL                                                                                              |
| SP-111                      | 2.3-E                        | QUALIFICATIONS OF CALIBRATION LAB                                                                                                |
| SP-111                      | 3.1                          | COUNTERPOISE TESTING                                                                                                             |
| SP-111                      | 3.1                          | EXOTHERMIC WELD TESTS                                                                                                            |
| SP-111                      | 3.2                          | GROUND ROD TESTING                                                                                                               |
| SP-111                      | 3.3                          | FIELD LIGHTNING ARRESTOR ASSEMBLY TESTING                                                                                        |
| SP-111                      | 3.4                          | EXISTING AIRFIELD LIGHTING CIRCUIT INSULATION RESISTANCE TESTING                                                                 |
| SP-111                      | 3.5                          | NEW AIRFIELD LIGHTING CIRCUIT CONTINUITY TESTING                                                                                 |
| SP-111                      | 3.6                          | NEW AIRFIELD LIGHTING CIRCUIT INSULATION RESISTANCE TESTING                                                                      |
| SP-111                      | 3.7                          | CONCLUDING AIRFIELD LIGHTING CIRCUIT TESTS                                                                                       |
| SP-111                      | 3.8                          | CONSTANT CURRENT REGULATOR CALIBRATION AND PERFORMANCE TEST PROCEDURE AND TEST EQUIPMENT                                         |
| SP-111                      | 3.9                          | FIXTURE WIRING SEQUENCE TESTING                                                                                                  |
| SP-111                      | 3.10                         | SHORT CIRCUIT/COORDINATION/DEVICE EVALUATION/ARC FLASH ANALYSIS                                                                  |
| SP-111                      | GENERIC REQUIREMENTS         | TESTING REQUIRED BY EQUIPMENT MANUF.                                                                                             |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SP-111                      | GENERIC REQUIREMENTS         | TORQUING OF ELECTRICAL TERMINATIONS                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| SP-111                      | GENERIC REQUIREMENTS         | TORQUING OF ANCHOR BOLTS                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L-115                       | 2.2                          | CONCRETE STRUCTURES                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| L-115                       | 2.4                          | L-867D JUNCTION CAN, L-867D JUNCTION CAN PLAZA – WITH ALL INSTALLATION COMPONENTS, ACCESSORIES AND INCIDENTALS.                                                                                                                                                                                                                                                                                                                                                               |
| L-115                       | 2.7                          | FRAMES AND COVERS                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| L-125                       | 1.1                          | INSTALLATION OF AIRPORT LIGHTING SYSTEMS                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L-125                       | 2.1                          | SHOP DRAWINGS OF EACH AIRFIELD LIGHTING COMPONENT, INDICATING FAA APPROVAL, SHALL BE SUBMITTED TO THE RPR FOR REVIEW AND APPROVAL AND BE APPROVED PRIOR TO ORDERING ANY MATERIALS FOR THIS ITEM. THIS SUBMITTAL SHALL INCLUDE THE PROPOSED METHOD OF INSTALLATION FOR ALL AIRFIELD LIGHTING COMPONENTS. THE SUBMITTAL SHALL INCLUDE DATA ON ALL COMPONENT PARTS OF THE ITEM OR SYSTEM, AND SHALL INCLUDE THE MANUFACTURERS LIST OR RECOMMENDED SPARE PARTS FOR ONE YEARS USE. |
| L-125                       | 2.1                          | AIRFIELD LIGHTING FIXTURE MANUFACTURER QUALIFICATIONS                                                                                                                                                                                                                                                                                                                                                                                                                         |
| L-125                       | 2.1                          | GUARANTEE                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| L-125                       | 2.7                          | LISTED LIGHTS IN TABLE – WITH ALL INSTALLATION COMPONENTS, ACCESSORIES AND INCIDENTALS.                                                                                                                                                                                                                                                                                                                                                                                       |
| L-125                       | 2.8                          | L-858(L) Y, R, L SIGN – WITH ALL INSTALLATION COMPONENTS, ACCESSORIES AND INCIDENTALS.                                                                                                                                                                                                                                                                                                                                                                                        |
| L-125                       | 2.8                          | MISC SIGN ITEMS                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| L-125                       | 2.12                         | LIGHT BASE AND TRANSFORMER HOUSING L-867                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| L-125                       | 2.13                         | L-830 ISOLATION TRANSFORMERS                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| L-125                       | 3.1                          | INSTALLATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| <b>Spec. Section Number</b> | <b>Spec Paragraph Number</b> | <b>Submittal Description</b>                                                                                                                             |
|-----------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| L-125                       | 5.1                          | EACH COMPONENT SHALL BE IDENTIFIED WITH THE SPECIFIC PAY ITEM OF WHICH IT IS A COMPONENT PART. COMPLETE ASSEMBLIES SHALL BE SUBMITTED FOR EACH PAY ITEM. |

PENSACOLA INTERNATIONAL AIRPORT  
RUNWAY 8-26 REHABILITATION

SUBMITTALS, RECORD DOCUMENTS  
AND MAINTENANCE MANUALS  
ITEM SP-106

**APPENDIX 1**

**“EXAMPLE OF COVER SHEET”**

**AIRPORT NAME**

**AIRPORT OWNER**

**PROJECT NAME**

**AIRPORT PROJECT NUMBER**

**AIP#**

**DOT#**

**AIRFIELD LIGHTING SYSTEM  
OPERATION AND MAINTENANCE MANUAL**

**OWNER MANUAL #1**

Note:

Submittal books and spline information similar.

**END OF ITEM SP-106**



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**ITEM SP-109S-16990 ALLOWANCES**

**I. RELATED DOCUMENTS**

1. Drawings and general provisions of the Contract, including Contractual Conditions and other Specification Sections, apply to this Section.

**II. SUMMARY**

1. This Section includes administrative and procedural requirements governing allowances.
2. All applicable allowances are listed on the Bid Form and contained with the noted specification.

**III. SELECTION AND PURCHASE**

1. At the earliest practical date after award of the Contract, advise the Engineer of the date when the final selection and purchase of each product or system described by an allowance must be completed to avoid delaying the Work.
2. At the Engineer's request, obtain proposals for each allowance item for use in making final selections. Include recommendations that are relevant to performing the Work.
3. Purchase products and systems selected by the Engineer from the designated supplier(s).

**IV. SUBMITTALS**

1. Submit proposals for purchase of products or systems included in allowances.
2. Submit invoices or delivery slips to show the actual services provided or quantities of materials delivered to the site for use in fulfillment of each allowance.

**V. ALLOWANCES**

1. Use the allowance only as directed for the Owner's purposes and only with the Engineer's written permission that indicate amounts to be charged to the allowance.
2. The Contractor's related costs in support for the allowance items and for products and equipment ordered by the Owner under the allowance including delivery, installation, taxes, insurance, equipment rental, and similar costs shall be considered an incidental part of the Contractors support pay item.

3. Allowance pay items are direct pass through invoices from the specified vendor, to contractor(s) to Owner. The contractor is not allowed any mark up or profit on the allowance pay items. The selected vendor shall detail the scope and work performed on the invoices.
4. Change Orders authorizing use of funds from the allowance shall include Contractor's related costs but shall not include overhead and profit margins, which shall be included in the Contractor's Base Bid allowance support pay item.

**VI. UNUSED ALLOWANCES**

1. At Project close-out, credit all unused allowance monies to the Owner by Change Order.

**VII. EXAMINATION**

1. Examine products covered by an allowance promptly upon delivery for damage or defects. Report findings to Engineer in writing.

**VIII. PREPARATION**

1. Coordinate services, materials and their installation for each allowance with related materials and installations to ensure that each allowance item is completely integrated and interfaced with related work.

**END OF ITEM SP-109S-16990**

**ITEM SP-110S-02310 INSTALLATION OF UNDERGROUND ELECTRICAL DIRECTIONAL  
BORE DUCT**

**PART 1 – GENERAL**

**02310-1.1 SECTION INCLUDES:**

**A.** The work specified in this section consists of furnishing and installing underground utilities using the method of installation commonly referred to as directional boring. This work shall include all supervision, services, equipment, materials, labor, area restoration, environmental protection and incidentals for the complete and functional installation of the directional bore duct as shown in the Contract Documents. A **2 AWG** solid copper counterpoise conductor shall be installed with and external to the directional bore duct. Quantity of counterpoise shall be per the details. **2 AWG** solid copper counterpoise shall be paid for by the linear foot in accordance with Item L-108.

**02310-1.2 RELATED DOCUMENTS AND SECTIONS:**

- A.** Item L-110; Airport Underground Electrical Duct Banks and Conduits.
- B. Section 555; Directional Bore - FDOT Specifications for Road and Bridge Construction**, latest edition.

**02310-1.3 SUBMITTALS:**

- A.** Work Plan: Prior to beginning work, the Contractor shall submit to the Engineer a general work plan outlining the procedure and schedule to be used to execute the work task. Plan shall document the thoughtful planning required to successfully complete the work.
- B.** Demonstration of compliance with Item L-110 and State DOT requirements.
- C.** Equipment: Contractor shall submit specifications on directional drilling equipment to be used in the work to ensure that the equipment will be adequate to complete the project.
- D.** Material: Specifications on material to be used shall be submitted to Engineer. Material shall include the pipe, conduit, fittings, splicing/coupling method, means of transition from HDPE to PVC and any other item, which is to be an installed component of the project.

**02310-1.4 WARRANTY:**

- A.** All equipment and materials furnished and installed under this section shall be guaranteed against defects in materials and workmanship for a period of twelve (12) months from final acceptance by the Owner. The defective materials and/or equipment shall be repaired or replaced, at the Owner's discretion, with no additional cost to the Owner.

## **PART 2 – EQUIPMENT REQUIREMENTS**

### **02310-2.1 GENERAL:**

**A.** The directional drilling equipment shall consist of a directional drilling rig of sufficient capacity to perform the bore and pullback the pipe, a drilling fluid mixing and delivery system of sufficient capacity to successfully complete the crossing, a guidance system to accurately guide boring operations and trained and competent personnel to operate the system. All equipment shall be in good, safe operating condition with sufficient supplies, materials and spare parts on hand to maintain the system in good working order for the duration of the project.

### **02310-2.2 GUIDANCE SYSTEM:**

**A.** The guidance system shall be of a proven type and shall be setup and operated by personnel trained and experienced with this system. The Operator shall be aware of any magnetic anomalies and shall consider such influences in the operation of the guidance system if using a magnetic system.

**B.** The locating and tracking system must provide information on:

1. Clock and pitch information.
2. Depth.
3. Transmitter temperature.
4. Battery status.
5. Position (X,Y).
6. Azimuth, where direct overhead readings (walkover) are not possible (i.e. subaqueous or limited access transportation facility).
7. Ensure proper calibration of all equipment before commencing directional drilling operation.

**C.** Take and record alignment readings or plot points such that elevations on top of and offset dimensions from the center of the product to a permanent fixed feature are provided. Such permanent fixed feature must have prior approval of the Engineer. Provide elevations and dimensions at all bore alignment corrections (vertical and horizontal) with a maximum distance of 5 feet between points. Provide additional points when requested by the Engineer. Provide a sufficient number of elevations and offset distances to accurately plot the vertical and horizontal alignment of the installed product.

### **02310-2.3 CONDUIT/DUCT:**

**A.** The directional bore conduit/duct shall be UL Listed Schedule 80 HDPE electrical conduit as described in NFPA 70, Article 353 suitable for placement with directional boring systems or accepted equivalent. HDPE conduit shall comply with UL Category EAZX and ANSI/UL 651A.

**B.** Conduit to meet all applicable ASTM testing and be UL listed. Conduit to be chamfered on inside diameter male ends to minimize cable damage. The bending radius for 4" PVC pipe should be no less than 100' as recommended by the PVC Pipe Industry. Installation bend radius should be kept at a minimum to reduce the tensile pull forces displaced onto the pipe. Pipe installation and joint assembly shall be per manufacturer recommendations.

**02310-2.4 CONCRETE:**

**C.** Concrete shall conform to item P-610, Structural Portland Cement Concrete or **FDOT "Standard Specification for Road and Bridge Construction" specification Section 346**, latest edition, concrete mix with Type I/II cement using 1-inch maximum size coarse aggregate (#57 stone) with a minimum 28 day compressive strength of 4,000 psi. Where reinforced duct banks are specified, reinforcing steel shall conform to ASTM A 615 grade 60. Concrete and reinforcing steel are incidental to the respective pay item of which they are a component part.

**A.** High early strength P-610 concrete will be required for use with ducts, conduits, base cans and other work as directed by the Engineer, in existing operational areas and within the safety areas of operational pavement to comply with the project phasing and scheduling requirements and where the pavement is required to open to aircraft at the end of the designated work period. The extra cost for the high early strength P-610 concrete shall be considered incidental to the respective pay item of which it is a component part.

**02310-2.5 COPPER COUNTERPOISE WIRE**

**A.** The 2 AWG solid copper counterpoise wire shall comply with Item L-108. The [2] AWG solid copper counterpoise wire shall be installed with and externally to the directionally bored duct.

**PART 3 – EXECUTION**

**02310-3.1 GENERAL:**

**A.** The Engineer shall be notified 48 hours in advance of starting work. The Directional Bore shall not begin until the Engineer is present at the job site and agrees that proper preparations for the operation have been made. The Engineer's approval for beginning the installation shall in no way relieve the Contractor of the ultimate responsibility for the satisfactory completion of the work as authorized under the Contract.

**02310-3.2 BORING PATH REPORT:**

**A.** Furnish a bore path report to the Engineer within seven days of the completion of each bore path. Include the following in the report:

1. Location of project and financial project number including the permit number when assigned.
2. Name of person collecting data, including title, position and company name.
3. Investigation site location (Contract plans station number or reference to a permanent structure within the project right-of-way).

4. Identification of the detection method used.
5. Elevations and offset dimensions as required in 02310-2.2.

**B.** Provide the Engineer a complete set of as-built plans showing all bores (successful and failed) within 30 calendar days of completing the work. As-built plans shall be PDF files or AutoCAD files, in the same scale as the Contract Plans, and formatted on **22 inch by 34 inch** sheets. Ensure that the plans are dimensionally correct copies of the Contract Plans and include pavement/area plan and profile, cross-section, boring location and subsurface conditions as directed by the Engineer. The plans must show elevations referenced to a permanent feature (edge of pavement, manhole, mast arm foundation, manhole inlet cover, head wall, etc). Specific plans content requirements include but may not be limited to the following:

1. Provide a horizontal plan and a profile plan for each bore path at 5 foot intervals, to an accuracy of 1 inch. Horizontal and vertical plan dimensions shall be referenced to fixed permanent monuments/references at the bore location.

2. In addition to the directional bore duct information, show the ground or pavement surface and crown elevation of each position along the bore path to an accuracy of 1 inch and at the ends and other exposed locations. Profile plans for bore paths crossing the runway, taxiway or roadway shall denote stationing of the crossing on the Contract Plans.

3. If, during boring, an obstruction is encountered which prevents completion of the installation in accordance with the design location and specification, and the product is left in place and taken out of service, show the failed bore path along with the final bore path on the plans. Note the failed bore path as "Failed Bore Path - Taken Out of Service". Also show the name of the utility owner, location and length of the drill head and any drill stems not removed from the bore path.

4. Show the top elevation, diameter and material type of all utilities encountered and physically observed during the work. For all other obstructions encountered during a subsoil investigation or the installation, show the type of material, horizontal and vertical location, top and lowest elevation observed, and note if the obstruction continues below the lowest point observed.

5. Include bore notes on each plan stating the final bore path diameter, product diameter, drilling fluid composition, composition of any other materials used to fill the annular void between the bore path and the product, or facility placed out of service. Note if the product is a casing as well as the size and type of carrier pipes placed within the casing as part of the Contract work.

### **02310-3.3 SPLICING, COUPLING AND TERMINATION:**

**A.** The HDPE duct shall "relax" for 48 hours after installation prior to splicing, coupling or termination.

**B.** Only Engineer accepted methods shall be used to splice, couple or terminate the HDPE duct.

- i. Dura-line Shur-Lok II Couplings,
- ii. Carlon HDPE to PVC Couplings
- iii. 8 inch minimum length Morris Compression Couplings
- iv. Fusion couplings
- v. OUC use: Carlon or Certainteed Restrained Joint.

**C.** All splicing, coupling and termination of HDPE or PVC ducts shall be performed and completed in strict accordance with the manufacturer's instructions and recommendations.

**D.** Splices and couplings shall be permitted only where connection to conventionally installed ducts or manholes are required. Fusion coupling is the only method of splicing ducts subject to installation pull tensions.

**E.** The duct shall be properly reamed in accordance with the applicable specification or details. prior to assembly.

**F.** All HDPE and PVC duct/conduit splicing, coupling and termination fittings and materials shall be listed or approved.

#### **02310-3.4 Ground Movement**

**A.** Before excavation, establish control points for measuring vertical movement of the runway, taxiway, roadway bore path under obstruction at 10 feet intervals along the centerline and 10 feet each side of the pipeline. A land surveyor licensed in the state in which the project is located shall monitor these points daily until construction is complete.

**B.** Cease trenchless operations when measured movement exceeds 0.02 feet. Determine cause of settlement and repair as necessary. Modify trenchless methods as needed.

#### **02310-3.5 Bore Depth Minimum Cover**

|           |                            |               |
|-----------|----------------------------|---------------|
| <b>A.</b> | Bore Diameter              | Minimum Cover |
|           | >2 inches to 6 inches      | 6 feet        |
|           | >6 inches up to 14 inches  | 12 feet       |
|           | >14 inches up to 24 inches | 15 feet       |
|           | >24 inches up to 48 inches | 25 feet       |

**B.** Contractor shall survey to verify bottom elevation of ditches, swales, ponds, rivers etc to determine bottom of obstruction. Bottom is considered natural earth less any settlement or sludge.

**C.** Contractor shall maintain 25-foot minimum cover between the bottom (not including sediments) of the pond, river or other body of water.

**D.** Proposed bore alignment shall maintain a minimum clearance of 10 foot horizontal and 5 foot vertical between bore and any obstruction.



**E.** Blow-out of any drill mud shall be immediately corrected and cleaned up by the Contractor at no additional expense to the Owner.

**PART 4 – METHOD OF MEASUREMENT**

**02310-4.1 METHOD OF MEASUREMENT:**

**A.** The items described in this item are incidental to other items and shall not be measured for payment.

**PART 5 – BASIS OF PAYMENT**

**02310-5.1 BASIS OF PAYMENT:**

**A.** No direct payment shall be made for the work described in this item. All directional bore costs shall be considered incidental to the respective L-108 or L-110 pay items.

**END OF ITEM SP-110S-02310**

## ITEM SP-111 AIRFIELD ELECTRICAL INSTALLATION TESTING

### PART 1 – DESCRIPTION

#### 111-1.1 GENERAL:

**a.** This item shall consist of performing all tasks necessary to prove proper performance of electrical and airfield lighting systems. Airfield lighting systems include airfield signage systems, approach lighting systems and other associated NAVAIDS.

**b.** Drawings, general provisions and special provisions of the Contract Documents, including Contractual Conditions and Division 1 Specification sections apply to this Item.

**c.** This section describes the electrical quality control testing furnished by the Contractor. The Contractor shall provide all testing as required by this item, including retesting of failed items. All items furnished and/or installed by the Contractor shall be tested in accordance with this item.

**d.** The Contractor shall furnish all equipment, labor, supervision, transportation, materials and appliances necessary or required for testing the airport lighting systems before and after installation. The Contractor shall perform all tests in the presence of the Engineer. The Contractor shall demonstrate the electrical characteristics to the satisfaction of the Engineer. All costs for testing are incidental to the pay item for the respective item being tested.

**e.** Any system installation errors or discrepancies of installation not in conformance with the Contract Documents shall be corrected at no additional cost to the Owner. Equipment, cables and other components which do not satisfy the test requirements of this Item shall be removed and replaced with new equipment, cables and other components which do satisfy the requirements of this item with no additional cost to the Owner. Any necessary retesting shall be performed at no additional cost to the Owner.

**f.** There are no approved "repair" procedures for items that have failed testing other than complete replacement. The Contractor shall replace all items that may be damaged during testing at no additional cost to the Owner.

**g.** Weather information necessary to complete the testing forms may be obtained from the listed website [http://www.faa.gov/air\\_traffic/weather/asos/](http://www.faa.gov/air_traffic/weather/asos/) by double clicking on the respective state and airport ID.

## **PART 2 – EQUIPMENT AND MATERIALS**

### **111-2.1 SUBMITTALS:**

**a.** Materials and equipment covered by this item shall be subject to acceptance through manufacturer's certification of compliance with the applicable specification when requested by the Engineer. All equipment, materials, methods and record keeping procedures shall be submitted to the Engineer for review.

**b.** The submittal shall include a list of the proposed test and measurement equipment to be used, the test and measurement equipment serial number, the current calibration certificate for each piece of equipment and a written description of the proposed test. A copy of the test and measurement equipment manufacturer's recommended test procedure is acceptable as a written description of the proposed test.

**c.** Submit a proposed field test plan, 60 days prior to testing the installed system. No field test shall be performed until the test plan is approved. The test plan shall consist of complete field test procedures including tests to be performed, test equipment required, and tolerance limits. The Contractor shall submit a complete list of required tests and a proposed schedule and duration of the tests. The schedule shall be integrated with the overall project schedule. The Contractor shall update the testing schedule on a monthly basis for the duration of the project.

**d.** All test results shall be submitted in the Operation and Maintenance (O&M) Manual. See Item L-106 for submittal requirements.

**e.** The Engineer reserves the right to reject any and all equipment, materials or procedures, which, in the Engineer's opinion, does not meet the system design and the standards and codes, specified herein.

### **111-2.2 SAFETY:**

**a.** Dangerous voltages are present during the system testing. The Contractor is solely responsible for the safety of all personnel involved in the testing activities and shall ensure that all testing areas are secured to prevent the entry of unqualified personnel.

**b.** Observe all safety instructions or precautions recommended by testing equipment manufacturer.

**c.** The Contractor shall perform a "Hazard/Risk Evaluation" of each test site and procedure. The Contractor shall provide a job briefing and install the necessary barriers and barricades at the respective approach boundaries prior to starting the testing.

**d.** The Contractor shall provide and use safety devices such as rubber gloves, protective barriers, and danger signs to protect and warn personnel in the test vicinity. The Contractor shall replace any device or equipment which is damaged due to improper test procedures or handling.

**e.** In addition to compliance with the Contractors safety program, the Contractor shall comply with the following documents/programs:

- 1.** OSHA, 29CFR1910 Occupational Safety and Health Standards

2. OSHA 29CFR1926 Safety and Health regulations for Construction
3. NFPA 70B Recommended Practice for Electrical Equipment Maintenance
4. NFPA 70E Standard for Electrical Safety in the Workplace
5. AC 150/5340-26 Maintenance of Airport Visual Aid Facilities
6. Owner's safety program.

### **111-2.3 QUALIFICATIONS:**

a. Each person performing testing shall have a minimum of 5 years (2,000 hours per year) of successful testing experience either constructing or maintaining airfield lighting systems. At least 3 of these years of experience shall have occurred in the last 5 years on projects of comparable size and complexity. Person performing testing shall be certified as a "Qualified Person". The definition of a Qualified Person is defined in OSHA 1910.399 and NFPA 70E (Article 100 & 110.2(1)). The certification shall be signed by the Contractor's Electrical License qualifier. Equivalent experience and training may be considered by the Engineer. Person performing testing may substitute one year of education in qualified electrical curriculum for one-half year electrical experience for a maximum of 2,000 hours of the required 10,000 hours of experience. Records of person performing testing experience demonstrating compliance with these requirements shall be submitted to the Engineer for verification and review prior to starting work. Experience records shall include contact information for references (name, title and phone number) from airports where the proposed person performing testing has completed the required electrical experience. Work shall not start prior to approval of the person performing testing, unless under the direct supervision of the Contractor's Electrical License qualifier.

b. The person performing the tests shall be a "qualified person" as defined in NFPA 70E and OSHA 29 CFR 1926.32(l).

c. The person performing the tests shall be thoroughly familiar with the test equipment used in the test performed and shall be sufficiently experienced to detect equipment abnormalities or questionable data obtained during the test procedure.

d. Each piece of test and measurement equipment utilized to demonstrate compliance with the Contract Documents shall be calibrated. The level of calibration shall be "*NIST traceable, Z540 Calibration with Data*". The test and measurement equipment calibration shall be valid for one year. At any time, the Engineer may require the Contractor to have a piece of test and measurement equipment recalibrated. Should the Engineer have any reason to doubt the accuracy of a piece of test and measurement equipment, even within current calibration, the test and measurement equipment shall be removed from the project. The Engineer shall determine which tests shall be redone due to the malfunctioning equipment.

e. The test and measurement equipment calibration lab as a minimum shall meet the requirements of ANSI/NCCL Z-540 and shall be A2LA accredited.

f. Electrical and photometric testing of airfield lighting systems shall be performed by a firm with demonstrated capability for field measurement of electrical characteristics of airfield lighting circuits and regulators and of the photometric performance of airfield lighting fixtures and signs. Because of the correlation between the electrical performance of airfield lighting circuits and the photometric performance of the airfield lighting fixtures, all work shall be done by a single firm. The firm shall have experience in evaluating the test results against FAA standards and manufacturers' performance criteria. See below for listed firms or approved equivalent.

- g. Two possible electrical and photometric testing firms are listed below:

Lean Photometrics, LLC  
113 S. 25th Street  
Phoenix, Arizona 85034  
Contact: Doron Lean  
(602) 244-1076  
[dlean@leanphotometrics.com](mailto:dlean@leanphotometrics.com)

MALMS Navaid  
481 Pleasant Hill Road  
Rutherfordton, NC 28139  
Contact: Bryan Fountain  
(855) 440-0541  
[bfountain@malms.aero](mailto:bfountain@malms.aero)

#### **111-2.4 TESTING COORDINATION:**

- a. After installation of all required components and before final acceptance, in accordance with project phasing and schedule make required tests to determine proper function of all circuits.
- b. Ten days prior to any testing the Contractor shall notify the Engineer of the need to perform testing. A mutually agreed upon time and date for each test will be set. All requirements under this item shall be coordinated with the Engineer.
- c. The Contractor shall perform the necessary inspection and tests for some items concurrently with the installation because of subsequent inaccessibility of some components.
- d. Photometric testing of a runway lighting system shall be conducted prior to opening the runway. Photometric testing of a taxiway lighting system may occur after all fixtures have been installed, provided the lighting system passes a visual inspection prior to opening to air traffic.
- e. The Contractor shall ensure that all test and measurement equipment, accessories and qualified personnel are available prior to scheduling tests.
- f. Forty-eight (48) hours prior to each test the Contractor shall send the Engineer a written confirmation that the scheduled test will occur at the proposed date and time. Any rescheduling of the test shall be coordinated with the Engineer.
- g. The Engineer shall be provided reasonable opportunity to observe all tests. Testing shall not proceed until the Engineer or designated representative is present to observe the testing, or until the Engineer has provided written instruction to proceed with the testing in the Engineer's absence.
- h. All test results shall be recorded by the Contractor and witnessed by the Engineer. Test results shall be submitted in the O&M Manuals in accordance with Item L-106.
- i. The Contractor is reminded that due to phasing constraints the photometric testing organization or any other testing organization/groups may have multiple mobilizations required to test airfield lighting systems as each area is complete. For projects completed in phases, the respective required testing must be completed by phase.

#### **111-2.5 TEST AND MEASUREMENT EQUIPMENT:**

- a. Electrical test and measurement equipment shall be an off the shelf item, not a one of a kind prototype. The test and measurement equipment shall be manufactured by a firm regularly engaged in the manufacture of precision electrical test and measurement equipment for the

industrial, commercial and utility marketplace. Typical accuracy for test equipment shall be  $\pm 2\%$  of instrument reading and shall be TRUE RMS.

**b.** The Engineer shall confirm that the test and measurement equipment proposed for use by the Contractor is suitable for the intended use. The Engineer shall have sole discretion to determine if the test and measurement equipment is suitable for the intended use.

**c.** The airfield lighting photometric test and measurement equipment is utilized in a specific airfield lighting application and is not required to meet Item A. above.

**d.** The ground point for the cable test equipment shall be the vault ground bus. The vault ground bus, vault ground system, airport ground grid and counterpoise system shall be complete prior to cable testing.

### **PART 3 – CONSTRUCTION METHODS**

#### **111-3.1 COUNTERPOISE TESTING:**

**a.** Continuity of counterpoise system shall be checked by visual inspection as construction progresses, prior to work being covered up.

**b.** Verification of counterpoise system continuity shall be checked by visual inspection at accessible locations during normal inspections.

**c.** Verification of counterpoise system continuity to the vault grounding system shall be tested with a micro-ohmmeter. Micro-Ohmmeter shall be Megger Ductor DLRO-10 or accepted equivalent. Resistance readings shall be corrected to standard temperature.

**d.** Should the counterpoise or ground grid conductors be damaged or are suspected to be damaged by construction activities (in the opinion of the Engineer) the Contractor shall test the conductors for continuity with a micro-ohmmeter. The conductors shall be isolated such that there is no parallel path. Alternatively, the Contractor may conduct tests and through mathematical computations prove the continuity of the conductors.

**e.** Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.

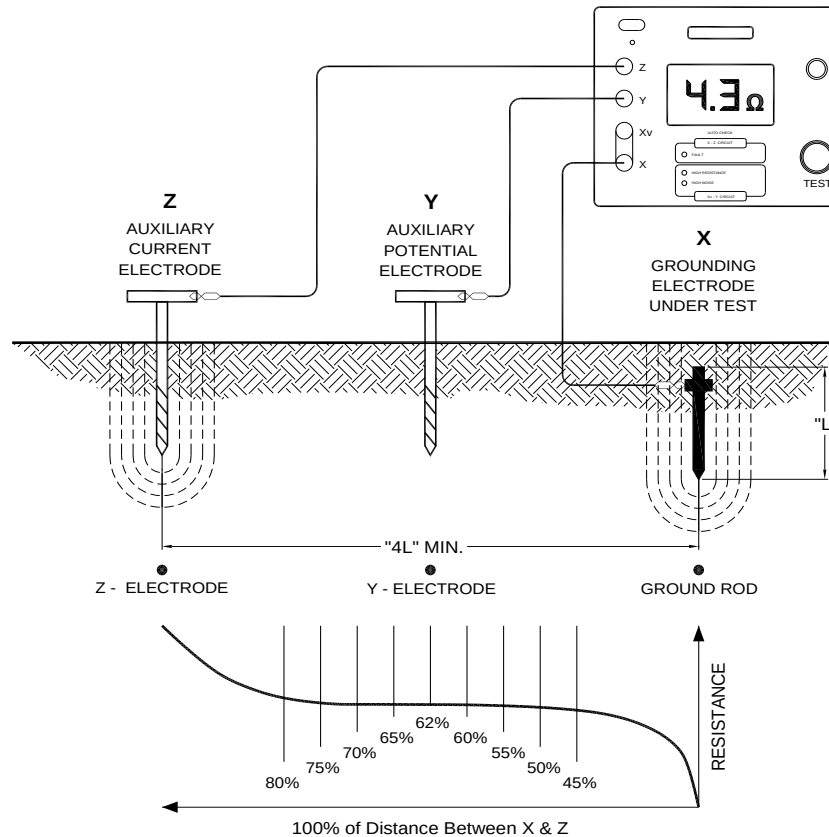
**f.** Record test results on the record drawing set indicating point to point readings.

#### **111-3.2 GROUND ROD TESTING:**

**a.** Contractor shall provide equipment and personnel to measure the resistance to earth of each ground rod installed. Earth resistance measurement tests shall adhere to recommendations of IEEE Standard 142, latest edition.

**b.** Earth resistance measurements shall be made in normally dry conditions not less than 48 hours after the last rainfall.

c. Initially the Contractor shall test each ground rod in accordance with the following instructions. The horizontal distance between the Ground Electrode Under Test (Electrode X) and the Auxiliary Current Electrode (Electrode Z) shall be 4 times the depth (length) of the ground electrode under test.



d. The three electrodes shall be installed in a straight line in a direction away from other underground metallic items. The test leads shall be separated.

e. The first test shall be conducted with the Auxiliary Potential Electrode (Electrode Y) at 45% of the distance between the X and Z electrodes. The earth resistance shall be measured and recorded at the 45% position. The Y electrode will then be moved 5% closer to the Z electrode to the 50% position and the earth resistance shall be measured and recorded. The previous steps will be repeated at 5% distance intervals through the 80% position. Graph the resultant points on the form provided at the end of this item.

f. The above test method shall be utilized at each new ground rod installation until otherwise determined by the Engineer. The Engineer will use the acquired data to establish a revised distance ratio between the X and Z electrode.

g. Once the new electrode distance ratio is set by the Engineer, the Contractor may perform the tests with a single test position for electrode Y.

h. Earth resistance tests shall be performed as each ground rod is installed. Earth resistance tests shall be conducted prior to the connection of any grounded, grounding or counterpoise

conductors. Ground rods not in compliance with the not to exceed maximum earth resistance shall have an additional 10' section(s) added until the required earth resistance is achieved.

i. Should the Contractor install 100 feet of sectional ground rod (ten rods) at a single location and not achieve the required earth resistance, the Contractor shall stop installing rods at this specific location and contact the Engineer for resolution.

### **111-3.3 FIELD LIGHTNING ARRESTOR ASSEMBLY TESTING:**

a. Each field lightning arrestor assembly (FLA) shall receive an inspection upon receipt by the Contractor.

b. The Contractor shall group the FLAs by circuit. Mark each FLA with circuit designation and unique number.

c. Each FLA shall be tested for continuity between the airfield lighting series circuit L-823 connectors. A reading of 200 milliohms or less is acceptable.

d. Each FLA shall have its internal resistance tested between the airfield lighting series circuit L-823 connector and the grounding screw. A reading of less than 10 gigohms is cause for rejection. Any FLA testing less than 10 gigohms shall be rejected.

e. The FLA internal resistance test shall be performed with a megohmmeter providing a test voltage of 1,000 volts DC. The output of the megohmmeter shall be direct reading in megohms and shall have an upper range limit in excess of 10 gigohms.

f. The Contractor shall then interconnect each FLA within that circuit to the other. The Contractor shall interconnect the ground screws on each unit with a #12AWG bare copper conductor. The Contractor shall then megger test the FLA circuit assembly and record the reading.

g. The Contractor shall repackage the FLA units, maintaining the circuit designations. The FLA shall be stored in a clean dry location until installation in the base cans.

h. Record data on insulation resistance test form provided at the end of this item.

### **111-3.4 EXISTING AIRFIELD LIGHTING CIRCUIT INSULATION RESISTANCE TESTING:**

a. Whenever the scope of work requires connection to an existing circuit, or should the scope of work involve working adjacent to or near an existing circuit, and the work could have any impact on the existing circuit, the existing circuit's insulation resistance shall be tested. *The Contractor shall measure and record the insulation resistance of ALL airfield lighting circuits prior to starting any work.* The test shall be performed prior to any activity which could possibly affect or impact the respective circuit. The Contractor shall record the results on forms included at the end of this item.

b. The existing airfield lighting circuit insulation resistance test shall be performed with a megohmmeter providing a test voltage of 1,000 volts DC. The output of the megohmmeter shall be direct reading in megohms and shall have an upper range limit in excess of 10 gigohms.

c. Observe required safety precautions.



d. Connect the test instrument in accordance with the manufacturer's recommendations. The insulation resistance test shall be conducted for two (2) minutes). Record data at 30 second intervals on the form provided at the end of this Item. The insulation resistance measurement results shall be corrected to 20° Centigrade (68° Fahrenheit).

e. When the work effecting the circuit is complete (or at end of shift prior to energization), the circuit's insulation resistance shall be checked again, in the presence of the Engineer. The Contractor shall record the results on forms included at the end of this item. The second reading shall be equal to or greater than the first reading or the Contractor shall make the necessary repairs to the circuit to bring the second reading above the first reading.

f. All repair costs including a complete replacement of the L-823 connectors, L-830 transformers and L-824 cable, if necessary, shall be borne by the Contractor.

g. Should the Contractor fail to perform the tests prior to starting work, the Contractor shall be responsible to bring the impacted circuits to a minimum 50 megohm insulation resistance value or higher value in conformance with any previous airport test records.

#### **111-3.5 NEW AIRFIELD LIGHTING CIRCUIT CONTINUITY TESTING:**

a. After new cable installation and before the operating test or connection to an existing system, the airfield lighting series circuit cable system shall be given a circuit internal resistance (continuity) test.

b. This test shall be conducted at the proposed point of connection to the Constant Current Regulator and shall be used to prove continuity of the entire circuit. The Contract Documents shall be reviewed to confirm proper circuit routing, connection and installation of necessary components prior to testing.

c. No Field Lighting Arrestor Assemblies shall be installed in the circuit during the internal resistance (continuity) test. No airfield lighting fixtures or signs shall be connected to the secondary of the L-830 isolation transformers internal resistance (continuity) test.

d. The L-824C series circuit cable, complete with all L-823 connectors and L-830 isolation transformers installed, shall be tested for continuity. The cable, connectors and transformers shall be in their final configuration, ready for connection to the constant current regulator (CCR).

e. Observe required safety precautions.

f. The circuit's internal resistance (continuity) test shall be performed using an ohmmeter with sufficient output to measure the circuit's internal resistance. The expected internal circuit resistance should be between 20 ohms and 70 ohms.

g. Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.

h. Upon successful completion of circuit's internal resistance (continuity) test, record the measured circuits internal resistance on the cable test form included at the end of this item.

#### **111-3.6 NEW AIRFIELD LIGHTING CIRCUIT INSULATION RESISTANCE TESTING:**

- a. Upon successful completion of the airfield lighting circuit's internal resistance (continuity) test perform an airfield lighting circuit insulation resistance test.
- b. In addition to successfully completing the airfield lighting circuit internal resistance (continuity) test, all prerequisite conditions required prior to performing the airfield lighting circuit internal resistance (continuity) test shall be performed prior to starting the airfield lighting circuit insulation resistance test.
- c. Airfield lighting circuit insulation resistance tests shall be made in normally dry conditions not less than 48 hours after the last rainfall.
- d. The airfield lighting circuit insulation resistance test shall be performed with a megohmmeter providing a test voltage of 1,000 volts DC. The output of the megohmmeter shall be direct reading in megohms and shall have an upper range limit in excess of 10 gigohms.
- e. Observe required safety precautions.
- f. Ground all other conductors within the conduits or ducts containing the conductor under test.
- g. Connect the test instrument in accordance with the manufacturer's recommendations. The insulation resistance test shall be conducted for ten (10) minutes). Record data at the required intervals on the form provided at the end of this Item. The insulation resistance measurement results shall be corrected to 20<sup>o</sup> Centigrade (68<sup>o</sup> Fahrenheit).
- h. The expected insulation resistance for the completed circuit should be in excess of 500 megohms.
- i. Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.
- j. Upon successful completion of circuit's insulation resistance test, record the measured circuits internal resistance on the cable test form included at the end of this item.

**111-3.7 CONCLUDING AIRFIELD LIGHTING CIRCUIT TESTS:**

- a. Upon successful completion of the airfield lighting circuit install and connect all airfield lighting fixtures, signs and perform an airfield lighting circuit insulation resistance test. Conduct test similar to test performed above (111-3.6).
- b. Airfield lighting circuit insulation resistance tests shall be made in normally dry conditions not less than 48 hours after the last rainfall.
- c. The airfield lighting circuit insulation resistance test shall be performed with a megohmmeter providing a test voltage of 1,000 volts DC. The output of the megohmmeter shall be direct reading in megohms and shall have an upper range limit in excess of 10 gigohms.
- d. Observe required safety precautions.

- e. Ground all other conductors within the conduits or ducts containing the conductor under test.
- f. Connect the test instrument in accordance with the manufacturer's recommendations. The insulation resistance test shall be conducted for ten (10) minutes). Record data at the required intervals on the form provided at the end of this Item. The insulation resistance measurement results shall be corrected to 20° Centigrade (68° Fahrenheit).
- g. The expected insulation resistance for the completed circuit should be in excess of 500 megohms.
- h. Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.
- i. Upon successful completion of circuit's insulation resistance test, record the measured circuits internal resistance on the cable test form included at the end of this item.
- j. Install respective circuit designated FLA assemblies. Use the sum of reciprocals rule to calculate the final circuit insulation resistance. Perform insulation resistance test on completed circuit in accordance with above criteria. Record final circuit insulation resistance value on form provided at the end of this Item.

#### **111-3.8      CONSTANT CURRENT REGULATOR (CCR) TESTS:**

- a. Each constant current regulator shall be examined to ensure that porcelain bushings are not cracked, no shipping damage has occurred, internal and external connections are correct, switches and relays operate freely and are not tied or blocked, fuses, if required, are correct, and liquid level of liquid-filled regulators is correct. Relay panel covers shall be removed only for this examination; it is not necessary to open the main tank of liquid-filled regulators. The Contractor shall comply with the manufacturer's startup and operational instructions. Covers shall be replaced tightly after completing examinations and tests.
- b. Verify the supply voltage and nameplate input voltage are the same. Verify CCR is properly grounded. With the loads disconnected, regulator shall be energized and the open circuit protector observed to ensure that it de-energizes the regulator within 2 seconds.
- c. Comply with the manufacturer's recommendation for calibration and set up of each new CCR and each existing CCR connected to a new or modified airfield lighting circuit. The regulator output current and voltage shall be calibrated using a true RMS meter. The CCR calibration shall be complete prior to making any other CCR measurements.
- d. Verify all airfield lighting fixtures and loads are operational.
- e. Complete an Airfield Lighting Circuit Data Sheet for each new CCR and each existing CCR connected to a new or modified airfield lighting circuit.
- f. A sample Airfield Lighting Circuit Data Sheet is included at the end of this Item.
- g. Record notable current, voltage or power wave forms of the circuits tested to demonstrate and identify normal operation and any unusual circuit characteristics.

**h.** Comply with the manufacturer's recommendation for calibration and set up of each new CCR and each existing CCR connected to a new or modified airfield lighting circuit. The regulator output current and voltage shall be calibrated using a true RMS meter. The CCR calibration shall be complete prior to opening the lighted pavement to aircraft and prior to performing photometric testing.

**i.** Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.

**j.** Perform required testing to complete information requested on the Airfield Lighting Circuit Data Sheet.

### **111-3.9 CONTROL SYSTEM TESTS:**

**a.** The equipment shall be demonstrated to operate in accordance with the requirements of the Contract Documents. Each applicable control device in the air traffic control tower (ATCT) airfield lighting control panel and other control locations shall be operated so that each control device position is engaged at least ten times. During this process, all lights and associated equipment shall be observed to determine that each control device/switch/indicator properly operates the corresponding component. Radio communication between the operator and the observers shall be provided by the Contractor.

**b.** The above tests shall be repeated for each individual circuit from the local control switches on the regulators. Each installed or revised lighting circuit shall be tested by operating the lamps throughout the range of applicable steps and shall be operated separately at step 3 or step 5 as appropriate for full intensity for not less than 4 hours. Visual examination shall be made at the beginning and at the end of this test to determine that the installed airfield light fixtures are illuminating at full intensity.

**c.** Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.

**d.** Systems tests shall confirm by demonstration that all airfield lighting circuits are in required operating condition in accordance with the Contract Documents.

### **111-3.11 FIXTURE WIRING SEQUENCE TEST:**

**a.** The fixture wiring sequence testing shall be performed at night with minimum interference to airport operations.

**b.** The Contract Documents require all centerline lighting fixtures be wired in numerical order. Touch down zone and edge lights shall be wired in numerical sequence without zig-zagging back and forth across the runway or taxiway.

**c.** To verify proper wiring sequence from fixture to fixture temporary intentional grounds shall be inserted into the 5 kV airfield lighting series circuits. The temporary intentional grounds shall be inserted in sequence, one location at a time, at each PC, PT or PI where the cable returns (loops in the same conduit) with the fixture conductor. The Engineer shall designate each point as circuiting is completed.

**d.** The Contractor is solely responsible for the safety of all personnel, materials and equipment within the construction area. In addition to the Contractor's requirements each temporary intentional ground location shall be barricaded. The barricade shall extend a minimum of twenty-five (25) feet in all directions around the temporary intentional ground. No one shall be allowed within the barricaded area during the test.

**e.** All wiring for the 5 kV airfield lighting series circuits shall be complete prior to the sequence testing. The test circuit shall consist of the #1 conductor connected between the constant current regulator (CCR) output terminal S1 and the lowest numbered airfield lighting fixture. The #3 conductor connected from fixture to fixture to the first test point noted in the Contract Documents. A temporary intentional ground shall be inserted at this point. The constant current regulator (CCR) output terminal S2 shall be connected to a temporary intentional ground. Each temporary intentional ground shall be barricaded. All personnel shall be outside the barricade. The CCR shall be energized remotely. A visual inspection of the airfield lighting fixtures shall be made to determine proper wiring sequence.

**f.** The temporary intentional ground shall be moved to the next test point and this procedure shall be repeated for each temporary intentional ground.

**g.** The final temporary intentional ground for each circuit shall test proper routing of the #1 wire from the CCR output terminal S1 to the lowest numbered fixture. The temporary intentional grounds shall be placed at the output terminal S1 of the CCR and on the #1 wire side of the lowest numbered fixture. Each temporary intentional ground shall be barricaded. All personnel shall be outside the barricade. The CCR shall be energized remotely. A visual inspection of the airfield lighting fixtures shall be made to determine proper wiring sequence.

**h.** Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents and retest.

**i.** Once proper wiring sequence has been verified remove all intentional temporary grounds and place 5 kV airfield lighting series circuits in proper working order.

**j.** Systems tests shall confirm by demonstration that all airfield lighting circuits are in required operating condition in accordance with the Contract Documents.

#### **111-3.12 AIRFIELD LIGHTING FIXTURE PHOTOMETRIC TESTING:**

**a.** Photometric testing shall be performed at night with minimum interference to airport operations. Not more than 24 hours prior to starting the test, the Contractor shall clean all the light fixture lens to assure that the system is ready for the photometric testing.

**b.** The photometric report shall include test results, general quality of installation in terms of photometric performance, identification of deficient lights, analysis to identify the cause of any deficient lights (e.g., fixture alignment, light fixture or specific primary circuit segment), and recommendation for correcting any deficiencies found. The test data shall be presented in the form of a log book with the first readings entered in the log book, and with provisions for entering future readings for each light.

**c.** The photometric test and measurement equipment shall have an array of sensors capable of taking simultaneous readings along the horizontal axis of the light output. (Reference FAA

Advisory Circular (AC) 150/5345-46, Latest Edition). Photometric testing shall include the measurement at each light fixture of the light distribution along the horizontal axis. The measurements shall be compared with FAA standards presented in the AC for each type of light fixture.

**d.** The highest sensor reading, together with the two adjacent readings, will be averaged, and the calculated average shall be at least 20% greater than the minimum average intensity specified in the AC or in the project specifications, whichever is greater, in order for that fixture to be considered acceptable.

**e.** In addition, all other readings within the specified ellipse shall be at least 50 percent of the specified minimum average intensity in order for the fixture to be considered acceptable. Results shall also be compared with the requirements set forth in FAA AC 150/5345-46.

**f.** The photometric equipment system software shall be capable of recording the data and analyzing that data to calculate:

- 1.** The average photometric output of the main beam of the fixture,
- 2.** The location of the maximum reading,
- 3.** The location of the minimum reading,
- 4.** The ratio of the maximum reading to the average output,
- 5.** The ratio of the minimum reading to the average output, and
- 6.** Comparisons of these values with FAA specified values.

**g.** In every group of fixtures, that is, touchdown zone, centerline, or other such group, ten percent (10%) of the fixtures shall be evaluated at three (3) different vertical angles, on the centerline of the light beam, and at two (2) degrees above and below the centerline of the beam.

**h.** All sensor readings for the light fixture being evaluated shall be displayed simultaneously for operator and Engineer review and evaluation. All sensor readings shall be recorded automatically through the computer and shall be printed out via a computer controlled printer. Hand written data will not be accepted.

**i.** For minimum impact on airport operations, the taking of photometric data at any one fixture shall be completed within thirty seconds.

**j.** Interim reports will be submitted periodically during the progress of the work so that corrective measures may be taken as required by test results.

**k.** Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.

**l.** Retesting shall be performed by the testing agency to verify that the deficient condition has been corrected.

**m.** The documented test results of each fixture's photometric condition shall be graded as:

- 1.** Passes/Meets FAA Requirements - This classification includes those new light fixtures which exceed the FAA requirement, or, in the case of existing lighting systems, those fixtures which exceed 70 percent of FAA requirements. (FAA

specifies that airfield lights must be replaced when the outputs are less than 70 percent of the required output.) In such cases, no further action is necessary other than periodic monitoring by the Owner. When nothing is indicated in the "Remarks" column of the photometric data, the light fixture is satisfactory.

2. Fail/Investigate - These light fixtures have not met the FAA required photometric output for their particular type of light fixture. These fixtures shall be investigated by the Contractor to determine the cause of unacceptable performance. Appropriate corrective measures shall be taken to bring the performance of the fixtures to FAA standards, and the fixtures shall then be retested to assure that the repairs/replacements are satisfactory.

- n. Photometric test data shall be tabulated with the following minimum information:

|                   |                                                                                      |
|-------------------|--------------------------------------------------------------------------------------|
| Fixture Number    | From the Identification sheets of the Contract Documents                             |
| Light Direction   | Direction of light beam                                                              |
| Max CD            | Maximum candela output in a point along the main beam                                |
| Avg CD            | Average candela output of the main beam                                              |
| Min Level (70%)   | 70 Percent of FAA specified output for type of fixture                               |
| Lens Color        | Color of lens on fixture being tested                                                |
| Con for 70%       | Condition compared with 70 percent of FAA standard (blank if fixture is acceptable). |
| Remarks           | Notes pertaining to the fixture being tested.                                        |
| Corrective Action | Notes action taken to correct deficiency.                                            |

### **111-3.13 AIRFIELD SIGN PHOTOMETRIC TESTING:**

a. All signs shall be tested in accordance with specification 150/5345-44. The photometric testing of the signs shall evaluate the performance of signs with respect to the following FAA criteria:

1. Luminance: The background of Type L-858Y signs and the legends of Type L-858R and L-858L signs shall have an average luminance of 10 to 30 ft lamberts.
2. Uniformity: Photometric measurements shall be made on a 3 inch (76 mm) grid over the entire face of the sign, with no measurement being closer than 3 inches (76 mm) to the sign frame. Adjacent measurements shall not exceed a 1.5:1 ratio.
3. The sign shall be discernable from eight hundred (800') away.

b. All new airfield guidance panels or signs installed on this project shall be tested at night. Style 2 and Style 3 signs shall be tested at each input current throughout the range on which the sign is to operate. Style 5 signs shall be tested at the highest rated input current (5.5 amperes). The photometric performance of the signs shall be evaluated using digital color images. These images shall be converted to gray-scale images that can be analyzed directly for photometric output. To provide calibration and control for the evaluation of the gray-scale image, direct photometric readings of the light output in foot-lamberts (FL) shall be taken at several locations on

the face of the sign using a calibrated photometer. Light emitted only from the sign shall be permitted to reach the photometer.

c. To document compliance with FAA requirements, the final report shall present the following information and data for each color for each sign tested:

1. Sign Designation & Date Sign was tested.
2. Digital & Gray-Scale Image of Sign.
3. Step - The step intensity of the regulator controlling the sign.
4. Arithmetic average of illumination levels (FL).
5. Maximum illumination level on sign face and/or message (FL).
6. Minimum illumination level sign face and/or message (FL).
7. Ratio of Maximum illumination to Minimum illumination.
8. Uniformity. The maximum ratio of the average illumination levels of adjacent 3-inch areas over the face of the sign.
9. Visual Discernability of the Sign including comments.
10. Statements whether or not sign meets FAA criteria.

d. Photometric measurements shall also be taken of the red background on mandatory signs (white message on red background) to evaluate the signs red to white contrast. (FAA does not have criteria for contrast).

e. Interim reports will be submitted periodically during the progress of the work so that corrective measures may be taken as required by test results.

f. The documented test results of each signs photometric condition shall be graded as Passes/Meets FAA Requirements or Fail/Investigate.

g. Investigate unsatisfactory results and make necessary corrections or replacements as required by the Contract Documents.

h. Retesting shall be performed by the testing agency to verify that the deficient condition has been corrected.

#### **111-3.14 SHORT CIRCUIT / COORDINATION / DEVICE EVALUATION / ARC FLASH ANALYSIS:**

a. A Short Circuit, Coordination, Device Evaluation and Arc Flash analysis shall consist of the tasks listed below:

1. This analysis will include: One Line Diagrams, Short Circuit Analysis, Coordination Analysis and Arc Flash Analysis and labels.
2. The selected firm's field service engineer will perform data gathering for analysis completion and device settings, perform device setting as recommended by the analysis and will provide Arc Flash labels with the Arc Flash Analysis. The components worst case value will be considered the available arc flash energy at that point in the system. The Arc Flash Labels will be similar to the example below. The Contractor shall install the new Arc Flash labels and remove all old Arc Flash labels being updated.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>WARNING</b>                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>#11 Arc Flash and Shock Hazard<br/>Appropriate PPE Required</b>                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>ARC FLASH HAZARD PROTECTION</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>SHOCK PROTECTION BOUNDARIES</b>                                                                                                                                                                                                                                                                                                                                                           |
| Incident Energy: <b>11 cal/cm<sup>2</sup></b><br>Arc Flash Boundary: <b>71 in</b><br>PPE: <ul style="list-style-type: none"><li>■ Cotton Underwear</li><li>■ FR Shirt and Pants (or Coverall)</li><li>■ Full Flash Suit and Hard Hat Liner</li><li>■ Hard Hat / Hood / Face Shield</li><li>■ Safety Glasses or Goggles</li><li>■ Hearing Protection</li><li>■ Arc-Rated Gloves and Leather Shoes</li><li>□ Non-Melting Shirt and Pants</li></ul> | Shock Hazard when cover is removed: <b>480 VAC</b><br>Limited Approach: <b>42 in</b><br>Restricted Approach: <b>12 in</b>                                                                                                                                                                                                                                                                    |
| <b>Bus: PANEL MP Prot: 1000A GENERATOR<br/>BREAKER</b>                                                                                                                                                                                                                                                                                                                                                                                           | <small>Warning: Changes in equipment settings or system configuration will invalidate the calculated values and PPE requirements. Arc Flash Hazard Analysis shall be reviewed periodically, not to exceed 5 years. See NFPA 70E 130.5. Contact AVCON INC. 407-599-1122.</small><br><br><b>APRIL 2016</b> |

#### ARC FLASH HAZARD LABEL

3. This work will NOT include analysis of any future equipment, or repair or replacement of any electrical equipment found to be damaged or broken.
  4. Seven copies of the written analysis and drawings shall be provided at the conclusion of the project. Two copies shall be assigned to the Owner, one copy assigned to the Engineer and one copy shall be inserted in each O&M Manual
- b. Two possible vendors/firms to perform the analysis are:
1. Eaton Electrical Services and Systems, Wayne Thompson, 863-287-3985, [waynethompson@eaton.com](mailto:waynethompson@eaton.com)
  2. Easy Power, ESA Incorporated, (503) 655-5059 (ask for Engineering Services) [services@EasyPower.com](mailto:services@EasyPower.com) or <http://www.easypower.com/services/engineering.php>

Additional firms may be determined as an accepted equivalent to the firms listed above.

#### PART 4 – METHOD OF MEASUREMENT

##### 111-4.1 METHOD OF MEASUREMENT:

a. With the exception of the Short Circuit/Coordination/Device Evaluation/Arc Flash Analysis, the items described in this section are incidental to other items and shall not be measured for payment.

b. A pay item has been established to provide for payment for the Short Circuit/Coordination/Device Evaluation/Arc Flash Analysis described above in Section 3.14. Each Short Circuit/Coordination/Device Evaluation/Arc Flash Analysis shall be measured by the completed unit installed, in place, completed, and accepted.

## **PART 5 – BASIS OF PAYMENT**

### **111-5.1 BASIS OF PAYMENT:**

a. No direct payment shall be made for the work described in this item, except for the Short Circuit/Coordination/Device Evaluation/Arc Flash Analysis, Photometric Testing and FAA Flight Check. The work described in this item is incidental to other pay items and shall be paid for in the respective bid item of which it is a component part.

b. Payment shall be made at the contract unit price for Short Circuit/Coordination/Device Evaluation/Arc Flash Analysis. This price shall be full compensation for furnishing all materials and for all preparation, assembly, and supervision, engineering, and for all labor, equipment, tools, and incidentals necessary, including but not limited to, site visits and investigation, where required, to complete this item in conformance with the Contract Documents.

c. Payment shall be made at the contract unit price for Photometric testing. This price shall be full compensation for furnishing all materials and for all preparation, assembly, and supervision, engineering, and for all labor, equipment, tools, and incidentals necessary, including but not limited to, site visits and investigation, where required, to complete this item in conformance with the Contract Documents.

d. Payment shall be made at the contract unit price for FAA Flight Check of Runway 8-26 Lighting. This price shall be full compensation for furnishing all materials and for all preparation, assembly, and supervision, engineering, and for all labor, equipment, tools, and incidentals necessary, including but not limited to, site visits and investigation, where required, to complete this item in conformance with the Contract Documents.

Payment will be made under:

Item SP-111-1 Short Circuit/Coordination/Device Evaluation/Arc Flash Analysis -- per Each (EA)

Item SP-111-2 Photometric Testing of Runway 8-26 Lighting and Signage -- Allowance (AL)

Item SP-111-3 FAA Flight Check of Runway 8-26 Lighting -- Allowance (AL)

**END OF ITEM SP-111**

## AIRFIELD LIGHTING CIRCUIT DATA SHEET

|                           |                   |                |      |                      |              |       |  |
|---------------------------|-------------------|----------------|------|----------------------|--------------|-------|--|
| Circuit Designation:      |                   |                |      | Date:                |              |       |  |
| Circuit Description:      |                   |                |      | Ph. Wire Size/Type:  |              |       |  |
| Source Panel/Ckt          |                   |                |      | GND. Wire Size/Type: |              |       |  |
| Panel Ø; CB Size:         |                   |                |      | Conduit Size/Type:   |              |       |  |
| CCR Size:                 |                   |                |      | CCR Output Amps:     |              |       |  |
| CCR Manufacturer:         |                   |                |      | Yr. Manufactured:    |              |       |  |
| CCR Type No.:             |                   |                |      | Part No.:            |              |       |  |
| Output Amp Steps:         | /                 | /              | /    | /                    |              |       |  |
| Input Volts:              | Ph.:              |                | Hz.: |                      | Input Amps:  |       |  |
| Control Voltage:          |                   |                |      | Serial Number:       |              |       |  |
| Oil Capacity:             | Gal.              | Oil Test Date: |      |                      | Weight:      | lbs.  |  |
| CCR Insp./Calib.:         | Date:             |                |      |                      | Circuit Age: | Years |  |
| Insul. Resist. at 1 min.: | $\Omega$ @ 1000 V |                |      |                      | Weather:     |       |  |
| Lamps in Ckt.:            | Lamp Watts:       |                |      |                      | Lamps Out:   |       |  |

Manufacturer's Set up and Calibration Completed by: \_\_\_\_\_

|         | Calibration | LOAD TEST   |                |            |           |            |             |           |          |
|---------|-------------|-------------|----------------|------------|-----------|------------|-------------|-----------|----------|
| Step    | Output Amps | Output Amps | Output Voltage | Output KVA | Output PF | Input Amps | Input Volts | Input KVA | Input PF |
| B1/B10  |             |             |                |            |           |            |             |           |          |
| B2/B30  |             |             |                |            |           |            |             |           |          |
| B3/B100 |             |             |                |            |           |            |             |           |          |
| B4      |             |             |                |            |           |            |             |           |          |
| B5      |             |             |                |            |           |            |             |           |          |

Comments:

Earth Resistance Test

Instrument Mfg. \_\_\_\_\_

Model \_\_\_\_\_

Serial Number \_\_\_\_\_

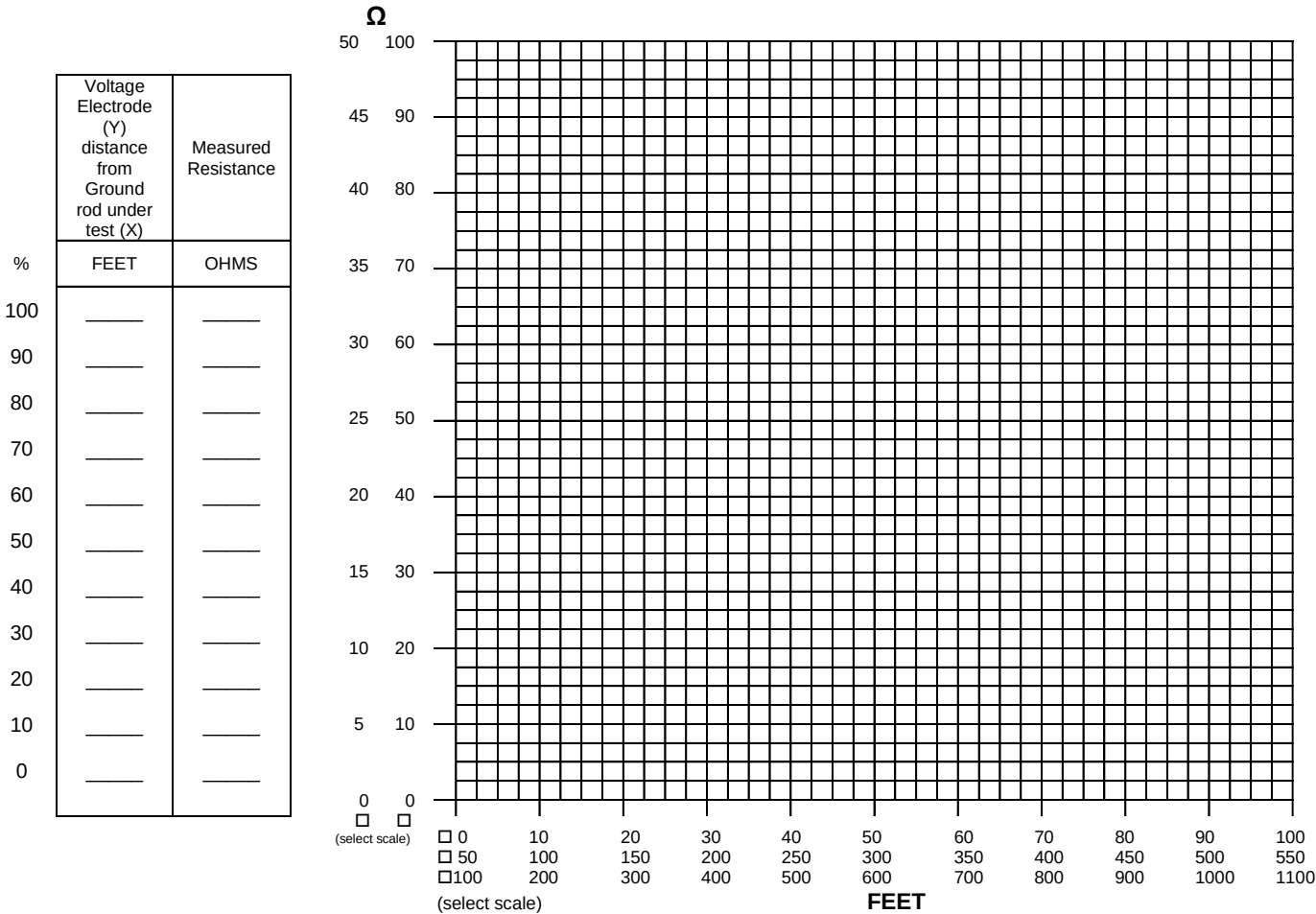
Test Date \_\_\_\_\_

Operator Name \_\_\_\_\_

Test Location \_\_\_\_\_

| Test Conditions |                                        |                                    |
|-----------------|----------------------------------------|------------------------------------|
| Soil Condition: | <input type="checkbox"/> Moist         | <input type="checkbox"/> Dry       |
|                 | Temperature _____ °F °C                |                                    |
| Soil Type:      | <input type="checkbox"/> Clay          | <input type="checkbox"/> Limestone |
| (choose one)    | <input type="checkbox"/> Granite       | <input type="checkbox"/> Shale     |
|                 | <input type="checkbox"/> Loam          | <input type="checkbox"/> Slate     |
|                 | <input type="checkbox"/> Sand & Gravel | <input type="checkbox"/> Sandstone |
|                 | <input type="checkbox"/> Other         |                                    |

| Weather Conditions                  |
|-------------------------------------|
| Relative Humidity: _____            |
| Precipitation Accumulation: _____   |
| Sky Conditions: _____               |
| Grounding System                    |
| <input type="checkbox"/> Single Rod |
| Rod Depth _____ ft                  |



Comments: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Airfield Lighting Circuit Continuity and Insulation Resistance Testing**

Date: \_\_\_\_\_

Start Time: \_\_\_\_\_ / Stop Time \_\_\_\_\_

|                                              |                    |                                                |                |                           |  |
|----------------------------------------------|--------------------|------------------------------------------------|----------------|---------------------------|--|
| Circuit Designation: _____                   |                    | Circuit Description _____                      |                | Circuit Length _____      |  |
| Circuit Operating Voltage _____              |                    | Circuit Isolation Transformers Qty/Size: _____ |                | _____                     |  |
| Hazard Risk Evaluation Performed?            |                    | Ambient Temperature: _____                     |                | Relative Humidity: _____  |  |
| Yes      No                                  |                    | Sky Conditions.: _____                         |                | Rain last 48hrs: _____    |  |
| Test Equipment: _____                        | Manufacturer _____ | Model # _____                                  | Serial# _____  | Upper Range Limit: _____  |  |
| Associated Equipment included in test: _____ |                    |                                                |                |                           |  |
| Circuit Continuity: _____                    |                    | Test Voltage: _____                            |                | Cable Manufacturer: _____ |  |
| Tested by: _____                             |                    |                                                | Witness: _____ |                           |  |

| Minutes    | 0.25     | 0.5 | 1.0 | 2.0 | 3.0 | 4.0 | 5.0 | 6.0 | 7.0 | 8.0 | 9.0 | 10.0 |
|------------|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| Reading    |          |     |     |     |     |     |     |     |     |     |     |      |
| Correction |          |     |     |     |     |     |     |     |     |     |     |      |
| Megohms    | Infinity |     |     |     |     |     |     |     |     |     |     |      |
|            | 10,000   |     |     |     |     |     |     |     |     |     |     |      |
|            | 5,000    |     |     |     |     |     |     |     |     |     |     |      |
|            | 3,000    |     |     |     |     |     |     |     |     |     |     |      |
|            | 2,000    |     |     |     |     |     |     |     |     |     |     |      |
|            | 1,000    |     |     |     |     |     |     |     |     |     |     |      |
|            | 800      |     |     |     |     |     |     |     |     |     |     |      |
|            | 600      |     |     |     |     |     |     |     |     |     |     |      |
|            | 400      |     |     |     |     |     |     |     |     |     |     |      |
|            | 300      |     |     |     |     |     |     |     |     |     |     |      |
|            | 200      |     |     |     |     |     |     |     |     |     |     |      |
|            | 150      |     |     |     |     |     |     |     |     |     |     |      |
|            | 100      |     |     |     |     |     |     |     |     |     |     |      |
|            | 80       |     |     |     |     |     |     |     |     |     |     |      |
|            | 60       |     |     |     |     |     |     |     |     |     |     |      |
|            | 40       |     |     |     |     |     |     |     |     |     |     |      |
|            | 30       |     |     |     |     |     |     |     |     |     |     |      |
|            | 20       |     |     |     |     |     |     |     |     |     |     |      |
|            | 15       |     |     |     |     |     |     |     |     |     |     |      |
|            | 10       |     |     |     |     |     |     |     |     |     |     |      |
|            | 6        |     |     |     |     |     |     |     |     |     |     |      |
|            | 4        |     |     |     |     |     |     |     |     |     |     |      |
|            | 2        |     |     |     |     |     |     |     |     |     |     |      |
|            | 1        |     |     |     |     |     |     |     |     |     |     |      |
|            | 0.6      |     |     |     |     |     |     |     |     |     |     |      |
|            | 0.2      |     |     |     |     |     |     |     |     |     |     |      |
|            | 0.1      |     |     |     |     |     |     |     |     |     |     |      |
|            | 0.06     |     |     |     |     |     |     |     |     |     |     |      |
|            | 0.02     |     |     |     |     |     |     |     |     |     |     |      |
|            | Zero     |     |     |     |     |     |     |     |     |     |     |      |

Comments: \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

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